Intersection Theory

— *EYNTKA* Series —

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Introduction

A reader who has worked through *Algebraic Geometry* [2] can already intersect. On a smooth projective surface two curves meet in a well-defined number of points, counted with multiplicity; that number depends only on the divisor classes, it is symmetric, and it computes things the naive count cannot, such as the self-intersection $E^2 = -1$ of an exceptional curve, which no set-theoretic count of points could produce. The construction was built by hand, and it was built for divisors on a surface. It says nothing about two surfaces inside a fourfold, nothing about a curve and a surface inside a threefold, and nothing about what happens when the two subvarieties share a component instead of meeting transversally.

Those are not exotic questions. They are the questions the subject was born from. How many lines meet four general lines in $\mathbb{P}^3$? How many conics are tangent to five general conics? Nineteenth century enumerative geometry answered such questions in quantity, by a principle of conservation of number: deform the configuration until the answer becomes visible, and trust that the count did not change along the way. The answers were, for the most part, right. The method was not a proof. Schubert's calculus of 1879 systematized it into a symbolic algebra of conditions so effective, and so unjustified, that Hilbert made its rigorous foundation the fifteenth of his 1900 problems.

The gap is precisely the one the surface theory papers over. Conservation of number needs an equivalence relation on cycles under which the count is invariant, and it needs the count to be defined even when the two cycles fail to meet properly. Rational equivalence supplies the first: two cycles are equivalent when one deforms to the other along a family parametrized by $\mathbb{P}^1$, and the quotient of the cycles by that relation is the Chow group. Deformation to the normal cone, introduced by Fulton and MacPherson in the 1970s, supplies the second: rather than moving the cycles until they meet properly, degenerate the ambient embedding to its normal cone, where the intersection can be read off the zero section of a bundle. That construction replaced the moving lemma, which had been the classical route and which is genuinely hard to prove in the generality one wants.

What comes out is more than a rigorous count. The Chow group is a functor, covariant for proper maps and contravariant for flat ones, and on a smooth variety it carries a ring structure, so questions about subvarieties become questions in a graded ring. Vector bundles acquire Chern classes valued in that ring, computed from Segre classes and the splitting principle. And Grothendieck's Riemann-Roch theorem relates the Chern character of a pushforward to the pushforward of a Chern character, subsuming Riemann-Roch for curves, which this reader has already seen [2, Riemann-Roch for curves], into a statement about arbitrary proper morphisms.

Goal

The book opens with cycles and rational equivalence, and establishes the Chow group $A_*(X)$ together with the two functorialities that make it usable: proper pushforward and flat pullback, with the localization sequence and homotopy invariance that compute it in the first examples ($\mathbb{A}^n$, $\mathbb{P}^n$, curves). Divisors come next, in the form the rest of the book needs them: not as the divisor class group of a normal variety, but as an operator $c_1(L) \cap -$ on cycle classes, commutative and compatible with both functorialities. That operator is the first Chern class, and it is the germ of everything that follows.

Segre classes of cones generalize it. They give the Chern classes of a vector bundle, the Whitney sum formula, the splitting principle, and with them the projective bundle formula, which computes the Chow groups of $\mathbb{P}(E)$ and closes the one debt the opening chapter leaves open. Deformation to the normal cone then delivers the Gysin map for a regular embedding, and the intersection product follows by reduction to the diagonal: on a smooth variety X, the product of two classes is the Gysin pullback along $X \hookrightarrow X \times X$. This is the point at which $A^*(X)$ becomes a ring, Bézout's theorem becomes a corollary, and the surface pairing built by hand upstream is recovered as a special case rather than an analogy.

The last movement turns outward. Comparing rational equivalence with algebraic, homological and numerical equivalence puts the Chow ring next to cohomology through the cycle class map, which is what makes algebraic cycles a Hodge-theoretic object and what the standard conjectures are about; correspondences, cycles on $X \times Y$ acting as operators, are the entry point to motives. Riemann-Roch is proved in Grothendieck's form and specialized back down to Hirzebruch-Riemann-Roch and Noether's formula. The book closes where it began, on enumerative geometry: the Chow ring of the Grassmannian, Schubert calculus done properly, degeneracy loci, and the classical counting problems that motivated the whole apparatus, now with proofs.

Sources

The treatment follows Fulton's *Intersection Theory* [8] as its main source, with Vakil's intersection-theory notes [11] for exposition, adapted to a reader who already has scheme theory and sheaf cohomology in full [2]. The standing conventions are collected at the start of Part I, where they first bind.

Part I

The Chow Group

Cycles are easy to write down and useless as they stand: the group of formal integer combinations of subvarieties is enormous, is not an invariant of anything, and admits no product. This part cuts it down to something that is. Rational equivalence identifies two cycles when one sweeps out to the other in a family over $\mathbb{P}^1$, and the quotient, the Chow group, turns out to carry exactly the structure the classical geometers were assuming when they deformed a configuration and trusted the count. Proper pushforward and flat pullback make it functorial, the localization sequence and homotopy invariance make it computable, and intersecting with a divisor gives the first operation that lowers dimension. That last operation, the first Chern class, is the seed of everything the rest of the book builds.

0.0.1 Convention: Standing Hypotheses Throughout the book a scheme is a *separated* scheme of finite type over a fixed algebraically closed field $\overline{k}$; such a scheme is called an *algebraic scheme*. A point is a closed point unless stated otherwise, a subvariety of a scheme is a closed integral subscheme, and a *variety* is a reduced and irreducible algebraic scheme, hence separated. Separatedness is folded in here rather than carried statement by statement because it is exactly the condition making the diagonal $\delta\colon X \to X \times X$ a closed immersion, and from chapter 5 onward every intersection product is built by pulling back along that diagonal. A non-separated example such as the line with a doubled origin is excluded by this convention, not by hypothesis.

0.0.2 Convention: $\mathbb{P}(E)$ Is the Bundle of Lines For a vector bundle E on X, this book writes $\mathbb{P}(E)$ for the bundle of one-dimensional *subspaces* of E: the fibre over x is the space of lines in E_x. This is Fulton's convention.
Algebraic Geometry [2] uses Grothendieck's opposite convention, under which $\mathbb{P}(\mathscr{E}) = \operatorname{Proj}(\operatorname{Sym}^\bullet \mathscr{E})$ parametrizes one-dimensional *quotients* of the fibres. The two are related by a dual: the bundle written $\mathbb{P}(E)$ here is the one written $\mathbb{P}(E^\vee)$ there, and a formula transported between the two must be dualized. The Segre classes of Chapter 1 onward are defined by pushing forward powers of $c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$, so reading such a formula in the wrong convention silently computes the classes of $E^\vee$ instead of E.

1

Rational Equivalence

Recall from [7] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x, y, z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x, y)], [g(x, y)] \in k[x, y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves f, g let F, G denote the resulting variety. The intersection of two (affine or projective) plane curves F, G at P was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}\mathbb{A}^2_k, p}{(f, g)} = \dim_K \mathcal{O}_{Z,P}$$

where $Z = \mathcal{Z}(f, g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$
2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) = i(P, F_2 \cdot G)$
3. (invariance of representation): if F' is given by $f + gh$ for some $h \in k[x, y]$:

   $$i(P, F' \cdot G) = i(P, F \cdot G)$$

4. If F, G have have a common component through P (i.e. the dimension of the irreducible variety in $F \cap G$ containing p is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.
5. If $P \notin F \cap G$:

   $$i(P, F \cdot G) = 0$$

6. if $f(x, y) = x - a$ and $g(x, y) = y - b$ (more generally $\frac{\partial(f,g)}{\partial(x,y)} \neq 0$ at p) then:

   $$i(P, F \cdot G) = 1$$

7. If F has no common component with G and H and P is a simple point of F:

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H)$$

1.1 Cycles and Rational Equivalence

The classical intersection number recalled above attaches an integer to a pair of plane curves meeting at a point. Turning that local count into a theory valid on any variety and in every dimension calls for a home for "formal combinations of subvarieties" together with an equivalence relation flexible enough to make intersection numbers well defined. That home is the Chow group, the higher-dimensional descendant of the divisor class group of [2, Divisor Class Group]: where the divisor class group records codimension-one subvarieties modulo the divisors of rational functions, the Chow group records subvarieties of *every* dimension modulo the same mechanism.

This section builds the three objects in turn, the group of cycles $Z_k(X)$, the subgroup $\mathrm{Rat}_k(X)$ of cycles rationally equivalent to zero, and the quotient $A_k(X)$. Throughout the chapter X is an algebraic scheme (a scheme of finite type over the algebraically closed field $\overline{k}$), a *subvariety* means a closed integral subscheme, and a point means a closed point unless stated otherwise.

The construction rests on the order of vanishing of a rational function along a codimension-one subvariety, introduced for divisors in [2, Divisors: Geometric Representation of Line Bundles]. On a variety W a prime divisor is a codimension-one subvariety V, and the local ring $\mathcal{O}_{W,V}$ is a one-dimensional local domain with fraction field the function field $k(W)$. When W is regular in codimension one this ring is a discrete valuation ring and its valuation ν_V records orders of vanishing ([2, Principal Divisor]); the length-theoretic definition below removes the regularity hypothesis, which matters as soon as W is an arbitrary subvariety of X.

Definition 1.1.1: Order Along a Subvariety

Let W be a variety and $V \subseteq W$ a codimension-one subvariety, and set $A = \mathcal{O}_{W,V}$, a one-dimensional local domain with fraction field $k(W)$. For nonzero $a \in A$ the quotient $A/(a)$ has finite length, and the *order of a along V* is

$$\mathrm{ord}_V(a) = \mathrm{length}_A \left(A/(a)\right)$$

Because $\mathrm{ord}_V(ab) = \mathrm{ord}_V(a) + \mathrm{ord}_V(b)$, this extends to a homomorphism $\mathrm{ord}_V : k(W)^* \to \mathbb{Z}$ by $\mathrm{ord}_V(a/b) = \mathrm{ord}_V(a) - \mathrm{ord}_V(b)$.

When $\mathcal{O}_{W,V}$ is a discrete valuation ring, in particular whenever W is normal, this length equals the valuation $\nu_V(a)$, so definition 1.1.1 agrees with the order function of [2, Divisors: Geometric Representation of Line Bundles]. For a fixed $r \in k(W)^*$ the integer $\mathrm{ord}_V(r)$ vanishes for all but finitely many V ([2, Almost All Valuations Vanish]), so each sum $\sum_V \mathrm{ord}_V(r)[V]$ below is finite.

Definition 1.1.2: Cycle

Let X be an algebraic scheme. A *k-cycle* on X is a finite formal $\mathbb{Z}$-linear combination $\sum_i n_i[V_i]$ of k-dimensional subvarieties $V_i \subseteq X$ with $n_i \in \mathbb{Z}$. The k-cycles form the free abelian group

$$Z_k(X) = \bigoplus_{\dim V = k} \mathbb{Z}\,[V]$$

on the set of k-dimensional subvarieties of X, and the group of all cycles is $Z_*(X) = \bigoplus_{k \geq 0} Z_k(X)$.

A cycle remembers only which subvarieties occur and with what integer multiplicity; it detects neither embedded nor nonreduced structure, a point made precise by the equality $Z_k(X) = Z_k(X_{\mathrm{red}})$. Two cases sit at the extremes. A 0-cycle is a finite formal sum of closed points. At the top, if X has dimension n with n-dimensional irreducible components $X_1, \ldots, X_r$, then $Z_n(X)$ is free on $[X_1], \ldots, [X_r]$.

The *fundamental cycle* of X ranges over *all* irreducible components $X_1, \ldots, X_t$, of whatever dimension:

$$[X] = \sum_{i=1}^{t} \mathrm{length}_{\mathcal{O}_{X,X_i}}\left(\mathcal{O}_{X,X_i}\right)[X_i] \;\in\; Z_*(X)$$

weighting each component by the multiplicity with which X carries it, and reducing to $\sum_i [X_i]$ when X is reduced. When X is equidimensional the sum is concentrated in degree n; in general it is not, and it must not be truncated to the top-dimensional components. A scheme such as the union of a plane and a line meeting it in a point carries a fundamental cycle with a 2-dimensional and a 1-dimensional term, and discarding the line would make the Segre classes of chapter 3 disagree with the standard ones on exactly such a base.

The one construction that produces relations among cycles is the divisor of a rational function. Given a $(k+1)$-dimensional subvariety $W \subseteq X$ and a nonzero $r \in k(W)^*$, its *divisor* is the k-cycle

$$[\mathrm{div}(r)] = \sum_V \mathrm{ord}_V(r)\,[V] \in Z_k(X)$$

the sum running over the codimension-one subvarieties V of W, each of which is a k-dimensional subvariety of X. The sum is finite by [2, Almost All Valuations Vanish], and $[\mathrm{div}(rs)] = [\mathrm{div}(r)] + [\mathrm{div}(s)]$ because ord_V is a homomorphism.

Definition 1.1.3: Rational Equivalence

A k-cycle $\alpha \in Z_k(X)$ is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are finitely many $(k+1)$-dimensional subvarieties $W_i \subseteq X$ and rational functions $r_i \in k(W_i)^*$ with

$$\alpha = \sum_i [\mathrm{div}(r_i)]$$

Two k-cycles are *rationally equivalent*, written $\alpha \sim \beta$, when $\alpha - \beta \sim 0$. The cycles rationally equivalent to zero form a subgroup $\mathrm{Rat}_k(X) \leq Z_k(X)$: it is closed under addition by definition and under negation because $[\mathrm{div}(r^{-1})] = -[\mathrm{div}(r)]$.

Geometrically, a subvariety W together with a rational function r presents $[\operatorname{div}(r)]$ as the difference between the zeros and the poles of r, that is, between the fibers over 0 and ∞ of the rational map $r\colon W \dashrightarrow \mathbb{P}^1$. Rational equivalence is thus the relation generated by sweeping one cycle to another through a family parametrized by the line. In the one codimension seen before, $k = n-1$ on an n-dimensional variety, a cycle is a Weil divisor and rational equivalence is exactly the linear equivalence of [2, Divisor Class Group]; the Chow group below carries $\operatorname{Cl} X$ to all dimensions at once.

Definition 1.1.4: Chow Group

The *k-th Chow group* of an algebraic scheme X is the quotient

$$A_k(X) = Z_k(X)/\mathrm{Rat}_k(X)$$

Its elements are *k-cycle classes*, and the class of a cycle α is written $[\alpha]$, or simply α when no confusion results. The *total Chow group* is

$$A_*(X) = \bigoplus_{k\geq 0} A_k(X) = Z_*(X)/\mathrm{Rat}_*(X), \qquad \mathrm{Rat}_*(X) = \bigoplus_{k\geq 0} \mathrm{Rat}_k(X)$$

Following the cohomological convention, much of the literature grades by codimension, writing $A^p(X) = A_{n-p}(X)$ or $CH^p(X)$ when X is equidimensional of dimension n; the two indexings carry the same information, and definition 1.1.4 is the promised general form of the construction previewed for surfaces in [2, Chow Rings and the General Intersection Product]. A cycle $\sum_i n_i[V_i]$ is *positive* (or *effective*) if it is nonzero with every $n_i \geq 0$, and a cycle class is positive if it has a positive representative. Positivity is the algebraic trace of an actual geometric subvariety, and deciding which classes are positive is among the central difficulties of the subject.

Passing to the quotient collapses exactly the ambiguity that a complete variety imposes on its points.

Example 1.1.5: Zero-Cycles on the Line

On $X = \mathbb{P}^1$ any two closed points are rationally equivalent. Write t for the affine coordinate and ∞ for the point at infinity. Distinct points $p, q \in \mathbb{A}^1 \subseteq \mathbb{P}^1$ are cut out by $t-p$ and $t-q$, and the rational function $r = (t-p)/(t-q)$ has

$$\operatorname{ord}_p(r) = 1, \qquad \operatorname{ord}_q(r) = -1, \qquad \operatorname{ord}_x(r) = 0 \text{ for } x \neq p, q$$

since r takes the finite nonzero value 1 at ∞. Hence $[\operatorname{div}(r)] = [p] - [q]$ and $[p] \sim [q]$; taking $r = t-p$ instead gives $[\operatorname{div}(t-p)] = [p] - [\infty]$, so $[\infty]$ lies in the same class. Every point of $\mathbb{P}^1$ therefore represents a single class h. A rational function on $\mathbb{P}^1$ has as many zeros as poles, so $\sum_V \operatorname{ord}_V(r) = 0$ for all r, and the total-coefficient map $\sum_i n_i[p_i] \mapsto \sum_i n_i$ descends to an isomorphism

$$A_0(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{Z}, \qquad h \mapsto 1$$

The affine line behaves oppositely. On $X = \mathbb{A}^1$ the function $t-p$ has the single zero p and no pole, so $[\operatorname{div}(t-p)] = [p]$ and every point satisfies $[p] \sim 0$; thus $A_0(\mathbb{A}^1) = 0$. On an affine line a point can be swept away to nothing, whereas on the complete line it cannot, and the surviving invariant of a class is its degree.

The contrast in example 1.1.5 is why completeness recurs throughout the theory: the degree of a zero-cycle is an invariant of its rational equivalence class only on a complete scheme, a point taken up in section 1.2. The remaining first properties of Chow groups are collected next; each is either immediate or follows from the localization sequence established later in the chapter.

Example 1.1.6: First Properties of Chow Groups
Let X be an algebraic scheme.

1. *Reduction.* Since cycles see only subvarieties, $A_k(X) = A_k(X_{\text{red}})$ for every k.
2. *Disjoint unions.* If $X = X' \sqcup X''$ is a disjoint union of open-and-closed subschemes, then $A_k(X) = A_k(X') \oplus A_k(X'')$, as every subvariety lies in one piece.
3. *Top-dimensional cycles.* If $\dim X = n$, there is no subvariety of dimension $n+1$, so $\mathrm{Rat}_n(X) = 0$ and $A_n(X) = Z_n(X)$ is free on the n-dimensional components of X. In particular the coefficient of such a component in a class $\alpha \in A_n(X)$ is well defined.
4. *Right-exactness.* For closed subschemes $X_1, X_2 \subseteq X$ there is an exact sequence
$$A_k(X_1 \cap X_2) \to A_k(X_1) \oplus A_k(X_2) \to A_k(X_1 \cup X_2) \to 0$$
with no left-exactness in general. This is a consequence of the localization sequence developed later in the chapter.

Exercise 1.1.1

1. Let X be the reduced scheme consisting of two distinct points. Compute $Z_k(X)$ and $A_k(X)$ for all k.
2. Show that $A_0(\mathbb{A}^n) = 0$ for every $n \geq 1$.
3. Prove from the definitions that $Z_k(X) = Z_k(X_{\text{red}})$ and $\mathrm{Rat}_k(X) = \mathrm{Rat}_k(X_{\text{red}})$, and deduce $A_k(X) = A_k(X_{\text{red}})$.
4. Let X have dimension n with n-dimensional irreducible components $X_1, \ldots, X_r$. Show that $A_n(X) = \bigoplus_{i=1}^r \mathbb{Z}\,[X_i]$.
5. Let C be a smooth projective curve over $\overline{k}$. Show that $A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$, and that the degree gives a short exact sequence $0 \to \mathrm{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$.

1.2 Proper Pushforward and the Degree

Cycles and their classes change under morphisms, and the first of these functorialities is covariant: a proper morphism pushes cycles forward, respects rational equivalence, and on a complete scheme produces the degree of a zero-cycle.

A proper morphism $f\colon X \to Y$ carries closed sets to closed sets, so the image of a subvariety $V \subseteq X$ is again a subvariety $W = f(V) \subseteq Y$: the image of an irreducible set is irreducible, and properness (universal closedness) makes it closed. Without properness the image can fail to be closed, as for the projection of [2, Non-Proper Morphisms], and pushforward is not defined. The inclusion of function fields $k(W) \hookrightarrow k(V)$ is a finite extension exactly when $\dim V = \dim W$, and its degree counts the points of V over a general point of W.

Definition 1.2.1: Proper Pushforward

Let $f\colon X \to Y$ be a proper morphism and $V \subseteq X$ a subvariety with image $W = f(V)$. Set

$$\deg(V/W) = \begin{cases} [k(V) : k(W)] & \dim V = \dim W \\ 0 & \dim V > \dim W \end{cases}$$

(the extension $k(W) \hookrightarrow k(V)$ is finite in the first case), and put $f_*[V] = \deg(V/W)\,[W]$. Extending $\mathbb{Z}$-linearly gives the *pushforward* $f_*\colon Z_k(X) \to Z_k(Y)$.

Pushforward is covariant: for composable proper morphisms $(g \circ f)_* = g_* \circ f_*$, since degrees of function-field extensions multiply in towers. When $\dim V = \dim W$ the number $\deg(V/W)$ counts the sheets of the generically finite cover $V \to W$, matching the topological degree over $\mathbb{C}$. That f_* descends to Chow groups is the content of this section, and rests on how it acts on the divisor of a rational function.

Proposition 1.2.2: Pushforward of Divisor

Let $f : X \to Y$ be a proper surjective morphisms between varieties, and let $r \in k(X)^*$. Then:

1. if $\dim(Y) < \dim(X)$, then
$$f_*[\operatorname{div}(r)] = 0$$
2. if $\dim(Y) = \dim(X)$, then
$$f_*[\operatorname{div}(r)] = [\operatorname{div}(N(r))]$$
where $N(r)$ is the *norm* of r (computed as the determinant of the $R(Y)$-linear endomorphism of $R(X)$ given by multiplying by r).

***Proof*:**
The order function and the norm are multiplicative, so both sides of each identity are homomorphisms in r. Both parts reduce to two model computations; write $n = \dim X$.

Step 1: the line over a point. Let $Y = \operatorname{Spec} k$ and $X = \mathbb{P}^1$, with f the structure map. Every closed point of $\mathbb{P}^1$ has residue field $k = \overline{k}$, so $f_*[p] = [Y]$ for each p and $f_*[\operatorname{div}(r)] = \big(\sum_V \operatorname{ord}_V(r)\big)[Y]$. Writing $r = c\prod_j (t - a_j)^{m_j}$ in the affine coordinate t, the order of r at a_j is m_j and its order at ∞ is $-\sum_j m_j$ (in $s = 1/t$ one has $r = s^{-\sum_j m_j} u$ with u a unit at $s = 0$), so $\sum_V \operatorname{ord}_V(r) = 0$ and $f_*[\operatorname{div}(r)] = 0$. The same count shows that on any proper curve C over a field K a principal divisor has degree zero: normalize C and choose a finite map $\widetilde{C} \to \mathbb{P}^1_K$, then apply Step 2 to that map and this computation to $\mathbb{P}^1_K \to \operatorname{Spec} K$.

Step 2: finite morphisms of equal dimension. Suppose f is finite with $\dim Y = \dim X$, so that $K = k(Y) \subseteq L = k(X)$ is a finite extension. Fix a codimension-one subvariety $W \subseteq Y$ and let $A = \mathcal{O}_{Y,W}$, a one-dimensional local domain with fraction field K. Its integral closure B in L is a finite A-module whose maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_s$ correspond to the codimension-one subvarieties $V_i \subseteq X$ over W, with $B_{\mathfrak{m}_i} = \mathcal{O}_{X,V_i}$ and residue degree $[k(V_i) : k(W)] = [B/\mathfrak{m}_i : A/\mathfrak{m}]$. Only these V_i contribute to the coefficient of $[W]$ in $f_*[\operatorname{div}(r)]$, which is therefore $\sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$, while the coefficient of $[W]$ in $[\operatorname{div}(N(r))]$ is $\operatorname{ord}_W(N(r))$. For $r \in B$ a nonzerodivisor these agree:

$$\operatorname{ord}_W(N(r)) = \operatorname{length}_A\bigl(A/N(r)A\bigr) = \operatorname{length}_A\bigl(B/rB\bigr) = \sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$$

the middle equality is the determinant-length formula, since $N(r)$ is the determinant of multiplication by r on the finite A-module B, and the last is additivity of length across the semilocal decomposition of B, weighted by residue degrees (see [8, Appendix A] for this length bookkeeping). As $L = \operatorname{Frac}(B)$ and both sides are homomorphisms, the identity holds for all $r \in L^*$, which is part (2) for finite f.

Step 3: the general case. A proper surjective f with $\dim Y = \dim X$ is generically finite, and the coefficient of a codimension-one $W \subseteq Y$ in $f_*[\operatorname{div}(r)]$ is computed over the generic point of W, where f becomes finite; Step 2 gives (2). For (1), let $\dim Y < n$. Since $f_*[\operatorname{div}(r)] \in Z_{n-1}(Y)$ vanishes when $\dim Y < n-1$, only $\dim Y = n-1$ remains, and there $f_*[\operatorname{div}(r)] = c\,[Y]$ with c its coefficient over the generic point of Y. Base-changing to $K = k(Y)$ replaces f by a proper curve $C = X_K$ over the field K and r by a function in $k(C)^*$, and $c = \sum_{P \in C} \operatorname{ord}_P(r)\,[k(P) : K]$ is the degree of its principal divisor, which vanishes by Step 1. Hence $f_*[\operatorname{div}(r)] = 0$, completing the proof.

Theorem 1.2.3: Pushforward Preserves Rational Equivalence

Let $f\colon X \to Y$ be proper. If $\alpha \in Z_k(X)$ is rationally equivalent to zero, then so is $f_*\alpha$. Consequently f_* descends to a homomorphism

$$f_*\colon A_k(X) \to A_k(Y)$$

and $A_*(-)$ equipped with proper pushforward is a covariant functor.

***Proof*:**
It suffices to push forward a generator $\alpha = [\operatorname{div}(r)]$, where $r \in k(W)^*$ for some $(k+1)$-dimensional subvariety $W \subseteq X$. Replacing X by W and Y by $f(W)$ makes f proper and surjective, and proposition 1.2.2 identifies $f_*[\operatorname{div}(r)]$ with 0 (when $\dim f(W) < \dim W$) or with $[\operatorname{div}(N(r))]$ (when $\dim f(W) = \dim W$). Either value is a principal divisor, hence rationally equivalent to zero, so $f_*\alpha \sim 0$. Functoriality on classes then follows from $(g \circ f)_* = g_* \circ f_*$ on cycles, as we sought to show.

Definition 1.2.4: Degree of a Zero-Cycle

Let X be complete (proper over $\operatorname{Spec} k$), with structure morphism $\pi\colon X \to \operatorname{Spec} k$, and let $\alpha = \sum_P n_P[P]$ be a zero-cycle on X. The *degree* of α is

$$\deg(\alpha) = \int_X \alpha = \sum_P n_P\,[k(P):k]$$

Since $\operatorname{Spec} k$ is a single point with no lower-dimensional subvarieties, $A_0(\operatorname{Spec} k) = Z_0(\operatorname{Spec} k) = \mathbb{Z}\,[\operatorname{Spec} k] \cong \mathbb{Z}$, and under this identification $\deg(\alpha) = \pi_*\alpha$. When the scheme is clear one writes $\int \alpha$ for $\int_X \alpha$.

By theorem 1.2.3 rationally equivalent zero-cycles have the same degree, so $\deg\colon A_0(X) \to \mathbb{Z}$ is well defined on a complete scheme. Extending by $\int_X \alpha = 0$ for $\alpha \in A_k(X)$ with $k > 0$ and using $\pi_X = \pi_Y \circ f$, any morphism $f\colon X \to Y$ of complete schemes satisfies

$$\int_X \alpha = \int_Y f_*\alpha$$

for every $\alpha \in A_*(X)$, a special case of the functoriality of pushforward.

1.2.5 Assuming Pushforward with subschemes Let $Y_1, \ldots, Y_r$ be closed subschemes of a scheme X. Let Y be a closed subscheme of X which contains all the Y_i. Given $\alpha_i \in A_*Y_i$, $i = 1, \ldots, r$, and $\beta \in A_*Y$, the notation "$\beta = \sum_{i=1}^r \alpha_i$ in A_*Y" stands in for the precise equation $\beta = \sum_{i=1}^r \varphi_{i*}(\alpha_i)$ where φ_i is the inclusion of Y_i in Y.

Example 1.2.6: Degrees, Bézout, and the Role of Properness

1. *Bézout's theorem.* Two curves $C, D \subseteq \mathbb{P}^2$ of degrees c, d with no common component meet in a zero-cycle $C \cdot D$ of degree cd. Pushing forward to a point, $\int_{\mathbb{P}^2}[C \cdot D] = cd$: the classical intersection number from the chapter opening is the degree of a zero-cycle, and its independence of the choice of C and D in their linear systems is rational equivalence together with theorem 1.2.3.

2. *Properness is necessary.* For the non-proper projection of [2, Non-Proper Morphisms] the image of a subvariety can fail to be closed, so f_* is not even defined; and on $\mathbb{A}^1$ every point is rationally equivalent to zero (example 1.1.5), so no degree survives. The degree is an invariant of a rational equivalence class only on a complete scheme.

3. *A finite cover of curves.* A finite morphism $\varphi\colon C \to C'$ of smooth projective curves satisfies $\varphi_*[P] = [k(P) : k(\varphi(P))]\,[\varphi(P)]$. For a point $Q \in C'$ away from the branch locus, $\varphi^{-1}(Q)$ consists of $\deg\varphi = [k(C) : k(C')]$ reduced points, so $\varphi_*\left[\varphi^{-1}(Q)\right] = (\deg\varphi)\,[Q]$; since $\deg(\varphi_*\alpha) = \deg(\alpha)$ on the complete curves, the sheet count is exactly the degree recorded by φ.

Exercise 1.2.1

1. For the finite morphism $\varphi\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (in characteristic $\neq 2$), compute $\varphi_*[p]$ for a closed point p and $\varphi_*[\varphi^{-1}(q)]$ for a closed point q. Verify that $\deg(\varphi_*\alpha) = \deg(\alpha)$ for every zero-cycle α, and explain where the degree $\deg\varphi = 2$ appears.
2. Let $\alpha = \sum_i n_i[P_i]$ be a zero-cycle on $\mathbb{P}^n$ over $\overline{k}$. Compute $\deg(\alpha)$ and prove it is invariant under rational equivalence.
3. Deduce from proposition 1.2.2 that a nonzero rational function on a smooth projective curve C has principal divisor of degree zero, and show by example that this fails on $\mathbb{A}^1$.
4. For the structure map $\pi\colon \mathbb{P}^1 \to \operatorname{Spec}\overline{k}$ and a rational function $r \in k(\mathbb{P}^1)^*$, verify $\int_{\mathbb{P}^1}[\operatorname{div}(r)] = \int_{\operatorname{Spec}\overline{k}} \pi_*[\operatorname{div}(r)]$, and say which hypothesis of definition 1.2.4 would fail were $\mathbb{P}^1$ replaced by $\mathbb{A}^1$.

1.3 Flat Pullback

Proper pushforward of section 1.2 is covariant: it moves cycles along a morphism in the direction of the arrow, lowering nothing and raising nothing in dimension. The second functoriality of the Chow group runs the other way. A morphism $f\colon X \to Y$ pulls a subvariety $V \subseteq Y$ back to its preimage $f^{-1}(V) \subseteq X$, and one would like to read that preimage as a cycle on X. For an arbitrary f this fails: the preimage can jump in dimension from fiber to fiber, can acquire embedded components, and carries no canonical multiplicities. The hypothesis that repairs every one of these defects at once is flatness. A flat morphism of relative dimension n has equidimensional fibers, so the preimage of a k-dimensional subvariety is pure of dimension $k+n$, and the scheme-theoretic preimage supplies the multiplicities with which its components must be counted. The resulting map $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ raises dimension by n, is contravariant, and descends to Chow groups.

This section constructs f^*, records its three structural properties (it is a graded homomorphism, it composes contravariantly, and it commutes with proper pushforward across a fiber square), and proves that it respects rational equivalence. The two guiding examples are the ones every later computation rests on: restriction to an open subscheme, and the projection $X \times \mathbb{A}^n \to X$ of a trivial affine bundle.

Recall that a morphism $f\colon X \to Y$ of algebraic schemes is flat if each local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. Flatness is the algebraic incarnation of the geometric require that the fibers of f vary without collapsing or jumping, and its precise dimensional consequence is what makes pullback of cycles possible.

Definition 1.3.1: Flat Morphism of Relative Dimension n

A morphism $f\colon X \to Y$ of algebraic schemes is *flat of relative dimension* n if f is flat and, for every subvariety $W \subseteq Y$ and every irreducible component Z of $f^{-1}(W)$,

$$\dim Z = \dim W + n$$

Equivalently, f is flat and every nonempty fiber of f has pure dimension n.

The equivalence stated in the definition is the mechanism the whole construction turns on, so it is worth isolating. Flatness forces the fiber dimension to be locally constant, and it forces the preimage of a subvariety to inherit that constant as an additive shift; neither holds without it.

Proposition 1.3.2: Flat Preimages Are Equidimensional

Let $f\colon X \to Y$ be a flat morphism of algebraic schemes with Y a variety, and suppose every nonempty fiber of f has pure dimension n. Then for every subvariety $W \subseteq Y$ each irreducible component of $f^{-1}(W)$ has dimension $\dim W + n$. In particular $f^{-1}(W)$ is either empty or pure of dimension $\dim W + n$.

***Proof*:**
Write $g\colon f^{-1}(W) \to W$ for the restriction of f; it is the base change of f along the closed immersion $W \hookrightarrow Y$, hence flat, and its fibers are fibers of f, so each nonempty fiber of g has pure dimension n. Thus it suffices to prove the statement for $W = Y$: that a flat morphism $g\colon X' \to W$ onto a variety W, with all nonempty fibers of pure dimension n, has $\dim Z = \dim W + n$ for every irreducible component Z of X'.

Let $Z \subseteq X'$ be an irreducible component, with generic point η. Flatness satisfies going-down (the going-down property of flat local homomorphisms, [9, p. III.9]), so the minimal prime η of X' contracts to the minimal prime of W; that is, $g(\eta)$ is the generic point of W, and g restricts to a dominant morphism $g\big|_Z\colon Z \to W$ of varieties. Both are integral of finite type over $\overline{k}$, so their dimensions are transcendence degrees of function fields, and the dimension of a fiber of a dominant morphism of varieties can be read at a closed point.
The fiber-dimension theorem for the dominant morphism $g\big|_Z\colon Z \to W$ of varieties ([9, Exercise II.3.22]) provides a dense open $U \subseteq Z$ such that for every closed point $z \in U$, the fiber $(g\big|_Z)^{-1}(g(z))$ has dimension exactly $\dim Z - \dim W$ at z. The complement in Z of the other irreducible components of X' is also a dense open, so their intersection is nonempty and contains a closed point z; write $w = g(z) \in W$. Since z lies off the other components, X' coincides with Z near z, so the fiber $X'_w = g^{-1}(w)$ coincides with $(g\big|_Z)^{-1}(w)$ near z, and its dimension at z is $\dim Z - \dim W$. By hypothesis X'_w is nonempty of pure dimension n, so this local fiber dimension equals n. Therefore

$$n = \dim Z - \dim W$$

hence $\dim Z = \dim W + n$, as we sought to show.

The proposition is the geometric license behind the definition of flat pullback. If $V \subseteq Y$ is a k-dimensional subvariety and f is flat of relative dimension n, then $f^{-1}(V)$ is a closed subscheme of X, pure of dimension $k+n$, and one takes its associated cycle, weighting each component by the length that measures the multiplicity with which the scheme carries it. This is exactly the fundamental cycle of definition 1.1.2 applied to the scheme-theoretic preimage.

Definition 1.3.3: Flat Pullback

Let $f\colon X \to Y$ be a flat morphism of relative dimension n and $V \subseteq Y$ a k-dimensional subvariety. The scheme-theoretic preimage $f^{-1}(V) = X \times_Y V$ is a closed subscheme of X, pure of dimension $k+n$, with irreducible components $Z_1, \dots, Z_r$. The *flat pullback* of $[V]$ is the $(k+n)$-cycle

$$f^*[V] = \left[f^{-1}(V)\right] = \sum_{i=1}^{r} \operatorname{length}_{\mathcal{O}_{f^{-1}(V),Z_i}}\left(\mathcal{O}_{f^{-1}(V),Z_i}\right)[Z_i]$$

the length being taken in the local ring of the preimage scheme at the generic point of Z_i. Extending $\mathbb{Z}$-linearly gives the flat pullback

$$f^*\colon Z_k(Y) \to Z_{k+n}(X)$$

The multiplicities are the point of the construction. The preimage as a topological set records only the components Z_i; the scheme structure records how many times f “covers” each one, and it is these lengths that make f^* compatible with divisors and hence with rational equivalence. When f is an open immersion the preimage is reduced and every length is 1, so $f^*[V] = [V \cap X]$; this is the source of the restriction map computed in the next example. The relative dimension n is the amount by which f^* raises the dimension of a cycle, so f^* is graded of degree $+n$.

Example 1.3.4: Open Restriction and the Projection $X \times \mathbb{A}^n \to X$

Two flat morphisms recur throughout the theory, and their pullbacks can be read off the definition.

1. *Open immersion.* Let $j\colon U \hookrightarrow Y$ be the inclusion of an open subscheme. Such j is flat of relative dimension 0: it is a localization at the level of local rings, and its fibers are single reduced points (over $u \in U$) or empty (over $y \notin U$). For a k-dimensional subvariety $V \subseteq Y$ the preimage $j^{-1}(V) = V \cap U$ is the open subscheme of V, which is reduced. If $V \cap U \neq \emptyset$ it is a k-dimensional subvariety of U and $j^*[V] = [V \cap U]$; if $V \subseteq Y \setminus U$ the preimage is empty and $j^*[V] = 0$. Thus flat pullback along an open immersion is the *restriction of cycles* $Z_k(Y) \to Z_k(U)$ that forgets the part of a cycle supported off U, the map at the heart of the localization sequence of section 1.4.

2. *Trivial affine bundle.* Let $p\colon X \times \mathbb{A}^n \to X$ be the projection. It is flat, being a base change of the flat morphism $\mathbb{A}^n_{\overline{k}} \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$, and every fiber is a copy of $\mathbb{A}^n$, pure of dimension n; so p is flat of relative dimension n. For a k-dimensional subvariety $V \subseteq X$ the preimage is $V \times \mathbb{A}^n$, which is integral of dimension $k+n$, so

$$p^*[V] = [V \times \mathbb{A}^n]$$

the cylinder over V. No multiplicity appears because $V \times \mathbb{A}^n$ is a variety. This is the computation that will make p^* an isomorphism on Chow groups (theorem 1.4.6).

Both examples exhibit the pattern the general theory generalizes: pullback restricts a cycle to an open part in the first case and thickens it uniformly along the fiber in the second. The structural properties below show that these behaviors are functorial.

The first property collects the ways in which f^* is well behaved as a map of cycle groups. It is a graded homomorphism raising dimension by the relative dimension, and it composes contravariantly, so that a tower of flat morphisms pulls back through the composite exactly as one expects.

Proposition 1.3.5: Basic Properties of Flat Pullback

Let $f\colon X \to Y$ be flat of relative dimension n.

1. *Homomorphism.* $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ is a homomorphism of abelian groups for each k, and assembles to a graded homomorphism $f^*\colon Z_*(Y) \to Z_*(X)$ of degree n.

2. *Fundamental cycle.* $f^*[Y] = [X]$ for any algebraic scheme Y: flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space.

3. *Composition.* If $g\colon Y \to Y'$ is flat of relative dimension m, then $g \circ f$ is flat of relative dimension $m + n$ and
$$(g \circ f)^* = f^* \circ g^*$$
as maps $Z_k(Y') \to Z_{k+m+n}(X)$.

4. *Identity.* $(\mathrm{id}_X)^* = \mathrm{id}$ on $Z_*(X)$.

***Proof*:**

1. *Homomorphism.* By construction f^* is defined on generators $[V]$ and extended $\mathbb{Z}$-linearly, so it is a homomorphism $Z_k(Y) \to Z_{k+n}(X)$ by fiat; proposition 1.3.2 is what guarantees the target group is $Z_{k+n}(X)$, since $f^{-1}(V)$ is pure of dimension $k + n$. Summing over k gives a degree-n graded homomorphism.

2. *Fundamental cycle.* Write $[Y] = \sum_i m_i[Y_i]$ over the irreducible components of Y, with $m_i = \mathrm{length}_{\mathcal{O}_{Y,Y_i}}(\mathcal{O}_{Y,Y_i})$. If Y is a variety this is the single generator $[Y]$ with multiplicity 1, and $f^{-1}(Y) = X$ gives $f^*[Y] = [X]$ at once.

   In general, $f^*[Y] = \sum_i m_i[f^{-1}(Y_i)]$ by definition of flat pullback, and since $Y = \bigcup_i Y_i$ we have $X = \bigcup_i f^{-1}(Y_i)$. The two sides are cycles, so it is enough to compare the coefficient of each subvariety of X, and the work is in matching up which subvarieties occur. Flatness enters through going-down: $\mathcal{O}_{Y,f(\eta)} \to \mathcal{O}_{X,\eta}$ is a flat local homomorphism for every $\eta \in X$, so every generization of $f(\eta)$ in Y lifts to a generization of η in X.

   Every component of X dominates a component of Y. Let X_j be a component of X, with generic point η_j, and put $y = f(\eta_j)$. A generization of y lifts to a generization of η_j; but η_j is a generic point of X, so its only generization in X is itself, and therefore y has no proper generization in Y. So y is the generic point of some component Y_i, and X_j is a component of $f^{-1}(Y_i)$. That i is unique: for $i' \neq i$ the generic point of Y_i does not lie in $Y_{i'}$, so $\eta_j \notin f^{-1}(Y_{i'})$ and only the i-th term of $\sum_i m_i[f^{-1}(Y_i)]$ contributes to the coefficient of $[X_j]$.

   Conversely, every component of every $f^{-1}(Y_i)$ is a component of X. Let W be one, with generic point η_W. Going-down applied at η_W lifts the generic point of Y_i to a generization of η_W inside $f^{-1}(Y_i)$, and η_W is maximal there, so W dominates Y_i, that is $f(\eta_W)$ is the generic point of Y_i. If W were not a component of X it would lie in some component $X_{j'}$, whose generic point generizes η_W and therefore maps to a generization of the generic point of Y_i; the latter being a generic point of Y, that forces $X_{j'}$ to dominate Y_i as well, so $\dim X_{j'} = \dim Y_i + n = \dim W$ by proposition 1.3.2, and $W = X_{j'}$. A component Y_i with $f^{-1}(Y_i) = \emptyset$ contributes to neither side.

Multiplicities. It remains to see that the multiplicity of X_j in $[X]$ equals m_i times its multiplicity in $[f^{-1}(Y_i)]$. This is a length computation: $\mathcal{O}_{Y,Y_i} \to \mathcal{O}_{X,X_j}$ is a flat local homomorphism of Noetherian local rings, and [4, Length Along a Flat Local Homomorphism] gives $\mathrm{length}(\mathcal{O}_{X,X_j}) = \mathrm{length}(\mathcal{O}_{Y,Y_i}) \cdot \mathrm{length}\left(\mathcal{O}_{X,X_j}/\mathfrak{m}_{Y_i}\mathcal{O}_{X,X_j}\right)$. Because Y_i is a subvariety and so reduced, $\mathcal{O}_{Y,Y_i}/\mathfrak{m}_{Y_i}$ is the function field of Y_i, whence $\mathcal{O}_{X,X_j}/\mathfrak{m}_{Y_i}\mathcal{O}_{X,X_j}$ is the local ring of $f^{-1}(Y_i)$ at X_j and the second factor is exactly the multiplicity of X_j in $[f^{-1}(Y_i)]$. Summing over i and j gives $f^*[Y] = [X]$, as we sought to show.

3. *Composition.* Composites of flat morphisms are flat, and relative dimensions add because fiber dimensions add along the tower: a fiber of $g \circ f$ over $y' \in Y'$ maps to the fiber $g^{-1}(y')$ of pure dimension m, with fibers of f of pure dimension n, so it has pure dimension $m+n$. Hence $g \circ f$ is flat of relative dimension $m+n$. For the cycle identity it suffices to check on a generator $[V]$ with $V \subseteq Y'$ a k-dimensional subvariety. Both sides are cycles supported on $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$, so the claim is that the multiplicities match component by component. The scheme-theoretic preimage is computed by iterated fiber product,

$$(g \circ f)^{-1}(V) = X \times_{Y'} V = X \times_Y (Y \times_{Y'} V) = f^{-1}\big(g^{-1}(V)\big)$$

so the underlying scheme of $(g \circ f)^{-1}(V)$ equals the underlying scheme of $f^{-1}(g^{-1}(V))$. It remains to see that the length multiplicities agree, that is, that applying f^* to the cycle $g^*[V] = \sum_i \ell_i[W_i]$ (where W_i are the components of $g^{-1}(V)$ with lengths ℓ_i) reproduces the multiplicities of $(g \circ f)^{-1}(V)$. Fix a component Z of the common preimage, with generic point z lying over the generic point w of some W_i. Flatness of f makes $\mathcal{O}_{X,z}$ flat over $\mathcal{O}_{Y,w}$, and length is additive and multiplies through a flat local homomorphism along the fiber: the length of $\mathcal{O}_{(g\circ f)^{-1}(V),z}$ equals ℓ_i times the length of $\mathcal{O}_{f^{-1}(W_i),z}$. Summing z over the components of $f^{-1}(W_i)$ and i over the components of $g^{-1}(V)$ gives $(g \circ f)^*[V] = f^*(g^*[V])$, as required.

4. *Identity.* The identity map is flat of relative dimension 0 with $(\mathrm{id})^{-1}(V) = V$ reduced, so $(\mathrm{id}_X)^*[V] = [V]$ on generators, completing the proof.

Contravariant functoriality (part 3) is what will let a Chow-group computation be assembled from simpler flat maps, exactly as covariant functoriality of pushforward let degrees be computed in towers. The fundamental-cycle identity (part 2) is the seed of every computation in section 1.5: it identifies the generator of the top Chow group of the total space of a flat family in terms of the base.

The remaining two properties are the interaction of flat pullback with the machinery already built. The first is its compatibility with proper pushforward. Pullback and pushforward move cycles in opposite directions, and they do not commute for a single morphism; but across a fiber square, where one has a flat map and a proper map meeting at right angles, the two operations do commute. This is the cycle-level form of the projection formula, and it is the technical engine behind essentially every later compatibility.

Proposition 1.3.6: Compatibility of Flat Pullback and Proper Pushforward

Consider a fiber square of algebraic schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

that is Cartesian, $X' = X \times_Y Y'$, with f proper and g flat of relative dimension n. Then f' is proper, g' is flat of relative dimension n, and

$$g^* \circ f_* = f'_* \circ g'^* : Z_k(X) \to Z_{k+n}(Y')$$

Consequently the identity holds on Chow groups.

***Proof*:**
Step 1: the maps in the base-changed square. Properness and flatness are stable under base change, so f' is proper and g' is flat; the fibers of g' over a point of X are fibers of g over its image in Y, so g' has the same pure relative dimension n as g. Both composites $g^* \circ f_*$ and $f'_* \circ g'^*$ therefore land in $Z_{k+n}(Y')$, and both are homomorphisms, so it suffices to check the identity on a generator $[V]$ with $V \subseteq X$ a k-dimensional subvariety.

Step 2: reduction to $X = V$. Replacing X by V, Y by $W = f(V)$, and forming the induced fiber squares, the four schemes become V, W, $V' = V \times_W W'$, $W' = W \times_Y Y'$, with $f : V \to W$ proper surjective and $g : W' \to W$ flat of relative dimension n. It suffices to prove

$$g^*(f_*[V]) = f'_*\big(g'^*[V]\big)$$

where now $f_*[V] = \deg(V/W)[W]$ by definition 1.2.1 and $g'^*[V] = [g'^{-1}(V)] = [V']$ (with its multiplicities).

Step 3: the two dimension regimes. There are two cases according to whether f drops dimension.

If $\dim V > \dim W$, then $\deg(V/W) = 0$, so the left side is $g^*(0) = 0$. On the right, $V' = V \times_W W'$ maps to W' under f' with fibers the fibers of $V \to W$, which are positive dimensional over the generic point; hence $\dim V' > \dim W'$ and every component of V' has image of strictly smaller dimension in W', so $f'_*[V'] = 0$ by the dimension clause of definition 1.2.1. Both sides vanish.

If $\dim V = \dim W$, write $d = \deg(V/W) = [k(V) : k(W)]$. The left side is $g^*(d\,[W]) = d\,[W']$, using $g^*[W] = [W'] = [g^{-1}(W)]$ (the base W is a variety and g is flat, so g^* of its fundamental cycle is the fundamental cycle of W' by proposition 1.3.5(2), which here is the total space $W' = g^{-1}(W)$). For the right side, $[V'] = g'^*[V]$ carries the pullback multiplicities flagged above, and $f'_*[V']$ must account for both those multiplicities and the residue-field degrees; flatness is what makes only the field degree survive. Fix a component W'_j of W', with generic point η_j. Localizing at η_j, the fiber of $f' : V' \to W'$ over η_j is Spec of $k(V) \otimes_{k(W)} k(W'_j)$: the generic fiber of $f : V \to W$ is $\operatorname{Spec} k(V)$, a $k(W)$-algebra of dimension d since V, W are varieties, and flat base change along $k(W) \hookrightarrow k(W'_j)$ preserves that dimension, so $\dim_{k(W'_j)}\big(k(V) \otimes_{k(W)} k(W'_j)\big) = d$. This dimension is exactly the total multiplicity that $f'_*[V']$ records over W'_j: summing the pullback multiplicity of each component of V' dominating W'_j against its residue degree over $k(W'_j)$ recovers the length d of the fiber algebra, because flatness of g (hence of g') forbids any length being created or lost in the base change. Thus $f'_*[V'] = \sum_j d\,[W'_j] = d\,[W']$, matching the left side. The two sides agree in both regimes, which establishes $g^* f_* = f'_* g'^*$ on cycles. Since each of g^*, f_*, f'_*, g'^* descends to Chow groups by theorem 1.2.3 and proposition 1.3.7 below, the identity descends as well, completing the proof.

The compatibility should be read as the statement that pushing a cycle forward and then restricting to a flat base change gives the same class as first restricting and then pushing forward within the base change. In the special case where g is an open immersion, it says that pushforward commutes with restriction to an open set, which is the reason the localization sequence of the next section is a sequence of A_*-functorial maps. In the case where f is the projection to a point, it recovers the invariance of the degree of a zero-cycle under a flat base change of the ground scheme.

Everything so far concerns cycles. To obtain a map of Chow groups, flat pullback must carry Rat_k into Rat_{k+n}. The mechanism is that flat pullback commutes with taking the divisor of a rational function, up to the multiplicities that flatness controls, so a cycle that is a sum of divisors pulls back to a sum of divisors.

Proposition 1.3.7: Flat Pullback Preserves Rational Equivalence

Let $f\colon X \to Y$ be flat of relative dimension n. If $\alpha \in Z_k(Y)$ is rationally equivalent to zero, then $f^*\alpha \in Z_{k+n}(X)$ is rationally equivalent to zero. Consequently f^* descends to a homomorphism

$$f^*\colon A_k(Y) \to A_{k+n}(X)$$

and $A_*(-)$ equipped with flat pullback is a contravariant functor on flat morphisms of algebraic schemes, raising dimension by the relative dimension.

***Proof*:**
It suffices to treat a generator of $\mathrm{Rat}_k(Y)$, namely $\alpha = [\mathrm{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq Y$ and a nonzero $r \in k(W)^*$. The claim is that $f^*[\mathrm{div}(r)]$ is a sum of divisors of rational functions on subvarieties of X.

Step 1: reduction to the total space over W. The cycle $[\mathrm{div}(r)]$ is supported on W, and $f^*[\mathrm{div}(r)]$ is supported on $f^{-1}(W)$. Let $W' = f^{-1}(W) = X \times_Y W$ with its scheme-theoretic (possibly non-reduced) structure, and let $f_W\colon W' \to W$ be the restriction of f, which is flat of relative dimension n by base change. Because the components of $f^{-1}(W)$ are the components of W' and the pullback multiplicities are computed inside W', it is enough to prove that $f_W^*[\mathrm{div}(r)] \sim 0$ in $Z_{k+n}(W')$; the class then remains rationally equivalent to zero in X under the proper (closed immersion) pushforward along $W' \hookrightarrow X$, which preserves rational equivalence by theorem 1.2.3. Thus one may assume $Y = W$ is a variety and $r \in k(Y)^*$.

Step 2: the key identity on components. With Y a variety and $r \in k(Y)^*$, the divisor $[\mathrm{div}(r)] = \sum_V \mathrm{ord}_V(r)[V]$ runs over the codimension-one subvarieties $V \subseteq Y$. The assertion to prove is the flat-pullback divisor formula

$$f^*[\mathrm{div}(r)] = \left[\mathrm{div}(f^\# r)\right] \quad \text{on each component of } X \tag{1.1}$$

interpreted componentwise as follows. Let $X_1, \ldots, X_s$ be the irreducible components of X, each of dimension $n + \dim Y$ by proposition 1.3.2, with generic points dominating Y (flatness sends generic points to the generic point of the base). On each X_j the pullback $f^\# r$ of r is a nonzero rational function, since $r \neq 0$ and X_j dominates Y. The claim is

$$f^*[\mathrm{div}(r)] = \sum_{j=1}^{s} m_j \,[\mathrm{div}(f^{\#}r\big|_{X_j})]$$

where $m_j = \mathrm{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$ is the multiplicity of the component X_j in the fundamental cycle $[X]$. Each term on the right is a divisor of a rational function on the subvariety $(X_j)_{\mathrm{red}}$, hence rationally equivalent to zero, so (1.1) gives $f^*[\mathrm{div}(r)] \sim 0$.

Step 3: verifying the identity. Fix a codimension-one subvariety $D \subseteq X_j$, a $(k+n)$-dimensional subvariety of X, and let $V = \overline{f(D)} \subseteq Y$. Two cases arise by the relative-dimension count of proposition 1.3.2 applied to $f\big|_{X_j}$.

If $\dim V = \dim Y$, then $V = Y$ and D dominates Y; the coefficient of $[D]$ in $[\mathrm{div}(f^{\#}r)]$ is $\mathrm{ord}_D(f^{\#}r)$, computed in the discrete-valuation-style local ring $\mathcal{O}_{X_j,D}$, while D does not appear in $f^*[\mathrm{div}(r)]$ because $[\mathrm{div}(r)]$ has no component equal to Y. But $\mathrm{ord}_D(f^{\#}r) = 0$. The point is that D dominates Y: its generic point ξ maps to the generic point of Y, at which r is a nonzero rational function, that is, a unit of $k(Y)$. Since $f^{\#}r$ is the pullback of r along the local homomorphism $\mathcal{O}_{Y,Y} = k(Y) \to \mathcal{O}_{X_j,D}$ induced on stalks at ξ, the image $f^{\#}r$ is a unit of $\mathcal{O}_{X_j,D}$, so $\mathrm{ord}_D(f^{\#}r) = 0$ and no such D contributes to either side. (A nonzero rational function need not be a unit along a codimension-one subvariety in general; here it is one precisely because D dominates the base and so sees r only at the generic point of Y.)

If $\dim V = \dim Y - 1$, then V is one of the codimension-one subvarieties appearing in $[\mathrm{div}(r)]$, and D is a component of $f^{-1}(V)$, lying on one or more of the components X_j. Two kinds of local ring enter the count, and keeping them distinct is the crux. Write

$$A = \mathcal{O}_{Y,V}, \qquad \widetilde{B} = \mathcal{O}_{X,D}$$

Here A is a one-dimensional local domain and $\mathrm{ord}_V(r) = \mathrm{length}_A(A/rA)$; the ring $\widetilde{B}$ is the local ring of the whole (possibly nonreduced) scheme X at the generic point of D, and it is flat over A because f is flat. The minimal primes of the one-dimensional ring $\widetilde{B}$ correspond to the components X_j passing through D; for such a component the quotient $B_j = \widetilde{B}/\mathfrak{p}_j = \mathcal{O}_{X_j,D}$ is the discrete-valuation-style local ring of the reduced component X_j at D, in which $\mathrm{ord}_D(f^{\#}r\big|_{X_j}) = \mathrm{length}_{B_j}(B_j/(f^{\#}r)B_j)$ is computed. When X is irreducible along D there is a single such j, but in general several components may meet along D.
The coefficient of $[D]$ in $f^*[\mathrm{div}(r)]$ is $\mathrm{ord}_V(r) \cdot e_D$, where

$$e_D = \mathrm{length}_{\mathcal{O}_{f^{-1}(V),D}}\big(\mathcal{O}_{f^{-1}(V),D}\big) = \mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big)$$

is the multiplicity of D in the pullback cycle $f^*[V]$, the second equality holding because $\mathcal{O}_{f^{-1}(V),D} = \widetilde{B} \otimes_A A/\mathfrak{m}_A = \widetilde{B}/\mathfrak{m}_A\widetilde{B}$. The coefficient of $[D]$ in $\sum_j m_j[\mathrm{div}(f^{\#}r\big|_{X_j})]$ is $\sum_{j\,:\,D\subseteq X_j} m_j \cdot \mathrm{ord}_D(f^{\#}r\big|_{X_j})$, the sum running over the components X_j that pass through D. That the two coefficients agree is two multiplicative facts chained together. First, flatness of $\widetilde{B}$ over A makes length multiply through the base change: filtering A/rA by its $\mathrm{ord}_V(r)$ composition factors $A/\mathfrak{m}_A$ and applying the exact functor $- \otimes_A \widetilde{B}$ gives

$$\mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/r\widetilde{B}\big) = \mathrm{length}_A(A/rA) \cdot \mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big) = \mathrm{ord}_V(r) \cdot e_D$$

Second, order in the one-dimensional local ring $\widetilde{B}$ distributes over its minimal primes with the generic multiplicities as weights: each minimal prime $\mathfrak{p}_j$ has residue ring $B_j = \mathcal{O}_{X_j,D}$ and localization $\widetilde{B}_{\mathfrak{p}_j} = \mathcal{O}_{X,X_j}$ of length $m_j = \operatorname{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$, so the additivity of order over the minimal primes of a one-dimensional local ring ([8, App. A.3]) gives, for the nonzerodivisor $f^\# r$,

$$\operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/(f^\# r)\widetilde{B}\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{length}_{B_j}\big(B_j/(f^\# r)B_j\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{ord}_D(f^\# r\big|_{X_j})$$

Since $f^\# r$ is the image of r in $\widetilde{B}$, the left sides of the two displays coincide, whence $\operatorname{ord}_V(r)\cdot e_D = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{ord}_D(f^\# r\big|_{X_j})$: the two coefficients of $[D]$ match. When X is irreducible along D the sum has a single term and reduces to $e_D = m_j \cdot e_{D/V}$, where $e_{D/V} = \operatorname{length}_{B_j}(B_j/\mathfrak{m}_A B_j)$ is the ramification-type multiplicity of D over V computed in the reduced component ring B_j; the pullback multiplicity e_D over the full preimage scheme then exceeds it by exactly the factor m_j. Summing over the divisors D and, on the right, over the components X_j reproduces both sides of (1.1) term by term. Hence $f^*[\operatorname{div}(r)] = \sum_j m_j[\operatorname{div}(f^\# r\big|_{X_j})]$ is a sum of divisors of rational functions, so $f^*[\operatorname{div}(r)] \sim 0$.

Descent to Chow groups follows: $f^*(\operatorname{Rat}_k(Y)) \subseteq \operatorname{Rat}_{k+n}(X)$, so f^* induces $A_k(Y) \to A_{k+n}(X)$. Functoriality on classes is inherited from proposition 1.3.5(3),(4) on cycles, giving a contravariant functor as claimed, completing the proof.

The proposition completes the package: flat pullback is a well-defined operation on Chow groups, contravariant and dimension-raising, dual to the covariant dimension-preserving pushforward of section 1.2. The pair (f_*, f^*) makes A_* into a functor covariant for proper maps and contravariant for flat maps, with the compatibility of proposition 1.3.6 tying the two together across fiber squares. This is the structure the localization sequence and homotopy invariance of section 1.4 exploit, and it is why the affine-bundle projection $p\colon X \times \mathbb{A}^n \to X$, whose pullback was computed in example 1.3.4, will turn out to induce an isomorphism on Chow groups.

Example 1.3.8: Pullback to a Product and to an Open Set

The two flat morphisms of example 1.3.4 descend to Chow groups explicitly.

1. *Cylinder map.* For $p\colon X \times \mathbb{A}^1 \to X$ the induced map $p^*\colon A_k(X) \to A_{k+1}(X \times \mathbb{A}^1)$ sends the class of a subvariety V to the class of the cylinder $V \times \mathbb{A}^1$. On $X = \operatorname{Spec}\overline{k}$ this reads $A_0(\operatorname{Spec}\overline{k}) = \mathbb{Z} \to A_1(\mathbb{A}^1)$, sending the class of the point to $[\mathbb{A}^1]$, the fundamental class of the affine line, consistent with $p^*[Y] = [X]$ from proposition 1.3.5(2).

2. *Restriction to an open set.* For an open immersion $j\colon U \hookrightarrow Y$ the map $j^*\colon A_k(Y) \to A_k(U)$ restricts a class to U, forgetting components supported on the complement $Y \setminus U$. On $Y = \mathbb{P}^1$ with $U = \mathbb{A}^1$ this is the surjection $A_0(\mathbb{P}^1) = \mathbb{Z} \twoheadrightarrow A_0(\mathbb{A}^1) = 0$ from example 1.1.5: the class of a point on $\mathbb{P}^1$ restricts to a point of $\mathbb{A}^1$, which is rationally trivial. The kernel is generated by the class of the point at infinity, the part of $\mathbb{P}^1$ removed. This surjectivity with the removed piece generating the kernel is the prototype of the localization sequence.

Exercise 1.3.1

1. Let $f\colon X \to Y$ be flat of relative dimension 0 (for instance a finite flat morphism, or an open

immersion). Show directly from definition 1.3.3 that f^* maps $Z_k(Y) \to Z_k(X)$ for every k, preserving dimension, and that $f^*[V] = \sum_i e_i[Z_i]$ where Z_i are the components of $f^{-1}(V)$ and e_i their multiplicities in the fiber over the generic point of V.

2. Let $\pi\colon \mathbb{A}^{n+1} \setminus \{0\} \to \mathbb{P}^n$ be the tautological $\mathbb{G}_m$-bundle. It is flat of relative dimension 1. Compute $\pi^*[\mathbb{P}^k]$ for a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$, identifying its class as a cycle on $\mathbb{A}^{n+1} \setminus \{0\}$.

3. Give an example of a flat morphism f for which $f^*\colon A_*(Y) \to A_*(X)$ is not injective, and one for which it is not surjective. (Hint: the projection $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ and the open immersion $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$.)

4. Let $q\colon X \times \mathbb{A}^m \times \mathbb{A}^n \to X \times \mathbb{A}^m$ and $p\colon X \times \mathbb{A}^m \to X$ be the projections. Verify the composition law proposition 1.3.5(3) for $p \circ (\mathrm{id} \times \text{-})$ directly on the class of a subvariety $V \subseteq X$, confirming both routes give $[V \times \mathbb{A}^m \times \mathbb{A}^n]$.

5. Let $f\colon X \to Y$ be proper and $j\colon U \hookrightarrow Y$ an open immersion, with fiber square $X_U = f^{-1}(U) \to X$ over j. Using proposition 1.3.6, prove that $(f|_{X_U})_*(\alpha|_{X_U}) = (f_*\alpha)|_U$ in $A_*(U)$ for every $\alpha \in A_*(X)$, that is, proper pushforward commutes with restriction to an open subset.

6. Let $E \to X$ be a vector bundle of rank r over a variety X of dimension d, with projection π flat of relative dimension r. Compute $\pi^*[X]$ and identify the top Chow group $A_{d+r}(E)$, first showing that E is irreducible.

1.4 The Fundamental Sequences

The two functorialities already in hand, proper pushforward (section 1.2) and flat pullback (section 1.3), combine into the two structural results that make Chow groups computable. The first is a long-exact-sequence surrogate: cutting a scheme into a closed piece and its open complement relates the three Chow groups by a right-exact sequence, the algebraic analogue of the excision property of a homology theory. The second is homotopy invariance in its algebraic form: an affine bundle over X, and in particular the trivial bundle $X \times \mathbb{A}^n$, has the same Chow groups as X, shifted in degree by the fibre dimension. Neither result computes a single Chow group in isolation, but together they reduce the computation of $A_*(X)$ for the spaces of the next section to bookkeeping: stratify X by locally closed pieces on which the answer is known, and assemble the pieces through the sequence.

Both proofs rest on a single principle. A cycle on X is a formal combination of subvarieties; a subvariety either meets the open set U or is contained in the closed complement Z, and rational equivalence is generated by divisors of rational functions, which live on subvarieties and so obey the same dichotomy. The localization sequence is the exact translation of this observation into a statement about the three groups, and affine-bundle invariance is its first substantial payoff, extracted by Noetherian induction.

Throughout, $Z \subseteq X$ is a closed subscheme with open complement $U = X \setminus Z$, and

$$i\colon Z \hookrightarrow X, \qquad j\colon U \hookrightarrow X$$

denote the inclusions. The closed immersion i is proper, so it pushes cycles forward by definition 1.2.1: a subvariety $V \subseteq Z$ is also a subvariety of X, and $i_*[V] = [V]$ since $\deg(V/V) = 1$. The open immersion j is flat of relative dimension 0, so it pulls cycles back by definition 1.3.3: for a subvariety

$V \subseteq X$ the restriction is $j^*[V] = [V \cap U]$ when V meets U and $j^*[V] = 0$ when $V \subseteq Z$. These two operations are what the sequence is built from.

Theorem 1.4.1: Localization Sequence

Let X be an algebraic scheme, $Z \subseteq X$ a closed subscheme, and $U = X \setminus Z$ its open complement, with $i\colon Z \hookrightarrow X$ and $j\colon U \hookrightarrow X$ the inclusions. For every k the sequence

$$A_k(Z) \xrightarrow{i_*} A_k(X) \xrightarrow{j^*} A_k(U) \longrightarrow 0$$

is exact: j^* is surjective, and $\ker(j^*) = \operatorname{im}(i_*)$.

***Proof*:**
The map i_* is proper pushforward along the closed immersion (theorem 1.2.3) and j^* is flat pullback along the open immersion (proposition 1.3.7); both are defined on Chow groups, so the sequence is a sequence of homomorphisms. Exactness is checked at the two nonzero spots.

Step 1: surjectivity of j^*. A subvariety $V' \subseteq U$ is a k-dimensional closed integral subscheme of the open set U. Its closure $V = \overline{V'}$ in X is a k-dimensional subvariety of X with $V \cap U = V'$, since closure adds only points lying in the closed set Z and does not raise the dimension. Thus $j^*[V] = [V']$, and every generator of $Z_k(U)$ is hit by a cycle on X. Hence $j^*\colon Z_k(X) \to Z_k(U)$ is already surjective at the level of cycles, and a fortiori $j^*\colon A_k(X) \to A_k(U)$ is surjective.

Step 2: the inclusion $\operatorname{im}(i_*) \subseteq \ker(j^*)$. For a subvariety $V \subseteq Z$ the intersection $V \cap U$ is empty, so $j^* i_*[V] = j^*[V] = 0$. By linearity $j^* \circ i_* = 0$ on $Z_k(Z)$, hence on $A_k(Z)$, which gives $\operatorname{im}(i_*) \subseteq \ker(j^*)$.

Step 3: the inclusion $\ker(j^*) \subseteq \operatorname{im}(i_*)$. This is the substance of the theorem. Let $\alpha \in Z_k(X)$ be a cycle whose class in $A_k(X)$ lies in $\ker(j^*)$, so $j^*\alpha \sim 0$ on U; the goal is to modify α by a cycle rationally equivalent to zero on X until it is supported on Z, that is, until it lies in the image of $i_*\colon Z_k(Z) \to Z_k(X)$.

Split the cycle by whether its components meet U. Group the subvarieties occurring in α as those contained in Z and those meeting U, and write

$$\alpha = \alpha_Z + \beta, \qquad \alpha_Z \in Z_k(Z), \quad \beta = \sum_i n_i[V_i]$$

where each V_i meets U. Since $\alpha_Z = i_*\alpha_Z$ already lies in the image of i_*, it may be discarded, and it suffices to treat β. Applying j^* gives $j^*\alpha = j^*\beta = \sum_i n_i[V_i']$ with $V_i' = V_i \cap U$ the distinct k-dimensional subvarieties of U, and by hypothesis this cycle is rationally equivalent to zero on U:

$$\sum_i n_i[V_i'] = \sum_j [\operatorname{div}(r_j')] \quad \text{in } Z_k(U)$$

for finitely many $(k+1)$-dimensional subvarieties $W_j' \subseteq U$ and rational functions $r_j' \in k(W_j')^*$.

The idea is now to lift each W_j' and r_j' from U to X and compare divisors. Let $W_j = \overline{W_j'} \subseteq X$ be the closure, a $(k+1)$-dimensional subvariety of X with $W_j \cap U = W_j'$. Restriction of functions to the dense open $W_j' \subseteq W_j$ is an isomorphism of function fields $k(W_j) \xrightarrow{\sim} k(W_j')$, since a variety and a dense open subset have the same function field; let $r_j \in k(W_j)^*$ be the function corresponding to r_j'. Form the divisor $[\operatorname{div}(r_j)] \in Z_k(X)$. Its restriction to U is governed by the flat-pullback compatibility of the divisor construction: for a codimension-one subvariety $V \subseteq W_j$ meeting U, the local ring $\mathcal{O}_{W_j,V} = \mathcal{O}_{W'_j,V\cap U}$ is unchanged by passing to the open set, so $\mathrm{ord}_V(r_j) = \mathrm{ord}_{V\cap U}(r'_j)$, while a codimension-one subvariety of W_j contained in Z contributes a component of $[\mathrm{div}(r_j)]$ that lies in $Z_k(Z)$. Splitting the sum over these two kinds of V,

$$[\mathrm{div}(r_j)] = [\mathrm{div}(r'_j)] + \gamma_j, \qquad \gamma_j \in Z_k(Z) \tag{1.2}$$

where the first term on the right is read inside $Z_k(X)$ via the identification $V \leftrightarrow \overline{V}$ of subvarieties of U with their closures, and γ_j collects the codimension-one subvarieties of W_j that lie in Z.

Step 3, conclusion. Sum (1.2) over j and use $\sum_j[\mathrm{div}(r'_j)] = \beta$ as cycles on X under the closure identification (both sides are the cycle $\sum_i n_i[V_i]$, since $V_i = \overline{V'_i}$). This yields

$$\sum_j[\mathrm{div}(r_j)] = \beta + \sum_j \gamma_j$$

so that

$$\beta = \sum_j[\mathrm{div}(r_j)] - \sum_j \gamma_j$$

The first sum is a $\mathbb{Z}$-combination of divisors of rational functions on X, hence lies in $\mathrm{Rat}_k(X)$; the second sum $\gamma := \sum_j \gamma_j$ lies in $Z_k(Z)$. Therefore, in $A_k(X)$,

$$[\beta] = -[\gamma] = i_*(-[\gamma]) \in \mathrm{im}(i_*)$$

Adding back the discarded $\alpha_Z = i_*[\alpha_Z]$ gives $[\alpha] = i_*\big([\alpha_Z] - [\gamma]\big)$, exhibiting the class of α in the image of i_*. This proves $\ker(j^*) \subseteq \mathrm{im}(i_*)$, and with Steps 1 and 2 the sequence is exact, completing the proof.

The mechanism of the proof is worth isolating, because it recurs whenever Chow groups are computed by stratification. Surjectivity on the right is elementary: a subvariety of the open set extends to its closure, so no class on U is unreachable from X. The content is in the middle, and it is entirely a statement about lifting rational equivalences. A relation $j^*\alpha \sim 0$ on U is witnessed by finitely many rational functions on subvarieties of U; each such function extends across Z to a rational function on a subvariety of X (function fields do not see the boundary), and the extended divisor differs from the original only in components supported on Z. The failure of the sequence to be exact on the left, which is left as it stands, measures exactly how much rational equivalence is created by the extra room X provides over Z: a cycle on Z can become rationally equivalent to zero in X without being so in Z, because X contains $(k+1)$-dimensional subvarieties running out of Z into U. The affine line makes this concrete below.

1.4.2 Remark: No Left-Exactness The map $i_*\colon A_k(Z) \to A_k(X)$ need not be injective. Take $X = \mathbb{A}^1$, $Z = \{0\}$, and $k = 0$. Then $A_0(Z) = \mathbb{Z}$, generated by $[0]$, but $[0] \sim 0$ in $A_0(\mathbb{A}^1)$ by example 1.1.5, so $i_*[0] = 0$ and i_* is the zero map on a nonzero group. The kernel of i_* records cycles on Z that bound in X but not in Z, and there is no functorial description of it. This is why the localization sequence is only a three-term right-exact sequence, not a long exact sequence: Bloch's higher Chow groups supply the missing terms, but that theory is outside the scope of this book.

The localization sequence is stated for one closed subscheme, but its most frequent use is for a pair. Two closed subschemes X_1, X_2 of an ambient X can be compared through their union and intersection, and the resulting sequence is the Mayer-Vietoris right-exactness that was promised without proof in example 1.1.6. It is a companion to the localization sequence rather than a corollary of it: the derivation below runs directly at the level of cycles, using only that a subvariety of the union lies in one of the two pieces, so no appeal to theorem 1.4.1 is needed.

Corollary 1.4.3: Right-Exactness for Two Closed Subschemes

Let X_1 and X_2 be closed subschemes of an algebraic scheme, and set $X_1 \cup X_2$ and $X_1 \cap X_2$ (with reduced or any fixed scheme structure; Chow groups see only the reduced structure by example 1.1.6). For every k the sequence

$$A_k(X_1 \cap X_2) \xrightarrow{(\iota_{1*}, -\iota_{2*})} A_k(X_1) \oplus A_k(X_2) \xrightarrow{j_{1*}+j_{2*}} A_k(X_1 \cup X_2) \longrightarrow 0$$

is exact, where $\iota_m \colon X_1 \cap X_2 \hookrightarrow X_m$ and $j_m \colon X_m \hookrightarrow X_1 \cup X_2$ are the inclusions. There is no exactness at $A_k(X_1 \cap X_2)$ in general.

***Proof*:**
Write $Y = X_1 \cup X_2$. The argument is cleanest at the level of cycles, using the one structural fact that an irreducible closed subset of $Y = X_1 \cup X_2$ is contained in X_1 or in X_2. Consequently a subvariety $V \subseteq Y$ lies in X_1 or in X_2 (or in both, in which case $V \subseteq X_1 \cap X_2$), and because $Z_k(Y)$ is free on subvarieties, a cycle expressible both by subvarieties of X_1 and by subvarieties of X_2 has all its components in $X_1 \cap X_2$.

Step 1: surjectivity of $j_{1*} + j_{2*}$. Every subvariety of Y lies in X_1 or X_2, so at the level of cycles $Z_k(Y) = j_{1*}Z_k(X_1) + j_{2*}Z_k(X_2)$; the map $Z_k(X_1) \oplus Z_k(X_2) \to Z_k(Y)$, $(a_1, a_2) \mapsto a_1 + a_2$, is surjective. Passing to classes, $j_{1*} + j_{2*} \colon A_k(X_1) \oplus A_k(X_2) \to A_k(Y)$ is surjective.

Step 2: the composite is zero. For a subvariety $V \subseteq X_1 \cap X_2$ the two pushforwards $j_{1*}\iota_{1*}[V]$ and $j_{2*}\iota_{2*}[V]$ both equal $[V]$ in $Z_k(Y)$, since each inclusion sends $[V]$ to $[V]$. Hence $(j_{1*} + j_{2*}) \circ (\iota_{1*}, -\iota_{2*}) = j_{1*}\iota_{1*} - j_{2*}\iota_{2*} = 0$, giving $\operatorname{im}(\iota_{1*}, -\iota_{2*}) \subseteq \ker(j_{1*} + j_{2*})$.

Step 3: exactness in the middle. Let $(\eta_1, \eta_2) \in A_k(X_1) \oplus A_k(X_2)$ satisfy $j_{1*}\eta_1 + j_{2*}\eta_2 = 0$ in $A_k(Y)$. Represent η_1 and η_2 by cycles $a_1 \in Z_k(X_1)$ and $a_2 \in Z_k(X_2)$. The hypothesis says $a_1 + a_2 \sim 0$ in $Z_k(Y)$, so there are finitely many $(k+1)$-dimensional subvarieties $W_j \subseteq Y$ and $r_j \in k(W_j)^*$ with

$$a_1 + a_2 = \sum_j [\operatorname{div}(r_j)] \quad \text{in } Z_k(Y)$$

Each W_j lies in X_1 or in X_2, and the divisor $[\operatorname{div}(r_j)]$ of a rational function on W_j is a cycle supported on W_j, hence on X_1 or on X_2 accordingly. Collect the terms:

$$b_1 = \sum_{W_j \subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_1), \qquad b_2 = \sum_{W_j \subseteq X_2,\, W_j \not\subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_2)$$

where a W_j lying in both is assigned to the first group; then $\sum_j [\operatorname{div}(r_j)] = b_1 + b_2$ and so $a_1 + a_2 = b_1 + b_2$ in $Z_k(Y)$. Rearranging,

$$c := a_1 - b_1 = b_2 - a_2$$

The left expression $a_1 - b_1$ is a cycle supported on X_1, and the right expression $b_2 - a_2$ is a cycle supported on X_2; being the same cycle in the free group $Z_k(Y)$, it is supported on $X_1 \cap X_2$, that is $c \in Z_k(X_1 \cap X_2)$. Now compute classes. In $A_k(X_1)$, $b_1 \in \mathrm{Rat}_k(X_1)$ is zero, so $\eta_1 = [a_1] = [a_1 - b_1] = \iota_{1*}[c]$; in $A_k(X_2)$, $b_2 \in \mathrm{Rat}_k(X_2)$ is zero, so $\eta_2 = [a_2] = [a_2 - b_2] = -\iota_{2*}[c]$. Therefore

$$(\eta_1, \eta_2) = (\iota_{1*}[c], -\iota_{2*}[c]) = (\iota_{1*}, -\iota_{2*})\big([c]\big)$$

lies in the image of $(\iota_{1*}, -\iota_{2*})$. This gives $\ker(j_{1*} + j_{2*}) \subseteq \mathrm{im}(\iota_{1*}, -\iota_{2*})$, and with Step 2 exactness in the middle.

No left-exactness. The map $(\iota_{1*}, -\iota_{2*})$ need not be injective. Take X_1 and X_2 two distinct lines through the origin in $\mathbb{A}^2$, so $X_1 \cap X_2 = \{0\}$, in degree $k = 0$: then $A_0(X_1 \cap X_2) = \mathbb{Z}$ is generated by $[0]$, while $\iota_{1*}[0] = [0] \sim 0$ in $A_0(X_1) = A_0(\mathbb{A}^1)$ and likewise $\iota_{2*}[0] = 0$ in $A_0(X_2)$, so $(\iota_{1*}, -\iota_{2*})[0] = 0$ on a nonzero group. Thus there is no exactness at $A_k(X_1 \cap X_2)$, as we sought to show.

The corollary is the exact sequence recorded, without proof, as item (4) of example 1.1.6; the present derivation discharges that promissory note. Its shape is the Chow-group form of the Mayer-Vietoris sequence familiar from topology, truncated on the left for the same reason the localization sequence is: the intersection $X_1 \cap X_2$ can carry cycles that become rationally equivalent to zero once they are free to move in the larger $X_1 \cup X_2$. Where it is used most is in dévissage arguments, where a scheme is written as a union of simpler closed pieces and its Chow groups are assembled from those of the pieces and their overlaps.

A single application of the localization sequence recovers, from scratch, the computation of $A_0(\mathbb{A}^1)$ recorded in example 1.1.5, this time without touching a rational function by hand. The point is that $\mathbb{A}^1$ is $\mathbb{P}^1$ with a point removed, and the localization sequence turns the known $A_0(\mathbb{P}^1) = \mathbb{Z}$ into the answer for the open complement.

Example 1.4.4: Recomputing $A_0(\mathbb{A}^1)$ by Localization

Take $X = \mathbb{P}^1$, $Z = \{\infty\}$ the point at infinity, and $U = \mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. The localization sequence in degree 0 reads

$$A_0(\{\infty\}) \xrightarrow{i_*} A_0(\mathbb{P}^1) \xrightarrow{j^*} A_0(\mathbb{A}^1) \longrightarrow 0$$

The left group is $A_0(\{\infty\}) = \mathbb{Z}$, generated by the class of the single point ∞, and $i_*[\infty] = [\infty] \in A_0(\mathbb{P}^1)$. By example 1.1.5 the group $A_0(\mathbb{P}^1) \cong \mathbb{Z}$ is generated by the class h of any point, and $[\infty] = h$ is a generator, so i_* is surjective. By exactness $\mathrm{im}(i_*) = \ker(j^*)$, and since i_* is onto, $\ker(j^*) = A_0(\mathbb{P}^1)$; the quotient by the whole group is trivial:

$$A_0(\mathbb{A}^1) = A_0(\mathbb{P}^1)/\ker(j^*) \cong \mathrm{coker}(i_*) = \mathbb{Z}/\mathbb{Z} = 0$$

Thus $A_0(\mathbb{A}^1) = 0$, in agreement with the direct computation. The single generator of $A_0(\mathbb{P}^1)$ is killed precisely because the point removed to form $\mathbb{A}^1$ was rationally equivalent to every remaining point; removing it collapses the whole group.

With the localization sequence available, the second structural result of the section follows by Noetherian induction. Homotopy invariance in algebraic geometry takes the form of affine-bundle invariance: pulling a cycle back along the projection of an affine bundle is an isomorphism on Chow groups. The motivating case is the trivial bundle $X \times \mathbb{A}^n \to X$, which asserts that appending an affine factor does not change the Chow group beyond the expected degree shift, the algebraic counterpart of the fact that $\mathbb{A}^n$ is contractible.

The result is stated for a general affine bundle, of which the trivial bundle is the leading case.

Definition 1.4.5: Affine Bundle of Relative Dimension n

An *affine bundle of relative dimension* n over X is a scheme $p\colon E \to X$ that is locally on X a product with affine space: there is an open cover $\{U_\lambda\}$ of X and isomorphisms $p^{-1}(U_\lambda) \cong U_\lambda \times \mathbb{A}^n$ over U_λ.

A vector bundle of rank n, with its transition maps in GL_n, is such a bundle, as is any torsor under a vector bundle; the trivial example is $X \times \mathbb{A}^n$ itself. In every case p is flat of relative dimension n, so flat pullback $p^*\colon A_k(X) \to A_{k+n}(E)$ is defined by definition 1.3.3.

Theorem 1.4.6: Affine-Bundle Invariance

Let $p\colon E \to X$ be an affine bundle of relative dimension n over an algebraic scheme X. Then the flat pullback

$$p^*\colon A_k(X) \xrightarrow{\sim} A_{k+n}(E)$$

is an isomorphism for every k. In particular, for the trivial bundle, $A_k(X) \xrightarrow{\sim} A_{k+n}(X \times \mathbb{A}^n)$.

***Proof*:**
The proof runs the argument directly at relative dimension n; the fibre dimension enters only through the trivial bundle, where a global tower is available. Note that a global tower of rank-one affine bundles is *not* claimed for a general bundle: the local identification $U\times\mathbb{A}^n = (U\times\mathbb{A}^{n-1})\times\mathbb{A}^1$ does not glue to a factorization $E \to X$ over X, since a general affine (or vector) bundle carries no global flag of subbundles. The trivial bundle $X \times \mathbb{A}^n$, being a literal product, *is* such a tower, $X \times \mathbb{A}^n = (X \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$; the construction below invokes the tower only there. Fix n and prove that $p^*\colon A_k(X) \to A_{k+n}(E)$ is an isomorphism for all k, by Noetherian induction on the closed subschemes of X. For a closed subscheme $W \subseteq X$ write $E_W = p^{-1}(W)$ and $p_W\colon E_W \to W$ for the restricted bundle, again an affine bundle of relative dimension n.

Step 1: reductions. The scheme X is Noetherian, so its closed subschemes satisfy the descending chain condition and it suffices to prove: if p_W^* is an isomorphism for every proper closed subscheme $W \subsetneq X$, then p^* is an isomorphism for X. Chow groups depend only on the reduced structure (example 1.1.6), so take X reduced. If X is reducible, write $X = X_1 \cup X_2$ with X_1 an irreducible component and X_2 the union of the remaining components, both proper closed subschemes; the pullbacks $p^*_{X_1}, p^*_{X_2}, p^*_{X_1\cap X_2}$ are isomorphisms by hypothesis. The Mayer-Vietoris sequences of corollary 1.4.3 for $X = X_1 \cup X_2$ and for $E = E_{X_1} \cup E_{X_2}$ (whose intersection is $E_{X_1\cap X_2}$) form a commutative ladder linked by the vertical pullbacks; a diagram chase using the three outer isomorphisms shows p^* is an isomorphism for X. So assume X is a variety (reduced and irreducible), and let $\eta = \operatorname{Spec} k(X)$ be its generic point, with generic fibre $E_\eta \cong \mathbb{A}^n_{k(X)}$.

Step 2: surjectivity of p^*. It suffices to realize the class of a single $(k+n)$-dimensional subvariety $V \subseteq E$ as a pullback modulo the image of $A_{k+n}(E_Z)$ for some proper closed $Z \subsetneq X$. Let $W = \overline{p(V)} \subseteq X$. If $W \subsetneq X$ is proper, then $V \subseteq E_W$ is supported over $Z = W$, and $[V] \in \operatorname{im}\big(A_{k+n}(E_W) \to A_{k+n}(E)\big)$; by the inductive hypothesis p_W^* is onto, and compatibility of p^* with the closed immersion (the fibre square for p over $W \hookrightarrow X$ gives $p^* i_* = i_* p_W^*$ by proposition 1.3.6) writes $[V]$ as a pullback from X. If instead $W = X$, so V dominates X, restrict to the generic fibre. There $V_\eta \subseteq \mathbb{A}^n_{k(X)}$ is a closed subvariety, and its class is controlled by the trivial-bundle case of the theorem over the field $K = k(X)$: the projection $\mathbb{A}^n_K \to \operatorname{Spec} K$ has surjective flat pullback. This last fact is elementary and needs no base induction, being the trivial bundle over a point: writing the tower $\mathbb{A}^n_K = (\mathbb{A}^{n-1}_K) \times \mathbb{A}^1$ and inducting on n, the base case $A_*(\mathbb{A}^1_K)$ is generated over $A_*(\operatorname{Spec} K)$ by the pullback (every closed point of $\mathbb{A}^1_K$ is the divisor of its minimal polynomial, so $A_0(\mathbb{A}^1_K) = 0$, exactly as in example 1.1.5 with K in place of $\overline{k}$), and the tower propagates this up the coordinates. Thus $[V_\eta]$ is rationally equivalent on $\mathbb{A}^n_{k(X)}$ to a pullback $p^*_\eta(a_\eta)$ of a class a_η on $\operatorname{Spec} k(X)$. Spreading out the finitely many rational functions that realize this equivalence, together with a_η, to an open $U \subseteq X$ shows $[V]|_{E_U} = p^*_U(\alpha_U)$ for a class $\alpha_U \in A_k(U)$ and hence $[V] - p^*(\alpha_V)$ (with $\alpha_V \in A_k(X)$ lifting α_U via surjectivity of j^*, theorem 1.4.1) restricts to 0 on E_U, so its support lies over the proper closed $Z = X \setminus U$. In every case

$$[V] = p^*(\alpha_V) + \gamma_V \quad \text{in } A_{k+n}(E), \qquad \gamma_V \in \operatorname{im}\big(A_{k+n}(E_Z) \to A_{k+n}(E)\big)$$

for some proper closed $Z \subsetneq X$ and $\alpha_V \in A_k(X)$. By the inductive hypothesis p^*_Z is onto $A_*(E_Z)$, so $\gamma_V = i_* p^*_Z(\delta_V) = p^*(i_*\delta_V)$ for a class δ_V on Z, again by proposition 1.3.6. Hence $[V] = p^*(\alpha_V + i_*\delta_V)$ and p^* is surjective.

Surjectivity of p^* is now in hand for every X; injectivity is the remaining point, and it is not a formal consequence of the localization ladder, since the two rows are only right-exact and provide no left term to chase against. It is proved by exhibiting a left inverse to p^* for the trivial bundle and reducing the general bundle to that case.

Step 3: a left inverse for the trivial bundle. The construction is carried out for a single affine coordinate over an arbitrary base Y; the trivial bundle $X \times \mathbb{A}^n$ is a tower of n such single coordinates, and the rank-n left inverse is the composite of the n single-coordinate left inverses (Step 3d). Take $E = Y \times \mathbb{A}^1$, with coordinate t on the second factor and projection $p\colon Y \times \mathbb{A}^1 \to Y$, and define a homomorphism $\sigma\colon Z_{k+1}(Y \times \mathbb{A}^1) \to Z_k(Y)$ on a generator $[V]$, $V \subseteq Y \times \mathbb{A}^1$ a $(k+1)$-dimensional subvariety, by

$$\sigma[V] = \begin{cases} p_*\big[\operatorname{div}_V(t|_V)\big] & t|_V \in k(V)^* \text{ nonconstant} \\ 0 & t|_V \text{ constant} \end{cases}$$

extended additively; here "$t|_V$ constant" is read to include the degenerate case $t|_V \equiv 0$ (when $V \subseteq Y \times \{0\}$), which likewise contributes 0, since then $t|_V \notin k(V)^*$ and no divisor $\operatorname{div}_V(t|_V)$ is formed. When $t|_V$ is nonconstant, $[\operatorname{div}_V(t|_V)]$ is a k-cycle on V; the pushforward p_* of this particular cycle is well-defined even though $p|_V\colon V \to \overline{p(V)} \subseteq Y$ need not be proper. The point is that t is a regular function on $Y \times \mathbb{A}^1$, so its restriction $t|_V$ is regular on V and $\operatorname{div}_V(t|_V)$ is effective, with support contained in the zero locus $V \cap (Y \times \{0\}) \subseteq Y \times \{0\}$. On $Y \times \{0\}$ the projection p is the closed immersion identifying $Y \times \{0\}$ with Y, in particular proper, so $p_*[\operatorname{div}_V(t|_V)]$ is a well-defined k-cycle on Y by definition 1.2.1, computed component by component over the locus where $t|_V$ vanishes. Geometrically $\sigma[V]$ is the fibre of V over $t = 0$, projected to Y: the specialization of V to the zero section.

Step 3a: $\sigma \circ p^* = \mathrm{id}$. For a k-dimensional subvariety $W \subseteq Y$ the pullback is $p^*[W] = [W \times \mathbb{A}^1]$, and t restricts to the coordinate on $W \times \mathbb{A}^1$, which is nonconstant with $[\operatorname{div}_{W\times\mathbb{A}^1}(t)] = [W \times \{0\}]$ (the coordinate t has its single zero along $W \times \{0\}$ and no pole on $W \times \mathbb{A}^1$). Pushing forward, $p_*[W \times \{0\}] = [W]$, so $\sigma(p^*[W]) = [W]$. By additivity $\sigma \circ p^* = \mathrm{id}$ on $Z_k(Y)$, hence on $A_k(Y)$ once σ is known to descend.

Step 3b: σ respects rational equivalence. A generator of $\mathrm{Rat}_{k+1}(Y \times \mathbb{A}^1)$ is $[\mathrm{div}_S(g)]$ for a $(k+2)$-dimensional subvariety $S \subseteq Y \times \mathbb{A}^1$ and $g \in k(S)^*$, and unwinding the definitions gives $\sigma[\mathrm{div}_S(g)] = p_*\big(\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t|_V)]\big)$, the outer sum over the codimension-one subvarieties $V \subseteq S$ on which $t|_V$ is nonconstant (a V with $t|_V$ constant, including $V \subseteq Y \times \{0\}$ where $t|_V \equiv 0$, drops out, contributing 0). This iterated divisor is symmetric in the two functions g and $t|_S$ modulo a principal divisor on S:

$$\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t)] - \sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)] \in \mathrm{Rat}_k(S) \tag{1.3}$$

The relation (1.3) is the commutativity of two divisor operations on the $(k+2)$-dimensional S, equivalently the well-definedness of intersecting a cycle with a Cartier divisor; its proof requires the divisor-theoretic machinery of the next chapter, and is supplied there by theorem 2.3.2 (its surface case is exactly (1.3)), so no tool used here depends on it circularly. Granting (1.3), $\sigma[\mathrm{div}_S(g)]$ differs from $p_*\big(\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]\big)$ by the pushforward of a principal divisor, and $\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]$ is itself a $\mathbb{Z}$-combination of principal divisors on the subvarieties V, so its pushforward lies in $\mathrm{Rat}_k(Y)$ by theorem 1.2.3. Either way $\sigma[\mathrm{div}_S(g)] \in \mathrm{Rat}_k(Y)$, so σ descends to

$$\sigma\colon A_{k+1}(Y \times \mathbb{A}^1) \to A_k(Y), \qquad \sigma \circ p^* = \mathrm{id}$$

In particular p^* is injective for the trivial rank-one bundle over any base Y, and being surjective by Step 2, it is an isomorphism there. The left inverse σ commutes with proper pushforward along closed immersions and with flat restriction to opens, since each of p_* and $\mathrm{div}(t)$ does; this compatibility is what carries the trivial-bundle result across the localization sequence in Step 3c.

Step 3d: the trivial rank-n bundle. For $E = X \times \mathbb{A}^n$ with coordinates $t_1, \dots, t_n$, the tower

$$X \times \mathbb{A}^n \xrightarrow{q_n} X \times \mathbb{A}^{n-1} \xrightarrow{q_{n-1}} \cdots \xrightarrow{q_1} X$$

factors $p = q_1 \circ \cdots \circ q_n$ into single-coordinate trivial rank-one bundles $q_m\colon (X \times \mathbb{A}^{m-1}) \times \mathbb{A}^1 \to X \times \mathbb{A}^{m-1}$, so $p^* = q_n^* \circ \cdots \circ q_1^*$ by functoriality of flat pullback (proposition 1.3.5). Applying Step 3b with $Y = X \times \mathbb{A}^{m-1}$ gives a left inverse σ_m to each q_m^*; the composite $\sigma := \sigma_1 \circ \cdots \circ \sigma_n\colon A_{k+n}(X \times \mathbb{A}^n) \to A_k(X)$ satisfies $\sigma \circ p^* = \mathrm{id}$, so p^* is injective, and with Step 2 it is an isomorphism for the trivial bundle. The compatibility of each σ_m with proper pushforward along closed immersions and with flat restriction to opens passes to the composite, so σ has the same compatibility used in the next step.

Step 3c: injectivity for a general bundle. Let $p\colon E \to X$ be an affine bundle of relative dimension n over the variety X and $\alpha \in A_k(X)$ with $p^*\alpha = 0$. Choose a dense open $U \subseteq X$ over which E trivializes, $E_U \cong U \times \mathbb{A}^n$, and set $Z = X \setminus U$, a proper closed subscheme. Flat base change (proposition 1.3.5) gives $p_U^*(j^*\alpha) = j^*(p^*\alpha) = 0$, and p_U^* is injective by Step 3d applied to the trivial bundle over U; hence $j^*\alpha = 0$. By the localization sequence for $Z \subseteq X$ (theorem 1.4.1), $\alpha = i_*\beta$ for some $\beta \in A_k(Z)$. Pushing β into E through the fibre square over $Z \hookrightarrow X$ (proposition 1.3.6) gives $p^*\alpha = p^*i_*\beta = i_*(p_Z^*\beta) = 0$, so $p_Z^*\beta$ lies in the kernel of $i_*\colon A_{k+n}(E_Z) \to A_{k+n}(E)$. Over the total space E the left inverse of Step 3d, restricted to the trivializing locus and extended by $\sigma_Z := (p_Z^*)^{-1}$ over E_Z (the inductive isomorphism), assembles into a retraction $r_E\colon A_{k+n}(E) \to A_k(X)$ of p^*; the compatibility noted at the end of Step 3d makes the square with i_* commute, so that $r_E \circ i_* = i_* \circ \sigma_Z$ on $A_{k+n}(E_Z)$. Evaluating at $p_Z^*\beta$,

$$i_*\beta = i_*\big(\sigma_Z p_Z^*\beta\big) = r_E\big(i_* p_Z^*\beta\big) = r_E(0) = 0$$

so $\alpha = i_*\beta = 0$. Hence p^* is injective.

> Steps 2 and 3 show p^* is an isomorphism for every X and every relative dimension n, completing the Noetherian induction; the tower entered only for the trivial bundle (Step 3d), where it is an actual global factorization, so no global tower was claimed for a general bundle, completing the proof.

The theorem is the algebraic-geometric form of a topological triviality: the projection $X \times \mathbb{A}^n \to X$ is a homotopy equivalence when $\mathbb{A}^n = \mathbb{C}^n$ is given the classical topology, so it induces isomorphisms on singular homology, and Chow groups behave the same way. The degree shift by n is the algebraic bookkeeping for the real dimension shift $2n$ on the topological side, since a k-cycle (complex dimension k) has real dimension $2k$ and A_{k+n} matches $H_{2(k+n)} = H_{2k+2n}$. The proof uses none of this: surjectivity of p^* needs only that every zero-cycle on an affine line over a field is rationally trivial ($A_0(\mathbb{A}^1_K) = 0$), propagated up the base by localization, while injectivity needs a left inverse, the specialization to the zero section. The one step left open, the divisor symmetry (1.3), is exactly the input that makes intersecting with a Cartier divisor well defined on rational equivalence, the payload of the next chapter; the isomorphism here is the first place that well-definedness is needed, and the forward reference is deliberate rather than a gap. It is discharged in theorem 2.3.2. The reason the argument needs Noetherian induction rather than a direct computation is the same reason the localization sequence is not left-exact: one cannot control what rational equivalences a cycle acquires on E without dismantling the base into pieces small enough that the bundle trivializes and the fibrewise vanishing takes over.

Exercise 1.4.1

1. Let $Z \subseteq X$ be closed with complement U. Show directly from theorem 1.4.1 that $A_k(U) \cong A_k(X)/\operatorname{im}\big(A_k(Z) \to A_k(X)\big)$, and use this to prove that if $A_k(Z) = 0$ then $j^*\colon A_k(X) \to A_k(U)$ is an isomorphism.
2. Let $X = \mathbb{A}^2$ and let $Z = \{0\}$ be the origin, so $U = \mathbb{A}^2 \setminus \{0\}$ is the punctured plane. Using the localization sequence together with the values $A_k(\mathbb{A}^2)$ (namely $A_2(\mathbb{A}^2) = \mathbb{Z}$ and $A_k(\mathbb{A}^2) = 0$ for $k < 2$, which you may take from section 1.5), compute $A_k(U)$ for all k.
3. Let X be a variety of dimension n and $U \subseteq X$ a nonempty open subset. Show that $A_n(X) \to A_n(U)$ is an isomorphism, and that both are $\mathbb{Z}$, generated by the fundamental class.
4. Compute $A_*(\mathbb{P}^n \times \mathbb{A}^m)$ in terms of $A_*(\mathbb{P}^n)$, assuming the value $A_k(\mathbb{P}^n) = \mathbb{Z}$ for $0 \le k \le n$ from section 1.5. State the answer as a graded group.
5. Let $p\colon E \to X$ be a vector bundle of rank r over an algebraic scheme X, with zero section $s\colon X \hookrightarrow E$. Affine-bundle invariance makes $p^*\colon A_k(X) \to A_{k+r}(E)$ an isomorphism. Explain why one cannot conclude $A_*(E) \cong A_*(X)$ by inverting p^* with a proper pushforward p_*, identifying the precise obstruction, and describe the degree by which p^* and the closed-immersion pushforward s_* each shift a cycle class.
6. Take X_1 and X_2 to be two distinct lines through the origin in $\mathbb{A}^2$, meeting only at $\{0\}$. Write out the sequence of corollary 1.4.3 in degree 0 and verify its exactness by hand, given that A_0 of a line is 0 and A_0 of a point is $\mathbb{Z}$.

1.5 First Computations

The machinery of the preceding sections is now assembled: cycles, rational equivalence, proper pushforward, flat pullback, and the two fundamental sequences. This section puts it to work on the spaces one meets first. We compute the Chow groups of affine space, of projective space, and of a smooth projective curve, and record the shape the answer takes for a projective bundle. These are the building blocks from which nearly every later computation is assembled, and they already display the two extremes of the theory: affine space, whose cycles almost all sweep away to nothing, and projective space, whose cycles are pinned down by a single integer in each dimension.

The pattern to watch is how the localization sequence and affine-bundle invariance from section 1.4 do the actual work. Affine space is an affine bundle over a point, so its Chow groups collapse; projective space is built from affine space by adding a hyperplane at infinity, so an induction along the localization sequence peels off one $\mathbb{Z}$ per dimension. Every computation below is an instance of one of these two moves.

Affine space is the first computation, and $\mathbb{A}^n$ retracts, in the sense of rational equivalence, all the way down to a point: every subvariety of dimension less than n can be swept off to zero, and only the fundamental class in top dimension survives. This is the higher-dimensional form of the observation $A_0(\mathbb{A}^1) = 0$ from example 1.1.5, and it follows in one line from affine-bundle invariance once one notices that $\mathbb{A}^n$ is an affine bundle over $\operatorname{Spec} \overline{k}$.

Proposition 1.5.1: Chow Groups of Affine Space

For every $n \geq 0$ the Chow groups of affine space are

$$A_k(\mathbb{A}^n) = \begin{cases} \mathbb{Z}\,[\mathbb{A}^n] & k = n \\ 0 & k \neq n \end{cases}$$

the group $A_n(\mathbb{A}^n)$ being free of rank one on the fundamental class $[\mathbb{A}^n]$.

***Proof*:**
The structure morphism $\pi\colon \mathbb{A}^n \to \operatorname{Spec} \overline{k}$ exhibits $\mathbb{A}^n$ as the trivial affine bundle of relative dimension n over a point. Affine-bundle invariance (theorem 1.4.6) states that flat pullback

$$\pi^*\colon A_k(\operatorname{Spec} \overline{k}) \xrightarrow{\sim} A_{k+n}(\mathbb{A}^n)$$

is an isomorphism for every k. The point $\operatorname{Spec} \overline{k}$ has a single subvariety, itself, of dimension 0, so $A_0(\operatorname{Spec} \overline{k}) = Z_0(\operatorname{Spec} \overline{k}) = \mathbb{Z}\,[\operatorname{Spec} \overline{k}]$ and $A_k(\operatorname{Spec} \overline{k}) = 0$ for $k \neq 0$. Transporting these through π^* gives $A_{k+n}(\mathbb{A}^n) = 0$ for $k \neq 0$, that is $A_j(\mathbb{A}^n) = 0$ for $j \neq n$, and $A_n(\mathbb{A}^n) = \pi^*\big(\mathbb{Z}\,[\operatorname{Spec} \overline{k}]\big)$. Flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space (definition 1.3.3), so $\pi^*[\operatorname{Spec} \overline{k}] = [\mathbb{A}^n]$, and $A_n(\mathbb{A}^n) = \mathbb{Z}\,[\mathbb{A}^n]$ is free of rank one, completing the proof.

Affine space thus has no cycles in low dimension worth remembering, a fact that reads two ways. From below it says that any k-dimensional subvariety $V \subseteq \mathbb{A}^n$ with $k < n$ bounds: there is a family of cycles, parametrized by the line, sweeping $[V]$ to zero. From above it isolates the one invariant that does survive, the coefficient of the fundamental class, which for a top cycle records the multiplicity with which $\mathbb{A}^n$ appears in it. The vanishing in low dimension is what makes affine space so poor a stage for enumerative geometry, and simultaneously what makes it the natural chart in which to localize a computation on a larger space. This second role is exactly how projective space is handled next.

The contrast to keep in mind is the affine line. A single point of $\mathbb{A}^1$ is a 0-cycle in dimension $0 < 1$, so proposition 1.5.1 predicts $A_0(\mathbb{A}^1) = 0$, recovering the computation of example 1.1.5: the coordinate function $t - p$ has $[\operatorname{div}(t-p)] = [p]$, presenting the point as a principal divisor and hence as zero in the Chow group. The proposition is the assertion that this collapse happens in every dimension below the top, uniformly and for the same reason.

Projective space is the opposite. Nothing sweeps to zero for free, because a rational function on a subvariety of $\mathbb{P}^n$ has as many zeros as poles, and the surviving invariant in each dimension is an integer. The computation is an induction that peels $\mathbb{P}^n$ apart along a hyperplane: the closed subscheme $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ at infinity, with open complement the affine chart $\mathbb{A}^n$. Affine space contributes nothing below the top by proposition 1.5.1, so the localization sequence forces every class of $\mathbb{P}^n$ in dimension $k < n$ to come from $\mathbb{P}^{n-1}$, and the induction runs.

The motivating principle is that projective space is *cellular*: it is stratified by the chain

$$\mathbb{P}^0 \subset \mathbb{P}^1 \subset \cdots \subset \mathbb{P}^{n-1} \subset \mathbb{P}^n$$

of linear subspaces, each obtained from the previous by adding an affine cell $\mathbb{A}^k = \mathbb{P}^k \setminus \mathbb{P}^{k-1}$. The Chow groups turn out to be free on the classes of these cells, one generator in each dimension. This is the algebraic form of the cell decomposition that computes the singular homology of $\mathbb{C}\mathrm{P}^n$ in [10], where the same chain of linear subspaces yields $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}$ and $H_{\text{odd}} = 0$.

Theorem 1.5.2: Chow Groups of Projective Space

For every $n \geq 0$ and every $0 \leq k \leq n$, the Chow group $A_k(\mathbb{P}^n)$ is free abelian of rank one, generated by the class $[\mathbb{P}^k]$ of any linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ of dimension k:

$$A_k(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^k], \qquad 0 \leq k \leq n$$

and $A_k(\mathbb{P}^n) = 0$ for $k > n$. Any two linear subspaces of the same dimension are rationally equivalent, so the generator does not depend on the choice.

***Proof*:**
The argument is an induction on n, run through projection from a point, which realizes the complement of a point in $\mathbb{P}^n$ as a line bundle over $\mathbb{P}^{n-1}$. Write h_k for the class $[\mathbb{P}^k]$ of a linear k-plane; the claim is that $A_k(\mathbb{P}^n) = \mathbb{Z}\,h_k$ is free of rank one for $0 \leq k \leq n$, and 0 otherwise. Groups $A_k(\mathbb{P}^n)$ with $k > n = \dim \mathbb{P}^n$ vanish because $\mathbb{P}^n$ has no subvariety of that dimension, so only $0 \leq k \leq n$ is at issue.

Step 1: base cases $n = 0$ and $k = 0$. For $n = 0$ the space $\mathbb{P}^0 = \operatorname{Spec} \overline{k}$ is a point, with $A_0(\mathbb{P}^0) = \mathbb{Z}\,[\mathbb{P}^0]$ and $A_k(\mathbb{P}^0) = 0$ for $k \neq 0$ by proposition 1.5.1, which is the claim. For general n and $k = 0$, every closed point of $\mathbb{P}^n$ is rationally equivalent to every other: two points p, q both lie on a line $\ell \cong \mathbb{P}^1 \subseteq \mathbb{P}^n$, and on ℓ they are rationally equivalent by example 1.1.5, so $[p] \sim [q]$ in $\mathbb{P}^n$. Thus $A_0(\mathbb{P}^n)$ is cyclic, generated by $h_0 = [p]$. Since $\mathbb{P}^n$ is complete, the degree map $\deg\colon A_0(\mathbb{P}^n) \to \mathbb{Z}$ of definition 1.2.4 is well defined and sends $h_0 = [p] \mapsto [k(p) : \overline{k}] = 1$, so h_0 has infinite order and $A_0(\mathbb{P}^n) = \mathbb{Z}\,h_0$ is free of rank one.

Step 2: projection from a point as a line bundle. Fix $n \geq 1$ and a closed point $P \in \mathbb{P}^n$. Choosing coordinates with $P = [0 : \cdots : 0 : 1]$, linear projection away from P is a morphism

$$q \colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}, \qquad [x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$$

whose fiber over a point $[a] \in \mathbb{P}^{n-1}$ is the line through P and $[a]$ with P removed, an affine line $\mathbb{A}^1$. This exhibits $\mathbb{P}^n \setminus \{P\}$ as a line bundle over $\mathbb{P}^{n-1}$, the total space of $\mathcal{O}_{\mathbb{P}^{n-1}}(1)$; in particular it is an affine bundle of relative dimension 1. Only the last clause is used below. Affine-bundle invariance (theorem 1.4.6) makes flat pullback

$$q^* \colon A_{k-1}(\mathbb{P}^{n-1}) \xrightarrow{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \tag{1.4}$$

an isomorphism for every k.

Step 3: removing the point via localization. The point $\{P\}$ is a closed subscheme of $\mathbb{P}^n$ with open complement $\mathbb{P}^n \setminus \{P\}$. The localization sequence (theorem 1.4.1) for this pair reads

$$A_k(\{P\}) \xrightarrow{i_*} A_k(\mathbb{P}^n) \xrightarrow{j^*} A_k(\mathbb{P}^n \setminus \{P\}) \to 0 \tag{1.5}$$

exact at $A_k(\mathbb{P}^n)$ and $A_k(\mathbb{P}^n \setminus \{P\})$. The point contributes $A_k(\{P\}) = 0$ for every $k \geq 1$ and $A_0(\{P\}) = \mathbb{Z}[P]$. Hence for $k \geq 1$ the left term vanishes and j^* is an isomorphism, which combined with (1.4) gives, for $k \geq 1$,

$$A_k(\mathbb{P}^n) \xrightarrow[j^*]{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \xleftarrow[q^*]{\sim} A_{k-1}(\mathbb{P}^{n-1}) \tag{1.6}$$

so $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$ as abelian groups.

Step 4: the induction. Fix k with $1 \leq k \leq n$. Iterating (1.6) k times descends both indices by k:

$$A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1}) \cong \cdots \cong A_0(\mathbb{P}^{n-k})$$

and $A_0(\mathbb{P}^{n-k}) = \mathbb{Z}$ is free of rank one by Step 1. Together with the case $k = 0$ of Step 1, this proves $A_k(\mathbb{P}^n) \cong \mathbb{Z}$ is free of rank one for every $0 \leq k \leq n$. For $k > n$ the group vanishes as noted at the outset.

Step 5: the generator is the class of a linear subspace. It remains to identify the free generator with $h_k = [\mathbb{P}^k]$. Trace a linear $\mathbb{P}^k \subseteq \mathbb{P}^n$ through (1.6). Choose the point $P \in \mathbb{P}^k$; projection of the linear subspace $\mathbb{P}^k$ from the point $P \in \mathbb{P}^k$ is again linear, of one lower dimension, so $q(\mathbb{P}^k \setminus \{P\})$ is a linear $\mathbb{P}^{k-1} \subseteq \mathbb{P}^{n-1}$. Restricting the bundle $q \colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}$ of Step 2 over this $\mathbb{P}^{k-1}$ realizes $\mathbb{P}^k \setminus \{P\}$ as the affine bundle over $\mathbb{P}^{k-1}$; concretely $q^{-1}(\mathbb{P}^{k-1}) = \mathbb{P}^k \setminus \{P\}$ as a reduced irreducible subvariety, so flat pullback gives $q^*[\mathbb{P}^{k-1}] = [q^{-1}(\mathbb{P}^{k-1})] = [\mathbb{P}^k \setminus \{P\}]$ with multiplicity 1. Hence by construction of (1.6) the class $h_k = [\mathbb{P}^k]$ satisfies $j^* h_k = [\mathbb{P}^k \setminus \{P\}] = q^*[\mathbb{P}^{k-1}]$, so h_k corresponds to h_{k-1} under the isomorphism $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$. Descending to $A_0(\mathbb{P}^{n-k})$, the class h_k maps to $h_0 = [\mathrm{pt}]$, which is the free generator by Step 1. Thus h_k is a generator of $A_k(\mathbb{P}^n) = \mathbb{Z}$. The top case $k = n$ is consistent with the fundamental-class computation $A_n(\mathbb{P}^n) = Z_n(\mathbb{P}^n) = \mathbb{Z}[\mathbb{P}^n]$ of example 1.1.6, since $\mathbb{P}^n$ is irreducible and $h_n = [\mathbb{P}^n]$.

Step 6: independence of the linear subspace. Any two linear k-planes $\Lambda, \Lambda' \subseteq \mathbb{P}^n$ have classes differing by an integer multiple of the generator; the previous step assigns each of them the generator h_0 under the descent to $A_0(\mathbb{P}^{n-k})$, so $[\Lambda] = [\Lambda']$ in $A_k(\mathbb{P}^n)$. Concretely a projective linear transformation carrying Λ to Λ' sweeps out a rational equivalence between them, so the generator h_k is independent of the chosen k-plane, completing the induction and the proof.

The theorem says that the Chow groups of $\mathbb{P}^n$ take the simplest possible form: one copy of $\mathbb{Z}$ in each dimension from 0 to n, with a canonical generator, the class of a linear subspace. Writing the total group additively,

$$A_*(\mathbb{P}^n) = \bigoplus_{k=0}^{n} \mathbb{Z}\, h_k, \qquad h_k = [\mathbb{P}^k]$$

so $A_*(\mathbb{P}^n)$ is free of rank $n+1$. In codimension indexing the generator $h_k = [\mathbb{P}^k]$ has codimension $n-k$, and it is common to write $h = [\mathbb{P}^{n-1}]$ for the hyperplane class and record h_k as the intersection h^{n-k} of $n-k$ hyperplanes; the multiplicative structure that makes this literal is the intersection product developed in a later chapter. For now the additive statement is what matters: on projective space, a k-cycle carries exactly one integer of information, its degree, the coefficient of h_k.

This is the precise sense in which projective space is rigid where affine space is soft. A subvariety $V \subseteq \mathbb{P}^n$ of dimension k and degree d (its degree as a projective variety, the number of points in which it meets a general linear $\mathbb{P}^{n-k}$) has class $[V] = d\, h_k$, and no amount of rational equivalence can change d. The degree is a complete invariant of the class, and rational equivalence on $\mathbb{P}^n$ is, for irreducible cycles, exactly the relation "same dimension and same degree" (and for arbitrary cycles, "same degree in each dimension"). The comparison with singular homology is exact: under the cycle class map to $H_{2k}(\mathbb{C}\mathrm{P}^n)$ studied later, h_k maps to the generator, and the freeness of $A_k(\mathbb{P}^n)$ mirrors the freeness of $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$ ([10]).

The one-dimensional case deserves its own statement, both because curves are where the theory began and because on a curve the Chow group is an object the reader already knows under another name. On a smooth projective curve C the only cycles are 0-cycles and the fundamental class, and the 0-cycles modulo rational equivalence are precisely the divisor classes. This ties the general construction back to the divisor class group $\mathrm{Cl}\, C$ and the Picard group $\mathrm{Pic}\, C$ of [2, Divisors: Geometric Representation of Line Bundles].

Proposition 1.5.3: Chow Groups of a Smooth Projective Curve

Let C be a smooth projective curve over $\overline{k}$. Then

$$A_1(C) = \mathbb{Z}\,[C], \qquad A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$$

and $A_k(C) = 0$ for $k \neq 0, 1$. Under the second isomorphism the degree of a zero-cycle corresponds to the degree of a divisor, giving a short exact sequence

$$0 \to \mathrm{Pic}^{\,0} C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$$

where $\mathrm{Pic}^{\,0} C$ is the group of divisor classes of degree zero.

***Proof*:**
Since $\dim C = 1$, the only nonzero groups are $A_0(C)$ and $A_1(C)$, and $A_k(C) = 0$ for $k \neq 0, 1$. The curve C is irreducible of dimension 1, so by example 1.1.6 its top group is $A_1(C) = Z_1(C) = \mathbb{Z}\,[C]$, free on the fundamental class.

The zero-cycles. A 0-cycle on C is a finite $\mathbb{Z}$-combination of closed points, so $Z_0(C)$ is the group $\mathrm{WDiv}\, C$ of Weil divisors on C. The subgroup $\mathrm{Rat}_0(C)$ of cycles rationally equivalent to zero is generated by the cycles $[\mathrm{div}(r)]$ for $r \in k(W)^*$ ranging over 1-dimensional subvarieties $W \subseteq C$; the only such W is C itself, so $\mathrm{Rat}_0(C)$ is generated by the principal divisors $[\mathrm{div}(r)]$ with

$r \in k(C)^*$. Thus $\mathrm{Rat}_0(C)$ is exactly the group $\mathrm{Prin}\, C$ of principal divisors in the sense of [2, Principal Divisor], and

$$A_0(C) = Z_0(C)/\mathrm{Rat}_0(C) = \mathrm{WDiv}\, C/\, \mathrm{Prin}\, C = \mathrm{Cl}\, C$$

is the divisor class group of [2, Divisor Class Group]. Because C is smooth, the divisor-to-line-bundle correspondence of [2, Divisors: Geometric Representation of Line Bundles] identifies $\mathrm{Cl}\, C$ with $\mathrm{Pic}\, C$.

The degree sequence. On a smooth projective curve linearly equivalent divisors have equal degree: if $D \sim D'$ then $\deg D = \deg D'$, since the Riemann-Roch theorem ([2, Riemann-Roch for curves]) gives $\chi(\mathcal{L}(D)) = \deg D + 1 - g$, a quantity depending only on the class of D, so $\deg D$ is determined by it. Hence deg descends to a homomorphism $\deg\colon A_0(C) \to \mathbb{Z}$, agreeing under the identification $A_0(C) = \mathrm{Cl}\, C$ with the degree of a divisor class. It is surjective because a single closed point has degree $[k(p) : \overline{k}] = 1$, and its kernel is the group $\mathrm{Pic}^0 C$ of degree-zero classes by definition. This gives the stated short exact sequence, completing the proof.

The proposition recovers, in the language of Chow groups, the entire divisor theory of a curve. The degree map splits the class group into a discrete part, the copy of $\mathbb{Z}$ recording degree, and a connected part $\mathrm{Pic}^0 C$, the Jacobian, carrying the continuous moduli of divisor classes of a fixed degree. For $C = \mathbb{P}^1$ the Jacobian is trivial and $A_0(\mathbb{P}^1) = \mathbb{Z}$, in agreement with example 1.1.5 and with the case $n = 1$ of theorem 1.5.2; for a curve of genus $g \geq 1$ the group $\mathrm{Pic}^0 C$ is a g-dimensional abelian variety and $A_0(C)$ is strictly larger than $\mathbb{Z}$, so the degree is no longer a complete invariant of a class. The Chow group of a curve is therefore the first place in the theory where an interesting continuous invariant appears, and it is the model on which the higher-dimensional theory is built.

The Chow group $A_0(C)$ also illustrates the failure of the total-degree map to see everything. On $\mathbb{P}^n$ the degree was a complete invariant of a k-cycle class; on a curve of positive genus it is not, because two points $p, q \in C$ need not be rationally equivalent even though $[p]$ and $[q]$ have the same degree. Their difference $[p] - [q]$ is a nonzero class in $\mathrm{Pic}^0 C$ precisely when p and q are not linearly equivalent, which for $g \geq 1$ is the generic situation. This is the geometric content of the statement that a curve of positive genus admits no nonconstant rational function of degree one.

The last computation of the section is stated but not proved, because its proof needs Segre classes, which are developed later. It generalizes the projective-space computation from the trivial bundle to an arbitrary vector bundle: the Chow groups of a projectivized vector bundle are a free module over the Chow groups of the base, on the powers of the tautological class. This is the projective-bundle formula, and it is the engine that will later define Chern classes.

Given a vector bundle E of rank $r + 1$ on a scheme X, write $\mathbb{P}(E) \to X$ for the associated projective bundle, whose fiber over a point x is the projective space $\mathbb{P}(E_x) \cong \mathbb{P}^r$ of one-dimensional subspaces of the fiber E_x. Let $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in A^1(\mathbb{P}(E))$ be the first Chern class of the Serre-twist line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ (the dual of the tautological subbundle, with the "one-dimensional subspaces" convention above), viewed as an operator on cycles. The projective-bundle formula states that the flat pullback π^* together with capping against powers of ξ makes $A_*(\mathbb{P}(E))$ a free module over $A_*(X)$ on the classes $1, \xi, \ldots, \xi^r$:

$$A_*(\mathbb{P}(E)) = \bigoplus_{i=0}^{r} \xi^i \cap \pi^* A_*(X)$$

Concretely, every class on $\mathbb{P}(E)$ is uniquely $\sum_{i=0}^{r} \xi^i \cap \pi^* \alpha_i$ with $\alpha_i \in A_*(X)$. Taking $X = \mathrm{Spec}\, \overline{k}$ and $E = \overline{k}^{n+1}$ recovers $\mathbb{P}(E) = \mathbb{P}^n$ and the free-module statement collapses to $A_*(\mathbb{P}^n) = \bigoplus_{i=0}^{n} \mathbb{Z}\, \xi^i$

of theorem 1.5.2, with ξ^i the class of a linear subspace of codimension i. The full statement, and the definition of the Chern class operators it rests on, are theorem 3.1.9 in section 3.1; the formula is recorded here because it is the natural common generalization of the two computations above, and because it is the shape every later Chow-group computation of a bundle takes.

Before the exercises, the three worked instances of the theory are collected side by side, so the common mechanism is visible in one place.

Example 1.5.4: The Chow Groups of $\mathbb{P}^1$, $\mathbb{P}^2$, and a Plane Curve

1. *The projective line.* For $n = 1$, theorem 1.5.2 gives $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$ and $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$, so
$$A_*(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$
The degree-0 part $A_0(\mathbb{P}^1) = \mathbb{Z}$ agrees with example 1.1.5, and with proposition 1.5.3 for the genus-zero curve $\mathbb{P}^1$, whose Jacobian $\operatorname{Pic}^0\mathbb{P}^1$ is trivial.

2. *The projective plane.* For $n = 2$,
$$A_*(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2] \oplus \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$
with $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^1]$ generated by the class of a line, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[\mathrm{pt}]$. A plane curve $D \subseteq \mathbb{P}^2$ of degree d has class $[D] = d\,[\mathbb{P}^1] = d\,h_1$ in $A_1(\mathbb{P}^2)$, since D is cut out by a form of degree d and hence is rationally equivalent to d lines (the pencil spanned by the form and ℓ^d for a linear form ℓ sweeps one to the other). Two plane curves of degrees c and d therefore have classes meeting, after the intersection product is available, in cd points, which is Bézout's theorem ([7]) read off from $h_1 \cdot h_1 = h_0$.

3. *A plane curve as a curve in its own right.* Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d, so C is a smooth projective curve of genus $g = \binom{d-1}{2}$. Two computations of its Chow groups coexist. Viewed inside $\mathbb{P}^2$, the class $[C] = d\,h_1 \in A_1(\mathbb{P}^2)$ records only its degree. Viewed intrinsically, proposition 1.5.3 gives $A_0(C) \cong \operatorname{Cl} C = \operatorname{Pic} C$ with its degree sequence $0 \to \operatorname{Pic}^0 C \to A_0(C) \to \mathbb{Z} \to 0$, and for $d \geq 3$ the genus is positive, so $\operatorname{Pic}^0 C$ is nontrivial and $A_0(C)$ is strictly larger than $\mathbb{Z}$. The pushforward $A_0(C) \to A_0(\mathbb{P}^2) = \mathbb{Z}$ along the inclusion is the degree map, collapsing all of $\operatorname{Pic}^0 C$ to zero: the intrinsic Chow group of the curve sees the moduli that the ambient projective plane cannot.

Exercise 1.5.1

1. Compute $A_*(\mathbb{A}^2 \setminus \{p\})$, the Chow groups of the affine plane with one closed point removed. (Use the localization sequence for the closed point inside $\mathbb{A}^2$ together with proposition 1.5.1.)

2. Let $H \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ and $U = \mathbb{P}^n \setminus H \cong \mathbb{A}^n$ its complement. Using the localization sequence, show directly that $j^*\colon A_k(\mathbb{P}^n) \to A_k(\mathbb{A}^n)$ is an isomorphism for $k = n$ and the zero map for $k < n$, and reconcile this with theorem 1.5.2.

3. Let $Q \subseteq \mathbb{P}^3$ be a smooth quadric surface. Assuming the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ and that $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on the classes of a point, the two rulings, and the fundamental class, determine the rank of each $A_k(Q)$ and describe generators. Compare with $A_*(\mathbb{P}^2)$ and explain why $A_1(Q)$ has rank 2 rather than 1.

4. For $\mathbb{P}^m$ and $\mathbb{P}^n$, use theorem 1.5.2 and the projective-bundle formula (with $\mathbb{P}^m \times \mathbb{P}^n = \mathbb{P}(\mathcal{O}^{\oplus(m+1)})$ over $\mathbb{P}^n$) to show that $A_k(\mathbb{P}^m \times \mathbb{P}^n)$ is free of rank equal to the number of pairs (i, j) with $0 \leq i \leq m$, $0 \leq j \leq n$, and $i + j = k$.

5. Let C be a smooth projective curve of genus 1 over $\overline{k}$. Show that $A_0(C)$ is not finitely generated, and identify the classes $[p] - [q]$ of differences of points inside $\operatorname{Pic}^0 C$. Contrast with $A_0(\mathbb{P}^1)$.

6. Let $X \subseteq \mathbb{A}^3$ be the affine cone $x^2 + y^2 = z^2$ over a conic, a surface singular at the origin. Using $A_k(X) = A_k(X_{\mathrm{red}})$ and the vanishing of $A_k(\mathbb{A}^n)$ in low degree as a guide, determine $A_2(X)$ and explain why the singular point does not contribute a class in $A_0(X)$ that survives rational equivalence.

2

Divisors, Line Bundles, and the First Chern Class

Chapter 1 built the Chow groups and their two functorialities, proper pushforward and flat pullback, but produced no way to multiply cycles: nothing so far intersects a k-cycle with anything to lower its dimension. The first and most basic such operation is intersection with a divisor. A Cartier divisor D on X, equivalently a line bundle together with a rational section, cuts each subvariety V not contained in its support down to a $(k-1)$-cycle $D \cdot [V]$, the divisor of the restricted section. This chapter makes that operation precise, proves it descends to rational equivalence, and establishes its one indispensable property, commutativity: intersecting with D and then with D' agrees with intersecting in the other order.

Commutativity is not a formality. It is the identity whose proof, resting on the vanishing degree of a principal divisor on a complete curve, supplies the well-definedness that the affine-bundle-invariance theorem of section 1.4 took on faith. Packaging intersection with a divisor as the *first Chern class* $c_1(L) \cap -$ of the associated line bundle turns these results into the seed of the entire theory of Chern classes, developed for bundles of higher rank in the chapter that follows.

2.1 Cartier Divisors, Line Bundles, and the Class Map

The Chow group of section 1.1 is graded by dimension, and every construction of sections 1.2 to 1.4 either preserves the grading (proper pushforward, flat pullback of the fundamental class) or shifts it by the fixed relative dimension of a flat map. What is still missing is an operation that *lowers* dimension by one, cutting a k-cycle down to a $(k-1)$-cycle by intersecting it with a hypersurface. The classical prototype is Bézout's theorem: a curve of degree d and a curve of degree e in $\mathbb{P}^2$ meet in de points. The intersection is a codimension-one operation, and the object one intersects with is a divisor. This section sets up the divisor side of that operation. It recalls, from *Algebraic Geometry* [2], the two faces of a divisor (the geometric Weil divisor and the sheaf-theoretic Cartier divisor) and the line bundle that packages the Cartier data, then constructs the first bridge from divisor theory to the Chow group: the class map $\mathrm{Pic}\,(X) \to A_{n-1}(X)$ sending a line bundle to the cycle of its rational sections. The intersection operation itself, and the proof that it is well defined on rational equivalence, occupy section 2.2; here the goal is to fix the input and record how it sits inside the Chow group of a fixed variety.

Throughout, X is an algebraic scheme over $\overline{k}$ and V ranges over subvarieties (closed integral subschemes); a point is a closed point, and $n = \dim X$.

Algebraic Geometry [2] developed two theories of divisors, and the reader who has that material in hand can move quickly here. A *Weil divisor* ([2, Weil Divisor]) on a Noetherian integral separated scheme regular in codimension one is a finite formal integer combination $\sum_i n_i[Y_i]$ of prime divisors, where a prime divisor is a codimension-one subvariety. This is precisely a codimension-one algebraic cycle in the sense of section 1.1: when $\dim X = n$, the group $\mathrm{WDiv}\,X$ of Weil divisors is exactly the cycle group $Z_{n-1}(X)$. A *principal divisor* $\mathrm{div}(f) = \sum_Y \nu_Y(f)[Y]$ ([2, Principal Divisor]) records the zeros and poles of a rational function $f \in K(X)^*$, weighted by the order ν_Y measured in the discrete valuation ring $\mathcal{O}_{X,\eta_Y}$; this is the same order-along-a-subvariety $\mathrm{ord}_Y(f)$ that defines rational equivalence in definition 1.1.1. The divisor class group $\mathrm{Cl}\,X = \mathrm{WDiv}\,X/\mathrm{PDiv}(X)$ of [2, Divisor Class Group] is therefore literally the top Chow group $A_{n-1}(X)$ in the case where the only cycles rationally equivalent to zero come from functions on X itself, that is, on a variety.

A *Cartier divisor* ([2, Cartier Divisor]) encodes the same data locally and sheaf-theoretically: it is a global section of $\mathscr{K}_X^*/\mathcal{O}_X^*$, represented by an open cover $\{U_i\}$ with rational functions $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ whose ratios f_i/f_j are units on overlaps. A Cartier divisor is *principal* when it comes from a single global rational function; two Cartier divisors are linearly equivalent when they differ by a principal one. The gluing data $\{(U_i, f_i)\}$ builds a line bundle $\mathcal{O}_X(D)$, the sub-$\mathcal{O}_X$-module of $\mathscr{K}_X$ generated locally by f_i^{-1}, and this assignment is the heart of the theory: $D \mapsto \mathcal{O}_X(D)$ is a homomorphism $\mathrm{CDiv}\,X \to \mathrm{Pic}\,X$ that carries addition of divisors to tensor product of line bundles, sends $-D$ to the dual, and vanishes exactly on principal divisors. On an integral scheme it induces an isomorphism from the Cartier divisor class group $\mathrm{CaCl}\,X$ of [2, Cartier Divisor Class Group] onto the whole Picard group (all of this is developed in [2, Divisors: Geometric Representation of Line Bundles]),

$$\mathrm{CaCl}\,X \xrightarrow{\sim} \mathrm{Pic}\,X$$

so a line bundle is the same thing as a Cartier divisor up to linear equivalence. This is the sense in which the section title lists “Cartier divisors” and “line bundles” as two names for one object.

The reader should keep both faces in view. The Cartier divisor is what one intersects with, because its local equations f_i give the equations to cut against; the line bundle $\mathcal{O}_X(D)$ is the coordinate-free invariant, the thing that survives when the choice of local equations is forgotten. The class map built below is precisely the passage from the line bundle back to a well-defined cycle class.

The two theories agree under a regularity hypothesis, and pinning down that hypothesis is what makes the class map land in the Chow group cleanly. The point is one of local principality: a Weil divisor is Cartier exactly when each of its prime divisors is locally cut out by a single equation. Regularity in codimension one is enough to send a Cartier divisor to a Weil divisor; local factoriality is what makes the map an isomorphism.

Proposition 2.1.1: Weil-Cartier Comparison

Let X be a variety of dimension n over $\overline{k}$.

1. If X is regular in codimension one (for instance X normal), there is a homomorphism
$$\operatorname{CDiv} X \longrightarrow \operatorname{WDiv} X = Z_{n-1}(X), \qquad D \longmapsto [D]$$
carrying principal Cartier divisors to principal Weil divisors. In local data $D = \{(U_i, f_i)\}$ the associated Weil cycle is $[D] = \sum_Y \nu_Y(f_i)[Y]$, the coefficient of a prime divisor Y being $\nu_Y(f_i)$ for any index i with $\eta_Y \in U_i$.

2. If X is locally factorial (every local ring $\mathcal{O}_{X,x}$ is a unique factorization domain; smooth varieties qualify), this map is an isomorphism $\operatorname{CDiv} X \xrightarrow{\sim} \operatorname{WDiv} X$ carrying principal divisors to principal divisors, hence an isomorphism of class groups
$$\operatorname{Pic} X \;\cong\; \operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;=\; A_{n-1}(X)$$

***Proof*:**
Both parts are proved in full in [2, Divisors: Geometric Representation of Line Bundles] and only recalled here. The homomorphism of part (1) is [2, Cartier Divisors to Weil Divisors], and its upgrade to an isomorphism under local factoriality (part (2)) is [2, Weil and Cartier Divisors Under Locally Factorial Hypotheses]; the resulting chain of class-group isomorphisms $\operatorname{Pic} X \cong \operatorname{CaCl} X \cong \operatorname{Cl} X$ is [2, Divisor Class Group and Picard Group]. Only the final equality $\operatorname{Cl} X = A_{n-1}(X)$ is new to this Part, and it is the identification made in the paragraph below: on a variety the codimension-one cycles modulo rational equivalence are exactly the Weil divisors modulo principal divisors, both quotients of $Z_{n-1}(X)$ by $\operatorname{PDiv}(X)$, as we sought to show.

The equality $\operatorname{Cl} X = A_{n-1}(X)$ deserves a moment's reflection, because it is easy to mistake for a definition. Rational equivalence on $Z_{n-1}(X)$ is generated by cycles $[\operatorname{div}(r)]$ where r is a rational function on a subvariety $W \subseteq X$ of dimension n. On a variety X, the only n-dimensional subvariety is X itself, so those generators are exactly the principal Weil divisors $\operatorname{div}(r)$ with $r \in K(X)^*$. Hence $A_{n-1}(X) = Z_{n-1}(X)/\operatorname{Rat}_{n-1}(X)$ and $\operatorname{Cl} X = \operatorname{WDiv} X/\operatorname{PDiv}(X)$ are the same quotient of the same group. Linear equivalence of divisors *is* rational equivalence of codimension-one cycles. This is what will let the class map below produce Chow classes, not just divisor classes.

Example 2.1.2: The Quadric Cone: Cartier Strictly Inside Weil
The comparison map need not be surjective when factoriality fails. On the affine quadric cone $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy-z^2)$, a normal surface singular at the vertex, [2, Cartier Divisors] computes that the ruling $Y = V(y, z)$ is a Weil divisor which is not Cartier at the vertex, while its double $2[Y] = \operatorname{div}(y)$ is, so
$$\operatorname{CaCl} X = 0 \subsetneq \operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z} = A_1(X)$$
the final equality being the identification $\operatorname{Cl} X = A_{n-1}(X)$ of proposition 2.1.1. Normality supplies part (1), but the vertex obstructs local principality, so part (2) fails; on a smooth variety no such gap exists and Weil, Cartier, Picard, and A_{n-1} all coincide.

On a smooth variety the isomorphism $\operatorname{Pic} X \cong A_{n-1}(X)$ of proposition 2.1.1 already provides a bridge from line bundles to Chow classes. To intersect and to compute Chern classes on *arbitrary* algebraic schemes, one needs this bridge without any regularity hypothesis, and one needs it stated at the level of line bundles rather than Cartier divisors, so that the input is a coordinate-free invariant. The construction is direct: a line bundle L has rational sections, a rational section has a divisor of zeros and poles, and that divisor is a codimension-one cycle whose class in $A_{n-1}(X)$ turns out not to depend on the section chosen.

Fix a variety X of dimension n with function field $K(X)$, and a line bundle L on X. A *rational section* of L is a nonzero section of $L \otimes_{\mathcal{O}_X} K(X)$ over X, equivalently a section of L over some nonempty open subset. Since X is a variety, $L \otimes K(X)$ is a one-dimensional $K(X)$-vector space, so rational sections exist and any two differ by multiplication by an element of $K(X)^*$. On the open set where s is defined and nonvanishing, s trivializes L; along each prime divisor Y one may compare s with a local trivialization of L near η_Y and read off an integer $\operatorname{ord}_Y(s)$, the order of vanishing (or of the pole) of s along Y. Concretely, if t is a local generator of L near η_Y and $s = g \cdot t$ with $g \in K(X)^*$, set $\operatorname{ord}_Y(s) := \nu_Y(g)$; a different generator $t' = ut$ with $u \in \mathcal{O}^*_{X,\eta_Y}$ changes g to $u^{-1}g$ and leaves $\nu_Y(g)$ unchanged, so $\operatorname{ord}_Y(s)$ is well defined. The *divisor of* s is the Weil divisor

$$\operatorname{div}(s) \;=\; \sum_Y \operatorname{ord}_Y(s)\,[Y] \;\in\; Z_{n-1}(X)$$

finite because s is a regular nonvanishing section over some nonempty open $U \subseteq X$, where it trivializes L and so $\operatorname{ord}_Y(s) = 0$ for every prime divisor Y meeting U; the only possible contributors are the finitely many prime divisors in the proper closed complement $X \setminus U$. This is the same finiteness that makes the principal divisor $\operatorname{div}(f)$ of [2, Principal Divisor] a well-defined cycle. If $L = \mathcal{O}_X(D)$ for a Cartier divisor D and s is the canonical rational section (the constant $1 \in \mathscr{K}_X$, which against the local generator f_i^{-1} of $\mathcal{O}_X(D)|_{U_i}$ is written $1 = f_i \cdot f_i^{-1}$, giving $\operatorname{ord}_Y(s) = \nu_Y(f_i)$), then $\operatorname{div}(s)$ is exactly the Weil cycle $[D]$ of proposition 2.1.1.

Proposition 2.1.3: The Class Map

Let X be a variety of dimension n. For a line bundle L on X, choose any nonzero rational section s of L and set

$$c(L) \;:=\; [\operatorname{div}(s)] \;\in\; A_{n-1}(X)$$

Then $c(L)$ is independent of the choice of s, and the resulting assignment

$$c\colon \operatorname{Pic} X \longrightarrow A_{n-1}(X), \qquad L \longmapsto c(L)$$

is a group homomorphism. When X is locally factorial (in particular smooth), c is the isomorphism $\operatorname{Pic} X \cong A_{n-1}(X)$ of proposition 2.1.1; on a general variety c need be neither injective nor surjective.

***Proof*:**
Step 1: independence of the section. Let s, s' be two nonzero rational sections of L. Both lie in the one-dimensional $K(X)$-vector space $L \otimes K(X)$, so $s' = gs$ for a unique $g \in K(X)^*$. Along each prime divisor Y, choose a local generator t of L near η_Y and write $s = at$, $s' = a't$ with $a, a' \in K(X)^*$; then $a' = ga$ and

$$\operatorname{ord}_Y(s') = \nu_Y(a') = \nu_Y(g) + \nu_Y(a) = \nu_Y(g) + \operatorname{ord}_Y(s)$$

because ν_Y is a valuation. Summing over Y gives $\operatorname{div}(s') = \operatorname{div}(g) + \operatorname{div}(s)$ in $Z_{n-1}(X)$, where $\operatorname{div}(g) = \sum_Y \nu_Y(g)[Y]$ is the principal Weil divisor of $g \in K(X)^*$. Since $\operatorname{div}(g)$ is a principal divisor on the variety X, it is rationally equivalent to zero in $A_{n-1}(X)$ by the identification $\mathrm{Rat}_{n-1}(X) = \mathrm{PDiv}(X)$ recorded after proposition 2.1.1. Hence $[\mathrm{div}(s')] = [\mathrm{div}(s)]$ in $A_{n-1}(X)$, and $c(L)$ is well defined.

Step 2: homomorphism property. Let L, L' be line bundles with rational sections s, s'. Then $s \otimes s'$ is a nonzero rational section of $L \otimes L'$, and along each prime divisor Y, choosing local generators t of L and t' of L' near η_Y and writing $s = at$, $s' = a't'$, the product $s \otimes s' = (aa')(t \otimes t')$ is expressed against the local generator $t \otimes t'$ of $L \otimes L'$. Thus

$$\mathrm{ord}_Y(s \otimes s') = \nu_Y(aa') = \nu_Y(a) + \nu_Y(a') = \mathrm{ord}_Y(s) + \mathrm{ord}_Y(s')$$

so $\mathrm{div}(s \otimes s') = \mathrm{div}(s) + \mathrm{div}(s')$, and passing to classes gives $c(L \otimes L') = c(L) + c(L')$. The trivial bundle $\mathcal{O}_X$ has the rational section 1 with $\mathrm{div}(1) = 0$, so $c(\mathcal{O}_X) = 0$, and $c(L^{-1}) = -c(L)$ follows from $L \otimes L^{-1} = \mathcal{O}_X$. Therefore c is a homomorphism.

Step 3: comparison with the smooth case. When X is locally factorial, every line bundle is $\mathcal{O}_X(D)$ for a Cartier divisor D, and the canonical rational section of $\mathcal{O}_X(D)$ has $\mathrm{div}(s) = [D]$, the Weil cycle of D. Under $\mathrm{Pic}\, X \cong \mathrm{CaCl}\, X$ ([2, Divisors: Geometric Representation of Line Bundles]) and $\mathrm{CaCl}\, X \cong \mathrm{Cl}\, X = A_{n-1}(X)$ (proposition 2.1.1), the assignment $L = \mathcal{O}_X(D) \mapsto [D]$ is exactly that isomorphism, so c is an isomorphism. On a variety where these hypotheses fail, c can lose injectivity or surjectivity, as example 2.1.2 shows for the class-group comparison. This completes the proof.

The class map is the first of the two ways line bundles act on the Chow group. It records *where* a line bundle is, as a codimension-one class $c(L) \in A_{n-1}(X)$, but it does not yet say how L intersects cycles of other dimensions. That is the job of the operator $c_1(L) \cap -$ built in section 2.4, whose value on the fundamental class $[X]$ recovers $c(L)$: writing ahead, $c_1(L) \cap [X] = c(L)$. The class map is thus the top-dimensional case of the full first Chern class, and everything in this chapter can be read as promoting c from a map on line bundles to an operator on all of $A_*(X)$.

The value of stating c at the level of line bundles, rather than resting on the isomorphism $\mathrm{Pic} \cong A_{n-1}$, is that c makes sense with no regularity hypothesis on X: a rational section and its order along a prime divisor require only that X be a variety, and the group $A_{n-1}(X)$ is defined for every algebraic scheme. The intersection theory of section 2.2 is built for arbitrary X, and it needs c in this general form.

Example 2.1.4: The Class Map on Projective Space

Projective space is where every ingredient can be computed by hand. Since $\mathbb{P}^n$ has all local rings unique factorization domains, it is locally factorial, so $\mathrm{Pic}\, \mathbb{P}^n \cong \mathrm{Cl}\, \mathbb{P}^n \cong \mathbb{Z}$, generated by $\mathcal{O}_{\mathbb{P}^n}(1)$, and by theorem 1.5.2 the top Chow group is $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^{n-1}]$, generated by the class of a hyperplane. The class map identifies the two.

A rational section of $\mathcal{O}_{\mathbb{P}^n}(1)$ is a nonzero linear form. Take $s = x_0$, the section whose zero scheme is the hyperplane $H = V_+(x_0)$, whose Cartier data $\{(U_i, x_0/x_i)\}$ was recorded in [2, Cartier Divisors]. On the standard chart $U_i = D_+(x_i) \cong \mathbb{A}^n$ the sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$ is trivialized by x_i, and $s = x_0 = (x_0/x_i)\, x_i$, so s corresponds to the rational function $x_0/x_i \in K(\mathbb{P}^n)$ against the generator x_i. Along the prime divisor H, the generic point lies in charts U_i with $i \neq 0$, where x_0/x_i is a local equation cutting out H with order one, so $\mathrm{ord}_H(s) = 1$; along every other prime divisor x_0/x_i is a unit near the generic point, contributing zero. Hence

$$\mathrm{div}(x_0) = [H] = [\mathbb{P}^{n-1}], \qquad c\big(\mathcal{O}_{\mathbb{P}^n}(1)\big) = [\mathbb{P}^{n-1}] \in A_{n-1}(\mathbb{P}^n)$$

Choosing instead $s = x_1$ gives the hyperplane $V_+(x_1)$, a different divisor but the same class: $x_1 = (x_1/x_0)\,x_0$ differs from x_0 by the rational function x_1/x_0, and $\operatorname{div}(x_1) - \operatorname{div}(x_0) = \operatorname{div}(x_1/x_0) \sim 0$, confirming the independence of proposition 2.1.3 in this case. More generally, $\mathcal{O}_{\mathbb{P}^n}(d)$ has rational sections given by degree-d forms; the section x_0^d has divisor $d[H]$, so $c(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,[\mathbb{P}^{n-1}]$, and $c\colon \mathbb{Z} \to \mathbb{Z}$ is the identity. The generator $\mathcal{O}(1)$ of Pic maps to the generator [hyperplane] of A_{n-1}, and the class map is the announced isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$.

There is one further refinement to record before intersecting, and it addresses a bookkeeping problem rather than a new geometric idea. When a Cartier divisor D is intersected with a subvariety V, one wants the resulting cycle to live on $V \cap |D|$, the part of V where D cuts, not just somewhere on V. Localizing the answer to the support is what makes the intersection product refined enough to carry information (for instance, an excess-intersection or a self-intersection formula) and what makes it functorial under the localization sequence of theorem 1.4.1. The device that carries this localized data is the *pseudo-divisor*: a line bundle, a chosen closed support, and a trivialization of the bundle away from that support.

Definition 2.1.5: Pseudo-Divisor

Let X be an algebraic scheme. A *pseudo-divisor* on X is a triple

$$\mathfrak{D} = (L, Z, s)$$

consisting of a line bundle L on X, a closed subset $Z \subseteq X$ (the *support*, written $|\mathfrak{D}|$), and a trivialization $s\colon \mathcal{O}_{X\setminus Z} \xrightarrow{\sim} L|_{X\setminus Z}$ of L over the complement of Z. The line bundle L is denoted $\mathcal{O}_X(\mathfrak{D})$.

A Cartier divisor D on X determines a pseudo-divisor $(\mathcal{O}_X(D),\ |D|,\ s_D)$ whose support is the support $|D|$ of D and whose trivialization s_D is the canonical rational section of $\mathcal{O}_X(D)$, which is a nonvanishing regular section away from $|D|$. Two Cartier divisors D, D' with the same support and the same associated line bundle give the same pseudo-divisor when their canonical sections agree off the support.

The three pieces of data are exactly what a divisor supplies for cutting. The line bundle L is the coordinate-free invariant whose class map value is $c(L)$; the support Z pins down where the cutting happens, so the product lands in $A_*(Z)$ rather than $A_*(X)$; and the trivialization s off Z is what lets one measure orders of vanishing, because on the locus where L is trivialized by s there is nothing to cut, and all the divisorial content is concentrated on Z. A Cartier divisor is the special case where $Z = |D|$ is as small as possible, but allowing Z to be larger is what gives the flexibility to restrict, pull back, and add pseudo-divisors while keeping supports under control. The next section takes a subvariety V not contained in the support and defines $\mathfrak{D} \cdot [V]$ as a class in $A_{k-1}(V \cap Z)$; when $V \subseteq Z$, the restricted line bundle $L|_V$ supplies the answer through its own class map. Here the pseudo-divisor is introduced only to fix the input; its use is the content of section 2.2.

Example 2.1.6: The Pseudo-Divisor of a Hyperplane

Let $X = \mathbb{P}^n$ and $H = V_+(x_0)$ the hyperplane at $x_0 = 0$. The associated pseudo-divisor is

$$\mathfrak{H} = \big(\mathcal{O}_{\mathbb{P}^n}(1),\ H,\ s\big)$$

with support the hyperplane H and trivialization s given by the section x_0 over the complement $\mathbb{P}^n \setminus H = D_+(x_0) \cong \mathbb{A}^n$, where x_0 is nonvanishing and so trivializes $\mathcal{O}(1)$. The line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$ carries the class $c(\mathcal{O}(1)) = [H] \in A_{n-1}(\mathbb{P}^n)$ computed in example 2.1.4; the support restricts any intersection against $\mathfrak{H}$ to H; and the trivialization records that off H the hyperplane cuts nothing. Intersecting $\mathfrak{H}$ with a subvariety $V \not\subseteq H$ will, in section 2.2, produce the cycle $[V \cap H]$ with its multiplicities, supported on $V \cap H$, which is the algebraic incarnation of Bézout's degree count.

The class map and the pseudo-divisor together fix the two ways a line bundle enters intersection theory: as a class $c(L) \in A_{n-1}(X)$, which is its total content, and as a localized cutting datum (L, Z, s), which is how it acts. With the input settled, the operation $\mathfrak{D} \cdot -$ can be defined and shown well defined on rational equivalence, which is the subject of section 2.2.

Exercise 2.1.1

1. Let $X = \mathbb{P}^n$ and let $Q = V_+(q)$ be a smooth quadric hypersurface, cut out by a nondegenerate quadratic form q. Compute $c(\mathcal{O}_{\mathbb{P}^n}(Q))$, the class map value of the line bundle of Q, in $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^{n-1}]$, and explain the answer in terms of the degree of Q.

2. Let X be a variety and $r \in K(X)^*$ a nonzero rational function. Show that the line bundle $\mathcal{O}_X(\operatorname{div}(r))$ associated to the principal Cartier divisor $\operatorname{div}(r)$ is trivial, and deduce that $c(\mathcal{O}_X(\operatorname{div}(r))) = 0$ directly from proposition 2.1.3. Which step of the class-map construction is this computing?

3. On the affine quadric cone $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$ of example 2.1.2, the class map $c\colon \operatorname{Pic} X \to A_1(X)$ is a homomorphism $\operatorname{CaCl} X \to \operatorname{Cl} X$. Using the values $\operatorname{CaCl} X = 0$ and $\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z}$ from that example, describe c explicitly and state whether it is injective, surjective, or neither. What does this say about the necessity of the factoriality hypothesis in proposition 2.1.1(2)?

4. Pseudo-divisors add: given $\mathfrak{D} = (L, Z, s)$ and $\mathfrak{D}' = (L', Z', s')$, define $\mathfrak{D} + \mathfrak{D}' := (L \otimes L',\ Z \cup Z',\ s \otimes s')$. Verify that this is a well-defined pseudo-divisor, that $\mathcal{O}_X(\mathfrak{D} + \mathfrak{D}') = \mathcal{O}_X(\mathfrak{D}) \otimes \mathcal{O}_X(\mathfrak{D}')$, and that on the line-bundle level this addition is compatible with the homomorphism property of the class map in proposition 2.1.3.

5. Let L be a line bundle on a variety X, s a nonzero rational section, and Y a prime divisor. Show that the order $\operatorname{ord}_Y(s) := \nu_Y(a)$, defined by writing $s = a\,t$ against a local generator t of L near η_Y with $a \in K(X)^*$, does not depend on the choice of generator t. Then show that if s is a regular nonvanishing section of L near η_Y, then $\operatorname{ord}_Y(s) = 0$.

2.2 Intersecting with a Divisor

The class map of section 2.1 sends a line bundle to a cycle class in codimension one, but it says nothing about how that cycle meets the other cycles already living on X. This section supplies the missing operation. A Cartier divisor D acts on cycles by cutting each subvariety down along the zeros and poles of a rational section, an operation that lowers dimension by one and records where the cut takes place. The construction is the first instance of a general principle of the subject, that a line bundle acts on the Chow group as an operator, and it is the operation whose associativity and commutativity the rest of the chapter is built to establish.

The geometry is the one already visible in codimension one. A Cartier divisor is locally the divisor of a single rational function f_i on an open U_i, and the transition ratios f_i/f_j are units, so the zeros and poles of the local equations patch to a well-defined codimension-one cycle on X. To intersect D with a k-dimensional subvariety V, restrict this local data to V and read off the resulting $(k-1)$-cycle on V. Two features force themselves on the construction. First, the answer is supported on the intersection $V \cap |D|$, since away from the support of D the local equations are units and contribute nothing. Second, the recipe stumbles when V lies inside $|D|$, where the local equations $f_i|_V$ may vanish identically and $\operatorname{div}(f_i|_V)$ is undefined; the trivialization off $|D|$ is exactly what is lost. The definition below treats the two cases separately and unifies them through the restricted line bundle.

Definition 2.2.1: Intersection with a Cartier Divisor

Let D be a Cartier divisor on X with support $|D|$ and associated line bundle $\mathcal{O}_X(D)$, and let $V \subseteq X$ be a k-dimensional subvariety. The *intersection class* $D \cdot [V] \in A_{k-1}(V \cap |D|)$ is defined as follows.

1. If $V \not\subseteq |D|$, then the local equations of D restrict to a nonzero rational function s_V on V: choosing a representative $\{(U_i, f_i)\}$ of D, the ratios $f_i|_V$ agree up to units on overlaps and glue to a rational section s_V of $\mathcal{O}_X(D)|_V$ that is nonzero on the dense open $V \setminus |D|$. Its divisor is a $(k-1)$-cycle supported on $V \cap |D|$, and one sets
$$D \cdot [V] = \big[\operatorname{div}(s_V)\big] \in A_{k-1}(V \cap |D|)$$
2. If $V \subseteq |D|$, then no such nonzero section exists, and one instead uses the restricted line bundle. The class map of proposition 2.1.3 applied to $\mathcal{O}_X(D)|_V$ produces a class in $A_{k-1}(V)$; since $V \subseteq |D|$ gives $V \cap |D| = V$, one sets
$$D \cdot [V] = c\big(\mathcal{O}_X(D)|_V\big) \in A_{k-1}(V) = A_{k-1}(V \cap |D|)$$

Extending $\mathbb{Z}$-linearly over the subvarieties appearing in a cycle $\alpha = \sum_i n_i [V_i]$ gives the *intersection cycle* $D \cdot \alpha = \sum_i n_i \,(D \cdot [V_i])$, a class in $A_{k-1}(|D| \cap \operatorname{Supp} \alpha)$, where $\operatorname{Supp} \alpha = \bigcup_i V_i$.

Two conventions in the definition deserve unpacking, since both recur throughout the theory. The first is that the intersection class is recorded on the intersection $V \cap |D|$, not on all of X. This is a refinement one does not want to discard prematurely: the intersection of a curve with a divisor is a set of points on the curve, and remembering that the points lie in $V \cap |D|$ is exactly what makes localized intersection numbers and excess-intersection formulas possible later. When only the class in $A_{k-1}(X)$ is wanted, push forward along the inclusion $V \cap |D| \hookrightarrow X$, a proper map. The second convention is the treatment of $V \subseteq |D|$. Here the naive recipe fails because the local equations restrict to zero, so there is no rational section to take the divisor of; but $\mathcal{O}_X(D)|_V$ is still a line bundle on V, and its class under the class map $c\colon \operatorname{Pic}(V) \to A_{k-1}(V)$ of proposition 2.1.3 is a $(k-1)$-cycle class. Both clauses are the same construction, taking the class of the restricted bundle $\mathcal{O}_X(D)|_V$. When $V \not\subseteq |D|$ the bundle comes with a preferred rational section s_V and the class is computed as $[\operatorname{div}(s_V)]$; when $V \subseteq |D|$ that section degenerates and only the bundle survives. The packaging of the whole operation as the first Chern class $c_1(\mathcal{O}_X(D)) \cap -$ is the subject of section 2.4.

That the operation lowers dimension by exactly one is built into the definition: the divisor of a rational section on a k-dimensional variety is a codimension-one cycle, hence $(k-1)$-dimensional. Geometrically, $D \cdot [V]$ is the divisor cut out on V by the equation defining D, the higher-dimensional form of substituting a curve into the equation of a hypersurface and reading off the intersection points with multiplicity. The next example makes this concrete in the case a reader already knows from the classical degree of a plane curve.

Example 2.2.2: A Hyperplane Meeting a Plane Curve

Let $X = \mathbb{P}^2$ with homogeneous coordinates $[x_0 : x_1 : x_2]$, let $H = \{x_0 = 0\}$ be the line at infinity, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(H) = \mathcal{O}_{\mathbb{P}^2}(1)$, and let $V \subseteq \mathbb{P}^2$ be a smooth conic, say the closure of $\{x_1^2 = x_0x_2\}$, a subvariety of dimension 1 not contained in H. The unique intersection point turns out to be $p = [0 : 0 : 1]$, which lies in the chart $U_2 = \{x_2 \neq 0\}$; work there, with affine coordinates $u = x_0/x_2$ and $v = x_1/x_2$. On U_2 the conic is $V = \{u = v^2\}$ (from $x_1^2 = x_0x_2$), the divisor $H = \{x_0 = 0\}$ is $\{u = 0\}$, and against the local generator x_2 of $\mathcal{O}_{\mathbb{P}^2}(1)$ the section x_0 representing H is the function $x_0/x_2 = u$. Its restriction to V is $s_V = u|_V = v^2$, which vanishes exactly where V meets H: $u = 0$ forces $v = 0$, so p is the only intersection point, and there V is tangent to H. In the local coordinate v on V near p, the function $s_V = v^2$ has a double zero (the conic meets its tangent line to order two), so

$$H \cdot [V] = \big[\operatorname{div}(s_V)\big] = 2\,[p] \in A_0(V \cap H)$$

Pushing forward to $\mathbb{P}^2$ and taking degrees gives $\deg(H \cdot [V]) = 2$, the degree of the conic. The intersection class remembers more than the number 2: it records that both units of intersection are concentrated at the single point p, information the bare degree discards.

The example illustrates the two things the intersection class is meant to capture, the total count and its distribution, and shows how the classical intersection number of two plane curves reappears as the degree of $D \cdot [V]$. For a curve V of degree d and a line H in general position the same computation gives d distinct simple points, and $\deg(H \cdot [V]) = d$ in every case; this is the statement, once the well-definedness below is in hand, that intersecting with a hyperplane computes the degree. Before turning to that, the second worked example records the complementary phenomenon, what happens when V lies inside the support of D, so that the second clause of definition 2.2.1 is in force.

Example 2.2.3: A Divisor Restricted to a Component of Its Support

Let $X = \mathbb{P}^2$ again and let $D = L$ be a line, say $L = \{x_0 = 0\}$, so $|D| = L$ and $\mathcal{O}_{\mathbb{P}^2}(L) = \mathcal{O}_{\mathbb{P}^2}(1)$. Take $V = L$ itself, a 1-dimensional subvariety contained in $|D|$. The first clause does not apply, since x_0 restricts to zero on L; the second clause instructs one to compute the class of the restricted line bundle $\mathcal{O}_{\mathbb{P}^2}(1)|_L$. Now $L \cong \mathbb{P}^1$, and $\mathcal{O}_{\mathbb{P}^2}(1)|_L = \mathcal{O}_{\mathbb{P}^1}(1)$ by the adjunction of the hyperplane bundle to a linear subspace. The class map $\operatorname{Pic}(\mathbb{P}^1) = \mathbb{Z} \to A_0(\mathbb{P}^1) = \mathbb{Z}$ of proposition 2.1.3 carries $\mathcal{O}_{\mathbb{P}^1}(1)$ to the class of a single point $[q]$, so

$$L \cdot [L] = c\big(\mathcal{O}_{\mathbb{P}^2}(1)|_L\big) = [q] \in A_0(L)$$

the class of one point on L. This is the self-intersection $L \cdot L = 1$ of a line in the plane, the positive-degree form of the fact that two distinct lines meet in one point: moving L to a nearby line L' and intersecting $L \cdot [L'] = [L \cap L']$ recovers the same class, now visibly one transverse point. The second clause of the definition is what lets self-intersection make sense at all, since a divisor cannot literally be substituted into a variety it defines.

The two examples exhibit the operation on both sides of the containment condition and already hint at the compatibility the operation must have with rational equivalence. In example 2.2.3 the value $L \cdot [L]$ was computed by restricting the bundle, but it was also claimed to agree with $L \cdot [L']$ for a nearby line L', obtained by the first clause. Since $L \sim L'$ as divisors, these two computations must give the same class if the operation is to depend only on the rational-equivalence class of the cycle, and only then does the self-intersection number deserve its name. Establishing that agreement in general is the content of the section, and it is the property that will let the intersection class be defined on the Chow group rather than on cycles alone.

The stake is higher than notational convenience. Intersection with a divisor is meant to be an operator on $A_*(X)$, so that $D \cdot -$ carries $A_k(X)$ to $A_{k-1}(X)$ and can be iterated; without invariance under rational equivalence the operation lives only on cycles, where iteration is not controlled and no ring structure can form. The following proposition is the foundation of everything downstream, and it is one half of the identity (1.3) that the affine-bundle-invariance theorem of section 1.4 deferred to this chapter, the half asserting that $D \cdot -$ is well defined; the commutativity half is section 2.3.

Proposition 2.2.4: Rational Invariance of Divisor Intersection

Let D be a Cartier divisor on X. If $\alpha \in Z_k(X)$ is rationally equivalent to zero and no component of α lies in $|D|$, then $D \cdot \alpha \sim 0$ in $A_{k-1}(|D|)$. Consequently, for any class $a \in A_k(X)$ the intersection $D \cdot a \in A_{k-1}(|D|)$ is well defined by choosing a representative cycle none of whose components lies in $|D|$, and $D \cdot -$ descends to a homomorphism

$$D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$$

independent of all choices.

The proof is the technical heart of the divisor calculus, and it is worth naming the obstacle before entering it. A cycle rationally equivalent to zero is a sum of divisors $[\mathrm{div}(r)]$ of rational functions r on $(k+1)$-dimensional subvarieties W. Intersecting with D replaces each such W-divisor by its trace under the section s_W of $\mathcal{O}_X(D)|_W$, and the claim is that this trace is again a boundary. On the surface W there are now two rational functions in play, the given r and the local equation of D, and the argument turns on the symmetry between them: the order of r measured along the zeros of D matches the order of the equation of D measured along the zeros of r, up to a principal divisor. That symmetry is a statement about a two-dimensional surface, and it is proved by pushing everything down to a complete curve, where a principal divisor has degree zero. Three inputs carry the argument: this degree-zero fact, proposition 1.2.2; the reciprocity lemma on the auxiliary curve that turns it into the two-function symmetry; and the localization sequence theorem 1.4.1 reducing the boundary strata to lower dimension. The degree-zero fact is why the argument could not be given in section 1.4 before the divisor machinery was in place.

***Proof*:**
By linearity it suffices to treat a single generator of $\mathrm{Rat}_k(X)$, so let $W \subseteq X$ be a $(k+1)$-dimensional subvariety and $r \in k(W)^*$, and set $\alpha = [\mathrm{div}(r)]$. The hypothesis that no component of α lies in $|D|$ is used only to make each $D \cdot [V]$ fall under the first clause of definition 2.2.1; the reduction below produces $D \cdot \alpha$ as the divisor of a rational function, and the containment cases are absorbed at the end.

Step 1: Reduction to the surface W. Replacing X by W and D by its restriction $D|_W$ changes neither side: the components V of $[\mathrm{div}(r)]$ are codimension-one subvarieties of W, the section s_V used to form $D \cdot [V]$ is the restriction to V of the section s_W of $\mathcal{O}_X(D)|_W$, and the inclusion $W \hookrightarrow X$ carries $A_{k-1}(|D|_W|) \to A_{k-1}(|D|)$ compatibly. It therefore suffices to prove the statement with $X = W$, so that D is a Cartier divisor on the $(k+1)$-dimensional variety W, $r \in k(W)^*$, and $D \cdot [\mathrm{div}(r)]$ is to be shown rationally equivalent to zero in $A_{k-1}(W)$. Restrict further to the dense open $W^\circ \subseteq W$ where D is principal, given by a single rational function $g \in k(W)^*$ with $\mathcal{O}_W(D)|_{W^\circ} = \mathcal{O}_{W^\circ} \cdot g^{-1}$; the codimension-one subvarieties of W meeting W°

are exactly those on which the computation is nontrivial, and those contained in the complement $W \setminus W^\circ$ contribute to both a principal divisor and the divisor of D and are handled by the localization sequence in Step 4. On W° the section s_V of clause (1) is $g|_V$, so

$$D \cdot [\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r)\, \big[\operatorname{div}_V(g|_V)\big]$$

the outer sum over the codimension-one subvarieties $V \subseteq W$ with r having a zero or pole along V, and the inner divisor formed on each such V.

Step 2: The two-function symmetry. The right-hand side is not visibly symmetric in r and g, but its symmetrization is the object that vanishes. Consider the two iterated sums

$$\Phi(r,g) = \sum_V \operatorname{ord}_V(r)\, \big[\operatorname{div}_V(g|_V)\big], \qquad \Phi(g,r) = \sum_V \operatorname{ord}_V(g)\, \big[\operatorname{div}_V(r|_V)\big]$$

both $(k-1)$-cycles on W, where in each the outer sum runs over codimension-one subvarieties $V \subseteq W$ and the inner divisor is formed on V of the restricted function. The claim of this step is the tame symmetry

$$\Phi(r,g) - \Phi(g,r) \sim 0 \quad \text{in } A_{k-1}(W)$$

Granting it, the proof is nearly complete: $\Phi(g,r) = \sum_V \operatorname{ord}_V(g)\, [\operatorname{div}_V(r|_V)]$ is a $\mathbb{Z}$-linear combination of divisors of the rational functions $r|_V$ on the k-dimensional subvarieties V, hence lies in $\operatorname{Rat}_{k-1}(W)$ by definition 1.1.3; combined with the symmetry, $\Phi(r,g) = D \cdot [\operatorname{div}(r)]$ is rationally equivalent to $\Phi(g,r) \sim 0$, which is the assertion.

Step 3a: Localizing the symmetry at a codimension-two locus. The symmetry is a statement whose content is concentrated in codimension two on W, so fix a codimension-two subvariety $Z \subseteq W$, a $(k-1)$-dimensional variety, and compare the coefficients of $[Z]$ on the two sides. A codimension-one $V \subseteq W$ contributes to the coefficient of $[Z]$ in $\Phi(r,g)$ only when $Z \subseteq V$, that is when Z is a prime divisor of V; write the coefficient of $[Z]$ in $\Phi(r,g)$ as

$$c_Z(r,g) = \sum_{Z \subseteq V} \operatorname{ord}_V(r) \operatorname{ord}_{Z,V}(g|_V)$$

the sum over codimension-one V with $Z \subseteq V \subseteq W$, and $\operatorname{ord}_{Z,V}$ the order along Z computed on V. The task is to show $c_Z(r,g) = c_Z(g,r)$ modulo the coefficient of $[Z]$ in a principal divisor, and this is a computation on the two-dimensional local geometry of W along Z. Passing to the local ring $\mathcal{O}_{W,Z}$, a two-dimensional local domain with fraction field $L = k(W)$ and residue field $K = k(Z)$, the codimension-one V containing Z correspond to the height-one primes $\mathfrak{p} \subseteq \mathcal{O}_{W,Z}$; each carries a discrete valuation $\operatorname{ord}_{\mathfrak{p}} = \operatorname{ord}_V$ on L, and the residue field $k(\mathfrak{p})$ of $\mathfrak{p}$ is the function field $k(V)$ of the divisor V. In these terms

$$c_Z(r,g) = \sum_{\mathfrak{p}} \operatorname{ord}_{\mathfrak{p}}(r) \operatorname{ord}_Z(g \bmod \mathfrak{p})$$

where $g \bmod \mathfrak{p}$ is the image of g in $k(\mathfrak{p}) = k(V)^*$ and ord_Z on $k(V)$ is the order along Z regarded as a prime divisor of V. The two factors play asymmetric roles: $\operatorname{ord}_{\mathfrak{p}}(r)$ is the transverse order of r along the entire divisor V, a single integer, while $\operatorname{ord}_Z(g \bmod \mathfrak{p})$ is the order of the residue $g|_V$ along the codimension-one subvariety $Z \subseteq V$. Swapping r and g interchanges these two roles rather than permuting equal factors, so $c_Z(r,g) = c_Z(g,r)$ is a genuine reciprocity between a transverse divisorial order and a residual order, not the commutativity of a product.

Step 3b: The exceptional curve and the tame symbol. The two integers in each term of $c_Z(r,g)$, the transverse order $\operatorname{ord}_{\mathfrak{p}}(r)$ and the residual order $\operatorname{ord}_Z(g \bmod \mathfrak{p})$, are the two ends of a single valuation datum, and completeness enters by resolving the two-dimensional geometry of $\mathcal{O}_{W,Z}$ into a curve. Choose a proper birational morphism $\pi\colon S \to \operatorname{Spec}\mathcal{O}_{W,Z}$ with S a regular two-dimensional scheme (a resolution of the localized surface, by normalization and blow-up), and let $E = \pi^{-1}(\mathfrak{m})$ be its exceptional fibre over the closed point, a complete curve over K whose integral components are $E_1,\dots,E_m$, each a complete curve over K with function field $k(E_i)$. The coefficient $c_Z(r,g)$ is invariant under π: since π is proper and birational, proposition 1.2.2 identifies the coefficient of $[Z]$ downstairs with the coefficient of the exceptional fibre upstairs, and among the prime divisors of S only the exceptional E_i map into Z, so the strict transforms of the V drop out and the coefficient is carried by $E_1,\dots,E_m$ alone. Along each exceptional component E_i the function r has a transverse order $a_i = \operatorname{ord}_{E_i}(r)$ and g a transverse order $b_i = \operatorname{ord}_{E_i}(g)$, both single integers, and once these orders are cleared r and g restrict to rational functions on E_i; their zeros and poles on E_i are exactly the points where the strict transforms $\widetilde{V}$ meet E_i. Attach to E_i the *tame symbol* of r and g, the residue element

$$\tau_i(r,g) = (-1)^{a_i b_i}\left(\frac{r^{b_i}}{g^{a_i}}\right)\Big|_{E_i} \in k(E_i)^*$$

which is a well-defined nonzero rational function on E_i because r^{b_i}/g^{a_i} has order zero along E_i. Reading off its order at a closed point $P \in E_i$ gives $\operatorname{ord}_P(\tau_i(r,g)) = b_i \operatorname{ord}_P(r|_{E_i}) - a_i \operatorname{ord}_P(g|_{E_i})$. The contribution of (i,P) to $c_Z(r,g)$ is the transverse order of r against the residual order of g, namely $a_i \operatorname{ord}_P(g|_{E_i})$, and its contribution to $c_Z(g,r)$ is $b_i \operatorname{ord}_P(r|_{E_i})$; their difference is thus $-\operatorname{ord}_P(\tau_i(r,g))$. Summing over S and using the π-invariance above, the two coefficients organize as

$$c_Z(r,g) - c_Z(g,r) = -\sum_{i=1}^{m}\sum_{P\in E_i}[k(P):K]\operatorname{ord}_P\bigl(\tau_i(r,g)\bigr)$$

the double sum over the components E_i of the exceptional curve and their closed points P. The pairing is asymmetric: the summand pairs the transverse order of one function along E_i against the residual order of the other at P, and swapping r and g replaces $\tau_i(r,g)$ by $\tau_i(r,g)^{-1}$ up to sign, so the difference is the sum of the point orders of the tame symbols, not a rearrangement of a symmetric product $\operatorname{ord}_P(r)\operatorname{ord}_P(g)$.

Step 3c: Assembling by degree zero. Each tame symbol $\tau_i(r,g)$ is a rational function on the complete curve E_i over K, so its divisor $\operatorname{div}_{E_i}(\tau_i(r,g))$ is a principal divisor on a complete curve. By proposition 1.2.2(1) applied to the proper structure map $E_i \to \operatorname{Spec} K$, its degree over K vanishes: $\deg_K \operatorname{div}_{E_i}(\tau_i(r,g)) = \sum_{P\in E_i}[k(P):K]\operatorname{ord}_P(\tau_i(r,g)) = 0$ for every i. Summing over the finitely many components,

$$c_Z(r,g) - c_Z(g,r) = -\sum_{i=1}^{m}\deg_K \operatorname{div}_{E_i}\bigl(\tau_i(r,g)\bigr) = 0$$

so $c_Z(r,g) = c_Z(g,r)$ for every codimension-two Z. This is the reciprocity that a naive term-by-term rearrangement would miss: it pairs the transverse order of one function against the residual order of the other, and its vanishing rests on completeness of each exceptional component E_i through the degree-zero property alone, with no external tame-symbol machinery imported. Consequently $\Phi(r,g) = \Phi(g,r)$ coefficient by coefficient on the locus where D is principal, any residual difference being a principal divisor supported off W°. This is the symmetry claimed in Step 2, as needed.

Step 4: The boundary components. It remains to account for the codimension-one subvarieties $V \subseteq W \setminus W^\circ$, where D was not represented by the single function g. Let $U = W^\circ$ and $Y = W \setminus W^\circ$ be its closed complement, of codimension at least one in W. The localization sequence (theorem 1.4.1) for $Y \subseteq W$,

$$A_{k-1}(Y) \to A_{k-1}(W) \to A_{k-1}(U) \to 0$$

is right exact, and Steps 1 through 3 established $D \cdot [\mathrm{div}(r)] \sim \Phi(g, r) \sim 0$ after restriction to U, that is, the class of $D \cdot [\mathrm{div}(r)]$ dies in $A_{k-1}(U)$. By exactness its class in $A_{k-1}(W)$ comes from $A_{k-1}(Y)$; but Y has dimension at most k, and the restriction $D|_Y$ is again a Cartier divisor, so the boundary class is computed by the same recipe on the lower-dimensional Y. To pin the inductive object down, decompose $Y = \bigcup_j Y_j$ into irreducible components, each of dimension at most k, with inclusions $\iota_j : Y_j \hookrightarrow W$. The class in $A_{k-1}(Y)$ produced by the boundary of the localization sequence is represented by $\sum_j (\iota_j)_*\big(D|_{Y_j} \cdot [\mathrm{div}(r_j)]\big)$, where $r_j := r|_{Y_j} \in k(Y_j)^*$ is the restriction of r to the generic point of Y_j, a nonzero rational function since the zero and pole components of r were split off in Steps 1 through 3, and $[\mathrm{div}(r_j)]$ is the principal cycle it cuts on the variety Y_j of dimension at most k. Each summand is thus $D|_{Y_j}$ intersected with a principal cycle on Y_j, the exact object to which the proposition applies one dimension lower. A Noetherian induction on $\dim W$ closes the argument: the base case $\dim W = 0$ is vacuous, and for the inductive step each summand $D|_{Y_j} \cdot [\mathrm{div}(r_j)]$ is rationally trivial on Y_j by the inductive hypothesis applied to the Cartier divisor $D|_{Y_j}$ and the principal cycle $[\mathrm{div}(r_j)]$ on the lower-dimensional Y_j; pushing forward along ι_j preserves rational equivalence, so the whole boundary class in $A_{k-1}(Y)$ vanishes. Hence $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(W)$, and pushing forward along $W \hookrightarrow X$ gives $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(|D|)$.

Step 5: Well-definedness of $D \cdot a$. Fix a class $a \in A_k(X)$. Every rational-equivalence class has a representative cycle none of whose finitely many components lies in $|D|$, since a component inside $|D|$ can be replaced by a rationally equivalent cycle whose components meet $|D|$ properly; on such a representative every $D \cdot [V]$ falls under the first clause of definition 2.2.1. Let $\alpha \sim \alpha'$ be two such representatives, so their difference lies in $\mathrm{Rat}_k(X)$, a finite $\mathbb{Z}$-combination of generators $[\mathrm{div}(r)]$ with $r \in k(W)^*$ on $(k+1)$-dimensional W. Steps 1 through 4 give $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(|D|)$ for each generator, so summing gives $D \cdot (\alpha - \alpha') \sim 0$, and $D \cdot a := [D \cdot \alpha]$ is independent of the chosen representative. Additivity in a is immediate from additivity of $D \cdot -$ on cycles, so $D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$ is a homomorphism, completing the proof.

The proposition is the license to speak of $D \cdot a$ for a class a, and with it the intersection with a divisor becomes an operator on the Chow group. Read backward, the argument explains why the well-definedness was out of reach in section 1.4: the entire content is the tame symmetry of Step 2, and that symmetry rests on the degree-zero property of a principal divisor on a complete curve, a fact about the divisor calculus that section did not yet have. The specialization map used there to invert affine-bundle pullback was built from exactly the operation $\mathrm{div}(t) \cdot -$ for the coordinate t, and its descent to Chow groups is this proposition applied to $D = \{t = 0\}$; the forward reference in that proof pointed here.

Two features of the argument recur and are worth isolating. The reduction of Step 1, replacing X by the $(k+1)$-dimensional W on which r lives, is the standard move that turns a statement about intersecting a principal cycle into a statement about a single surface with two functions on it, and it will reappear when commutativity is proved in section 2.3. The symmetry of Step 3, that pairing the zeros of r against g equals pairing the zeros of g against r, is the algebraic form of Weil reciprocity, here proved from the ground up rather than cited: the tame symbol is constructed by hand along each component of the exceptional curve, and its reciprocity is reduced to the vanishing degree of a principal divisor on that complete curve, so no external tame-symbol machinery is imported. The same reciprocity, promoted one dimension higher, is what makes $D \cdot D' = D' \cdot D$, and the two proofs share their engine.

The operator now available, $D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$, invites the computation that motivated the whole construction. Intersecting repeatedly with a hyperplane on projective space should descend through the dimensions and terminate in a point, recovering the degree; the next example carries this out and previews the Chern-class computation of section 2.4.

Example 2.2.5: Iterated Hyperplane Sections on Projective Space

Let $X = \mathbb{P}^n$ and $H = \{x_0 = 0\}$ a hyperplane, so $\mathcal{O}_{\mathbb{P}^n}(H) = \mathcal{O}_{\mathbb{P}^n}(1)$ and $|H| \cong \mathbb{P}^{n-1}$. By theorem 1.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ with h_k the class of a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$. To compute $H \cdot h_k$, represent h_k by a linear $\mathbb{P}^k$ meeting H transversally, that is by an $L \cong \mathbb{P}^k$ not contained in H; then $L \cap H$ is a linear $\mathbb{P}^{k-1}$, and the restricted section x_0/x_j on L has its zero divisor equal to $L \cap H$ with multiplicity one. Hence

$$H \cdot h_k = [L \cap H] = h_{k-1} \in A_{k-1}(\mathbb{P}^{n-1})$$

pushed into $\mathbb{P}^n$, the class of a linear $\mathbb{P}^{k-1}$. Intersecting n times starting from the fundamental class $h_n = [\mathbb{P}^n]$,

$$\underbrace{H \cdot H \cdots H}_{k} \cdot [\mathbb{P}^n] = h_{n-k}$$

and in particular $H^n \cdot [\mathbb{P}^n] = h_0 = [\mathrm{pt}]$, a single reduced point of degree one. The iterated intersection walks down the tower of linear subspaces one step at a time, and its terminal value is the class of a point, whose degree 1 is the degree of $\mathbb{P}^n$ under the embedding $\mathcal{O}(1)$. The choice of a transversal representative L at each stage is legitimated precisely by proposition 2.2.4: the answer h_{k-1} does not depend on which $\mathbb{P}^k$ in the class h_k is used, so long as it meets H properly.

The computation is the prototype of every enumerative count the theory produces. Reinterpreted through the first Chern class in section 2.4, it reads $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = h_{n-k}$, and its terminal instance $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathrm{pt}]$ is the statement that the top self- intersection of the hyperplane class on $\mathbb{P}^n$ is one, the algebraic form of the assertion that n general hyperplanes meet in a single point. That the intermediate steps are independent of the representatives chosen is the payoff of this section, and it is what turns a naive count of intersection points into an invariant.

Exercise 2.2.1

1. On $X = \mathbb{P}^2$ let $D = \{x_0 x_1 = 0\}$ be the union of two lines, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(D) = \mathcal{O}_{\mathbb{P}^2}(2)$, and let V be the conic $\{x_2^2 = x_0 x_1\}$. Compute $D \cdot [V] \in A_0(V \cap |D|)$ explicitly as a sum of points with multiplicities, and verify that its degree equals 4. Explain how the multiplicities distribute among the intersection points.

2. Let $V \subseteq \mathbb{P}^2$ be a smooth curve of degree d and H a line meeting V transversally in d distinct points. Using definition 2.2.1, show that $H \cdot [V]$ is the sum of those d points each with multiplicity one, and deduce $\deg(H \cdot [V]) = d$. Then explain why the value $\deg(H \cdot [V]) = d$ holds for every line H, transversal or not, without recomputing, citing proposition 2.2.4.

3. Let $D = \operatorname{div}(f)$ be a principal Cartier divisor on a variety X, so $\mathcal{O}_X(D) \cong \mathcal{O}_X$ is trivial. Show that $D \cdot a = 0$ in $A_{k-1}(X)$ for every class $a \in A_k(X)$. (Push $D \cdot a$ forward from $|D|$ into X;

the point is that a global trivialization of $\mathcal{O}_X(D)$ makes the restricted section extend to all of V.)

4. On $\mathbb{P}^3$ let H be a hyperplane and let V be a smooth cubic surface, regarded through its inclusion as a subvariety, with the induced very ample bundle $\mathcal{O}_V(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_V$. Compute $H \cdot [V]$ as a class in $A_1(V \cap H)$, identify $V \cap H$ geometrically, and compute the degree of the further intersection $H \cdot (H \cdot [V])$.

5. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$ and let $D = f_1 = \{a\} \times \mathbb{P}^1$ be a ruling of the first kind, with $\mathcal{O}_Q(D) = \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(1)$. Compute $D \cdot [f_1]$ and $D \cdot [f_2]$, where $f_2 = \mathbb{P}^1 \times \{b\}$ is a ruling of the second kind, and interpret the results as the self-intersection f_1^2 and the cross-intersection $f_1 \cdot f_2$. Which case uses the second clause of definition 2.2.1?

6. Give an explicit example on $X = \mathbb{P}^1$ of a Cartier divisor D and two rationally equivalent zero-cycles $\alpha \sim \alpha'$ for which the naive formula would assign different answers if one ignored the requirement that components avoid $|D|$, and explain how definition 2.2.1 and proposition 2.2.4 together resolve the apparent discrepancy. (Here $k = 0$, so $D \cdot \alpha$ lands in $A_{-1} = 0$; use this to frame what the proposition is and is not asserting in the lowest dimension.)

2.3 Commutativity and the Projection Formula

Intersection with a divisor, made precise in section 2.2, is an operation $D \cdot -$ that lowers dimension by one and, by proposition 2.2.4, descends to rational equivalence. Nothing so far says these operations may be applied in either order. Given two Cartier divisors D and D' on X and a k-cycle α, one can form $D \cdot (D' \cdot \alpha)$ and $D' \cdot (D \cdot \alpha)$, each a class in $A_{k-2}(X)$; that they agree is the single algebraic fact on which the entire intersection calculus rests. Its proof does not follow from the definitions by formal manipulation, because the two orders of intersection produce different cycles before passing to rational equivalence. The content is that the difference is always a boundary, and the reason it is a boundary is the vanishing of the degree of a principal divisor on a complete curve, established back in proposition 1.2.2.

This is also the point at which a debt from chapter 1 comes due. The proof of affine-bundle invariance (theorem 1.4.6) deferred one identity, the divisor symmetry (1.3), on the grounds that the tools were not yet in place; that identity is exactly the commutativity proved here. This section discharges it in full and then draws the two functorial consequences, compatibility of $D \cdot -$ with proper pushforward and with flat pullback, that make the operation part of the same covariant/contravariant package as pushforward and pullback themselves.

The obstruction to commutativity lives on a surface, and the mechanism that removes it is a symmetry between two rational functions. Before the theorem, the mechanism deserves to be isolated as a lemma. Let S be a complete variety of dimension two and let $f, g \in k(S)^*$ be nonzero rational functions. Each has a divisor $[\mathrm{div}(f)] = \sum_W \mathrm{ord}_W(f)\,[W]$, the sum over the codimension-one subvarieties (the curves) $W \subseteq S$, and on each such curve W the other function g restricts, wherever it is neither zero nor a pole identically along W, to a rational function $g|_W$ with its own divisor of points. The two-function symmetry lemma compares the two zero-cycles obtained by cutting first with f and then with g, against the same two cuts in the opposite order.

The statement needs the order of one function measured against the divisor of another. For a curve $W \subseteq S$ with $\mathrm{ord}_W(g) = 0$, so that g is a unit at the generic point of W and restricts to a nonzero rational function $g|_W \in k(W)^*$, write $[\mathrm{div}_W(g)] = \sum_{P\in W} \mathrm{ord}_P(g|_W)\,[P]$ for the divisor of that restriction, a zero-cycle on W and hence on S. The lemma is a reciprocity between the iterated cuts $\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)]$ and $\sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)]$.

Lemma 2.3.1: Two-Function Reciprocity on a Surface

Let S be a complete variety of dimension two and let $f, g \in k(S)^*$. Then

$$\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)] \;\sim\; \sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)] \quad \text{in } A_0(S)$$

where each sum runs over the curves $W \subseteq S$ on which the relevant restriction is defined, and the divisors $[\mathrm{div}_W(g)]$, $[\mathrm{div}_W(f)]$ are the zero-cycles of the restricted functions. In fact both sides have the same degree, and their difference is a sum of principal divisors of rational functions on the curves $W \subseteq S$, hence rationally trivial.

The lemma is the two-dimensional heart of commutativity, and its proof is the one place where the degree-zero property of principal divisors on complete curves does the decisive work. The strategy is to reduce the equality of two zero-cycles to an equality of degrees along each curve, and then to recognize each such degree as the value at a point of the *tame symbol*, a bimultiplicative pairing on pairs of rational functions whose reciprocity across a complete curve is precisely the statement that a principal divisor has degree zero.

***Proof*:**
Step 1: the tame symbol at a curve. Fix a curve $W \subseteq S$ and let $A = \mathcal{O}_{S,W}$ be the local ring of S at the generic point of W. Because S is a variety and W has codimension one, A is a one-dimensional local domain with fraction field $k(S)$ and residue field $k(W)$; its normalization is a discrete valuation ring, and after replacing S by its normalization along W (which changes neither side of the asserted identity, both being computed by orders and residues that are birational invariants) one may assume A is itself a discrete valuation ring with valuation $v_W = \mathrm{ord}_W$ and uniformizer π. Define the *tame symbol* of f, g at W to be the residue class

$$\partial_W(f,g) \;=\; (-1)^{v_W(f)\,v_W(g)}\,\frac{f^{\,v_W(g)}}{g^{\,v_W(f)}}\Big|_W \;\in\; k(W)^*$$

the restriction to W of the indicated unit, which lies in A and is a unit there because its valuation

$$v_W\Big((-1)^{v_W(f)v_W(g)}\,f^{v_W(g)}g^{-v_W(f)}\Big) = v_W(g)\,v_W(f) - v_W(f)\,v_W(g) = 0$$

vanishes. The construction is bimultiplicative in f and g and antisymmetric, $\partial_W(g,f) = \partial_W(f,g)^{-1}$, both directly from the formula. The point of the sign is that this antisymmetry holds on the nose, not just up to squares.

Step 2: the tame symbol computes the difference of iterated divisors, pointwise. Fix a closed point $P \in S$ and a curve W through P. When $\mathrm{ord}_W(g) = 0$ the restriction $g|_W$ is a nonzero function on W and the coefficient of $[P]$ in $\mathrm{ord}_W(f)\,[\mathrm{div}_W(g)]$ is $\mathrm{ord}_W(f)\,\mathrm{ord}_P(g|_W)$; symmetrically for the other cut. Comparing the two iterated cuts at P amounts to comparing, over all curves $W \ni P$,

$$\sum_{W\ni P} \mathrm{ord}_W(f)\,\mathrm{ord}_P(g|_W) \quad \text{against} \quad \sum_{W\ni P} \mathrm{ord}_W(g)\,\mathrm{ord}_P(f|_W)$$

The tame symbol reconciles them, but the reconciliation is delicate at the curves where both functions have nonzero order, and getting it right is the whole content of the step. For a single curve W with $\operatorname{ord}_W(f) = a$ and $\operatorname{ord}_W(g) = b$, write $f = \pi^a u$ and $g = \pi^b w$ with u, w units at W. The tame symbol of *Step 1* then simplifies to

$$\partial_W(f, g) = (-1)^{ab}\, \bar{u}^{\,b}\, \bar{w}^{\,-a} \;\in\; k(W)^*$$

the powers of π cancelling, so its order at P is always

$$\operatorname{ord}_P\bigl(\partial_W(f, g)\bigr) = b \operatorname{ord}_P(\bar{u}) - a \operatorname{ord}_P(\bar{w}) \tag{2.1}$$

regardless of whether a or b vanishes. The iterated cuts, on the other hand, see only the restricted functions, and $f|_W = \bar{u}$ is defined precisely when $a = 0$ while $g|_W = \bar{w}$ is defined precisely when $b = 0$. Compare the two curve-by-curve.

If $a = 0$ and $b \neq 0$: then $\operatorname{ord}_W(g)\,[\operatorname{div}_W(f)]$ contributes $b \operatorname{ord}_P(\bar{u})$ at P while $\operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = 0$, so the iterated-cut difference is $b \operatorname{ord}_P(\bar{u})$, which is exactly (2.1) since $a = 0$ kills the second term. *If $a \neq 0$ and $b = 0$:* symmetrically the difference is $-a \operatorname{ord}_P(\bar{w})$, again (2.1). *If $a = b = 0$:* both restrictions are units, both sides vanish. In all three of these *non-mixed* cases,

$$\Bigl(\operatorname{ord}_W(g) \operatorname{ord}_P(f|_W) - \operatorname{ord}_W(f) \operatorname{ord}_P(g|_W)\Bigr) = \operatorname{ord}_P\bigl(\partial_W(f, g)\bigr)$$

the left side being the contribution of W to the iterated-cut difference at P, where a term whose restricted function is undefined is read as zero because its coefficient $\operatorname{ord}_W(g)$ or $\operatorname{ord}_W(f)$ then vanishes. The remaining case is the *mixed* one, $a \neq 0$ and $b \neq 0$: here *neither* $f|_W$ nor $g|_W$ is defined, so W contributes nothing to either iterated cut, yet $\partial_W(f, g) = (-1)^{ab}\bar{u}^{\,b}\bar{w}^{\,-a}$ is a genuine nonzero rational function on W whose divisor is generally nonzero. Thus on a mixed curve the iterated-cut difference and $\operatorname{ord}_P(\partial_W)$ do *not* agree; the tame symbol overcounts by exactly the mixed contribution. Summing the non-mixed agreement over the curves through P,

$$\Bigl(\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)]\Bigr)_P = \sum_{\substack{W \ni P \\ \text{non-mixed}}} \operatorname{ord}_P\bigl(\partial_W(f, g)\bigr) \tag{2.2}$$

the sum on the right running only over curves $W \ni P$ on which at least one of f, g has order zero, and the left side denoting the coefficient of $[P]$ in the difference of the two iterated cuts. The mixed curves are deliberately excluded from the right side; they carry no iterated-cut contribution, and Step 3 will show their tame symbols are individually harmless.

Step 3: reciprocity along each curve kills the total. Fix a curve $W \subseteq S$. The tame symbol $\partial_W(f, g)$ is a nonzero rational function on the complete curve W (it is complete because S is complete and W is closed in S). Its divisor $[\operatorname{div}(\partial_W(f, g))] = \sum_P \operatorname{ord}_P(\partial_W(f, g))\,[P]$ is a principal divisor on W, so by proposition 1.2.2(1) applied to the structure map $W \to \operatorname{Spec} \overline{k}$, which is proper surjective with target of strictly smaller dimension, its pushforward vanishes: $\deg[\operatorname{div}(\partial_W(f, g))] = 0$. This holds for *every* curve W, mixed or not. Summing (2.2) over all closed points P and reorganizing the double sum by curves,

$$\deg\Bigl(\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)]\Bigr) = \sum_{\text{non-mixed } W} \deg[\operatorname{div}(\partial_W(f, g))] = 0$$

each summand vanishing individually. Thus the difference of the two iterated zero-cycles has degree zero on S.

Step 4: from degree zero to rational equivalence. A zero-cycle of degree zero on the complete variety S need not be rationally equivalent to zero in general, but the difference here is more than a degree-zero cycle: (2.2), summed over all closed points P and reorganized by curves, exhibits it as a sum of principal divisors,

$$\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(f,g))]$$

the right side running over the curves on which one of f, g has order zero, each term the divisor of the rational function $\partial_W(f,g)$ on the complete curve $W \subseteq S$. Each $[\operatorname{div}(\partial_W(f,g))]$ is by definition rationally equivalent to zero on W, hence on S under the proper pushforward along $W \hookrightarrow S$, which preserves rational equivalence by theorem 1.2.3. Therefore the difference is a finite sum of cycles each ~ 0, so it is itself ~ 0 in $A_0(S)$, which is the asserted reciprocity, as we sought to show.

The lemma is the algebraic incarnation of a classical reciprocity, but the mechanism it uses is elementary: on each curve W the tame symbol $\partial_W(f,g)$ is a single rational function, its divisor on the complete curve W is principal and so rationally trivial, and the difference of the two iterated cuts is a finite sum of such divisors. No appeal to Weil reciprocity or to the point-level tame symbol $\prod_P \partial_P(f,g) = 1$ as an external input is needed, and indeed none is available in this book; the entire force of the lemma comes from the single fact that a rational function on a complete curve has as many zeros as poles, counted with multiplicity and residue degree, applied one curve at a time. That fact is proposition 1.2.2(1), proved in chapter 1 by the model computation on $\mathbb{P}^1$. The reader should notice that the surface S never needed to be smooth; normalizing along each curve reduced every local computation to a discrete valuation ring, which is all the tame symbol requires.

With the reciprocity lemma in hand, commutativity is a matter of reducing the general assertion to the surface case the lemma governs. The reduction is entirely formal, following the pattern used throughout chapter 1: the operation $D \cdot -$ is a homomorphism, so it suffices to check the identity on the class of a single subvariety, and the intersection with D and D' is supported where the two divisors meet the subvariety, so one may replace the ambient scheme by the subvariety itself.

Theorem 2.3.2: Commutativity of Divisor Intersection

Let D and D' be Cartier divisors on an algebraic scheme X, and let $\alpha \in A_k(X)$. Then

$$D \cdot (D' \cdot \alpha) = D' \cdot (D \cdot \alpha) \quad \text{in } A_{k-2}(X)$$

More precisely, both sides are represented by well-defined classes in $A_{k-2}\big(|D| \cap |D'| \cap \operatorname{supp}\alpha\big)$, and they coincide there. This identity is the divisor symmetry (1.3) deferred in the proof of theorem 1.4.6.

***Proof*:**
Step 1: reduction to a variety. Both $D \cdot -$ and $D' \cdot -$ are homomorphisms $A_k \to A_{k-1}$ by definition 2.2.1 and proposition 2.2.4, so both iterated operations are homomorphisms $A_k(X) \to A_{k-2}(X)$. It therefore suffices to verify the identity on a generator $\alpha = [V]$, the class of a k-dimensional subvariety $V \subseteq X$. Restricting the line bundles $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ and their rational sections to V, the entire computation of $D \cdot (D' \cdot [V])$ and $D' \cdot (D \cdot [V])$ takes place

on V using the restricted divisors $D|_V$ and $D'|_V$. Replacing X by V, one is reduced to proving

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X])$$

for X itself a k-dimensional variety and D, D' Cartier divisors on it, with the common value a class in $A_{k-2}(|D| \cap |D'|)$.

Step 2: choosing sections and unwinding the two orders. Let s and s' be the rational sections of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ cutting out D and D', so that on the variety X,

$$D \cdot [X] = [\mathrm{div}(s)] = \sum_W \mathrm{ord}_W(s)\,[W], \qquad D' \cdot [X] = [\mathrm{div}(s')] = \sum_{W'} \mathrm{ord}_{W'}(s')\,[W']$$

the sums over the codimension-one subvarieties $W, W' \subseteq X$. To compute the double intersection one then intersects each $[W']$ with D and each $[W]$ with D'. On a codimension-one subvariety W not contained in $|D'|$, the section s' restricts to a rational section $s'|_W$ of $\mathcal{O}_X(D')|_W$, whose divisor is $[\mathrm{div}_W(s')]$, and by definition 2.2.1,

$$D' \cdot (D \cdot [X]) = \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')], \qquad D \cdot (D' \cdot [X]) = \sum_{W'} \mathrm{ord}_{W'}(s')\,[\mathrm{div}_{W'}(s)]$$

as cycles supported on $|D| \cap |D'|$. The two expressions are the two iterated cuts of the reciprocity lemma, with the roles of the two divisors interchanged.

Step 3: the difference as a sum of tame symbols. The difference

$$\Delta \;=\; \sum_{W'} \mathrm{ord}_{W'}(s')\,[\mathrm{div}_{W'}(s)] \;-\; \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')]$$

is a $(k-2)$-cycle supported on $|D| \cap |D'|$, and the claim is $\Delta \sim 0$ in $A_{k-2}(|D| \cap |D'|)$. The strategy is to write Δ as a $\mathbb{Z}$-combination of divisors of rational functions on $(k-1)$-dimensional subvarieties of X, using the tame symbol of lemma 2.3.1. For a $(k-1)$-dimensional subvariety $W \subseteq X$ on which one of the two sections has nonzero order, choose a common local trivialization of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ near the generic point of W (a line bundle is locally trivial, and one open set trivializes both), turning s, s' into rational functions in the local ring $\mathcal{O}_{X,W}$, a two-dimensional local domain. The tame symbol

$$\partial_W(s, s') = (-1)^{\mathrm{ord}_W(s)\mathrm{ord}_W(s')}\, \frac{s^{\,\mathrm{ord}_W(s')}}{s'^{\,\mathrm{ord}_W(s)}}\bigg|_W \;\in\; k(W)^*$$

is then a nonzero rational function on the $(k-1)$-dimensional variety W, defined exactly as in the lemma. Localizing X along W removes the trivialization ambiguity: two trivializations differ by a unit, which does not change $\partial_W(s, s')$ up to a rational function that is regular and nonvanishing at the generic point of W, so its divisor on W is well defined.

Step 4: applying reciprocity, one codimension-two point at a time. Fix a $(k-2)$-dimensional subvariety $Z \subseteq |D| \cap |D'|$ and compute the coefficient of $[Z]$ on both sides of the asserted identity $\Delta = \sum_{\text{non-mixed } W} [\mathrm{div}(\partial_W(s, s'))]$, the sum running over $(k-1)$-dimensional W on which one of s, s' has order zero. That coefficient is determined at the generic point ζ of Z, where $\mathcal{O}_{X,\zeta}$ has dimension two; the two-dimensional local situation there is governed by lemma 2.3.1, whose (2.2) reads, at ζ,

$$\Big(\sum_W \mathrm{ord}_W(s')\,[\mathrm{div}_W(s)] - \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')]\Big)_\zeta \;=\; \sum_{\substack{W \ni \zeta \\ \text{non-mixed}}} \mathrm{ord}_\zeta\big(\partial_W(s, s')\big)$$

the right sum over the $(k-1)$-dimensional subvarieties W through ζ on which one of s, s' has order zero, exactly as in the lemma; the mixed W, on which both sections have nonzero order, are excluded because they contribute to neither iterated cut. This is the coefficient of $[Z]$ in Δ set equal to the coefficient of $[Z]$ in $\sum_{\text{non-mixed } W}[\operatorname{div}(\partial_W(s,s'))]$. Since the two sides agree at every such Z, they agree as cycles:

$$\Delta = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(s,s'))]$$

Each term $[\operatorname{div}(\partial_W(s,s'))]$ is the divisor of a rational function on the $(k-1)$-dimensional subvariety W, hence rationally equivalent to zero on W, and its proper pushforward to $|D| \cap |D'|$ along $W \hookrightarrow |D| \cap |D'|$ preserves that equivalence by theorem 1.2.3. As a finite sum of cycles each ~ 0, therefore $\Delta \sim 0$, so

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X]) \quad \text{in } A_{k-2}(|D| \cap |D'|)$$

Pushing forward to X along the inclusion, the identity holds in $A_{k-2}(X)$ for $\alpha = [X]$, and by Step 1 for every $\alpha \in A_k(X)$. The surface case $X = S$, in which the difference of the two orders is exhibited as a sum of principal cycles, is precisely the deferred divisor symmetry (1.3), with the two rational functions $s|_S$ and $s'|_S$ here playing the roles of g and the coordinate t there, completing the proof.

Commutativity is the property that makes the phrase “intersection number” unambiguous. Without it, the number attached to a chain of divisors would depend on the order in which they are imposed, and no well-defined product on cycles could exist. With it, one may write $D_1 \cdot D_2 \cdots D_r \cdot \alpha$ for an iterated intersection and speak of it without specifying an order, and the class of a complete intersection $D_1 \cap \cdots \cap D_r$ in A_* is symmetric in the defining divisors. The proof also settles the debt from chapter 1: the identity (1.3) that theorem 1.4.6 took on faith is the surface case here, so affine-bundle invariance is now unconditional. The tame symbol was the only tool required beyond the definitions, and its reciprocity was nothing more than the degree-zero property of principal divisors, propagated from curves to surfaces.

A first reading of the theorem should note what it does *not* require. The divisors need not be effective, the sections need not be regular, and X need not be smooth or even normal; the entire argument localizes to one-dimensional local rings, where a divisor is controlled by a single valuation. This is why intersection with a Cartier divisor is available on every algebraic scheme, in contrast to the intersection product of arbitrary cycles, which will require the deformation-to-the-normal-cone machinery of a later chapter and works cleanly only on smooth varieties.

The second half of the section records how $D \cdot -$ interacts with the two functorialities of the Chow group. Both statements are what one would guess from the topological analogy, in which cap product with a cohomology class is natural for maps: pushforward absorbs the pullback of a divisor, and flat pullback commutes with intersection outright. The proofs run through the same reduction to a single subvariety and the same input, the pushforward of a principal divisor, that powered commutativity.

For a proper morphism $f\colon X \to Y$ and a Cartier divisor D on Y, the pullback f^*D is defined whenever no component of X maps into $|D|$, by pulling back the line bundle $\mathcal{O}_Y(D)$ and its rational section; it is the Cartier divisor $\operatorname{div}(f^\# s)$ of the pulled-back section. The projection formula compares intersecting upstairs and pushing down against pushing down and intersecting downstairs.

Proposition 2.3.3: Projection Formula for Divisors

Let D be a Cartier divisor on Y.

1. *Proper pushforward.* If $f\colon X \to Y$ is proper and $\alpha \in A_k(X)$ is a cycle no component of whose support maps into $|D|$, so that $f^*D \cdot \alpha$ is defined, then

$$f_*\big(f^*D \cdot \alpha\big) = D \cdot f_*\alpha \quad \text{in } A_{k-1}(Y)$$

and, more precisely, both sides are classes in $A_{k-1}\big(|D| \cap f(\mathrm{supp}\alpha)\big)$ and agree there, f being proper so that it carries $f^{-1}(|D|) \cap \mathrm{supp}\alpha$ onto $|D| \cap f(\mathrm{supp}\alpha)$.

2. *Flat pullback.* If $g\colon X' \to Y$ is flat of relative dimension n and $\beta \in A_k(Y)$, then

$$g^*\big(D \cdot \beta\big) = g^*D \cdot g^*\beta \quad \text{in } A_{k+n-1}(X')$$

where g^*D is the pullback Cartier divisor $\mathrm{div}(g^\# s)$; more precisely both sides are classes in $A_{k+n-1}\big(g^{-1}(|D| \cap \mathrm{supp}\beta)\big)$ and agree there.

***Proof*:**
Both statements are linear in α and β, so it suffices to verify each on the class of a subvariety, and both reduce, by the argument that opened the commutativity proof, to a computation with the divisor of a rational function that is settled by proposition 1.2.2 and proposition 1.3.6. The refined statements come for free: definition 2.2.1 records $D \cdot [V]$ on $V \cap |D|$ to begin with, and each identity below is an equality of cycles supported there, so no step passes to the ambient group. This refinement is what chapter 4 consumes.

1. *Proper pushforward.* Take $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$, and set $W = f(V) \subseteq Y$, so $f_*[V] = \deg(V/W)\,[W]$ if $\dim W = k$ and $f_*[V] = 0$ if $\dim W < k$, by definition 1.2.1. Restricting to V and W, one may assume $X = V$, $Y = W$, and $f\colon V \to W$ proper surjective with V a variety. Let s be the rational section of $\mathcal{O}_Y(D)$ cutting out D; its pullback $f^\# s$ is a rational section of $f^*\mathcal{O}_Y(D)$ cutting out f^*D, and $f^*D \cdot [V] = [\mathrm{div}(f^\# s)]$ while $D \cdot [W] = [\mathrm{div}(s)]$, both by definition 2.2.1. There are two cases.

   If $\dim W < k$, then $f_*[V] = 0$, so the right side $D \cdot f_*[V]$ vanishes. On the left, $[\mathrm{div}(f^\# s)]$ is a $(k-1)$-cycle on V, and f pushes it into Y; but $f(V) = W$ has dimension $< k$, so $f_*[\mathrm{div}(f^\# s)]$ is a cycle of dimension $k-1$ supported on the $(< k)$-dimensional W, and proposition 1.2.2(1) applied to $f\colon V \to W$ (proper surjective, target of strictly smaller dimension) gives $f_*[\mathrm{div}(f^\# s)] = 0$. Both sides vanish.

   If $\dim W = k$, then $f\colon V \to W$ is proper surjective of equal dimension, generically finite of degree $d = \deg(V/W) = [k(V) : k(W)]$. Now $f^\# s$ is a rational section whose local expression, in a trivialization of $\mathcal{O}_Y(D)$ near a curve of W, is a rational function $r \in k(V)^*$ pulled back from $k(W)$, and proposition 1.2.2(2) gives $f_*[\mathrm{div}(f^\# s)] = [\mathrm{div}(N(f^\# s))]$ with N the norm along $k(V)/k(W)$. Because $f^\# s$ is pulled back from s, its norm is $N(f^\# s) = s^d$ up to the local trivialization, so $[\mathrm{div}(N(f^\# s))] = d\,[\mathrm{div}(s)] = d\,(D \cdot [W])$. Hence

$$f_*\big(f^*D \cdot [V]\big) = f_*[\mathrm{div}(f^\# s)] = d\,(D \cdot [W]) = D \cdot (d\,[W]) = D \cdot f_*[V]$$

as required.

2. *Flat pullback.* Take $\beta = [V]$ for a k-dimensional subvariety $V \subseteq Y$. Both sides are supported over $V \cap |D|$, so restricting to V one may assume $Y = V$ is a variety and D a Cartier divisor on it with cutting section s, so $D \cdot [V] = [\mathrm{div}(s)]$. Then $g^*(D \cdot [V]) = g^*[\mathrm{div}(s)]$, and by the flat-pullback divisor formula established in proposition 1.3.7, flat pullback commutes with taking the divisor of a rational function: $g^*[\mathrm{div}(s)] = [\mathrm{div}(g^\# s)]$ componentwise, with the multiplicities of the components of $g^{-1}(V)$ accounted by flatness. On the other side, g^*D is the Cartier divisor $\mathrm{div}(g^\# s)$ on X' by definition, and $g^*[V] = [g^{-1}(V)] = \sum_j m_j [X'_j]$ over the components X'_j of the preimage with their flat multiplicities m_j by proposition 1.3.5. Intersecting,
$$g^*D \cdot g^*[V] = g^*D \cdot \sum_j m_j [X'_j] = \sum_j m_j \left[\mathrm{div}((g^\# s)|_{X'_j})\right] = [\mathrm{div}(g^\# s)] = g^*(D \cdot [V])$$
the middle equality being definition 2.2.1 applied on each component and the last the same componentwise divisor formula. This is the claim; the compatibility of these classes with the base change is proposition 1.3.6, which guarantees the two routes land in the same class of $A_{k+n-1}(X')$. This establishes both parts, completing the proof.

The projection formula is the mechanism by which a computation on a resolved or covering space transfers to the base. In its pushforward form, intersecting the pullback of D upstairs and pushing the answer down is the same as pushing the cycle down and intersecting with D there, which is exactly what one does when computing an intersection number on Y by passing to a modification $X \to Y$ where the geometry is transparent. The scalar $d = \deg(V/W)$ in the equal-dimensional case is the reason degrees multiply under finite covers, and it is the algebraic source of formulas like $\deg(f^*D) = (\deg f)(\deg D)$ for a finite map of curves. In its flat form, the formula says intersection with a divisor is a local construction, insensitive to flat base change: pulling back the divisor and pulling back the cycle and then intersecting reproduces the pullback of the intersection. Together the two halves make $D \cdot -$ compatible with the covariant/contravariant package (f_*, g^*) of chapter 1, the bimodule structure that all of intersection theory is built to respect.

The abstractions above are best seen at work on the projective plane, where every class is a known multiple of a linear space and the intersection numbers can be read off by hand. The following example verifies commutativity concretely and recovers the intersection number of two lines.

Example 2.3.4: Two Lines on the Projective Plane

Let $X = \mathbb{P}^2$ over $\overline{k}$, with homogeneous coordinates $[x_0 : x_1 : x_2]$. By theorem 1.5.2, $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[L]$ generated by the class of a line L, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[P]$ generated by the class of a point. Take the two Cartier divisors $D = \{x_0 = 0\}$ and $D' = \{x_1 = 0\}$, each a line, with $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(D') \cong \mathcal{O}(1)$ and cutting sections $s = x_0$, $s' = x_1$ (rational sections of $\mathcal{O}(1)$ in the trivialization on the chart $U_2 = \{x_2 \neq 0\}$, where the affine coordinates are $u = x_0/x_2$, $v = x_1/x_2$).

First order. Intersecting $[\mathbb{P}^2]$ with D gives $D \cdot [\mathbb{P}^2] = [\mathrm{div}(s)] = [D] = [L]$, the line $\{x_0 = 0\}$. Now intersect with D'. Since $D = \{x_0 = 0\}$ is not contained in $|D'| = \{x_1 = 0\}$, the section $s'|_D = x_1|_D$ is a rational section of $\mathcal{O}(1)|_D$; on the line $D \cong \mathbb{P}^1$ with coordinate $[x_1 : x_2]$ it vanishes to order one at the single point $x_1 = 0$, namely $[0 : 0 : 1]$. Thus
$$D' \cdot (D \cdot [\mathbb{P}^2]) = D' \cdot [L] = [\mathrm{div}_D(s')] = \big[\,[0 : 0 : 1]\,\big]$$
the class of the point where the two lines meet.

Reverse order. Intersecting $[\mathbb{P}^2]$ with D' first gives $D' \cdot [\mathbb{P}^2] = [\operatorname{div}(s')] = [D'] = [L]$, the line $\{x_1 = 0\}$. Intersecting with D, the section $s|_{D'} = x_0|_{D'}$ on $D' \cong \mathbb{P}^1$ with coordinate $[x_0 : x_2]$ vanishes to order one at $x_0 = 0$, again $[0 : 0 : 1]$, so

$$D \cdot (D' \cdot [\mathbb{P}^2]) = D \cdot [L] = [\operatorname{div}_{D'}(s)] = \big[\,[0 : 0 : 1]\,\big]$$

The two orders produce the same zero-cycle, the reduced point $[0 : 0 : 1]$, on the nose, which is theorem 2.3.2 made visible: here no rational equivalence was even needed, because the two lines are transverse and the tame symbol $\partial_W(x_0, x_1)$ has no zeros or poles. Taking degrees on the complete $\mathbb{P}^2$,

$$\deg\big(D \cdot D' \cdot [\mathbb{P}^2]\big) = 1$$

the intersection number of two distinct lines: they meet in exactly one point. This is the case $c = d = 1$ of Bézout's theorem ([7]), recovered from the divisor calculus.

Where reciprocity would enter. Replacing D' by D itself, one must intersect $[L]$ with the same line, which requires moving D within its linear system: since $\mathcal{O}(1)$ has many sections, choose instead $s'' = x_1$, whose divisor $\{x_1 = 0\}$ is linearly equivalent to $\{x_0 = 0\}$, and the previous computation gives $D \cdot D \cdot [\mathbb{P}^2] = \big[\,[0 : 0 : 1]\,\big]$ of degree 1, so the self-intersection $L^2 = 1$ on $\mathbb{P}^2$. The passage from $s = x_0$ to the linearly equivalent $s'' = x_1$ changes the cutting section by the rational function x_1/x_0, and that the answer is unchanged is precisely proposition 2.2.4, the well-definedness whose commutativity companion is proved in this section.

The example is the enumerative payoff in miniature. The number 1 is the degree of a zero-cycle, the two lines meet where their defining sections simultaneously vanish, and the independence of the answer from the order of intersection and from the choice of representative in each linear system is exactly what theorem 2.3.2 and proposition 2.2.4 guarantee. Iterating, r general hyperplanes in $\mathbb{P}^n$ intersect in $c_1(\mathcal{O}(1))^r \cap [\mathbb{P}^n] = [\mathbb{P}^{n-r}]$, and taking $r = n$ recovers a single reduced point of degree one, the statement that n general hyperplanes meet in one point. This is the computation the next section packages as the first Chern class.

Exercise 2.3.1

1. On $\mathbb{P}^2$, let $D = \{x_0x_1 = 0\}$ be the union of two coordinate lines (a divisor in the class $2L$) and let $D' = \{x_2 = 0\}$ be a third line. Compute $D \cdot (D' \cdot [\mathbb{P}^2])$ and $D' \cdot (D \cdot [\mathbb{P}^2])$ as zero-cycles, verify they agree, and compute the degree of the product. Confirm the answer is $2 = \deg D \cdot \deg D'$, consistent with Bézout.

2. Let $S = \mathbb{P}^2$ with affine coordinates $u = x_0/x_2$, $v = x_1/x_2$ on the chart $x_2 \neq 0$. For the rational functions $f = u$ and $g = v$, compute the tame symbol $\partial_W(f, g)$ at each coordinate curve $W \in \{u = 0\}, \{v = 0\}$ and at the line at infinity, and verify directly that $\sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = \sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)]$ as zero-cycles.

3. Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be the degree-two map $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (characteristic $\neq 2$) and let $D = [\infty]$ be the divisor of the point at infinity on the target. Using proposition 2.3.3(1), compute both $f_*(f^*D \cdot [\mathbb{P}^1])$ and $D \cdot f_*[\mathbb{P}^1]$, identify the scalar $d = \deg f$, and confirm they agree.

4. Let $g\colon \mathbb{A}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ be the projection, flat of relative dimension one, and let $D = [0]$ be the origin on $\mathbb{P}^1$. Compute $g^*(D \cdot [\mathbb{P}^1])$ and $g^*D \cdot g^*[\mathbb{P}^1]$ and verify proposition 2.3.3(2) directly.

5. Deduce from theorem 2.3.2 that for any Cartier divisor D on a complete variety X of dimension

n, the iterated intersection $D^n \cdot [X] \in A_0(X)$ is symmetric under any reordering of the n copies of D (trivially, but state precisely why the number $\deg(D^n \cdot [X])$ is well defined), and use theorem 1.5.2 to compute $H^n \cdot [\mathbb{P}^n]$ for H a hyperplane.

6. Explain why the proof of theorem 2.3.2 does not require X to be normal, identifying the exact step where normalization is used and why it changes neither side of the identity. (One sentence on each.)

2.4 The First Chern Class of a Line Bundle

The three previous sections built a single operation and studied it thoroughly: a Cartier divisor D on X acts on cycles by $\alpha \mapsto D \cdot \alpha$, this action descends to rational equivalence (proposition 2.2.4), and two such actions commute (theorem 2.3.2). The divisor D, however, is not the most invariant object in sight. What acts on $A_*(X)$ is not D itself but its linear-equivalence class, since D and $D + \operatorname{div}(r)$ produce the same operator, and a linear-equivalence class of Cartier divisors is exactly a line bundle. The right name for the operator is therefore the *first Chern class* of the associated line bundle $\mathcal{O}_X(D)$, written $c_1(\mathcal{O}_X(D)) \cap -$.

The renaming records that the operator depends only on $L = \mathcal{O}_X(D)$, so it is defined for a line bundle whether or not a divisor is at hand; it makes the additive structure transparent, because tensoring line bundles adds their Chern classes; and it is the first instance of a construction that in the next chapter will attach to a vector bundle E of rank r a whole family of operators $c_1(E), \ldots, c_r(E)$, the higher Chern classes. Everything the first Chern class does is already contained in the divisor calculus of the preceding sections; this section repackages that calculus into its final, functorial form and reads off the consequences.

The starting point is that the operator $D \cdot -$ sees only the line bundle $\mathcal{O}_X(D)$. Two Cartier divisors with isomorphic associated bundles are linearly equivalent, that is, they differ by the divisor of a rational function, and a principal divisor intersects every subvariety in a principal divisor, hence acts as zero on $A_*(X)$; proposition 2.4.2 below records this and is the independence needed to base the definition on L rather than on D.

Definition 2.4.1: The First Chern Class of a Line Bundle

Let L be a line bundle on an algebraic scheme X, and suppose $L \cong \mathcal{O}_X(D)$ for some Cartier divisor D on X. The *first Chern class* of L is the operator on Chow groups

$$c_1(L) \cap -\colon A_k(X) \longrightarrow A_{k-1}(X), \qquad c_1(L) \cap \alpha := D \cdot \alpha$$

defined for every k, where $D \cdot \alpha$ is the intersection with the Cartier divisor D of definition 2.2.1. The class $c_1(L)$ is the operator itself; the notation $c_1(L) \cap \alpha$ reads "$c_1(L)$ capped with α".

The definition names D but must not depend on it: a line bundle admits many divisors of sections, and the Chern class has to be the same for all of them. The following proposition is the price of the definition, and its proof is short because the work was done in section 2.2.

Proposition 2.4.2: Independence of the Divisor

Let L be a line bundle on X and let D, D' be Cartier divisors with $\mathcal{O}_X(D) \cong L \cong \mathcal{O}_X(D')$. Then $D \cdot \alpha = D' \cdot \alpha$ in $A_{k-1}(X)$ for every $\alpha \in A_k(X)$. Consequently $c_1(L) \cap -$ of definition 2.4.1 is well defined, independent of the choice of D.

***Proof*:**
An isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ of line bundles is exactly a linear equivalence $D \sim D'$ of Cartier divisors: the tautological rational section of $\mathcal{O}_X(D)$ and that of $\mathcal{O}_X(D')$, compared through the isomorphism, differ by a global invertible section r of the sheaf of total quotient rings $\mathcal{K}_X^*$, so $D - D' = \operatorname{div}(r)$ is principal (on a variety $\mathcal{K}_X^* = k(X)^*$ and r is a rational function; on a general algebraic scheme r is a section of the total-quotient sheaf, which is all the definition of a principal Cartier divisor requires). It therefore suffices to show that a principal Cartier divisor $\operatorname{div}(r)$ acts as zero on rational equivalence classes.

Fix $\alpha \in A_k(X)$ and represent it by a cycle $\sum_i n_i[V_i]$. Since $\operatorname{div}(r) \cdot -$ is defined component by component and extended $\mathbb{Z}$-linearly (definition 2.2.1), it suffices to show $\operatorname{div}(r) \cdot [V] = 0$ in $A_{k-1}(X)$ for a single k-dimensional subvariety V. The associated bundle of the principal divisor $\operatorname{div}(r)$ is $\mathcal{O}_X$, and the rational section of $\mathcal{O}_X$ that represents $\operatorname{div}(r)$ is r itself; unwinding definition 2.2.1 for this section gives

$$\operatorname{div}(r) \cdot [V] = \big[\operatorname{div}\big(r|_V\big)\big] \in A_{k-1}(V)$$

where $r|_V \in k(V)^*$ is the restriction of r to V (a nonzero rational function when $V \not\subseteq |\operatorname{div}(r)|$, and otherwise computed through the restricted trivial bundle $\mathcal{O}_X|_V \cong \mathcal{O}_V$, whose section is again a rational function on V). In either case $\operatorname{div}(r|_V)$ is the divisor of a rational function on the variety V, hence a *principal* Weil divisor on V, which is rationally equivalent to zero by the very definition of rational equivalence (definition 1.1.3). Its class in $A_{k-1}(V)$, and so its image in $A_{k-1}(X)$, is therefore 0: intersecting a subvariety with a principal divisor produces a principal divisor, which dies in the Chow group. This is the same computation that drives the well-definedness of proposition 2.2.4, read now with the roles reversed, the divisor rather than the cycle being principal. Summing over the components of α gives $\operatorname{div}(r) \cdot \alpha = 0$, and so

$$D \cdot \alpha - D' \cdot \alpha = (D - D') \cdot \alpha = \operatorname{div}(r) \cdot \alpha = 0$$

giving $D \cdot \alpha = D' \cdot \alpha$, as we sought to show.

The proposition is the reason the notation $c_1(L)$ is legitimate at all. It shows the divisor served only to build the operator: once the operator exists, it is recorded by the line bundle alone. From here on a divisor is used to compute $c_1(L)$, never to define it, and the choice of divisor is a matter of convenience. On a variety, every line bundle admits a Cartier divisor, so the definition applies to all line bundles; on a general scheme one takes L to be of the form $\mathcal{O}_X(D)$, which covers every line bundle whose class lies in the image of the Cartier divisor class group. The definition and the properties below are stated for a general algebraic scheme, but every worked example and exercise that follows runs on a variety, where a rational section and its function field $k(V)$ are always at hand; the general-scheme statements specialize to these illustrations by taking X integral.

A first sanity check grounds the definition before its formal properties: the Chern class of the trivial bundle is the zero operator, and the two constructions available for a hyperplane bundle agree.

Example 2.4.3: The Trivial Bundle and the Hyperplane Bundle

1. *The trivial bundle.* For $L = \mathcal{O}_X$ one may take $D = 0$, the empty divisor. Then $c_1(\mathcal{O}_X) \cap \alpha = 0 \cdot \alpha = 0$ for every α, so $c_1(\mathcal{O}_X)$ is the zero operator. This matches the reading of c_1 as a measure of the twisting of L: a trivial bundle has no twisting to record.

2. *The hyperplane bundle on $\mathbb{P}^n$.* Let $L = \mathcal{O}_{\mathbb{P}^n}(1)$ and let $H \subseteq \mathbb{P}^n$ be a hyperplane, cut out by a linear form; then $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$. The class map of proposition 2.1.3 sends $\mathcal{O}_{\mathbb{P}^n}(1) \mapsto [H] \in A_{n-1}(\mathbb{P}^n)$, the generator of theorem 1.5.2 in codimension one. Capping with $[\mathbb{P}^n]$ recovers this class,

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^n] = [H]$$

since $\mathbb{P}^n \not\subseteq H$ and the restricted section of $\mathcal{O}_{\mathbb{P}^n}(1)$ to $\mathbb{P}^n$ has divisor exactly H. The first Chern class of $\mathcal{O}(1)$, capped against the fundamental class, is the hyperplane class.

With the operator anchored, its formal properties follow. Each is a translation of a divisor identity already proved, and the point of collecting them is to see the first Chern class as a well-behaved endomorphism of the graded group $A_*(X)$: additive in α, commuting with its own kind, additive in L, compatible with pushforward and pullback, and eventually nilpotent. The statement below is the working summary of the theory of c_1, and every clause is used repeatedly in the chapters that follow.

Proposition 2.4.4: Properties of the First Chern Class

Let X be an algebraic scheme, L, L' line bundles on X, and $\alpha \in A_k(X)$.

1. *Homomorphism.* $c_1(L) \cap -\colon A_k(X) \to A_{k-1}(X)$ is a group homomorphism: $c_1(L) \cap (\alpha + \beta) = c_1(L) \cap \alpha + c_1(L) \cap \beta$.
2. *Commutativity.* For line bundles L, L',
$$c_1(L) \cap \big(c_1(L') \cap \alpha\big) = c_1(L') \cap \big(c_1(L) \cap \alpha\big)$$
in $A_{k-2}(X)$.
3. *Additivity.* $c_1(L \otimes L') = c_1(L) + c_1(L')$ as operators; that is, $c_1(L \otimes L') \cap \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$. In particular $c_1(L^\vee) = -c_1(L)$.
4. *Projection formula.* For a proper morphism $f\colon X' \to X$, a line bundle L on X, and $\alpha' \in A_k(X')$,
$$f_*\big(c_1(f^*L) \cap \alpha'\big) = c_1(L) \cap f_*\alpha'$$
and for a flat morphism $g\colon X'' \to X$ of relative dimension m and $\beta \in A_k(X)$,
$$c_1(g^*L) \cap g^*\beta = g^*\big(c_1(L) \cap \beta\big)$$
in $A_{k+m-1}(X'')$.
5. *Nilpotence.* If X is a variety of dimension n, then for any line bundle L,
$$c_1(L)^{n+1} \cap [X] = 0$$
where $c_1(L)^{n+1} \cap [X]$ means $c_1(L)$ capped $n+1$ times with the fundamental class $[X]$.

***Proof*:**
Choose Cartier divisors D, D' with $\mathcal{O}_X(D) \cong L$ and $\mathcal{O}_X(D') \cong L'$; by proposition 2.4.2 the choices do not affect any of the claims. Each part translates a divisor identity from section 2.2 or section 2.3.

1. *Homomorphism.* The intersection $D \cdot -$ is defined on cycles component by component and extended $\mathbb{Z}$-linearly (definition 2.2.1), so it is additive on $Z_k(X)$, and proposition 2.2.4 makes it descend to a well-defined additive map on $A_k(X)$. Hence $c_1(L) \cap - = D \cdot -$ is a homomorphism $A_k(X) \to A_{k-1}(X)$.
2. *Commutativity.* By definition $c_1(L) \cap (c_1(L') \cap \alpha) = D \cdot (D' \cdot \alpha)$ and $c_1(L') \cap (c_1(L) \cap \alpha) = D' \cdot (D \cdot \alpha)$. These are equal in $A_{k-2}(X)$ by theorem 2.3.2, the commutativity of intersection with two Cartier divisors.
3. *Additivity.* The associated-bundle construction of [2, Divisors: Geometric Representation of Line Bundles] converts the sum of Cartier divisors into the tensor product of line bundles, $\mathcal{O}_X(D+D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$, so $D+D'$ is a Cartier divisor with $\mathcal{O}_X(D+D') \cong L \otimes L'$. Using this divisor to compute the Chern class of $L \otimes L'$,
$$c_1(L \otimes L') \cap \alpha = (D + D') \cdot \alpha = D \cdot \alpha + D' \cdot \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$$

where the middle equality is the additivity of $- \cdot \alpha$ in the divisor, and it holds already at the level of cycles, on *every* subvariety V, with no repositioning of α required. Since $D \cdot -$ is defined component by component and extended $\mathbb{Z}$-linearly (definition 2.2.1), it suffices to check $(D+D')\cdot[V] = D\cdot[V]+D'\cdot[V]$ for a single k-dimensional subvariety V. Restricting the tensor-product isomorphism above to V gives $\mathcal{O}_X(D+D')|_V \cong \mathcal{O}_X(D)|_V \otimes_{\mathcal{O}_V} \mathcal{O}_X(D')|_V$, so a rational section s_V of $\mathcal{O}_X(D+D')|_V$ representing $(D+D')|_V$ is the product $s_V = s_V^D \cdot s_V^{D'}$ of representing rational sections of $\mathcal{O}_X(D)|_V$ and $\mathcal{O}_X(D')|_V$. By definition 2.2.1 the cap is the class of the divisor of the restricted section computed inside V, and the order function along a codimension-one subvariety W of V is additive on a product of rational sections, $\operatorname{ord}_W(s_V^D \cdot s_V^{D'}) = \operatorname{ord}_W(s_V^D)+\operatorname{ord}_W(s_V^{D'})$, since ord_W is a valuation. Summing over such W gives $[\operatorname{div}(s_V)] = [\operatorname{div}(s_V^D)]+[\operatorname{div}(s_V^{D'})]$ in $A_{k-1}(V)$, that is $(D+D')\cdot[V] = D\cdot[V]+D'\cdot[V]$; pushing forward to X and summing over the components of α gives the middle equality. This is a cycle-level identity holding for each V, so no moving lemma repositioning α off $|D|$ and $|D'|$ is needed, and none exists in general. Taking $L' = L^\vee$, so that $\mathcal{O}_X \cong L \otimes L^\vee$ and $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, gives $c_1(L) + c_1(L^\vee) = 0$, hence $c_1(L^\vee) = -c_1(L)$.

4. *Projection formula.* The pullback along f of the Cartier divisor D, in the pseudo-divisor formalism of definition 2.1.5, is a pseudo-divisor f^*D with $f^*\mathcal{O}_X(D) \cong \mathcal{O}_{X'}(f^*D)$; so $c_1(f^*L) \cap \alpha' = f^*D \cdot \alpha'$. The proper case is the divisor projection formula $f_*(f^*D \cdot \alpha') = D \cdot f_*\alpha'$ of proposition 2.3.3, which rewritten with Chern-class notation is exactly $f_*(c_1(f^*L) \cap \alpha') = c_1(L) \cap f_*\alpha'$. The flat case is the compatibility $g^*(D \cdot \beta) = g^*D \cdot g^*\beta$ of the same proposition, and since $c_1(g^*L) \cap g^*\beta = g^*D \cdot g^*\beta$ this reads $c_1(g^*L) \cap g^*\beta = g^*(c_1(L) \cap \beta)$.

5. *Nilpotence.* Each cap with $c_1(L)$ lowers dimension by one, since $c_1(L) \cap -\colon A_j(X) \to A_{j-1}(X)$ by definition 2.4.1, so repeatedly capping $[X] \in A_n(X)$ gives

$$c_1(L)^{n+1} \cap [X] \in A_{n-(n+1)}(X) = A_{-1}(X)$$

and $A_j(X) = 0$ for $j < 0$, there being no subvarieties of negative dimension. Hence the class is zero.

Each clause has been reduced to a result of the preceding sections, completing the proof.

The well-definedness proved above lets $D \cdot -$ be evaluated on a representative chosen to avoid $|D|$, which is all the divisor calculus of this chapter needs. Later constructions cannot make that choice: in chapter 4 a cycle is intersected with the fibre at infinity of a degeneration, and *every* component of the relevant representative lies in that fibre. The following strengthening removes the avoidance hypothesis, and it is what makes those constructions legitimate.

Corollary 2.4.5: Divisor Intersection on a Closed Subscheme

Let D be a Cartier divisor on X and $T \subseteq X$ a closed subscheme, with no hypothesis relating T to $|D|$ and none making $D|_T$ a Cartier divisor on T. Then intersection with D descends to a homomorphism

$$D \cdot - \;:\; A_k(T) \longrightarrow A_{k-1}\big(T \cap |D|\big)$$

computed on any representative cycle, and compatible with the inclusion of T in X: for a cycle on T the two readings of definition 2.2.1, one in T and one in X, agree.

***Proof*:**
At the level of cycles there is nothing to prove: definition 2.2.1 computes $D \cdot [V]$ for a subvariety V from the restricted line bundle $\mathcal{O}_X(D)|_V$ and from whether $V \subseteq |D|$, and neither datum changes when V is regarded inside T rather than X. This is the reason the definition was framed through the pseudo-divisor $(\mathcal{O}_X(D), |D|, s_D)$ of definition 2.1.5: a pseudo-divisor restricts to any subscheme, whereas a Cartier divisor need not, its local equations being free to become zero divisors.

What needs an argument is that the operation kills cycles rationally equivalent to zero *on* T, and lands in the smaller group. Such a cycle is a sum of $\operatorname{div}(r)$ for $(k+1)$-dimensional subvarieties $W \subseteq T$ and $r \in k(W)^*$ (definition 1.1.3). The proof of proposition 2.4.6 treats exactly these generators, and in both of its cases the vanishing it produces is already recorded on W: in the case $W \subseteq |D|$ the class is computed in $A_{k-1}(W)$ before being pushed forward, and in the case $W \not\subseteq |D|$ the appeal to theorem 2.3.2 places the identity in $A_{k-1}(|D| \cap |\operatorname{div}(r)| \cap W)$. Both groups map to $A_{k-1}(T \cap |D|)$, so $D \cdot \operatorname{div}(r) = 0$ there, as we sought to show.

Proposition 2.4.6: Divisor Intersection on an Arbitrary Representative

Let D be a Cartier divisor on an algebraic scheme X and let $\alpha \in Z_k(X)$ be a cycle rationally equivalent to zero, with *no* hypothesis on its components. Then

$$D \cdot \alpha = 0 \quad \text{in } A_{k-1}(|D|)$$

Consequently $D \cdot -$ may be computed from any representative of a class in $A_k(X)$, including one all of whose components lie in $|D|$, and every representative gives the same class in $A_{k-1}(|D|)$.

***Proof*:**
Both sides are additive, so it suffices to treat $\alpha = [\operatorname{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq X$ and a rational function $r \in k(W)^*$, these being the generators of the subgroup of cycles rationally equivalent to zero (definition 1.1.3). Write $\iota \colon W \hookrightarrow X$ for the inclusion.

Case 1: $W \subseteq |D|$. Every component of $\operatorname{div}(r)$ lies in W, hence in $|D|$, so clause (2) of definition 2.2.1 computes $D \cdot [\operatorname{div}(r)]$ from the restricted line bundle. By definition 2.4.1 that value is $c_1\big(\mathcal{O}_X(D)|_W\big) \cap [\operatorname{div}(r)]$, computed in $A_{k-1}(W)$. Now $[\operatorname{div}(r)] = 0$ in $A_k(W)$ by the definition of rational equivalence, and $c_1(\mathcal{O}_X(D)|_W) \cap -$ is a homomorphism on Chow *classes* (proposition 2.4.4(1)), so it sends 0 to 0. Pushing forward along the closed immersion $W \hookrightarrow |D|$ gives $D \cdot [\operatorname{div}(r)] = 0$ in $A_{k-1}(|D|)$.

Case 2: $W \not\subseteq |D|$. Then D restricts to a Cartier divisor $D|_W$ on W, and $[\operatorname{div}(r)] = \operatorname{div}(r) \cdot [W]$ exhibits α as the result of intersecting the fundamental class of W with the principal Cartier divisor $\operatorname{div}(r)$ on W. Commutativity of divisor intersection (theorem 2.3.2), applied on W to the pair $D|_W$ and $\operatorname{div}(r)$, gives

$$D|_W \cdot \big(\operatorname{div}(r) \cdot [W]\big) \;=\; \operatorname{div}(r) \cdot \big(D|_W \cdot [W]\big)$$

and that theorem records the identity in the refined group $A_{k-1}\big(|D| \cap |\operatorname{div}(r)| \cap W\big)$, so it may be read there. The right-hand side vanishes: the line bundle of a principal divisor is trivial, $\mathcal{O}_W(\operatorname{div}(r)) \cong \mathcal{O}_W$, so $\operatorname{div}(r) \cdot -$ is $c_1(\mathcal{O}_W) \cap -$, and c_1 of a trivial bundle is the zero operator by proposition 2.4.4(3) applied to $L = L' = \mathcal{O}_W$. Hence $D \cdot [\operatorname{div}(r)] = 0$ in $A_{k-1}(|D|)$, completing the proof.

Nothing in the argument is circular: theorem 2.3.2 is proved from the avoidance-hypothesis form proposition 2.2.4 alone, so it may be used here to remove that hypothesis.

Parts (1) through (4) say that c_1 is a homomorphism of graded groups compatible with all the structure already on A_*, and that $L \mapsto c_1(L)$ turns tensor product into addition, so $c_1 \colon \mathrm{Pic}\,(X) \to \mathrm{End}(A_*(X))$ is a homomorphism from the Picard group into the operators lowering degree by one. Part (2), the commutativity, is the fact that makes these operators generate a commutative subring of $\mathrm{End}(A_*(X))$; it is the discharged form of the divisor symmetry that theorem 1.4.6 in section 1.4 took on faith, now proved through theorem 2.3.2. Part (5), the nilpotence, is a dimension count: it says a single line bundle can only cut down the fundamental class n times before the result must vanish, and it is what makes the top self-intersection number $c_1(L)^n \cap [X]$, a zero-cycle, the natural degree-type invariant of L on a complete variety.

The nilpotence has a converse worth flagging: while $c_1(L)^{n+1} \cap [X]$ always vanishes, the intermediate power $c_1(L)^n \cap [X] \in A_0(X)$ need not, and its degree $\int_X c_1(L)^n$ is the numerical invariant of a line bundle on an n-dimensional complete variety. For $n = 1$ this is the degree of a line bundle on a curve; for a surface it is the self-intersection L^2. These invariants are the raw material of the intersection numbers computed throughout this book.

The first substantial computation is the self-intersection of $\mathcal{O}(1)$ on projective space, which reproduces the entire linear-subspace basis of theorem 1.5.2 from a single line bundle. It shows the first Chern class already generates the Chow ring of $\mathbb{P}^n$, and it is the model computation the higher Chern classes generalize.

Example 2.4.7: Powers of the Hyperplane Class on $\mathbb{P}^n$

Let $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$ act on $A_*(\mathbb{P}^n)$. By theorem 1.5.2 the Chow group $A_*(\mathbb{P}^n) = \bigoplus_{j=0}^{n} \mathbb{Z}\,[\mathbb{P}^j]$ is free on the classes of linear subspaces, with $[\mathbb{P}^j]$ the class of a j-plane. The claim is

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \qquad 0 \le k \le n$$

so the k-th self-cap of the hyperplane bundle is the class of a linear subspace of codimension k.

The proof is induction on k. For $k = 0$ the left side is $[\mathbb{P}^n]$, the fundamental class, and $[\mathbb{P}^{n-0}] = [\mathbb{P}^n]$. Assume $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for some $k < n$, so the class is represented by the subvariety $\mathbb{P}^{n-k} \subseteq \mathbb{P}^n$, a linear subspace of dimension $n-k$. Cap once more with H. There is a hyperplane H_0 with $\mathbb{P}^{n-k} \not\subseteq H_0$, because some linear form does not vanish identically on the nonempty $\mathbb{P}^{n-k}$ (equivalently, $\mathbb{P}^{n-k}$ is not contained in every hyperplane); its zero locus is such an H_0, and $H_0 \cap \mathbb{P}^{n-k}$ is then a hyperplane inside $\mathbb{P}^{n-k}$, of dimension $n-k-1$. Thus $\mathbb{P}^{n-k} \not\subseteq |H_0|$, and by definition 2.2.1 the cap $H \cdot [\mathbb{P}^{n-k}]$ is the divisor of the restricted linear form on $\mathbb{P}^{n-k}$, namely the hyperplane section $H_0 \cap \mathbb{P}^{n-k}$, a linear subspace $\mathbb{P}^{n-k-1}$ of dimension $n-k-1$:

$$c_1(\mathcal{O}(1))^{k+1} \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^{n-k}] = [H_0 \cap \mathbb{P}^{n-k}] = [\mathbb{P}^{n-k-1}]$$

completing the induction. In particular $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathbb{P}^0] = [\mathrm{pt}]$ is the class of a single point, and its degree is

$$\int_{\mathbb{P}^n} c_1(\mathcal{O}_{\mathbb{P}^n}(1))^n = 1$$

the statement that n general hyperplanes in $\mathbb{P}^n$ meet in a single reduced point. Capping once more, $c_1(\mathcal{O}(1))^{n+1} \cap [\mathbb{P}^n] = H \cdot [\mathrm{pt}] = 0$, in agreement with the nilpotence of proposition 2.4.4(5). Every generator $[\mathbb{P}^j]$ of $A_*(\mathbb{P}^n)$ is thus a power of $c_1(\mathcal{O}(1))$ applied to $[\mathbb{P}^n]$, so the first Chern class of the hyperplane bundle generates the Chow group of projective space as a module over the operators.

The computation is the first appearance of the ring structure that intersection theory is heading toward: on $\mathbb{P}^n$ the operators H^k multiply the way linear subspaces intersect, $[\mathbb{P}^a] \cdot [\mathbb{P}^b] = [\mathbb{P}^{a+b-n}]$ when $a + b \geq n$ and zero otherwise, and $\int_{\mathbb{P}^n} H^n = 1$ is the normalization. The Chow group $A_*(\mathbb{P}^n)$ is the truncated polynomial ring $\mathbb{Z}[H]/(H^{n+1})$ with $H = c_1(\mathcal{O}(1))$, and the nilpotence relation $H^{n+1} = 0$ is exactly proposition 2.4.4(5). This ring will be built in general later in this Part, once the intersection product is available; here it is visible for projective space through the single line bundle $\mathcal{O}(1)$.

The second computation identifies c_1 on a curve with the classical degree, tying the abstract operator to the invariant every reader already knows. On a smooth projective curve the first Chern class is the degree map, and capping with $c_1(L)$ is subtraction of that degree from the point count.

Example 2.4.8: The First Chern Class on a Curve is the Degree

Let C be a smooth projective curve over $\overline{k}$ and L a line bundle on C. Since $\dim C = 1$, the only nonzero cap is on $A_1(C) = \mathbb{Z}\,[C]$, and it lands in $A_0(C)$. Write $L \cong \mathcal{O}_C(D)$ for a Cartier divisor $D = \sum_P n_P\,[P]$, which on a smooth curve is the same as a Weil divisor, a finite $\mathbb{Z}$-combination of closed points. Then

$$c_1(L) \cap [C] = D \cdot [C] = \sum_P n_P\,[P] \in A_0(C)$$

by definition 2.2.1, the section of $\mathcal{O}_C(D)$ having divisor D on the curve C itself. Pushing this zero-cycle to a point through $\pi\colon C \to \operatorname{Spec} \overline{k}$ and taking degrees (definition 1.2.4),

$$\int_C c_1(L) \cap [C] = \sum_P n_P\,[\overline{k}(P) : \overline{k}] = \sum_P n_P = \deg D = \deg L$$

since every closed point of a variety over the algebraically closed field $\overline{k}$ has residue field $\overline{k}$. The number $\int_C c_1(L)$ is the degree of the line bundle. Under the identification $A_0(C) \cong \operatorname{Cl} C = \operatorname{Pic} C$ of proposition 1.5.3, capping with $c_1(L)$ against $[C]$ is the class map $L \mapsto [D]$, and its composite with the degree $A_0(C) \xrightarrow{\deg} \mathbb{Z}$ recovers the degree of the divisor class, fitting into the sequence $0 \to \operatorname{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$ of proposition 1.5.3.

The two computations frame the whole theory. On projective space $c_1(\mathcal{O}(1))$ generates the Chow ring and its top power counts a single point; on a curve $c_1(L)$ is the degree, the invariant that governs Riemann-Roch through $\chi(L) = \deg L + 1 - g$ ([2, Riemann-Roch for curves]). Both come from one construction: the first Chern class turns a line bundle into an operator on cycles, and its powers are the numerical invariants that intersection theory computes. The chapter that follows generalizes the construction from line bundles to vector bundles of arbitrary rank: a rank-r bundle E carries Chern classes $c_1(E), \ldots, c_r(E)$, built so that c_1 of a line bundle is the operator of this section, and so that the total Chern class is multiplicative in short exact sequences. The machine that produces them, Segre classes of the projectivized bundle $\mathbb{P}(E)$, is the subject of the next chapter, and the divisor calculus of this chapter is the rank-one base case on which it is built.

Exercise 2.4.1

1. Compute $\int_{\mathbb{P}^1} c_1(\mathcal{O}_{\mathbb{P}^1}(d))$ for every integer d, both by the curve-degree formula of example 2.4.8 and directly from a divisor of a section of $\mathcal{O}_{\mathbb{P}^1}(d)$. Confirm that $c_1 \colon \operatorname{Pic}(\mathbb{P}^1) \to A_0(\mathbb{P}^1) = \mathbb{Z}$ is an isomorphism.

2. Let S be a smooth projective surface and $L = \mathcal{O}_S(C)$ for a curve $C \subseteq S$. Express $c_1(L)^2 \cap [S] \in A_0(S)$ and its degree $\int_S c_1(L)^2$ in terms of the divisor C, and identify $\int_S c_1(L)^2$ with the self-intersection number $C \cdot C$ of the surface theory of [2, Intersection Theory on Surfaces]. Why does proposition 2.4.4(5) say $c_1(L)^3 \cap [S] = 0$?

3. Using proposition 2.4.4(3), show that $c_1 \colon \mathrm{Pic}\,(X) \to \mathrm{End}(A_*(X))$ is a group homomorphism from the Picard group to the additive group of degree-lowering operators. Deduce that $c_1(L^{\otimes m}) = m\, c_1(L)$ for every integer m, and that c_1 factors through the numerical class of L in the sense that linearly equivalent line bundles have equal first Chern class.

4. On $\mathbb{P}^1 \times \mathbb{P}^1$, let p_1, p_2 be the two projections and $L = p_1^*\mathcal{O}(a) \otimes p_2^*\mathcal{O}(b)$ the line bundle of bidegree (a, b). Using the projection formula proposition 2.4.4(4) and additivity, compute $\int_{\mathbb{P}^1 \times \mathbb{P}^1} c_1(L)^2$ in terms of a and b. (You may use $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ free on the classes of a point, the two rulings, and the fundamental class, with each ruling of self-intersection zero and the two rulings meeting in one point.)

5. Let $f \colon X \to Y$ be a proper surjective morphism of varieties with $\dim X = \dim Y = n$ and generic degree δ, and let L be a line bundle on Y. Show that $\int_X c_1(f^*L)^n = \delta \int_Y c_1(L)^n$. (Cap the projection formula n times and use $f_*[X] = \delta\,[Y]$.) Then explain why $c_1(f^*L)^{n+1} \cap [X] = 0$ holds regardless of δ.

Part II

Chern Classes and the Intersection Product

Everything so far has been rank one. The Chow group was cut out of the cycle group by rational equivalence, made functorial by proper pushforward and flat pullback, and then acted on by a single operator: intersecting with a Cartier divisor, which is to say capping with the first Chern class of a line bundle. That operator lowers dimension by one, commutes with itself, and satisfies a projection formula. It is also the only operator available, and a line bundle is the only bundle it sees.

This part removes both restrictions. A vector bundle of rank $r+1$ carries $r+1$ operators rather than one, and the projective bundle of its lines is the space on which they can be constructed: pushing powers of the tautological class down to the base produces the Segre classes, inverting the resulting series produces the Chern classes, and the geometry of the bundle forces the relations among them. With operators of every codimension in hand, the remaining question is whether two cycles can be multiplied rather than operated on. They can, and the mechanism is deformation to the normal cone, which replaces the classical manoeuvre of moving cycles into general position with a degeneration of the ambient embedding. The product it yields makes $A^*(X)$ a ring for smooth X, recovers Bézout's theorem, and subsumes the surface intersection pairing that *Algebraic Geometry* [2] built by hand.

3

Cones, Segre Classes, and Chern Classes of Vector Bundles

The construction of this chapter rests on one geometric observation. A vector bundle E on X has no interesting cycles of its own, being an affine bundle and so contributing nothing beyond $A_*(X)$ by theorem 1.4.6. Its projectivization $\mathbb{P}(E)$ does: it is proper over X, it carries the tautological line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$, and the first Chern class of that line bundle is an operator of the kind Chapter 2 already supplies. Pushing its powers back down to X therefore produces operators on $A_*(X)$ out of nothing but the rank-one theory, and those operators are the Segre classes.

The chapter develops this in four movements. The Segre classes come first, along with the projective-bundle formula that computes $A_*(\mathbb{P}(E))$ completely and which was promised without proof in chapter 1. Chern classes follow by inverting the Segre series, and the splitting principle, made legitimate by the flag bundle, reduces their identities to the rank-one case already in hand. The Chern character and Todd class are then assembled as polynomial expressions in these operators, with the rational coefficients their denominators force. The chapter closes by extending Segre classes from vector bundles to cones, and so to arbitrary closed subschemes, which is the input chapter 4 needs.

3.1 Segre Classes and the Projective Bundle

A vector bundle E on X determines a projective bundle $\pi\colon \mathbb{P}(E) \to X$ carrying a tautological line bundle. Pushing powers of that line bundle's first Chern class down to X produces a family of operators on $A_*(X)$, the Segre classes, and the structural theorem behind them computes $A_*(\mathbb{P}(E))$ from $A_*(X)$ together with the powers of the tautological class. Both are built here, the second in full, discharging the formula recorded without proof in section 1.5.

The projective bundle itself is not built here. *Algebraic Geometry* [2] constructs Proj of a graded quasi-coherent algebra ([2, graded quasi-coherent algebra]) and its twisting sheaf ([2, Relative Proj]), specializes the construction to a locally free sheaf ([2, the projective bundle]), and proves there that the resulting morphism is proper and flat ([2, A Projective Bundle is Proper and Flat]) and locally trivial over a trivializing cover ([2, Local Triviality of a Projective Bundle]). One translation is needed at the point of import. What that volume writes $\mathbb{P}(\mathscr{E})$ parametrizes one-dimensional *quotients* of the fibres, while this book's $\mathbb{P}(E)$ parametrizes one-dimensional *subspaces* (box 0.0.2). Writing $\mathscr{E}$ for the sheaf of sections of E, the two are related by a dual:

$$\mathbb{P}(E) \;=\; \operatorname{Proj}\left(\operatorname{Sym}^\bullet \mathscr{E}^\vee\right)$$

so the bundle this book calls $\mathbb{P}(E)$ is the one [2] calls $\mathbb{P}(\mathscr{E}^\vee)$. Under that identification the twisting sheaf $\mathcal{O}(1)$ of the relative Proj is the universal rank-one quotient of $\pi^*\mathscr{E}^\vee$ ([2, The Tautological Quotient]); dualizing that surjection exhibits $\mathcal{O}(1)^\vee$ as the universal rank-one *subsheaf* of $\pi^*\mathscr{E}$, which is the tautological line. So $\mathcal{O}_{\mathbb{P}(E)}(1)$ is the dual of the tautological subbundle, which is what the subspace convention requires. Every formula below is read in that convention.

Throughout, E has rank $r+1$, so the fibre $\mathbb{P}(E_x)$ is a projective space of dimension r. All fibres of π are pure of dimension r, and π is flat, so π is flat of relative dimension r in the sense of definition 1.3.1; flat pullback $\pi^*\colon A_k(X) \to A_{k+r}(\mathbb{P}(E))$ is therefore defined by definition 1.3.3, and π being proper, pushforward $\pi_*\colon A_k(\mathbb{P}(E)) \to A_k(X)$ is defined by definition 1.2.1. These two maps and the divisor calculus of chapter 2 are all this section uses.

Definition 3.1.1: The Tautological Class of a Projective Bundle

Let E be a vector bundle of rank $r+1$ on an algebraic scheme X, let $\pi\colon \mathbb{P}(E) \to X$ be the associated bundle of lines in E, and let $\mathcal{O}_{\mathbb{P}(E)}(1)$ be the dual of the tautological subbundle. The *tautological class* of E is the first Chern class operator

$$\xi \;=\; c_1\left(\mathcal{O}_{\mathbb{P}(E)}(1)\right)$$

of definition 2.4.1, so that $\xi \cap -\colon A_k(\mathbb{P}(E)) \to A_{k-1}(\mathbb{P}(E))$ for every k. Its powers $\xi^i \cap -$ are the i-fold iterates of this operator, with $\xi^0 \cap -$ the identity.

The class ξ is an operator on the Chow groups of the total space, not an element of a ring: there is no product on $A_*(\mathbb{P}(E))$ until the intersection product of a later chapter supplies one. What makes ξ usable is that the operators $\xi \cap -$ commute among themselves and with every other first Chern class, by proposition 2.4.4(2), so an expression such as $\xi^i \cap \pi^*\alpha$ is unambiguous however it is bracketed.

Example 3.1.2: The Tautological Class on $\mathbb{P}^n$ and on a Line Bundle

Two extreme cases fix the meaning of ξ.

1. *A trivial bundle over a point.* Let $X = \operatorname{Spec} \overline{k}$ and $E = \overline{k}^{n+1}$, so $\mathbb{P}(E) = \mathbb{P}^n$ is the space of lines through the origin in $\overline{k}^{n+1}$ and $\mathcal{O}_{\mathbb{P}(E)}(1) = \mathcal{O}_{\mathbb{P}^n}(1)$ is the Serre twist. Here $\xi = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, and example 2.4.7 computes its powers outright:

$$\xi^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \qquad 0 \le k \le n$$

with $\xi^{n+1} \cap [\mathbb{P}^n] = 0$. The tautological class of the trivial bundle over a point is the hyperplane class.

2. *A line bundle.* Let $E = L$ have rank one, so $r = 0$. A one-dimensional subspace of a one-dimensional fibre is the whole fibre, so $\mathbb{P}(L) = X$ and $\pi = \mathrm{id}$; in the Proj description, $\mathrm{Sym}^\bullet$ of an invertible sheaf has $\mathrm{Proj}\,(\mathrm{Sym}^\bullet \mathscr{E}^\vee) = X$ and $\mathcal{O}(1) = \mathscr{E}^\vee$, where $\mathscr{E}$ is the invertible sheaf of sections of L. The tautological subbundle is L itself, hence $\mathcal{O}_{\mathbb{P}(L)}(1) = L^\vee$ and, by proposition 2.4.4(3),
$$\xi = c_1(L^\vee) = -c_1(L)$$
The sign is forced by the convention: the tautological object is a *sub*bundle, and $\mathcal{O}(1)$ is its dual.

The tautological class lives on $\mathbb{P}(E)$, and the Chow groups one wants to compute with live on X. Transporting a class α on X up to $\mathbb{P}(E)$, cutting it with powers of ξ, and pushing the result back down produces an operator on $A_*(X)$ built from E alone. The bookkeeping is forced: π^* raises dimension by r, capping with ξ^m lowers it by m, and π_* preserves it, so the composite lowers dimension by $m - r$. Writing $m = r + i$ makes i the amount of dimension lost, and these composites are the Segre classes.

Definition 3.1.3: Segre Class Operators of a Vector Bundle

Let E be a vector bundle of rank $r + 1$ on an algebraic scheme X, with $\pi\colon \mathbb{P}(E) \to X$ and tautological class ξ. For an integer $i \geq -r$ the *i-th Segre class* of E is the operator defined on $\alpha \in A_k(X)$ by
$$s_i(E) \cap \alpha \;=\; \pi_*\big(\xi^{r+i} \cap \pi^*\alpha\big)$$
and $s_i(E) \cap \alpha = 0$ for $i < -r$. Each is a homomorphism $s_i(E) \cap -\colon A_k(X) \to A_{k-i}(X)$. The *total Segre class* of E is the sum
$$s(E) \;=\; \sum_{i \geq -r} s_i(E)$$
which is a finite sum of operators on any fixed class, since $s_i(E)$ lowers dimension by i and $A_j(X) = 0$ for $j < 0$.

That $s_i(E) \cap -$ is a homomorphism is inherited from its three constituents: π^* is additive by proposition 1.3.5(1), capping with ξ is additive by proposition 2.4.4(1), and π_* is additive by definition 1.2.1. Each of the three also descends to rational equivalence, by proposition 1.3.7, proposition 2.2.4 and theorem 1.2.3 respectively, so $s_i(E) \cap -$ is well defined on Chow groups without further argument. The content of the definition is not that these operators exist but that they are computable and that they encode E; the rank-one case already shows what they look like.

Example 3.1.4: The Segre Classes of a Line Bundle

Let L be a line bundle on X, so $r = 0$ and, by example 3.1.2(2), $\mathbb{P}(L) = X$ with $\pi = \mathrm{id}$ and $\xi = -c_1(L)$. The definition reads
$$s_i(L) \cap \alpha = \xi^{\,i} \cap \alpha = (-1)^i\, c_1(L)^i \cap \alpha, \qquad i \geq 0$$
there being no indices below $-r = 0$. The total Segre class is therefore the alternating series
$$s(L) = 1 - c_1(L) + c_1(L)^2 - c_1(L)^3 + \cdots$$
which on any fixed class terminates, since $c_1(L)^m \cap \alpha$ lies in $A_{k-m}(X)$ and this group vanishes once $m > k$. The series is the inverse of $1 + c_1(L)$: capping $s(L) \cap \alpha$ with $1 + c_1(L)$ telescopes,

every term cancelling against its successor, and leaves α.
For a concrete instance take $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(1)$. Then $c_1(L)^i \cap [\mathbb{P}^n] = [\mathbb{P}^{n-i}]$ by example 2.4.7, so

$$s_i\big(\mathcal{O}_{\mathbb{P}^n}(1)\big) \cap [\mathbb{P}^n] = (-1)^i\,[\mathbb{P}^{n-i}], \qquad 0 \le i \le n$$

and $s_i(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = 0$ for $i > n$. The Segre classes of a line bundle carry exactly the information in c_1, packaged with alternating signs.

The alternating signs in example 3.1.4 indicate that $s(E)$ is the wrong normalization for a class one wants to be additive on direct sums; the repair, carried out in section 3.2, is to invert the series. Before that, the operators need their formal properties: which of them vanish, which one is the identity, whether they commute, and how they behave under the two functorialities of the Chow group. All four are consequences of the corresponding properties of c_1 and of the flat-proper calculus of chapter 1.

Proposition 3.1.5: Basic Properties of Segre Classes

Let E be a vector bundle of rank $r+1$ on an algebraic scheme X.

1. *Vanishing.* $s_i(E) \cap \alpha = 0$ for every $i < 0$ and every $\alpha \in A_*(X)$.
2. *Normalization.* $s_0(E) \cap \alpha = \alpha$ for every $\alpha \in A_*(X)$; that is, $s_0(E)$ is the identity operator. Consequently $s(E) = 1 + s_1(E) + s_2(E) + \cdots$.
3. *Commutativity.* If F is a second vector bundle on X, then for all i, j
$$s_i(E) \cap \big(s_j(F) \cap \alpha\big) = s_j(F) \cap \big(s_i(E) \cap \alpha\big)$$
4. *Functoriality.* If $f\colon X' \to X$ is proper and $\alpha' \in A_k(X')$, then
$$f_*\big(s_i(f^*E) \cap \alpha'\big) = s_i(E) \cap f_*\alpha'$$
and if $g\colon X'' \to X$ is flat of relative dimension m and $\beta \in A_k(X)$, then
$$s_i(g^*E) \cap g^*\beta = g^*\big(s_i(E) \cap \beta\big)$$

***Proof*:**
Formation of $\mathbb{P}(E)$ and of $\mathcal{O}_{\mathbb{P}(E)}(1)$ from $\mathrm{Sym}^\bullet \mathscr{E}^\vee$ commutes with base change along an arbitrary morphism ([2, Relative Proj Commutes with Base Change]), so for any morphism $h\colon Y \to X$ there is a fibre square

$$\begin{array}{ccc} \mathbb{P}(h^*E) & \xrightarrow{h'} & \mathbb{P}(E) \\ \downarrow \pi_Y & & \downarrow \pi \\ Y & \xrightarrow{h} & X \end{array}$$

with $\mathbb{P}(h^*E) = \mathbb{P}(E) \times_X Y$ and $\mathcal{O}_{\mathbb{P}(h^*E)}(1) = h'^*\mathcal{O}_{\mathbb{P}(E)}(1)$, hence $\xi_Y = h'^*\xi$ for the tautological classes. This square is used in every part.

1. *Vanishing.* By additivity it suffices to treat $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$, with closed immersion $h\colon V \hookrightarrow X$. Since h is proper and π is flat, proposition 1.3.6 applied

to the square above gives $\pi^* h_* = h'_* \pi_V^*$, so

$$\pi^*[V] = h'_*\, \pi_V^*[V]$$

and proposition 2.4.4(4) for the proper h', applied $r+i$ times with $\xi_V = h'^*\xi$, gives

$$\xi^{\,r+i} \cap \pi^*[V] = h'_*\big(\xi_V^{r+i} \cap \pi_V^*[V]\big)$$

Pushing forward by π and using $\pi \circ h' = h \circ \pi_V$ with functoriality of proper pushforward,

$$s_i(E) \cap [V] = h_*\, \pi_{V*}\big(\xi_V^{r+i} \cap \pi_V^*[V]\big) \tag{3.1}$$

The class inside h_* lies in $A_{k-i}(V)$, and for $i < 0$ one has $k - i > k = \dim V$, so that group is zero: V has no subvariety of dimension exceeding its own. Hence $s_i(E) \cap [V] = 0$.

2. *Normalization.* The inner class in (3.1) is $s_i(E|_V) \cap [V]$ computed on V, so that identity reduces the claim to the case $\alpha = [X]$ with X a variety of dimension k, where it reads $\pi_*(\xi^{\,r} \cap [\mathbb{P}(E)]) = [X]$; note that $\pi^*[X] = [\mathbb{P}(E)]$ by proposition 1.3.5(2). The total space $\mathbb{P}(E)$ is a variety: it is covered by open sets isomorphic to $U_\lambda \times \mathbb{P}^r$ for a trivializing cover $\{U_\lambda\}$ of X ([2, Local Triviality of a Projective Bundle]), each of which is irreducible and reduced, and any two of them meet because their images in the irreducible X do. So $\dim \mathbb{P}(E) = k + r$ and $\pi_*(\xi^r \cap [\mathbb{P}(E)])$ lies in $A_k(X) = Z_k(X) = \mathbb{Z}\,[X]$ by example 1.1.6(3); write it as $m\,[X]$ with $m \in \mathbb{Z}$.

   The integer m is computed after restriction to a trivializing open. Choose a nonempty open $U \subseteq X$ with $E|_U$ trivial, and let $j\colon U \hookrightarrow X$ and $j'\colon \mathbb{P}(E|_U) \hookrightarrow \mathbb{P}(E)$ be the inclusions, both flat of relative dimension zero. Flat pullback commutes with π_* across the fibre square by proposition 1.3.6 and with capping by ξ by proposition 2.4.4(4), and $j'^*[\mathbb{P}(E)] = [\mathbb{P}(E|_U)]$ because restriction along an open immersion carries a fundamental cycle to a fundamental cycle (example 1.3.4(1)), so

$$j^*\big(m\,[X]\big) = \pi_{U*}\big(\xi_U^r \cap [\mathbb{P}(E|_U)]\big)$$

   Under a trivialization $\mathbb{P}(E|_U) \cong U \times \mathbb{P}^r$ the twisting sheaf becomes $\rho^*\mathcal{O}_{\mathbb{P}^r}(1)$ for the second projection $\rho\colon U \times \mathbb{P}^r \to \mathbb{P}^r$ ([2, Twisting Sheaf]), and ρ is flat of relative dimension $\dim U$, so $\rho^*[\mathbb{P}^r] = [U \times \mathbb{P}^r]$ by proposition 1.3.5(2). Hence, by the flat case of proposition 2.4.4(4) and by example 2.4.7,

$$\xi_U^r \cap [U \times \mathbb{P}^r] = \rho^*\big(c_1(\mathcal{O}_{\mathbb{P}^r}(1))^r \cap [\mathbb{P}^r]\big) = \rho^*[\mathrm{pt}] = [U \times \{\mathrm{pt}\}]$$

   The projection π_U restricts to an isomorphism $U \times \{\mathrm{pt}\} \to U$, so $\pi_{U*}[U \times \{\mathrm{pt}\}] = [U]$ by definition 1.2.1. Thus $m\,[U] = [U]$ in $A_k(U) = \mathbb{Z}\,[U]$, which is free on the fundamental class by example 1.1.6(3), and therefore $m = 1$. This gives $s_0(E) \cap [X] = [X]$, and with part (1) the total class has no terms in negative degree.

3. *Commutativity.* Let F have rank $f + 1$, with $q\colon \mathbb{P}(F) \to X$ and tautological class η, and form the fibre product $W = \mathbb{P}(E) \times_X \mathbb{P}(F)$ with its two projections $q'\colon W \to \mathbb{P}(E)$ and $p'\colon W \to \mathbb{P}(F)$. Since q is proper and π is flat, q' is proper and p' is flat of relative dimension r; write $\rho = \pi \circ q' = q \circ p'\colon W \to X$, which is proper, and flat of relative

dimension $r+f$ by proposition 1.3.5(3). Put $\tilde{\xi} = q'^*\xi$ and $\tilde{\eta} = p'^*\eta$. Then

$$\begin{aligned} s_i(E) \cap \big(s_j(F) \cap \alpha\big) &= \pi_*\Big(\xi^{\,r+i} \cap \pi^* q_*\big(\eta^{\,f+j} \cap q^*\alpha\big)\Big) \\ &= \pi_*\Big(\xi^{\,r+i} \cap q'_* p'^*\big(\eta^{\,f+j} \cap q^*\alpha\big)\Big) \\ &= \pi_*\Big(\xi^{\,r+i} \cap q'_*\big(\tilde{\eta}^{\,f+j} \cap \rho^*\alpha\big)\Big) \\ &= \pi_* q'_*\Big(\tilde{\xi}^{\,r+i} \cap \tilde{\eta}^{\,f+j} \cap \rho^*\alpha\Big) = \rho_*\Big(\tilde{\xi}^{\,r+i} \cap \tilde{\eta}^{\,f+j} \cap \rho^*\alpha\Big) \end{aligned}$$

the second line by proposition 1.3.6 for the fibre square defining W, the third by the flat case of proposition 2.4.4(4) together with $p'^*q^* = \rho^*$ (proposition 1.3.5(3)), and the fourth by the proper case of proposition 2.4.4(4) for q'. Running the same computation with the roles of E and F exchanged uses the same scheme W with its two projections interchanged, and expresses $s_j(F) \cap (s_i(E) \cap \alpha)$ as $\rho_*(\tilde{\eta}^{\,f+j} \cap \tilde{\xi}^{\,r+i} \cap \rho^*\alpha)$. The two expressions differ only in the order of two first Chern class cap operators, which commute by proposition 2.4.4(2), so they agree; this is the asserted commutativity.

4. *Functoriality.* For the proper case take $h = f$ in the fibre square above, so $f' \colon \mathbb{P}(f^*E) \to \mathbb{P}(E)$ is proper and $\pi_{X'}$ is proper and flat of relative dimension r. Then

$$f_*\big(s_i(f^*E) \cap \alpha'\big) = f_*\, \pi_{X'*}\big(\xi_{X'}^{r+i} \cap \pi_{X'}^*\alpha'\big) = \pi_*\, f'_*\big(f'^*\xi^{\,r+i} \cap \pi_{X'}^*\alpha'\big)$$

using $f \circ \pi_{X'} = \pi \circ f'$ and $\xi_{X'} = f'^*\xi$. The proper case of proposition 2.4.4(4) moves the cap outside, and proposition 1.3.6 for the same square gives $f'_*\pi_{X'}^* = \pi^* f_*$, so the right side is $\pi_*(\xi^{\,r+i} \cap \pi^* f_*\alpha') = s_i(E) \cap f_*\alpha'$.

For the flat case take $h = g$, so $g' \colon \mathbb{P}(g^*E) \to \mathbb{P}(E)$ is flat of relative dimension m. By proposition 1.3.5(3), $\pi_{X''}^* g^* = (g \circ \pi_{X''})^* = (\pi \circ g')^* = g'^*\pi^*$, and by the flat case of proposition 2.4.4(4) with $\xi_{X''} = g'^*\xi$,

$$\xi_{X''}^{r+i} \cap \pi_{X''}^* g^*\beta = g'^*\big(\xi^{\,r+i} \cap \pi^*\beta\big)$$

Applying $\pi_{X''*}$ and using $\pi_{X''*}g'^* = g^*\pi_*$, again proposition 1.3.6, gives $s_i(g^*E) \cap g^*\beta = g^*(s_i(E) \cap \beta)$, completing the proof.

Parts (1) and (2) say that the total Segre class is an operator of the form 1+(dimension-lowering terms), hence invertible: a series $1 + s_1 + s_2 + \cdots$ whose higher terms strictly lower dimension has a two-sided inverse of the same shape, computed by the usual recursion, and section 3.2 names that inverse the total Chern class. Part (3) is what makes any such formal manipulation legitimate, since inverting a series requires its terms to commute. Part (4) says the Segre classes are natural: they are computed in whichever space is most convenient and transported by the functorialities already available. The normalization in part (2) has an immediate structural consequence for π^* itself.

Corollary 3.1.6: Flat Pullback along a Projective Bundle is Split Injective

Let E be a vector bundle of rank $r+1$ on algebraic scheme X, with $\pi \colon \mathbb{P}(E) \to X$ and tautological class ξ. Then for every k the map $\pi^* \colon A_k(X) \to A_{k+r}(\mathbb{P}(E))$ is injective, and $\beta \mapsto \pi_*(\xi^{\,r} \cap \beta)$ is a left inverse for it.

***Proof*:**
For $\alpha \in A_k(X)$ the composite is $\pi_*(\xi^r \cap \pi^*\alpha) = s_0(E) \cap \alpha = \alpha$ by definition 3.1.3 and proposition 3.1.5(2). A map admitting a left inverse is injective, as we sought to show.

No cycle class on the base is lost on passing to the projective bundle, and the loss can be undone by an explicit formula rather than an abstract argument. This is the mechanism that will make the flag bundle a legitimate device for proving identities among Chern classes rather than a heuristic; the iteration is carried out in section 3.2. A second consequence of proposition 3.1.5 is a computation, and it is the one that calibrates the whole normalization.

Example 3.1.7: The Segre Classes of a Trivial Bundle

Let $E = \mathcal{O}_X^{\oplus(r+1)}$ be trivial of rank $r+1$, so $\mathbb{P}(E) = X \times \mathbb{P}^r$ with π the first projection, ρ the second, and $\mathcal{O}_{\mathbb{P}(E)}(1) = \rho^*\mathcal{O}_{\mathbb{P}^r}(1)$, whence $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$. Then

$$s_0(E) = 1, \qquad s_i(E) = 0 \ \text{ for } i \neq 0$$

so $s(E) = 1$.

The case $i < 0$ is proposition 3.1.5(1) and the case $i = 0$ is proposition 3.1.5(2), so only $i \geq 1$ needs an argument. By (3.1) it suffices to take $X = V$ a variety of dimension k and $\alpha = [V]$. Then $\pi^*[V] = [V \times \mathbb{P}^r]$, and $[V \times \mathbb{P}^r] = \rho^*[\mathbb{P}^r]$ because ρ is flat of relative dimension k over the variety $\mathbb{P}^r$ (proposition 1.3.5(2)). The flat case of proposition 2.4.4(4) then moves the whole cap into the base:

$$\xi^{r+i} \cap \pi^*[V] = \rho^*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^r}(1)\big)^{r+i} \cap [\mathbb{P}^r]\Big)$$

For $i \geq 1$ the exponent exceeds $r = \dim \mathbb{P}^r$, so the class in brackets vanishes by proposition 2.4.4(5), and therefore $s_i(E) \cap [V] = \pi_*(0) = 0$. A trivial bundle has trivial total Segre class, which is the normalization that makes $s(E)$, and with it the Chern classes built from it, an invariant of the twisting of E rather than of its rank.

The computation in example 3.1.7 used only the structure of $\mathbb{P}^r$ over a point, transported along a flat projection. The same transport, applied to a hyperplane rather than to the whole space, produces the identity that drives the main theorem: on a trivial projective bundle, the tautological class is the class of a sub-bundle of hyperplanes, so capping with ξ is the same as pushing forward from a smaller projective bundle.

Lemma 3.1.8: The Hyperplane Divisor on a Trivial Projective Bundle

Let X be an algebraic scheme, let $r \geq 1$, and let $P = X \times \mathbb{P}^r$ with projections $\pi\colon P \to X$ and $\rho\colon P \to \mathbb{P}^r$, and $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$. Fix a hyperplane $H \subseteq \mathbb{P}^r$, put $Q = X \times H$ with its closed immersion $\iota\colon Q \hookrightarrow P$, and let $\pi_Q : Q \to X$ be the projection. Then for every $\alpha \in A_k(X)$,

$$\xi \cap \pi^*\alpha = \iota_*\,\pi_Q^*\alpha \quad \text{in } A_{k+r-1}(P)$$

***Proof*:**
Since H is cut out by a linear form, $\mathcal{O}_{\mathbb{P}^r}(H) \cong \mathcal{O}_{\mathbb{P}^r}(1)$ by example 2.4.3(2). No component of P maps into H, so the local equations of H pull back to local equations of a Cartier divisor Q on P, and $\mathcal{O}_P(Q) \cong \rho^*\mathcal{O}_{\mathbb{P}^r}(1)$ because $\mathcal{O}(D)$ is formed from those same local equations. (This is

the pullback of a Cartier divisor along an arbitrary morphism meeting the support properly, not the flat pullback of proposition 2.3.3, whose relative-dimension hypothesis ρ need not satisfy when X has components of different dimensions.) By definition 2.4.1 this identifies the operator: $\xi \cap \pi^*\alpha = Q \cdot \pi^*\alpha$.

Both sides of the asserted identity are additive homomorphisms $A_k(X) \to A_{k+r-1}(P)$, the left by proposition 1.3.7 and proposition 2.2.4, the right by proposition 1.3.7 and theorem 1.2.3, so it suffices to check them on the class of a k-dimensional subvariety $W \subseteq X$. The morphism π is flat of relative dimension r, being the base change of $\mathbb{P}^r \to \operatorname{Spec} \overline{k}$, so $\pi^*[W] = [W \times \mathbb{P}^r]$ by definition 1.3.3, the preimage $W \times \mathbb{P}^r$ being integral of dimension $k + r$ and hence carrying multiplicity one. This subvariety is not contained in $|Q| = X \times H$, so the first clause of definition 2.2.1 applies: writing s for the rational section of $\mathcal{O}_P(Q)$ cutting out Q, which is the pullback along ρ of the linear form defining H, its restriction to $W \times \mathbb{P}^r$ has divisor

$$\left[\operatorname{div}\left(s|_{W \times \mathbb{P}^r}\right)\right] = [W \times H]$$

with multiplicity one, since a linear form vanishes to order one along the hyperplane it defines. Hence $Q \cdot \pi^*[W] = [W \times H]$. On the other side π_Q is flat of relative dimension $r - 1$, so $\pi_Q^*[W] = [W \times H]$ by the same reasoning, and ι_* carries this class to $[W \times H]$ in $Z_{k+r-1}(P)$, the closed immersion having degree one on each subvariety (definition 1.2.1). The two sides therefore agree on every generator, completing the proof.

The lemma converts a divisor operation into a pushforward, and pushforward is what the localization sequence controls. That is the whole strategy for the theorem below: peel $\mathbb{P}^r$ apart into a hyperplane and its affine complement, use theorem 1.4.6 on the complement, use the lemma to recognize the hyperplane contribution as one more power of ξ, and induct. The general bundle is then reduced to the trivial one by a second induction, over the closed subschemes of the base, since a bundle is trivial over some dense open set. Injectivity, by contrast, needs no induction at all: the Segre classes already invert the construction.

Theorem 3.1.9: The Projective Bundle Formula

Let E be a vector bundle of rank $r + 1$ on an algebraic scheme X, with $\pi\colon \mathbb{P}(E) \to X$ and tautological class ξ. For every k the homomorphism

$$\Psi\colon \bigoplus_{i=0}^{r} A_{k-r+i}(X) \longrightarrow A_k\big(\mathbb{P}(E)\big), \qquad (\alpha_0, \dots, \alpha_r) \longmapsto \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$$

is an isomorphism. Every class $\beta \in A_k(\mathbb{P}(E))$ is thus uniquely of the form $\beta = \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$ with $\alpha_i \in A_{k-r+i}(X)$, and the components are recovered from β by the triangular system

$$\pi_*\big(\xi^{\,m} \cap \beta\big) = \alpha_{r-m} + \sum_{j=1}^{m} s_j(E) \cap \alpha_{r-m+j}, \qquad 0 \le m \le r \tag{3.2}$$

***Proof*:**
Each summand of Ψ lands in $A_k(\mathbb{P}(E))$: for $\alpha_i \in A_{k-r+i}(X)$ the class $\pi^*\alpha_i$ lies in $A_{k+i}(\mathbb{P}(E))$ and capping with ξ^i lowers dimension by i. The map is additive because π^* and $\xi \cap -$ are.

Step 1: the triangular system, and injectivity. Let $\beta = \sum_{i=0}^{r} \xi^i \cap \pi^*\alpha_i$ and fix $m \geq 0$. Then

$$\pi_*\big(\xi^m \cap \beta\big) = \sum_{i=0}^{r} \pi_*\big(\xi^{m+i} \cap \pi^*\alpha_i\big) = \sum_{i=0}^{r} s_{m+i-r}(E) \cap \alpha_i$$

directly from definition 3.1.3. By proposition 3.1.5(1) the terms with $m+i-r<0$ vanish, and by proposition 3.1.5(2) the term with $m+i-r=0$ is α_{r-m}; writing $j = m+i-r$ for the surviving terms, which satisfy $1 \leq j \leq m$ since $i \leq r$, gives exactly (3.2).

Suppose now $\Psi(\alpha_0, \dots, \alpha_r) = 0$. Taking $m = 0$ in (3.2) gives $\alpha_r = 0$. Assume inductively that $\alpha_r = \cdots = \alpha_{r-m+1} = 0$ for some $m \leq r$; then every correction term in (3.2) involves one of these vanishing classes, so the identity reduces to $\alpha_{r-m} = 0$. Descending from $m = 0$ to $m = r$ kills every component, so Ψ is injective. This argument used no hypothesis on X beyond those standing.

Step 2: surjectivity for a trivial bundle. Let $E = \mathcal{O}_X^{\oplus(r+1)}$, so $\mathbb{P}(E) = X \times \mathbb{P}^r$ and $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$ for the second projection ρ. The claim is proved by induction on r, uniformly in the base X.

For $r = 0$ one has $\mathbb{P}(E) = X$ with $\pi = \mathrm{id}$, so $\Psi(\alpha_0) = \alpha_0$ is the identity map of $A_k(X)$, which is surjective. Let $r \geq 1$ and assume the claim for $r-1$ over every base. Fix a hyperplane $H \subseteq \mathbb{P}^r$, put $Q = X \times H$ with closed immersion $\iota\colon Q \hookrightarrow P = X \times \mathbb{P}^r$, and let $j\colon X \times \mathbb{A}^r \hookrightarrow P$ be the inclusion of the open complement, $\mathbb{A}^r = \mathbb{P}^r \setminus H$. The localization sequence (theorem 1.4.1) reads

$$A_k(Q) \xrightarrow{\iota_*} A_k(P) \xrightarrow{j^*} A_k(X \times \mathbb{A}^r) \longrightarrow 0$$

Let $q = \pi \circ j\colon X \times \mathbb{A}^r \to X$ be the projection. By affine-bundle invariance (theorem 1.4.6) the pullback $q^*\colon A_{k-r}(X) \to A_k(X \times \mathbb{A}^r)$ is an isomorphism, and $q^* = j^*\pi^*$ by proposition 1.3.5(3). Given $\beta \in A_k(P)$, there is therefore $\alpha_0 \in A_{k-r}(X)$ with $j^*\beta = q^*\alpha_0 = j^*(\pi^*\alpha_0)$, whence $\beta - \pi^*\alpha_0$ lies in $\ker j^* = \operatorname{im} \iota_*$ by exactness: say

$$\beta = \pi^*\alpha_0 + \iota_*\gamma, \qquad \gamma \in A_k(Q)$$

Now $Q = X \times H \cong X \times \mathbb{P}^{r-1}$ is the trivial projective bundle of one lower fibre dimension, and its tautological class is $\xi_Q = \iota^*\xi$: choosing a second hyperplane $H' \neq H$, the restriction $\mathcal{O}_{\mathbb{P}^r}(1)|_H \cong \mathcal{O}_{\mathbb{P}^r}(H')|_H$ is the line bundle of the hyperplane $H' \cap H$ of H, which is $\mathcal{O}_{\mathbb{P}^{r-1}}(1)$ under the identification $H \cong \mathbb{P}^{r-1}$. By the inductive hypothesis,

$$\gamma = \sum_{i=0}^{r-1} \xi_Q^{\,i} \cap \pi_Q^*\alpha_i', \qquad \alpha_i' \in A_{k-(r-1)+i}(X)$$

Pushing forward, the proper case of proposition 2.4.4(4) applied i times with $\xi_Q = \iota^*\xi$ gives $\iota_*(\xi_Q^{\,i} \cap \pi_Q^*\alpha_i') = \xi^i \cap \iota_*\pi_Q^*\alpha_i'$, and lemma 3.1.8 identifies $\iota_*\pi_Q^*\alpha_i' = \xi \cap \pi^*\alpha_i'$. Hence

$$\iota_*\gamma = \sum_{i=0}^{r-1} \xi^{i+1} \cap \pi^*\alpha_i'$$

and setting $\alpha_{i+1} = \alpha_i' \in A_{k-r+(i+1)}(X)$ for $0 \leq i \leq r-1$ exhibits $\beta = \sum_{i=0}^{r} \xi^i \cap \pi^*\alpha_i$. This completes the induction on r, so Ψ is surjective for every trivial bundle.

Step 3: surjectivity in general. The scheme X is of finite type over $\overline{k}$, hence Noetherian, so its closed subschemes satisfy the descending chain condition and the following suffices: if $\Psi_{E|_W}$ is surjective in every degree for every proper closed subscheme $W \subsetneq X$, then Ψ_E is surjective. The empty scheme starts the induction.

First produce a dense trivializing open. The scheme X has finitely many irreducible components, with generic points $\eta_1, \dots, \eta_s$. For each t choose an open $V_t \ni \eta_t$ over which E is trivial, and set $U_t = V_t \setminus \bigcup_{u \neq t} \overline{\{\eta_u\}}$, still an open neighbourhood of η_t since η_t lies on no other component. The U_t are pairwise disjoint: each U_u is contained in $\overline{\{\eta_u\}}$, because the components cover X, while U_t misses $\overline{\{\eta_u\}}$ for $u \neq t$. Their union $U = \bigsqcup_t U_t$ is an open set containing every generic point, hence dense, and $E|_U$ is trivial of rank $r+1$, being trivial on each of the disjoint pieces. If $U = X$ then E is trivial and Step 2 applies; otherwise let $Z = X \setminus U$ with its reduced structure, a proper closed subscheme.

Write $j\colon U \hookrightarrow X$ and $i_Z\colon Z \hookrightarrow X$ for the inclusions, and $j'\colon \mathbb{P}(E|_U) \hookrightarrow \mathbb{P}(E)$, $\iota\colon \mathbb{P}(E|_Z) \hookrightarrow \mathbb{P}(E)$ for the induced ones; these are the open and closed pieces of $\mathbb{P}(E)$, since forming $\mathbb{P}(E)$ commutes with base change. Let $\beta \in A_k(\mathbb{P}(E))$. Restricting to the trivializing part and applying Step 2 over U,

$$j'^*\beta = \sum_{i=0}^{r} \xi_U^{\,i} \cap \pi_U^* \alpha_i^U, \qquad \alpha_i^U \in A_{k-r+i}(U)$$

By theorem 1.4.1 the restriction $j^*\colon A_{k-r+i}(X) \to A_{k-r+i}(U)$ is surjective, so each α_i^U lifts to some $\alpha_i \in A_{k-r+i}(X)$. Flat pullback along j' commutes with π^*, since $\pi \circ j' = j \circ \pi_U$ gives $j'^*\pi^* = \pi_U^* j^*$ by proposition 1.3.5(3), and with capping by ξ by the flat case of proposition 2.4.4(4), using $\xi_U = j'^*\xi$. Hence $j'^*\big(\beta - \Psi_E(\alpha_0, \dots, \alpha_r)\big) = 0$, and the localization sequence for the closed $\mathbb{P}(E|_Z) \subseteq \mathbb{P}(E)$ with open complement $\mathbb{P}(E|_U)$ produces $\delta \in A_k(\mathbb{P}(E|_Z))$ with

$$\beta - \Psi_E(\alpha_0, \dots, \alpha_r) = \iota_*\delta$$

Since $Z \subsetneq X$ is a proper closed subscheme, the inductive hypothesis writes $\delta = \sum_{i=0}^{r} \xi_Z^{\,i} \cap \pi_Z^* \epsilon_i$ with $\epsilon_i \in A_{k-r+i}(Z)$. The proper case of proposition 2.4.4(4) with $\xi_Z = \iota^*\xi$, followed by proposition 1.3.6 for the fibre square over i_Z (whose horizontal map is proper and whose vertical map π is flat), gives

$$\iota_*\big(\xi_Z^{\,i} \cap \pi_Z^*\epsilon_i\big) = \xi^{\,i} \cap \iota_*\pi_Z^*\epsilon_i = \xi^{\,i} \cap \pi^*\big(i_{Z*}\epsilon_i\big)$$

Therefore $\beta = \Psi_E(\alpha_0 + i_{Z*}\epsilon_0, \dots, \alpha_r + i_{Z*}\epsilon_r)$ lies in the image of Ψ_E. This closes the Noetherian induction, so Ψ is surjective for every vector bundle, and with Step 1 it is an isomorphism, completing the proof.

The theorem computes the Chow groups of a projective bundle from those of its base, and it does so with an explicit basis and an explicit inversion. The classes $1, \xi, \dots, \xi^r$ behave like a basis of $A_*(\mathbb{P}(E))$ over $A_*(X)$, and it is standard to say that $A_*(\mathbb{P}(E))$ is a free $A_*(X)$-module on them; that phrasing anticipates a ring structure on $A_*(X)$ that does not exist yet, so the content here is the isomorphism Ψ of graded groups together with the fact that its components are the operators $\xi^i \cap \pi^*(-)$. The inversion (3.2) is what makes the statement usable in practice: to identify the components of a given class one pushes forward its caps with successive powers of ξ, and the Segre classes appear as the correction terms in a triangular system that can be solved from the top down. This is also the precise sense in which the Segre classes measure the failure of ξ to behave like the hyperplane class on a product, since a trivial bundle has all corrections zero by example 3.1.7.

Taking $X = \operatorname{Spec} \overline{k}$ and $E = \overline{k}^{\,n+1}$ recovers $A_k(\mathbb{P}^n) = \mathbb{Z}\left(\xi^{\,n-k} \cap [\mathbb{P}^n]\right)$, the statement of theorem 1.5.2: the group $A_j(\operatorname{Spec} \overline{k})$ is $\mathbb{Z}$ for $j = 0$ and zero otherwise, so only the index $i = n - k$ survives in the sum, and $\pi^*[\operatorname{Spec}\overline{k}] = [\mathbb{P}^n]$. The theorem is thus the exact generalization of the projective-space computation from a point to an arbitrary base, which is how it was announced in section 1.5, and its proof runs by reducing to that computation twice: once through example 2.4.7 in the normalization $s_0(E) = 1$, and once through affine-bundle invariance in Step 2. The corresponding statement in singular cohomology, that $H^*(\mathbb{P}(E))$ is free over $H^*(X)$ on the powers of the class of $\mathcal{O}(1)$, holds as well, but it enters the theory at a different point: in [10] the topological Chern classes are defined as pullbacks of the polynomial generators of $H^*(BU(n))$, and the projective-bundle statement is a consequence, proved once they are in hand. Here the dependence runs the other way, the formula above coming first and the Chern operators of section 3.2 being defined by inverting the total Segre class; and the argument owes nothing to topology, being valid over any algebraically closed field and on singular schemes.

Exercise 3.1.1

1. Let L be a line bundle on an algebraic scheme X. Using example 3.1.2(2), show that $s_i(L) \cap \alpha = (-1)^i c_1(L)^i \cap \alpha$ for every $i \geq 0$ and every $\alpha \in A_k(X)$, and deduce that the operators $s(L) \cap -$ and $(1 + c_1(L)) \cap -$ are mutually inverse on $A_k(X)$. Explain why the series terminates on any fixed class.

2. Let E have rank $r+1$ on X, with $\pi\colon \mathbb{P}(E) \to X$. Show that $\pi_*\pi^*\alpha = 0$ for every $\alpha \in A_*(X)$ when $r \geq 1$, while $\pi_*\pi^* = \mathrm{id}$ when $r = 0$. Show also that $\beta \mapsto \pi_*(\xi^r \cap \beta)$ is a left inverse of π^* for every r, and reconcile the two computations.

3. Take $X = \mathbb{P}^n$ and $E = \mathcal{O}_{\mathbb{P}^n}^{\oplus(m+1)}$, so that $\mathbb{P}(E) = \mathbb{P}^n \times \mathbb{P}^m$ with π the first projection. Use theorem 3.1.9 and theorem 1.5.2 to show that $A_k(\mathbb{P}^n \times \mathbb{P}^m)$ is free abelian of rank equal to the number of pairs (a, b) with $0 \leq a \leq m$, $0 \leq b \leq n$ and $a + b = k$, and identify a basis geometrically.

4. Let X be a complete variety of dimension n and E a vector bundle of rank $r+1$ on X. Show that
$$\int_X s_n(E) \cap [X] = \int_{\mathbb{P}(E)} \xi^{n+r} \cap [\mathbb{P}(E)]$$
and verify that both sides vanish for E trivial and $n \geq 1$.

5. Let E have rank $r+1$ on X and let $\alpha \in A_k(X)$. By theorem 3.1.9 there are unique classes $\beta_i \in A_{k-1-r+i}(X)$ with $\xi^{r+1} \cap \pi^*\alpha = \sum_{i=0}^{r} \xi^i \cap \pi^*\beta_i$. Determine β_r and β_{r-1} in terms of the Segre operators of E applied to α, and describe the recursion that determines the remaining β_i.

3.2 Chern Classes

The Segre operators of section 3.1 record everything a vector bundle contributes to the Chow group, but in a form nothing can be computed with. The list $s_0(E), s_1(E), s_2(E), \ldots$ does not terminate, so no finite collection of them is an invariant attached to E; already for a line bundle every $s_i(L)$ is nonzero as long as $c_1(L)^i$ is. Worse, the operators attached to a direct sum are not the product of those attached to the summands, so a bundle assembled from simpler pieces has Segre operators that cannot be read off the pieces. Inverting the total Segre operator repairs both defects at once, and the inverse is what the rest of the book uses.

The section establishes four properties of the resulting Chern operators: they vanish above the rank of the bundle, they are natural for flat pullback and satisfy a projection formula, they are multiplicative on short exact sequences, and in rank one they reduce to the operator $c_1(L) \cap -$ of section 2.4. The last two are proved by the splitting principle, which replaces X by a tower of projective bundles over it on which every bundle in sight is filtered by line bundles. That replacement is legitimate rather than heuristic because the projective-bundle formula makes π^* injective, so an identity verified upstairs descends.

Throughout, E is a vector bundle of rank $e \geq 1$ on an algebraic scheme X, the projection $\pi\colon \mathbb{P}(E) \to X$ has relative dimension $r = e - 1$, and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$ is the tautological class of section 3.1, so that

$$s_i(E) \cap \alpha = \pi_*\big(\xi^{r+i} \cap \pi^*\alpha\big)$$

for $\alpha \in A_*(X)$, with $s_0(E)$ the identity and $s_i(E) = 0$ for $i < 0$. Every vector bundle named below is assumed of positive rank: the operators are manufactured from $\mathbb{P}(E)$, which is empty for the zero bundle, so neither $s(0)$ nor $c(0)$ is defined.

The first point to settle is that the inversion makes sense. A Segre operator lowers the dimension of a cycle class, and on a scheme of finite dimension an operator that lowers dimension by too much has nowhere to land; that is the whole reason the total operator is invertible.

Lemma 3.2.1: Dimension-Lowering Operators Vanish Above the Dimension

Let X be an algebraic scheme of dimension n. Call an operator on $A_*(X)$ *homogeneous of degree d* if it carries $A_k(X)$ into $A_{k-d}(X)$ for every k. Then every operator on $A_*(X)$ homogeneous of degree $d > n$ is the zero operator.

***Proof*:**
A k-dimensional cycle class satisfies $0 \leq k \leq n$, because X has no subvarieties of negative dimension and none of dimension exceeding n; hence $A_j(X) = 0$ for $j < 0$ and for $j > n$. An operator homogeneous of degree $d > n$ carries every $A_k(X)$ with $0 \leq k \leq n$ into $A_{k-d}(X)$ with $k - d < 0$, whose target vanishes, completing the proof.

Each $s_i(E)$ is homogeneous of degree i (section 3.1): π^* raises dimension by r and capping with ξ lowers it by one (definition 2.4.1). The lemma therefore applies to them and to every composite of them.

Lemma 3.2.2: Invertibility of the Total Segre Operator

Let X be an algebraic scheme of dimension n and E a vector bundle on X. For $\alpha \in A_k(X)$ only finitely many of the classes $s_i(E) \cap \alpha$ are nonzero, so

$$s(E) := \sum_{i \geq 0} s_i(E)$$

is a well-defined operator on $A_*(X)$. It is invertible: there is a unique operator μ on $A_*(X)$ with

$$\mu \circ s(E) = s(E) \circ \mu = \mathrm{id}$$

and it is given by the finite sum $\mu = \sum_{j=0}^{n} (-1)^j \nu^j$, where $\nu = s(E) - \mathrm{id} = \sum_{i \geq 1} s_i(E)$.

***Proof*:**
By lemma 3.2.1, $s_i(E) = 0$ for $i > n$, so the sum defining $s(E)$ has at most $n+1$ nonzero terms and is a well-defined operator.

The same lemma makes $\nu = \sum_{i\geq 1} s_i(E)$ nilpotent. Expanding ν^{n+1} produces a sum of composites $s_{i_1}(E) \circ \cdots \circ s_{i_{n+1}}(E)$ with every $i_t \geq 1$, each homogeneous of degree $i_1 + \cdots + i_{n+1} \geq n+1 > n$, hence each zero by lemma 3.2.1. Thus $\nu^{n+1} = 0$, and the finite sum $\mu = \sum_{j=0}^{n} (-1)^j \nu^j$ satisfies

$$\mu\,(\mathrm{id} + \nu) = (\mathrm{id} + \nu)\,\mu = \sum_{j=0}^{n} (-1)^j \nu^j + \sum_{j=0}^{n} (-1)^j \nu^{j+1} = \mathrm{id} + (-1)^n \nu^{n+1} = \mathrm{id}$$

the two sums telescoping. Since $s(E) = \mathrm{id} + \nu$, the operator μ is a two-sided inverse. Uniqueness is the usual argument for inverses in a ring: if μ' is another, then $\mu' = \mu' \circ (s(E) \circ \mu) = (\mu' \circ s(E)) \circ \mu = \mu$, completing the proof.

Nilpotence, not any positivity or finiteness of E, is what powers the inversion. The bundle may have arbitrary rank and X may be singular, reducible, or non-reduced; all that is used is that a cycle class cannot be cut down more times than the dimension of the ambient scheme allows. The inverse is therefore available on every algebraic scheme, which is the reason Chern classes in the Chow setting need no smoothness hypothesis.

Definition 3.2.3: Chern Classes of a Vector Bundle

Let E be a vector bundle on an algebraic scheme X. The *total Chern class* of E is the operator

$$c(E) := s(E)^{-1}$$

on $A_*(X)$ furnished by lemma 3.2.2, and the *i-th Chern class* $c_i(E)$ is its homogeneous component of degree i, an operator

$$c_i(E) \cap - : A_k(X) \longrightarrow A_{k-i}(X)$$

for every k. Equivalently, the $c_i(E)$ are determined by $c_0(E) = \mathrm{id}$ together with

$$\sum_{i+j=m} c_i(E) \circ s_j(E) = 0 \qquad \text{for every } m \geq 1$$

Unwinding the recursion $c_m(E) = -\sum_{j=1}^{m} c_{m-j}(E) \circ s_j(E)$ expresses each Chern class as an integer polynomial in the Segre classes:

$$\begin{aligned} &c_0(E) = \mathrm{id}, \qquad c_1(E) = -s_1(E), \qquad c_2(E) = s_1(E)^2 - s_2(E), \\ &c_3(E) = -s_1(E)^3 + 2\,s_1(E)s_2(E) - s_3(E) \end{aligned}$$

where a product of operators means composition. Each $c_m(E)$ is thus a $\mathbb{Z}$-combination of composites of $s_1(E), \ldots, s_m(E)$, a fact used below to transfer identities from Segre classes to Chern classes.

3.2.4 Note: Chern Classes Are Operators, Not Elements A Chern class here is an endomorphism of the graded group $A_*(X)$, not a member of it. There is no ring $A^*(X)$ yet: the intersection product that turns cycle classes into a ring for smooth X rests on the deformation construction of chapter 4, and until it is available the only multiplication in sight is composition of operators. Expressions such as $c_1(E)^2$ or $c(E)c(F)$ always mean composites, and an equation between them is an equation of maps $A_*(X) \to A_*(X)$, asserted for every class α. Once the product exists, capping against the fundamental class converts each operator into an element of $A^*(X)$ and the identities below become identities in a ring.

Every computation in this section runs through one imported object. The projective bundle carries a tautological line inside the pullback of E, and because this book and the scheme-theory volume use opposite conventions for $\mathbb{P}(-)$, that import has to be dualized in the open before it is used.

3.2.5 Note: The Tautological Subbundle and Its Quotient By box 0.0.2, the bundle of lines $\mathbb{P}(E)$ used here is $\operatorname{Proj}\left(\operatorname{Sym}^\bullet E^\vee\right)$, which is the projective bundle [2, the projective bundle $\mathbb{P}(\mathcal{E})$] of the *dual* sheaf. The universal surjection $\pi^*(E^\vee) \twoheadrightarrow \mathcal{O}_{\mathbb{P}(E)}(1)$ supplied there dualizes to an inclusion of vector bundles

$$\mathcal{O}_{\mathbb{P}(E)}(-1) \hookrightarrow \pi^* E$$

whose fibre over a point of $\mathbb{P}(E)$ lying above $x \in X$ and corresponding to a line $\ell \subseteq E_x$ is that line. Its cokernel Q is locally free of rank $\operatorname{rank} E - 1$, since the inclusion is a subbundle inclusion: over an open $U \subseteq X$ trivializing E the total space is $\mathbb{P}^r \times U$ and the inclusion is the tautological one on the first factor. Failing to dualize here replaces E by $E^\vee$ throughout and negates every odd Chern class, which is why the convention is restated at each import.

The definition inverts a series and so says nothing recognizable on its face. The rank-one case makes it concrete and settles the compatibility the notation presumes: the symbol c_1 has already been used in section 2.4 for the divisor operator of a line bundle, and the two meanings must agree.

Proposition 3.2.6: Chern Classes of a Line Bundle

Let L be a line bundle on an algebraic scheme X. Then

$$c(L) = \mathrm{id} + c_1(L)$$

where $c_1(L)$ on the right is the first Chern class operator of definition 2.4.1. In particular $c_1(L)$ in the sense of definition 3.2.3 is that operator, and $c_i(L) = 0$ for $i \geq 2$.

***Proof*:**
Both inputs are already computed in section 3.1: example 3.1.2(2) identifies $\mathbb{P}(L) = X$ with π the identity and $\mathcal{O}_{\mathbb{P}(L)}(1) = L^\vee$, whence $\xi = c_1(L^\vee) = -c_1(L)$, and example 3.1.4 evaluates the resulting Segre operators as $s_i(L) \cap \alpha = (-1)^i c_1(L)^i \cap \alpha$, so $s(L) = \sum_{i \geq 0} (-1)^i c_1(L)^i$. The operator $c_1(L)$ is nilpotent by the lemma 3.2.1, so this sum is the inverse of $\mathrm{id} + c_1(L)$: indeed

$$\Big(\sum_{i \geq 0} (-1)^i c_1(L)^i\Big)\big(\mathrm{id} + c_1(L)\big) = \mathrm{id}$$

the terms cancelling in pairs and the tail vanishing by nilpotence. Inverses being unique (lemma 3.2.2), $c(L) = s(L)^{-1} = \mathrm{id} + c_1(L)$. Reading off homogeneous components gives $c_1(L)$ in degree one and zero in every degree $i \geq 2$, as we sought to show.

The agreement is what licenses a single symbol for both constructions, and it fixes the sign convention of the whole theory: had $\mathbb{P}(E)$ been taken as the bundle of hyperplanes rather than of lines, the tautological class would be $c_1(L)$ instead of $-c_1(L)$ and the inversion would produce $c_1(L^\vee)$ where $c_1(L)$ is wanted. Box 0.0.2 therefore cannot be skipped. The rank-one case also shows that Chern classes terminate while Segre classes do not: $c_i(L) = 0$ beyond $i = 1$, while $s_i(L)$ is nonzero for every i with $c_1(L)^i \neq 0$.

Example 3.2.7: The Twisting Sheaves on Projective Space

Let $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(d)$ for an integer d. Write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, the hyperplane operator of example 2.4.7. Additivity of the first Chern class (proposition 2.4.4(3)) gives $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,H$, so by proposition 3.2.6

$$c(\mathcal{O}_{\mathbb{P}^n}(d)) = \mathrm{id} + d\,H$$

Capping against the fundamental class and using $H^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ from example 2.4.7,

$$c(\mathcal{O}_{\mathbb{P}^n}(d)) \cap [\mathbb{P}^n] = [\mathbb{P}^n] + d\,[\mathbb{P}^{n-1}]$$

so the total Chern class of $\mathcal{O}(d)$ records the single integer d as the multiple of the hyperplane class it produces. The corresponding Segre operator is the full geometric series $s(\mathcal{O}_{\mathbb{P}^n}(d)) = \sum_{i \geq 0} (-d)^i H^i$, whose cap against $[\mathbb{P}^n]$ is $\sum_{i=0}^{n} (-d)^i [\mathbb{P}^{n-i}]$: every dimension carries a nonzero coefficient when $d \neq 0$, while the Chern class stops after one step. This is the terminating behaviour that theorem 3.2.9 establishes in general.

Termination in rank one came from an explicit computation. In general it comes from the projective-bundle formula, which pins down exactly one relation among the powers of ξ and identifies its coefficients as the Chern classes. The following criterion extracts from that formula the tool used twice below: a class on $\mathbb{P}(E)$ is detected by finitely many pushforwards.

Lemma 3.2.8: A Vanishing Criterion on the Projective Bundle

Let E be a vector bundle of rank $e = r + 1$ on X, with $\pi \colon \mathbb{P}(E) \to X$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$. A class $\beta \in A_*(\mathbb{P}(E))$ vanishes if and only if

$$\pi_*\big(\xi^m \cap \beta\big) = 0 \qquad \text{for } 0 \leq m \leq r$$

***Proof*:**

If $\beta = 0$ then every $\pi_*(\xi^m \cap \beta)$ vanishes, since capping and pushforward are homomorphisms.

Conversely, suppose the $r + 1$ pushforwards vanish. By the projective-bundle formula (theorem 3.1.9) there are unique classes $\alpha_0, \ldots, \alpha_r \in A_*(X)$ with

$$\beta = \sum_{j=0}^{r} \xi^j \cap \pi^* \alpha_j$$

Capping with ξ^m and pushing forward, the definition of the Segre operators gives

$$\pi_*\big(\xi^m \cap \beta\big) = \sum_{j=0}^{r} \pi_*\big(\xi^{m+j} \cap \pi^*\alpha_j\big) = \sum_{j=0}^{r} s_{m+j-r}(E) \cap \alpha_j$$

Take $m = 0$. Every index $j < r$ contributes $s_{j-r}(E) = 0$, since $j - r < 0$, so the sum reduces to $s_0(E) \cap \alpha_r = \alpha_r$, and the hypothesis forces $\alpha_r = 0$. Proceed by descending induction: suppose $\alpha_r = \cdots = \alpha_{r-m+1} = 0$ for some $1 \leq m \leq r$. In the displayed sum for that m, the indices $j < r - m$ contribute zero because $m + j - r < 0$, the indices $j > r - m$ contribute zero because $\alpha_j = 0$ by the inductive hypothesis, and the single index $j = r - m$ contributes $s_0(E) \cap \alpha_{r-m} = \alpha_{r-m}$. The hypothesis forces $\alpha_{r-m} = 0$, and the induction runs down to $m = r$, giving $\alpha_0 = 0$. All coefficients vanish, so $\beta = 0$, completing the proof.

With a criterion for vanishing in hand, the relation defining the Chern classes can be produced. The statement below is the one that makes them computable: on $\mathbb{P}(E)$ the power ξ^e is not independent of the lower powers, and the coefficients expressing it in terms of them are exactly the Chern classes of E. Everything else in this section is extracted from it.

Theorem 3.2.9: The Chern Relation on the Projective Bundle

Let E be a vector bundle of rank $e = r + 1$ on an algebraic scheme X, with $\pi\colon \mathbb{P}(E) \to X$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$.

1. For every $\alpha \in A_*(X)$,
$$\sum_{i=0}^{e} \xi^{e-i} \cap \pi^*\big(c_i(E) \cap \alpha\big) = 0 \quad \text{in } A_*(\mathbb{P}(E))$$
2. $c_i(E) = 0$ for every $i > e$; that is, the Chern classes of E terminate at its rank.
3. The operators $c_0(E) = \mathrm{id}, c_1(E), \ldots, c_e(E)$ are the unique operators on $A_*(X)$, homogeneous of degrees $0, 1, \ldots, e$ respectively, whose degree-zero member is the identity and which satisfy the relation in (1).

***Proof*:**
Write $\gamma_\alpha = \sum_{i=0}^{e} \xi^{e-i} \cap \pi^*(c_i(E) \cap \alpha)$ for the class in question. Capping with ξ^m and pushing forward, the definition of the Segre operators gives, for every $m \geq 0$,

$$\pi_*\big(\xi^m \cap \gamma_\alpha\big) = \sum_{i=0}^{e} \pi_*\Big(\xi^{m+e-i} \cap \pi^*\big(c_i(E) \cap \alpha\big)\Big) = \sum_{i=0}^{e} s_{m+1-i}(E) \cap \big(c_i(E) \cap \alpha\big) \qquad (3.3)$$

using $m + e - i = r + (m + 1 - i)$, valid because $m + e - i \geq 0$ for $0 \leq i \leq e$ and $m \geq 0$, with the convention $s_j(E) = 0$ for $j < 0$ absorbing the terms where $i > m + 1$.

1. Fix m with $0 \leq m \leq r$ and set $n = m + 1$, so $1 \leq n \leq e$. In (3.3) the terms with $i > n$ vanish, so the right side is $\sum_{i=0}^{n} s_{n-i}(E) c_i(E) \cap \alpha$, which is the degree-$n$ homogeneous component of $s(E) \circ c(E) = \mathrm{id}$ applied to α. Since $n \geq 1$ that component is zero. Thus $\pi_*(\xi^m \cap \gamma_\alpha) = 0$ for $0 \leq m \leq r$, and lemma 3.2.8 gives $\gamma_\alpha = 0$.

2. By (1) the class γ_α vanishes, hence $\pi_*(\xi^m \cap \gamma_\alpha) = 0$ for *every* $m \geq 0$, not only for $m \leq r$. Reading (3.3) in that range and writing $n = m+1$, this says

$$\sum_{i=0}^{e} s_{n-i}(E) \circ c_i(E) = 0 \qquad \text{for every } n \geq 1$$

since α was arbitrary. Set $\widetilde{c} := \sum_{i=0}^{e} c_i(E)$, the truncation of the total Chern class at degree e. The displayed identities, together with $s_0(E) \circ c_0(E) = \mathrm{id}$ in degree zero, say precisely that $s(E) \circ \widetilde{c} = \mathrm{id}$. The inverse of $s(E)$ is two-sided and unique (lemma 3.2.2), so $\widetilde{c} = c(E)$, and comparing homogeneous components gives $c_i(E) = 0$ for $i > e$.

3. Let $c'_0 = \mathrm{id}, c'_1, \ldots, c'_e$ be operators, homogeneous of the stated degrees, satisfying the relation of (1) for every α. Subtracting the two relations, the terms with $i = 0$ cancel and

$$\sum_{i=1}^{e} \xi^{e-i} \cap \pi^*\Big(\big(c_i(E) - c'_i\big) \cap \alpha\Big) = 0$$

The exponents $e - i$ run over $0, 1, \ldots, e-1 = r$, so this is a representation of the zero class in the form supplied by theorem 3.1.9. Uniqueness of that representation forces $(c_i(E) - c'_i) \cap \alpha = 0$ for each i, and since α was arbitrary, $c'_i = c_i(E)$ for $1 \leq i \leq e$, completing the proof.

Part (2) is what gives a vector bundle a finite amount of Chern data: a rank-e bundle has exactly e potentially nonzero Chern classes, $c_1(E), \ldots, c_e(E)$, and the top one $c_e(E)$ carries the obstruction to finding a nowhere-vanishing section, as exercise 3.2.1 (3) below makes precise. Part (3) is the more useful of the three in practice. It says the Chern classes need not be computed by inverting a Segre series at all: any operators satisfying the relation on $\mathbb{P}(E)$ *are* the Chern classes. Every identity proved below is obtained by producing the relation from geometry and reading off its coefficients, and the Segre definition is never returned to.

The functoriality of the Segre operators transfers to the Chern classes for the same reason it held for them: the projective bundle of a pulled-back bundle is the pullback of the projective bundle. Two bundles on the same base are compared on the fibre product of their projective bundles, which is what makes their Chern classes commute.

Proposition 3.2.10: Properties of the Chern Class Operators

Let X be an algebraic scheme and E, F vector bundles on X.

1. *Commutativity.* For all $i, j \geq 0$ and $\alpha \in A_*(X)$,
$$c_i(E) \cap \big(c_j(F) \cap \alpha\big) = c_j(F) \cap \big(c_i(E) \cap \alpha\big)$$
2. *Projection formula.* If $f\colon X' \to X$ is proper and $\alpha' \in A_*(X')$, then
$$f_*\big(c_i(f^*E) \cap \alpha'\big) = c_i(E) \cap f_*\alpha'$$
3. *Flat pullback.* If $g\colon X'' \to X$ is flat of relative dimension m and $\alpha \in A_*(X)$, then
$$c_i(g^*E) \cap g^*\alpha = g^*\big(c_i(E) \cap \alpha\big)$$

***Proof*:**
Each $c_i(E)$ is a $\mathbb{Z}$-combination of composites of $s_1(E), \ldots, s_i(E)$ by the recursion following definition 3.2.3, so each of the three statements follows from the corresponding statement for Segre classes: expand the polynomials and apply the Segre statement to each factor of each monomial in turn. All three Segre statements are already proved in section 3.1.

1. *Commutativity.* proposition 3.1.5(3) gives $s_i(E) \cap (s_j(F) \cap \alpha) = s_j(F) \cap (s_i(E) \cap \alpha)$. Polynomials in commuting operators commute, so $c_i(E)$ and $c_j(F)$ commute as well.
2. *Proper pushforward.* This is the first identity of proposition 3.1.5(4), $f_*(s_i(f^*E) \cap \alpha') = s_i(E) \cap f_*\alpha'$, together with the fact that f_* is additive, so the identity passes through any $\mathbb{Z}$-combination of composites.
3. *Flat pullback.* This is the second identity of proposition 3.1.5(4), $s_i(g^*E) \cap g^*\beta = g^*(s_i(E) \cap \beta)$, and the same additivity argument, completing the proof.

Commutativity is what allows the Chern classes of all bundles on X to be handled as though they were elements of a commutative ring, long before any such ring exists; every polynomial identity written below is an identity among commuting operators, and its meaning is unambiguous. The other two parts say that the Chern classes are attached to the bundle and not to the base: they survive restriction to an open set, pullback along a projective bundle, and pushforward along a proper map, which is exactly what a proof carried out after a base change needs in order to descend.

The identities that remain, multiplicativity on exact sequences chief among them, are proved by making E as simple as possible, and the simplest a bundle can be is filtered by line bundles. For such a bundle the relation of theorem 3.2.9 can be written down by hand, because the tautological line at a point of $\mathbb{P}(E)$ must have a nonzero image in one of the successive quotients. That observation is the whole proof of the next lemma, and through it of the Whitney formula.

Lemma 3.2.11: Chern Classes of a Filtered Bundle

Let E be a vector bundle of rank e on an algebraic scheme X, and suppose E admits a filtration by subbundles

$$0 = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_e = E$$

with each quotient $L_i = E_i/E_{i-1}$ a line bundle. Then

$$c(E) = \prod_{i=1}^{e} \bigl(\mathrm{id} + c_1(L_i)\bigr)$$

as operators on $A_*(X)$; equivalently $c_m(E)$ is the m-th elementary symmetric expression

$$\sigma_m = \sum_{i_1<\cdots<i_m} c_1(L_{i_1})\cdots c_1(L_{i_m})$$

in the commuting operators $c_1(L_1), \ldots, c_1(L_e)$, for $0 \le m \le e$.

***Proof*:**
Write $P = \mathbb{P}(E)$ with projection π and tautological class ξ, and for $1 \le i \le e$ let $P_i = \mathbb{P}(E_i) \subseteq P$ with projection π_i and inclusion $\iota_i \colon P_i \hookrightarrow P$; set $P_0 = \mathbb{P}(E_0) = \emptyset$, so $P_e = P$. Each P_i is a closed subscheme of P, because in a local trivialization of the filtration over $U \subseteq X$ the inclusion $P_i|_U \subseteq P|_U$ is the linear inclusion $\mathbb{P}^{i-1} \times U \subseteq \mathbb{P}^{e-1} \times U$, and these glue. Put

$$\theta_i = c_1\bigl(\mathcal{O}_P(1) \otimes \pi^* L_i\bigr) = \xi + c_1(\pi^* L_i)$$

the equality by additivity of the first Chern class (proposition 2.4.4(3)); the θ_i commute with each other by proposition 2.4.4(2).

Step 1: each θ_i pushes a class on P_i down onto P_{i-1}. The tautological subbundle of box 3.2.5 restricts along ι_i to the tautological subbundle of P_i, since a point of P_i is a line $\ell \subseteq (E_i)_x \subseteq E_x$ and the line attached to it is the same either way; thus $\iota_i^* \mathcal{O}_P(1) = \mathcal{O}_{P_i}(1)$ and $\mathcal{O}_{P_i}(-1) \subseteq \pi_i^* E_i$. Composing that inclusion with the quotient map $\pi_i^* E_i \to \pi_i^* L_i$ gives a homomorphism $\mathcal{O}_{P_i}(-1) \to \pi_i^* L_i$, that is, a global section τ_i of the line bundle

$$M_i := \mathcal{O}_{P_i}(1) \otimes \pi_i^* L_i = \iota_i^*\bigl(\mathcal{O}_P(1) \otimes \pi^* L_i\bigr)$$

At a point of P_i above x corresponding to a line $\ell \subseteq (E_i)_x$, the section τ_i vanishes exactly when ℓ maps to zero in $(L_i)_x$, that is when $\ell \subseteq (E_{i-1})_x$. Its zero locus is therefore P_{i-1}. In a local trivialization as above, with homogeneous fibre coordinates $y_1, \ldots, y_i$ chosen so that E_{i-1} is cut out by $y_i = 0$, the section τ_i is the linear form y_i, a nonzerodivisor; hence $\mathrm{div}(\tau_i)$ is an effective Cartier divisor on P_i with support P_{i-1} and with $\mathcal{O}_{P_i}(\mathrm{div}\tau_i) \cong M_i$.

Now let $\beta = \iota_{i*}\beta_i$ for some $\beta_i \in A_*(P_i)$. The map ι_i is a closed immersion, hence proper, so the projection formula (proposition 2.4.4(4)) gives

$$\theta_i \cap \iota_{i*}\beta_i = \iota_{i*}\bigl(c_1(M_i) \cap \beta_i\bigr)$$

using $\iota_i^*(\mathcal{O}_P(1) \otimes \pi^* L_i) = M_i$. By definition 2.4.1 the class $c_1(M_i) \cap \beta_i$ is $\mathrm{div}(\tau_i) \cdot \beta_i$, which by proposition 2.2.4 is a well-defined class in $A_*\bigl(|\mathrm{div}\tau_i|\bigr) = A_*(P_{i-1})$. Pushing forward along $P_{i-1} \hookrightarrow P_i \hookrightarrow P$, the class $\theta_i \cap \beta$ is the image of a class on P_{i-1}, which is the assertion of this step. For $i = 1$ the filtration gives $E_1 = L_1$, the section τ_1 is the identity of L_1 under $\mathcal{O}_{P_1}(-1) = L_1 = \pi_1^* L_1$, its divisor is empty, and $A_*(P_0) = A_*(\emptyset) = 0$, so θ_1 annihilates every class coming from P_1.

Step 2: the product of the θ_i is zero. Let $\beta \in A_*(P) = A_*(P_e)$. Applying Step 1 with $i = e$ exhibits $\theta_e \cap \beta$ as coming from $A_*(P_{e-1})$; applying it again with $i = e-1$ exhibits $\theta_{e-1} \cap \theta_e \cap \beta$ as coming from $A_*(P_{e-2})$; after e steps the class $\theta_1 \cap \cdots \cap \theta_e \cap \beta$ comes from $A_*(P_0) = 0$ and so vanishes. The θ_i commute, so the product may be taken in any order:

$$\Big(\prod_{i=1}^{e} \theta_i\Big) \cap \beta = 0 \qquad \text{for every } \beta \in A_*(P)$$

Step 3: reading off the coefficients. Take $\beta = \pi^*\alpha$ for $\alpha \in A_*(X)$ and expand the product, which is legitimate because the factors commute:

$$0 = \Big(\prod_{i=1}^{e} \big(\xi + c_1(\pi^* L_i)\big)\Big) \cap \pi^*\alpha = \sum_{m=0}^{e} \xi^{\,e-m} \cap \Big(\sigma_m\big(c_1(\pi^* L_1), \dots, c_1(\pi^* L_e)\big) \cap \pi^*\alpha\Big)$$

The projection π is flat, so the flat compatibility of proposition 2.4.4(4) gives $c_1(\pi^* L) \cap \pi^*\alpha = \pi^*(c_1(L) \cap \alpha)$ for each L; iterating over the factors of each monomial,

$$\sigma_m\big(c_1(\pi^* L_1), \dots, c_1(\pi^* L_e)\big) \cap \pi^*\alpha = \pi^*\big(\sigma_m \cap \alpha\big)$$

with $\sigma_m = \sigma_m(c_1(L_1), \dots, c_1(L_e))$ formed on X. The displayed vanishing is therefore

$$\sum_{m=0}^{e} \xi^{\,e-m} \cap \pi^*\big(\sigma_m \cap \alpha\big) = 0$$

for every α, and $\sigma_0 = \mathrm{id}$. This is the relation of theorem 3.2.9(1) with σ_m in place of $c_m(E)$, so the uniqueness statement theorem 3.2.9(3) gives $c_m(E) = \sigma_m$ for $0 \le m \le e$, while $c_m(E) = 0$ for $m > e$ by part (2). Summing over m gives $c(E) = \prod_{i=1}^{e}(\mathrm{id} + c_1(L_i))$, as we sought to show.

The proof is geometric where it matters. A line inside E cannot map to zero in every quotient L_i of the filtration, and each factor θ_i cuts the support of a class down one step of the filtration, so after e factors nothing is left; the algebra afterwards only names the coefficients. Note also what the lemma does not require: no hypothesis on X, and no splitting of the filtration. A filtration with line-bundle quotients is enough, and an extension that does not split has the same Chern classes as the direct sum of its pieces.

Example 3.2.12: Split Bundles on Projective Space

1. *The trivial bundle.* The bundle $\mathcal{O}_X^{\oplus e}$ is filtered by the coordinate subbundles $\mathcal{O}_X^{\oplus i}$, with all quotients equal to $\mathcal{O}_X$. Each factor of lemma 3.2.11 is then $\mathrm{id} + c_1(\mathcal{O}_X) = \mathrm{id}$, since $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, so
$$c\big(\mathcal{O}_X^{\oplus e}\big) = \mathrm{id}$$
and every trivial bundle has trivial total Chern class, of any rank and on any X.
2. *A sum of twists on $\mathbb{P}^n$.* Let $E = \mathcal{O}_{\mathbb{P}^n}(a) \oplus \mathcal{O}_{\mathbb{P}^n}(b)$ and write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$. The summand

$\mathcal{O}_{\mathbb{P}^n}(a)$ is a subbundle with quotient $\mathcal{O}_{\mathbb{P}^n}(b)$, so the lemma gives

$$c(E) = \big(\mathrm{id} + aH\big)\big(\mathrm{id} + bH\big) = \mathrm{id} + (a+b)H + ab\,H^2$$

and hence $c_1(E) = (a+b)H$ and $c_2(E) = ab\,H^2$. Capping with the fundamental class and using example 2.4.7,

$$c_1(E) \cap [\mathbb{P}^n] = (a+b)\,[\mathbb{P}^{n-1}], \qquad c_2(E) \cap [\mathbb{P}^n] = ab\,[\mathbb{P}^{n-2}]$$

for $n \geq 2$. On $\mathbb{P}^1$ the second class vanishes for a reason unrelated to the rank: H^2 lowers dimension by two on a one-dimensional scheme, so it is the zero operator by the degree count in lemma 3.2.2, and $c_2(E) = 0$ even though $\operatorname{rank} E = 2$.

Most bundles carry no filtration by line bundles. The projective bundle manufactures one, at the cost of replacing X by a larger scheme, and iterating the construction strips off one line bundle at a time until nothing is left. The scheme at the end of the tower is the following.

Definition 3.2.13: The Flag Bundle

Let E be a vector bundle of rank e on an algebraic scheme X and let $0 \leq m \leq e$. Set $Y_0 = X$ and $E^{(0)} = E$, and for $1 \leq i \leq m$ let

$$p_i \colon Y_i = \mathbb{P}\big(E^{(i-1)}\big) \longrightarrow Y_{i-1}$$

be the projective bundle of lines in $E^{(i-1)}$, with tautological line subbundle $\Lambda_i \subseteq p_i^* E^{(i-1)}$ and quotient $E^{(i)} = p_i^* E^{(i-1)}/\Lambda_i$ of rank $e - i$ (box 3.2.5). The *m-step flag bundle* of E is

$$q \colon \mathrm{Fl}_m(E) := Y_m \longrightarrow X,$$

the composite of the m projections. Its fibre over $x \in X$ is the variety of flags $S_1 \subset \cdots \subset S_m \subseteq E_x$ with $\dim S_i = i$; write $S_i \subseteq q^*E$ for the corresponding rank-i subbundle, so that $S_i/S_{i-1} \cong \Lambda_i$.
The *full* flag bundle is the case $m = e-1$, written $\mathrm{Fl}(E)$; its fibre is the variety of complete flags in E_x, and for a bundle of rank two the tower has one stage, so $\mathrm{Fl}(E) = \mathbb{P}(E)$. The case $m = 0$ is X itself. The top case $m = e$ adds nothing: $E^{(e-1)}$ has rank one, so p_e is an isomorphism and $\mathrm{Fl}_e(E) = \mathrm{Fl}_{e-1}(E)$, with $S_e = q^*E$. It is allowed because the degeneracy-locus arguments of chapter 8 reach it whenever the rank drops to zero.

The projective-bundle formula guarantees that nothing is lost in passing to the flag bundle: the pullback map on Chow groups is injective at each stage, so an identity of operators verified upstairs is an identity downstairs. That is the content of the splitting principle, and it is a theorem rather than a slogan precisely because of the injectivity.

Theorem 3.2.14: The Splitting Principle

Let X be an algebraic scheme and $E_1, \dots, E_t$ vector bundles on X. There is a scheme X' and a proper, flat morphism $f\colon X' \to X$ such that

1. $f^*\colon A_*(X) \to A_*(X')$ is injective, and
2. each f^*E_j admits a filtration by subbundles whose successive quotients are line bundles.

***Proof*:**
Suppose first that $t = 1$, and write $E = E_1$ of rank e; the morphism produced will be the structure morphism of the flag bundle $\mathrm{Fl}(E)$ of definition 3.2.13. Induct on e. If $e = 1$ the bundle is already a line bundle and $f = \mathrm{id}_X$ serves. Let $e > 1$ and let $\pi\colon \mathbb{P}(E) \to X$ be the projective bundle, proper and flat by [2, A Projective Bundle is Proper and Flat]. The Flat pullback along it is injective by corollary 3.1.6, which also exhibits the left inverse $\beta \mapsto \pi_*(\xi^{e-1} \cap \beta)$. On $\mathbb{P}(E)$ the tautological subbundle of box 3.2.5 gives an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}(E)}(-1) \longrightarrow \pi^*E \longrightarrow Q \longrightarrow 0$$

with Q locally free of rank $e-1$. By the inductive hypothesis there is a proper flat $g\colon X' \to \mathbb{P}(E)$ with g^* injective and g^*Q filtered by subbundles with line-bundle quotients $N_2, \dots, N_e$. Set $f = \pi \circ g$, proper and flat as a composite of such, with $f^* = g^* \circ \pi^*$ injective as a composite of injections (proposition 1.3.5(3) identifies the composite pullback with the pullback of the composite). Pulling the displayed sequence back along g, which is exact because pullback of locally free sheaves is exact, gives a line subbundle $F_1 = g^*\mathcal{O}_{\mathbb{P}(E)}(-1) \subseteq f^*E$ with quotient g^*Q. Taking F_{j+1} to be the preimage in f^*E of the j-th step of the filtration of g^*Q produces

$$0 = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_e = f^*E$$

with F_1 of rank one and F_{j+1}/F_j isomorphic to the j-th quotient of the filtration of g^*Q, hence a line bundle. This proves the case $t = 1$.

For general t, apply the case $t = 1$ to E_1, obtaining $f_1\colon X_1 \to X$; then to $f_1^*E_2$ on X_1, obtaining $f_2\colon X_2 \to X_1$; and so on, and take $f = f_1 \circ \cdots \circ f_t$. Each stage is proper and flat with injective pullback, so the composite is too. A filtration by subbundles with line-bundle quotients pulls back to a filtration of the same shape, again because pullback is exact on locally free sheaves, so the filtration of $f_j^*E_j$ constructed at stage j survives all later stages. Hence every f^*E_j is filtered with line-bundle quotients, completing the proof.

Two features of the statement carry the weight. Injectivity of f^* is what turns the construction into a proof technique, since an identity of Chern operators is a family of identities of classes, and each descends. Flatness is what makes f^* available at all at this stage of the theory, there being no pullback for a general morphism until chapter 4. Both come from the projective-bundle formula, which is why that formula had to be proved before any of this.

The reduction the splitting principle enables is the multiplicativity of Chern classes on exact sequences. This is the property that makes them computable in practice: bundles are almost always presented by exact sequences, and the formula converts such a presentation directly into Chern data.

Theorem 3.2.15: The Whitney Sum Formula

Let

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

be an exact sequence of vector bundles on an algebraic scheme X. Then

$$c(E) = c(E')\, c(E'')$$

as operators on $A_*(X)$; that is, for every $m \geq 0$ and every $\alpha \in A_*(X)$,

$$c_m(E) \cap \alpha = \sum_{i+j=m} c_i(E') \cap \big(c_j(E'') \cap \alpha\big)$$

***Proof*:**
Let $e' = \operatorname{rank} E'$ and $e'' = \operatorname{rank} E''$, so $\operatorname{rank} E = e' + e''$.

Step 1: passing to a splitting cover. By theorem 3.2.14 applied to the pair E', E'' there is a proper flat $f\colon X' \to X$ with f^* injective, a filtration of f^*E' with line-bundle quotients $L'_1, \ldots, L'_{e'}$, and a filtration of f^*E'' with line-bundle quotients $L''_1, \ldots, L''_{e''}$. Pulling the given sequence back along f keeps it exact, pullback being exact on locally free sheaves, so

$$0 \longrightarrow f^*E' \longrightarrow f^*E \longrightarrow f^*E'' \longrightarrow 0$$

is exact on X'. Splice the two filtrations: take the filtration of f^*E' as the bottom e' steps of a filtration of f^*E, and above it take the preimages in f^*E of the steps of the filtration of f^*E''. The successive quotients are $L'_1, \ldots, L'_{e'}$ followed by $L''_1, \ldots, L''_{e''}$, all line bundles, so f^*E is filtered with line-bundle quotients.

Step 2: the identity upstairs. Lemma 3.2.11 applied to the three filtered bundles on X' gives

$$c(f^*E) = \prod_{i=1}^{e'} \big(\mathrm{id} + c_1(L'_i)\big) \cdot \prod_{j=1}^{e''} \big(\mathrm{id} + c_1(L''_j)\big) = c(f^*E')\, c(f^*E'')$$

the middle expression being the product over the quotients of the spliced filtration, regrouped using commutativity of first Chern class operators (proposition 2.4.4(2)). Comparing homogeneous components of degree m,

$$c_m(f^*E) = \sum_{i+j=m} c_i(f^*E')\, c_j(f^*E'')$$

Step 3: descending to X. Fix m and $\alpha \in A_*(X)$. Flat pullback naturality (proposition 3.2.10(3)) gives $f^*(c_m(E) \cap \alpha) = c_m(f^*E) \cap f^*\alpha$, and applying it twice,

$$f^*\Big(c_i(E') \cap \big(c_j(E'') \cap \alpha\big)\Big) = c_i(f^*E') \cap \big(c_j(f^*E'') \cap f^*\alpha\big)$$

Subtracting and using Step 2,

$$f^*\Big(c_m(E) \cap \alpha - \sum_{i+j=m} c_i(E') \cap \big(c_j(E'') \cap \alpha\big)\Big) = \Big(c_m(f^*E) - \sum_{i+j=m} c_i(f^*E')\, c_j(f^*E'')\Big) \cap f^*\alpha = 0$$

Since f^* is injective, the class inside vanishes. As α and m were arbitrary, the two operators agree, completing the proof.

The formula makes the total Chern class a homomorphism from short exact sequences to products, and this is what the Grothendieck group $K^\circ(X)$ of vector bundles will formalize when Riemann-Roch is taken up: the total Chern class is insensitive to the extension class, seeing only the sub and the quotient. Two immediate consequences deserve recording. First, $c(E)$ is invertible for every bundle E, with inverse $s(E)$: by definition 3.2.3 it is the inverse of $s(E)$, and that inverse is two-sided by lemma 3.2.2. So an exact sequence determines the Chern classes of any one of its three terms from the other two: $c(E'') = c(E)c(E')^{-1}$. Second, a trivial sub or quotient bundle contributes nothing, since $c(\mathcal{O}_X^{\oplus t}) = \mathrm{id}$ by example 3.2.12. Both are used repeatedly below.

Corollary 3.2.16: Chern Classes of a Dual Bundle

Let E be a vector bundle on an algebraic scheme X. Then

$$c_i(E^\vee) = (-1)^i\, c_i(E)$$

as operators on $A_*(X)$, for every $i \geq 0$.

***Proof*:**
Write $e = \mathrm{rank} E$ and let $f\colon X' \to X$ be as in theorem 3.2.14 for the single bundle E, with f^*E filtered by subbundles $0 = F_0 \subseteq \cdots \subseteq F_e = f^*E$ having line-bundle quotients $L_i = F_i/F_{i-1}$. Dualizing the filtration gives subbundles $(f^*E/F_i)^\vee \subseteq (f^*E)^\vee = f^*(E^\vee)$, increasing as i decreases, with successive quotients

$$\big(f^*E/F_{i-1}\big)^\vee \big/ \big(f^*E/F_i\big)^\vee \cong \big(F_i/F_{i-1}\big)^\vee = L_i^\vee$$

each a line bundle, so $f^*(E^\vee)$ is filtered with line-bundle quotients $L_e^\vee, \ldots, L_1^\vee$. By lemma 3.2.11 and $c_1(L^\vee) = -c_1(L)$ (proposition 2.4.4(3)),

$$c\big(f^*(E^\vee)\big) = \prod_{i=1}^{e} \big(\mathrm{id} - c_1(L_i)\big)$$

whose degree-i component is $(-1)^i \sigma_i(c_1(L_1), \ldots, c_1(L_e)) = (-1)^i c_i(f^*E)$, again by lemma 3.2.11. Naturality (proposition 3.2.10(3)) rewrites both sides as pullbacks of operators on X, and injectivity of f^* gives $c_i(E^\vee) = (-1)^i c_i(E)$, as we sought to show.

The remaining ingredient of a working calculus is a supply of computed examples. The tangent bundle of projective space is the one every later computation leans on, and it is produced from the Euler sequence by a single application of the Whitney formula.

Example 3.2.17: The Tangent Bundle of Projective Space
The Euler sequence [2, The Euler Sequence] relates the sheaf of differentials of $\mathbb{P}^n$ to the twisting sheaves through the exact sequence

$$0 \longrightarrow \Omega_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow 0$$

Projective space is smooth, so $\Omega_{\mathbb{P}^n}$ is locally free of rank n and the sequence is a sequence of vector bundles; dualizing it, which preserves exactness, gives the sequence in the form used here,

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

with $T_{\mathbb{P}^n} = \Omega^\vee_{\mathbb{P}^n}$ the tangent bundle. Write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$. The Whitney formula (theorem 3.2.15) applied to this sequence reads

$$c\big(\mathcal{O}_{\mathbb{P}^n}\big)\, c\big(T_{\mathbb{P}^n}\big) = c\big(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}\big)$$

The left factor is the identity by example 3.2.12(1), and the right side is $(\mathrm{id}+H)^{n+1}$, the direct sum of $n+1$ copies of $\mathcal{O}(1)$ being filtered with all quotients $\mathcal{O}_{\mathbb{P}^n}(1)$ and each factor contributing $\mathrm{id}+H$ by lemma 3.2.11. Hence

$$c\big(T_{\mathbb{P}^n}\big) = (\mathrm{id}+H)^{n+1}, \qquad c_i\big(T_{\mathbb{P}^n}\big) = \binom{n+1}{i} H^i$$

for $0 \le i \le n$, with $c_i(T_{\mathbb{P}^n}) = 0$ for $i > n$ in accordance with theorem 3.2.9(2). The binomial coefficient $\binom{n+1}{n+1} = 1$ in degree $n+1$ is not a contradiction: H^{n+1} is the zero operator on $A_*(\mathbb{P}^n)$ by the degree count of lemma 3.2.2, matching the nilpotence recorded in proposition 2.4.4(5).

Capping the top class against the fundamental class and using $H^n \cap [\mathbb{P}^n] = [\mathbb{P}^0]$ from example 2.4.7,

$$c_n\big(T_{\mathbb{P}^n}\big) \cap [\mathbb{P}^n] = (n+1)\,[\mathbb{P}^0], \qquad \int_{\mathbb{P}^n} c_n\big(T_{\mathbb{P}^n}\big) = n+1$$

the degree being taken with definition 1.2.4. For $n = 1$ this gives $\int_{\mathbb{P}^1} c_1(T_{\mathbb{P}^1}) = 2$, the degree of the tangent bundle of a rational curve, consistent with $\deg T_C = 2 - 2g$ for $g = 0$ through example 2.4.8. For $n = 2$ it gives $\int_{\mathbb{P}^2} c_2(T_{\mathbb{P}^2}) = 3$.

The integers $n+1$ produced by the top Chern class are the Euler characteristics of the underlying spaces, which is the first sign that $c_n(T_X)$ counts something. The general statement, that the degree of the top Chern class of the tangent bundle of a smooth complete variety is its topological Euler characteristic, needs the comparison between Chow groups and cohomology through the cycle class map and is not available yet; what is available is the computation itself, and it is already enough to run enumerative arguments once the intersection product exists.

3.2.18 Remark: The Topological Chern Classes A reader who has met Chern classes topologically, as the characteristic classes of a complex vector bundle in *Algebraic Topology* [10], will recognize the shape of this section: the same axioms, the same Whitney formula, the same splitting principle through a flag bundle. The two theories are not the same construction. The classes there live in singular cohomology and are built for complex bundles on topological spaces; the classes here live in the Chow group of an algebraic scheme over an arbitrary algebraically closed field, and the scheme is allowed to be singular, reducible, and non-reduced. Neither proof above uses the other theory, and none could: no topology is available over a field of positive characteristic. The two are compared, for a smooth complex variety, by the cycle class map built later in this book, which carries the algebraic classes to the topological ones.

The last construction of this section is the calculus that makes such identities routine, together with the rational coefficients several of them require.

The total Chern class of section 3.2 is multiplicative on short exact sequences: the Whitney formula turns an extension into a product. Riemann-Roch needs the opposite bookkeeping. It computes the Euler characteristic of a coherent sheaf, and Euler characteristics are *additive* on short exact sequences, so the characteristic class that Riemann-Roch pairs against must be additive too. This section builds that class, the Chern character, and alongside it the Todd class, the multiplicative correction factor the theorem attaches to the tangent bundle.

Both are universal power series in the Chern classes, and both differ from anything in section 3.2 in two ways. Their coefficients are rational, forced by denominators that no normalization removes, and they are infinite series rather than finite sums, which is legitimate only because a composite of more than $\dim X$ dimension-lowering operators is zero. Those two points come first, then the formal calculus of Chern roots that makes symmetric expressions in r formal symbols into operators, and then the two classes and the identities they satisfy.

The first point is a change of coefficients. Nothing in section 3.2 required it: Segre and Chern classes are operators on $A_*(X)$ with integer coefficients. The Chern character does require it, because its degree-two component is $\frac{1}{2}(c_1^2 - 2c_2)$, and no rescaling removes that $\frac{1}{2}$ while keeping additivity.

Definition 3.2.19: Chow Groups with Rational Coefficients

For an algebraic scheme X, set

$$A_k(X)_{\mathbb{Q}} := A_k(X) \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad A_*(X)_{\mathbb{Q}} := \bigoplus_k A_k(X)_{\mathbb{Q}}$$

A class $\alpha \in A_k(X)$ is identified with $\alpha \otimes 1$, and every operator $A_k(X) \to A_{k-i}(X)$ extends uniquely to a $\mathbb{Q}$-linear operator $A_k(X)_{\mathbb{Q}} \to A_{k-i}(X)_{\mathbb{Q}}$. The degree map of definition 1.2.4 on a complete X is extended $\mathbb{Q}$-linearly and still written $\int_X : A_0(X)_{\mathbb{Q}} \to \mathbb{Q}$.

Since $\mathbb{Q}$ is flat over $\mathbb{Z}$, an injective map of Chow groups stays injective after this change of coefficients, so the injectivity statements of section 3.2 are available rationally without further comment. What is lost is torsion: the natural map $A_*(X) \to A_*(X)_{\mathbb{Q}}$ has kernel the torsion subgroup, so any torsion class becomes invisible. On $\mathbb{P}^n$ nothing is lost, since $A_*(\mathbb{P}^n) = \bigoplus_j \mathbb{Z}\,[\mathbb{P}^j]$ is free (theorem 1.5.2) and $A_*(\mathbb{P}^n)_{\mathbb{Q}} = \bigoplus_j \mathbb{Q}\,[\mathbb{P}^j]$; the class $\frac{3}{2}\,[\mathrm{pt}] \in A_0(\mathbb{P}^2)_{\mathbb{Q}}$, which will appear below as a component of $\mathrm{ch}(T_{\mathbb{P}^2})$, is a legitimate element of the rational Chow group and of nothing smaller.

The second point is that infinite series of operators are already finite sums. An operator lowering dimension can only be applied $\dim X$ times before the target group vanishes, which is the nilpotence recorded for a single line bundle in proposition 2.4.4(5), stated here for an arbitrary commuting family.

Lemma 3.2.20: Power Series in Dimension-Lowering Operators

Let X be an algebraic scheme of dimension n and let $\theta_1, \ldots, \theta_m$ be pairwise commuting $\mathbb{Q}$-linear endomorphisms of $A_*(X)_{\mathbb{Q}}$ with $\theta_j\big(A_k(X)_{\mathbb{Q}}\big) \subseteq A_{k-d_j}(X)_{\mathbb{Q}}$ for integers $d_j \geq 1$. Write $\theta^\nu = \theta_1^{\nu_1} \cdots \theta_m^{\nu_m}$ and $|\nu| = \nu_1 + \cdots + \nu_m$.

1. $\theta^\nu = 0$ whenever $|\nu| > n$.
2. The assignment
$$\sum_\nu \lambda_\nu\, a^\nu \longmapsto \sum_{|\nu| \leq n} \lambda_\nu\, \theta^\nu$$
is a homomorphism of $\mathbb{Q}$-algebras $\mathbb{Q}[[a_1, \ldots, a_m]] \to \mathrm{End}\big(A_*(X)_{\mathbb{Q}}\big)$ carrying a_j to θ_j.

***Proof*:**

1. A k-cycle is a combination of k-dimensional subvarieties, so $A_k(X)_{\mathbb{Q}} = 0$ unless $0 \le k \le n$. The operator θ^ν carries $A_k(X)_{\mathbb{Q}}$ into $A_{k-e}(X)_{\mathbb{Q}}$ where $e = \sum_j \nu_j d_j \ge |\nu|$. If $|\nu| > n$ then for every k with $A_k(X)_{\mathbb{Q}} \ne 0$ one has $k - e \le n - |\nu| < 0$, so the target vanishes and $\theta^\nu = 0$.

2. Write T for the assignment. It is $\mathbb{Q}$-linear by construction, and $T(1) = \mathrm{id}$. For multiplicativity, let $\Phi = \sum_\mu \lambda_\mu a^\mu$ and $\Psi = \sum_\rho \kappa_\rho a^\rho$. The product $\Phi\Psi$ has coefficients $\sum_{\mu+\rho=\nu} \lambda_\mu \kappa_\rho$, so
$$T(\Phi\Psi) = \sum_{|\nu|\le n} \Big(\sum_{\mu+\rho=\nu} \lambda_\mu \kappa_\rho \Big) \theta^\nu$$
On the other side, since the θ_j commute, $T(\Phi)T(\Psi) = \sum_{|\mu|\le n,\, |\rho|\le n} \lambda_\mu \kappa_\rho\, \theta^{\mu+\rho}$. Every term with $|\mu + \rho| > n$ vanishes by part (1), and every pair (μ, ρ) with $|\mu + \rho| \le n$ automatically satisfies $|\mu| \le n$ and $|\rho| \le n$, so the two expressions have the same nonzero terms, completing the proof.

The Chern classes $c_1(E), \ldots, c_r(E)$ of a bundle of rank r are exactly such a family: they commute with one another by section 3.2, ultimately because two divisor operators commute (theorem 2.3.2), and $c_i(E)$ lowers dimension by i. A formal power series in r variables therefore evaluates on them to a well-defined operator, given by a finite sum. This is the mechanism behind every definition below.

What remains is to say which power series. Both the Chern character and the Todd class are described most efficiently as symmetric expressions in r formal symbols $a_1, \ldots, a_r$, the Chern roots, which behave as though E were a direct sum of line bundles with those first Chern classes. The splitting principle of section 3.2 is what makes the symbols legitimate.

Definition 3.2.21: Chern Roots and Splitting Maps

Let E be a vector bundle of rank r on an algebraic scheme X. A *splitting map* for E is a flat proper morphism $f\colon X' \to X$ such that

1. $f^*\colon A_*(X) \to A_*(X')$ is injective, and
2. f^*E admits a filtration by subbundles $0 = E_0 \subset E_1 \subset \cdots \subset E_r = f^*E$ whose successive quotients $L_p = E_p/E_{p-1}$ are line bundles.

The operators $a_p := c_1(L_p)$ on $A_*(X')_{\mathbb{Q}}$, for $p = 1, \ldots, r$, are the *Chern roots* of E relative to f.

The splitting principle of section 3.2 produces a splitting map for any bundle: the flag bundle of E is a tower of projective bundles, proper and flat over X by [2, A Projective Bundle is Proper and Flat], and the projective-bundle formula of section 3.1 makes each stage of the tower inject on Chow groups. By flatness of $\mathbb{Q}$ over $\mathbb{Z}$, condition (1) also gives injectivity of $f^*\colon A_*(X)_{\mathbb{Q}} \to A_*(X')_{\mathbb{Q}}$. Writing $a_p = c_1(L_p)$ for the Chern roots, lemma 3.2.11 gives

$$c(f^*E) = \prod_{p=1}^{r} \big(\mathrm{id} + a_p\big), \qquad \text{that is} \qquad c_i(f^*E) = e_i(a_1, \ldots, a_r) \tag{3.4}$$

where e_i is the i-th elementary symmetric polynomial. The roots are not canonical, and they are not operators on X at all, only on X'. What descends to X is any symmetric expression in them, and the reason is the classical structure theorem for symmetric functions.

Any symmetric formal power series in the Chern roots is a power series in the elementary symmetric polynomials, hence a genuine operator: this is [6, Symmetric Power Series in the Chern Roots], the formal-power-series refinement of the classical fundamental theorem on symmetric functions proved there.

The two lemmas combine into the calculus this section runs on: a symmetric power series in r variables defines an operator attached to any rank-r bundle, and that operator may be computed as though the bundle were split.

Proposition 3.2.22: The Chern Root Calculus

Let E be a vector bundle of rank r on an algebraic scheme X and let $\Phi \in \mathbb{Q}[[a_1, \ldots, a_r]]$ be symmetric, with $\widetilde{\Phi}$ as in [6, Symmetric Power Series in the Chern Roots]. Define

$$\Phi(E) := \widetilde{\Phi}\big(c_1(E), \ldots, c_r(E)\big)$$

Then:

1. $\Phi(E)$ is a well-defined operator on $A_*(X)_{\mathbb{Q}}$, equal to a finite sum of composites of Chern classes of E, and it commutes with $\Psi(F)$ for every bundle F and symmetric Ψ.
2. For a flat morphism $g\colon X'' \to X$ and $\alpha \in A_*(X)_{\mathbb{Q}}$,
$$g^*\big(\Phi(E) \cap \alpha\big) = \Phi(g^*E) \cap g^*\alpha$$
3. If E itself admits a filtration by subbundles with line-bundle quotients $L_1, \ldots, L_r$, then $\Phi(E) = \Phi\big(c_1(L_1), \ldots, c_1(L_r)\big)$, the right-hand side being the evaluation of lemma 3.2.20.
4. If $f\colon X' \to X$ is a splitting map for E with Chern roots $a_1, \ldots, a_r$, then $f^*\big(\Phi(E) \cap \alpha\big) = \Phi(a_1, \ldots, a_r) \cap f^*\alpha$ for all $\alpha \in A_*(X)_{\mathbb{Q}}$.

***Proof*:**

1. The classes $c_1(E), \ldots, c_r(E)$ commute pairwise and $c_i(E)$ lowers dimension by i, so lemma 3.2.20 applies and $\widetilde{\Phi}$ evaluates to a well-defined operator, equal to the sum of the terms of $\widetilde{\Phi}$ of weight at most $\dim X$. Uniqueness of $\widetilde{\Phi}$ in [6, Symmetric Power Series in the Chern Roots] is what makes the value depend on Φ alone. Chern classes of any two bundles commute by section 3.2, so the images of the two evaluation homomorphisms commute.
2. Chern classes are natural under flat pullback, $c_i(g^*E) \cap g^*\beta = g^*(c_i(E) \cap \beta)$ by section 3.2, generalizing proposition 2.4.4(4). Applying this once for each factor of a monomial in the $c_i(E)$, and summing over the finitely many monomials of $\widetilde{\Phi}$ that survive, gives the claim.
3. By (3.4) applied to the filtration of E itself, $c_i(E) = e_i(a_1, \ldots, a_r)$ with $a_p = c_1(L_p)$. The evaluation homomorphism $\mathbb{Q}[[a_1, \ldots, a_r]] \to \operatorname{End}(A_*(X)_{\mathbb{Q}})$ of lemma 3.2.20 for the commuting family $a_1, \ldots, a_r$ is a ring homomorphism, so it may be applied to the identity

$\Phi = \widetilde{\Phi}(e_1, \ldots, e_r)$ of [6, Symmetric Power Series in the Chern Roots] term by term, giving

$$\Phi(a_1, \ldots, a_r) = \widetilde{\Phi}\big(e_1(a), \ldots, e_r(a)\big) = \widetilde{\Phi}\big(c_1(E), \ldots, c_r(E)\big) = \Phi(E)$$

as operators.

4. Combining parts (2) and (3): the morphism f is flat, so $f^*(\Phi(E) \cap \alpha) = \Phi(f^*E) \cap f^*\alpha$, and f^*E carries a filtration with line-bundle quotients L_p, so $\Phi(f^*E) = \Phi(a_1, \ldots, a_r)$ by part (3). This completes the proof.

Part (4) is the working form. To prove an identity between operators built from bundles on X, pass to a splitting map, verify the corresponding identity of symmetric power series in the roots, and descend by injectivity of f^*. The roots never appear in a statement, only in a proof.

Exercise 3.2.1

1. Using only definition 3.2.3, express $s_1(E)$, $s_2(E)$ and $s_3(E)$ in terms of $c_1(E), c_2(E), c_3(E)$, and check that the resulting formulas are obtained from the expressions for c_1, c_2, c_3 in terms of s_1, s_2, s_3 by exchanging the letters c and s. Explain in one sentence why that symmetry is forced by the definition.

2. Let $E = \mathcal{O}_{\mathbb{P}^n}(a) \oplus \mathcal{O}_{\mathbb{P}^n}(b)$ with $n \geq 2$. Compute $c(E)$, $c(E^\vee)$ and $c(E \otimes \mathcal{O}_{\mathbb{P}^n}(1))$ as operators on $A_*(\mathbb{P}^n)$, and compute $c_2(E) \cap [\mathbb{P}^n]$. Then take $n = 1$ and explain why $c_2(E) = 0$ there, identifying which of the two possible reasons (the rank of E or the dimension of the base) is responsible.

3. Let E be a vector bundle of rank $e \geq 2$ on an algebraic scheme X admitting a nowhere-vanishing global section τ, so that $\tau \colon \mathcal{O}_X \to E$ is an inclusion of a subbundle with locally free quotient Q of rank $e - 1 \geq 1$. Show that $c(E) = c(Q)$ and deduce $c_e(E) = 0$. Show separately that a line bundle with a nowhere-vanishing section is trivial, so that the conclusion holds in the excluded case $e = 1$ as well. Conversely, explain why the computation $\int_{\mathbb{P}^n} c_n(T_{\mathbb{P}^n}) = n + 1 \neq 0$ of example 3.2.17 shows that $T_{\mathbb{P}^n}$ has no nowhere-vanishing global section.

4. Let E be a vector bundle of rank two on an algebraic scheme X and L a line bundle on X. Using theorem 3.2.14 and lemma 3.2.11, prove

$$c_1(E \otimes L) = c_1(E) + 2\,c_1(L), \qquad c_2(E \otimes L) = c_2(E) + c_1(E)\,c_1(L) + c_1(L)^2$$

as operators on $A_*(X)$. (You may use that tensoring a filtration by a line bundle again gives a filtration, with each quotient tensored by that line bundle.)

5. Compute $c(\Omega_{\mathbb{P}^n})$ as an operator on $A_*(\mathbb{P}^n)$ in closed form, and evaluate $\int_{\mathbb{P}^2} c_2(\Omega_{\mathbb{P}^2})$ and $\int_{\mathbb{P}^2} c_1(\Omega_{\mathbb{P}^2})^2$. Compare the first with $\int_{\mathbb{P}^2} c_2(T_{\mathbb{P}^2})$ and account for the agreement.

6. Let C be a smooth projective curve and E a vector bundle of rank two on C. Show that $c_2(E) = 0$ as an operator on $A_*(C)$, and that $c(E)$ is determined by the single class $c_1(E) \cap [C] \in A_0(C)$. Show further that if E sits in an exact sequence $0 \to L_1 \to E \to L_2 \to 0$ of bundles on C, then $\int_C c_1(E) = \deg L_1 + \deg L_2$.

3.3 Segre Classes of Cones and Subschemes

The Segre operators of section 3.1 came out of one recipe: projectivize, cap with powers of the tautological class, push down. Local triviality was never used in that recipe. What it used was a graded sheaf of algebras on X whose relative Proj is proper over X and carries a canonical line bundle, and a vector bundle supplies one such algebra among many. This section runs the recipe on the general input, a *cone*, and then on the case that matters here: the normal cone of a closed subscheme $Z \subseteq X$. The resulting class $s(Z, X)$ is an invariant of the embedding defined with no smoothness, no regularity and no dimension hypothesis, and it is the object that chapter 4 will deform X onto.

The algebra that produces a cone is required to look like a polynomial algebra in the two respects that matter for Proj: it starts in degree zero at $\mathcal{O}_X$, and it is generated by its degree-one part, which is coherent.

Definition 3.3.1: Cone

Let X be an algebraic scheme. A *cone* over X is the relative spectrum

$$C = \operatorname{Spec}\left(S^\bullet\right) \xrightarrow{p} X$$

of a sheaf $S^\bullet = \bigoplus_{n \geq 0} S_n$ of graded quasi-coherent $\mathcal{O}_X$-algebras ([2, graded quasi-coherent algebra]) such that

1. $S_0 = \mathcal{O}_X$;
2. S_1 is coherent;
3. $S^\bullet$ is generated as an $\mathcal{O}_X$-algebra by S_1.

The relative spectrum is the construction of [2, Relative Spectrum], glued from the affine pieces $\operatorname{Spec}\Gamma(U, S^\bullet)$ over an affine open cover of X, exactly as the relative Proj of [2, Relative Proj] is. Writing $S_+ = \bigoplus_{n\geq 1} S_n$, the surjection $S^\bullet \to S^\bullet / S_+ = \mathcal{O}_X$ defines a closed immersion $X \hookrightarrow C$, the *vertex* of the cone.

Conditions (2) and (3) force every S_n to be coherent, so p is affine and of finite type; the grading is what distinguishes a cone from an arbitrary affine morphism, and it is the grading that supplies the vertex and, below, the compactification. A cone is not required to be flat over X, nor to have fibres of constant dimension, and both failures are the point: the cones that arise from geometry are singular, jump in dimension along strata, and carry exactly the information those defects encode.

Example 3.3.2: Bundles, the Zero Cone, and the Affine Cone Over a Projective Variety

1. *A vector bundle.* Let E be a vector bundle of rank e on X with sheaf of sections $\mathcal{E}$. The functions on the total space of E are the symmetric algebra on the dual, so $S^\bullet = \operatorname{Sym}^\bullet(\mathcal{E}^\vee)$ has $\operatorname{Spec}(S^\bullet) = E$, and conditions (1) to (3) hold with $S_1 = \mathcal{E}^\vee$ locally free of rank e. The vertex $X \hookrightarrow E$ is the zero section. The dual is exactly what makes $\operatorname{Proj}(\operatorname{Sym}^\bullet \mathcal{E}^\vee)$ the bundle of *lines* in E used throughout this book (box 0.0.2), and not the bundle of rank-one quotients of $\mathcal{E}$ that [2, the projective bundle] writes as $\mathbb{P}(\mathcal{E})$.
2. *The zero cone.* Taking $S^\bullet = \mathcal{O}_X$, so that $S_+ = 0$, gives $C = X$ with its vertex the identity.

The degenerate case earns its place: it is the normal cone of X in itself.

3. *An affine cone.* Let $Y = \operatorname{Proj}(R) \subseteq \mathbb{P}^n$ be a closed subscheme, where $R = \overline{k}[x_0, \dots, x_n]/I$ for a homogeneous ideal I, and take $X = \operatorname{Spec} \overline{k}$. Then R itself satisfies (1) to (3), and $C = \operatorname{Spec}(R)$ is the affine cone over Y: the union of the lines of $\mathbb{A}^{n+1}$ through the origin lying over the points of Y, together with the origin, which is the vertex. For $I = 0$ this is $\mathbb{A}^{n+1}$ itself, the trivial bundle of rank $n+1$ over a point, recovering item (1).

A cone is affine over X, so p is proper only in the degenerate case where $S^\bullet$ is coherent as an $\mathcal{O}_X$-module and p is therefore finite; the zero cone of example 3.3.2(2), with $p = \mathrm{id}$, is such a case. The cones that carry geometry are not: a vector bundle of positive rank has fibres $\mathbb{A}^e$, and an affine scheme of positive dimension over a field is not proper. So p_* is unavailable in general, and pushing a cycle from C down to X is not an option. The remedy is the one projective space applies to affine space: compactify by adding a hyperplane at infinity. Adjoining one variable in degree one does this, and does it in a way that keeps the answer a relative Proj, so the canonical line bundle comes along for free.

Definition 3.3.3: Projective Cone and Projective Completion

Let $C = \operatorname{Spec}(S^\bullet)$ be a cone over X. Adjoin a variable z in degree one, giving the graded $\mathcal{O}_X$-algebra $S^\bullet[z]$ with $\left(S^\bullet[z]\right)_n = \bigoplus_{j=0}^n S_j\, z^{\,n-j}$, which again satisfies the conditions of definition 3.3.1. The *projective cone* of C and the *projective completion* of C are the relative Proj schemes

$$\mathbb{P}(C) = \operatorname{Proj}\left(S^\bullet\right), \qquad \mathbb{P}(C \oplus 1) = \operatorname{Proj}\left(S^\bullet[z]\right)$$

Both are proper over X ([2, Relative Proj]). Write $q\colon \mathbb{P}(C \oplus 1) \to X$ for the structure morphism and $\mathcal{O}(1) = \mathcal{O}_{\mathbb{P}(C\oplus 1)}(1)$ for the canonical line bundle.

Three identifications justify the name, and each is read straight off the graded algebra. First, on the locus where z is invertible, $D_+(z) = \operatorname{Spec}\left((S^\bullet[z])_{(z)}\right)$, and the assignment $s/z^n \mapsto s$ for $s \in S_n$ is an isomorphism of $\mathcal{O}_X$-algebras $(S^\bullet[z])_{(z)} \cong S^\bullet$; so the open subscheme of $\mathbb{P}(C \oplus 1)$ where $z \neq 0$ is C itself. Second, z is a nonzerodivisor in $S^\bullet[z]$ and a global section of $\mathcal{O}(1)$, and its zero scheme is $\operatorname{Proj}\left(S^\bullet[z]/(z)\right) = \mathbb{P}(C)$; so $\mathbb{P}(C)$ is an effective Cartier divisor ([2, Effective Cartier Divisor]) on $\mathbb{P}(C \oplus 1)$ with complement C, the hyperplane at infinity. Third, the surjection $S^\bullet[z] \to \mathcal{O}_X[z]$ killing S_+ realizes $X = \operatorname{Proj}(\mathcal{O}_X[z])$ as a closed subscheme of $\mathbb{P}(C \oplus 1)$ contained in the open piece C, where it is the vertex of definition 3.3.1.

So $\mathbb{P}(C \oplus 1)$ is C with $\mathbb{P}(C)$ glued on at infinity, and it is proper over X whereas C, outside the degenerate case just noted, is not. That properness is why it is introduced: it makes q_* available (definition 1.2.1 and theorem 1.2.3), and with $\mathcal{O}(1)$ in hand the construction of section 3.1 can be run verbatim.

The model case is the one the terminology is borrowed from. Take $X = \operatorname{Spec} \overline{k}$ and $S^\bullet = \overline{k}[t_1, \dots, t_n]$, so that $C = \mathbb{A}^n$ is the trivial bundle of rank n over a point. Then $S^\bullet[z] = \overline{k}[t_1, \dots, t_n, z]$ and $\mathbb{P}(C \oplus 1) = \mathbb{P}^n$, with $\mathbb{P}(C) = \mathbb{P}^{n-1}$ the hyperplane $z = 0$ at infinity, the open complement $D_+(z)$ the standard affine chart $\mathbb{A}^n$, and the vertex the origin of that chart. Compactifying a cone is compactifying affine space by a hyperplane, carried out fibrewise over X and allowing the fibres to be as singular as they like.

Definition 3.3.4: Segre Class of a Cone

Let C be a cone over an algebraic scheme X, with projective completion $q\colon \mathbb{P}(C \oplus 1) \to X$ and canonical line bundle $\mathcal{O}(1)$. The *Segre class* of C is the cycle class

$$s(C) \;=\; q_*\Big(\sum_{i\geq 0} c_1(\mathcal{O}(1))^i \cap \big[\mathbb{P}(C\oplus 1)\big]\Big) \;\in\; A_*(X)$$

where $[\mathbb{P}(C \oplus 1)]$ is the fundamental cycle of definition 1.1.2 and $c_1(\mathcal{O}(1)) \cap -$ is the operator of definition 2.4.1. Write $s(C)_k$ for the component of $s(C)$ in $A_k(X)$.

The sum is finite. Each cap lowers dimension by one, so the term indexed by i lies in $A_{d-i}\big(\mathbb{P}(C\oplus 1)\big)$ for $d = \dim \mathbb{P}(C \oplus 1)$, and this group vanishes once $i > d$ because there are no cycles of negative dimension. Two features of the definition deserve emphasis. The Segre class of a cone is a *class*, not an operator: the capping has already happened, against the fundamental cycle of the compactification. And it is not homogeneous; $s(C)$ has components in several dimensions at once, and the interesting content is usually spread across them.

For the model cone $C = \mathbb{A}^n$ over $X = \operatorname{Spec} \overline{k}$ the class can be read off. The completion is $\mathbb{P}^n$, and $q_*\big(c_1(\mathcal{O}(1))^i \cap [\mathbb{P}^n]\big)$ lies in $A_{n-i}(\operatorname{Spec} \overline{k})$, which vanishes unless $i = n$. For $i = n$, example 2.4.7 gives $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathrm{pt}]$, whose pushforward to the point is the fundamental class. Hence $s(\mathbb{A}^n) = [\operatorname{Spec} \overline{k}]$: the trivial bundle has trivial Segre class, as it must.

Before the definition can be trusted it has to be checked against the case it generalizes. A vector bundle is a cone, and its Segre class as a cone had better be its total Segre operator evaluated on the fundamental class of the base. The bundle has to have positive rank for that comparison to mean anything: the operators $s_i(E)$ of section 3.1 are manufactured from $\mathbb{P}(E)$, which is empty for the zero bundle, so neither $s(E)$ nor $c(E)$ is defined there. The rank-zero cone is the zero cone of example 3.3.2(2), whose Segre class is computed directly below.

Proposition 3.3.5: The Segre Class of a Vector Bundle as a Cone

Let E be a vector bundle of rank $e \geq 1$ on an algebraic scheme X, and let C be its total space regarded as a cone (example 3.3.2(1)). Then

$$s(C) \;=\; s(E) \cap [X]$$

where $s(E) = \sum_{j\geq 0} s_j(E)$ is the total Segre operator of section 3.1.

***Proof*:**
Step 1: the projective completion is a projective bundle. With $S^\bullet = \operatorname{Sym}^\bullet(\mathcal{E}^\vee)$ one has $S^\bullet[z] = \operatorname{Sym}^\bullet\big(\mathcal{E}^\vee \oplus \mathcal{O}_X\big) = \operatorname{Sym}^\bullet\big((\mathcal{E} \oplus \mathcal{O}_X)^\vee\big)$, the variable z being the coordinate dual to the added trivial summand. Hence $\mathbb{P}(C \oplus 1) = \mathbb{P}(E \oplus 1)$ is the bundle of lines in the rank-$(e+1)$ bundle $E \oplus 1$, its fibres are copies of $\mathbb{P}^e$, and its canonical bundle $\mathcal{O}(1)$ is the one whose first Chern class is the tautological class of section 3.1. In particular q is proper and flat of relative dimension e ([2, A Projective Bundle is Proper and Flat]).

Step 2: the fundamental cycle is a flat pullback. Write $n = \dim X$ and let $X_1, \ldots, X_r$ be the n-dimensional irreducible components of X, so that the fundamental cycle is $[X] = \sum_\lambda m_\lambda [X_\lambda]$

with m_λ the length of $\mathcal{O}_{X,X_\lambda}$. Each preimage $q^{-1}(X_\lambda)$ is a projective bundle over the variety X_λ, hence integral, so $q^*[X_\lambda] = [q^{-1}(X_\lambda)]$ by proposition 1.3.5(2); and the length multiplicity of $q^{-1}(X_\lambda)$ in $[\mathbb{P}(E \oplus 1)]$ is again m_λ, because the local homomorphism $\mathcal{O}_{X,X_\lambda} \to \mathcal{O}_{\mathbb{P}(E\oplus 1),\, q^{-1}(X_\lambda)}$ is flat with fibre the function field of a projective space, so the two lengths agree. Summing over λ,

$$q^*[X] = \big[\mathbb{P}(E \oplus 1)\big]$$

Step 3: the terms with $i \geq e$. Definition 3.1.3 defines the Segre operators of a bundle F of rank f by $s_j(F) \cap \alpha = \pi_*\big(\xi^{\,f-1+j} \cap \pi^*\alpha\big)$, where $\pi\colon \mathbb{P}(F) \to X$ has relative dimension $f-1$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(F)}(1))$. Applied to $F = E \oplus 1$, of rank $e+1$, this reads

$$s_j(E \oplus 1) \cap \alpha = q_*\big(c_1(\mathcal{O}(1))^{\,e+j} \cap q^*\alpha\big), \qquad j \geq 0$$

so by Step 2 the terms of definition 3.3.4 indexed by $i = e+j$ with $j \geq 0$ sum to $\sum_{j\geq 0} s_j(E \oplus 1) \cap [X] = s(E \oplus 1) \cap [X]$.

Step 4: the terms with $i < e$ vanish. Flat pullback raises dimension by e and each cap lowers it by one, so $q_*\big(c_1(\mathcal{O}(1))^i \cap q^*[X]\big)$ lies in $A_{n+e-i}(X)$. For $i < e$ this index exceeds $n = \dim X$, and X carries no subvariety of dimension greater than its own, so the group is zero and every such term vanishes.

Step 5: dropping the trivial summand. This is the only step that uses $e \geq 1$: both E and $E \oplus 1$ then have positive rank, so both have invertible total Segre operators (lemma 3.2.2) and Chern operators to compare (definition 3.2.3). The split exact sequence $0 \to E \to E \oplus 1 \to \mathcal{O}_X \to 0$ and the Whitney formula (theorem 3.2.15) give $c(E \oplus 1) = c(E)\,c(\mathcal{O}_X)$. A line bundle has no Chern classes above the rank (proposition 3.2.6), so $c(\mathcal{O}_X) = \mathrm{id} + c_1(\mathcal{O}_X)$, and $c_1(\mathcal{O}_X) = 0$ by example 2.4.3; hence $c(E \oplus 1) = c(E)$ and, inverting, $s(E \oplus 1) = s(E)$. Combining with Steps 3 and 4,

$$s(C) = s(E \oplus 1) \cap [X] = s(E) \cap [X]$$

as we sought to show.

The proposition licenses the notation: for the cone of a vector bundle the class $s(C)$ of definition 3.3.4 and the operator $s(E)$ of section 3.1 determine one another, and writing s for both loses nothing. What the proposition also shows is where the new content will sit. On a bundle, the compactification $\mathbb{P}(E \oplus 1)$ is flat over X with fibres of constant dimension, the Segre operators exist, and $s(C)$ is a formal consequence of them. On a general cone none of that is available, and $s(C)$ does not come from any operator; it is a class computed once, by pushing forward from a compactification that may be singular and may jump in dimension over X.

The cone this was built for measures how a closed subscheme sits inside an ambient scheme. When Z is a smooth subvariety of a smooth variety, the correct linear model is the normal bundle, and its rank is the codimension. When either smoothness hypothesis fails there is no bundle, but the ideal sheaf still has an associated graded algebra, and that algebra is always a cone.

Definition 3.3.6: Normal Cone of a Closed Subscheme

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. The graded sheaf of algebras $\bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$ has degree-zero part $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$, coherent degree-one part $\mathcal{I}/\mathcal{I}^2$, and is generated in degree one, so it satisfies the conditions of definition 3.3.1. The *normal cone* of Z in X is the cone over Z

$$C_Z X \;=\; \operatorname{Spec}\Big(\bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \;\longrightarrow\; Z$$

On an affine open $U = \operatorname{Spec} A$ of X with $Z \cap U = \operatorname{Spec} A/I$, this is $\operatorname{Spec}\big(\bigoplus_{n \geq 0} I^n/I^{n+1}\big)$.

The normal cone is intrinsic to the pair (Z, X) and depends on the scheme structure of Z, not just on its underlying set: the ideal $\mathcal{I}$ is the input, and thickening Z changes it. When Z is a single reduced point P of a variety X, the normal cone is the classical tangent cone $\operatorname{Spec}\big(\operatorname{gr}_{\mathfrak{m}} \mathcal{O}_{X,P}\big)$, the cone of limiting directions of branches of X through P, and it is where the local singularity of X at P becomes visible.

Example 3.3.7: The Tangent Cone at a Node

Assume $\operatorname{char} \overline{k} \neq 2$ and let $X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2$ be the affine nodal cubic, with P the origin and $\mathfrak{m} = (x,y)$ its maximal ideal.[a] The associated graded ring of $\overline{k}[x,y]$ for the filtration by powers of (x,y) is again $\overline{k}[x,y]$, a domain, so initial forms multiply: the initial form of gh is the product of those of g and h. Consequently the initial forms of the elements of the principal ideal $(y^2 - x^2 - x^3)$ are generated by the initial form of its generator, namely $y^2 - x^2$. Hence

$$\bigoplus_{n \geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1} \;\cong\; \overline{k}[x,y]/(y^2 - x^2)$$

and $C_P X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^2)$ is the union of the two lines $y = x$ and $y = -x$: precisely the two branches of the node, straightened to their tangent directions. The normal cone is one-dimensional, matching $\dim X = 1$, but it is reducible where X is irreducible, and it is not a vector bundle over the point P (a rank-one bundle over a point is a single line). The failure is exactly the singularity.

[a] The hypothesis on the characteristic cannot be dropped. In characteristic 2 the substitution $u = y + x$ turns the equation into $u^2 = x^3$, so the curve is a cuspidal cubic with a single branch, and $y^2 - x^2 = (y+x)^2$ makes the ring computed below the coordinate ring of a single non-reduced double line, which is irreducible; the contrast drawn at the end of the example then fails.

The normal cone is a cone over Z, so it has a Segre class, and that class is what records how Z sits inside X when no bundle is available to record it.

Definition 3.3.8: Segre Class of a Closed Subscheme

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme. The *Segre class of Z in X* is the Segre class of the normal cone,

$$s(Z, X) \;=\; s\big(C_Z X\big) \;\in\; A_*(Z)$$

Two points about the target group. The class lives on Z, not on X: like the intersection classes of definition 2.2.1, it is recorded where the geometry happens, and one pushes forward along $Z \hookrightarrow X$

only when the cruder class in $A_*(X)$ is wanted. And no hypothesis whatsoever was imposed on the pair (Z, X): X may be singular, reducible, or nonreduced, Z may be any closed subscheme including X itself, and $s(Z, X)$ is defined regardless. That generality is what makes the class useful and also what makes it hard to compute; the two computations below are the cases where it can be read off.

The extreme case is immediate. Taking $Z = X$ makes $\mathcal{I} = 0$, so the normal cone is the zero cone of example 3.3.2(2), with projective completion $\mathrm{Proj}\,(\mathcal{O}_X[z]) = X$ and with $\mathcal{O}(1)$ free on the generator z, hence trivial. Every term of definition 3.3.4 beyond the first therefore vanishes, since $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, and $s(X, X) = [X]$. A subscheme sitting inside itself has no normal directions, and its Segre class is its own fundamental cycle.

3.3.9 Foreshadowing: Why the Normal Cone The normal cone is not introduced here for its own sake. Chapter 4 constructs a family over $\mathbb{P}^1$ whose general fibre is the embedding $Z \hookrightarrow X$ and whose special fibre is the embedding of Z as the vertex of $C_Z X$; intersecting with Z inside X is then replaced by intersecting with the vertex inside the cone, where the answer can be read off. When $C_Z X$ is a vector bundle that replacement produces the Gysin map of a regular embedding; when it is not, the Segre class $s(Z, X)$ is what survives, and it is the term that appears in the excess intersection formula.

The first computable case is the one where the normal cone degenerates to a bundle. The condition on the ideal is the classical one: it should be generated, locally, by the right number of equations imposing independent conditions.

Regular embeddings and their normal bundles are scheme theory, and they belong to, and are defined in, *Algebraic Geometry*: a closed immersion whose ideal sheaf is locally generated by a regular sequence of length d is a *regular embedding of codimension* d ([2, Regular Embedding]), its conormal sheaf $\mathcal{I}/\mathcal{I}^2$ is then locally free of rank d ([2, The Conormal Sheaf of a Regular Embedding]), and its *normal bundle* is the rank-d vector bundle

$$N_Z X \;=\; \underline{\mathrm{Spec}}_Z\Big(\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)\Big) \;\longrightarrow\; Z$$

with sheaf of sections $(\mathcal{I}/\mathcal{I}^2)^\vee$ ([2, Conormal Sheaf and Normal Bundle]). What this book adds is the cycle-level consequence below.

The definition is local on Z, and the number d is forced: the rank of $\mathcal{I}/\mathcal{I}^2$ is a local invariant, so a regular embedding has a well-defined codimension on each connected component of Z. The two extremes are familiar. The origin in $\mathbb{A}^n$, whose ideal $(x_1, \ldots, x_n)$ is a regular sequence, is a regular embedding of codimension n with trivial normal bundle of rank n; an effective Cartier divisor, whose ideal is locally generated by a single nonzerodivisor, is the codimension-one case, with normal bundle a line bundle. What makes the notion useful here is that the whole associated graded algebra, not its degree-one part, becomes a symmetric algebra.

Lemma 3.3.10: The Associated Graded Algebra of a Regular Sequence

Let A be a ring and let $a_1, \ldots, a_d$ be a regular sequence generating an ideal $I \subseteq A$. Then the homomorphism of graded A/I-algebras

$$(A/I)[T_1, \ldots, T_d] \longrightarrow \bigoplus_{n \geq 0} I^n/I^{n+1}, \qquad T_j \longmapsto a_j \bmod I^2$$

is an isomorphism. In particular I/I^2 is a free A/I-module of rank d on the classes of $a_1, \ldots, a_d$, and the canonical surjection $\mathrm{Sym}^\bullet_{A/I}(I/I^2) \to \bigoplus_{n\geq 0} I^n/I^{n+1}$ is an isomorphism.

***Proof*:**
This is [4, Associated Graded Ring of a Regular Sequence], proved there by induction on the length of the sequence. It is pure commutative algebra and is not re-derived here.

Proposition 3.3.11: Segre Class of a Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, with normal bundle $N = N_Z X$. Then the normal cone is the normal bundle,

$$C_Z X \;=\; N_Z X$$

and the Segre class of Z in X is

$$s(Z, X) \;=\; s(N) \cap [Z] \;=\; c(N)^{-1} \cap [Z]$$

where $s(N)$ and $c(N)$ are the total Segre and Chern operators of sections 3.1 and 3.2.

***Proof*:**
Work on an affine open $U = \operatorname{Spec} A$ of X on which $I = \mathcal{I}(U)$ is generated by a regular sequence $a_1, \ldots, a_d$. There is always a canonical surjection of graded A/I-algebras

$$\mathrm{Sym}^\bullet_{A/I}(I/I^2) \longrightarrow \bigoplus_{n \geq 0} I^n/I^{n+1}$$

sending a product of degree-one classes to the class of the corresponding product in I^n; it is surjective because the target is generated in degree one. Lemma 3.3.10 says that it is injective as well, and that I/I^2 is free of rank d on the classes of the a_j, so it is an isomorphism over U.

The surjection above is canonical, so the local isomorphisms are compatible on overlaps and glue to an isomorphism of graded $\mathcal{O}_Z$-algebras $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \xrightarrow{\sim} \bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$. Taking relative spectra and comparing definition 3.3.6 with [2, Regular Embedding] gives $C_Z X = N_Z X$.

The normal cone is therefore the cone of a vector bundle of rank $d \geq 1$ over Z, so proposition 3.3.5 applies and gives $s(Z, X) = s(C_Z X) = s(N) \cap [Z]$. Finally $s(N) = c(N)^{-1}$, the Chern operators of definition 3.2.3 being defined by inverting the Segre series, completing the proof.

The normal cone is functorial in the subscheme, and for a regular embedding it sits inside a pulled-back normal bundle. Both facts are needed by chapter 4, and both are statements about cones as such, so they belong here rather than there.

Lemma 3.3.12: The Normal Cone of a Closed Subscheme

Let $Z, V \subseteq X$ be closed subschemes of an algebraic scheme and let $W = V \cap Z$ be their scheme-theoretic intersection. Then W is the scheme-theoretic preimage of Z under $V \hookrightarrow X$, and there is a canonical closed immersion

$$C_W V \ \hookrightarrow \ C_Z X$$

lying over the inclusion $W \hookrightarrow Z$.

***Proof*:**
The ideal sheaf of W in V is $\mathcal{I}_W = \mathcal{I}\mathcal{O}_V$, which is exactly the statement that W is the scheme-theoretic preimage of Z under the closed immersion $V \hookrightarrow X$. Consequently $\mathcal{I}^n\mathcal{O}_V = \mathcal{I}_W^n$ for every n, so the homomorphism of graded $\mathcal{O}_W$-algebras

$$\Big(\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \otimes_{\mathcal{O}_Z} \mathcal{O}_W \longrightarrow \bigoplus_{n\geq 0} \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$$

is surjective in each degree, the image of $\mathcal{I}^n \otimes \mathcal{O}_V \to \mathcal{O}_V$ being $\mathcal{I}_W^n$. On an affine open this is a surjection of graded rings, so by [2, Relative Spectrum](1) it induces a closed immersion of relative spectra; the local closed immersions are induced by a globally defined homomorphism of sheaves of algebras, so they glue. This exhibits $C_W V$ as a closed subscheme of the relative spectrum of the left side, which is $C_Z X \times_Z W$, again by the affine description of the relative spectrum. Composing with the closed immersion $C_Z X \times_Z W \hookrightarrow C_Z X$, the base change of $W \hookrightarrow Z$, gives the asserted closed immersion, completing the proof.

Lemma 3.3.13: The Normal Cone of a Preimage Embeds in the Pulled-Back Normal Bundle

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with ideal sheaf $\mathcal{I}$ and normal bundle $N = N_Z X$ ([2, Regular Embedding]). Let $g\colon X' \to X$ be a morphism of algebraic schemes, let $Z' = g^{-1}(Z)$ be the scheme-theoretic preimage, and let $h\colon Z' \to Z$ be the induced morphism. Write $N' = h^*N$, a vector bundle of rank d on Z'. Then there is a canonical closed immersion of cones over Z'

$$j\colon C_{Z'}X' \hookrightarrow N'$$

Moreover:

1. j is an isomorphism whenever $Z' \hookrightarrow X'$ is itself a regular embedding of codimension d;

2. for $g = \mathrm{id}_X$ it is the identification $C_Z X = N$ of proposition 3.3.11.

***Proof*:**
Write $\mathcal{I}' \subseteq \mathcal{O}_{X'}$ for the ideal sheaf of Z'.

Step 1: the conormal surjection. By definition of the scheme-theoretic preimage, $\mathcal{I}'$ is the image of the map $g^*\mathcal{I} \to \mathcal{O}_{X'}$; that is, $\mathcal{I}'$ is generated by the pullbacks of local sections of $\mathcal{I}$. Composing $g^*\mathcal{I} \to \mathcal{I}'$ with the quotient $\mathcal{I}' \to \mathcal{I}'/\mathcal{I}'^2$ gives an $\mathcal{O}_{X'}$-linear surjection onto $\mathcal{I}'/\mathcal{I}'^2$. It kills the image of $g^*(\mathcal{I}^2)$, a product of two local sections of $\mathcal{I}$ pulling back to a product of two local sections of $\mathcal{I}'$. Since g^* is right exact, $g^*(\mathcal{I}/\mathcal{I}^2)$ is the cokernel of $g^*(\mathcal{I}^2) \to g^*\mathcal{I}$, so the surjection factors through it; and $g^*(\mathcal{I}/\mathcal{I}^2)$ is annihilated by $\mathcal{I}'$, hence is the $\mathcal{O}_{Z'}$-module $h^*(\mathcal{I}/\mathcal{I}^2)$. This gives a surjection of $\mathcal{O}_{Z'}$-modules

$$h^*\big(\mathcal{I}/\mathcal{I}^2\big) \twoheadrightarrow \mathcal{I}'/\mathcal{I}'^2$$

Step 2: the surjection of graded algebras. The symmetric algebra functor preserves surjections, so Step 1 gives a surjection $\mathrm{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \to \mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ of graded $\mathcal{O}_{Z'}$-algebras. The graded algebra $\bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$ of definition 3.3.6 is generated in degree one over $\mathcal{O}_{Z'}$, so the canonical map from $\mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ onto it is surjective as well. Composing,

$$\mathrm{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \twoheadrightarrow \bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$$

is a surjection of graded quasi-coherent $\mathcal{O}_{Z'}$-algebras.

Step 3: taking relative spectra. On an affine open $U \subseteq Z'$ the displayed surjection is a surjection of rings $\Gamma(U, \mathrm{Sym}^\bullet h^*(\mathcal{I}/\mathcal{I}^2)) \to \Gamma(U, \bigoplus \mathcal{I}'^n/\mathcal{I}'^{n+1})$, which induces a closed immersion of affine schemes in the opposite direction; these are compatible on overlaps because the surjection is a map of sheaves, so they glue to a closed immersion over Z' by [2, Relative Spectrum]. The source of that closed immersion is $C_{Z'}X'$ by definition 3.3.6. Its target is, by [2, Regular Embedding] and example 3.3.2(1), the total space of the vector bundle on Z' whose sheaf of sections is $\big(h^*(\mathcal{I}/\mathcal{I}^2)\big)^\vee = h^*\big((\mathcal{I}/\mathcal{I}^2)^\vee\big)$, that is $N' = h^*N$; regularity of i enters here, and only here, through the local freeness of $\mathcal{I}/\mathcal{I}^2$ of rank d (lemma 3.3.10), which makes $h^*(\mathcal{I}/\mathcal{I}^2)$ locally free of rank d on Z'. The immersion is a morphism of cones because the algebra surjection is graded.

Step 4: part (1). Suppose $Z' \hookrightarrow X'$ is a regular embedding of codimension d. Then $\mathcal{I}'/\mathcal{I}'^2$ is locally free of rank d and the canonical map $\mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2) \to \bigoplus_n \mathcal{I}'^n/\mathcal{I}'^{n+1}$ is an isomorphism, both by lemma 3.3.10 applied to X'. So j is the morphism induced by the surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ of locally free $\mathcal{O}_{Z'}$-modules of the same rank d. Such a surjection is an isomorphism: the target is locally free, hence projective, so the surjection splits and the source decomposes as the target plus the kernel; the kernel is then a direct summand of a locally free module of finite rank, hence itself locally free, and comparing ranks locally makes that rank zero, so the kernel vanishes. Therefore j is an isomorphism of cones.

Step 5: part (2). For $g = \mathrm{id}_X$ one has $\mathcal{I}' = \mathcal{I}$, $Z' = Z$, $h = \mathrm{id}_Z$, and the surjection of Step 2 is the canonical map $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \to \bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$. That map is the isomorphism exhibited in the proof of proposition 3.3.11, so j is the identification $C_Z X = N$ recorded there, completing the proof.

Both immersions above are induced by surjections of graded algebras out of $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$, and when they are both available they agree. This is used whenever a normal cone is read inside a normal bundle.

Lemma 3.3.14: The Two Immersions Agree

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d with normal bundle N, and let $V \subseteq X$ be a closed subscheme with $W = V \cap Z$. Then the composite

$$C_W V \;\hookrightarrow\; N|_W \;\hookrightarrow\; N$$

of the immersion j of lemma 3.3.13 (for $g\colon V \hookrightarrow X$) with the restriction of N to W coincides with the immersion $C_W V \hookrightarrow C_Z X = N$ of lemma 3.3.12.

***Proof*:**
Both immersions are the map on relative Spec induced by a surjection of graded $\mathcal{O}_W$-algebras onto $\bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$, so it is enough to see that the two surjections coincide. The immersion of lemma 3.3.12 is induced by

$$\Big(\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \otimes_{\mathcal{O}_Z} \mathcal{O}_W \;\twoheadrightarrow\; \bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$$

and j is induced by $\operatorname{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \twoheadrightarrow \bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$. Because i is a regular embedding, proposition 3.3.11 identifies $\operatorname{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$ with $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$ as graded $\mathcal{O}_Z$-algebras, and that identification is the identity in degree one. Pulling it back along h turns the second surjection into the first, since both are determined by their degree-one component, which in each case is the map $\mathcal{I}/\mathcal{I}^2 \otimes \mathcal{O}_W \to \mathcal{I}_W/\mathcal{I}_W^2$ induced by $\mathcal{I}\mathcal{O}_V = \mathcal{I}_W$, as we sought to show.

For a regular embedding the Segre class carries no information beyond the normal bundle. The hypothesis $d \geq 1$ costs nothing: a regular embedding of codimension zero has zero ideal sheaf, so it is $Z = X$, and that case was settled directly above by the zero cone. The inverse $c(N)^{-1}$ is computed by the terminating geometric series $\mathrm{id} - c_1(N) + \big(c_1(N)^2 - c_2(N)\big) - \cdots$, terminating because every $c_j(N)$ with $j \geq 1$ lowers dimension and $A_k(Z) = 0$ for $k < 0$. The content of $s(Z, X)$ is therefore entirely in the irregular case: it is the invariant that continues to exist, and to enter the intersection formulas, when the ideal of Z needs more equations than the codimension allows.

The codimension-one case is the one already visible in chapter 2, and it recovers the tower of self-intersections of a divisor.

Example 3.3.15: An Effective Cartier Divisor and a Plane Conic

Let $D \subseteq X$ be an effective Cartier divisor ([2, Effective Cartier Divisor]), so its ideal sheaf $\mathcal{I} = \mathcal{O}_X(-D)$ is invertible and locally generated by a single nonzerodivisor. This is a regular embedding of codimension one, with conormal sheaf $\mathcal{I}/\mathcal{I}^2 = \mathcal{O}_X(-D)|_D$ and normal bundle the line bundle $N_D X = \mathcal{O}_X(D)|_D$. A line bundle L has $c(L) = \mathrm{id} + c_1(L)$, so $c(L)^{-1} = \sum_{i\geq 0}(-1)^i c_1(L)^i$, and proposition 3.3.11 gives

$$s(D, X) \;=\; \sum_{i\geq 0}(-1)^i\, c_1\big(\mathcal{O}_X(D)|_D\big)^i \cap [D] \;=\; [D] - c_1\big(\mathcal{O}_X(D)|_D\big) \cap [D] + \cdots \tag{3.5}$$

the series terminating in dimension zero.
Take $X = \mathbb{P}^2$ and D a smooth conic, so $D \cong \mathbb{P}^1$ and $A_*(D) = \mathbb{Z}\,[D] \oplus \mathbb{Z}\,[\mathrm{pt}]$ by theorem 1.5.2; only the first two terms of (3.5) can be nonzero. The conic is cut out by a quadratic form, so $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(2)$ (example 2.1.4). To evaluate the second term, push forward along the

closed immersion $\iota\colon D \hookrightarrow \mathbb{P}^2$ and use the projection formula proposition 2.4.4(4), additivity proposition 2.4.4(3) for $\mathcal{O}(2) = \mathcal{O}(1) \otimes \mathcal{O}(1)$, the class $\iota_*[D] = 2\,[\mathbb{P}^1]$ of a conic (example 1.5.4), and example 2.4.7:

$$\iota_*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^2}(2)|_D\big) \cap [D]\Big) = c_1\big(\mathcal{O}_{\mathbb{P}^2}(2)\big) \cap \iota_*[D] = 2\,c_1\big(\mathcal{O}_{\mathbb{P}^2}(1)\big) \cap 2\,[\mathbb{P}^1] = 4\,[\mathrm{pt}]$$

Both $A_0(D)$ and $A_0(\mathbb{P}^2)$ are infinite cyclic on the class of a point and ι_* carries generator to generator, so $c_1(\mathcal{O}_{\mathbb{P}^2}(2)|_D) \cap [D] = 4\,[\mathrm{pt}]$ already in $A_0(D)$. Hence

$$s(D, \mathbb{P}^2) \;=\; [D] - 4\,[\mathrm{pt}]$$

The coefficient 4 is the self-intersection number of the conic, the degree $2 \cdot 2$ predicted by Bézout, and it enters the Segre class with a sign. For a divisor, then, $s(D, X)$ is a bookkeeping device for the alternating tower of self-intersections of D, and it contains exactly the information the line bundle $\mathcal{O}_X(D)|_D$ contains.

The irregular case is where the Segre class stops being a repackaging. The next computation is a closed subscheme whose normal cone is not a bundle, and the class detects the multiplicity of a singularity.

Example 3.3.16: The Vertex of the Quadric Cone

Let $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$ be the affine quadric cone of example 2.1.2, a normal surface with an isolated singularity at the vertex, and let $Z = \{P\}$ be the reduced vertex, with ideal $\mathfrak{m} = (x, y, z)$.

The normal cone. Write $A = \overline{k}[x, y, z]/(xy - z^2)$. The defining polynomial is homogeneous, so A is a graded ring with A_n its degree-n piece and $\mathfrak{m}^n = \bigoplus_{m \geq n} A_m$; therefore $\mathfrak{m}^n/\mathfrak{m}^{n+1} = A_n$ and $\bigoplus_{n \geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1} = A$. The quadric cone is its own tangent cone at the vertex, and $C_P X = \operatorname{Spec} A = X$. It is not a vector bundle over the point P: a rank-r bundle over a point is $\mathbb{A}^r$, whereas X is singular at the origin. Equivalently, the embedding $\{P\} \hookrightarrow X$ is not regular, since $\mathfrak{m}$ needs three generators while $\dim X = 2$.

The projective completion. Adjoining w in degree one gives $A[w] = \overline{k}[x, y, z, w]/(xy - z^2)$, so

$$\mathbb{P}(C_P X \oplus 1) \;=\; \operatorname{Proj}\big(\overline{k}[x, y, z, w]/(xy - z^2)\big) \;=\; Q \subseteq \mathbb{P}^3$$

the quadric surface $\{xy = z^2\}$, which is integral because $xy - z^2$ is irreducible, and $\mathcal{O}(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_Q$ by the surjection $\overline{k}[x, y, z, w] \to A[w]$.

The class. Since Z is a single point, $A_k(Z) = 0$ for $k \neq 0$, so only the term of definition 3.3.4 landing in A_0 survives, namely $i = \dim Q = 2$:

$$s(P, X) \;=\; \Big(\int_Q c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)|_Q\big)^2 \cap [Q]\Big)\,[P]$$

Let $\iota\colon Q \hookrightarrow \mathbb{P}^3$ be the inclusion. The quadric is the divisor of the section $xy - z^2$ of $\mathcal{O}_{\mathbb{P}^3}(2)$, so $\iota_*[Q] = 2\,[\mathbb{P}^2]$ in $A_2(\mathbb{P}^3)$ by the class map (proposition 2.1.3 and example 2.1.4). By the projection formula proposition 2.4.4(4) and the fact that capping a linear-subspace class with $c_1(\mathcal{O}_{\mathbb{P}^3}(1))$ drops its dimension by one (example 2.4.7),

$$\iota_*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)|_Q\big)^2 \cap [Q]\Big) = c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)\big)^2 \cap 2\,[\mathbb{P}^2] = 2\,[\mathrm{pt}]$$

Degrees are preserved by proper pushforward (definition 1.2.4), so the integral equals 2 and

$$s(P, X) \;=\; 2\,[P]$$

The contrast. Take instead the origin $P' \in \mathbb{A}^2$, where $\mathfrak{m} = (x, y)$ is a regular sequence and $\{P'\} \hookrightarrow \mathbb{A}^2$ is a regular embedding of codimension two. Proposition 3.3.11 gives $s(P', \mathbb{A}^2) = c(N)^{-1} \cap [P']$, and every $c_j(N) \cap [P']$ with $j \geq 1$ lies in $A_{-j}(P') = 0$, so $s(P', \mathbb{A}^2) = [P']$. At a smooth point of a surface the Segre class of the point is the point itself; at the vertex of the quadric cone it is twice the point. The integer it produces is the multiplicity of the singularity.

Exercise 3.3.1

1. Let X be an algebraic scheme and let $C = X$ be the zero cone of example 3.3.2(2), that is $S^\bullet = \mathcal{O}_X$. Show that $\mathbb{P}(C) = \emptyset$, that $\mathbb{P}(C \oplus 1) = X$ with q the identity, and that $\mathcal{O}(1)$ is the trivial bundle. Deduce $s(X, X) = [X]$, and then explain why proposition 3.3.11 cannot be used to reach the same answer, even though $X \hookrightarrow X$ is a regular embedding of codimension zero.
2. Let $Z = \mathbb{P}^1 \subseteq \mathbb{P}^3$ be a line, cut out by two of the homogeneous coordinates. Show that this is a regular embedding of codimension two with conormal sheaf $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, hence with normal bundle $N = \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$, and compute $s(\mathbb{P}^1, \mathbb{P}^3)$.
3. Let $D \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d. Show that $s(D, \mathbb{P}^2) = [D] - \beta$ for a class $\beta \in A_0(D)$ of degree d^2. Then explain why, for $d \geq 3$, one may *not* conclude $\beta = d^2\,[\mathrm{pt}]$, whereas for $d \leq 2$ one may.
4. Let $f \in \overline{k}[x, y, z]$ be a homogeneous polynomial of degree d defining a smooth plane curve $Y \subseteq \mathbb{P}^2$, let $X = \operatorname{Spec} \overline{k}[x, y, z]/(f)$ be the affine cone over Y, and let $P \in X$ be the vertex. Compute $s(P, X)$, generalizing example 3.3.16.
5. Let $X = \mathbb{A}^1$ and let $Z = \operatorname{Spec} \overline{k}[x]/(x^2)$ be the double point at the origin, with $Z_{\mathrm{red}} = \{P\}$ the reduced point. Compute both $s(Z, X)$ and $s(Z_{\mathrm{red}}, X)$, and say what the comparison shows about the dependence of the Segre class on the scheme structure of Z.

4

Deformation to the Normal Cone

Classical enumerative geometry intersected two cycles by moving them until they met properly and trusting that the count did not change. Making that rigorous requires a moving lemma, and moving lemmas are hard, restrictive, and unavailable on a singular ambient scheme. The construction of this chapter avoids the manoeuvre entirely: instead of moving the cycles, it degenerates the embedding.

Given a closed subscheme $Z \subseteq X$, there is a family over $\mathbb{P}^1$ whose fibre away from ∞ is the given embedding $Z \subseteq X$ and whose fibre at ∞ is the embedding of Z as the zero section of its normal cone (definition 3.3.6). Nothing has been perturbed; the ambient geometry has been deformed to a cone, where it is linear along the fibres and where intersecting is a matter of restricting to a zero section. Specializing a cycle to ∞ therefore converts an arbitrary intersection problem into one on a cone, and when the embedding is regular the cone is a vector bundle ([2, Regular Embedding]) and the specialization can be inverted against theorem 1.4.6 to land back on Z.

That composite is the Gysin map, and it is the last piece of machinery the intersection product needs. This chapter builds the deformation space, proves that specializing is well defined on rational equivalence, and assembles the Gysin map for a regular embedding together with the refined version that chapter 5 will apply to the diagonal.

4.1 The Deformation to the Normal Cone

This section builds the family and establishes the three facts everything later rests on: it is flat over $\mathbb{P}^1$, its fibre away from ∞ is the given X, and its fibre at ∞ is the normal cone.

Throughout, $Z \subseteq X$ is a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. Fix a coordinate t on $\mathbb{P}^1$, so that $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ is the affine line with coordinate t, and write $U = \mathbb{P}^1 \setminus \{0\}$ for the complementary chart, an affine line with coordinate $u = t^{-1}$ vanishing exactly at ∞. Since X is of finite type over $\overline{k}$ (box 0.0.1), the product $X \times \mathbb{P}^1$ is Noetherian, so the blow-up construction of

[2, Blow-up] applies to it.

The object to be built is a scheme over $\mathbb{P}^1$ whose fibre over every point of $\mathbb{A}^1$ is X and whose fibre over ∞ is $C_Z X$. The mechanism that produces such a family is visible in the local algebra. A function a in the ideal $\mathcal{I}$ vanishes on Z; dividing it by the parameter u rescales the normal directions to Z by the factor u^{-1}, and as u tends to 0 the rescaling magnifies an arbitrarily small neighbourhood of Z until only the behaviour of a to leading order in $\mathcal{I}$ survives. Adjoining all the functions au^{-1} with $a \in \mathcal{I}$ to $\mathcal{O}_X[u]$ therefore builds a family whose limit at $u = 0$ is manufactured from the successive quotients $\mathcal{I}^n/\mathcal{I}^{n+1}$, which is what $C_Z X$ is manufactured from (definition 3.3.6).

Adjoining $\mathcal{I}u^{-1}$ is exactly what makes the ideal $\mathcal{I} + (u)$ of $Z \times \{\infty\}$ principal, generated by u; and the universal solution to that require is the blow-up ([2, Basic Properties of Blow-ups], part (4)). The construction accordingly starts from a blow-up, and the rescaled algebra above is recovered as one of its charts.

Definition 4.1.1: The Deformation Space

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, and let $\mathcal{J} \subseteq \mathcal{O}_{X\times\mathbb{P}^1}$ be the ideal sheaf of the closed subscheme $Z \times \{\infty\}$. The *deformation space* of Z in X is the blow-up

$$M_Z X \;=\; \mathrm{Bl}_{Z\times\{\infty\}}\big(X \times \mathbb{P}^1\big) \;=\; \mathrm{Proj}\Big(\bigoplus_{n\geq 0} \mathcal{J}^n\Big) \xrightarrow{\;p\;} X \times \mathbb{P}^1$$

of [2, Blow-up]. Write $\pi\colon M_Z X \to \mathbb{P}^1$ for the composite of p with the projection $X \times \mathbb{P}^1 \to \mathbb{P}^1$.

Nothing has been achieved yet: $M_Z X$ is a blow-up like any other, and its fibre over ∞ is larger than the normal cone. The next lemma computes the part of $M_Z X$ lying over the chart U, splitting it into the piece that carries the cone and a piece that does not. The splitting is the reason the deformation is defined on an open subscheme of $M_Z X$ rather than on all of it.

Lemma 4.1.2: The Chart at Infinity

Let $\mathcal{R} \subseteq \mathcal{O}_X[u, u^{-1}]$ be the sheaf of $\mathcal{O}_X[u]$-subalgebras generated by $\mathcal{I}u^{-1}$, the *extended Rees algebra* of $\mathcal{I}$; grading by powers of u,

$$\mathcal{R} \;=\; \bigoplus_{m\geq 0} \mathcal{O}_X\, u^m \;\oplus\; \bigoplus_{n\geq 1} \mathcal{I}^n u^{-n} \tag{4.1}$$

Write $D_+(u) \subseteq p^{-1}(X \times U)$ for the open subscheme attached to the degree-one section u of $\bigoplus_n \mathcal{J}^n$. Then:

1. $D_+(u)$ is the relative spectrum $\mathrm{Spec}\,(\mathcal{R})$ over $X \times U$;
2. its closed complement $p^{-1}(X \times U) \setminus D_+(u)$ is canonically isomorphic to $\mathrm{Bl}_Z X$, carried by p into $X \times \{\infty\}$ by the blow-down morphism.

The closed subscheme $\mathrm{Bl}_Z X$ so obtained is closed in $M_Z X$.

***Proof*:**
Step 1: the ideal and its powers. Work on an affine open $\operatorname{Spec} A \subseteq X$ with $\mathcal{I}(\operatorname{Spec} A) = I$, and set $B = A[u]$, so that $\operatorname{Spec} A \times U = \operatorname{Spec} B$ and the ideal of $Z \times \{\infty\}$ there is $J = IB + uB$. Expanding the product of the two ideals gives

$$J^n \;=\; \sum_{j=0}^{n} I^j B \cdot u^{n-j}$$

Two consequences are read off this expansion. Every term with $j < n$ is divisible by u, so

$$J^n + uB \;=\; I^n B + uB \tag{4.2}$$

And for $n \geq 1$,

$$J^n \cap uB \;=\; u\, J^{n-1} \tag{4.3}$$

Indeed, write $x = \sum_{j=0}^{n} a_j u^{n-j}$ with $a_j \in I^j B$. The terms with $j < n$ all lie in uJ^{n-1}, since $I^j B\, u^{n-1-j} \subseteq J^{n-1}$. The remaining term $a_n \in I^n B$ is a polynomial in u with coefficients in I^n, so it lies in uB precisely when its constant coefficient vanishes, that is when $a_n \in uI^n B \subseteq uJ^{n-1}$. The reverse inclusion is immediate from $uJ^{n-1} \subseteq J^n$.

Step 2: the chart. Write $S = \bigoplus_{n\geq 0} J^n$, so that $p^{-1}(\operatorname{Spec} A \times U) = \operatorname{Proj}(S)$ by definition 4.1.1 and $D_+(u) = \operatorname{Spec}\left(S_{(u)}\right)$ is the affine chart of the relative Proj attached to the degree-one element $u \in J = S_1$ ([2, Relative Proj]). Because u is a nonzerodivisor on B, the assignment $x/u^n \mapsto xu^{-n}$ is a well-defined injection of $S_{(u)}$ into $A[u, u^{-1}]$, and its image is $\bigcup_{n\geq 0} J^n u^{-n}$, which by Step 1 is the B-subalgebra R generated by Iu^{-1}. To see the grading (4.1), note $R = \sum_{n\geq 0} I^n B\, u^{-n}$, so the coefficient of u^m in R is $\sum_{n\geq 0,\, n+m\geq 0} I^n$, which is A for $m \geq 0$ and I^{-m} for $m < 0$. Hence $D_+(u) = \operatorname{Spec}(R)$ as a scheme over $\operatorname{Spec} B$, which is (1).

Step 3: the complement. Reduction modulo u gives a surjection of graded B-algebras $S \to \overline{S}$, where $\overline{S}_n = (J^n + uB)/uB$; by (4.2) this is the image of $I^n B$ in $B/uB = A$, namely I^n. So $\overline{S} = \bigoplus_{n\geq 0} I^n$ is the Rees algebra of I, its relative Proj is $\mathrm{Bl}_Z X$ over $\operatorname{Spec} A$ by [2, Blow-up], and the surjection exhibits it as a closed subscheme of $\operatorname{Proj}(S)$ lying over $V(u) = \operatorname{Spec} A \times \{\infty\}$, with structure morphism to $\operatorname{Spec} A$ the blow-down. Since u maps to 0 in $\overline{S}_1$, this closed subscheme is disjoint from $D_+(u)$.
For the reverse inclusion, the closed complement of $D_+(u)$ is $\operatorname{Proj}(S/uS)$, where uS is the homogeneous ideal generated by $u \in S_1$, whose degree-n part is uJ^{n-1} for $n \geq 1$ and 0 in degree 0. By (4.3) the ideal uJ^{n-1} is exactly the kernel of $J^n \to I^n$, so S/uS and $\overline{S}$ agree in every positive degree and differ only in degree 0, where $A[u] \to A$ kills u. In S/uS the degree-zero element u annihilates every positive degree, so for a homogeneous $f \in S_1$ the image of u in the localization $(S/uS)_{(f)}$ is $uf/f = 0$; hence $(S/uS)_{(f)} = \overline{S}_{(f)}$ for every such f, and the two relative Proj schemes coincide. The closed complement of $D_+(u)$ is therefore $\mathrm{Bl}_Z X$, which is (2).

Step 4: globalization. The algebra $\mathcal{R}$ and the surjection of Step 3 are defined without choices, so the local identifications agree on overlaps of an affine cover of X and glue. For the final assertion, $\mathrm{Bl}_Z X$ is closed in the open subscheme $p^{-1}(X \times U)$ and is contained in $p^{-1}(X \times \{\infty\})$, which is closed in $M_Z X$ and contained in $p^{-1}(X \times U)$; writing $\mathrm{Bl}_Z X = C \cap p^{-1}(X \times U)$ with C closed in $M_Z X$, one gets $\mathrm{Bl}_Z X = C \cap p^{-1}(X \times \{\infty\})$, an intersection of two closed subsets and hence closed in $M_Z X$, completing the proof.

The lemma is what makes the construction computable: over the chart at infinity the deformation is not a Proj at all but an affine scheme over $X \times U$, given by the single algebra (4.1), written R when $X = \operatorname{Spec} A$ is affine, and its reduction modulo u is the associated graded algebra of $\mathcal{I}$. The smallest instance already shows the mechanism.

Example 4.1.3: A Point on a Smooth Curve

Let X be a variety of dimension one and $P \in X$ a closed point at which X is smooth, and take $Z = \{P\}$ with its reduced structure. The local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring; choosing a uniformizer s makes $\mathcal{I}$ locally generated by the single nonzerodivisor s, so $Z \hookrightarrow X$ is a regular embedding of codimension one ([2, Regular Embedding]) and $\operatorname{gr}_{\mathfrak{m}_P} \mathcal{O}_{X,P} \cong \overline{k}[\overline{s}]$ is a polynomial ring in one variable. Hence $C_P X = \mathbb{A}^1$: the deformation replaces the curve by its tangent line at P and the point by the origin of that line.

The computation is explicit for $X = \mathbb{A}^1 = \operatorname{Spec} \overline{k}[x]$ and P the origin, where $I = (x)$. The algebra of (4.1) is $R = \overline{k}[x, u][xu^{-1}]$, and setting $v = xu^{-1}$ gives $x = uv$, so R is generated by u and v alone. These two are algebraically independent over $\overline{k}$: they lie in $\overline{k}(x, u)$, which has transcendence degree two, and they generate it, so they form a transcendence basis. Hence

$$R \;\cong\; \overline{k}[u, v], \qquad \operatorname{Spec}(R) = \mathbb{A}^2 \longrightarrow \mathbb{A}^1_u$$

is the trivial family with fibre $\mathbb{A}^1$. Over $u = c \neq 0$ the fibre is identified with X by $x = cv$, and over $u = 0$ it is $\operatorname{Spec}\big(R/uR\big) = \operatorname{Spec} \overline{k}[v]$, which is $C_P X$. The family is the rescaling $x \mapsto x/u$ of the line, and the closed subscheme $\{v = 0\}$ is the point P travelling along it.

The lemma isolates a piece of $M_Z X$ over ∞, namely $\mathrm{Bl}_Z X$, which the rescaling picture does not see and which is discarded.

Definition 4.1.4: The Deformation to the Normal Cone

With $Z \subseteq X$ and $M_Z X$ as in definition 4.1.1, and with $\mathrm{Bl}_Z X \subseteq M_Z X$ the closed subscheme of lemma 4.1.2(2), the *deformation to the normal cone* of Z in X is the open subscheme

$$M_Z^\circ X \;=\; M_Z X \setminus \mathrm{Bl}_Z X$$

together with the restriction of $\pi \colon M_Z X \to \mathbb{P}^1$, again written π, and the restriction of p, whose composite with the projection $X \times \mathbb{P}^1 \to X$ is written $\rho \colon M_Z^\circ X \to X$.

Everything the chapter needs is contained in the behaviour of π over the two charts, and lemma 4.1.2 has already computed one of them. Collecting that computation with the elementary behaviour over $\mathbb{A}^1$ gives the properties that section 4.2 and section 4.3 consume.

Theorem 4.1.5: The Fibres of the Deformation

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, with $\pi\colon M_Z^\circ X \to \mathbb{P}^1$ as in definition 4.1.4.

1. Over $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ the blow-down restricts to an isomorphism $\pi^{-1}(\mathbb{A}^1) \xrightarrow{\sim} X \times \mathbb{A}^1$ commuting with the projections to $\mathbb{A}^1$.
2. The fibre over ∞ is the normal cone, $\pi^{-1}(\infty) = C_Z X$ of definition 3.3.6, and it is an effective Cartier divisor on $M_Z^\circ X$, cut out by the function u.
3. π is flat.
4. There is a closed immersion $Z \times \mathbb{P}^1 \hookrightarrow M_Z^\circ X$ over $\mathbb{P}^1$ which over $\mathbb{A}^1$ is the inclusion $Z \times \mathbb{A}^1 \subseteq X \times \mathbb{A}^1$ of (1), and over ∞ is the vertex $Z \hookrightarrow C_Z X$ of definition 3.3.1.

***Proof*:**
Step 1: the chart over $\mathbb{A}^1$. The centre $Z \times \{\infty\}$ is disjoint from the open set $X \times \mathbb{A}^1$, so [2, Basic Properties of Blow-ups](2) makes p an isomorphism $p^{-1}(X \times \mathbb{A}^1) \to X \times \mathbb{A}^1$. By lemma 4.1.2(2) the deleted subscheme $\mathrm{Bl}_Z X$ lies over $X \times \{\infty\}$, so it does not meet $p^{-1}(X \times \mathbb{A}^1)$, and therefore $\pi^{-1}(\mathbb{A}^1) = p^{-1}(X \times \mathbb{A}^1) \cong X \times \mathbb{A}^1$ compatibly with the projections to $\mathbb{A}^1$. This is (1).

Step 2: the fibre at infinity. Over the other chart, lemma 4.1.2 gives $\pi^{-1}(U) = \mathrm{Spec}\,(\mathcal{R})$ with $\mathcal{R}$ the extended Rees algebra, so the scheme-theoretic fibre over $\infty = \{u = 0\}$ is $\mathrm{Spec}\,\big(\mathcal{R}/u\mathcal{R}\big)$. Compute the quotient degree by degree in the grading (4.1). Writing $\mathcal{R}_m$ for the coefficient of u^m, so that $\mathcal{R}_m = \mathcal{O}_X$ for $m \geq 0$ and $\mathcal{R}_{-n} = \mathcal{I}^n$ for $n \geq 1$, the coefficient of u^m in $u\mathcal{R}$ is $\mathcal{R}_{m-1}$. Hence $(\mathcal{R}/u\mathcal{R})_m$ vanishes for $m \geq 1$, equals $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$ for $m = 0$, and equals $\mathcal{I}^n/\mathcal{I}^{n+1}$ for $m = -n$ with $n \geq 1$. Therefore

$$\mathcal{R}/u\mathcal{R} \;\cong\; \bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$$

which is the graded algebra of definition 3.3.6, so $\pi^{-1}(\infty) = C_Z X$.

Step 3: the fibre at infinity is Cartier. The element u acts invertibly on $\mathcal{O}_X[u, u^{-1}]$ and hence injectively on the subsheaf $\mathcal{R}$, so u is a nonzerodivisor and $\pi^{-1}(\infty) \subseteq \pi^{-1}(U)$ is cut out by a single nonzerodivisor. On the complementary open $\pi^{-1}(\mathbb{A}^1)$ the fibre is empty, cut out by the unit 1. So $\pi^{-1}(\infty)$ is locally principal, generated at each point by a nonzerodivisor, which is an effective Cartier divisor ([2, Effective Cartier Divisor]). With Step 2 this is (2).

Step 4: flatness. Flatness may be checked locally on source and target, so it suffices to treat the two charts separately. Over $\mathbb{A}^1$ the morphism is the projection $X \times \mathbb{A}^1 \to \mathbb{A}^1$ by Step 1, which is the base change of $X \to \mathrm{Spec}\,\overline{k}$ along $\mathbb{A}^1 \to \mathrm{Spec}\,\overline{k}$; every scheme is flat over a field and flatness is stable under base change, so this projection is flat. Over U the morphism is $\mathrm{Spec}\,(\mathcal{R}) \to U = \mathrm{Spec}\,\overline{k}[u]$. The ring $\overline{k}[u]$ is a principal ideal domain, over which a module is flat exactly when it is torsion-free. On an affine open $\mathrm{Spec}\,A \subseteq X$ one has $\mathcal{R}(\mathrm{Spec}\,A) = R \subseteq A[u, u^{-1}]$, and $A[u, u^{-1}]$ is torsion-free over $\overline{k}[u]$: given $f \in \overline{k}[u]$ nonzero of degree d with leading coefficient $c \in \overline{k}^*$, and a nonzero $x = \sum_{i \leq M} a_i u^i$ with $a_M \neq 0$, the coefficient of u^{M+d} in fx is $c\,a_M \neq 0$, so $fx \neq 0$. A submodule of a torsion-free module is torsion-free, so R is torsion-free and hence flat over $\overline{k}[u]$. This is (3).

Step 5: the constant subscheme. Let $\mathfrak{q} \subseteq \mathcal{R}$ be the ideal generated by $\mathcal{I}u^{-1}$; it contains $\mathcal{I}$, since

$\mathcal{I} = (\mathcal{I}u^{-1}) \cdot u$ and $u \in \mathcal{R}$. Its coefficient of u^m is $\mathcal{I} \cdot \mathcal{R}_{m+1}$, which for $m \geq 0$ is $\mathcal{I}$ and for $m = -n < 0$ is $\mathcal{I} \cdot \mathcal{I}^{n-1} = \mathcal{I}^n = \mathcal{R}_{-n}$. So

$$\mathcal{R}/\mathfrak{q} \;\cong\; \bigoplus_{m \geq 0} (\mathcal{O}_X/\mathcal{I})\, u^m \;=\; \mathcal{O}_Z[u]$$

and $\mathfrak{q}$ defines a closed immersion $Z \times U \hookrightarrow \operatorname{Spec}(\mathcal{R})$ over $X \times U$. Inverting u makes u^{-1} a unit, so $\mathfrak{q}$ generates $\mathcal{I}$ there and the immersion restricts over $U \setminus \{\infty\}$ to $Z \times (U \setminus \{\infty\}) \subseteq X \times (U \setminus \{\infty\})$; it therefore glues with the inclusion $Z \times \mathbb{A}^1 \subseteq X \times \mathbb{A}^1$ of Step 1 to a closed immersion $Z \times \mathbb{P}^1 \hookrightarrow M_Z^\circ X$ over $\mathbb{P}^1$. Modulo u the ideal $\mathfrak{q}$ becomes the ideal generated by the degree-one part $\mathcal{I}/\mathcal{I}^2$ of $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$, whose quotient is $\mathcal{O}_Z$; by definition 3.3.1 that is the vertex of $C_Z X$. So the immersion restricts over ∞ to the vertex, which is (4), completing the proof.

The theorem produces a single scheme over $\mathbb{P}^1$ carrying two embeddings at once. Away from ∞ the pair is the given $Z \subseteq X$, unchanged; at ∞ it is the vertex inside $C_Z X$. What moves in the family is the ambient scheme, not the subscheme: by (4) the subscheme is the constant family $Z \times \mathbb{P}^1$, and the ambient scheme degenerates around it until it becomes a cone with Z as its vertex.

The morphism $\rho\colon M_Z^\circ X \to X$ records how each fibre maps back to the original scheme. Over $\mathbb{A}^1$ it is the projection $X \times \mathbb{A}^1 \to X$ by Step 1 of the proof, and over ∞ it is the composite $C_Z X \to Z \hookrightarrow X$, since the structure map $\mathcal{O}_X \to \mathcal{R}$ becomes $\mathcal{O}_X \to \mathcal{O}_Z$ in degree zero after reducing modulo u. One property the family does not have is properness: its fibre over ∞ is affine over Z (definition 3.3.1), and an affine scheme of positive dimension over a field is not proper, so π is not a proper morphism as soon as $C_Z X$ has positive-dimensional fibres. Cycles will therefore be moved across the family by restriction to the Cartier divisor $\pi^{-1}(\infty)$ rather than by pushforward.

4.1.6 Remark: Why the Blow-Up Is Not Enough The subscheme deleted in definition 4.1.4 is not a technicality. Lemma 4.1.2 splits the part of $M_Z X$ lying over the chart U into $D_+(u)$ and $\mathrm{Bl}_Z X$, and the fibre over ∞ lies entirely in that part; its intersection with $D_+(u)$ is $C_Z X$ by Step 2 of the proof of theorem 4.1.5, while $\mathrm{Bl}_Z X$ maps into $X \times \{\infty\}$ and so lies in the fibre. The special fibre of the unmodified blow-up is therefore the disjoint union of $C_Z X$ and $\mathrm{Bl}_Z X$ as a set, and only the first piece records the normal directions along Z; the second is a copy of X modified along Z, carrying no information the construction is after. Deleting it costs the properness of the blow-down p over $X \times \mathbb{P}^1$ and gives a special fibre equal to $C_Z X$ on the nose.

The two extremes of the construction are the case where $C_Z X$ is a vector bundle and the case where it is not. The first is the case the Gysin map of section 4.3 will need, and codimension one is where it is easiest to see.

Example 4.1.7: An Effective Cartier Divisor and a Line in the Plane

Let $D \subseteq X$ be an effective Cartier divisor ([2, Effective Cartier Divisor]). Its ideal sheaf is invertible and locally generated by one nonzerodivisor, so $D \hookrightarrow X$ is a regular embedding of codimension one with normal bundle $N_D X = \mathcal{O}_X(D)|_D$ (example 3.3.15), and $C_D X = N_D X$ by proposition 3.3.11. By theorem 4.1.5 the deformation therefore degenerates $D \subseteq X$ into D sitting inside the total space of a line bundle on itself, as the zero section, which is the vertex of that cone (example 3.3.2(1)). Take $X = \mathbb{P}^2$ and $D = L$ a line. Then $\mathcal{O}_{\mathbb{P}^2}(L) \cong \mathcal{O}_{\mathbb{P}^2}(1)$ by example 2.1.4, and its restriction to $L \cong \mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(1)$ (example 2.2.3), so $C_L \mathbb{P}^2$ is the total space of $\mathcal{O}_{\mathbb{P}^1}(1)$ over L. That total space is $\mathbb{P}^2$ with a point removed: linear projection away from a point $P \notin L$ exhibits $\mathbb{P}^2 \setminus \{P\}$ as a line bundle over L whose zero section is L itself, and the degree of that bundle is the self-intersection $L \cdot L = 1$ of example 2.2.3, so it is $\mathcal{O}_{\mathbb{P}^1}(1)$. (The proof of theorem 1.5.2 notes the projection but uses only that it is an affine bundle, which is all its own argument needs.) So the family degenerates the pair $L \subseteq \mathbb{P}^2$ into the pair $L \subseteq \mathbb{P}^2 \setminus \{P\}$, with L the zero section of a line bundle. On the special fibre, intersecting a cycle with L has become intersecting it with the zero section of a line bundle, and flat pullback along the projection of that bundle is an isomorphism on Chow groups (theorem 1.4.6).

When the embedding is not regular the cone is not a bundle, and the deformation exhibits the singularity of the pair rather than a linear model of it. The node is the case already computed in example 3.3.7, and the whole family can be written down.

Example 4.1.8: The Node and Its Tangent Cone

Assume char$\overline{k} \neq 2$ and let $X = \operatorname{Spec} A$ with $A = \overline{k}[x,y]/(y^2 - x^2 - x^3)$ be the affine nodal cubic, P the origin, $Z = \{P\}$ and $I = (x,y)$. Set $v = xu^{-1}$ and $w = yu^{-1}$ in $A[u,u^{-1}]$, so that $x = uv$ and $y = uw$; the algebra $R = A[u][Iu^{-1}]$ of (4.1) is generated by u, v, w. Substituting into the defining relation gives $u^2w^2 = u^2v^2 + u^3v^3$, and u is a nonzerodivisor on the domain $A[u,u^{-1}]$, so

$$w^2 \;=\; v^2 + uv^3 \tag{4.4}$$

The relation is the only one. The ring A is free over $\overline{k}[x]$ on $1, y$, because its defining relation is monic of degree two in y; hence $A[u,u^{-1}]$ is free over $\overline{k}[x,u,u^{-1}]$ on $1, y$. Using (4.4) to reduce powers of w, the ring R is generated as a $\overline{k}[u,v]$-module by 1 and w. Those two are independent: if $f(u,v) + g(u,v)w = 0$, then, reading off the coefficients of the basis $1, y$, one gets $f(u, xu^{-1}) = 0$ and $g(u, xu^{-1}) = 0$ in $\overline{k}[x,u,u^{-1}]$, and u and xu^{-1} are algebraically independent over $\overline{k}$ (they generate $\overline{k}(x,u)$, of transcendence degree two), so $f = g = 0$. Therefore

$$R \;\cong\; \overline{k}[u,v,w]/\big(w^2 - v^2 - uv^3\big)$$

The fibres. Over $u = c \neq 0$ the fibre is $\operatorname{Spec} \overline{k}[v,w]/(w^2 - v^2 - cv^3)$, carried isomorphically onto X by $x = cv$, $y = cw$, as substituting confirms. Over $u = 0$ the fibre is $\operatorname{Spec} \overline{k}[v,w]/(w^2 - v^2)$, the union of the two lines $w = \pm v$: the tangent cone of example 3.3.7, in agreement with theorem 4.1.5(2). The family straightens the two branches of the node into their tangent lines while keeping the singular point fixed, and the cubic term uv^3, which distinguishes the nodal cubic from a pair of lines, is exactly the term that the rescaling drives to zero.

Exercise 4.1.1

1. Compute the deformation in the two degenerate cases. First take $Z = X$, so that $\mathcal{I} = 0$: identify $\mathcal{R}$, show that $M_X^\circ X = X \times \mathbb{P}^1$ with π the projection, and check that the fibre over ∞ is the zero cone of example 3.3.2(2). Then take $Z = \emptyset$, so that $\mathcal{I} = \mathcal{O}_X$: identify $\mathcal{R}$, show that $M_\emptyset^\circ X = X \times \mathbb{A}^1$, and reconcile the empty fibre over ∞ with definition 3.3.6.
2. Let $X = \mathbb{A}^n$ and let $Z = \{0\}$ be the origin, with $I = (x_1, \ldots, x_n)$. Show that the algebra of (4.1) is a polynomial ring $R \cong \overline{k}[u, v_1, \ldots, v_n]$ with $v_i = x_iu^{-1}$, so that $\pi^{-1}(U) \cong \mathbb{A}^{n+1} \to \mathbb{A}^1_u$ is the trivial family. Identify the isomorphism of the fibre over $u = c \neq 0$ with $\mathbb{A}^n$, identify $C_0\mathbb{A}^n$, and locate the closed subscheme $Z \times U$ of theorem 4.1.5(4) inside $\mathbb{A}^{n+1}$.

3. Let $X = \mathbb{A}^1 = \operatorname{Spec} \overline{k}[x]$ and let $Z = \operatorname{Spec} \overline{k}[x]/(x^2)$ be the double point at the origin, so $I = (x^2)$. Show that $\operatorname{Spec}(\mathcal{R})$ is the affine quadric cone of example 2.1.2, and identify the fibre over ∞. Comment on what the computation shows about the smoothness of the deformation space.

4. Let $D \subseteq \mathbb{P}^2$ be a smooth conic. Using example 3.3.15 and proposition 3.3.11, identify $C_D\mathbb{P}^2$ as the total space of a line bundle on $D \cong \mathbb{P}^1$, and determine which line bundle it is.

5. Let $X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^3)$ be the cuspidal cubic and P the origin. Compute $C_P X$ and its fundamental cycle $[C_P X]$ in the sense of definition 1.1.2, and contrast the answer with the node of example 4.1.8.

4.2 Specialization of Cycles

A cycle on X is spread over $\mathbb{A}^1$, its closure is followed to the fibre at infinity, and what that closure cuts on the fibre is a cycle on the normal cone. The resulting operation $\sigma\colon A_k(X) \to A_k(C_Z X)$ is the specialization homomorphism. Three things have to be proved about it: that it is well defined on rational equivalence, that on a subvariety V it returns the normal cone of $V \cap Z$ in V, and that it is compatible with proper pushforward and flat pullback.

Throughout, $Z \subseteq X$ is a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I}$, and $\pi\colon M_Z^\circ X \to \mathbb{P}^1$ is the projection of definition 4.1.4, which by theorem 4.1.5(1) to (3) is flat with fibres

$$\pi^{-1}(\mathbb{A}^1) \;=\; X \times \mathbb{A}^1, \qquad \pi^{-1}(\infty) \;=\; C_Z X$$

Write $j\colon X \times \mathbb{A}^1 \hookrightarrow M_Z^\circ X$ for the open immersion of the first fibre, $i_\infty\colon C_Z X \hookrightarrow M_Z^\circ X$ for the closed immersion of the second, and $q\colon X \times \mathbb{A}^1 \to X$ for the projection. On $\mathbb{P}^1$ fix the coordinate u vanishing to order one at ∞, so that $\mathbb{P}^1 \setminus \{0\} = \operatorname{Spec} \overline{k}[u]$ and ∞ is the closed subscheme $\{u = 0\}$; on the complementary chart $\mathbb{A}^1$ the coordinate is $t = u^{-1}$.

The fibre at infinity is cut out by a single equation (theorem 4.1.5(2)), which is what makes definition 2.2.1 applicable. One further fact is needed before anything can be intersected: the line bundle carrying that equation is trivial along the fibre itself. It looks like an accident of the construction, and it is what will make the whole operation independent of choices.

Lemma 4.2.1: The Line Bundle of the Fibre at Infinity

Let D_∞ be the effective Cartier divisor of theorem 4.1.5(2), whose support is the fibre $\pi^{-1}(\infty) = C_Z X$. Then

$$\mathcal{O}_{M_Z^\circ X}(D_\infty) \;=\; \pi^*\mathcal{O}_{\mathbb{P}^1}(\infty), \qquad \mathcal{O}_{M_Z^\circ X}(D_\infty)\big|_{C_Z X} \;\cong\; \mathcal{O}_{C_Z X}$$

so the line bundle of D_∞ restricts to the trivial bundle on the fibre it cuts out.

***Proof*:**
On $\mathbb{P}^1$ the point ∞ is an effective Cartier divisor ([2, Effective Cartier Divisor]): it is cut out by the nonzerodivisor u on $\operatorname{Spec} \overline{k}[u]$ and by the unit 1 on the complementary chart $\mathbb{A}^1$, and its line bundle is $\mathcal{O}_{\mathbb{P}^1}(\infty)$.

By theorem 4.1.5(2) the divisor D_∞ is cut out by $\pi^{\#}u$ over $\pi^{-1}(\mathbb{P}^1 \setminus \{0\})$, and over $\pi^{-1}(\mathbb{A}^1)$ it is

empty, cut out by the unit 1. Its local data $\{(\pi^{-1}(\mathbb{A}^1), 1), (\pi^{-1}(\mathbb{P}^1 \setminus \{0\}), \pi^{\#}u)\}$ are therefore the pullbacks along π of the local data just recorded for ∞ on $\mathbb{P}^1$, so $D_\infty = \pi^*(\infty)$ and the associated line bundle is $\pi^*\mathcal{O}_{\mathbb{P}^1}(\infty)$.
For the restriction, the composite $\pi \circ i_\infty \colon C_Z X \to \mathbb{P}^1$ factors through the closed subscheme $\{\infty\} = \operatorname{Spec} \overline{k}$, since i_∞ is the inclusion of the fibre over ∞, so

$$\mathcal{O}_{M^\circ_Z X}(D_\infty)\big|_{C_Z X} \;=\; i_\infty^* \pi^* \mathcal{O}_{\mathbb{P}^1}(\infty) \;=\; \big(\pi \circ i_\infty\big)^* \mathcal{O}_{\mathbb{P}^1}(\infty)$$

is the pullback of $\mathcal{O}_{\mathbb{P}^1}(\infty)|_{\{\infty\}}$ along the structure morphism $C_Z X \to \operatorname{Spec} \overline{k}$. A line bundle on the spectrum of a field is free of rank one, so that pullback is trivial, completing the proof.

The first consequence of the triviality is what the definition of σ needs: the operator $D_\infty \cdot -$ annihilates every class that already lives on the fibre at infinity. Geometrically, a cycle sitting inside the special fibre is not cut further by the equation of that fibre.

Lemma 4.2.2: Vanishing on Classes Supported at Infinity

For every $\gamma \in A_{k+1}(C_Z X)$,

$$D_\infty \cdot (i_\infty)_* \gamma \;=\; 0 \quad \text{in } A_k(C_Z X)$$

***Proof*:**
By proposition 2.2.4 the operator $D_\infty \cdot - \colon A_{k+1}(M^\circ_Z X) \to A_k(|D_\infty|) = A_k(C_Z X)$ is a homomorphism independent of all choices, so it may be evaluated on any cycle representing the class. Represent γ by a cycle $\sum_l n_l[T_l]$ with each $T_l \subseteq C_Z X$ a $(k+1)$-dimensional subvariety; then $(i_\infty)_*\gamma$ is represented by the same cycle, read on $M^\circ_Z X$, and every T_l is contained in $|D_\infty| = C_Z X$. The second clause of definition 2.2.1 is therefore in force on each term, and

$$D_\infty \cdot [T_l] \;=\; c\Big(\mathcal{O}_{M^\circ_Z X}(D_\infty)\big|_{T_l}\Big) \;\in\; A_k(T_l)$$

By lemma 4.2.1 the restricted bundle is trivial, and the class map of proposition 2.1.3 is a group homomorphism, so it sends the trivial bundle, the identity element of $\operatorname{Pic}(T_l)$, to zero. Hence $D_\infty \cdot [T_l] = 0$ for every l, and summing gives $D_\infty \cdot (i_\infty)_*\gamma = 0$, as we sought to show.

With these two facts the specialization map can be built. The recipe has three moves: thicken a class on X into a class on $X \times \mathbb{A}^1$, extend it across the family, and cut with the fibre at infinity. The middle move is the only one that involves a choice, since a class on the open part of $M^\circ_Z X$ has many extensions; the localization sequence measures the ambiguity exactly, and lemma 4.2.2 kills it.

Definition 4.2.3: Specialization of Cycles

Let $Z \subseteq X$ be a closed subscheme and $\alpha \in A_k(X)$. A *lift* of α is a class $\beta \in A_{k+1}(M^\circ_Z X)$ with

$$j^*\beta \;=\; q^*\alpha \quad \text{in } A_{k+1}\big(X \times \mathbb{A}^1\big)$$

The *closure lift* attached to a cycle $\sum_l n_l[V_l]$ representing α is the class of

$$\overline{\alpha \times \mathbb{A}^1} \;=\; \sum_l n_l\big[\overline{V_l \times \mathbb{A}^1}\big]$$

the closures being taken in $M^\circ_Z X$. The *specialization* of α is

$$\sigma(\alpha) \;=\; D_\infty \cdot \beta \;\in\; A_k\big(C_Z X\big)$$

for any lift β, the intersection being that of definition 2.2.1 with the Cartier divisor of lemma 4.2.1, whose support is $C_Z X$.

The definition presupposes two things, that a lift exists and that the answer does not depend on which one is chosen, and both come from the localization sequence for the closed fibre $C_Z X$ inside $M^\circ_Z X$. It also presupposes that the recipe respects rational equivalence on X, which is where flat pullback does its work.

Theorem 4.2.4: The Specialization Homomorphism

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme. Every class $\alpha \in A_k(X)$ admits a lift, and $D_\infty \cdot \beta$ is the same for all lifts β of α. The assignment of definition 4.2.3 is therefore a well-defined homomorphism

$$\sigma\colon A_k(X) \longrightarrow A_k\big(C_Z X\big)$$

for every k.

***Proof*:**
Step 1: existence of a lift. The fibre $C_Z X$ is closed in $M^\circ_Z X$ with open complement $X \times \mathbb{A}^1$, so theorem 1.4.1 gives an exact sequence

$$A_{k+1}\big(C_Z X\big) \xrightarrow{(i_\infty)_*} A_{k+1}\big(M^\circ_Z X\big) \xrightarrow{j^*} A_{k+1}\big(X \times \mathbb{A}^1\big) \longrightarrow 0 \tag{4.5}$$

Surjectivity of j^* produces a lift of $q^*\alpha$, so lifts exist.

Step 2: independence of the lift. Let β and β' both lift α. Then $j^*(\beta - \beta') = 0$, so by exactness of (4.5) there is $\gamma \in A_{k+1}(C_Z X)$ with $\beta - \beta' = (i_\infty)_*\gamma$. Since $D_\infty \cdot -$ is a homomorphism on $A_{k+1}(M^\circ_Z X)$ (proposition 2.2.4),

$$D_\infty \cdot \beta - D_\infty \cdot \beta' \;=\; D_\infty \cdot (i_\infty)_*\gamma \;=\; 0$$

by lemma 4.2.2. So the value depends only on α.

Step 3: additivity. Flat pullback q^* and open restriction j^* are homomorphisms (proposition 1.3.7), so the sum of a lift of α and a lift of α' is a lift of $\alpha + \alpha'$; and $D_\infty \cdot -$ is a homomorphism. Hence $\sigma(\alpha+\alpha') = \sigma(\alpha)+\sigma(\alpha')$, and σ is a homomorphism $A_k(X) \to A_k(C_Z X)$, completing the proof.

Nothing in the argument required α to be represented by any particular kind of cycle, and that is the point of routing the construction through Chow groups rather than through cycles: rational equivalence on X is absorbed by q^* before the family is entered, and the ambiguity created by the passage to $M_Z^\circ X$ is absorbed by lemma 4.2.2 after it. In practice, one lift is more useful than the others. Given a cycle $\sum_l n_l[V_l]$ representing α, the flat pullback $q^*[V_l] = [V_l \times \mathbb{A}^1]$ (example 1.3.4(2)) extends across the family by taking closures in $M_Z^\circ X$, and the closure lift of definition 4.2.3 restricts along the open immersion j to $q^*\alpha$ by example 1.3.4(1), so it is indeed a lift. Every one of its components dominates $\mathbb{P}^1$, hence lies in no fibre, and in particular none is contained in $|D_\infty|$; the first clause of definition 2.2.1 therefore governs every term. This is the representative used in every computation below.

Example 4.2.5: Specializing at a Point of the Projective Line

Let $X = \mathbb{P}^1$ and let $Z = \{P\}$ be a reduced closed point. The ideal $\mathcal{I}$ is generated at P by a local coordinate, so $\mathcal{I}/\mathcal{I}^2$ is one-dimensional over $\overline{k}$ and every $\mathcal{I}^n/\mathcal{I}^{n+1}$ is one-dimensional; the normal cone $C_P\mathbb{P}^1$ of definition 3.3.6 is the affine line $\mathbb{A}^1$ over the point P. The deformation space is concrete here: $X \times \mathbb{P}^1 = \mathbb{P}^1 \times \mathbb{P}^1$, the centre is the single point (P, ∞), and $M_Z X$ is the blow-up of that surface at that point, with exceptional divisor a copy of $\mathbb{P}^1$. Removing $\mathrm{Bl}_P\mathbb{P}^1 = \mathbb{P}^1$, the strict transform of $\mathbb{P}^1 \times \{\infty\}$, leaves $M_Z^\circ X$, whose fibre over ∞ is the exceptional $\mathbb{P}^1$ with one point deleted, namely $\mathbb{A}^1 = C_P\mathbb{P}^1$.

Two computations are available, one in each dimension. In dimension one, $A_1(\mathbb{P}^1) = \mathbb{Z}[\mathbb{P}^1]$ by theorem 1.5.2, and the closure of $\mathbb{P}^1 \times \mathbb{A}^1$ is all of $M_Z^\circ X$, an integral surface, so the first clause of definition 2.2.1 gives $\sigma[\mathbb{P}^1] = [\mathrm{div}(\pi^\# u)]$. The zero scheme of $\pi^\# u$ is the fibre over ∞, which is the reduced curve $\mathbb{A}^1$, so by definition 1.1.1 the order of $\pi^\# u$ along it is the length of the local ring of a reduced curve at its generic point, namely one. Hence

$$\sigma\big[\mathbb{P}^1\big] \;=\; \big[\mathbb{A}^1\big] \;=\; \big[C_P\mathbb{P}^1\big] \;\in\; A_1\big(C_P\mathbb{P}^1\big) = \mathbb{Z}$$

and σ is an isomorphism in this dimension. In dimension zero the behaviour is the opposite. Let $Q \neq P$ be a closed point. The closure of $\{Q\} \times \mathbb{A}^1$ in $M_Z X$ is the strict transform of $\{Q\} \times \mathbb{P}^1$, which misses the centre (P, ∞) entirely and therefore meets the fibre over ∞ in the single point (Q, ∞); that point lies on the strict transform of $\mathbb{P}^1 \times \{\infty\}$, which was deleted. So the closure lift of $[Q]$ in $M_Z^\circ X$ does not meet $|D_\infty|$ at all, and definition 2.2.1 records the intersection class on the empty intersection:

$$\sigma[Q] \;=\; 0 \;\in\; A_0\big(C_P\mathbb{P}^1\big)$$

consistent with $A_0(\mathbb{A}^1) = 0$ (proposition 1.5.1). Since $A_0(\mathbb{P}^1) = \mathbb{Z}$ by theorem 1.5.2, specialization is not injective. A cycle disjoint from Z escapes to infinity as the family degenerates, and the operation records nothing of it.

The example computed σ on the class of a subvariety by taking the closure of its cylinder and reading the fibre at infinity, and the answer was in both cases a normal cone: the normal cone of Z in $\mathbb{P}^1$ in the first computation, and the empty cone $C_\emptyset\{Q\}$ in the second. That this always happens is the main theorem of the section, and it is what makes σ computable. Proving it requires knowing that the deformation space of a closed subscheme sits inside the deformation space of the ambient one, which is the content of the next two results.

The first is a statement about normal cones alone. If $V \subseteq X$ is a closed subscheme, the normal cone of $V \cap Z$ in V has an intrinsic map to the normal cone of Z in X, obtained by restricting the ideal $\mathcal{I}$ to V.

The second is the corresponding statement one level up, for the whole family rather than its special fibre. It is stated for an arbitrary morphism because the two compatibilities at the end of the section need the proper and the flat cases, while the theorem below needs the case of a closed immersion.

Lemma 4.2.6: Functoriality of the Deformation Space

Let $f\colon X' \to X$ be a morphism of algebraic schemes, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$ its scheme-theoretic preimage. Write $\pi'\colon M^{\circ}_{Z'}X' \to \mathbb{P}^1$ for the projection of the corresponding deformation space. Then there is a unique morphism $\widetilde{f}\colon M_{Z'}X' \to M_Z X$ over $f \times \mathrm{id}_{\mathbb{P}^1}$, and it satisfies

$$\widetilde{f}^{-1}\big(M^{\circ}_Z X\big) \;=\; M^{\circ}_{Z'}X'$$

so that it restricts to a morphism $\widetilde{f}\colon M^{\circ}_{Z'}X' \to M^{\circ}_Z X$. Moreover:

1. $\pi \circ \widetilde{f} = \pi'$; over $\mathbb{A}^1$ the morphism $\widetilde{f}$ restricts to $f \times \mathrm{id}_{\mathbb{A}^1}$, and over ∞ it restricts to the morphism of normal cones $C(f)\colon C_{Z'}X' \to C_Z X$ induced by the identity $\mathcal{I}\mathcal{O}_{X'} = \mathcal{I}'$.
2. If f is proper, then $\widetilde{f}$ and $C(f)$ are proper.
3. If f is flat of relative dimension n, then $\widetilde{f}$ and $C(f)$ are flat of relative dimension n, and the square formed by $\widetilde{f}$, $f \times \mathrm{id}_{\mathbb{P}^1}$ and the two blow-up morphisms is Cartesian.

***Proof*:**
Write $g = f \times \mathrm{id}_{\mathbb{P}^1}$, let $\mathcal{J} \subseteq \mathcal{O}_{X \times \mathbb{P}^1}$ be the ideal sheaf of $Z \times \{\infty\}$, and let $\mathcal{J}'$ be the ideal sheaf of $Z' \times \{\infty\}$ in $X' \times \mathbb{P}^1$.

Step 1: the ideals correspond. Scheme-theoretically $Z \times \{\infty\} = (Z \times \mathbb{P}^1) \cap (X \times \{\infty\})$, so $\mathcal{J}$ is the sum of the ideal sheaves of the two factors, and the same holds upstairs. The inverse image ideal of a sum is the sum of the inverse image ideals; the ideal of $Z \times \mathbb{P}^1$ pulls back to that of $Z' \times \mathbb{P}^1$ because $\mathcal{I}\mathcal{O}_{X'} = \mathcal{I}'$ by definition of the scheme-theoretic preimage, and the ideal of $X \times \{\infty\}$, generated by u, pulls back to the ideal of $X' \times \{\infty\}$, generated by the same u. Hence

$$\mathcal{J}\mathcal{O}_{X' \times \mathbb{P}^1} \;=\; \mathcal{J}'$$

Step 2: existence and uniqueness. On $M_{Z'}X'$ the ideal $\mathcal{J}'\mathcal{O}_{M_{Z'}X'}$ is invertible by [2, Basic Properties of Blow-ups](3), and by Step 1 it is the inverse image ideal of $\mathcal{J}$ under the composite $M_{Z'}X' \to X' \times \mathbb{P}^1 \xrightarrow{g} X \times \mathbb{P}^1$. The universal property, [2, Basic Properties of Blow-ups](4), therefore factors that composite uniquely through the blow-down $p\colon M_Z X \to X \times \mathbb{P}^1$, and the factorization is the asserted $\widetilde{f}$. Uniqueness of the factorization says precisely that $\widetilde{f}$ is the only morphism $M_{Z'}X' \to M_Z X$ lying over g. Since π is the second projection composed with p, and $\widetilde{f}$ lies over g, which is the identity in the second factor, $\pi \circ \widetilde{f} = \pi'$.

Step 3: the two open subschemes correspond. The assertion is local on X and on $\mathbb{P}^1$. Over $\mathbb{A}^1$ the blow-ups are isomorphisms, since the centres lie over ∞ ([2, Basic Properties of Blow-ups](2)), and both deformation spaces contain the whole preimage of $\mathbb{A}^1$; there $\widetilde{f}^{-1}(X \times \mathbb{A}^1) = X' \times \mathbb{A}^1$ because $\pi \circ \widetilde{f} = \pi'$. Over $\mathbb{P}^1 \setminus \{0\} = \operatorname{Spec} \overline{k}[u]$, take an affine open $U = \operatorname{Spec} A$ of X with

$\mathcal{I}|_U$ given by an ideal $I \subseteq A$, and put $J = IA[u] + (u) \subseteq A[u]$, the ideal of $Z \times \{\infty\}$ over $U \times (\mathbb{P}^1 \setminus \{0\})$, so that by [2, Blow-up] the deformation space is there the relative Proj of the Rees algebra $\bigoplus_{n \geq 0} J^n$, in which u is an element of degree one. By lemma 4.1.2 the open subscheme $M_Z^\circ X = M_Z X \setminus \mathrm{Bl}_Z X$ is over $U \times (\mathbb{P}^1 \setminus \{0\})$ exactly the basic open $D_+(u)$ of that relative Proj, with coordinate ring the extended Rees algebra $A[u][I/u]$, its closed complement there being $\mathrm{Bl}_Z X$.
The same description applies upstairs over any affine open $U' = \operatorname{Spec} A'$ of $f^{-1}(U)$, with $I' = IA'$ and $J' = I'A'[u] + (u)$, giving $M_{Z'}^\circ X' = D_+(u)$ there. By Step 1, $J^n A'[u] = J'^n$ for every n, so the graded homomorphism $\bigoplus_n J^n \otimes_{A[u]} A'[u] \to \bigoplus_n J'^n$ is surjective in every degree; it therefore induces a morphism of relative Proj schemes defined on all of $M_{Z'}X'$, and that morphism lies over g; by the uniqueness of Step 2 it is $\widetilde{f}$. A morphism of Proj schemes induced by a graded homomorphism φ pulls the basic open $D_+(x)$ back to $D_+(\varphi(x))$, and $\varphi(u) = u$, so $\widetilde{f}^{-1}(D_+(u)) = D_+(u)$. Hence $\widetilde{f}^{-1}(M_Z^\circ X) = M_{Z'}^\circ X'$.

The three numbered assertions remain.

1. *The restrictions to the two fibres.* Over $\mathbb{A}^1$ both blow-ups are isomorphisms onto $X \times \mathbb{A}^1$ and $X' \times \mathbb{A}^1$, and $\widetilde{f}$ lies over g, so it restricts there to $f \times \mathrm{id}_{\mathbb{A}^1}$. Over ∞, the chart of Step 3 has coordinate ring $A[u][I/u]$, and the fibre over ∞ is cut out by u; on that chart the identification of the fibre with the normal cone given by theorem 4.1.5(2) reads
$$A[u][I/u]/(u) \;\cong\; \bigoplus_{n \geq 0} I^n/I^{n+1}$$
carrying the class of a/u, for $a \in I$, to the class of a in degree one. The morphism $\widetilde{f}$ is induced by $A[u] \to A'[u]$ and sends a/u to $f^\#(a)/u$, so on fibres it induces the homomorphism of graded algebras determined in degree one by $\bar{a} \mapsto \overline{f^\# a}$. That is the homomorphism defining $C(f)$.

2. *Properness.* Suppose f is proper. The blow-up $M_{Z'}X' \to X' \times \mathbb{P}^1$ is projective, hence proper, by [2, Basic Properties of Blow-ups](1), and g is proper as a base change of f, so the composite $M_{Z'}X' \to X \times \mathbb{P}^1$ is proper. Factor $\widetilde{f}$ as
$$M_{Z'}X' \xrightarrow{(\mathrm{id}, \widetilde{f})} M_{Z'}X' \times_{X \times \mathbb{P}^1} M_Z X \xrightarrow{\mathrm{pr}_2} M_Z X$$
The first map is the graph of $\widetilde{f}$ over $X \times \mathbb{P}^1$, a closed immersion because $M_Z X \to X \times \mathbb{P}^1$ is separated, and the second is a base change of the proper morphism $M_{Z'}X' \to X \times \mathbb{P}^1$, hence proper. So $\widetilde{f}$ is proper. Restricting over the open subscheme $M_Z^\circ X$, whose preimage is $M_{Z'}^\circ X'$ by Step 3, preserves properness, and $C(f)$ is the base change of $\widetilde{f}$ along the closed immersion $C_Z X \hookrightarrow M_Z^\circ X$, hence proper.

3. *Flatness.* Suppose f is flat of relative dimension n; then so is g, being a base change of f. Flatness gives $\mathcal{J}^n \mathcal{O}_{X' \times \mathbb{P}^1} = g^* \mathcal{J}^n$ for every n, so the Rees algebra of $\mathcal{J}'$ is the pullback g^* of the Rees algebra of $\mathcal{J}$, and [2, Relative Proj Commutes with Base Change] identifies
$$M_{Z'}X' \;\cong\; M_Z X \times_{X \times \mathbb{P}^1} \left(X' \times \mathbb{P}^1\right)$$
over $X' \times \mathbb{P}^1$. Under this identification the second projection is a morphism over g, hence equals $\widetilde{f}$ by the uniqueness of Step 2, and the square is Cartesian. So $\widetilde{f}$ is a base change of g, therefore flat with the same fibres, that is, flat of relative dimension n. The same holds after restricting to the two open subschemes, which correspond by Step 3, and $C(f)$ is again a base change of $\widetilde{f}$, so it too is flat of relative dimension n, completing the proof.

The computation of σ on a subvariety can now be carried out. The statement is the one the construction was designed to produce: specializing a subvariety returns the normal cone of its intersection with Z, which is the linearized model of how V meets Z.

Theorem 4.2.7: Specialization of a Subvariety

Let $Z \subseteq X$ be a closed subscheme, let $V \subseteq X$ be a k-dimensional subvariety, and let $W = V \cap Z$ be the scheme-theoretic intersection. Then

$$\sigma[V] \;=\; \big[C_W V\big] \;\in\; A_k\big(C_Z X\big)$$

the fundamental cycle of the normal cone $C_W V$, regarded as a closed subscheme of $C_Z X$ through lemma 3.3.12. In particular $C_W V$ is pure of dimension k.

***Proof*:**

Step 1: the closure lift. Let $\widetilde{V}$ be the closure of $V \times \mathbb{A}^1$ in $M_Z^\circ X$, with its reduced structure, a subvariety of dimension $k+1$. Since $V \times \mathbb{A}^1$ is closed in $X \times \mathbb{A}^1$, the open restriction of $[\widetilde{V}]$ is $j^*[\widetilde{V}] = [V \times \mathbb{A}^1] = q^*[V]$ by example 1.3.4, so $[\widetilde{V}]$ is a lift of $[V]$ and $\sigma[V] = D_\infty \cdot [\widetilde{V}]$.

Step 2: the smaller deformation space. Apply lemma 4.2.6 to the closed immersion $\iota\colon V \hookrightarrow X$, whose scheme-theoretic preimage of Z is W by lemma 3.3.12. It supplies a proper morphism

$$\widetilde{\iota}\colon M_W^\circ V \longrightarrow M_Z^\circ X$$

commuting with the projections to $\mathbb{P}^1$ and restricting over $\mathbb{A}^1$ to the closed immersion $V \times \mathbb{A}^1 \hookrightarrow X \times \mathbb{A}^1$. Over ∞ it restricts to the morphism $C(\iota)\colon C_W V \to C_Z X$, which is induced by the graded homomorphism attached to $\mathcal{I}\mathcal{O}_V = \mathcal{I}_W$ and is therefore the closed immersion of lemma 3.3.12.

The scheme $M_W V$ is integral. Indeed $V \times \mathbb{P}^1$ is a variety, and on an affine open $\operatorname{Spec} B \subseteq V \times \mathbb{P}^1$ the ideal $\mathfrak{b}$ of $W \times \{\infty\}$ is either the unit ideal, in which case the corresponding piece of the blow-up is $\operatorname{Spec} B$, or a nonzero proper ideal, in which case the charts of $\operatorname{Proj} \bigoplus_n \mathfrak{b}^n$ have coordinate rings $B[\mathfrak{b}/b]$ for $0 \neq b \in \mathfrak{b}$, subrings of the function field $k(V \times \mathbb{P}^1)$ containing B. Every chart is thus the spectrum of a domain with fraction field $k(V)(t)$, so $M_W V$ is reduced and irreducible, and so is its open subscheme $M_W^\circ V$. The latter has dimension $k+1$, since it contains $V \times \mathbb{A}^1$ as a nonempty, hence dense, open subscheme.

Step 3: the image and the degree. The morphism $\widetilde{\iota}$ is proper, hence closed, and $V \times \mathbb{A}^1$ is dense in $M_W^\circ V$, so the image is the closure of $\widetilde{\iota}(V \times \mathbb{A}^1) = V \times \mathbb{A}^1$, which is $\widetilde{V}$. Over $\mathbb{A}^1$ the morphism is an isomorphism onto $V \times \mathbb{A}^1$, which is dense in both $M_W^\circ V$ and $\widetilde{V}$, so the two schemes have the same function field and $\deg\big(M_W^\circ V/\widetilde{V}\big) = 1$. By definition 1.2.1,

$$\widetilde{\iota}_*\big[M_W^\circ V\big] \;=\; \big[\widetilde{V}\big]$$

Step 4: cutting the smaller family. The projection $\pi'\colon M_W^\circ V \to \mathbb{P}^1$ is flat by theorem 4.1.5(3), applied to the closed subscheme $W \subseteq V$. Each nonempty fibre is cut out locally by one equation, so every component has dimension at least k by Krull's principal ideal theorem, and each fibre is a proper closed subset of the integral $M_W^\circ V$, so every component has dimension at most k. Thus π' is flat of relative dimension k in the sense of definition 1.3.1, and its fibre over ∞, which is $C_W V$, is pure of dimension k.

Since $\pi \circ \widetilde{\iota} = \pi'$, the pullback Cartier divisor is $\widetilde{\iota}^* D_\infty = \pi'^*(\infty)$, cut out by $\pi'^\# u$; it is defined because $M^\circ_W V$ is dominant over $\mathbb{P}^1$, its restriction over $\mathbb{A}^1$ being the nonempty $V \times \mathbb{A}^1$, and so is not contained in $|D_\infty|$. The first clause of definition 2.2.1 therefore gives

$$\widetilde{\iota}^* D_\infty \cdot \big[M^\circ_W V\big] \;=\; \big[\mathrm{div}\big(\pi'^\# u\big)\big] \;=\; \sum_T \mathrm{ord}_T\big(\pi'^\# u\big)\,[T]$$

the sum over the codimension-one subvarieties T of $M^\circ_W V$. By definition 1.1.1 the coefficient $\mathrm{ord}_T(\pi'^\# u)$ is the length of $\mathcal{O}_{M^\circ_W V, T}/(\pi'^\# u)$, which is the local ring at T of the zero scheme of $\pi'^\# u$, that is, of the fibre $C_W V$. The coefficient vanishes unless $T \subseteq C_W V$, and the k-dimensional subvarieties of $C_W V$ are exactly its irreducible components, by the purity just established. Comparing with the fundamental cycle of definition 1.1.2,

$$\widetilde{\iota}^* D_\infty \cdot \big[M^\circ_W V\big] \;=\; \big[C_W V\big]$$

Step 5: descent to $C_Z X$. Combining the previous steps with the projection formula proposition 2.3.3(1) for the proper morphism $\widetilde{\iota}$,

$$\sigma[V] \;=\; D_\infty \cdot \widetilde{\iota}_*\big[M^\circ_W V\big] \;=\; \widetilde{\iota}_*\Big(\widetilde{\iota}^* D_\infty \cdot \big[M^\circ_W V\big]\Big) \;=\; \widetilde{\iota}_*\big[C_W V\big]$$

as classes in $A_k(C_Z X)$. By Step 2 the morphism $\widetilde{\iota}$ restricts over ∞ to the closed immersion of lemma 3.3.12, and a closed immersion carries a fundamental cycle to the fundamental cycle of its image, every degree in definition 1.2.1 being 1. Hence $\widetilde{\iota}_*[C_W V] = [C_W V]$ read inside $C_Z X$, as we sought to show.

The theorem is what turns σ from a construction into a computation. It also explains the two features of example 4.2.5: for $V = \mathbb{P}^1$ the intersection $V \cap Z$ is Z itself and the answer is $[C_P \mathbb{P}^1]$, while for $V = \{Q\}$ with $Q \notin Z$ the intersection is empty, the normal cone of the empty subscheme is empty, and the answer is 0. Specialization does not preserve the geometry of V, only its normal geometry along Z: the class $[C_W V]$ remembers how V meets Z and forgets everything about V away from Z. The multiplicities carry information: the fundamental cycle of definition 1.1.2 weights each component of $C_W V$ by the length of the local ring there, so a component along which the cone is non-reduced contributes with multiplicity greater than one. In the example below the cone is reduced and every multiplicity is one, the two components recording the two tangent directions of a node; the cuspidal cubic of exercise 4.2.1 (3) is the opposite case, a single tangent line doubled. And the purity statement is not an extra hypothesis but a by-product: the normal cone of any closed subscheme in a variety is pure of the dimension of that variety, because it is a fibre of a flat family whose other fibres are the variety itself.

Example 4.2.8: Specializing the Nodal Cubic to its Tangent Cone

Assume $\mathrm{char}\,\overline{k} \neq 2$, let $X = \mathrm{Spec}\,\overline{k}[x,y]/(y^2 - x^2 - x^3)$ be the affine nodal cubic, and let $Z = \{P\}$ be the reduced origin. The curve X is a variety of dimension one, so $A_1(X) = \mathbb{Z}\,[X]$ by example 1.1.6(3). Taking $V = X$ in theorem 4.2.7 gives $W = X \cap Z = Z$ and

$$\sigma[X] \;=\; \big[C_P X\big]$$

and example 3.3.7 computed the normal cone: $C_P X = \mathrm{Spec}\,\overline{k}[x,y]/(y^2 - x^2)$, the union of the two lines $L_+ = \{y = x\}$ and $L_- = \{y = -x\}$. The polynomial $y^2 - x^2 = (y-x)(y+x)$ is squarefree, so the cone is reduced and each component carries multiplicity one:

$$\sigma[X] \;=\; [L_+] + [L_-] \;\in\; A_1(C_P X)$$

which is free of rank two on the two lines, again by example 1.1.6(3). The specialization has replaced an irreducible curve by a reducible cone, one line for each branch of the node, and the two summands are exactly the two tangent directions along which X approaches P. The class of X in $A_1(X) = \mathbb{Z}$ carried no information about the singularity; its specialization, in a group of rank two, carries the branch structure.

What remains is to check that σ is compatible with the two functorialities of chapter 1. Both statements are forced by the shape of the construction: σ was built from the flat pullback q^*, the localization sequence, and the divisor operator, and each of the three commutes with proper pushforward and with flat pullback. The compatibilities are what allow σ, and hence the Gysin map of section 4.3, to be computed by moving to a cover or restricting to an open set.

Proposition 4.2.9: Compatibility with Proper Pushforward

Let $f\colon X' \to X$ be a proper morphism, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$. Write σ' for the specialization homomorphism of $Z' \subseteq X'$ and $C(f)\colon C_{Z'}X' \to C_Z X$ for the induced proper morphism of normal cones. Then for every $\alpha' \in A_k(X')$,

$$\sigma\big(f_*\alpha'\big) \;=\; C(f)_*\big(\sigma'(\alpha')\big) \quad \text{in } A_k\big(C_Z X\big)$$

***Proof*:**

Let $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$ be the proper morphism of lemma 4.2.6, and let β' be the closure lift of α', so that $j'^*\beta' = q'^*\alpha'$ and every component of β' dominates $\mathbb{P}^1$.

Step 1: the pushed-forward lift. Since $\pi \circ \widetilde{f} = \pi'$, the square with horizontal maps $\widetilde{f}$ and $f \times \mathrm{id}_{\mathbb{A}^1}$ and vertical maps the two open immersions j', j is Cartesian: the preimage $\widetilde{f}^{-1}(X \times \mathbb{A}^1)$ is $\pi'^{-1}(\mathbb{A}^1) = X' \times \mathbb{A}^1$, and $\widetilde{f}$ restricts there to $f \times \mathrm{id}_{\mathbb{A}^1}$ by lemma 4.2.6(1). As j is flat of relative dimension 0 and $\widetilde{f}$ is proper, proposition 1.3.6 gives

$$j^*\widetilde{f}_*\beta' \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)_* j'^*\beta' \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)_* q'^*\alpha'$$

The square with horizontal maps $f \times \mathrm{id}_{\mathbb{A}^1}$ and f and vertical maps q' and q is Cartesian as well, since $X' \times \mathbb{A}^1 = X' \times_X (X \times \mathbb{A}^1)$, so a second application of proposition 1.3.6, now with q flat of relative dimension 1 and f proper, gives $(f \times \mathrm{id}_{\mathbb{A}^1})_* q'^*\alpha' = q^* f_*\alpha'$. Hence $\widetilde{f}_*\beta'$ is a lift of $f_*\alpha'$.

Step 2: the projection formula. Every component of β' dominates $\mathbb{P}^1$, and $\pi \circ \widetilde{f} = \pi'$, so no component of the support of β' maps into $|D_\infty| = C_Z X$; the pullback divisor $\widetilde{f}^* D_\infty$ is therefore defined, and it equals $\pi'^*(\infty) = D'_\infty$. By proposition 2.3.3(1),

$$D_\infty \cdot \widetilde{f}_*\beta' \;=\; \widetilde{f}_*\big(D'_\infty \cdot \beta'\big)$$

as classes in $A_k(C_Z X)$. The left side is $\sigma(f_*\alpha')$ by Step 1 and theorem 4.2.4, and $D'_\infty \cdot \beta' = \sigma'(\alpha')$ is a class on $C_{Z'}X'$, on which $\widetilde{f}_*$ acts as $C(f)_*$ by lemma 4.2.6(1). Therefore $\sigma(f_*\alpha') = C(f)_*(\sigma'(\alpha'))$, completing the proof.

Proposition 4.2.10: Compatibility with Flat Pullback

Let $f\colon X' \to X$ be flat of relative dimension n, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$. Then the induced morphism $C(f)\colon C_{Z'}X' \to C_Z X$ is flat of relative dimension n, and for every $\alpha \in A_k(X)$,

$$\sigma'\big(f^*\alpha\big) \;=\; C(f)^*\big(\sigma(\alpha)\big) \quad \text{in } A_{k+n}\big(C_{Z'}X'\big)$$

***Proof*:**

Flatness of $C(f)$ is lemma 4.2.6(3), which also makes $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$ flat of relative dimension n with $\widetilde{f}^{-1}(C_Z X) = C_{Z'}X'$ and $\widetilde{f}^{-1}(X \times \mathbb{A}^1) = X' \times \mathbb{A}^1$.

Step 1: the pulled-back lift. Let β be the closure lift of α. Contravariant functoriality of flat pullback (proposition 1.3.5(3)) applied to the two equal composites $j \circ (f \times \mathrm{id}_{\mathbb{A}^1}) = \widetilde{f} \circ j'$ and $q \circ (f \times \mathrm{id}_{\mathbb{A}^1}) = f \circ q'$ gives

$$j'^*\widetilde{f}^*\beta \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)^* j^*\beta \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)^* q^*\alpha \;=\; q'^* f^*\alpha$$

so $\widetilde{f}^*\beta$ is a lift of $f^*\alpha$. Its components are components of flat preimages of the components of β, each of which dominates $\mathbb{P}^1$, so they dominate $\mathbb{P}^1$ as well and none lies in $|D'_\infty|$.

Step 2: the projection formula. The pullback Cartier divisor is $\widetilde{f}^*D_\infty = \pi'^*(\infty) = D'_\infty$, again because $\pi \circ \widetilde{f} = \pi'$. By proposition 2.3.3(2),

$$\widetilde{f}^*\big(D_\infty \cdot \beta\big) \;=\; D'_\infty \cdot \widetilde{f}^*\beta$$

as classes in $A_{k+n}(\widetilde{f}^{-1}(C_Z X)) = A_{k+n}(C_{Z'}X')$. The right side is $\sigma'(f^*\alpha)$ by Step 1 and theorem 4.2.4. On the left, $D_\infty \cdot \beta = \sigma(\alpha)$ is a class on $C_Z X$, and the restriction of $\widetilde{f}^*$ to classes on $C_Z X$ is the flat pullback along the base change $C(f)$, since flat pullback is computed on preimages and $\widetilde{f}^{-1}(C_Z X) = C_{Z'}X'$. Hence $\sigma'(f^*\alpha) = C(f)^*(\sigma(\alpha))$, as we sought to show.

The two propositions say that specialization is a natural transformation, covariant for proper morphisms and contravariant for flat ones, between the Chow groups of X and those of the normal cone. The hypothesis common to both, $Z' = f^{-1}(Z)$, is not a restriction imposed for convenience: the normal cone of Z' in X' maps to that of Z in X precisely because the ideal of Z pulls back to the ideal of Z', and without that identity there is no morphism $C(f)$ to state the compatibility with.

Exercise 4.2.1

1. Take $Z = X$. Exercise 4.1.1 (1) identifies the deformation space as $M^\circ_X X = X \times \mathbb{P}^1$ with special fibre $C_X X = X$. Deduce from theorem 4.2.7 that $\sigma\colon A_k(X) \to A_k(X)$ is the identity.
2. Let $X = \mathbb{P}^2$ and let $Z = L$ be a line, an effective Cartier divisor on X. Using example 3.3.15, identify $C_L\mathbb{P}^2$ with the total space of the line bundle $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$ on $L \cong \mathbb{P}^1$, and compute $A_k(C_L\mathbb{P}^2)$ for all k by theorem 1.4.6. Then compute $\sigma[\mathbb{P}^2]$, $\sigma[L']$ for a line $L' \neq L$, and $\sigma[Q]$ for a closed point Q, and say in which dimensions σ is an isomorphism.
3. Let $X = \operatorname{Spec} \overline{k}[x, y]/(y^2 - x^3)$ be the affine cuspidal cubic and $Z = \{P\}$ the reduced origin. Taking the cone $C_P X = \operatorname{Spec} \overline{k}[x, y]/(y^2)$ and its fundamental cycle $2\,[L]$ from exercise 4.1.1 (5),

compute $\sigma[X]$ as a cycle. Compare the answer with example 4.2.8 and say what distinguishes a cusp from a node in this computation. Does the answer depend on char$\overline{k}$?

4. Let $\iota\colon Z \hookrightarrow C_Z X$ be the vertex of the normal cone (definition 3.3.1) and $i_Z\colon Z \hookrightarrow X$ the inclusion. Show that $\sigma \circ (i_Z)_* = \iota_*$ as homomorphisms $A_k(Z) \to A_k(C_Z X)$. (Hint: apply theorem 4.2.7 to a subvariety $V \subseteq Z$ and identify $C_V V$.)

5. Show that if $V \subseteq X$ is a subvariety with $V \cap Z = \emptyset$, then $\sigma[V] = 0$. Deduce that for $X = \mathbb{P}^n$ with $n \geq 1$ and $Z = \{P\}$ a closed point, σ vanishes on all of $A_0(\mathbb{P}^n)$, and check that this is consistent with a direct computation of $A_0(C_P\mathbb{P}^n)$.

4.3 The Gysin Map for a Regular Embedding

The specialization homomorphism of section 4.2 accepts an arbitrary closed subscheme, and its target is the Chow group of a cone that is usually not a bundle. This section takes the case in which it is. For a regular embedding the normal cone is the normal bundle (proposition 3.3.11), so specialization lands in $A_k(N)$, and flat pullback along $N \to Z$ is an isomorphism (theorem 1.4.6) that can be inverted. The composite lowers dimension by the codimension and lands on Z.

Two things have to be established beyond that one-line recipe. The first is that the recipe computes what a reader expects when the intersection is proper: the cycle $[V \cap Z]$, with the multiplicities the scheme structure supplies. The second is that the construction survives base change. For any morphism $X' \to X$ the normal cone of the preimage of Z still sits inside the pulled-back normal bundle, and the same recipe produces a homomorphism landing on that preimage rather than on Z. It is this refined form, not the plain one, that chapter 5 applies to the diagonal.

Both rest on the following comparison between the normal cone of a pullback and the pullback of the normal bundle. Regularity of i is what makes the comparison available: it is exactly the hypothesis under which the conormal sheaf is locally free, so that the symmetric algebra it generates is a bundle into which the associated graded algebra downstairs maps.

The lemma says that regularity of i propagates one step: it is not inherited by $Z' \hookrightarrow X'$, which can be badly irregular, but the failure is confined inside a bundle of the correct rank d. The cone $C_{Z'}X'$ may be singular, may be reducible, and may have components of several dimensions, and none of that prevents it from being a closed subscheme of a rank-d bundle on Z'. That containment is what lets the same recipe be applied over X' as over X, and the discrepancy between $C_{Z'}X'$ and all of N' is what the excess intersection formula of chapter 5 measures.

With $g = \mathrm{id}_X$ the lemma reduces to the case already in hand, and the construction can be stated.

Definition 4.3.1: The Gysin Map of a Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 0$ with normal bundle $\pi\colon N \to Z$. For $d = 0$ the ideal sheaf is zero, i is an open and closed immersion, N is the zero bundle over Z, and the formula below reads as flat pullback along i; the construction is stated for $d \geq 1$ below and the degenerate case is recorded here so that statements about codimension-zero embeddings, which occur in chapter 5, have a definition to refer to. By proposition 3.3.11 the normal cone $C_Z X$ is N, so the specialization homomorphism of definition 4.2.3 is a homomorphism $\sigma\colon A_k(X) \to A_k(N)$ (theorem 4.2.4). The *Gysin map* of i is

$$i^! \;=\; (\pi^*)^{-1} \circ \sigma \;:\; A_k(X) \longrightarrow A_{k-d}(Z)$$

where $\pi^*\colon A_{k-d}(Z) \to A_k(N)$ is flat pullback along π.

The inverse in the definition exists because N is a vector bundle of rank d, hence an affine bundle of relative dimension d (definition 1.4.5), so π^* is an isomorphism by theorem 1.4.6. The codimension-zero case is excluded because it is empty of content: an ideal generated locally by the empty regular sequence is the zero ideal, so a regular embedding of codimension zero is $Z = X$.

The smallest case can be run through at once. Take $X = \mathbb{A}^1$ and $Z = \{0\}$, whose ideal is generated by the single nonzerodivisor t, so that i is a regular embedding of codimension one and N is a one-dimensional vector space over the point Z, with total space $\mathbb{A}^1$. By theorem 4.2.7, $\sigma[\mathbb{A}^1] = [C_Z\mathbb{A}^1] = [N]$, and $\pi^*[Z] = [N]$ by proposition 1.3.5(2), so $i^![\mathbb{A}^1] = [Z]$ in $A_0(Z) = \mathbb{Z}$: the origin with multiplicity one.

Proposition 4.3.2: The Gysin Map is a Homomorphism of Degree $-d$

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$. Then:

1. $i^!\colon A_k(X) \to A_{k-d}(Z)$ is a well-defined homomorphism of abelian groups for every k, and it is the unique map satisfying $\pi^* \circ i^! = \sigma$;
2. $i^! = 0$ on $A_k(X)$ for $k < d$.

***Proof*:**

1. A vector bundle of rank d is an affine bundle of relative dimension d in the sense of definition 1.4.5, a local trivialization of the bundle being in particular a local product decomposition with $\mathbb{A}^d$. Theorem 1.4.6 therefore makes $\pi^*\colon A_{k-d}(Z) \to A_k(N)$ an isomorphism for every k. The specialization homomorphism $\sigma\colon A_k(X) \to A_k(N)$ is a homomorphism defined on rational equivalence classes by theorem 4.2.4, and $C_Z X = N$ by proposition 3.3.11, so the composite $(\pi^*)^{-1} \circ \sigma$ is a homomorphism $A_k(X) \to A_{k-d}(Z)$. If $\tau\colon A_k(X) \to A_{k-d}(Z)$ satisfies $\pi^* \circ \tau = \sigma$, then $\pi^*\tau = \pi^* i^!$, and injectivity of π^* gives $\tau = i^!$.
2. For $k < d$ the target $A_{k-d}(Z)$ is a Chow group in negative dimension. No scheme has a subvariety of negative dimension, so $Z_{k-d}(Z) = 0$ and hence $A_{k-d}(Z) = 0$, forcing $i^! = 0$ there, as we sought to show.

The characterization $\pi^* \circ i^! = \sigma$ is how the map is used in practice: rather than inverting π^*, one produces a candidate class on Z and checks that its pullback to N is the specialization. The degree $-d$ matches the naive expectation that intersecting a k-dimensional cycle with a codimension-d subscheme leaves something of dimension $k-d$. What the construction adds to that expectation is that the answer is defined for every class, with no hypothesis on how Z and a representing cycle meet, and that it is a rational-equivalence invariant.

The first thing to verify is that the machinery reproduces the naive answer when the naive answer makes sense. The condition for that is exactly regularity of the induced embedding.

Proposition 4.3.3: The Gysin Map of a Regular Intersection

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, let $V \subseteq X$ be a k-dimensional subvariety, and let $W = V \cap Z$ be the scheme-theoretic intersection, that is the preimage of Z under the inclusion $V \hookrightarrow X$. If $W \hookrightarrow V$ is a regular embedding of codimension d, then W is pure of dimension $k-d$ and

$$i^![V] \;=\; \iota_*[W] \;\in\; A_{k-d}(Z)$$

where $[W]$ is the fundamental cycle of definition 1.1.2 and $\iota\colon W \hookrightarrow Z$ is the inclusion.

***Proof*:**
Apply lemma 3.3.13 to the closed immersion $g\colon V \hookrightarrow X$, whose preimage of Z is W: it produces a closed immersion $j\colon C_W V \hookrightarrow N|_W$ of cones over W, and by part (1) of that lemma the hypothesis on $W \hookrightarrow V$ makes j an isomorphism. Thus $C_W V = N|_W$, the restriction of N to W, viewed as a closed subscheme of $C_Z X = N$.

By theorem 4.2.7, $\sigma[V] = [C_W V]$ in $A_k(N)$, the fundamental cycle of the normal cone of V along W read inside $C_Z X$. The previous paragraph identifies that cycle as $[N|_W]$. Now $\pi_W\colon N|_W \to W$ is flat of relative dimension d, being a vector bundle of rank d, so proposition 1.3.5(2) gives $\pi_W^*[W] = [N|_W]$; and flat pullback of a cycle is computed componentwise from scheme-theoretic preimages (definition 1.3.3), so the cycle $\pi_W^*[W]$ on $N|_W$ is the cycle $\pi^*\iota_*[W]$ on N. Hence

$$\pi^*\big(\iota_*[W]\big) = [N|_W] = \sigma[V]$$

and proposition 4.3.2(1) gives $i^![V] = \iota_*[W]$.

Finally, $\sigma[V]$ lies in $A_k(N)$, so $[N|_W]$ is a k-cycle; since π_W raises dimension by exactly d on each component, every irreducible component of W has dimension $k-d$, completing the proof.

The hypothesis is the scheme-theoretic form of "V meets Z properly and without excess tangency". It is not a condition on dimensions alone: a subvariety can meet Z in the expected dimension $k-d$ and still fail it, if the intersection is not cut out by d equations forming a regular sequence, and then $i^![V]$ carries multiplicities the set $V \cap Z$ does not display. When the hypothesis does hold, the Gysin map returns the fundamental cycle of the intersection scheme, so the multiplicities are the lengths that scheme records and nothing more.

Example 4.3.4: Linear Subspaces of Projective Space

Fix homogeneous coordinates $x_0, \dots, x_n$ on $\mathbb{P}^n$ and let $Z = \{x_0 = \cdots = x_{d-1} = 0\} \cong \mathbb{P}^{n-d}$ with $1 \le d \le n$. On each standard affine chart $\{x_r \neq 0\}$ meeting Z, the ideal of Z is generated by the d ratios $x_0/x_r, \dots, x_{d-1}/x_r$, which form a regular sequence in the polynomial ring of that chart: each is a nonzerodivisor modulo its predecessors, the quotients being polynomial rings in the remaining coordinates. So $i\colon Z \hookrightarrow \mathbb{P}^n$ is a regular embedding of codimension d and $i^!\colon A_m(\mathbb{P}^n) \to A_{m-d}(Z)$ is defined.

By theorem 1.5.2, $A_m(\mathbb{P}^n) = \mathbb{Z}\, h_m$ with h_m the class of a linear subspace $\mathbb{P}^m \subseteq \mathbb{P}^n$, and likewise $A_{m-d}(Z) = \mathbb{Z}\, h^Z_{m-d}$ for $Z \cong \mathbb{P}^{n-d}$. Take $m \ge d$ and represent h_m by $L = \{x_{m+1} = \cdots = x_n = 0\}$, the span of the coordinate points $e_0, \dots, e_m$. Then

$$L \cap Z \;=\; \{x_0 = \cdots = x_{d-1} = 0\} \cap L$$

is the span of $e_d, \dots, e_m$, a linear $\mathbb{P}^{m-d}$, and its ideal in $L \cong \mathbb{P}^m$ is generated by the restrictions of $x_0, \dots, x_{d-1}$, again a regular sequence of length d on every chart of L meeting it. So $L \cap Z \hookrightarrow L$ is a regular embedding of codimension d, and proposition 4.3.3 gives

$$i^!(h_m) \;=\; \left[\mathbb{P}^{m-d}\right] \;=\; h^Z_{m-d}$$

the intersection scheme being reduced and irreducible, so that its fundamental cycle carries multiplicity one. For $m < d$ the target $A_{m-d}(Z)$ vanishes and $i^!(h_m) = 0$ by proposition 4.3.2(2). The Gysin map of a linear subspace is therefore the shift $h_m \mapsto h^Z_{m-d}$, carrying generator to generator in every degree where the target is nonzero.

The computation is the expected one, and its value is that no general position was invoked to get it: the representative L was chosen for convenience, and proposition 4.3.2 guarantees that any other representative of h_m, including one contained in Z or tangent to it, returns the same class. That is the difference between this construction and a count of intersection points.

The codimension-one case can be compared with the operator built by hand in chapter 2, and the two agree wherever the first clause of definition 2.2.1 applies.

Proposition 4.3.5: Agreement with Intersection with a Divisor

Let D be an effective Cartier divisor on an algebraic scheme X with $D \neq X$, and let $i\colon D \hookrightarrow X$ be the inclusion, a regular embedding of codimension one with normal bundle $N = \mathcal{O}_X(D)|_D$ (example 3.3.15). Let $V \subseteq X$ be a k-dimensional subvariety with $V \not\subseteq |D|$, and let $W = V \cap D$ be the scheme-theoretic intersection, a closed subscheme with support $V \cap |D|$. Then W is pure of dimension $k-1$, the fundamental cycle of W and the intersection class of definition 2.2.1 agree,

$$[W] \;=\; D \cdot [V] \;\in\; A_{k-1}\big(V \cap |D|\big)$$

already as cycles and not as classes, and the Gysin class of V is their pushforward,

$$i^![V] \;=\; \iota_*\big(D \cdot [V]\big) \;\in\; A_{k-1}(D)$$

where $\iota\colon W \hookrightarrow D$ is the inclusion.

***Proof*:**
Recall that $W = V \cap D$ is the closed subscheme of V cut out by the ideal $\mathcal{I} \cdot \mathcal{O}_V$, where $\mathcal{I}$ is the ideal sheaf of D.

Step 1: the hypothesis of proposition 4.3.3 holds. Since D is an effective Cartier divisor, $\mathcal{I}$ is locally generated by a single nonzerodivisor $f \in \mathcal{O}_X(U)$. As V is integral and $V \not\subseteq |D|$, the restriction $f|_V$ is a nonzero element of the domain $\mathcal{O}_V(U \cap V)$, hence a nonzerodivisor, and it generates $\mathcal{I} \cdot \mathcal{O}_V$ there. So $W \hookrightarrow V$ is a regular embedding of codimension one, and proposition 4.3.3 gives $i^![V] = \iota_*[W]$ in $A_{k-1}(D)$, with W pure of dimension $k-1$.

Step 2: the fundamental cycle of W is the divisor of the section. The intersection class of definition 2.2.1 is computed by its first clause, V not being contained in $|D|$: the local equations of D restrict to a rational section s_V of $\mathcal{O}_X(D)|_V$, given on $U \cap V$ by $f|_V$, and $D \cdot [V] = [\operatorname{div}(s_V)] = \sum_T \operatorname{ord}_T(s_V)\,[T]$, the sum over codimension-one subvarieties $T \subseteq V$. Fix such a T and let $A = \mathcal{O}_{V,T}$, a one-dimensional local domain. The local ring of W at the generic point of T is $A/(f|_V)$, so the coefficient of $[T]$ in the fundamental cycle $[W]$ of definition 1.1.2 is $\operatorname{length}_A\big(A/(f|_V)\big)$, which is $\operatorname{ord}_T(s_V)$ by definition 1.1.1. The coefficient is zero exactly when $f|_V$ is a unit in A, that is when T is not a component of W, so the two cycles have the same components as well as the same multiplicities. Hence $[W] = [\operatorname{div}(s_V)]$ as cycles on $V \cap D$, so $[W] = D \cdot [V]$ in $A_{k-1}(V \cap |D|)$; pushing that identity forward along ι and combining it with Step 1 gives $i^![V] = \iota_*[W] = \iota_*\big(D \cdot [V]\big)$ in $A_{k-1}(D)$, completing the proof.

The proposition is the consistency check in codimension one: the operator $D \cdot -$, built in chapter 2 out of local equations and orders of vanishing, is the specialization construction in disguise. The restriction $V \not\subseteq |D|$ is exactly the first clause of definition 2.2.1; the second clause, where V lies inside the support and $D \cdot [V]$ is computed from the restricted line bundle, corresponds on the Gysin side to a cycle supported on the zero section of N, and evaluating it needs the self-intersection formula of chapter 5.

Everything so far records the answer on Z. The construction gives more: it records it on the part of Z that the cycle meets, and it does so for a cycle living on any scheme over X, not only on X itself. Lemma 3.3.13 is what makes this possible, since it supplies a bundle of rank d on the preimage of Z regardless of how singular that preimage is.

Definition 4.3.6: The Refined Gysin Homomorphism

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N, and let $g\colon X' \to X$ be a morphism of algebraic schemes. Form the fibre square

$$\begin{tikzcd} Z' \arrow[r, "i'"] \arrow[d, "h"'] & X' \arrow[d, "g"] \\ Z \arrow[r, "i"] & X \end{tikzcd}$$

with $Z' = g^{-1}(Z)$ and i' the induced closed immersion, and write $\pi'\colon N' = h^*N \to Z'$. Let $\sigma'\colon A_k(X') \to A_k(C_{Z'}X')$ be the specialization homomorphism of definition 4.2.3 for the pair (Z', X'), well defined by theorem 4.2.4, and let $j\colon C_{Z'}X' \hookrightarrow N'$ be the closed immersion of lemma 3.3.13. The *refined Gysin homomorphism* of i along g is

$$i^! \;=\; (\pi'^*)^{-1} \circ j_* \circ \sigma' \;:\; A_k(X') \longrightarrow A_{k-d}(Z')$$

where j_* is proper pushforward along the closed immersion j and $\pi'^*\colon A_{k-d}(Z') \to A_k(N')$ is the isomorphism of theorem 1.4.6.

The notation deliberately reuses $i^!$: the map depends on g, but the source and target groups record which g is meant, and no ambiguity arises in practice. Each of the three maps is defined for the reasons already in place. Specialization for (Z', X') needs no hypothesis on Z', which is why the construction tolerates a preimage that is singular or of the wrong dimension; j_* is defined because a closed immersion is proper, and proper pushforward preserves the dimension of a cycle (definition 1.2.1); and π'^* is invertible because N' is a bundle of rank d on Z', by definition 1.4.5 and theorem 1.4.6 as before. The only step that used regularity of i is lemma 3.3.13, and it used it to guarantee that the rank is d on the nose.

Proposition 4.3.7: The Refined Map Extends the Gysin Map

In the situation of definition 4.3.6 with $X' = X$ and $g = \mathrm{id}_X$, the refined Gysin homomorphism is the Gysin map $i^!$ of definition 4.3.1.

***Proof*:**
With $g = \mathrm{id}_X$ one has $Z' = Z$, $h = \mathrm{id}_Z$ and $N' = N$. By lemma 3.3.13(2) the closed immersion j is the identification $C_Z X = N$ of proposition 3.3.11, so j_* is the identity on $A_k(N)$ and $\sigma' = \sigma$. The composite $(\pi^*)^{-1} \circ \mathrm{id} \circ \sigma$ is definition 4.3.1, as claimed.

Example 4.3.8: Two Lines in the Plane
Let $X = \mathbb{P}^2$ and $Z = L = \{x_0 = 0\}$, an effective Cartier divisor and hence a regular embedding of codimension one with $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$ (example 3.3.15).
A transverse line. Take $g\colon X' = L' \hookrightarrow \mathbb{P}^2$ with $L' = \{x_1 = 0\}$. Then $Z' = g^{-1}(L)$ is cut out on L' by x_0, so $Z' = \{p\}$ with $p = [0:0:1]$, a reduced point, and $Z' \hookrightarrow L'$ is a regular embedding of codimension one. By lemma 3.3.13(1) the immersion j is an isomorphism, so $C_{Z'}X' = N' = N|_p$, a one-dimensional vector space over p. Specialization gives $\sigma'[L'] = [C_{Z'}L'] = [N']$ by theorem 4.2.7,

and $\pi'^*[p] = [N']$ by proposition 1.3.5(2), so

$$i^![L'] \;=\; [p] \;\in\; A_0(L \cap L') = \mathbb{Z}$$

The class is recorded on the single point $L \cap L'$, not on L, and since $\{p\}$ is the scheme-theoretic intersection $L' \cap L$, proposition 4.3.5 identifies it there with the intersection class $L \cdot [L'] \in A_0(L' \cap |L|)$. Pushing it forward along $\{p\} \hookrightarrow L$ returns the class of a point on L, which by the same proposition is the plain Gysin class of L' in $A_0(L)$.

The same line. Take instead $g = i$, so $X' = L$ and $Z' = L$. The normal cone of L in itself is the zero cone of example 3.3.2(2), so $C_{Z'}X' = L$ and j is the vertex of $N' = N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$, that is the zero section $\zeta \colon L \to N$. Hence

$$i^![L] \;=\; (\pi^*)^{-1}\big(\zeta_*[L]\big) \;\in\; A_0(L)$$

Nothing in this section evaluates the right-hand side. What it must equal is known from chapter 2: example 2.2.3 computes $L \cdot [L] = [q]$, the class of one point, so the class of the zero section in $A_1(N)$ has to be $\pi^*[q]$. Identifying it is the self-intersection formula, proved in chapter 5.

The two halves of the example are the two regimes the refined map is built to handle uniformly. In the first, the preimage Z' has the expected dimension and the answer is a cycle on it. In the second, Z' is far too large, the cone degenerates onto the zero section, and the answer is carried entirely by the normal bundle. No hypothesis distinguishes the cases in definition 4.3.6; the difference shows up only in how much of N' the cone $C_{Z'}X'$ fills.

4.3.9 Foreshadowing: What the Gysin Map Is Not Yet The Gysin map is a homomorphism of Chow groups, not a product. Three things it is expected to satisfy are not proved here and are the business of chapter 5: that $i^!$ applied to the diagonal $X \hookrightarrow X \times X$ of a smooth variety turns $A_*(X)$ into a commutative ring; that $i^!$ of a class supported on Z is computed by capping with the top Chern class of N, which is the self-intersection formula and the missing step in example 4.3.8; and that two refined Gysin maps for two regular embeddings commute, which is what makes the resulting product associative. Everything in the present section is input to those proofs rather than a consequence of them.

The third item is settled before this chapter ends, in section 4.4; the first two belong to chapter 5.

The properties chapter 5 consumes are the two functorialities. They say that the refined Gysin map is a morphism of the theory over X: it commutes with pushing a cycle forward along a proper X-morphism and with pulling it back along a flat one, in each case through the induced morphism of preimages.

Theorem 4.3.10: Compatibility of the Refined Gysin Homomorphism

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$. Let $X' \to X$ and $X'' \to X$ be morphisms of algebraic schemes and $p\colon X'' \to X'$ a morphism over X; write $Z' \subseteq X'$ and $Z'' \subseteq X''$ for the preimages of Z and $q\colon Z'' \to Z'$ for the morphism induced by p.

1. *Proper pushforward.* If p is proper then so is q, and for every $\alpha \in A_k(X'')$,

$$i^!\big(p_*\alpha\big) \;=\; q_*\big(i^!\alpha\big) \quad \text{in } A_{k-d}(Z')$$

2. *Flat pullback.* If p is flat of relative dimension n then so is q, and for every $\beta \in A_k(X')$,

$$i^!\big(p^*\beta\big) \;=\; q^*\big(i^!\beta\big) \quad \text{in } A_{k+n-d}(Z'')$$

***Proof*:**
Write $C' = C_{Z'}X'$ and $C'' = C_{Z''}X''$, let $h'\colon Z' \to Z$ and $h''\colon Z'' \to Z$ be the structure morphisms, and set $N' = h'^*N$ and $N'' = h''^*N$ with projections π' and π''. Let $j'\colon C' \hookrightarrow N'$ and $j''\colon C'' \hookrightarrow N''$ be the closed immersions of lemma 3.3.13.

Since $X'' \to X$ factors through p, the preimage of Z in X'' is $Z'' = p^{-1}(Z')$, so q is the base change of p along $Z' \hookrightarrow X'$; properness and flatness of relative dimension n are both stable under base change, which proves the first assertion of each part. From $h'' = h' \circ q$ one gets $N'' = q^*N'$, so

$$\begin{tikzcd} N'' \arrow[r, "\bar{q}"] \arrow[d, "\pi''"'] & N' \arrow[d, "\pi'"] \\ Z'' \arrow[r, "q"] & Z' \end{tikzcd}$$

is a fibre square, called the *bundle square* below, with $\bar{q}$ the projection. The ideal sheaves satisfy $\mathcal{I}'\mathcal{O}_{X''} = \mathcal{I}''$ because $Z'' = p^{-1}(Z')$, so p induces a morphism of graded algebras and hence a morphism of cones $\bar{p}\colon C'' \to C'$ over q; and the *cone square*

$$\begin{tikzcd} C'' \arrow[r, "j''"] \arrow[d, "\bar{p}"'] & N'' \arrow[d, "\bar{q}"] \\ C' \arrow[r, "j'"] & N' \end{tikzcd}$$

commutes, since j' and j'' are induced by the canonical surjections of lemma 3.3.13 and those surjections are compatible with pullback along p, being built from the single conormal surjection of $\mathcal{I}$ on X. Concretely, both composites in the corresponding square of graded algebras are maps out of $\mathrm{Sym}^\bullet\big(h'^*(\mathcal{I}/\mathcal{I}^2)\big)$, hence are determined by their degree-one parts, and in degree one both send the class of a local section a of $\mathcal{I}$ to the class of its pullback in $\mathcal{I}''/\mathcal{I}''^2$, because $\mathcal{I}\mathcal{O}_{X''} = \mathcal{I}''$.

1. Let p be proper. Then $\bar{q}$ is proper, being the base change of q along π', and $\bar{p}$ is proper as well: it is the composite $C'' \hookrightarrow N'' \to N'$ of a closed immersion and a proper morphism, corestricted to the closed subscheme $C' \subseteq N'$. The compatibility of specialization with proper pushforward (proposition 4.2.9) gives

$$\sigma'\big(p_*\alpha\big) = \bar{p}_*\big(\sigma''\alpha\big)$$

Pushing forward along j' and using the cone square together with functoriality of proper pushforward (definition 1.2.1),

$$j'_*\sigma'\big(p_*\alpha\big) = j'_*\bar{p}_*\big(\sigma''\alpha\big) = \bar{q}_*j''_*\big(\sigma''\alpha\big)$$

It remains to move $\bar{q}_*$ past the inverse of π'^*. The bundle square has q proper and π' flat, so proposition 1.3.6 gives $\pi'^*q_* = \bar{q}_*\pi''^*$. Applying this to $\gamma = (\pi''^*)^{-1}j''_*\sigma''\alpha = i^!\alpha$,

$$\pi'^*\big(q_*(i^!\alpha)\big) = \bar{q}_*\big(\pi''^*\gamma\big) = \bar{q}_*j''_*\big(\sigma''\alpha\big) = j'_*\sigma'\big(p_*\alpha\big) = \pi'^*\big(i^!(p_*\alpha)\big)$$

Since π'^* is injective, $q_*(i^!\alpha) = i^!(p_*\alpha)$.

2. Let p be flat of relative dimension n. Flatness gives $\mathcal{I}'^m\mathcal{O}_{X''} = p^*\mathcal{I}'^m$ for every m and makes the formation of the quotients $\mathcal{I}'^m/\mathcal{I}'^{m+1}$ commute with pullback, so $C'' = C'\times_{Z'}Z''$ and the induced $\bar{p}\colon C'' \to C'$ is flat of relative dimension n. Consequently the cone square is a fibre square as well, both vertical maps being base change along $Z'' \to Z'$. The compatibility of specialization with flat pullback (proposition 4.2.10) gives

$$\sigma''\big(p^*\beta\big) = \bar{p}^*\big(\sigma'\beta\big)$$

Applying j''_* and using proposition 1.3.6 for the cone square, whose horizontal maps j', j'' are proper and whose vertical maps $\bar{p}, \bar{q}$ are flat,

$$j''_*\sigma''\big(p^*\beta\big) = j''_*\bar{p}^*\big(\sigma'\beta\big) = \bar{q}^*j'_*\big(\sigma'\beta\big)$$

Finally $\pi'\circ\bar{q} = q\circ\pi''$ is a composite of flat morphisms in two ways, so proposition 1.3.5(3) gives $\bar{q}^*\pi'^* = \pi''^*q^*$. Applying this to $\delta = (\pi'^*)^{-1}j'_*\sigma'\beta = i^!\beta$,

$$\pi''^*\big(q^*(i^!\beta)\big) = \bar{q}^*\big(\pi'^*\delta\big) = \bar{q}^*j'_*\big(\sigma'\beta\big) = j''_*\sigma''\big(p^*\beta\big) = \pi''^*\big(i^!(p^*\beta)\big)$$

and injectivity of π''^* gives $q^*(i^!\beta) = i^!(p^*\beta)$, completing the proof.

Part (1) is what makes the refined map a refinement rather than a separate construction: a class computed on a small piece of X and then pushed forward agrees with the class computed on X directly. Part (2) says that the map commutes with restriction to an open subscheme, that being flat pullback of relative dimension zero: $(i^!\alpha)|_{Z\cap U} = i^!(\alpha|_U)$. It does *not* say that $i^!\alpha$ can be reassembled from its restrictions to an open cover, and it could not: Chow groups are not a Zariski sheaf, and $A_0(\mathbb{P}^1) = \mathbb{Z}$ restricts to zero on each of the two standard charts (theorem 1.5.2, proposition 1.5.1). What the compatibility does give is the useful one-way statement of exercise 4.3.1 (4): a class vanishing on an open set containing Z has vanishing Gysin image. Both are also the form in which chapter 5 uses the map: the projection formula and the compatibility of the intersection product with proper pushforward are read off them.

The first corollary is the statement that the plain Gysin map of a subvariety is always the pushforward of a class living on the intersection.

Corollary 4.3.11: The Gysin Class of a Subvariety Lives on the Intersection

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq X$ be a k-dimensional subvariety, with inclusion $\iota_V\colon V \hookrightarrow X$ and $W = V \cap Z = \iota_V^{-1}(Z)$. Then

$$i^![V] \;=\; \iota_*\big(i_V^![V]\big)$$

where $i_V^!\colon A_k(V) \to A_{k-d}(W)$ is the refined Gysin homomorphism along ι_V and $\iota\colon W \hookrightarrow Z$ is the inclusion.

***Proof*:**
Apply theorem 4.3.10(1) with $X' = X$, $X'' = V$ and $p = \iota_V$, which is proper as a closed immersion. The preimage of Z in V is W and the induced morphism q is ι. By definition 1.2.1 the class $[V] \in A_k(V)$ has $p_*[V] = [V] \in A_k(X)$, the extension $k(V) \subseteq k(V)$ having degree one. On the left-hand side the refined Gysin homomorphism along id_X is the Gysin map of definition 4.3.1 by proposition 4.3.7, so the identity reads $i^![V] = \iota_*(i_V^![V])$, as we sought to show.

The class $i^![V] \in A_{k-d}(Z)$ discards information: it records a cycle on all of Z, whereas the geometry produced it on $V \cap Z$, which may be much smaller. The refined class $i_V^![V] \in A_{k-d}(V \cap Z)$ retains that localization, and it is the object every localized intersection number is a degree of. Proposition 4.3.3 is the special case in which the refined class is the fundamental cycle $[V \cap Z]$ itself, and example 4.3.8 exhibits both a case where it is and a case where it is not.

Exercise 4.3.1

1. Let $Z = \{x_2 = x_3 = 0\} \subseteq \mathbb{P}^3$, a line. Show that $Z \hookrightarrow \mathbb{P}^3$ is a regular embedding of codimension two, and compute $i^!\colon A_m(\mathbb{P}^3) \to A_{m-2}(Z)$ on the generator of each group, for $m = 0, 1, 2, 3$.

2. Let X be a variety of dimension n and let $D \subseteq X$ be an effective Cartier divisor with $D \neq X$, with inclusion i. Show that $i^![X] = [D]$, the fundamental cycle of the scheme D. Then take $X = \mathbb{P}^2$ and D a smooth plane cubic, and compute the image of $i^![\mathbb{P}^2]$ in $A_1(\mathbb{P}^2)$.

3. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq X$ be a k-dimensional subvariety with $V \cap Z = \emptyset$. Show that $i^![V] = 0$. Then let $Z = \{0\}$ be the origin in $X = \mathbb{A}^n$ with $n \geq 1$, verify that this is a regular embedding of codimension n, and determine $i^!$ on every group $A_k(\mathbb{A}^n)$.

4. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $u\colon U \hookrightarrow X$ be an open immersion. Show that the preimage of Z in U is $Z \cap U$ and that $i^!\big(u^*\alpha\big) = \big(i^!\alpha\big)\big|_{Z\cap U}$ for every $\alpha \in A_k(X)$. Deduce that if $Z \subseteq U$ and α restricts to zero on U, then $i^!\alpha = 0$.

5. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq Z$ be a k-dimensional subvariety of Z. Show that $\sigma[V] = \zeta_*[V]$, where $\zeta\colon Z \to N$ is the zero section, and hence that $i^![V] = (\pi^*)^{-1}\zeta_*[V]$. Now take $X = \mathbb{A}^2$, $Z = \{y = 0\}$ and $V = Z$, and evaluate both $i^![Z]$ and $D \cdot [Z]$ of definition 2.2.1, checking that they agree.

4.4 Commutativity, Composition, and Sections

The Gysin map is built; the algebra it obeys is not yet available. Three identities are used downstream, and all three are properties of the construction of section 4.3 rather than of anything later, so they belong here.

They are not independent, and the dependency runs in a direction worth naming before starting. Commutativity of two Gysin maps looks the hardest and is in fact a consequence of the composition rule: two regular embeddings determine a single regular embedding of the product, that embedding factors through the two intermediate stages in two ways, and the composition rule applied to both factorizations equates the two orders. The work is therefore concentrated in composition, and the section is arranged accordingly. Composition in turn needs a commutativity statement of its own, but only the case of a *divisor*, and that case the construction hands over directly: specialization is intersection with a Cartier divisor, so two of them commute because divisors do.

Lemma 4.4.1: The Product of Two Regular Embeddings

Let $i\colon Z \hookrightarrow P$ and $i'\colon Z' \hookrightarrow P'$ be regular embeddings of codimensions d and d'. Then

$$i \times i' \;:\; Z \times Z' \;\hookrightarrow\; P \times P'$$

is a regular embedding of codimension $d + d'$, and it factors in two ways as a composite of regular embeddings,

$$Z \times Z' \xhookrightarrow{\mathrm{id}_Z \times i'} Z \times P' \xhookrightarrow{i \times \mathrm{id}_{P'}} P \times P', \qquad Z \times Z' \xhookrightarrow{i \times \mathrm{id}_{Z'}} P \times Z' \xhookrightarrow{\mathrm{id}_P \times i'} P \times P'$$

the four factors being regular of codimensions d', d, d and d' respectively.

***Proof*:**
Step 1: each factor is regular. The morphism $i \times \mathrm{id}_{P'}$ is the base change of i along the projection $\mathrm{pr}_1\colon P \times P' \to P$, which is flat, and the scheme-theoretic preimage of Z under that projection is $Z \times P'$. A regular embedding pulls back along a flat morphism to a regular embedding of the same codimension ([2, Regular Embeddings Pull Back Along Flat Morphisms]), so $i \times \mathrm{id}_{P'}$ is regular of codimension d. The same argument along pr_2 makes $\mathrm{id}_P \times i'$ and $\mathrm{id}_Z \times i'$ regular of codimension d', the latter as the base change of i' along $Z \times P' \to P'$; and $i \times \mathrm{id}_{Z'}$ is regular of codimension d, being the base change of i along the flat projection $P \times Z' \to P$. That accounts for all four factors named in the statement.

Step 2: the composite is regular. Work on affine opens $\operatorname{Spec} A \subseteq P$ and $\operatorname{Spec} A' \subseteq P'$ over which the ideals of Z and Z' are generated by regular sequences $f_1, \ldots, f_d$ and $g_1, \ldots, g_{d'}$. On $\operatorname{Spec}(A \otimes_{\overline{k}} A')$ the ideal of $Z \times Z'$ is generated by the images of the f's and the g's together, and that concatenation is a regular sequence. Indeed $A \to A \otimes_{\overline{k}} A'$ is flat, so the f's remain a regular sequence after it by the argument of Step 1; and modulo them the ring is $(A/(f)) \otimes_{\overline{k}} A'$, which is a flat base change of A', so the g's are regular there by the same argument with the two factors exchanged. Hence $i \times i'$ is a regular embedding of codimension $d + d'$.

The two factorizations are immediate from the definitions of the four morphisms, and Step 1 identified each factor as a regular embedding of the stated codimension, completing the proof.

The first identity says that the Gysin map of a section undoes the flat pullback along the morphism it is a section of. It is the only one of the three that can be proved directly from the proper-intersection computation already available.

Theorem 4.4.2: Gysin Maps of Sections

Let $\pi\colon P \to S$ be smooth of relative dimension $e \geq 1$ and let $s\colon S \to P$ be a section, so that $\pi \circ s = \mathrm{id}_S$. Then s is a regular embedding of codimension e and

$$s^! \circ \pi^* \;=\; \mathrm{id} \quad \text{on } A_k(S) \text{ for every } k$$

***Proof*:**
That s is a regular embedding of codimension e is [2, A Section of a Smooth Morphism Is a Regular Embedding]. Both $s^!$ and π^* are homomorphisms, so it suffices to prove the identity on the class of a k-dimensional subvariety $V \subseteq S$.

Step 1: the preimage and its components. The morphism π is flat of relative dimension e by [2, Smooth Morphisms Are Flat], so π^* is defined and $\pi^*[V] = [P_V]$ for $P_V = \pi^{-1}(V)$ the scheme-theoretic preimage (definition 1.3.3). The projection $\pi_V\colon P_V \to V$ is smooth of relative dimension e, being the base change of π along $V \hookrightarrow S$ ([2, Stability of Smooth Morphisms](1)), hence flat by the same citation as above; since its base V is a variety, proposition 1.3.2 applies to π_V and P_V is pure of dimension $k+e$. Because $\pi \circ s = \mathrm{id}_S$, the section restricts to a section $s_V\colon V \to P_V$ of π_V, and indeed $s^{-1}(P_V) = (\pi \circ s)^{-1}(V) = V$ scheme-theoretically, so $s(S) \cap P_V = s_V(V)$.

Flatness of π_V makes every irreducible component of P_V dominate V, by the going-down argument already used in the proof of proposition 1.3.2. So the generic point of each component lies in the generic fibre, which is smooth over the field $k(V)$ and therefore regular ([2, Smooth Implies Regular]); its local rings are consequently domains. Hence P_V is generically reduced along every component, so $[P_V] = \sum_j [V_j]$ with all multiplicities one (definition 1.1.2), and no two components share a generic point of the generic fibre. The subvariety $s_V(V)$ dominates V, so its generic point lies in the generic fibre and therefore in exactly one component, which is moreover the unique component containing $s_V(V)$; call it V_1, and let $V_2, \ldots, V_r$ be the others.

Step 2: the other components miss the section altogether. Let $j \geq 2$ and put $W_j = V_j \cap s(S) = V_j \cap s_V(V)$, using Step 1's identification of $s(S) \cap P_V$. Suppose $W_j \neq \emptyset$. Applying theorem 4.2.7 to the subvariety $V_j \subseteq P_V$, of dimension $k+e$, its normal cone $C_{W_j}V_j$ is pure of dimension $k+e$, and lemma 3.3.12 embeds it in $C_{s_V(V)}P_V$ over the inclusion $W_j \hookrightarrow s_V(V)$. Now $s_V(V) \hookrightarrow P_V$ is a section of the smooth morphism π_V, hence a regular embedding of codimension e ([2, A Section of a Smooth Morphism Is a Regular Embedding]), so that cone is the normal bundle N_V, of rank e over $s_V(V) \cong V$ (proposition 3.3.11). The part of N_V lying over W_j has dimension $\dim W_j + e$, so purity forces $\dim W_j = k$. But W_j is then a k-dimensional closed subscheme of the k-dimensional variety $s_V(V)$, hence all of it, so $V_j \supseteq s_V(V)$, contradicting the uniqueness of V_1 in Step 1. Therefore $W_j = \emptyset$ for every $j \geq 2$, and in particular $s^![V_j] = 0$, the Chow groups of the empty scheme vanishing.

Step 3: the remaining component. Step 2 has a consequence worth isolating, because the naive statement it replaces is false: an irreducible component of a smooth V-scheme need not itself be smooth over V, so one may not argue that $V_1 \to V$ is smooth because V_1 is the component through the section. What is true, and all that is needed, is that

$$U \;=\; P_V \setminus \bigcup_{j \geq 2} V_j$$

is an open subscheme of P_V containing $s_V(V)$, by Step 2, and $U \subseteq V_1$. Being a regular embedding of a given codimension is a local condition along the subscheme ([2, Regular Embedding]), and $s_V(V) \hookrightarrow P_V$ is regular of codimension e as recorded in Step 2. Restricting to the open U and using that U is also open in V_1, the embedding $s_V(V) \hookrightarrow V_1$ is regular of codimension e.

The scheme-theoretic intersection $V_1 \cap s(S)$ is $s_V(V)$: this is a scheme-theoretic and not a set-theoretic claim, so it needs an argument rather than a comparison of points. By Step 1, $s^{-1}(P_V) = V$ scheme-theoretically; $s^{-1}(V_1)$ is a closed subscheme of V because the ideal sheaf of V_1 contains that of P_V; it has full support since $V_1 \supseteq s_V(V)$; and V is reduced, so $s^{-1}(V_1) = V$.

Proposition 4.3.3 therefore applies to the regular embedding s of codimension e and the $(k+e)$-dimensional subvariety V_1, and gives

$$s^![V_1] \;=\; \iota_*\big[s_V(V)\big] \;\in\; A_k\big(s(S)\big)$$

Identifying $s(S)$ with S through the isomorphism s, the right side is $[V]$.

Summing Steps 2 and 3 gives $s^!\big(\pi^*[V]\big) = s^![P_V] = [V]$, as we sought to show.

The remaining two identities concern a pair of Gysin maps rather than a single one, and both rest on one observation about definition 4.3.6 that is worth isolating first. The refined map is built from the specialization of the pair (Z', X'), the cone $C_{Z'}X'$, and the bundle N' into which that cone embeds. Nothing else about the ambient X enters. So an embedding may be enlarged, or crossed with a further factor, without changing the resulting homomorphism, provided the preimage and the normal bundle come along unchanged.

Lemma 4.4.3: The Refined Gysin Map Does Not See the Ambient Scheme

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with ideal sheaf $\mathcal{I}$, and let $\varphi\colon \widetilde{X} \to X$ be a morphism whose preimage $\widetilde{Z} = \varphi^{-1}(Z)$, with ideal sheaf $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$, is again a regular embedding $\widetilde{i}\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ *of the same codimension* d. Then for every morphism $g\colon X' \to \widetilde{X}$, writing $Z' = g^{-1}(\widetilde{Z}) = (\varphi \circ g)^{-1}(Z)$,

$$\widetilde{i}^!_g \;=\; i^!_{\varphi\circ g} \;:\; A_k(X') \longrightarrow A_{k-d}(Z')$$

***Proof*:**
First, the hypothesis on codimensions already forces the conormal sheaf to be the pullback. The canonical map

$$\varphi_Z^*\big(\mathcal{I}/\mathcal{I}^2\big) \;\longrightarrow\; \widetilde{\mathcal{I}}/\widetilde{\mathcal{I}}^2, \qquad \varphi_Z\colon \widetilde{Z} \to Z,$$

is surjective, since $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ is generated by the image of $\mathcal{I}$. Both sides are locally free of rank d ([2, The Conormal Sheaf of a Regular Embedding]), the source as the pullback of a rank-d bundle and the target because $\widetilde{i}$ is regular of codimension d. A surjection of locally free sheaves of equal rank is an isomorphism, its kernel being locally free of rank zero. So the canonical map is an isomorphism, and we may use it below.

The two closed subschemes $g^{-1}(\widetilde{Z})$ and $(\varphi \circ g)^{-1}(Z)$ of X' coincide because $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ and pullback of ideals is transitive, so the target group is the same and both homomorphisms are the composite of definition 4.3.6 for one and the same pair (Z', X'). In particular they use the same specialization σ' and the same cone $C_{Z'}X'$, neither of which mentions the ambient scheme at all.

What must be compared is the bundle and the embedding of the cone in it. Write $h\colon Z' \to Z$ and $\widetilde{h}\colon Z' \to \widetilde{Z}$ for the structure morphisms, so $h = \varphi_Z \circ \widetilde{h}$. The hypothesis gives a canonical isomorphism $\widetilde{h}^*(\widetilde{\mathcal{I}}/\widetilde{\mathcal{I}}^2) \cong \widetilde{h}^*\varphi_Z^*(\mathcal{I}/\mathcal{I}^2) = h^*(\mathcal{I}/\mathcal{I}^2)$, that is $\widetilde{N}' \cong N'$ as bundles on Z', and hence π' is the same projection in both constructions.

It remains to see that the closed immersion j of lemma 3.3.13 is carried to the other by this isomorphism. That immersion is induced by a surjection of graded $\mathcal{O}_{Z'}$-algebras out of $\mathrm{Sym}^\bullet$ of the pullback conormal sheaf, and such a morphism is determined by its degree-one part. For i the degree-one part sends the class of a local section α of $\mathcal{I}$ to the class of $\alpha\mathcal{O}_{X'}$ in $\mathcal{I}'/\mathcal{I}'^2$; for $\widetilde{i}$ it sends the class of a local section of $\widetilde{\mathcal{I}}$ to the class of its extension to X'. The hypothesis isomorphism identifies the class of α with the class of $\alpha\mathcal{O}_{\widetilde{X}}$, and $(\alpha\mathcal{O}_{\widetilde{X}})\mathcal{O}_{X'} = \alpha\mathcal{O}_{X'}$ by transitivity of extension of ideals along $X' \to \widetilde{X} \to X$. So the two degree-one parts correspond, the two immersions j agree, and the three factors $(\pi'^*)^{-1}$, j_*, σ' agree one by one.

The hypothesis is satisfied whenever φ is flat, by [2, Regular Embeddings Pull Back Along Flat Morphisms], and that is the only case used below: φ will be a projection off a product.

Three lemmas stand between here and the composition rule. The first is the only place a genuine commutativity statement is needed, and it is needed only for a divisor, where the construction delivers it directly: specialization *is* intersection with a Cartier divisor (definition 4.2.3), so two such intersections commute by theorem 2.3.2.

Lemma 4.4.4: The Gysin Map Commutes with Intersection by a Divisor

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, let $g\colon X' \to X$ be a morphism with $Z' = g^{-1}(Z)$, and let D be a Cartier divisor on X'. Then for every $\alpha \in A_k(X')$,

$$i^!\big(D \cdot \alpha\big) \;=\; D \cdot \big(i^!\alpha\big) \quad \text{in } A_{k-d-1}(Z')$$

both sides being read through corollary 2.4.5, on X' for the left and on the closed subscheme Z' for the right.

***Proof*:**
Write $M = M^\circ_{Z'}X'$ for the deformation space of definition 4.1.4, with blow-down $\rho\colon M \to X'$, fibre $D_\infty = C_{Z'}X'$ over ∞, and $\kappa\colon C_{Z'}X' \hookrightarrow N'$, $\pi'\colon N' \to Z'$ as in definition 4.3.6, so that $i^! = (\pi'^*)^{-1} \circ \kappa_* \circ \sigma'$. Throughout, a Cartier divisor is intersected against classes on closed subschemes of the ambient through corollary 2.4.5; the relevant restrictions are in general *not* Cartier on those subschemes, and it is to avoid needing them to be that the corollary was stated.

Step 0: ρ^*D is a Cartier divisor on M. The blow-down is not flat, its fibre at infinity being a cone, so this needs an argument. Over $\mathbb{A}^1$ the map is the projection $X' \times \mathbb{A}^1 \to X'$, which is flat, and flat pullback of a Cartier divisor is Cartier. Over the chart at infinity, lemma 4.1.2 and (4.1) present $\mathcal{O}_M$ as the extended Rees algebra, a graded subsheaf of $\mathcal{O}_{X'}[u, u^{-1}]$. A local equation f of D is a non-zero divisor on $\mathcal{O}_{X'}$, hence on the free module $\mathcal{O}_{X'}[u, u^{-1}]$, hence on any subsheaf of it; so $\rho^\# f$ is a non-zero divisor and ρ^*D is Cartier.

Step 1: specialization. Let $\beta \in A_{k+1}(M)$ be a lift of α in the sense of definition 4.2.3, so that β restricts to $q^*\alpha$ on $X' \times \mathbb{A}^1$. Then $\rho^*D \cdot \beta$ is a lift of $D \cdot \alpha$: restriction along that open immersion is flat pullback of relative dimension zero, proposition 2.3.3(2) applies with no hypothesis on supports, and ρ^*D restricts to q^*D there by theorem 4.1.5(1). Hence, by theorem 4.2.4 and by theorem 2.3.2 applied on M to the Cartier divisors D_∞ and ρ^*D, the first by theorem 4.1.5(2) and the second by Step 0,

$$\sigma'\big(D \cdot \alpha\big) \;=\; D_\infty \cdot \big(\rho^*D \cdot \beta\big) \;=\; \rho^*D \cdot \big(D_\infty \cdot \beta\big) \;=\; \rho^*D \cdot \sigma'(\alpha)$$

an identity of classes on the cone, the refined form of theorem 2.3.2 placing both sides in the Chow group of $|D_\infty| \cap |\rho^*D| \cap \operatorname{supp}\beta$.

Step 2: pushforward along the cone. On N' put $D_{N'} = \pi'^*D$, the flat pullback along π' of D restricted to Z'; it is a Cartier divisor there, π' being flat. Its restriction to the cone and the restriction of ρ^*D agree, both being computed from the line bundle $\mathcal{O}_{X'}(D)$ pulled back to the cone with its canonical section. A closed immersion carries a cycle to itself, so corollary 2.4.5, applied to $C_{Z'}X' \subseteq N'$, gives

$$\kappa_*\big(\rho^*D \cdot \gamma\big) \;=\; D_{N'} \cdot \kappa_*\gamma$$

for every class γ on the cone, with no hypothesis on its support. This is the step at which the projection formula would otherwise be invoked, and proposition 2.3.3(1) could not be: its hypothesis is that no component of the support maps into $|D|$, which $\sigma'(\alpha)$ need not satisfy.

Step 3: the bundle inverse. Since π' is flat, proposition 2.3.3(2) gives $\pi'^*\big(D \cdot \delta\big) = D_{N'} \cdot \pi'^*\delta$ for every $\delta \in A_*(Z')$, again with no support hypothesis. Applying this to $\delta = i^!\alpha$ and using Steps 1 and 2,

$$\pi'^*\Big(D \cdot i^!\alpha\Big) = D_{N'} \cdot \pi'^*\big(i^!\alpha\big) = D_{N'} \cdot \kappa_*\sigma'(\alpha) = \kappa_*\sigma'\big(D \cdot \alpha\big) = \pi'^*\Big(i^!\big(D \cdot \alpha\big)\Big)$$

and π'^* is injective by theorem 1.4.6, which gives the identity.

The second lemma settles the composition rule when the outer embedding is the zero section of a vector bundle. This is the case in which the normal bundle of the composite genuinely splits, and it is where the conormal sequence of a chain does its work.

Lemma 4.4.5: Composition Through a Zero Section

Let E be a vector bundle of rank $e \geq 1$ on an algebraic scheme Y with zero section $\zeta\colon Y \hookrightarrow E$, and let $i\colon Z \hookrightarrow Y$ be a regular embedding of codimension $d \geq 1$. Let $g\colon S \to Y$ be a morphism, $E' = g^*E$ with projection $\pi'\colon E' \to S$, and let $\mathrm{pr}\colon E' \to E$ be the induced morphism. Then for every $\alpha \in A_k(E')$,

$$(\zeta \circ i)^!_{\mathrm{pr}}(\alpha) \;=\; i^!_g\Big(\zeta^!_{\mathrm{pr}}(\alpha)\Big) \quad \text{in } A_{k-d-e}\big(g^{-1}(Z)\big)$$

***Proof*:**
Write $S_Z = g^{-1}(Z)$. The preimage of Y under pr is the zero section of E', which we identify with S; the preimage of Z is S_Z. Both refined maps are therefore defined and land as stated. That $\zeta^!_{\mathrm{pr}} = (\pi'^*)^{-1}$ is not merely the observation that the normal cone of a zero section is the bundle itself, which is the case $\mathcal{I} = 0$ of (4.6) below: that identifies the cone and the bundle

and makes the immersion of lemma 3.3.13 an identity, but leaves $\zeta^!$ equal to $(\pi'^*)^{-1} \circ \sigma'$, and specialization is emphatically not the identity in general (example 4.2.5). What settles it is that the preimage of Y under pr is the zero section of E', a regular embedding of codimension e, so lemma 4.4.3 identifies $\zeta^!_{\mathrm{pr}}$ with the refined Gysin map of that zero section along id, which is its plain Gysin map (proposition 4.3.7); and $\pi' \colon E' \to S$ is smooth of relative dimension e with the zero section as a section, so theorem 4.4.2 gives $\zeta^! \circ \pi'^* = \mathrm{id}$, that is $\zeta^! = (\pi'^*)^{-1}$.

Step 1: the cone of the composite splits off the bundle. Let $\mathcal{I}$ be the ideal sheaf of S_Z in S and let $\mathcal{S}^\bullet = \mathrm{Sym}^\bullet(\mathcal{E}^\vee)$ present $E' = \mathrm{Spec}_S \mathcal{S}^\bullet$, so that $\mathcal{S}^0 = \mathcal{O}_S$ and $\mathcal{S}^1 = \mathcal{E}^\vee$. The ideal $\mathcal{J}$ of S_Z in E' is generated by $\mathcal{I}$ in degree zero together with all of $\mathcal{S}^1$: a function vanishes on the zero section over S_Z exactly when its degree-zero part lies in $\mathcal{I}$. Consequently

$$\mathcal{J}^n = \bigoplus_{q \le n} \mathcal{I}^{n-q} \mathcal{S}^q \oplus \bigoplus_{q > n} \mathcal{S}^q, \qquad \mathcal{J}^n/\mathcal{J}^{n+1} = \bigoplus_{p+q=n} \left(\mathcal{I}^p/\mathcal{I}^{p+1}\right) \otimes \mathcal{S}^q$$

so that the total graded algebra factors, $\bigoplus_n \mathcal{J}^n/\mathcal{J}^{n+1} = \left(\bigoplus_p \mathcal{I}^p/\mathcal{I}^{p+1}\right) \otimes_{\mathcal{O}_{S_Z}} \mathcal{S}^\bullet$. Taking Spec and using definition 3.3.6 this reads

$$C_{S_Z} E' \;\cong\; C_{S_Z} S \times_{S_Z} E'|_{S_Z} \tag{4.6}$$

Applied with $S = Y$ and $g = \mathrm{id}$ it identifies the normal bundle of the composite as $N_Z E = N_Z Y \oplus i^* E$, which is the splitting of [2, The Conormal Sequence of a Chain of Regular Embeddings] for the chain $Z \subseteq Y \subseteq E$; the splitting is the one induced by the bundle projection.

Step 2: reduction. By theorem 1.4.6 every class on E' is $\pi'^*\beta$ for a unique $\beta \in A_{k-e}(S)$, and by linearity we may take $\beta = [V]$ for $V \subseteq S$ a subvariety. Both sides of the assertion commute with proper pushforward along $V \hookrightarrow S$ and its base change $V \times_S E' \hookrightarrow E'$, by theorem 4.3.10(1), so we may replace S by V. Then S is a variety, $\alpha = [E']$, and $\zeta^! \alpha = [S]$.

Step 3: comparison of fundamental cycles. Write $h \colon S_Z \to Z$ for the induced morphism, $r \colon h^* N_Z Y \to S_Z$ and $q \colon h^* N_Z E \to h^* N_Z Y$ for the projections, the latter using the splitting of Step 1, so that $r \circ q$ is the projection of $h^* N_Z E$. Both S and E' are varieties, so theorem 4.2.7 gives $\sigma[S] = [C_{S_Z} S]$ and $\sigma[E'] = [C_{S_Z} E']$. By (4.6) the second cone is the preimage of the first under q, so $q^*[C_{S_Z} S] = [C_{S_Z} E']$ by definition 1.3.3. Now definition 4.3.6 reads the two Gysin classes as

$$r^*\big(i^![S]\big) = [C_{S_Z} S], \qquad (r \circ q)^*\big((\zeta i)^![E']\big) = [C_{S_Z} E']$$

Combining the three displays, $(r \circ q)^*\big((\zeta i)^![E']\big) = q^* r^*\big(i^![S]\big) = (r \circ q)^*\big(i^![S]\big)$ by proposition 1.3.5(3), and $(r \circ q)^*$ is injective by theorem 1.4.6. Hence $(\zeta i)^![E'] = i^![S] = i^! \zeta^! [E']$, which by Step 2 is the assertion in general.

The third lemma is the only input that is not about Gysin maps at all. It says that cutting a class on $X \times \mathbb{P}^1$ by a fibre gives an answer that does not depend on which fibre.

Lemma 4.4.6: Cutting by a Fibre of $X \times \mathbb{P}^1$ Is Independent of the Fibre

Let X be an algebraic scheme and for $t \in \mathbb{P}^1$ a closed point let D_t denote the Cartier divisor $X \times \{t\}$ on $X \times \mathbb{P}^1$. Then for every $\beta \in A_k(X \times \mathbb{P}^1)$ the class

$$D_t \cdot \beta \;\in\; A_{k-1}\big(X \times \{t\}\big) \;=\; A_{k-1}(X)$$

is the same for all t, the identification being the projection $X \times \{t\} \to X$.

***Proof*:**
Write $X \times \mathbb{P}^1 = \mathbb{P}(\mathcal{O}_X^{\oplus 2})$ with projection p and tautological class ξ; for the trivial bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ is pulled back from $\mathbb{P}^1$, and D_t is the divisor of the section of it cutting out t, so $\mathcal{O}(D_t) = \mathcal{O}_{\mathbb{P}(E)}(1)$ for every t. By theorem 3.1.9 with $r = 1$ there are unique classes $\alpha_0 \in A_{k-1}(X)$ and $\alpha_1 \in A_k(X)$ with

$$\beta \;=\; p^*\alpha_0 \;+\; \xi \cap p^*\alpha_1$$

It suffices to evaluate $D_t \cdot$ on each summand. For the first, take $\alpha_0 = [V]$ with $V \subseteq X$ a subvariety; then $p^*[V] = [V \times \mathbb{P}^1]$, which is not contained in $|D_t|$, so definition 2.2.1(1) computes $D_t \cdot p^*[V] = [V \times \{t\}]$, the divisor of the restricted section, and under $X \times \{t\} \cong X$ this is $[V]$. For the second, $D_t \cdot (\xi \cap p^*\alpha_1) = \xi^2 \cap p^*\alpha_1$ once pushed into $A_{k-2}(X \times \mathbb{P}^1)$, and $\xi^2 = 0$ by theorem 3.2.9, the bundle being trivial. To descend that vanishing to the refined class on $X \times \{t\}$, note that the first computation identifies the pushforward $\iota_{t*} \colon A_m(X) = A_m(X \times \{t\}) \to A_m(X \times \mathbb{P}^1)$ with $\gamma \mapsto \xi \cap p^*\gamma$, and that map is injective by the uniqueness clause of theorem 3.1.9, being the restriction of the isomorphism Ψ to the summand $\alpha_0 = 0$. Hence the refined class vanishes as well, and $D_t \cdot \beta = \alpha_0$ for every t, as we sought to show.

Theorem 4.4.7: Composition of Gysin Maps

Let $i \colon Z \hookrightarrow Y$ and $j \colon Y \hookrightarrow P$ be regular embeddings of codimensions $d \geq 1$ and $e \geq 1$. Then $j \circ i$ is a regular embedding of codimension $d + e$, and for every morphism $a \colon W \to P$, writing $W_Y = a^{-1}(Y)$, $W_Z = a^{-1}(Z)$ and $a' \colon W_Y \to Y$ for the induced morphism,

$$(j \circ i)^!_a \;=\; i^!_{a'} \circ j^!_a \;:\; A_k(W) \longrightarrow A_{k-d-e}(W_Z)$$

***Proof*:**
That $j \circ i$ is a regular embedding of codimension $d + e$ is the first assertion of [2, The Conormal Sequence of a Chain of Regular Embeddings]. Both sides commute with proper pushforward (theorem 4.3.10(1)), and $A_k(W)$ is generated by the classes of k-dimensional subvarieties, so it suffices to prove the identity for W a variety and the class $[W]$.

The two deformations. Let $M = M^{\circ}_Y P$ and $M' = M^{\circ}_{W_Y} W$ be the deformation spaces of definition 4.1.4, with projections to $\mathbb{P}^1$ written π and π'. The morphism a induces $\widetilde{a} \colon M' \to M$ over $\mathbb{P}^1$ by lemma 4.2.6, whose hypothesis $W_Y = a^{-1}(Y)$ holds by definition. Let

$$\chi \;:\; Z \times \mathbb{P}^1 \xhookrightarrow{i \times \mathrm{id}} Y \times \mathbb{P}^1 \hookrightarrow M$$

be the composite of $i \times \mathrm{id}$ with the closed immersion of theorem 4.1.5(4). Its first factor is a regular embedding of codimension d, being the base change of i along the flat projection

$Y \times \mathbb{P}^1 \to Y$ ([2, Regular Embeddings Pull Back Along Flat Morphisms]); its second is a regular embedding of codimension e. This last point needs the charts rather than the fibres: $\pi^{-1}(\mathbb{A}^1)$ is open, and there the immersion is $j \times \mathrm{id}$, but $\pi^{-1}(\infty)$ is closed, so regularity of the immersion into it says nothing about a neighbourhood. On the complementary chart $\pi^{-1}(\mathbb{P}^1 \setminus \{0\})$, lemma 4.1.2 presents the coordinate ring as an extended Rees algebra $\mathcal{R}$, in which the ideal of $Y \times \mathbb{P}^1$ is generated by $g_1/u, \ldots, g_e/u$ for $g_1, \ldots, g_e$ a local regular sequence cutting out Y in P. Here u is a non-zero divisor on $\mathcal{R}$ and $\mathcal{R}/u\mathcal{R} = \mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$, in which the images of the g_m/u are the coordinates and so form a regular sequence; hence $u, g_1/u, \ldots, g_e/u$ is a regular sequence, and permuting a regular sequence in a Noetherian local ring leaves it regular, so $g_1/u, \ldots, g_e/u$ is one. So χ is a regular embedding of codimension $d + e$, again by the chain result cited above, and $\widetilde{a}^{-1}(Z \times \mathbb{P}^1) = W_Z \times \mathbb{P}^1$.

Cutting by a fibre. For a closed point $t \in \mathbb{P}^1$ let D_t be the fibre $\pi'^{-1}(t)$ on M'. It is an effective Cartier divisor: for $t = \infty$ by theorem 4.1.5(2), and for $t \neq \infty$ because M' is a variety, as the proof of theorem 4.2.7 records, so that any nonzero local equation is a non-zero divisor. Theorem 4.1.5(1),(2) identify the fibres. Over $\mathbb{A}^1$ the fibre is $W \times \{t\}$, which is reduced, so definition 2.2.1(1) computes $D_t \cdot [M']$ as its fundamental cycle with multiplicity one; over ∞ the same intersection is by definition the specialization, which theorem 4.2.7 evaluates. So

$$D_t \cdot [M'] \;=\; [M'_t] \;=\; \begin{cases} [W] & t \neq \infty \\ [C_{W_Y} W] & t = \infty \end{cases} \tag{4.7}$$

Now apply $\chi^!$ along $\widetilde{a}$. The scheme M'_t maps to M_t, the inclusion $M_t \hookrightarrow M$ pulls $Z \times \mathbb{P}^1$ back to $Z \times \{t\}$, and the induced embedding is $j \circ i$ for $t \neq \infty$ and $\widetilde{\jmath} \circ i$ for $t = \infty$, where $\widetilde{\jmath}$ is the zero section of $N_Y P$; both are regular of codimension $d + e$. So lemma 4.4.3, applied with φ the inclusion of the fibre, gives

$$\chi^!\big(D_t \cdot [M']\big) \;=\; \begin{cases} (j \circ i)^!\,[W] & t \neq \infty \\ (\widetilde{\jmath} \circ i)^!\,[C_{W_Y} W] & t = \infty \end{cases} \tag{4.8}$$

The value at infinity. The cone $C_{W_Y} W$ is a closed subscheme of the pullback bundle $(a')^* N_Y P$ over W_Y (lemma 3.3.13). Lemma 4.4.5, stated for an arbitrary class on that bundle, therefore applies with $E = N_Y P$ and $S = W_Y$ to the pushforward of $[C_{W_Y} W]$; and theorem 4.3.10(1), for the closed immersion $C_{W_Y} W \hookrightarrow (a')^* N_Y P$, says that computing before or after that pushforward gives the same answer, the induced morphism on the preimages of Z being the identity of W_Z because the vertex of $C_{W_Y} W$ is W_Y, the cone algebra being generated in degree one. This gives

$$(\widetilde{\jmath} \circ i)^!\,[C_{W_Y} W] \;=\; i^!\Big(\widetilde{\jmath}^!\,[C_{W_Y} W]\Big) \;=\; i^!_{a'}\Big(j^!_a[W]\Big)$$

the last equality because $\widetilde{\jmath}^!$ is the inverse of the bundle pullback and $\sigma[W] = [C_{W_Y} W]$, which is exactly how definition 4.3.6 computes $j^!_a[W]$.

Independence of t. By lemma 4.4.4, applied to the regular embedding χ and the divisor D_t, whose restriction to $W_Z \times \mathbb{P}^1$ is the Cartier divisor $W_Z \times \{t\}$,

$$\chi^!\big(D_t \cdot [M']\big) \;=\; D_t \cdot \big(\chi^![M']\big)$$

and $\chi^![M']$ is a class on $W_Z \times \mathbb{P}^1$ not depending on t, so lemma 4.4.6 makes the right-hand side independent of t.

One point of bookkeeping remains. Lemma 4.4.4 records its identity in $A_*(W_Z \times \mathbb{P}^1)$, whereas (4.8) produces classes on $W_Z \times \{t\}$; the two are related by the pushforward ι_{t*} along the closed immersion $W_Z \times \{t\} \hookrightarrow W_Z \times \mathbb{P}^1$, through theorem 4.3.10(1) applied to $M'_t \hookrightarrow M'$. That costs nothing, because the proof of lemma 4.4.6 identifies ι_{t*} with $\gamma \mapsto \xi \cap p^*\gamma$, which is one and the same map for every t and is injective. Comparing the two lines of (4.8) at $t \neq \infty$ and at $t = \infty$, their images under ι_{t*} agree, hence so do the classes themselves, and $(j \circ i)^![W] = i^!_{a'}\big(j^!_a[W]\big)$, as we sought to show.

Commutativity now follows, and the route is the one announced at the start of the section: the two embeddings are made into a single embedding of a product, which factors in two ways.

Corollary 4.4.8: Commutativity of Gysin Maps

Let $i\colon Z \hookrightarrow P$ and $i'\colon Z' \hookrightarrow P'$ be regular embeddings of codimensions d and d', let W be an algebraic scheme, and let $a\colon W \to P$ and $b\colon W \to P'$ be morphisms. Put $W_1 = a^{-1}(Z)$, $W_2 = b^{-1}(Z')$ and $W_{12} = W_1 \cap W_2$. Then for every $\alpha \in A_k(W)$,

$$i^!_{a|_{W_2}}\big(i'^!_b\alpha\big) \;=\; i'^!_{b|_{W_1}}\big(i^!_a\alpha\big) \quad \text{in } A_{k-d-d'}(W_{12})$$

***Proof*:**
Let $c = (a,b)\colon W \to P \times P'$. Lemma 4.4.1 makes $i \times i'$ a regular embedding of codimension $d + d'$ and exhibits its two factorizations through $Z \times P'$ and through $P \times Z'$, all four factors being regular embeddings. The preimages under c are

$$c^{-1}(Z \times P') = W_1, \qquad c^{-1}(P \times Z') = W_2, \qquad c^{-1}(Z \times Z') = W_{12},$$

because the ideal of $Z \times P'$ is $\mathcal{I}_Z\mathcal{O}_{P\times P'}$ and $\mathrm{pr}_1 \circ c = a$, and symmetrically. The third follows from the first two: the ideal of $Z \times Z'$ is the sum $\mathcal{I}_Z\mathcal{O}_{P\times P'} + \mathcal{I}_{Z'}\mathcal{O}_{P\times P'}$, so $Z \times Z'$ is the scheme-theoretic intersection of $Z \times P'$ and $P \times Z'$, and extension of ideals along c carries a sum of ideals to the sum of the extensions.

Each of the four factors is the preimage of i or of i' along a projection off a product. Such a projection is the base change of a structure morphism to $\operatorname{Spec}\overline{k}$, hence flat, so lemma 4.4.3 applies to each factor and identifies its refined Gysin map with that of i or of i' along the corresponding composite. Explicitly, with $\varphi = \mathrm{pr}_1$ one gets $(i \times \mathrm{id}_{P'})^!_c = i^!_a$ and $(i \times \mathrm{id}_{Z'})^!_{c''} = i^!_{a|_{W_2}}$ for $c''\colon W_2 \to P \times Z'$, and with $\varphi = \mathrm{pr}_2$ one gets $(\mathrm{id}_P \times i')^!_c = i'^!_b$ and $(\mathrm{id}_Z \times i')^!_{c'} = i'^!_{b|_{W_1}}$ for $c'\colon W_1 \to Z \times P'$. Here c' and c'' are the morphisms induced by c on the preimages W_1 and W_2, so that $\mathrm{pr}_2 \circ c' = b|_{W_1}$ and $\mathrm{pr}_1 \circ c'' = a|_{W_2}$; they are exactly the morphisms that theorem 4.4.7 calls a', for the two factorizations respectively.

Apply theorem 4.4.7 to each factorization in turn, with c as the morphism to $P \times P'$. The first gives $(i \times i')^!_c = i'^!_{b|_{W_1}} \circ i^!_a$ and the second gives $(i \times i')^!_c = i^!_{a|_{W_2}} \circ i'^!_b$. The left sides agree, so the right sides do, which is the assertion.

Exercise 4.4.1

1. Theorem 4.4.2 assumes $e \geq 1$, and so does definition 4.3.6. Show that for $e = 0$ a section s of a smooth $\pi\colon P \to S$ of relative dimension zero is an open and closed immersion, so that

the only sensible reading of $s^!$ is flat pullback along s, and that with that reading $s^! \circ \pi^* = \mathrm{id}$ follows from proposition 1.3.5(3) with no deformation at all.

2. In lemma 4.4.1, show that the normal bundle of $i \times i'$ is $\mathrm{pr}_1^* N \oplus \mathrm{pr}_2^* N'$, and that the two factorizations exhibit the two summands as the normal bundles of the two intermediate stages. Which of the two summands does lemma 4.4.3 match to N when applied with $\varphi = \mathrm{pr}_1$?

3. Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension d and take $P' = P$ and $i' = i$, but keep two distinct morphisms $a, b\colon W \to P$. Write out what corollary 4.4.8 asserts, and identify the regular embedding of $P \times P$ to which lemma 4.4.1 applies. Then show that for $a = b$ the identity degenerates to a tautology, and explain why the corollary is in no case a statement about $i^!$ commuting with itself as an endomorphism.

4. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d and $\varphi\colon U \hookrightarrow X$ an open immersion with $Z \subseteq U$. Check that φ satisfies the hypotheses of lemma 4.4.3, and deduce that the refined Gysin map of i may be computed in any open neighbourhood of Z.

5. In theorem 4.4.7 verify directly, from the definition of a regular embedding, that a local regular sequence for Z in Y lifted to P and appended to one for Y in P generates the ideal of Z in P, and that the concatenation is regular. Then write out the sequence $0 \to N_Z Y \to N_Z P \to N_Y P|_Z \to 0$ of [2, The Conormal Sequence of a Chain of Regular Embeddings] for $Z = \{0\} \subseteq Y = \mathbb{A}^1 \subseteq P = \mathbb{A}^2$, identifying all three bundles and the splitting.

5

Intersection Products

The refined Gysin map of chapter 4 intersects a cycle with a regularly embedded subscheme. It is not yet a product: it takes a cycle on the ambient scheme and a regular embedding, not two cycles. The step that converts it into one is a piece of bookkeeping that costs nothing and gives everything.

For X smooth, the diagonal $\delta\colon X \hookrightarrow X \times X$ is a regular embedding, and two cycles α, β on X determine a single cycle $\alpha \times \beta$ on $X \times X$. Applying $\delta^!$ to it lands back on X and, as this chapter shows, the result is bilinear, commutative, associative, and has $[X]$ as a unit. That is a ring, and it is the Chow ring $A^*(X)$: the object the classical geometers were computing in when they counted, and which *Algebraic Geometry* [2] could construct only on a surface, by hand.

Nothing here needs a moving lemma. The cycles are never perturbed; the diagonal is what gets degenerated, and chapter 4 already did that work. The moving lemma is stated in its place, both because it is true and because seeing what it would have been used for is the clearest way to see why the deformation replaced it.

The chapter closes by paying two debts. The self-intersection formula computes i^*i_* as capping with the top Chern class, which recovers the $E^2 = -1$ of a blown-up surface as a special case of a general theorem; and the agreement theorem shows that on a smooth projective surface this product is the one built cohomologically upstream, so the surface theory that book developed by hand is subsumed rather than paralleled.

5.1 The Intersection Product

The first ingredient is a way of assembling a cycle on X and a cycle on Y into a single cycle on $X \times Y$. On cycles the rule is forced, $[V] \times [W] = [V \times W]$, and the content is that it survives rational equivalence. The mechanism that makes it survive is flat pullback along the two projections, which is already known to preserve rational equivalence (proposition 1.3.7), so the whole construction is arranged to be a composite of operations that do.

Lemma 5.1.1: Projections of a Product Are Flat

Let X and Y be algebraic schemes with Y pure of dimension l, and let $p\colon X \times Y \to X$ be the projection. Then p is flat of relative dimension l, and for every k-dimensional subvariety $V \subseteq X$ the scheme-theoretic preimage is $p^{-1}(V) = V \times Y$, so that

$$p^*[V] \;=\; [V \times Y]$$

the fundamental cycle of $V \times Y$, which is pure of dimension $k + l$.

***Proof*:**
The product is taken over $\operatorname{Spec} \overline{k}$, so p is the base change of the structure morphism $Y \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$. Every module over a field is free, hence flat, so $Y \to \operatorname{Spec} \overline{k}$ is flat, and flatness is stable under base change; thus p is flat. The fibre of p over a point $x \in X$ is $Y \times_{\operatorname{Spec} \overline{k}} \operatorname{Spec} k(x)$, which is pure of dimension l, extension of the ground field preserving the dimension of a scheme of finite type. So p is flat of relative dimension l in the sense of definition 1.3.3.

The scheme-theoretic preimage of V is $p^{-1}(V) = (X \times Y) \times_X V = V \times Y$, and definition 1.3.3 computes $p^*[V]$ as the fundamental cycle of that preimage. For the dimension, restrict p to V: the induced morphism $V \times Y \to V$ is again flat with all fibres of pure dimension l, and its base V is a variety, so proposition 1.3.2 applies and every irreducible component of $V \times Y$ has dimension $k + l$, as we sought to show.

The lemma supplies both the cycle and the guarantee that it has the expected dimension, so the exterior product can be defined by the geometric formula and its bookkeeping checked afterwards.

Definition 5.1.2: The Exterior Product of Cycles

Let X and Y be algebraic schemes. For a k-dimensional subvariety $V \subseteq X$ and an l-dimensional subvariety $W \subseteq Y$, the product $V \times W$ is a closed subscheme of $X \times Y$, pure of dimension $k + l$ by lemma 5.1.1. The *exterior product* of $[V]$ and $[W]$ is its fundamental cycle (definition 1.1.2),

$$[V] \times [W] \;=\; [V \times W] \;\in\; Z_{k+l}(X \times Y)$$

Extending bilinearly gives the exterior product of cycles

$$\times \;:\; Z_k(X) \times Z_l(Y) \longrightarrow Z_{k+l}(X \times Y)$$

Example 5.1.3: The Two Rulings of a Quadric Surface
Take $X = Y = \mathbb{P}^1$. By theorem 1.5.2, $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$ and $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$, the second generated by the class of any closed point. Fix closed points $p, q \in \mathbb{P}^1$. The four exterior products of generators are

$$[\mathbb{P}^1] \times [\mathbb{P}^1] = [\mathbb{P}^1 \times \mathbb{P}^1], \qquad [\mathbb{P}^1] \times [q] = [\mathbb{P}^1 \times \{q\}], \qquad [p] \times [\mathbb{P}^1] = [\{p\} \times \mathbb{P}^1], \qquad [p] \times [q] = [(p, q)]$$

each product of varieties being a variety here: $\mathbb{P}^1 \times \mathbb{P}^1$ is covered by four copies of $\mathbb{A}^2$, any two of which meet, so it is irreducible, and it is reduced because those charts are.

The geometry is visible through the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, $([a:b],[c:d]) \mapsto [ac:ad:bc:bd]$, whose image is the quadric surface $\{x_0x_3 = x_1x_2\}$. The middle two classes are the classes of the two families of lines on that quadric: $\mathbb{P}^1 \times \{q\}$ and $\{p\} \times \mathbb{P}^1$ each map to a line, and two members of the same family are disjoint while members of opposite families meet in one point. The first class is the fundamental class of the surface and the last is the class of a point.
Not every subvariety of a product is a product. The diagonal $\Delta = \{([a:b],[a:b])\} \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ is a curve meeting both families, so it is neither of the form $V \times \mathbb{P}^1$ nor of the form $\mathbb{P}^1 \times W$; its class is nonetheless a sum of exterior products, $[\Delta] = [\mathbb{P}^1] \times [q] + [p] \times [\mathbb{P}^1]$, which exercise 5.1.1 (2) verifies by exhibiting a rational function.

Rational equivalence is not visible in definition 5.1.2, which is written on cycles. The route to it is to recognize the exterior product as a composite of a flat pullback and a proper pushforward, one factor at a time.

Proposition 5.1.4: The Exterior Product Descends to Rational Equivalence

Let X and Y be algebraic schemes.

1. *Flat descriptions.* Let $W \subseteq Y$ be an l-dimensional subvariety, let $p_W : X \times W \to X$ be the projection and $\iota_W : X \times W \hookrightarrow X \times Y$ the closed immersion. Then for every $\alpha \in Z_k(X)$,
$$\alpha \times [W] \;=\; (\iota_W)_*\big(p_W^*\alpha\big)$$
Symmetrically, if $V \subseteq X$ is a k-dimensional subvariety, $q_V : V \times Y \to Y$ the projection and $\iota_V : V \times Y \hookrightarrow X \times Y$ the closed immersion, then $[V] \times \beta = (\iota_V)_*(q_V^*\beta)$ for every $\beta \in Z_l(Y)$.

2. *Descent.* The exterior product carries $\mathrm{Rat}_k(X) \times Z_l(Y)$ and $Z_k(X) \times \mathrm{Rat}_l(Y)$ into $\mathrm{Rat}_{k+l}(X \times Y)$, so it induces a bilinear map
$$\times \;:\; A_k(X) \times A_l(Y) \longrightarrow A_{k+l}(X \times Y)$$

3. *Closed immersions.* If $f : X' \hookrightarrow X$ and $g : Y' \hookrightarrow Y$ are closed immersions of algebraic schemes, then $(f \times g)_*(\alpha \times \beta) = f_*\alpha \times g_*\beta$ for all $\alpha \in A_k(X')$ and $\beta \in A_l(Y')$.

***Proof*:**

1. Both sides are additive in α, so it suffices to take $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$. The subscheme W is a variety, hence pure of dimension l, so lemma 5.1.1 applies to p_W and gives $p_W^*[V] = [V \times W]$, the fundamental cycle of $V \times W$ computed inside $X \times W$. Its components are the components of $V \times W$, each carrying the same length whether $V \times W$ is regarded in $X \times W$ or in $X \times Y$, and each maps isomorphically to itself under the closed immersion ι_W; so $(\iota_W)_*[V \times W] = [V \times W]$ by definition 1.2.1, the field extension in each degree being trivial. This is $[V] \times [W]$. The symmetric statement is the same argument with the roles of the two factors exchanged.

2. Let $\alpha \in \mathrm{Rat}_k(X)$ and $\beta \in Z_l(Y)$. Writing $\beta = \sum_j n_j[W_j]$ and using bilinearity, it is enough to show $\alpha \times [W] \in \mathrm{Rat}_{k+l}(X \times Y)$ for a single l-dimensional subvariety W. By part (1) this class is $(\iota_W)_*(p_W^*\alpha)$. Flat pullback carries cycles rationally equivalent to zero

to cycles rationally equivalent to zero (proposition 1.3.7), and so does proper pushforward along the closed immersion ι_W (theorem 1.2.3); hence $\alpha \times [W] \in \mathrm{Rat}_{k+l}(X \times Y)$. The symmetric half is identical, using the second formula of part (1) and additivity in the first variable.
For classes, let $\alpha \sim \alpha'$ in $Z_k(X)$ and $\beta \sim \beta'$ in $Z_l(Y)$. Then

$$\alpha \times \beta - \alpha' \times \beta' \;=\; (\alpha - \alpha') \times \beta \;+\; \alpha' \times (\beta - \beta')$$

and both terms on the right lie in $\mathrm{Rat}_{k+l}(X \times Y)$ by the previous paragraph. So the product of the classes is well defined, and it is bilinear because it is bilinear on cycles.

3. Both sides are bilinear, so it suffices to check on $\alpha = [V']$ and $\beta = [W']$ for subvarieties $V' \subseteq X'$ and $W' \subseteq Y'$. A closed immersion identifies V' with its image $f(V')$ and W' with $g(W')$, so $V' \times W'$ and $f(V') \times g(W')$ are the same closed subscheme of $X \times Y$; their fundamental cycles therefore agree, and each component of $V' \times W'$ maps isomorphically under $f \times g$. Hence $(f \times g)_*[V' \times W'] = [f(V') \times g(W')] = f_*[V'] \times g_*[W']$ by definition 1.2.1, completing the proof.

The proof is short because the exterior product was built out of the two functorialities of chapter 1 rather than defined by an independent construction, and that is the only reason it respects rational equivalence: nothing about a product of schemes is used beyond the flatness of its projections. Part (3) is the compatibility that lets a product computed on small closed subschemes be compared with the same product computed on the ambient scheme, and it is what the refined form of the intersection product below rests on.

What the exterior product does not do is produce a class on X. A class on $X \times X$ is returned to X by pulling back along the diagonal, and chapter 4 supplies the pullback provided the diagonal is a regular embedding. The diagonal morphism $\delta \colon X \to X \times X$ exists for every scheme, and it is a closed immersion exactly when X is separated, that being the definition of separatedness; every variety appearing from here on is separated, and the hypothesis is carried in the statements that use it.

The first step is to identify the conormal sheaf of the diagonal. It requires no hypothesis on X beyond separatedness, and it is what converts smoothness into the regularity of δ.

Lemma 5.1.5: The Conormal Sheaf of the Diagonal

Let X be an algebraic scheme, let $\delta \colon X \to X \times X$ be the diagonal and let $\mathcal{J} \subseteq \mathcal{O}_{X \times X}$ be the ideal sheaf of the closed subscheme $\delta(X)$. Regard $\mathcal{J}/\mathcal{J}^2$ as a sheaf of $\mathcal{O}_X$-modules through the isomorphism $\delta \colon X \to \delta(X)$. Then there is a canonical isomorphism

$$\mathcal{J}/\mathcal{J}^2 \;\cong\; \Omega_X$$

of $\mathcal{O}_X$-modules, under which the class of $1 \otimes a - a \otimes 1$ corresponds to da.

***Proof*:**
There is nothing to prove: this is how the sheaf of differentials is defined upstream. The cotangent sheaf is $\Omega_{X/Y} := \Delta^*(\mathscr{I}/\mathscr{I}^2)$ for $\mathscr{I}$ the ideal sheaf of the diagonal, with universal derivation $d(b) = 1 \otimes b - b \otimes 1$ modulo $\mathscr{I}^2$ ([2, Sheaf of Differentials]); taking $Y = \operatorname{Spec} \overline{k}$ and using that δ is a closed immersion gives the statement. The affine content, that I/I^2 with this d satisfies the universal property of $\Omega_{B/A}$, is [2, Differentials via the Diagonal], as we sought to show.

The identification uses no smoothness, and it is the reason the sheaf of differentials can be constructed from the diagonal in the first place: Ω_X is the conormal sheaf of δ, so it records the first infinitesimal neighbourhood of the diagonal and nothing more. What smoothness adds is that Ω_X is then locally free of the right rank, and local freeness of the conormal sheaf is the visible half of being a regular embedding. The other half, that the ideal is generated by a regular sequence and not by the right number of elements, is where the local rings of $X \times X$ enter.

Theorem 5.1.6: The Diagonal of a Smooth Variety Is a Regular Embedding

Let X be a separated smooth variety of dimension n over $\overline{k}$ and let $\delta\colon X \to X \times X$ be the diagonal. Then δ is a regular embedding of codimension n ([2, Regular Embedding]), and its normal bundle is the tangent bundle of X,

$$N_\delta \;=\; T_X$$

***Proof*:**
Write $\Delta = \delta(X)$ and let $\mathcal{J}$ be its ideal sheaf.

Step 1: the diagonal is closed of codimension n. Separatedness of X says exactly that δ is a closed immersion, and δ is an isomorphism onto Δ, so Δ is a closed subscheme of $X \times X$ isomorphic to X and of dimension n. Smoothness is stable under products and dimensions add, so $X \times X$ is smooth of dimension $2n$ ([2, Stability of Smooth Morphisms](3)); moreover $X \times X$ is pure of dimension $2n$ by lemma 5.1.1 applied with $V = X$. Its local rings are therefore regular, of dimension $2n$ at every closed point. The codimension of Δ in $X \times X$ is $2n - n = n$.

Step 2: the ideal is generated by n elements along the diagonal. Fix an affine open $U = \operatorname{Spec} A$ of X and keep the notation of lemma 5.1.5, so that $U \times U = \operatorname{Spec} B$ with $B = A \otimes_{\overline{k}} A$, and $J = \ker \mu$ is the ideal of $\delta(U)$. By lemma 5.1.5 there is an isomorphism $\Omega_A \cong J/J^2$ of A-modules, and smoothness of X makes Ω_A locally free of rank n.

The differentials da generate Ω_A, so after shrinking U around a chosen closed point there are $t_1, \ldots, t_n \in A$ whose differentials generate Ω_A: pick elements whose differentials span the fibre of Ω_A at the point, which has dimension n, and apply Nakayama together with coherence to spread the generation to a neighbourhood. The resulting surjection $A^n \to \Omega_A$ onto a locally free module of rank n is an isomorphism, since after localizing at any prime it becomes a surjective endomorphism of a finitely generated module over a commutative ring and every such endomorphism is injective. So $dt_1, \ldots, dt_n$ is a basis of Ω_A.

Transporting through lemma 5.1.5, the classes of

$$u_i \;=\; t_i \otimes 1 - 1 \otimes t_i, \qquad i = 1, \ldots, n$$

form a basis of J/J^2 over A. In particular $J = (u_1, \ldots, u_n) + J^2$. Let $z \in \delta(U)$ and put $R = \mathcal{O}_{X\times X,z}$ with maximal ideal $\mathfrak{m}$. Localizing the last equality gives $JR = (u_1, \ldots, u_n)R + J^2R$, and $J \subseteq \mathfrak{m}$ forces $J^2R \subseteq \mathfrak{m}\,JR$; so the finitely generated R-module $N = JR/(u_1, \ldots, u_n)R$ satisfies $N = \mathfrak{m}N$ and vanishes by Nakayama. Hence

$$JR \;=\; (u_1, \ldots, u_n)R \quad \text{for every } z \in \delta(U)$$

The coherent sheaf $\mathcal{J}/(u_1, \ldots, u_n)\mathcal{O}_{U\times U}$ therefore has vanishing stalk at every point of $\delta(U)$, so its support is a closed subset of $U \times U$ disjoint from $\delta(U)$, and on the open complement, an open neighbourhood of $\delta(U)$ in $X \times X$, the ideal sheaf of Δ is generated by $u_1, \ldots, u_n$.

Step 3: the generators form a regular sequence. Let $z \in \delta(U)$ be a closed point, corresponding to the closed point $x \in U$, and keep $R = \mathcal{O}_{X\times X,z}$. By Step 1 the ring R is regular local of dimension $2n$, and

$$R/JR \;=\; \mathcal{O}_{\Delta,z} \;=\; \mathcal{O}_{X,x}$$

is regular local of dimension n, again by smoothness of X. Regularity makes the embedding dimensions equal to the Krull dimensions, so $\dim_{\overline{k}} \mathfrak{m}/\mathfrak{m}^2 = 2n$ while the quotient $\mathfrak{m}/(\mathfrak{m}^2 + JR)$, which is the corresponding space for R/JR, has dimension n. The image of JR in $\mathfrak{m}/\mathfrak{m}^2$ is therefore n-dimensional, and it is spanned by the images of $u_1, \dots, u_n$ by Step 2; those n images are consequently linearly independent in $\mathfrak{m}/\mathfrak{m}^2$, that is, $u_1, \dots, u_n$ is part of a regular system of parameters of R. A regular system of parameters of a regular local ring is a regular sequence, and so is any part of one: extend $u_1, \dots, u_n$ to a regular system of parameters $u_1, \dots, u_{2n}$ of R and put $R_i = R/(u_1, \dots, u_i)$, whose maximal ideal is generated by the images of $u_{i+1}, \dots, u_{2n}$. Its embedding dimension is therefore at most $2n - i$, while its dimension is at least $2n - i$, since killing one element drops the dimension by at most one; the two agree, so R_i is regular local and hence a domain, and the image of u_{i+1} in it is nonzero, since otherwise $\mathfrak{m}R_i$ would be generated by $2n - i - 1$ elements. Each u_{i+1} is thus a nonzerodivisor on R_i, that is, $u_1, \dots, u_n$ is a regular sequence in R.

This was proved at closed points, and it propagates to the rest of Δ by localization. If $z' \in \Delta$ is arbitrary, its closure meets the closed points of $X \times X$, since closed points are dense in a closed subset of a scheme of finite type over a field; choosing a closed point z in that closure, $\mathcal{O}_{X\times X,z'}$ is a localization of $\mathcal{O}_{X\times X,z}$ at a prime containing J. Localization is exact, so a nonzerodivisor stays a nonzerodivisor and a regular sequence stays one, the quotient remaining nonzero because $z' \in \Delta$. Thus every point of Δ has a neighbourhood in $X \times X$ on which $\mathcal{J}$ is generated by n sections whose images in the local rings along Δ form a regular sequence, which is the local condition of [2, Regular Embedding]. Since Δ has codimension n by Step 1, δ is a regular embedding of codimension n.

Step 4: the normal bundle. The normal bundle of a regular embedding is the vector bundle whose sheaf of sections is the dual of the conormal sheaf ([2, Conormal Sheaf and Normal Bundle]), and lemma 5.1.5 identifies that conormal sheaf $\mathcal{J}/\mathcal{J}^2$ with Ω_X. The dual of Ω_X is the tangent sheaf ([2, Tangent Sheaf]), whose associated bundle is T_X. Hence $N_\delta = T_X$, completing the proof.

Two features of the statement are worth separating. The codimension is n, which is what makes $\delta^!$ lower dimension by exactly n and so makes the product of a k-cycle and an l-cycle a cycle of dimension $k + l - n$, the dimension a transverse intersection would have. The normal bundle is T_X, which is where the geometry of X enters the product: every formula below in which a Chern class of the tangent bundle appears is tracing back to this identification, and the self-intersection formula of section 5.3 is the first of them. Smoothness is used twice and cannot be weakened to either half alone: the local rings of $X \times X$ must be regular for Step 3, and Ω_X must be locally free for Step 2.

Example 5.1.7: The Diagonal of Affine Space

Let $X = \mathbb{A}^n = \operatorname{Spec} \overline{k}[t_1, \dots, t_n]$. Then $X \times X = \operatorname{Spec} \overline{k}[t_1, \dots, t_n, s_1, \dots, s_n] = \mathbb{A}^{2n}$, and the ideal of the diagonal is $J = (t_1 - s_1, \dots, t_n - s_n)$, so the generators of theorem 5.1.6 are visible globally: $u_i = t_i - s_i$. That they form a regular sequence needs no local algebra here, since $\overline{k}[t, s] = \overline{k}[t_1, \dots, t_n, u_1, \dots, u_n]$ is again a polynomial ring and each successive quotient $\overline{k}[t, u]/(u_1, \dots, u_i)$ is a polynomial ring, hence a domain, in which u_{i+1} is a nonzero element and therefore a nonzerodivisor. The conormal sheaf is free on the classes of the u_i, matching the basis $dt_1, \dots, dt_n$ of $\Omega_{\mathbb{A}^n}$ under lemma 5.1.5, and N_δ is the trivial bundle of rank n, which is $T_{\mathbb{A}^n}$.

The Chow groups are as small as possible: $A_k(\mathbb{A}^m) = 0$ for $k < m$ and $A_m(\mathbb{A}^m) = \mathbb{Z}[\mathbb{A}^m]$ by proposition 1.5.1. So the only product that can be nonzero is $[\mathbb{A}^n] \cdot [\mathbb{A}^n]$, and it is computed by the exterior product $[\mathbb{A}^n] \times [\mathbb{A}^n] = [\mathbb{A}^{2n}]$ followed by $\delta^!$. The intersection $\mathbb{A}^{2n} \cap \Delta$ is Δ itself, and $\Delta \hookrightarrow \mathbb{A}^{2n}$ is a regular embedding of codimension n, so proposition 4.3.3 gives $\delta^![\mathbb{A}^{2n}] = [\Delta]$, that is

$$[\mathbb{A}^n] \cdot [\mathbb{A}^n] \;=\; [\mathbb{A}^n]$$

the fundamental class reproducing itself.

The exterior product turns a pair of classes on X into one class on $X \times X$, and the diagonal is a regular embedding, so the Gysin map of definition 4.3.1 returns a class on X.

Definition 5.1.8: The Intersection Product

Let X be a separated smooth variety of dimension n over $\overline{k}$, with diagonal $\delta \colon X \hookrightarrow X \times X$, a regular embedding of codimension n by theorem 5.1.6. For $\alpha \in A_k(X)$ and $\beta \in A_l(X)$ the *intersection product* is

$$\alpha \cdot \beta \;=\; \delta^!\big(\alpha \times \beta\big) \;\in\; A_{k+l-n}(X)$$

where $\alpha \times \beta \in A_{k+l}(X \times X)$ is the exterior product of proposition 5.1.4 and $\delta^! \colon A_{k+l}(X \times X) \to A_{k+l-n}(X)$ is the Gysin map of definition 4.3.1.

The definition is the whole of the construction, and every property of the product is a property of one of its two inputs. It is defined for arbitrary classes, with no hypothesis on how representing cycles meet, because the Gysin map has none; and it depends only on the rational equivalence classes of the two factors, because both the exterior product and the Gysin map do. The dimension bookkeeping is the expected one: $k + l - n$ is the dimension of a transverse intersection of a k-dimensional and an l-dimensional subvariety of an n-dimensional variety, so in codimension the product adds. That grading, and the ring axioms it grades, are established in section 5.2; until then the product is exactly what definition 5.1.8 says it is.

Bilinearity and the vanishing below the expected dimension are inherited from the two inputs, and are recorded here for the computations that use them.

Proposition 5.1.9: Bilinearity and Vanishing in Low Dimension

Let X be a separated smooth variety of dimension n.

1. The product $A_k(X) \times A_l(X) \to A_{k+l-n}(X)$ is bilinear.
2. If $k + l < n$ then $\alpha \cdot \beta = 0$ for all $\alpha \in A_k(X)$ and $\beta \in A_l(X)$.

***Proof*:**

1. The exterior product is bilinear by proposition 5.1.4(2) and $\delta^!$ is a homomorphism of abelian groups by proposition 4.3.2(1), so the composite is bilinear in each variable.
2. The class $\alpha \times \beta$ lies in $A_{k+l}(X \times X)$ with $k + l < n$, and the codimension of δ is n, so proposition 4.3.2(2) gives $\delta^!(\alpha \times \beta) = 0$. Equivalently the target group $A_{k+l-n}(X)$ is a Chow group in negative dimension and vanishes, as we sought to show.

The product of definition 5.1.8 records its answer on all of X, and that discards information the construction has available. If α is carried by a closed subscheme V and β by a closed subscheme W, the intersection ought to be carried by $V \cap W$, and the refined Gysin homomorphism of definition 4.3.6 delivers exactly that, because the preimage of $V \times W$ under the diagonal is the scheme-theoretic intersection. Indeed the ideal sheaf of $V \times W$ in $X \times X$ is generated by the pullbacks of $\mathcal{I}_V$ and $\mathcal{I}_W$ along the two projections, and composing either projection with δ is the identity, so that ideal pulls back to $\mathcal{I}_V + \mathcal{I}_W$, the ideal of $V \cap W$.

Definition 5.1.10: The Refined Intersection Product

Let X be a separated smooth variety of dimension n and let $V, W \subseteq X$ be closed subschemes, with $g\colon V \times W \to X \times X$ the closed immersion. Since $\delta^{-1}(V \times W) = V \cap W$, the refined Gysin homomorphism of definition 4.3.6 along g is a homomorphism

$$\delta^! \;:\; A_m(V \times W) \longrightarrow A_{m-n}(V \cap W)$$

For $\alpha \in A_k(V)$ and $\beta \in A_l(W)$ the *refined intersection product* is

$$\alpha \cdot_X \beta \;=\; \delta^!\big(\alpha \times \beta\big) \;\in\; A_{k+l-n}(V \cap W)$$

Proposition 5.1.11: The Refined Product Determines the Product

Let X be a separated smooth variety of dimension n, let $V, W \subseteq X$ be closed subschemes with inclusions i_V and i_W, and let $\iota\colon V \cap W \hookrightarrow X$ be the inclusion. For all $\alpha \in A_k(V)$ and $\beta \in A_l(W)$,

$$\iota_*\big(\alpha \cdot_X \beta\big) \;=\; \big(i_{V*}\alpha\big) \cdot \big(i_{W*}\beta\big) \;\in\; A_{k+l-n}(X)$$

***Proof*:**
Apply theorem 4.3.10(1) to the regular embedding δ with $X' = X \times X$, $X'' = V \times W$ and $p = g$ the closed immersion, which is proper. The preimage of $\delta(X)$ in X' is $\delta(X)$ itself and the preimage in X'' is $V \cap W$, so the induced morphism q of preimages is ι, read through the isomorphism $\delta\colon X \to \delta(X)$. The theorem gives

$$\delta^!\big(g_*\gamma\big) \;=\; \iota_*\big(\delta^!\gamma\big)$$

for $\gamma \in A_{k+l}(V \times W)$, where the left-hand $\delta^!$ is the refined map along $\mathrm{id}_{X\times X}$, which is the Gysin map of definition 4.3.1 by proposition 4.3.7. Take $\gamma = \alpha \times \beta$. By proposition 5.1.4(3) applied to the closed immersions i_V and i_W,

$$g_*(\alpha \times \beta) \;=\; i_{V*}\alpha \times i_{W*}\beta$$

so the left-hand side is $\delta^!\big(i_{V*}\alpha \times i_{W*}\beta\big)$, which is $(i_{V*}\alpha) \cdot (i_{W*}\beta)$ by definition 5.1.8, and the right-hand side is $\iota_*(\alpha \cdot_X \beta)$ by definition 5.1.10, as we sought to show.

The pattern is the one recorded in corollary 4.3.11 for a single subvariety: the class computed on the ambient scheme is the pushforward of a class computed on the intersection. The refined product is the finer object, and every localized intersection number in the classical sense is a degree taken of it; the product of definition 5.1.8 is its pushforward to X.

Example 5.1.12: Two Lines in the Projective Plane

Let $X = \mathbb{P}^2$, which is a smooth variety of dimension 2, and let $L = \{x_0 = 0\}$ and $L' = \{x_1 = 0\}$ be two distinct lines, meeting in the single reduced point $P = [0 : 0 : 1]$.

The exterior product. By definition 5.1.2, $[L] \times [L'] = [L \times L']$, and $L \times L' \cong \mathbb{P}^1 \times \mathbb{P}^1$ is a variety of dimension 2 by the argument of example 5.1.3, so the exterior product is the class of a single subvariety of $\mathbb{P}^2 \times \mathbb{P}^2$ with multiplicity one.

The Gysin pullback. The intersection of $L \times L'$ with the diagonal is $\delta^{-1}(L \times L') = L \cap L' = \{P\}$, a reduced point. In the affine chart of $L \times L'$ around (P, P), with coordinates y on $L \setminus \{x_2 = 0\}$ and z on $L' \setminus \{x_2 = 0\}$, that point is the origin of $\mathbb{A}^2$, cut out by the regular sequence y, z; so $\{P\} \hookrightarrow L \times L'$ is a regular embedding of codimension $2 = \mathrm{codim}(\delta)$. Proposition 4.3.3 therefore applies to δ and the subvariety $L \times L'$ and gives

$$[L] \cdot [L'] \;=\; \delta^! \big[L \times L'\big] \;=\; [P] \;\in\; A_0(\mathbb{P}^2)$$

The refined form of definition 5.1.10 records a class on $L \cap L' = \{P\}$, and that class is the generator of $A_0(\{P\}) = \mathbb{Z}$: by proposition 5.1.11 its image in $A_0(\mathbb{P}^2)$ is $[P]$, and the pushforward $A_0(\{P\}) \to A_0(\mathbb{P}^2)$ carries generator to generator.

The self-intersection. By theorem 1.5.2 the two lines have the same class, $[L] = [L'] = h_1$ in $A_1(\mathbb{P}^2)$. Since the product is defined on classes, the computation just made also evaluates

$$[L] \cdot [L] \;=\; h_1 \cdot h_1 \;=\; [P]$$

No cycle was perturbed to get this, and no representative of h_1 was required to meet L properly: the two factors were replaced by equal classes, which proposition 5.1.4(2) and proposition 4.3.2(1) permit. The refined product, by contrast, does depend on the representatives: computed on L against L it lives on $A_0(L)$ rather than on $A_0(\{P\})$, and evaluating it there is the business of section 5.3.

Exercise 5.1.1

1. Let $P = \operatorname{Spec} \overline{k}$ and let X be an algebraic scheme. Show that $A_0(P) = \mathbb{Z}[P]$, and that under the canonical isomorphism $X \times P \cong X$ the exterior product of proposition 5.1.4 sends $(\alpha, m[P])$ to $m\alpha$. Then show that the exterior product of cycles is associative: for algebraic schemes X, Y, T and cycles α, β, γ on them, $(\alpha \times \beta) \times \gamma = \alpha \times (\beta \times \gamma)$ under the identification $(X \times Y) \times T = X \times (Y \times T)$.

2. Let $S = \mathbb{P}^1 \times \mathbb{P}^1$ with coordinates $([a : b], [c : d])$, let $\Delta \subseteq S$ be the diagonal, and let $F = \mathbb{P}^1 \times \{[1 : 0]\}$ and $G = \{[0 : 1]\} \times \mathbb{P}^1$. Show that

$$[\Delta] \;=\; [\mathbb{P}^1] \times [\mathrm{pt}] + [\mathrm{pt}] \times [\mathbb{P}^1] \quad \text{in } A_1(S)$$

   by computing the divisor of the rational function $f = (ad - bc)/(ad)$ on S chart by chart.

3. Let $X = \operatorname{Spec} \overline{k}[x, y]/(y^2 - x^3)$, a curve with a cusp at the origin p. Using lemma 5.1.5 and the presentation $\Omega_X = \big(\mathcal{O}_X\, dx \oplus \mathcal{O}_X\, dy\big)/\big(d(y^2 - x^3)\big)$, show that the conormal sheaf of the diagonal of X is not locally free of rank one, and deduce that $\delta \colon X \to X \times X$ is not a regular embedding of codimension one. Where does the proof of theorem 5.1.6 break down for this X?

4. Let X be a separated smooth variety of dimension n and let $V, W \subseteq X$ be closed subschemes with $V \cap W = \emptyset$. Show that $[V] \cdot [W] = 0$ in $A_*(X)$, where $[V]$ and $[W]$ denote the

images in $A_*(X)$ of the fundamental cycles. Give a second proof of the vanishing when $\dim V + \dim W < n$ that does not use disjointness.

5. Fix $0 \leq a, b \leq n$ with $a + b \geq n$, and let $L = \{x_{a+1} = \cdots = x_n = 0\}$ and $M = \{x_0 = \cdots = x_{n-b-1} = 0\}$ be coordinate subspaces of $\mathbb{P}^n$, so that $L \cong \mathbb{P}^a$, $M \cong \mathbb{P}^b$ and $L \cap M \cong \mathbb{P}^{a+b-n}$. Show that $L \cap M \hookrightarrow L \times M$ is a regular embedding of codimension n, and deduce from proposition 4.3.3 that $h_a \cdot h_b = h_{a+b-n}$ in $A_*(\mathbb{P}^n)$, where h_m is the class of a linear $\mathbb{P}^m$ as in example 4.3.4.

5.2 The Chow Ring

Section 5.1 produced a biadditive pairing on the Chow groups of a smooth variety. This section proves that the pairing obeys the three laws that make it a ring, and it indexes the groups by codimension so that the degrees add rather than subtract. The ring is then made contravariant: a morphism of smooth varieties pulls classes back, and so does a local complete intersection morphism between arbitrary algebraic schemes, which is the form Grothendieck-Riemann-Roch will need.

Two ingredients are imported rather than built. One is a pair of statements about regular embeddings that belong to scheme theory; the other is the commutativity of two Gysin maps, which chapter 4 flagged and did not prove. Both are isolated below, so that everything resting on them is visible at a glance.

5.2.1 Convention: Subscripts on Refined Gysin Maps When several refinements of one Gysin map occur in the same formula, the morphism along which the map is refined is written as a subscript: $i^!_g$ is the refined homomorphism of definition 4.3.6 for the morphism g, and an unadorned $i^!$ is the Gysin map of definition 4.3.1.

One reading has to be fixed first. Definition 4.3.1 is stated in codimension $d \geq 1$, whereas a regular embedding of codimension 0 is an open and closed immersion, its ideal sheaf vanishing locally. For such an $i\colon Z \hookrightarrow P$ both $i^!$ and its refinements $i^!_g$ mean flat pullback throughout this section, along i and along the induced immersion $g^{-1}(Z) \hookrightarrow P'$ for $g\colon P' \to P$; this is the reading definition 5.2.11 uses again. When the case arises from a diagonal or a graph the smooth variety in question has dimension 0, hence is $\operatorname{Spec} \overline{k}$: the embedding is then the identity under the canonical identification of $W \times \operatorname{Spec} \overline{k}$ with W, its flat pullback is the identity on Chow groups, and every statement below in which that happens holds by inspection. The arguments are written for the remaining case.

A cycle class on a smooth n-dimensional variety carries two equally good indices, its dimension and its codimension, and the product adds the second while subtracting the first from $2n$. A ring wants its grading to be additive, so the codimension index is the one to record.

Definition 5.2.2: The Chow Ring

Let X be a smooth variety of dimension n. For $0 \le p \le n$ write

$$A^p(X) \;=\; A_{n-p}(X)$$

and call $A^p(X)$ the group of cycle classes of *codimension* p. The graded abelian group

$$A^*(X) \;=\; \bigoplus_{p=0}^{n} A^p(X)$$

equipped with the intersection product $\alpha \cdot \beta = \delta^!(\alpha \times \beta)$ of section 5.1, is the *Chow ring* of X. That the product makes it a ring is theorem 5.2.7.

The two indices are interchangeable data, and both are used: a statement about the geometry of cycles is usually cleaner in dimension, a statement about the ring is cleaner in codimension. The class $[X]$ lies in $A^0(X)$, a divisor class lies in $A^1(X)$, and the class of a point on a complete X lies in $A^n(X)$.

Projective space is the first ring that can be written down, and it can be written down before the ring axioms are available, because each product in it is a single application of proposition 4.3.3.

Example 5.2.3: The Chow Ring of Projective Space

Let $X = \mathbb{P}^n$ with homogeneous coordinates $x_0, \ldots, x_n$. By theorem 1.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ with h_k the class of a linear subspace $\mathbb{P}^k$, so $A^p(\mathbb{P}^n) = \mathbb{Z}\, h_{n-p}$ is infinite cyclic for $0 \le p \le n$ and zero otherwise.

Fix $p, q \ge 0$ with $p + q \le n$ and take the linear subspaces

$$L = \{x_0 = \cdots = x_{p-1} = 0\}, \qquad L' = \{x_p = \cdots = x_{p+q-1} = 0\}$$

of dimensions $n-p$ and $n-q$, so that $[L] = h_{n-p}$ and $[L'] = h_{n-q}$. Their product is $\delta^!\big([L] \times [L']\big) = \delta^![L \times L']$, and proposition 4.3.3 evaluates it once the scheme-theoretic intersection $W = (L \times L') \cap \Delta$ is known to be regularly embedded in $L \times L'$ with codimension n. Set-theoretically W consists of the pairs (z, z) with $z \in L \cap L'$, so $W \cong L \cap L' = \{x_0 = \cdots = x_{p+q-1} = 0\} \cong \mathbb{P}^{n-p-q}$.

Work on the chart $\{x_n \neq 0\}$ of each factor, which meets W because the coordinate point e_n lies in $L \cap L'$. Writing $u_i = x_i/x_n$ for the affine coordinates on the first factor and $v_i = x_i/x_n$ for those on the second, the chart of $L \times L'$ is the affine space with coordinate ring

$$\overline{k}\big[u_p, \ldots, u_{n-1};\ v_0, \ldots, v_{p-1},\ v_{p+q}, \ldots, v_{n-1}\big]$$

the coordinates $u_0, \ldots, u_{p-1}$ vanishing on L and $v_p, \ldots, v_{p+q-1}$ on L'. The diagonal is cut out on $\mathbb{P}^n \times \mathbb{P}^n$ in this chart by the n equations $u_i = v_i$ with $0 \le i \le n-1$, and restricting them to $L \times L'$ leaves

$$v_0,\ \ldots,\ v_{p-1}, \qquad u_p,\ \ldots,\ u_{p+q-1}, \qquad u_i - v_i \ \ (p+q \le i \le n-1)$$

a list of n linear forms in distinct variables. Each is a nonzerodivisor modulo its predecessors, every successive quotient being a polynomial ring, so the list is a regular sequence and $W \hookrightarrow L \times L'$ is a regular embedding of codimension n; the quotient is the polynomial ring on the remaining $n-p-q$ variables, so W is reduced and irreducible with fundamental cycle $[\mathbb{P}^{n-p-q}]$. The same computation runs on every chart meeting W. Proposition 4.3.3 therefore gives

$$h_{n-p} \cdot h_{n-q} \;=\; h_{n-p-q}$$

For $p+q > n$ the product lands in $A^{p+q}(\mathbb{P}^n) = A_{n-p-q}(\mathbb{P}^n) = 0$ and is zero, by proposition 4.3.2(2). Writing $h = h_{n-1}$ for the class of a hyperplane, induction on p gives $h^p = h_{n-p}$, and

$$A^*(\mathbb{P}^n) \;\cong\; \mathbb{Z}[h]/(h^{n+1})$$

once the ring axioms below license the power notation.

No general position argument entered the computation. The two subspaces were chosen in special position relative to the coordinates but in general position relative to each other, and proposition 4.3.2 guarantees that any other pair of representatives, including two equal ones, returns the same product. What the example does not show is that the answer depends on the two classes through a ring structure, and that is the content of the rest of the section.

The first import is scheme-theoretic. Both of its statements concern regular embeddings and smooth morphisms rather than cycles, so both belong to the scheme theory upstream, where they are now proved. Neither is re-derived here.

Lemma 5.2.4: Sections, Flat Base Change, and Graphs

1. Let $\pi\colon P \to S$ be a smooth morphism of relative dimension $e \geq 0$ and $s\colon S \to P$ a section of it. Then s is a regular embedding of codimension e with normal bundle $s^*T_{P/S}$.
2. Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension d and $g\colon P' \to P$ a flat morphism. Then $g^{-1}(Z) \hookrightarrow P'$ is a regular embedding of codimension d, with normal bundle the pullback of the normal bundle of i.
3. Let X be a smooth variety of dimension $n \geq 0$, let V be an algebraic scheme, and let $f\colon V \to X$ be a morphism. Then the scheme-theoretic preimage of the diagonal under $f \times \mathrm{id}_X\colon V \times X \to X \times X$ is the graph Γ_f, the first projection restricts to an isomorphism $\Gamma_f \to V$, and $\gamma_f\colon V \to V \times X$ is a regular embedding of codimension n with normal bundle f^*T_X.

***Proof*:**
Parts (1) and (2) are upstream results and are cited rather than re-derived, both being statements about regular embeddings as such: part (1) is [2, A Section of a Smooth Morphism Is a Regular Embedding] and part (2) is [2, Regular Embeddings Pull Back Along Flat Morphisms]. Part (3) follows from part (1). A T-valued point of $(f \times \mathrm{id}_X)^{-1}(\Delta)$ is a pair (t_V, t_X) of T-points of V and X with $f \circ t_V = t_X$, so the functor of points of the preimage is that of V, and the isomorphism is induced by the first projection, with inverse γ_f. The projection $\mathrm{pr}_1\colon V \times X \to V$ is the base change of $X \to \operatorname{Spec} \overline{k}$, hence smooth of relative dimension n, and γ_f is a section of it, so part (1) applies. Its relative tangent bundle is $\mathrm{pr}_2^* T_X$, whose pullback along γ_f is f^*T_X, completing the proof.

Part (3) extends the diagonal computation of section 5.1 from the identity morphism to an arbitrary one, and it is the reason the intersection product can be transported along a map. Taking $V = X$ and $f = \mathrm{id}_X$ returns the diagonal.

The second import is about Gysin maps. That two regular embeddings can be intersected against in either order is what makes the product associative, and that the Gysin map of a composite of two regular embeddings factors is what makes the pullback along a local complete intersection morphism well defined. Both are proved in section 4.4, as corollary 4.4.8 and theorem 4.4.7, and are used here without further comment; the section identity theorem 4.4.2 of the same section is the third.

This theorem and parts (1) and (2) of lemma 5.2.4 are the only results the arguments below use whose proofs are cited rather than given. On the theorem rest the associativity of the intersection product, the identity of theorem 5.2.10(1) between intersecting with the graph and pulling back, and with it the projection formula, the multiplicativity and functoriality of pullback, and the independence of the local complete intersection pullback from its factorization. On the lemma rests everything that passes through a graph: its part (3) is deduced from part (1), so the unit law, definition 5.2.8 and all four parts of theorem 5.2.10 rest on it, and Step 1 of theorem 5.2.12(1) uses parts (1) and (2) directly. Commutativity of the product itself and the grading rest on neither.

The last preparation is bookkeeping. A Gysin map applied to an exterior product in which one factor is inert should return that factor untouched.

Lemma 5.2.5: Gysin Maps and Exterior Products

Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$, let W be an algebraic scheme, and let $\mathrm{pr}\colon P \times W \to P$ and $\mathrm{pr}'\colon W \times P \to P$ be the projections, whose scheme-theoretic preimages of Z are $Z \times W$ and $W \times Z$. Then for all $\xi \in A_k(P)$ and $\gamma \in A_l(W)$,

$$i^!_{\mathrm{pr}}(\xi \times \gamma) = (i^!\xi) \times \gamma, \qquad i^!_{\mathrm{pr}'}(\gamma \times \xi) = \gamma \times (i^!\xi)$$

***Proof*:**
Both sides are additive in γ, so it suffices to treat $\gamma = [W']$ for a subvariety $W' \subseteq W$ of dimension l, with inclusion $\iota\colon W' \hookrightarrow W$. Write $u = \mathrm{id}_P \times \iota\colon P \times W' \to P \times W$, a closed immersion and hence proper, and $\rho\colon P \times W' \to P$ for the projection. As W' is a variety of dimension l over $\overline{k}$, the structure morphism $W' \to \operatorname{Spec} \overline{k}$ is flat with l-dimensional fibre, so its base change ρ is flat of relative dimension l; and $\rho = \mathrm{pr} \circ u$, so u is a morphism over P.

Step 1: the exterior product as a pullback followed by a pushforward. For a subvariety $V \subseteq P$ the preimage $\rho^{-1}(V)$ is $V \times W'$, so $\rho^*[V] = [V \times W']$ by definition 1.3.3, and $u_*[V \times W'] = [V \times W']$ by definition 1.2.1, the immersion u carrying $V \times W'$ isomorphically onto its image. Hence $\xi \times [W'] = u_*(\rho^*\xi)$ for every ξ. The same computation on Z gives $q^*\eta = \eta \times [W']$ for the projection $q\colon Z \times W' \to Z$ and every $\eta \in A_*(Z)$.

Step 2: the two compatibilities. Theorem 4.3.10(2), applied with $X' = P$ and the identity, $X'' = P \times W'$ and ρ, and with ρ itself as the morphism over P, gives

$$i^!_{\rho}(\rho^*\xi) \;=\; q^*(i^!\xi) \;=\; (i^!\xi) \times [W']$$

where proposition 4.3.7 was used to read the refined map along the identity as $i^!$. Theorem 4.3.10(1), applied with $X' = P \times W$ and pr, $X'' = P \times W'$ and ρ, and with the proper morphism u over P, gives

$$i^!_{\mathrm{pr}}(u_*\eta) \;=\; q'_*(i^!_{\rho}\eta)$$

for the closed immersion $q' \colon Z \times W' \hookrightarrow Z \times W$ induced by u. Combining the two with Step 1, and using $q'_*\big(\zeta \times [W']\big) = \zeta \times [W']$ by definition 1.2.1 again, yields the first identity. Exchanging the two factors throughout gives the second, as we sought to show.

One comparison is used repeatedly below: when the preimage of Z is as well behaved as Z itself, the refined map is the plain one.

Proposition 5.2.6: Refined Gysin Maps with a Regular Preimage

Let $i \colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$ and $g \colon P' \to P$ a morphism of algebraic schemes such that $Z' = g^{-1}(Z)$ is regularly embedded in P' with codimension d. Then the refined Gysin homomorphism $i^!_g$ of definition 4.3.6 is the Gysin map $(i')^!$ of definition 4.3.1 for $i' \colon Z' \hookrightarrow P'$.

***Proof*:**
Write $C = C_{Z'}P'$ and $N' = h^*N$ as in definition 4.3.6, and let $j \colon C \hookrightarrow N'$ be the immersion of lemma 3.3.13. By part (1) of that lemma j is an isomorphism, and it is a morphism of cones over Z', so the projections satisfy $\pi_{N'} \circ j = \pi_C$. Since i' is a regular embedding, proposition 3.3.11 identifies C with the normal bundle of i', and definition 4.3.1 reads $(i')^! = (\pi_C^*)^{-1} \circ \sigma'$. Flat pullback along the isomorphism j is inverse to proper pushforward along it, each carrying a subvariety to its preimage or image with multiplicity one, so $j_*\pi_C^* = j_*j^*\pi_{N'}^* = \pi_{N'}^*$ by proposition 1.3.5(3). Hence $(\pi_{N'}^*)^{-1} \circ j_* = (\pi_C^*)^{-1}$, and composing with σ' turns definition 4.3.6 into definition 4.3.1, as we sought to show.

Theorem 5.2.7: The Chow Ring of a Smooth Variety

Let X be a smooth variety of dimension n, with diagonal $\delta \colon X \hookrightarrow X \times X$. Then, for all $\alpha, \beta, \gamma \in A^*(X)$:

1. the product is biadditive and $A^p(X) \cdot A^q(X) \subseteq A^{p+q}(X)$;
2. $\alpha \cdot \beta = \beta \cdot \alpha$;
3. $(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$;
4. $[X] \cdot \alpha = \alpha$.

Thus $A^*(X)$ is a commutative graded ring with unit $[X]$.

***Proof*:**

1. Biadditivity is proposition 5.1.9(1). For the degrees, let $\alpha \in A^p(X) = A_{n-p}(X)$ and $\beta \in A^q(X) = A_{n-q}(X)$. Their exterior product lies in $A_{2n-p-q}(X \times X)$, and δ is a regular embedding of codimension n (theorem 5.1.6), so $\delta^!$ lowers dimension by n by proposition 4.3.2(1). The product therefore lies in $A_{n-p-q}(X) = A^{p+q}(X)$.
2. Let $\tau \colon X \times X \to X \times X$ exchange the two factors. It carries a product of subvarieties $V \times W$ isomorphically onto $W \times V$, so $\tau_*(\alpha \times \beta) = \beta \times \alpha$ by definition 1.2.1, the exchange inducing an isomorphism of function fields. Since τ is an involution fixing the diagonal pointwise,

$\tau^{-1}(\Delta) = \Delta$, which is regularly embedded in $X \times X$ with codimension n; proposition 5.2.6 therefore identifies $\delta^!_\tau$ with $\delta^!$. Applying theorem 4.3.10(1) with $X' = X \times X$ and the identity, $X'' = X \times X$ and τ, and with τ itself as the morphism over $X \times X$, which is proper because it is an isomorphism, and noting that the induced morphism of preimages is $\tau|_\Delta = \mathrm{id}_\Delta$,

$$\delta^!\big(\tau_*\xi\big) \;=\; \delta^!_\tau(\xi) \;=\; \delta^!(\xi)$$

for every $\xi \in A_*(X \times X)$. With $\xi = \alpha \times \beta$ this reads $\beta \cdot \alpha = \alpha \cdot \beta$.

3. Write $\Xi = \alpha \times \beta \times \gamma \in A_*(X \times X \times X)$, which is unambiguous because the exterior product is associative on cycles, and let $\mathrm{pr}_{12}, \mathrm{pr}_{23} \colon X \times X \times X \to X \times X$ be the two projections. Their preimages of the diagonal are $W_1 = \Delta \times X$ and $W_2 = X \times \Delta$, each identified with $X \times X$ by the two projections that are not constrained; under those identifications $\mathrm{pr}_{23}|_{W_1}$ and $\mathrm{pr}_{12}|_{W_2}$ are the identity of $X \times X$, so the refined maps along them are $\delta^!$ by proposition 4.3.7. By lemma 5.2.5, applied to $i = \delta$ with $W = X$ on each side in turn,

$$(\alpha \cdot \beta) \times \gamma = \delta^!_{\mathrm{pr}_{12}}(\Xi), \qquad \alpha \times (\beta \cdot \gamma) = \delta^!_{\mathrm{pr}_{23}}(\Xi)$$

Applying $\delta^!$ to each and using corollary 4.4.8 with $i = i' = \delta$, $W = X \times X \times X$ and the two projections,

$$(\alpha \cdot \beta) \cdot \gamma = \delta^!\big(\delta^!_{\mathrm{pr}_{12}}\Xi\big) = \delta^!\big(\delta^!_{\mathrm{pr}_{23}}\Xi\big) = \alpha \cdot (\beta \cdot \gamma)$$

both sides being computed on $W_{12} = (\Delta \times X) \cap (X \times \Delta)$, the small diagonal, which the first projection identifies with X.

4. By additivity it suffices to take $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$, and by part (2) it suffices to evaluate $[V] \cdot [X] = \delta^![V \times X]$, the exterior product of the two fundamental cycles being the fundamental cycle of the variety $V \times X$ of dimension $k + n$ (definition 5.1.2). Apply lemma 5.2.4(3) to the inclusion $f \colon V \hookrightarrow X$: the scheme-theoretic intersection $(V \times X) \cap \Delta$, which is the preimage of Δ under $f \times \mathrm{id}_X$, is the graph Γ_f, isomorphic to V and regularly embedded in $V \times X$ with codimension n. Proposition 4.3.3 therefore gives $\delta^![V \times X] = \iota_*[\Gamma_f] = [V]$ in $A_k(\Delta) = A_k(X)$, where ι is the inclusion of the graph in the diagonal, which the projections identify with the inclusion of V in X, completing the proof.

The ring exists, and its two ends are already familiar: $A^0(X) = \mathbb{Z}\,[X]$ because a variety has only itself as a subvariety of top dimension, and $A^1(X) = A_{n-1}(X)$ is the group of divisor classes on which chapter 2 built the operator $D \cdot -$ by hand. The operators of section 3.2 can now be recorded as elements: for a vector bundle E on X the class $c_i(E) \cap [X]$ lies in $A^i(X)$, so Chern classes become members of a ring rather than endomorphisms of a graded group, and an identity of operators may be read as an identity in $A^*(X)$ by evaluating both sides on $[X]$.

The grading also explains an absence. On $\mathbb{P}^3$ two distinct lines meeting in a point have classes in A^2, so their product lies in $A^4(\mathbb{P}^3) = 0$: the ring records no trace of the point in which they meet. The computation is not defective. The intersection is not proper, one of the two lines can be moved off the other within its rational equivalence class, and the invariant answer to “how much do they meet” is zero.

A morphism $f \colon X \to Y$ of smooth varieties does not act on cycles in either direction in any naive way, but its graph is a regular embedding by lemma 5.2.4(3), and intersecting with the graph is a pullback.

Definition 5.2.8: Pullback Along a Morphism of Smooth Varieties

Let $f\colon X \to Y$ be a morphism of smooth varieties, of dimensions m and n, and let $\gamma_f\colon X \to X \times Y$ be its graph, a regular embedding of codimension n by lemma 5.2.4(3); for $n = 0$, where $Y = \operatorname{Spec} \overline{k}$ and γ_f is an isomorphism, $\gamma_f^!$ is flat pullback along it. The *pullback* along f is

$$f^*\colon A_l(Y) \longrightarrow A_{l+m-n}(X), \qquad f^*\beta \;=\; \gamma_f^!\big([X] \times \beta\big)$$

Equivalently $f^*\beta = \gamma_f^!\big(\mathrm{pr}_Y^*\beta\big)$, since $\mathrm{pr}_Y\colon X \times Y \to Y$ is flat of relative dimension m with $\mathrm{pr}_Y^*\beta = [X] \times \beta$.

In codimension the map preserves degree: $\beta \in A^q(Y)$ means $l = n - q$, and then $l + m - n = m - q$, so $f^*\beta \in A^q(X)$. The definition is the one the geometry suggests. To pull a cycle on Y back to X, place it on $X \times Y$, where it is inert in the first factor, and intersect with the graph, which is the locus where the second coordinate is the image of the first.

Example 5.2.9: Pullback to a Linear Subspace

Let $i\colon \mathbb{P}^m \hookrightarrow \mathbb{P}^n$ be the linear subspace $\{x_0 = \cdots = x_{n-m-1} = 0\}$ with $1 \le m < n$, and write h and h' for the hyperplane classes of $\mathbb{P}^n$ and $\mathbb{P}^m$. On the chart where $x_n \neq 0$ in both factors, with affine coordinates u_j on $\mathbb{P}^m$ for $n - m \le j \le n - 1$ and v_j on $\mathbb{P}^n$ for $0 \le j \le n - 1$, the graph $\Gamma_i = \{(z, z) \big| z \in \mathbb{P}^m\}$ is cut out of $\mathbb{P}^m \times \mathbb{P}^n$ by the n equations

$$v_j \quad (j < n - m), \qquad v_j - u_j \quad (j \ge n - m)$$

Fix $1 \le p \le m$ and represent h_{n-p} by $L = \{x_{n-m} = \cdots = x_{n-m+p-1} = 0\}$, a linear subspace of dimension $n - p$ meeting $\mathbb{P}^m$ in the linear subspace $\mathbb{P}^{m-p}$. Then $[\mathbb{P}^m] \times h_{n-p}$ is represented by the subvariety $\mathbb{P}^m \times L$, of dimension $m + n - p$, whose intersection with Γ_i is $\{(z, z) \big| z \in \mathbb{P}^{m-p}\}$. On the chart above, the coordinates $v_{n-m}, \ldots, v_{n-m+p-1}$ vanish on L, so restricting the n equations of the graph to $\mathbb{P}^m \times L$ leaves

$$v_j \quad (j < n - m), \qquad u_j \quad (n - m \le j < n - m + p), \qquad v_j - u_j \quad (j \ge n - m + p)$$

again n coordinates and coordinate differences in distinct variables, hence a regular sequence exactly as in example 5.2.3. The intersection is therefore regularly embedded of codimension n, and reduced and irreducible, so proposition 4.3.3 gives

$$i^*h_{n-p} \;=\; \big[\mathbb{P}^{m-p}\big] \;=\; h'_{m-p}$$

For $p > m$ the target $A_{m-p}(\mathbb{P}^m)$ vanishes and the pullback is zero. The restriction of the class of a linear subspace is the class of a linear subspace of the same codimension, which is what a subspace meeting $\mathbb{P}^m$ properly should give; in the notation of example 5.2.3 this reads $i^*(h^p) = (h')^p$ for $p \le m$.

Theorem 5.2.10: Properties of Pullback

Let $f\colon X \to Y$ be a morphism of smooth varieties, of dimensions m and n.

1. For $\alpha \in A_*(X)$ and $\beta \in A_*(Y)$,
$$\gamma_f^!\big(\alpha \times \beta\big) \;=\; \alpha \cdot f^*\beta$$
and in particular $f^*[Y] = [X]$.
2. If f is proper then $f_*\big(\alpha \cdot f^*\beta\big) = f_*\alpha \cdot \beta$ for all $\alpha \in A_*(X)$ and $\beta \in A_*(Y)$.
3. $f^*\colon A^*(Y) \to A^*(X)$ is a homomorphism of graded rings.
4. $\mathrm{id}_X^* = \mathrm{id}$, and $(g \circ f)^* = f^* \circ g^*$ for a further morphism $g\colon Y \to Z$ of smooth varieties.

***Proof*:**
Throughout, $\Gamma_f \subseteq X \times Y$ is the graph and $\gamma_f\colon X \to X \times Y$ the regular embedding of codimension n onto it (lemma 5.2.4(3)).

1. By lemma 5.2.5, applied to γ_f with an inert factor X on the left,
$$\alpha \times f^*\beta \;=\; \alpha \times \gamma_f^!\big([X] \times \beta\big) \;=\; (\gamma_f)^!_{\mathrm{pr}_{23}}\big(\alpha \times [X] \times \beta\big)$$
where $\mathrm{pr}_{23}\colon X \times X \times Y \to X \times Y$ is the projection, whose preimage of Γ_f is $X \times \Gamma_f$, identified with $X \times X$ by the first two projections. Applying $\delta^!$ and invoking corollary 4.4.8 for the regular embeddings δ and γ_f with the projections pr_{12} and pr_{23},
$$\alpha \cdot f^*\beta \;=\; \delta^!\Big((\gamma_f)^!_{\mathrm{pr}_{23}}\big(\alpha \times [X] \times \beta\big)\Big) \;=\; (\gamma_f)^!_\rho\Big(\delta^!_{\mathrm{pr}_{12}}\big(\alpha \times [X] \times \beta\big)\Big)$$
On the left the outer map is refined along $\mathrm{pr}_{12}|_{X \times \Gamma_f}$, which is the identity of $X \times X$ under the identification above, so it is $\delta^!$ by proposition 4.3.7. On the right the preimage of Δ under pr_{12} is $\Delta \times Y$, identified with $X \times Y$, and ρ is the restriction of pr_{23} to it, which is the identity of $X \times Y$; so the outer map is $\gamma_f^!$, again by proposition 4.3.7. The inner map is evaluated by lemma 5.2.5 and the unit law theorem 5.2.7(4):
$$\delta^!_{\mathrm{pr}_{12}}\big((\alpha \times [X]) \times \beta\big) = \big(\alpha \cdot [X]\big) \times \beta = \alpha \times \beta$$
which gives the stated identity. Taking $\alpha = [X]$ and $\beta = [Y]$, the left side is $\gamma_f^![X \times Y]$; the variety $X \times Y$ meets Γ_f in Γ_f itself, a regular embedding of codimension n, so proposition 4.3.3 gives $\gamma_f^![X \times Y] = [\Gamma_f] = [X]$. Hence $[X] \cdot f^*[Y] = [X]$, and $f^*[Y] = [X]$ by the unit law.
2. Let f be proper. On cycles, $(f \times \mathrm{id}_Y)_*\big(\alpha \times \beta\big) = f_*\alpha \times \beta$: for subvarieties $V \subseteq X$ and $W \subseteq Y$ the image of $V \times W$ is $f(V) \times W$, and the field extension $k(f(V) \times W) \subseteq k(V \times W)$ is obtained from $k(f(V)) \subseteq k(V)$ by tensoring with $k(W)$ over $\overline{k}$ and passing to fraction fields, so the two degrees agree and definition 1.2.1 applies. Now $(f \times \mathrm{id}_Y)^{-1}(\Delta_Y) = \Gamma_f$ by lemma 5.2.4(3), and Γ_f is regularly embedded of codimension n, so proposition 5.2.6 identifies $(\delta_Y)^!_{f \times \mathrm{id}}$ with $\gamma_f^!$. Applying theorem 4.3.10(1) with $X' = Y \times Y$ and the identity,

$X'' = X \times Y$ and $f \times \mathrm{id}_Y$, and with $f \times \mathrm{id}_Y$ as the proper morphism over $Y \times Y$, whose induced morphism of preimages is f itself,

$$f_*\alpha \cdot \beta \;=\; \delta_Y^!\big((f \times \mathrm{id}_Y)_*(\alpha \times \beta)\big) \;=\; f_*\Big(\gamma_f^!\big(\alpha \times \beta\big)\Big) \;=\; f_*\big(\alpha \cdot f^*\beta\big)$$

the last step by part (1).

3. Degrees were checked after definition 5.2.8, additivity is clear from the definition, and $f^*[Y] = [X]$ is part (1). For multiplicativity, let $\beta, \beta' \in A_*(Y)$ and put $\Xi = [X] \times \beta \times \beta' \in A_*(X \times Y \times Y)$. By lemma 5.2.5,

$$[X] \times \big(\beta \cdot \beta'\big) = (\delta_Y)^!_{\mathrm{pr}_{23}}(\Xi), \qquad \big(f^*\beta\big) \times \beta' = (\gamma_f)^!_{\mathrm{pr}_{12}}(\Xi)$$

with $\mathrm{pr}_{23} \colon X \times Y \times Y \to Y \times Y$ and $\mathrm{pr}_{12} \colon X \times Y \times Y \to X \times Y$. Applying $\gamma_f^!$ to the first identity gives $f^*(\beta \cdot \beta')$, the map being refined along $\mathrm{pr}_{12}|_{X \times \Delta_Y}$, which is the identity of $X \times Y$. Corollary 4.4.8, for the regular embeddings γ_f and δ_Y with the projections pr_{12} and pr_{23}, exchanges the order:

$$f^*\big(\beta \cdot \beta'\big) = \gamma_f^!\Big((\delta_Y)^!_{\mathrm{pr}_{23}}\Xi\Big) = (\delta_Y)^!_\psi\Big((\gamma_f)^!_{\mathrm{pr}_{12}}\Xi\Big)$$

where $\psi \colon \Gamma_f \times Y \to Y \times Y$ is the restriction of pr_{23}. The first and third projections identify $\Gamma_f \times Y$ with $X \times Y$, and under that identification ψ is $f \times \mathrm{id}_Y$, whose preimage of Δ_Y is Γ_f; so $(\delta_Y)^!_\psi = \gamma_f^!$ by proposition 5.2.6. Hence

$$f^*\big(\beta \cdot \beta'\big) = \gamma_f^!\big(f^*\beta \times \beta'\big) = f^*\beta \cdot f^*\beta'$$

the last equality by part (1) with $\alpha = f^*\beta$.

4. For $f = \mathrm{id}_X$ the graph is the diagonal, and part (1) with $\alpha = [X]$ gives $\mathrm{id}^*\beta = [X] \cdot \beta = \beta$. For the composite, let $\zeta \in A_*(Z)$ and put $\Theta = [X \times Y] \times \zeta \in A_*(X \times Y \times Z)$. By lemma 5.2.5 and $[X \times Y] = [X] \times [Y]$,

$$[X] \times g^*\zeta = [X] \times \gamma_g^!\big([Y] \times \zeta\big) = (\gamma_g)^!_{\mathrm{pr}_{23}}(\Theta)$$

with $\mathrm{pr}_{23} \colon X \times Y \times Z \to Y \times Z$, so $f^*g^*\zeta = \gamma_f^!\big((\gamma_g)^!_{\mathrm{pr}_{23}}\Theta\big)$, the outer map being read on $X \times \Gamma_g \cong X \times Y$. Corollary 4.4.8, for the regular embeddings γ_f and γ_g with the projections pr_{12} and pr_{23}, exchanges the order, giving $(\gamma_g)^!_\rho\big((\gamma_f)^!_{\mathrm{pr}_{12}}\Theta\big)$ with $\rho \colon \Gamma_f \times Z \to Y \times Z$ the restriction of pr_{23}. By lemma 5.2.5 and part (1),

$$(\gamma_f)^!_{\mathrm{pr}_{12}}\Theta = \gamma_f^![X \times Y] \times \zeta = [X] \times \zeta$$

Identifying $\Gamma_f \times Z$ with $X \times Z$, the morphism ρ becomes $f \times \mathrm{id}_Z$, whose preimage of Γ_g is $\{(x, z) \big| z = g(f(x))\} = \Gamma_{g \circ f}$, regularly embedded of codimension $\dim Z$ by lemma 5.2.4(3). So proposition 5.2.6 identifies $(\gamma_g)^!_\rho$ with $\gamma^!_{g \circ f}$, and

$$f^*g^*\zeta \;=\; \gamma^!_{g \circ f}\big([X] \times \zeta\big) \;=\; (g \circ f)^*\zeta$$

completing the proof.

Part (1) says that intersecting with the graph is the same as pulling back and then multiplying, which is what makes part (2) a statement about a ring rather than about a Gysin map. The projection formula is the identity used most often downstream: it converts a computation on X into a computation on Y whenever a class on X arose by restriction, and it is how an intersection number computed on a subvariety is compared with one computed on the ambient variety.

Smoothness of the source is more than the construction needs. What was used about X is that its graph in $X \times Y$ is regularly embedded, and by lemma 5.2.4(3) that holds for any algebraic scheme mapping to a smooth variety. Dropping smoothness of the target as well requires a hypothesis on the morphism instead: it must factor as a regular embedding followed by a smooth morphism.

Definition 5.2.11: Gysin Pullback for a Local Complete Intersection Morphism

Throughout this definition f is assumed to admit a *global* factorization $f = p \circ i$ with i a regular embedding and p smooth. The upstream definition ([2, Local Complete Intersection Morphism]) requires such a factorization only locally on the source, which is strictly weaker and does not suffice to fix a single i and p; the global hypothesis is Fulton's convention and is what the construction below needs. Let $f\colon X \to Y$ be a local complete intersection morphism of algebraic schemes ([2, Local Complete Intersection Morphism]) which admits a factorization on all of X, and fix one: $f = p \circ i$ with $i\colon X \hookrightarrow P$ a regular embedding of codimension d and $p\colon P \to Y$ smooth of relative dimension e. The *Gysin pullback* of f is

$$f^! \;=\; i^! \circ p^* \;:\; A_k(Y) \longrightarrow A_{k+e-d}(X)$$

where p^* is flat pullback (definition 1.3.3) and $i^!$ is the Gysin map of definition 4.3.1, read for $d = 0$ as flat pullback along i, which is then an open and closed immersion. The integer $e - d$ is the *virtual relative dimension* of f.

The factorization on all of X is a hypothesis, not a consequence. The definition just cited asks only that every point of X have a neighbourhood on which f factors in this way, and local factorizations need not be glued into one on all of X; without one there is no p^* and no $i^!$ to compose, and $f^!$ is not defined. Wherever $f^!$ is written a factorization on all of X is therefore part of the data. The cases used below carry an evident one: a regular embedding and a smooth morphism are their own factorizations, and a morphism of smooth varieties is factored by its graph.

Theorem 5.2.12: The Gysin Pullback Is Well Defined

Let $f\colon X \to Y$ be a local complete intersection morphism admitting a factorization on all of X.

1. $f^!$ and the virtual relative dimension $e - d$ are independent of the chosen factorization.
2. If f is smooth of relative dimension e then $f^!$ is flat pullback, and if f is a regular embedding then $f^!$ is its Gysin map.
3. If X and Y are smooth varieties then every morphism $f\colon X \to Y$ meets the hypotheses above, its graph supplying a factorization on all of X, and $f^!$ is the pullback f^* of definition 5.2.8.

***Proof*:**

1. Let $f = p \circ i = p' \circ i'$ be two factorizations, with data (P, i, p, d, e) and (P', i', p', d', e'). Form $P'' = P \times_Y P'$ with projections $q\colon P'' \to P$ and $q'\colon P'' \to P'$, and set $p'' = p \circ q = p' \circ q'$. Being base changes of p' and p, the morphisms q and q' are smooth of relative dimensions e' and e, so p'' is smooth of relative dimension $e + e'$. It suffices to prove $i^! \circ p^* = (i'')^! \circ (p'')^*$ for $i'' = (i, i')\colon X \to P''$, since the same argument with the roles of the two factorizations exchanged gives $i'^! \circ p'^* = (i'')^! \circ (p'')^*$.

   Step 1: factoring i''. Write $W_1 = q^{-1}(X) = X \times_Y P'$, a closed subscheme of P'', and let $\tilde{i}\colon W_1 \hookrightarrow P''$ be its inclusion; since q is flat, lemma 5.2.4(2) makes $\tilde{i}$ a regular embedding of codimension d. The projection $\bar{q}\colon W_1 \to X$ is the base change of p' along f, hence smooth of relative dimension e', and $\gamma = (\mathrm{id}_X, i')\colon X \to W_1$ is a section of it, hence a regular embedding of codimension e' by lemma 5.2.4(1). As $i'' = \tilde{i} \circ \gamma$, theorem 4.4.7 gives $(i'')^! = \gamma^! \circ \tilde{i}^!$ and makes i'' a regular embedding of codimension $d + e'$.

   Step 2: the comparison. Let $\beta \in A_k(Y)$. Flat pullback is functorial by proposition 1.3.5(3), so $(p'')^*\beta = q^*\big(p^*\beta\big)$. Applying theorem 4.3.10(2) to the regular embedding i with $X' = P$ and the identity, $X'' = P''$ and q, and with the flat q as the morphism over P, and using proposition 5.2.6 to read the refined map along q as $\tilde{i}^!$,

   $$\tilde{i}^!\big((p'')^*\beta\big) \;=\; \bar{q}^*\big(i^! p^*\beta\big)$$

   Since γ is a section of the smooth morphism $\bar{q}$, theorem 4.4.2 gives $\gamma^! \circ \bar{q}^* = \mathrm{id}$, whence

   $$(i'')^!(p'')^*\beta \;=\; \gamma^!\bar{q}^*\big(i^! p^*\beta\big) \;=\; i^! p^*\beta$$

   Step 3: the degree. Step 1 exhibits i'' as a regular embedding of codimension $d + e'$, and the same step applied to the other factorization exhibits it as one of codimension $d' + e$. The codimension of a regular embedding is determined by the rank of its conormal sheaf, so $d + e' = d' + e$ and therefore $e - d = e' - d'$.

2. If f is smooth, take $P = X$ with $i = \mathrm{id}_X$, a regular embedding of codimension zero, and $p = f$; then $f^! = \mathrm{id}^* \circ f^* = f^*$. If f is a regular embedding, take $P = Y$ with $p = \mathrm{id}_Y$, smooth of relative dimension zero, and $i = f$; then $f^!$ is the Gysin map of f. Part (1) makes both readings legitimate.

3. Let X and Y be smooth of dimensions m and n. Factor f as

   $$X \xrightarrow{\gamma_f} X \times Y \xrightarrow{\mathrm{pr}_Y} Y$$

   The first map is a regular embedding of codimension n by lemma 5.2.4(3), and the second is smooth of relative dimension m, being the base change of $X \to \operatorname{Spec} \overline{k}$. So f is a local complete intersection morphism with a factorization on all of X, and by part (1) its Gysin pullback may be computed from this factorization: $f^!\beta = \gamma_f^!\big(\mathrm{pr}_Y^*\beta\big)$, which is definition 5.2.8, completing the proof.

The virtual relative dimension is the bookkeeping device the name suggests: for a smooth morphism it is the actual relative dimension, for a regular embedding it is minus the codimension, and for a general local complete intersection morphism it is the difference of the two, which need not be realized by any fibre. That generality is what Grothendieck-Riemann-Roch is stated in, the morphisms there being proper local complete intersection morphisms between possibly singular schemes.

One classical statement remains to be placed. The construction in this chapter never moved a cycle; the ambient embedding was degenerated instead. The classical construction did the opposite, and it rested on the following.

Theorem 5.2.13: The Moving Lemma

Let X be a smooth quasi-projective variety of dimension n, let α be a k-cycle on X and β an l-cycle on X. Then there is a k-cycle α' rationally equivalent to α such that α' and β intersect properly: every irreducible component of $|\alpha'| \cap |\beta|$ has dimension $k+l-n$. Moreover the cycle class obtained by counting the components of $|\alpha'| \cap |\beta|$ with their intersection multiplicities depends only on the classes of α and β.

No proof is given, and none is used. The lemma is stated because it is what the intersection product looked like before chapter 4: a product defined for cycles in general position and extended to all pairs by first moving one of them, the whole construction resting on the second assertion above, that the answer does not depend on how the cycle was moved. That second assertion is the difficult half, and it is what makes the moving lemma a theorem rather than a definition; the argument goes back to Chow, and a complete treatment is in Fulton [8].

Two costs were paid on that route and are not paid here. The lemma requires quasi-projectivity, so the classical product exists only on quasi-projective varieties, whereas theorem 5.2.7 holds on every smooth variety. And a product defined by moving is defined only up to the choice made, so every property of it, commutativity included, has to be proved compatible with all choices. The deformation construction makes no choice: $\delta^!$ is a homomorphism on classes from the outset, and the proofs above use only its functorialities.

Exercise 5.2.1

1. Let L and L' be distinct lines in $\mathbb{P}^3$ meeting in a point. Compute $[L] \cdot [L']$ in $A^*(\mathbb{P}^3)$ from the grading alone, and compute $[L] \cdot [H]$ for a plane H containing neither line. Then explain why the first answer is not a defect of the theory, by exhibiting a cycle rationally equivalent to $[L]$ whose support misses L'.

2. Let X be a smooth variety and let $V, W \subseteq X$ be subvarieties with $V \cap W = \emptyset$. Show that $[V] \cdot [W] = 0$ in $A^*(X)$, working from the definition of the product rather than from the grading.

3. Let $f\colon X \to Y$ be a proper morphism of smooth varieties of the same dimension n, with $\deg f$ the degree of definition 1.2.1. Show that $f_*\big(f^*\beta\big) = (\deg f)\,\beta$ for every $\beta \in A^*(Y)$, and deduce that f^* is injective on $A^*(Y) \otimes \mathbb{Q}$ when $\deg f \neq 0$.

4. Let $p\colon P \to Y$ be smooth of relative dimension e and let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$. Show that $p \circ i$ is a local complete intersection morphism of virtual relative dimension $e - d$. Then take $Y = \operatorname{Spec} \overline{k}$, $P = \mathbb{P}^n$ and $Z = \mathbb{P}^{n-d}$ a linear subspace, and compute the image of the generator of $A_0(\operatorname{Spec} \overline{k})$ under $(p \circ i)^!$.

5. Let $i\colon \mathbb{P}^2 \hookrightarrow \mathbb{P}^3$ be a plane and let $D \subseteq \mathbb{P}^3$ be a surface of degree r not containing it, so that $[D] = r\,h$ in $A^1(\mathbb{P}^3)$ by example 1.5.4. Compute $i^*[D]$ in $A^1(\mathbb{P}^2)$, and verify the projection

formula $i_*\big(i^*[D]\big) = [D] \cdot [\mathbb{P}^2]$ by computing both sides in $A^*(\mathbb{P}^3)$.

5.3 Self-Intersection and Excess

The Gysin map returns the fundamental cycle of the intersection scheme when the intersection is as proper as it can be (proposition 4.3.3). At the opposite extreme, when the cycle being intersected already lies inside Z, nothing proved so far evaluates it at all: the second half of example 4.3.8 left exactly that computation open, and for the same reason proposition 4.3.5 reached only the first clause of definition 2.2.1. This section closes the gap. The answer is the top Chern class of the normal bundle, and the statement interpolating between the two extremes, in which the intersection has the wrong dimension by a prescribed amount, is the excess intersection formula.

Everything below is stated for the Gysin map of a regular embedding, so by section 5.1 it is at the same time a statement about the product: on a smooth variety the product is $\delta^!$ of an exterior product, and δ is a regular embedding like any other. The formulas are then spent on blow-ups, which is where a subvariety is forced to meet the locus it is intersected with in the worst possible way.

Throughout, $i\colon Z \hookrightarrow X$ is a regular embedding of codimension $d \geq 1$ of algebraic schemes, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ and normal bundle $\pi\colon N \to Z$.

The reduction to a single computation is immediate from the construction. Apply definition 4.3.6 to the morphism $g = i$ itself: the preimage of Z in Z is Z, so the normal cone in question is $C_Z Z$, the zero cone of example 3.3.2(2), and the immersion of that cone into N is the vertex, that is the zero section. Specialization is therefore the identity, and the whole matter is the identification of the class of the zero section inside the Chow group of the total space of the bundle.

Lemma 5.3.1: The Class of the Zero Section

Let $\pi\colon N \to Z$ be a vector bundle of rank $d \geq 1$ on an algebraic scheme Z, with zero section $\zeta\colon Z \to N$. Then for every $\alpha \in A_k(Z)$,

$$\zeta_*\alpha \;=\; \pi^*\big(c_d(N) \cap \alpha\big) \quad \text{in } A_k(N)$$

equivalently $(\pi^*)^{-1}\zeta_*\alpha = c_d(N) \cap \alpha$, the top Chern class of N applied to α.

***Proof*:**
Write $P = \mathbb{P}(N \oplus 1)$ for the projective completion of definition 3.3.3, where $N \oplus 1 = N \oplus \mathcal{O}_Z$ has rank $d+1$, with projection $p\colon P \to Z$ and tautological class $\eta = c_1(\mathcal{O}_P(1))$. Let $u\colon N \hookrightarrow P$ be the open immersion, so that $p \circ u = \pi$, and let $\bar{\zeta} = u \circ \zeta\colon Z \to P$, a section of p and hence a closed immersion. Fix $\alpha \in A_k(Z)$ and put $\beta = \bar{\zeta}_*\alpha \in A_k(P)$.

Step 1: two trivializations. Writing $S^\bullet = \mathrm{Sym}^\bullet(N^\vee)$, the completion is $P = \mathrm{Proj}\big(S^\bullet[z]\big)$ with z the degree-one variable dual to the trivial summand, and N is the chart $D_+(z)$. On that chart the sheaf $\mathcal{O}_P(1)$ is free on the generator z, so

$$\mathcal{O}_P(1)\big|_N \;\cong\; \mathcal{O}_N$$

Since $\bar{\zeta}$ factors through N, it follows that $\bar{\zeta}^*\mathcal{O}_P(1) \cong \mathcal{O}_Z$ as well. Finally, the exact sequence $0 \to N \to N \oplus 1 \to \mathcal{O}_Z \to 0$ and theorem 3.2.15 give $c(N \oplus 1) = c(N)\,c(\mathcal{O}_Z) = c(N)$, the total Chern class of a trivial bundle being the identity (example 3.2.12(1)); in particular $c_i(N \oplus 1) = c_i(N)$ for every i, and this vanishes for $i > d$ by theorem 3.2.9(2).

Step 2: η annihilates β. The morphism $\bar\zeta$ is proper, being a closed immersion, so the projection formula (proposition 2.4.4(4)) and Step 1 give

$$\eta \cap \bar\zeta_* \alpha \;=\; \bar\zeta_*\Big(c_1\big(\bar\zeta^* \mathcal{O}_P(1)\big) \cap \alpha\Big) \;=\; \bar\zeta_*\big(c_1(\mathcal{O}_Z) \cap \alpha\big) \;=\; 0$$

the last equality because c_1 of a trivial line bundle is the zero operator (example 2.4.3).

Step 3: the top coefficient. The bundle $N \oplus 1$ has rank $d+1$, so theorem 3.1.9 writes β uniquely as

$$\beta \;=\; \sum_{j=0}^{d} \eta^j \cap p^* \alpha_j, \qquad \alpha_j \in A_{k-d+j}(Z) \tag{5.1}$$

By definition 3.1.3 applied to $N \oplus 1$, whose projective bundle has relative dimension d, one has $p_*\big(\eta^j \cap p^*\gamma\big) = s_{j-d}(N \oplus 1) \cap \gamma$, which vanishes for $j < d$ and is γ for $j = d$ by proposition 3.1.5(1),(2). Applying p_* to (5.1) therefore gives $p_*\beta = \alpha_d$. On the other hand $p \circ \bar\zeta = \mathrm{id}_Z$, so functoriality of proper pushforward (definition 1.2.1) gives $p_*\beta = \alpha$. Hence $\alpha_d = \alpha$.

Step 4: the remaining coefficients. Theorem 3.2.9(1) applied to $N \oplus 1$, together with the identification of its Chern classes in Step 1, reads

$$\sum_{i=0}^{d} \eta^{d+1-i} \cap p^*\big(c_i(N) \cap \gamma\big) \;=\; 0$$

for every $\gamma \in A_*(Z)$, so that $\eta^{d+1} \cap p^*\gamma = -\sum_{i=1}^{d} \eta^{d+1-i} \cap p^*(c_i(N) \cap \gamma)$. Capping (5.1) with η, using Step 2 on the left and this relation on the term $j = d$, and reindexing the resulting sum by $j = d - i$,

$$0 \;=\; \sum_{j=0}^{d-1} \eta^{j+1} \cap p^* \alpha_j \;-\; \sum_{j=0}^{d-1} \eta^{j+1} \cap p^*\big(c_{d-j}(N) \cap \alpha_d\big) \;=\; \sum_{m=1}^{d} \eta^m \cap p^*\big(\alpha_{m-1} - c_{d-m+1}(N) \cap \alpha_d\big)$$

The exponents occurring lie in $\{0, 1, \ldots, d\}$, so this is a representation of the zero class in the form of theorem 3.1.9, and uniqueness forces every coefficient to vanish: $\alpha_{m-1} = c_{d-m+1}(N) \cap \alpha_d$ for $1 \le m \le d$. Taking $m = 1$ and using Step 3,

$$\alpha_0 \;=\; c_d(N) \cap \alpha$$

Step 5: restriction to the bundle. The open immersion u is flat of relative dimension zero. The square with $\bar\zeta$ and ζ over u is a fibre square, $\bar\zeta$ factoring through N, so proposition 1.3.6 gives $u^*\beta = \zeta_*\alpha$. Applying u^* to (5.1), every term with $j \ge 1$ contains a factor $\eta \cap -$, whose pullback is $c_1(\mathcal{O}_P(1)|_N) \cap u^*(-) = 0$ by proposition 2.4.4(4) and Step 1; and the term $j = 0$ is $u^* p^* \alpha_0 = \pi^* \alpha_0$ by proposition 1.3.5(3). Hence $\zeta_* \alpha = \pi^* \alpha_0 = \pi^*(c_d(N) \cap \alpha)$, as we sought to show.

The zero section meets every fibre of N in exactly one point, and yet the class it carries is not the pullback of α but the pullback of α cut down by the top Chern class. On a trivial bundle of rank $d \ge 1$ the top Chern class vanishes and the zero section therefore carries the zero class, and the same holds whenever N admits a nowhere-vanishing section: such a section spans a trivial line subbundle with locally free quotient Q of rank $d-1$, so theorem 3.2.15 gives $c(N) = c(Q)$ and $c_d(N) = 0$, no bundle carrying a nonzero Chern class above its rank. Nonvanishing of $\zeta_*\alpha$ is therefore an obstruction to the existence of such a section and not a criterion for one. On $\mathbb{P}^1$ the bundle $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$ has c_2 the zero operator, the groups $A_{k-2}(\mathbb{P}^1)$ vanishing for every $k \leq 1$, and yet $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no nonzero section, so every section of the sum vanishes wherever its second coordinate does. Since π^* is an isomorphism (theorem 1.4.6), the identity is equivalent to the evaluation of $(\pi^*)^{-1}$ on classes supported on the zero section, and that is what the Gysin map of chapter 4 was waiting for.

Theorem 5.3.2: The Self-Intersection Formula

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N. Then for every $\alpha \in A_k(Z)$:

1. the refined Gysin homomorphism of i along i itself is the operator $c_d(N) \cap -$, that is $i^!_i\alpha = c_d(N) \cap \alpha$ in $A_{k-d}(Z)$, in the notation of box 5.2.1;
2. the Gysin map of definition 4.3.1 satisfies

$$i^!(i_*\alpha) \;=\; c_d(N) \cap \alpha \quad \text{in } A_{k-d}(Z)$$

***Proof*:**

1. Apply definition 4.3.6 with $X' = Z$ and $g = i$. The ideal sheaf of Z in Z is zero, so the preimage $Z' = i^{-1}(Z)$ is Z itself, $h = \mathrm{id}_Z$ and $N' = h^*N = N$. The associated graded algebra of the zero ideal is $\mathcal{O}_Z$ concentrated in degree zero, so $C_{Z'}X' = C_Z Z$ is the zero cone of example 3.3.2(2), whose total space is Z; and the immersion $j\colon C_Z Z \hookrightarrow N$ of lemma 3.3.13 is induced by the surjection of graded algebras $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \to \mathcal{O}_Z$ killing every positive degree, which is the vertex of definition 3.3.1, that is the zero section ζ. For the specialization homomorphism $\sigma'\colon A_k(Z) \to A_k(C_Z Z) = A_k(Z)$ of the pair (Z, Z), theorem 4.2.7 gives $\sigma'[V] = [C_V V] = [V]$ for every k-dimensional subvariety $V \subseteq Z$, the preimage of Z in V being V. Such classes generate $A_k(Z)$ (definition 1.1.4), so σ' is the identity. Therefore

$$i^!_i\alpha \;=\; (\pi^*)^{-1} j_*\sigma'\alpha \;=\; (\pi^*)^{-1}\zeta_*\alpha \;=\; c_d(N) \cap \alpha$$

the last equality by lemma 5.3.1.

2. Apply theorem 4.3.10(1) with $X' = X$, $X'' = Z$ and $p = i$, which is proper as a closed immersion. The preimage of Z in Z is Z and the induced morphism q is id_Z, so the identity reads $i^!(i_*\alpha) = i^!_i\alpha$, where the left-hand side is the refined map along id_X, which is the Gysin map of definition 4.3.1 by proposition 4.3.7, and the right-hand side is the refined map along i evaluated in part (1). Combining the two gives $i^!(i_*\alpha) = c_d(N) \cap \alpha$, completing the proof.

The theorem says what it means to intersect a subvariety with itself. Pushing α into X and pulling it back records no information about how Z sits inside X except through the normal bundle, and the whole of that information is the single operator $c_d(N)$. Two consequences are worth separating. The composite $i^!i_*$ is not the identity, and it is not even injective: on a subvariety with trivial normal bundle it is zero. And the sign of a self-intersection number, which has no counterpart in a naive count of intersection points, is the sign of a Chern number of N; this is why the exceptional curve of a blow-up can meet itself a negative number of times. Since the Gysin pullback of an l.c.i. morphism restricts to the Gysin map on a regular embedding (theorem 5.2.12(2)), part (2) is at the same time a statement about the pullback of definition 5.2.11: for the l.c.i. morphism i the composite $i^! \circ i_*$ is the operator $c_d(N) \cap -$.

In codimension one the theorem completes the comparison begun in proposition 4.3.5, which matched the Gysin map with the operator $D \cdot -$ of chapter 2 only for cycles not contained in the support of D.

Corollary 5.3.3: The Gysin Map on a Divisor Class

Let D be an effective Cartier divisor on an algebraic scheme X with $D \neq X$, and let $i\colon D \hookrightarrow X$ be the inclusion. Then for every $\alpha \in A_k(D)$,

$$i^!(i_*\alpha) \;=\; c_1\big(\mathcal{O}_X(D)|_D\big) \cap \alpha$$

In particular, for a k-dimensional subvariety $V \subseteq D$ the class $i^![V]$ is the image in $A_{k-1}(D)$ of the intersection class $D \cdot [V]$ computed by the second clause of definition 2.2.1.

***Proof*:**
The inclusion of an effective Cartier divisor is a regular embedding of codimension one with normal bundle $N = \mathcal{O}_X(D)|_D$ (example 3.3.15), and $c_1(N)$ is the top Chern class of a line bundle, so theorem 5.3.2(2) gives the displayed identity. For the second assertion, let $V \subseteq D$ be a k-dimensional subvariety with inclusion ι_V into D. Since V is contained in $|D|$, the second clause of definition 2.2.1 computes $D \cdot [V] = c_1(\mathcal{O}_X(D)|_V) \cap [V]$ in $A_{k-1}(V)$. Pushing that class forward along ι_V and applying the projection formula (proposition 2.4.4(4)) to the proper morphism ι_V, whose pullback of $\mathcal{O}_X(D)|_D$ is $\mathcal{O}_X(D)|_V$, gives $\iota_{V*}(D \cdot [V]) = c_1(\mathcal{O}_X(D)|_D) \cap [V]$, which is $i^!(i_*[V])$ by the first assertion, as we sought to show.

Example 5.3.4: A Line Meeting Itself in the Plane
Let $X = \mathbb{P}^2$ and $L = \{x_0 = 0\}$, an effective Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(L) \cong \mathcal{O}_{\mathbb{P}^2}(1)$ (example 2.1.4), so that $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (example 4.1.7). The second half of example 4.3.8 produced the class $i_i^![L] = (\pi^*)^{-1}\zeta_*[L]$, the refined Gysin map along i applied to $[L]$, and could go no further. Theorem 5.3.2(1) evaluates it:

$$i_i^![L] \;=\; c_1\big(\mathcal{O}_{\mathbb{P}^1}(1)\big) \cap [L] \;=\; [\mathrm{pt}]$$

by example 2.4.7, a single point of L with multiplicity one. This is the value predicted there from example 2.2.3, where $L \cdot [L] = [q]$ was computed from the restricted line bundle, and corollary 5.3.3 identifies the two computations in general rather than in this instance.
For a smooth conic $C \subseteq \mathbb{P}^2$ the same argument gives $N = \mathcal{O}_{\mathbb{P}^2}(2)|_C$, and example 3.3.15 evaluated $c_1(\mathcal{O}_{\mathbb{P}^2}(2)|_C) \cap [C] = 4\,[\mathrm{pt}]$, so $i^!(i_*[C]) = 4\,[\mathrm{pt}]$. The self-intersection number of a plane curve of degree m is m^2, and the Gysin map records it on the curve itself rather than in $A_0(\mathbb{P}^2)$.

The examples so far are divisors, where the normal bundle is a line bundle and $c_d(N)$ is the operator of chapter 2. The construction that supplies higher codimension, and with it the geometry the formula was built for, is the blow-up: its exceptional divisor is a projectivized normal bundle, and it meets itself in every point.

Lemma 5.3.5: The Exceptional Divisor of a Blow-up Along a Regular Centre

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with $Z \neq \emptyset$, ideal sheaf $\mathcal{I}$ and normal bundle N. Let $f\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up ([2, Blow-up]) and let $\widetilde{Z} = f^{-1}(Z)$ be its exceptional divisor, the closed subscheme cut out by $\mathcal{I}' = \mathcal{I}\mathcal{O}_{\widetilde{X}}$. Then:

1. $\widetilde{Z} = \mathbb{P}(N)$, with the restriction $g\colon \mathbb{P}(N) \to Z$ of f as its bundle projection;
2. $j\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ is a regular embedding of codimension one whose conormal surjection $g^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is the universal surjection $g^*(N^\vee) \to \mathcal{O}_{\mathbb{P}(N)}(1)$ of box 3.2.5. In particular the normal bundle of j is $\mathcal{O}_{\mathbb{P}(N)}(-1)$, sitting inside g^*N as the tautological subbundle with quotient the universal quotient bundle Q of rank $d-1$.

***Proof*:**

Realize the Rees algebra as the subalgebra $S = \bigoplus_{n\geq 0} \mathcal{I}^n t^n$ of $\mathcal{O}_X[t]$, so that $\widetilde{X} = \mathrm{Proj}\,(S)$ over X by [2, Blow-up]. The algebra S is generated in degree one, so the affine charts $D_+(at)$ attached to local sections a of $\mathcal{I}$ cover $\widetilde{X}$; on such a chart $\widetilde{X}$ is $\mathrm{Spec}\,(S_{(at)})$, whose elements are the fractions y/a^n with y a local section of $\mathcal{I}^n$.

1. The homogeneous ideal $\mathcal{I}S \subseteq S$ has degree-n part $\mathcal{I} \cdot \mathcal{I}^n t^n = \mathcal{I}^{n+1} t^n$, so
$$S/\mathcal{I}S \;=\; \bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$$
which is $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$ by lemma 3.3.10, glued over an affine cover as in the proof of proposition 3.3.11. Taking relative Proj of the surjection $S \to S/\mathcal{I}S$ exhibits $\mathrm{Proj}\,\big(\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)\big)$ as the closed subscheme of $\widetilde{X}$ cut out by the ideal generated by $\mathcal{I}$, that is by $\mathcal{I}'$, and its structure morphism to Z is the restriction of f. Since N has $(\mathcal{I}/\mathcal{I}^2)^\vee$ as its sheaf of sections ([2, Conormal Sheaf and Normal Bundle]), that relative Proj is $\mathbb{P}(N)$ in the convention of box 0.0.2.
2. *The ideal is invertible.* On the chart $D_+(at)$ the ideal $\mathcal{I}'$ is generated by the image of a, since every local section y of $\mathcal{I}$ satisfies $y = a \cdot (y/a)$ with y/a in $S_{(at)}$. That generator is a nonzerodivisor: if $a \cdot (y/a^n) = 0$ in $S_{(at)}$ with y a local section of $\mathcal{I}^n$, then $(at)^m\, a\, y\, t^n = 0$ in $\mathcal{O}_X[t]$ for some $m \geq 0$, that is $a^{m+1} y = 0$, and the same relation forces $y/a^n = 0$ in $S_{(at)}$. Hence $\widetilde{Z}$ is an effective Cartier divisor and j is a regular embedding of codimension one (example 3.3.15), which also recovers [2, Basic Properties of Blow-ups](3); its conormal sheaf is $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{I}' \otimes \mathcal{O}_{\widetilde{Z}}$, invertible on $\widetilde{Z}$.
Which invertible sheaf. The ideal $\mathcal{I}'$ is generated by the image of $\mathcal{I}$, so the canonical map $g^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is surjective. Compare it on $D_+(at) \cap \widetilde{Z}$ with the universal surjection of $\mathbb{P}(N) = \mathrm{Proj}\,\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$. There $\mathcal{I}'/\mathcal{I}'^2$ is free on the class of a by the previous paragraph, while $\mathcal{O}_{\mathbb{P}(N)}(1)$ is free on the image $\bar{a}$ of the degree-one element at. A local section b of $\mathcal{I}$ satisfies $b = a \cdot (b/a)$ upstairs, and its class satisfies $\bar{b} = \bar{a} \cdot \overline{b/a}$ downstairs, the reduction $S \to S/\mathcal{I}S$ carrying b/a to $\overline{b/a}$. Under the two trivializations the two surjections therefore have the same coefficient on every local section, so they agree and $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{O}_{\mathbb{P}(N)}(1)$.

Dualizing gives the inclusion $\mathcal{O}_{\mathbb{P}(N)}(-1) \hookrightarrow g^*N$, which is the tautological subbundle of box 3.2.5, whose quotient Q is locally free of rank $d-1$, completing the proof.

The lemma is the reason the blow-up is the standard source of examples for the self-intersection formula: the exceptional divisor is a divisor whose normal bundle is the tautological line bundle, with first Chern class $-\xi$ for the tautological class ξ of section 3.1. In the smallest case the resulting number was computed by hand upstream, and the general formula reproduces it.

Example 5.3.6: The Exceptional Curve of a Blown-up Surface

Let $X = \mathbb{P}^2$ and let $Z = \{P\}$ with $P = [0:0:1]$. On the affine chart $\{x_2 \neq 0\}$ the ideal of P is $(x_0/x_2, x_1/x_2)$, a regular sequence of length two in the polynomial ring of the chart, so $Z \hookrightarrow X$ is a regular embedding of codimension two whose normal bundle N is a rank-two vector bundle over a point, that is a two-dimensional vector space. Write $f\colon \widetilde{X} = \mathrm{Bl}_P\mathbb{P}^2 \to \mathbb{P}^2$ for the blow-up, $\widetilde{Z}$ for its exceptional divisor, written E in [2, Intersection Theory on the Blow-up], and $j\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ for its inclusion. By lemma 5.3.5,

$$\widetilde{Z} \;=\; \mathbb{P}(N) \;\cong\; \mathbb{P}^1, \qquad N_{\widetilde{Z}}\widetilde{X} \;=\; \mathcal{O}_{\mathbb{P}^1}(-1)$$

The self-intersection formula now evaluates the intersection of $\widetilde{Z}$ with itself. By corollary 5.3.3 and additivity of the first Chern class (proposition 2.4.4(3)),

$$j^!\big(j_*[\widetilde{Z}]\big) \;=\; c_1\big(\mathcal{O}_{\mathbb{P}^1}(-1)\big) \cap [\mathbb{P}^1] \;=\; -\,c_1\big(\mathcal{O}_{\mathbb{P}^1}(1)\big) \cap [\mathbb{P}^1] \;=\; -[\mathrm{pt}]$$

using example 2.4.7. Its degree (definition 1.2.4) is -1, which is the self-intersection number $E^2 = -1$ recorded upstream. The same computation applies at any point of any smooth surface, differing only in the verification that the maximal ideal at that point is generated by a regular sequence of length two; what the general formula contributes is that the number -1 is not an accident of the plane but the degree of the tautological line bundle on the projectivized normal space.

The negative sign is where the Gysin map departs furthest from counting points. A cycle contained in Z has no intersection points with Z to count that are not already in it, and the number the formula returns is a Chern number of the normal bundle, with whatever sign that bundle has. The upstream proof of $E^2 = -1$ argued from the surface pairing and the geometry of the blow-down; the argument here uses no property of surfaces and no property of blow-ups beyond lemma 5.3.5.

Between the proper case, where the answer is the fundamental cycle of $V \cap Z$, and the self-intersection case, where it is a top Chern class, lies everything else: the preimage Z' has some dimension between the expected one and $\dim X'$, and the normal cone fills only part of the pulled-back normal bundle. When the preimage is itself regularly embedded the discrepancy is a vector bundle, and it is the only new input the general formula needs.

The comparison rests on the conormal sheaves. For a regular embedding of codimension d' the conormal sheaf $\mathcal{I}'/\mathcal{I}'^2$ is locally free of rank d' ([2, The Conormal Sheaf of a Regular Embedding]), and since $\mathcal{I}' = \mathcal{I}\mathcal{O}_{X'}$ is generated by the image of $\mathcal{I}$, the canonical map $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is a surjection of vector bundles. A surjection of vector bundles has locally free kernel, being locally split, so dualizing gives an inclusion of vector bundles with locally free cokernel; in particular $d' \leq d$.

Definition 5.3.7: The Excess Bundle

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N and ideal sheaf $\mathcal{I}$, let $g\colon X' \to X$ be a morphism of algebraic schemes, and form the fibre square of definition 4.3.6, with $Z' = g^{-1}(Z)$, ideal sheaf $\mathcal{I}' = \mathcal{I}\mathcal{O}_{X'}$, and $h\colon Z' \to Z$. Suppose $i'\colon Z' \hookrightarrow X'$ is a regular embedding of codimension d' with normal bundle N'. The dual of the conormal surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is an inclusion of vector bundles $N' \hookrightarrow h^*N$ on Z', and the *excess bundle* of the square is its cokernel

$$E \;=\; h^*N/N'$$

a vector bundle of rank $e = d - d'$ on Z'. When $e = 0$ the excess bundle is the zero bundle, and the operator $c_0(E)$ is then read as the identity, definition 3.2.3 being stated for bundles of positive rank.

The excess bundle measures how much of the normal directions to Z fail to be normal directions to Z' after base change. It vanishes exactly when $d' = d$, that is when the preimage has the expected codimension and the two normal bundles agree; at the other extreme, when $X' = Z$ and $g = i$, the subbundle N' is zero and the excess bundle is all of N. Its rank is the number of dimensions by which the intersection is too large.

The identity that converts the excess bundle into a correction term is the following, which contains lemma 5.3.1 as the case $N' = 0$ and is deduced from it.

Lemma 5.3.8: The Class of a Subbundle

Let $0 \to N' \xrightarrow{j} N \to E \to 0$ be an exact sequence of vector bundles on an algebraic scheme Z, with $\operatorname{rank} E = e$ and $\operatorname{rank} N' = d'$, and write $\pi\colon N \to Z$ and $\pi'\colon N' \to Z$ for the projections. Then for every $\gamma \in A_m(Z)$,

$$j_*\pi'^*\gamma \;=\; \pi^*\big(c_e(E) \cap \gamma\big) \quad \text{in } A_{m+d'}(N)$$

***Proof*:**
If $e = 0$ then j is an isomorphism of bundles and both sides are $\pi^*\gamma$, the operator $c_0(E)$ being the identity by the convention of definition 5.3.7. So assume $e \geq 1$, write $\pi_E\colon E \to Z$ for the projection and $\zeta_E\colon Z \to E$ for the zero section, and let $\rho\colon N \to E$ be the morphism of total spaces induced by the surjection $N \to E$.

Step 1: the square is a fibre square with ρ flat. Both assertions are local on Z. Over an affine open on which N', N and E are free, the surjection $N \to E$ splits, so there $N \cong N' \times E$ over the base with ρ the second projection, which is flat of relative dimension d', and the scheme-theoretic preimage $\rho^{-1}\big(\zeta_E(Z)\big)$ is N' embedded by j. Hence ρ is flat of relative dimension d' and

$$\begin{tikzcd} N' \arrow[r, "j"] \arrow[d, "\pi'"'] & N \arrow[d, "\rho"] \\ Z \arrow[r, "\zeta_E"] & E \end{tikzcd}$$

is a fibre square, with proper horizontal maps and flat vertical maps.

Step 2: transport the zero-section identity. Proposition 1.3.6 applied to that square gives $\rho^*\zeta_{E*}\gamma = j_*\pi'^*\gamma$. By lemma 5.3.1 applied to the bundle E, the left-hand side is $\rho^*\pi_E^*(c_e(E)\cap\gamma)$, and $\pi_E \circ \rho = \pi$, so proposition 1.3.5(3) rewrites it as $\pi^*(c_e(E)\cap\gamma)$. Combining the two displays gives the assertion, completing the proof.

The refined Gysin map is built from a specialization followed by a pushforward into the pulled-back normal bundle, and when the preimage is regularly embedded the specialization is already the Gysin map of the preimage; the pushforward is then the subbundle inclusion, and lemma 5.3.8 evaluates it.

Theorem 5.3.9: The Excess Intersection Formula

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N, let $g\colon X' \to X$ be a morphism of algebraic schemes, and suppose the preimage $Z' = g^{-1}(Z)$ is regularly embedded in X' of codimension $d' \geq 1$. Let E be the excess bundle of definition 5.3.7, of rank $e = d - d'$. Then for every $\alpha \in A_k(X')$,

$$i_g^!\alpha \;=\; c_e(E) \cap i'^!\alpha \quad \text{in } A_{k-d}(Z')$$

where $i_g^!$ is the refined Gysin homomorphism of i along g and $i'^!$ is the Gysin map of i', in the notation of box 5.2.1.

***Proof*:**
Write $\mathcal{I}$ and $\mathcal{I}'$ for the ideal sheaves of Z in X and of Z' in X', let $N' = N_{Z'}X'$ with projection π', and let $\varpi\colon h^*N \to Z'$ be the projection of the pulled-back normal bundle.

Step 1: the immersion of the cone is the subbundle inclusion. By lemma 3.3.13 the closed immersion $j\colon C_{Z'}X' \hookrightarrow h^*N$ is induced by the canonical surjection of graded $\mathcal{O}_{Z'}$-algebras

$$\operatorname{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \longrightarrow \bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$$

Since i' is a regular embedding, lemma 3.3.10 identifies the target with $\operatorname{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ by the canonical surjection, which is the identity in degree one, and identifies $C_{Z'}X'$ with N' (proposition 3.3.11). A homomorphism of graded algebras out of a symmetric algebra is determined by its degree-one part, so the displayed map is $\operatorname{Sym}^\bullet(\varphi)$ for $\varphi\colon h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ the conormal surjection. Taking relative spectra, $j\colon N' \to h^*N$ is the inclusion of vector bundles dual to φ, that is the inclusion of definition 5.3.7, with cokernel E of rank e.

Step 2: evaluation. Let $\sigma'\colon A_k(X') \to A_k(C_{Z'}X') = A_k(N')$ be the specialization homomorphism of the pair (Z', X'). By definition 4.3.1 applied to i', the Gysin map is $i'^! = (\pi'^*)^{-1} \circ \sigma'$, so $\sigma'\alpha = \pi'^*(i'^!\alpha)$. Hence, using definition 4.3.6 and then lemma 5.3.8 with $\gamma = i'^!\alpha$,

$$\varpi^*\big(i_g^!\alpha\big) \;=\; j_*\sigma'\alpha \;=\; j_*\pi'^*\big(i'^!\alpha\big) \;=\; \varpi^*\Big(c_e(E)\cap i'^!\alpha\Big)$$

The pullback ϖ^* is injective, being an isomorphism by theorem 1.4.6, so $i_g^!\alpha = c_e(E)\cap i'^!\alpha$, as we sought to show.

The formula says that an intersection of the wrong dimension is computed by intersecting properly on the preimage and then correcting by the top Chern class of what was left over. When $e = 0$ the correction is the identity and the two Gysin maps agree, which is the regime of proposition 4.3.3:

the preimage has the expected codimension and the answer is read off it directly. When $e = d$, so that $d' = 0$ and $Z' = X'$, the hypothesis $d' \geq 1$ puts the case outside the statement, though only nominally: definition 5.2.11 reads the Gysin map of a codimension-zero embedding as flat pullback along it, which for $i' = \mathrm{id}_{X'}$ is the identity, so with that reading the uncorrected term is α itself, the correction is all of $c_d(h^*N)$, and for $g = i$ the formula returns theorem 5.3.2(1). The formula also explains the role of the normal cone in the general construction: without a regularity hypothesis on Z' there is no excess bundle, and the discrepancy between $C_{Z'}X'$ and h^*N is recorded by a Segre class instead.

Example 5.3.10: A Plane Through a Line in Three-Space

Let $X = \mathbb{P}^3$ with coordinates $x_0, \ldots, x_3$, let $Z = L = \{x_2 = x_3 = 0\}$ be a line and let $X' = H = \{x_3 = 0\}$ be a plane containing it, with $g\colon H \hookrightarrow \mathbb{P}^3$ the inclusion. By exercise 3.3.1 (2) the embedding i is regular of codimension two with $N = \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}$. The expected dimension of $H \cap L$ is $\dim H - 2 = 0$, but the preimage is $Z' = g^{-1}(L) = L$, of dimension one.

The excess bundle. The line L is an effective Cartier divisor on $H \cong \mathbb{P}^2$ cut out by x_2, so i' is a regular embedding of codimension $d' = 1$ with $N' = \mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (example 2.1.4, example 3.3.15). The excess bundle has rank $e = 2 - 1 = 1$, and the Whitney formula (theorem 3.2.15) applied to $0 \to N' \to N \to E \to 0$ gives, with $h = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$,

$$c(E) \;=\; c(N)\,c(N')^{-1} \;=\; (\mathrm{id} + h)^2(\mathrm{id} + h)^{-1} \;=\; \mathrm{id} + h$$

so $c_1(E) = h$.

The class. The plane H is a variety not contained in $|L|$, and its scheme-theoretic intersection with L is L, so $i'^![H] = [L]$ by proposition 4.3.5, which is the computation of exercise 4.3.1 (2). Theorem 5.3.9 then gives

$$i_g^![H] \;=\; c_1(E) \cap [L] \;=\; h \cap [L] \;=\; [\mathrm{pt}]$$

in $A_0(L)$, of degree one. The uncorrected answer $[L]$ has the wrong dimension and the wrong degree; the correction cuts it down by one dimension and returns the intersection number a plane and a line in general position would have. What the refined map adds is that the class is recorded on L, the locus where the two meet, and not only as a number.

Applying the formula to a blow-up produces the statement that lets Chow groups be computed after blowing up. The square in question is the defining one for the exceptional divisor, and lemma 5.3.5 has already identified all of its data.

Chern classes were built in chapter 3 as *operators* on Chow groups, and box 3.2.4 was careful to say that they are not elements. On a smooth variety they may be read as elements after all, because the ring supplies a canonical one to read them as: the class $c_i(E) \cap [X]$. That the operator and multiplication by that class agree is not a definition and not formal, and chapter 8 leans on it whenever a Schubert class is identified with a Chern class.

Before that, one criterion for reading a product as a count. It is the precise form of the classical principle that transverse intersections may simply be counted, and chapter 8 appeals to it in every enumerative computation.

Proposition 5.3.11: Multiplicity One at a Transverse Point

Let X be a smooth variety and let $V_1, V_2 \subseteq X$ be closed subvarieties with $e = \dim V_1 + \dim V_2 - \dim X$, and suppose $Z = V_1 \cap V_2$ is irreducible of dimension e. If there is a closed point $p \in Z$ at which V_1 and V_2 are both smooth and $\dim(T_pV_1 \cap T_pV_2) = e$, then

$$[V_1] \cdot [V_2] = [Z]$$

***Proof*:**
V_1 and V_2 meet properly, Z being their only component, so definition 5.4.8 gives $[V_1] \cdot [V_2] = i(Z; V_1 \cdot V_2; X)\,[Z]$, and the claim is that the multiplicity is 1. The ideal of the scheme $V_1 \cap V_2$ is $\mathcal{I}_{V_1} + \mathcal{I}_{V_2}$, so $T_p(V_1 \cap V_2) = T_pV_1 \cap T_pV_2$ has dimension e, while $\dim \mathcal{O}_{V_1 \cap V_2, p} = e$ because Z is irreducible of dimension e; a local ring whose Zariski tangent space has dimension equal to its Krull dimension is regular, hence a domain. Let η be the generic point of Z and $\mathcal{O} = \mathcal{O}_{X,\eta}$, a regular local ring of dimension $\dim X - e$. Localizing the domain $\mathcal{O}_{V_1 \cap V_2, p}$ at its minimal prime gives a field, so

$$\mathcal{O}/(\mathcal{I}_{V_1} + \mathcal{I}_{V_2}) \;=\; \mathcal{O}_{V_1 \cap V_2, \eta}$$

is a field, that is, $\mathcal{I}_{V_1} + \mathcal{I}_{V_2} = \mathfrak{m}$, the maximal ideal of $\mathcal{O}$. Each V_t is smooth at p, hence on an open set meeting Z, hence at η; so $\mathcal{O}/\mathcal{I}_{V_t}$ is regular of dimension $\dim V_t - e$, and the image of $\mathcal{I}_{V_t}$ in $\mathfrak{m}/\mathfrak{m}^2$ has dimension $(\dim X - e) - (\dim V_t - e) = \dim X - \dim V_t$, the codimension d_t of V_t. Choose $x_1, \ldots, x_{d_t} \in \mathcal{I}_{V_t}$ lifting a basis of that image. Then $\mathcal{O}/(x_1, \ldots, x_{d_t})$ is regular of dimension $\dim X - e - d_t$, hence a domain, and it surjects onto $\mathcal{O}/\mathcal{I}_{V_t}$, a ring of the same dimension. A quotient of a regular local ring is catenary and equidimensional, so a nonzero ideal in it lowers the dimension; the kernel of the surjection is therefore zero and $\mathcal{I}_{V_t} = (x_1, \ldots, x_{d_t})$. Now $d_1 + d_2 = \dim X - e = \dim \mathcal{O}$, and the two chosen sets of elements together generate $\mathfrak{m}$, so they form a regular system of parameters of $\mathcal{O}$ and in particular a regular sequence. The Koszul complex on the generators of $\mathcal{I}_{V_2}$ is therefore a free resolution of $\mathcal{O}/\mathcal{I}_{V_2}$ over $\mathcal{O}$, and tensoring it with $\mathcal{O}/\mathcal{I}_{V_1}$ gives the Koszul complex on the images of those generators in the regular local ring $\mathcal{O}/\mathcal{I}_{V_1}$, where they again form a regular sequence, being a system of $d_2 = \dim \mathcal{O}/\mathcal{I}_{V_1}$ generators of its maximal ideal. Hence $\mathrm{Tor}_j^{\mathcal{O}}(\mathcal{O}/\mathcal{I}_{V_1}, \mathcal{O}/\mathcal{I}_{V_2}) = 0$ for $j \geq 1$, while the term $j = 0$ is $\mathcal{O}/\mathfrak{m}$, of length one. By theorem 5.4.9 the multiplicity is 1.

Proposition 5.3.12: Degree of a Transverse Intersection

Let X be a smooth *complete* variety of dimension N, so that the degree of a zero-cycle is defined (definition 1.2.4), and let $Z_1, \ldots, Z_s \subseteq X$ be closed subvarieties. If $\sum_i \operatorname{codim} Z_i > N$ then $[Z_1] \cdots [Z_s] = 0$, the product lying in a Chow group of negative dimension. If $\sum_i \operatorname{codim} Z_i = N$, the intersection $Z_1 \cap \cdots \cap Z_s$ is finite, and at each point P of it every Z_i is smooth with $T_P Z_1 \cap \cdots \cap T_P Z_s = 0$, then

$$\deg\big([Z_1] \cdots [Z_s]\big) = \#\big(Z_1 \cap \cdots \cap Z_s\big)$$

each point counting once.

***Proof*:**
Iterating the two-factor case (theorem 4.4.7), the product is the refined Gysin class (definition 4.3.6) of the small diagonal $\delta\colon X \to X^s$, a regular embedding whose normal bundle is $T^{\oplus(s-1)}$ for T the tangent bundle, applied to $[Z_1 \times \cdots \times Z_s]$. At a point P of the intersection the intersection scheme has tangent space $\bigcap_i T_P Z_i = 0$ and so is the reduced point P there; the normal cone to it in $Z_1 \times \cdots \times Z_s$ is the tangent space $T_P Z_1 \oplus \cdots \oplus T_P Z_s$, of dimension $\sum_i (N - \operatorname{codim} Z_i) = (s-1)N$; and the map $(u_1, \ldots, u_s) \mapsto (u_1 - u_2, \ldots, u_{s-1} - u_s)$ carrying it into the fibre $T_P^{\oplus(s-1)}$ has kernel $\bigcap_i T_P Z_i = 0$, hence is an isomorphism. So the cone fills the normal space and the Gysin class is $[P]$. Summing over the finitely many points gives the count; when the codimensions overshoot, the same computation places the class in a Chow group of negative dimension, which vanishes, as we sought to show.

Proposition 5.3.13: Capping with a Chern Class Is Multiplication

Let X be a smooth variety and E a vector bundle on X. Then for every $\alpha \in A_*(X)$ and every i,
$$c_i(E) \cap \alpha \;=\; \big(c_i(E) \cap [X]\big) \cdot \alpha$$
the product on the right being that of theorem 5.2.7. Consequently the total Chern class may be treated as the element $c(E) \cap [X]$ of $A^*(X)$, and an identity of Chern class operators is an identity in the ring.

***Proof*:**
Step 1: reduction to a line bundle. Both sides are additive in α and, as functions of E, are determined by the Chern classes of line bundles: the splitting principle (theorem 3.2.14) produces a flat morphism $f\colon X' \to X$ with f^* injective on Chow groups and f^*E filtered with line bundle quotients, and both sides commute with f^*, the left by proposition 3.2.10(3) and the right because f^* is a ring homomorphism (theorem 5.2.10). Since the Chern classes of E are the elementary symmetric functions of the Chern roots (lemma 3.2.11), it suffices to treat $c_1(L)$ for a line bundle L.

Step 2: reduction to an effective divisor. Write $L = \mathcal{O}_X(D)$ for a Cartier divisor D, possible because X is smooth, hence locally factorial, so that $\operatorname{Pic} X \to A^1(X)$ is the isomorphism of proposition 2.1.3. On a smooth variety every divisor is a difference $D = D_+ - D_-$ of effective ones, and both sides of the claim are additive: the left because c_1 is additive on tensor products (proposition 2.4.4(3)), the right by biadditivity of the product (theorem 5.2.7(1)). So it suffices to treat D effective with $D \neq X$.

Step 3: the effective case. Let $i\colon D \hookrightarrow X$ be the inclusion, a regular embedding of codimension one, and let $V \subseteq X$ be a k-dimensional subvariety; by additivity it is enough to prove $D \cdot [V] = [D] \cdot [V]$. By proposition 5.1.11, applied with the two closed subschemes D and X, the ring product $[D] \cdot [V]$ is the pushforward along $D \cap V \hookrightarrow X$ of the refined product, which for the class of a divisor is the refined Gysin map $i^![V]$ of definition 4.3.6.

It remains to identify $i^![V]$ with $D \cdot [V]$, and the two clauses of definition 2.2.1 are covered by two separate results. If $V \not\subseteq |D|$, then proposition 4.3.5 gives $i^![V] = \iota_*(D \cdot [V])$ directly, its hypothesis $V \not\subseteq |D|$ being exactly this case. If $V \subseteq |D|$, then corollary 5.3.3 applies instead and identifies $i^![V]$ with the class that the second clause of definition 2.2.1 computes from the restricted line bundle, which is $D \cdot [V]$ by definition. Since $c_1(L) \cap [V] = D \cdot [V]$ by definition 2.4.1

and $c_1(L) \cap [X] = [D]$, this is the assertion for a line bundle.

Step 4. Steps 1 to 3 give the identity for every E and every α. The last sentence follows by evaluating an operator identity on $[X]$ and reading the result through the identity just proved, as we sought to show.

A section of a bundle whose zero scheme has the expected codimension cuts out a class that the top Chern class already knows. This is the form in which Chern classes do enumerative work, and chapter 8 uses it throughout.

Lemma 5.3.14: The Class of a Zero Scheme

Let Y be a smooth variety, G a vector bundle of rank h on Y, and $\psi \in \Gamma(Y, G)$ a section whose zero scheme $Z = Z(\psi)$ is empty or of pure codimension h. Writing $\iota\colon Z \to Y$ for the inclusion,

$$\iota_*[Z] = c_h(G) \cap [Y]$$

***Proof*:**
Write $\pi\colon G \to Y$ also for the total space of the bundle, $z\colon Y \to G$ for the zero section and $\psi\colon Y \to G$ for the section. By definition $Z = \psi^{-1}(z(Y))$ as schemes.

Step 1: Z is regularly embedded with normal bundle $G|_Z$. Over an open set trivializing G, the section ψ is an h-tuple of regular functions and Z is their common zero scheme. The local rings of Y are regular, hence Cohen-Macaulay, and in a Cohen-Macaulay local ring h elements generating an ideal of height h form a regular sequence ([5, Height-Many Elements in a Cohen-Macaulay Ring Form a Regular Sequence]); the height is h precisely because Z has pure codimension h. So $Z \subseteq Y$ is a regular embedding of codimension h ([2, Regular Embedding]) with conormal sheaf $\mathcal{I}/\mathcal{I}^2$ locally free of rank h. The bundle map $G^\vee \to \mathcal{I}/\mathcal{I}^2$ sending a functional ℓ to the class of $\ell(\psi)$ is surjective, since the components of ψ generate $\mathcal{I}$; a surjection between locally free sheaves of the same finite rank is an isomorphism, so the normal bundle is $G|_Z$.

Step 2: the two sections carry $[Y]$ to the same class on G. Let $T \subseteq G \times \mathbb{P}^1$ be the closure of the image of the morphism $Y \times \mathbb{A}^1 \to G \times \mathbb{A}^1$, $(y, \lambda) \mapsto (\lambda\psi(y), \lambda)$. That morphism is a section of the projection $G \times \mathbb{A}^1 \to Y \times \mathbb{A}^1$, hence the pullback of a diagonal and so a closed immersion; thus $T \cap (G \times \mathbb{A}^1) \cong Y \times \mathbb{A}^1$ is a variety of dimension $\dim Y + 1$ and T is a variety dominating $\mathbb{P}^1$. On T the coordinate λ is a rational function whose divisor, computed on $Y \times \mathbb{A}^1$ where λ is the second coordinate, is $Y \times \{0\}$ with multiplicity one together with a part supported over $\lambda = \infty$; the divisor of $\lambda - 1$ is $Y \times \{1\}$ with multiplicity one together with the same part over $\lambda = \infty$, since a rational function and its translate by 1 have equal orders along any subvariety where that order is negative. Subtracting the two divisors and applying definition 1.1.4,

$$[T \cap (G \times \{0\})] - [T \cap (G \times \{1\})] = [\mathrm{div}(\lambda)] - [\mathrm{div}(\lambda - 1)] \sim 0$$

in $A_{\dim Y}(G \times \mathbb{P}^1)$. Pushing forward along the projection $G \times \mathbb{P}^1 \to G$, which is proper because $\mathbb{P}^1$ is complete and which carries the two fibres isomorphically onto $z(Y)$ and $\psi(Y)$, gives $z_*[Y] = \psi_*[Y]$ in $A_{\dim Y}(G)$.

Step 3: computation of the Gysin pullback. The zero section z is a regular embedding of codimension h with normal bundle G, and Z is the fibre product of z and ψ. By Step 1 the induced embedding $Z \subseteq Y$ is regular of codimension h as well, so the excess bundle of that fibre square is the zero bundle and theorem 5.3.9 identifies the refined Gysin class of definition 4.3.6 as $z^![Y] = [Z]$ in $A_{\dim Y-h}(Z)$. Its image in $A_{\dim Y-h}(Y)$ is $z^*(\psi_*[Y])$ by theorem 4.3.10. Combining with Step 2 and then with the self-intersection formula theorem 5.3.2 applied to z, whose normal bundle is G,

$$\iota_*[Z] = z^*(\psi_*[Y]) = z^*(z_*[Y]) = c_h(G) \cap [Y]$$

as we sought to show. When Z is empty the same chain reads $0 = c_h(G) \cap [Y]$.

Theorem 5.3.15: The Blow-up Formula

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with $Z \neq \emptyset$ and normal bundle N, let $f\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up with exceptional divisor $j\colon \widetilde{Z} = \mathbb{P}(N) \hookrightarrow \widetilde{X}$ and $g\colon \widetilde{Z} \to Z$, and let Q be the universal quotient bundle of rank $d-1$ on $\mathbb{P}(N)$. Then for every $\beta \in A_k(\widetilde{X})$:

1. $i^!_f\beta = c_{d-1}(Q) \cap j^!\beta$ in $A_{k-d}(\widetilde{Z})$, where $i^!_f$ is the refined Gysin homomorphism of i along f and $j^!$ is the Gysin map of j;
2. $i^!(f_*\beta) = g_*\Big(c_{d-1}(Q) \cap j^!\beta\Big)$ in $A_{k-d}(Z)$, where the left-hand $i^!$ is the Gysin map of definition 4.3.1.

***Proof*:**

1. The exceptional divisor is the scheme-theoretic preimage $f^{-1}(Z)$, so the square formed by j, f, i and g is the square of definition 4.3.6 for the morphism f. By lemma 5.3.5(2) the preimage $\widetilde{Z}$ is regularly embedded in $\widetilde{X}$ of codimension $d' = 1$, and its conormal surjection, which is the one definition 5.3.7 dualizes, is there identified with the universal surjection; so the excess bundle of the square is the universal quotient Q, of rank $e = d-1$. Theorem 5.3.9 therefore gives $i^!_f\beta = c_{d-1}(Q) \cap j^!\beta$.
2. The blow-down f is proper ([2, Basic Properties of Blow-ups](1)), so theorem 4.3.10(1) applies with $X' = X$, $X'' = \widetilde{X}$ and $p = f$. The preimage of Z in $\widetilde{X}$ is $\widetilde{Z}$ and the induced morphism on preimages is g, so $i^!(f_*\beta) = g_*(i^!_f\beta)$, the left-hand side being the Gysin map of i by proposition 4.3.7. Substituting part (1) completes the proof.

Part (1) computes an intersection with Z upstairs, where the geometry is a divisor and therefore elementary, at the cost of one Chern class of the universal quotient bundle. Part (2) is the form used in practice: it computes the Gysin image of any class pushed down from the blow-up, and it is the identity from which the Chow groups of $\widetilde{X}$ are assembled from those of X and of $\mathbb{P}(N)$. The smallest instance is a consistency check that can be run in both directions.

Example 5.3.16: The Fundamental Class of a Blow-up

Let X be a variety of dimension n and $Z \subseteq X$ a nonempty subvariety with $Z \neq X$, regularly embedded of codimension $d \geq 1$, and take $\beta = [\widetilde{X}]$. The Rees algebra is a graded subalgebra of $\mathcal{O}_X[t]$ and hence a sheaf of domains, so $\widetilde{X}$ is integral, that is a variety; and f restricts to an isomorphism over $X \setminus Z$ ([2, Basic Properties of Blow-ups](2)), so the two function fields agree and $f_*[\widetilde{X}] = [X]$ by definition 1.2.1. Also $i^![X] = [Z]$ by proposition 4.3.3, the intersection of X with Z being Z itself, and $j^![\widetilde{X}] = [\widetilde{Z}]$ by proposition 4.3.5 applied to the subvariety $\widetilde{X}$ of itself. Substituting into theorem 5.3.15(2),

$$[Z] \;=\; g_*\Big(c_{d-1}(Q) \cap \big[\mathbb{P}(N)\big]\Big)$$

The identity can be checked directly, and the check is a use of the Segre operators rather than of anything proved in this section. The tautological sequence $0 \to \mathcal{O}_{\mathbb{P}(N)}(-1) \to g^*N \to Q \to 0$ and the Whitney formula (theorem 3.2.15) give $c(Q) = c(g^*N)\,(\mathrm{id} - \xi)^{-1}$ with $\xi = c_1(\mathcal{O}_{\mathbb{P}(N)}(1))$, the inverse being the terminating series $\sum_{m\geq 0} \xi^{\,m}$, so

$$c_{d-1}(Q) \;=\; \sum_{m=0}^{d-1} \xi^{\,m} \circ c_{d-1-m}\big(g^*N\big)$$

Capping with $[\mathbb{P}(N)] = g^*[Z]$ and pushing forward, naturality of the Chern operators for the flat morphism g (proposition 3.2.10(3)) and the definition of the Segre operators (definition 3.1.3), for which $\mathbb{P}(N) \to Z$ has relative dimension $d-1$, give

$$g_*\Big(\xi^{\,m} \cap g^*\big(c_{d-1-m}(N) \cap [Z]\big)\Big) \;=\; s_{m-d+1}(N) \cap \big(c_{d-1-m}(N) \cap [Z]\big)$$

for each m. Every term with $m < d-1$ vanishes, the Segre operator having negative index, and the term $m = d-1$ is $s_0(N) \cap [Z] = [Z]$; summing over m gives $g_*\big(c_{d-1}(Q) \cap [\mathbb{P}(N)]\big) = [Z]$, so the two computations agree.

Exercise 5.3.1

1. Let $Y \subseteq \mathbb{P}^2$ be a smooth plane curve of degree m, with inclusion i. Show that $i\colon Y \hookrightarrow \mathbb{P}^2$ is a regular embedding of codimension one, identify its normal bundle, and compute the degree of $i^!(i_*[Y]) \in A_0(Y)$. Then explain why the answer determines $i^!(i_*[Y])$ itself when $m \leq 2$ but not when $m \geq 3$.

2. Let $Z = \{x_0 = 0\} \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ with $n \geq 1$, with inclusion i. Identify the normal bundle of i and compute $i^!(i_*\alpha)$ for α the generator h^Z_m of $A_m(Z)$ for each $0 \leq m \leq n-1$. Compare the answer with the Gysin map $i^!$ of example 4.3.4 in the case $d = 1$.

3. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ whose normal bundle is trivial, $N \cong \mathcal{O}_Z^{\oplus d}$. Show that $i^! \circ i_* = 0$ on $A_k(Z)$ for every k. Then let $X = \mathbb{P}^1 \times \mathbb{P}^1$, let $Z = \{q\} \times \mathbb{P}^1$ for a point $q \in \mathbb{P}^1$, and verify the hypothesis, deducing that the ruling has self-intersection zero.

4. In $\mathbb{P}^4$ with coordinates $x_0, \ldots, x_4$, let $Z = \{x_3 = x_4 = 0\}$ and $X' = \{x_2 = x_4 = 0\}$ be two planes, so that $Z \cap X' = \{x_2 = x_3 = x_4 = 0\}$ is a line L, and let $g\colon X' \hookrightarrow \mathbb{P}^4$ be the inclusion. Show that $Z \hookrightarrow \mathbb{P}^4$ is a regular embedding of codimension two with $N = \mathcal{O}_{\mathbb{P}^2}(1)^{\oplus 2}$, identify the preimage Z' of Z in X' and its normal bundle, compute the excess bundle, and evaluate $i^!_g[X']$. Interpret the degree of the answer.

5. Let $\widetilde{X} = \mathrm{Bl}_P\mathbb{P}^2$ as in example 5.3.6, with exceptional divisor $\widetilde{Z} \cong \mathbb{P}^1$ and the remaining notation that of theorem 5.3.15. Identify the universal quotient bundle Q on $\widetilde{Z}$ and compute $c_1(Q)$. Then verify theorem 5.3.15(2) for $\beta = [\widetilde{X}]$ by computing both sides, and compute $i^!\big(f_*j_*[\widetilde{Z}]\big)$.

5.4 Bézout and the Agreement Theorem

Two counts were recorded on credit. Example 1.2.6 asserted that two plane curves of degrees c and d meet in a zero-cycle of degree cd, and example 1.5.4 read that assertion off an identity $h_1 \cdot h_1 = h_0$ in a ring that did not then exist. Both are settled here, in the sharp form: not only the total count, but its distribution over the points of $C \cap D$, which is what an intersection multiplicity is.

Every enumerative count in projective space reduces to one computation, the product of the classes of two linear subspaces, because theorem 1.5.2 leaves nothing else in $A_*(\mathbb{P}^n)$ to compute, and that computation was carried out in example 5.2.3. Written in dimension rather than codimension, with $h_k = [\mathbb{P}^k] \in A_k(\mathbb{P}^n)$ the class of a linear subspace of dimension k and $h = h_{n-1}$ the hyperplane class, it reads

$$h_r \cdot h_s \;=\; \begin{cases} h_{r+s-n} & \text{if } r+s \geq n \\ 0 & \text{if } r+s < n \end{cases}$$

for $0 \leq r, s \leq n$, together with $h^p = h_{n-p}$ for $0 \leq p \leq n$, the vanishing $h^{n+1} = 0$, and the presentation

$$A^*(\mathbb{P}^n) \;\cong\; \mathbb{Z}[h]/(h^{n+1})$$

as a graded ring with h in codimension one. Nothing else about $\mathbb{P}^n$ is needed below.

The presentation converts every question about cycles on $\mathbb{P}^n$ into a question about polynomials in one variable. A class in $A^*(\mathbb{P}^n)$ is a polynomial in h of degree at most n; multiplying two of them and truncating above degree n is the whole intersection calculus of projective space. The truncation is geometric: $h^p h^q = 0$ for $p + q > n$ says that two subvarieties whose codimensions exceed the ambient dimension can be moved apart, and rational equivalence records that they have been.

The single integer attached to a subvariety by theorem 1.5.2 now deserves a name, since it is the only input Bézout's theorem takes.

Definition 5.4.1: Degree of a Subvariety of Projective Space

Let $V \subseteq \mathbb{P}^n$ be a closed subvariety of dimension k. Since $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ is free of rank one (theorem 1.5.2), there is a unique integer $\deg V$ with

$$[V] \;=\; (\deg V)\, h_k \;\in\; A_k(\mathbb{P}^n)$$

called the *degree* of V. The same definition applies to a closed subscheme pure of dimension k, using its fundamental cycle.

The definition is a tautology until the integer is computed, and the two computations that matter are the classical ones: a hypersurface has the degree of its equation, and the degree of anything is the number of points it cuts on a complementary linear space.

Proposition 5.4.2: Degrees of Hypersurfaces and Hyperplane Sections

Let $H \subseteq \mathbb{P}^n$ be a hyperplane, so that $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$.

1. If $V \subseteq \mathbb{P}^n$ is a closed subvariety of dimension k, then $H^k \cdot [V] = (\deg V)\, h_0$ in $A_0(\mathbb{P}^n)$, so $\deg V = \int_{\mathbb{P}^n} \left(H^k \cdot [V]\right)$, the iterated divisor operator of definition 2.2.1 applied k times.
2. A linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ has degree 1.
3. If F is a nonzero homogeneous form of degree $d \geq 1$ on $\mathbb{P}^n$, the hypersurface $V(F)$ has degree d.

***Proof*:**

1. Example 2.2.5 computes $H \cdot h_j = h_{j-1}$ for $1 \leq j \leq n$. The operator $H \cdot -$ is a homomorphism of Chow groups (proposition 2.4.4(1)), so from $[V] = (\deg V)\, h_k$ one gets $H \cdot [V] = (\deg V)\, h_{k-1}$, and iterating k times gives $H^k \cdot [V] = (\deg V)\, h_0$. The class h_0 is that of a single closed point, of degree $[\overline{k} : \overline{k}] = 1$ by definition 1.2.4, whence $\int_{\mathbb{P}^n}(H^k \cdot [V]) = \deg V$.
2. By theorem 1.5.2 the class of a linear $\mathbb{P}^k$ is the generator h_k itself, so the coefficient is 1.
3. Write $Z = V(F)$ for the closed subscheme cut out by F, a hypersurface, and let D be the associated effective Cartier divisor, with $\mathcal{O}_{\mathbb{P}^n}(D) \cong \mathcal{O}_{\mathbb{P}^n}(d)$ because F is a section of $\mathcal{O}_{\mathbb{P}^n}(d)$. Applying proposition 4.3.5 with $V = \mathbb{P}^n$, which is legitimate since $\mathbb{P}^n$ is a variety not contained in $|D|$, the fundamental cycle of Z is $[Z] = D \cdot [\mathbb{P}^n]$. By definition 2.4.1 this is $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) \cap [\mathbb{P}^n]$, and $\mathcal{O}_{\mathbb{P}^n}(d) \cong \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d}$, so additivity of the first Chern class (proposition 2.4.4(3)) gives
$$[Z] \;=\; d\left(c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n]\right) \;=\; d\, h_{n-1}$$
the last equality being example 2.4.3(2). Hence $\deg Z = d$, completing the proof.

Part (1) is the reason the word "degree" is the same word as in the classical theory: cutting a k-dimensional subvariety by k hyperplanes leaves a zero-cycle whose degree is the invariant, and by proposition 2.4.6 the hyperplanes need not be chosen in general position for the answer to come out right. Part (3) recovers the plane-curve count of example 1.5.4(2) as the case $n = 2$. Degrees of subvarieties that are not hypersurfaces are computed by pushing forward from a parametrization, as the next example illustrates.

Example 5.4.3: The Twisted Cubic

Let $\nu\colon \mathbb{P}^1 \to \mathbb{P}^3$ be the third Veronese embedding, $[s : t] \mapsto [s^3 : s^2t : st^2 : t^3]$, and let $C = \nu(\mathbb{P}^1)$ be the twisted cubic, a closed subvariety of $\mathbb{P}^3$ of dimension one. Since ν is a closed immersion, $\nu_*[\mathbb{P}^1] = [C]$ by definition 1.2.1, the extension of function fields being trivial.
Let $H = \{x_0 = 0\} \subseteq \mathbb{P}^3$. The pullback ν^*H is the Cartier divisor on $\mathbb{P}^1$ cut by s^3, that is $3\,[0:1]$, so $\nu^*H \cdot [\mathbb{P}^1] = 3\left[[0:1]\right]$ by definition 2.2.1. The projection formula for divisors (proposition 2.3.3(1)) applied to the proper morphism ν gives
$$H \cdot [C] \;=\; \nu_*\left(\nu^*H \cdot [\mathbb{P}^1]\right) \;=\; 3\,[P], \qquad P = \nu([0:1]) = [0:0:0:1]$$

a zero-cycle of degree 3. By proposition 5.4.2(1) with $k = 1$, $\deg C = 3$: the twisted cubic has degree three, all of it concentrated at the single point where H is tangent to C to order three. A general plane, by contrast, meets C in three distinct points, and proposition 2.4.6 is what makes the two computations agree without either being privileged.

With degrees in hand, Bézout's theorem is a restatement of example 5.2.3.

Theorem 5.4.4: Bézout's Theorem

Let $V, W \subseteq \mathbb{P}^n$ be closed subvarieties of dimensions r and s. If $r + s \geq n$, then

$$[V] \cdot [W] \;=\; (\deg V)(\deg W)\, h_{r+s-n} \;\in\; A_{r+s-n}(\mathbb{P}^n)$$

and if $r + s < n$ then $[V] \cdot [W] = 0$. In the boundary case $r + s = n$,

$$\int_{\mathbb{P}^n} \big([V] \cdot [W]\big) \;=\; (\deg V)(\deg W)$$

***Proof*:**
By definition 5.4.1 the classes are $[V] = (\deg V)\, h_r$ and $[W] = (\deg W)\, h_s$. The intersection product is $\mathbb{Z}$-bilinear (proposition 5.1.9(1)), so

$$[V] \cdot [W] \;=\; (\deg V)(\deg W)\, (h_r \cdot h_s)$$

and example 5.2.3 evaluates $h_r \cdot h_s$ as h_{r+s-n} when $r + s \geq n$ and as 0 otherwise. For $r + s = n$ the class h_0 is that of a single reduced closed point, of degree one by definition 1.2.4, and the degree map is a homomorphism, so $\int_{\mathbb{P}^n}([V] \cdot [W]) = (\deg V)(\deg W)$, as we sought to show.

The theorem is stated for classes, and in that form it costs nothing beyond the ring structure. What it does not yet say is where the count sits. The product $[V] \cdot [W]$ is defined as $\delta^!([V] \times [W])$, and the refined form of the Gysin map records that class on the preimage of the diagonal, that is on $V \cap W$ (corollary 4.3.11 and definition 5.1.10). Reading the refined class off requires knowing that $V \cap W$ sits inside $V \times W$ as a regular embedding, which is a hypothesis and not a theorem. The following lemma verifies it in the case where the verification is cheapest and most used: two curves on a smooth surface.

Lemma 5.4.5: Distinct Curves on a Smooth Surface Meet Regularly

Let S be a smooth variety of dimension two over $\overline{k}$ and let $C, D \subseteq S$ be distinct one-dimensional closed subvarieties. Then:

1. $C \cap D$ is a finite set of closed points, and for $P \in C \cap D$ with local equations $f, g \in \mathcal{O}_{S,P}$ of C and D, the local ring of the scheme $C \cap D$ at P is $\mathcal{O}_{S,P}/(f,g)$, of length equal to its dimension $\dim_{\overline{k}} \mathcal{O}_{S,P}/(f,g)$ as a $\overline{k}$-vector space;

2. the closed immersion $C \cap D \hookrightarrow C \times D$ induced by the diagonal is a regular embedding of codimension two;

3. consequently

$$[C] \cdot [D] \;=\; \sum_{P \in C \cap D} \dim_{\overline{k}} \big(\mathcal{O}_{S,P}/(f,g)\big)\,[P] \;\in\; A_0(S)$$

Proof:
Since S is smooth its local rings are regular, hence factorial, so each of C and D is an effective Cartier divisor and the local equations f, g exist and are nonzerodivisors.

1. $C \cap D$ is a closed subscheme of the irreducible curve C not equal to C, because $D \neq C$ and both are irreducible of dimension one; a proper closed subscheme of a curve is finite. Scheme-theoretically $C \cap D$ is cut out on S by the ideal (f,g), so its local ring at P is $\mathcal{O}_{S,P}/(f,g)$. That ring is Artinian local with residue field $\overline{k}$, since $\overline{k}$ is algebraically closed and P is a closed point, so each factor of a composition series is a copy of $\overline{k}$ and its length equals its $\overline{k}$-dimension.

2. Fix $P \in C \cap D$ and write $\mathcal{O} = \mathcal{O}_{S,P}$, a regular local ring of dimension two with residue field $\overline{k}$, and $A = \mathcal{O}/(f)$, $B = \mathcal{O}/(g)$, the local rings of C and D at P. Let R be the local ring of $C \times D$ at (P,P), that is the localization of $A \otimes_{\overline{k}} B$ at the maximal ideal corresponding to (P,P).
Step 1: R is Cohen-Macaulay of dimension two. The product of two curves over $\overline{k}$ is a surface, so $\dim R = 2$. Because $\mathcal{O}$ is regular of dimension two and f is a nonzerodivisor, A has depth one, so $\mathfrak{m}_A$ contains a nonzerodivisor a; likewise $\mathfrak{m}_B$ contains a nonzerodivisor b. Now B is free as a $\overline{k}$-module, so $A \to A \otimes_{\overline{k}} B$ is flat and $a \otimes 1$ is a nonzerodivisor, with quotient $(A/aA) \otimes_{\overline{k}} B$; and A/aA is again free over $\overline{k}$, so $B \to (A/aA) \otimes_{\overline{k}} B$ is flat and $1 \otimes b$ is a nonzerodivisor on that quotient. Localizing at the maximal ideal, the pair $a \otimes 1$, $1 \otimes b$ is a regular sequence in R of length two, so $\operatorname{depth} R \geq 2 = \dim R$ and R is Cohen-Macaulay.
Step 2: the diagonal cuts R by a system of parameters. The diagonal $\delta\colon S \hookrightarrow S \times S$ is a regular embedding of codimension two (theorem 5.1.6), so its ideal is generated near (P,P) by two elements θ_1, θ_2. The scheme-theoretic preimage of $\delta(S)$ under $C \times D \hookrightarrow S \times S$ is $C \cap D$, whose ideal in R is therefore generated by the images of θ_1 and θ_2. By part (1) that quotient is Artinian, so the two images form a system of parameters of the two-dimensional local ring R. In a Cohen-Macaulay local ring every system of parameters is a regular sequence ([5, A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence]), so $C \cap D \hookrightarrow C \times D$ is a regular embedding of codimension two.

3. The subvarieties C and D are integral over the algebraically closed field $\overline{k}$, so $C \times D$ is a subvariety of $S \times S$ of dimension two, with $[C] \times [D] = [C \times D]$ (definition 5.1.2). Part (2) supplies the hypothesis of proposition 4.3.3 for the regular embedding δ of codimension two and the subvariety $C \times D$, so

$$[C] \cdot [D] \;=\; \delta^![C \times D] \;=\; [C \cap D]$$

the fundamental cycle of the scheme $C \cap D$ read on S through δ. By definition 1.1.2 that cycle is $\sum_P \operatorname{length}(\mathcal{O}_{C\cap D,P})\,[P]$, and part (1) identifies each length with $\dim_{\overline{k}} \mathcal{O}_{S,P}/(f,g)$, completing the proof.

The lemma is the bridge from a statement about classes to a statement about points. In $\mathbb{P}^2$ every curve is a hypersurface, so its degree is the degree of its equation by proposition 5.4.2(3); the lemma settles a pair of distinct integral curves, and additivity of the local count in each argument extends it to arbitrary curves without a common component. Bézout's theorem then acquires its classical form.

Corollary 5.4.6: Bézout's Theorem for Plane Curves

Let F and G be nonzero forms on $\mathbb{P}^2$ of degrees c and d with no common irreducible factor, and let $C = V(F)$ and $D = V(G)$ be the plane curves they cut out, so that C and D have no common component; neither is assumed irreducible or reduced. Then $C \cap D$ is finite and

$$\sum_{P \in C\cap D} \dim_{\overline{k}} \big(\mathcal{O}_{\mathbb{P}^2,P}/(f_P, g_P)\big) \;=\; cd$$

where f_P and g_P are the local equations of C and D at P, obtained by dehomogenizing F and G.

***Proof*:**
Factor $F = \prod_i F_i^{m_i}$ and $G = \prod_j G_j^{n_j}$ into powers of pairwise nonassociate irreducible forms; by hypothesis no F_i is an associate of any G_j. The curves $C_i = V(F_i)$ and $D_j = V(G_j)$ are one-dimensional closed subvarieties of $\mathbb{P}^2$, of degrees $\deg F_i$ and $\deg G_j$ by proposition 5.4.2(3), and $C_i \neq D_j$ for all i, j. For a form u and a point P write $u_P \in \mathcal{O}_{\mathbb{P}^2,P}$ for the dehomogenization of u at P, so that $f_P = F_P$ and $g_P = G_P$.

Step 1: the count for a pair of components. The plane $\mathbb{P}^2$ is a smooth variety of dimension two and $C_i \neq D_j$ are distinct one-dimensional closed subvarieties, so lemma 5.4.5 applies to the pair (C_i, D_j): the intersection $C_i \cap D_j$ is finite and $[C_i] \cdot [D_j] = \sum_P \dim_{\overline{k}}(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P}))\,[P]$ in $A_0(\mathbb{P}^2)$. Each point is closed with residue field $\overline{k}$, so taking degrees (definition 1.2.4) and comparing with theorem 5.4.4 for $n = 2$, $r = s = 1$ gives

$$\sum_{P \in C_i\cap D_j} \dim_{\overline{k}} \big(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P})\big) \;=\; (\deg F_i)(\deg G_j)$$

Since $C \cap D = \bigcup_{i,j} C_i \cap D_j$ as sets, $C \cap D$ is finite.

Step 2: the local count is additive in each argument. Fix P and write $\mathcal{O} = \mathcal{O}_{\mathbb{P}^2,P}$, a regular local ring of dimension two. Let u be a divisor of F and let v, w be divisors of G, or the same with the roles of F and G exchanged; then $V(u) \cap V(vw) \subseteq C \cap D$ is finite by Step 1, so $\mathcal{O}/(u_P, (vw)_P)$ and its two companions below are Artinian as in lemma 5.4.5(1), of finite $\overline{k}$-dimension. We may assume u_P and w_P lie in the maximal ideal, since otherwise one of the three quotients is zero and the additivity claimed below is trivial. Being Artinian, $\mathcal{O}/(u_P, w_P)$ exhibits u_P, w_P as a system of parameters of $\mathcal{O}$, hence as a regular sequence ([5, A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence]); in particular w_P is a nonzerodivisor on $A = \mathcal{O}/(u_P)$. The sequence

$$0 \longrightarrow A/(v_P) \xrightarrow{w_P} A/(v_P w_P) \longrightarrow A/(w_P) \longrightarrow 0$$

is then exact: surjectivity on the right and exactness in the middle are immediate, and if $w_P h \in (v_P w_P)A$ then $w_P(h - v_P a) = 0$ in A for some a, so $h \in (v_P)A$ because w_P is a nonzerodivisor on A. Counting $\overline{k}$-dimensions gives

$$\dim_{\overline{k}} \mathcal{O}/(u_P, v_P w_P) \;=\; \dim_{\overline{k}} \mathcal{O}/(u_P, v_P) \;+\; \dim_{\overline{k}} \mathcal{O}/(u_P, w_P)$$

that is, additivity in the second argument, and the exchanged case gives additivity in the first. Iterating over the two factorizations,

$$\dim_{\overline{k}} \left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P, g_P)\right) \;=\; \sum_{i,j} m_i n_j \dim_{\overline{k}} \left(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P})\right)$$

Step 3: summation. Sum the last display over the finitely many points of $C \cap D$; a point outside $C_i \cap D_j$ contributes nothing to the (i,j) term, since there $F_{i,P}$ or $G_{j,P}$ is a unit, so Step 1 evaluates each term and

$$\sum_{P \in C \cap D} \dim_{\overline{k}} \left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P, g_P)\right) \;=\; \sum_{i,j} m_i n_j (\deg F_i)(\deg G_j) \;=\; \Big(\sum_i m_i \deg F_i\Big)\Big(\sum_j n_j \deg G_j\Big) \;=\; cd$$

the last equality because $\deg F = \sum_i m_i \deg F_i$ and $\deg G = \sum_j n_j \deg G_j$, completing the proof.

This is the count promised in example 1.2.6 and the identity $h_1 \cdot h_1 = h_0$ anticipated in example 1.5.4(2), now with the local distribution attached: the integer $\dim_{\overline{k}} \mathcal{O}_{\mathbb{P}^2,P}/(f_P, g_P)$ says how many of the cd units of intersection are spent at P. Nothing in the argument assumed that the curves are irreducible, reduced, smooth, or that they meet transversally; only that they share no component, which is exactly what makes the intersection finite.

Example 5.4.7: A Cuspidal Cubic and Its Cuspidal Tangent

Let $C = \{y^2 z = x^3\} \subseteq \mathbb{P}^2$, a cubic curve with a cusp at $P = [0:0:1]$, and let $D = \{y = 0\}$, the tangent line to C at the cusp. The cubic is irreducible and is not the line, so the two share no component and corollary 5.4.6 predicts a total of $3 \cdot 1 = 3$.

Set-theoretically, $y = 0$ together with $y^2 z = x^3$ forces $x^3 = 0$, hence $x = 0$: the only intersection point is P. Working in the chart $z = 1$ with $\mathcal{O} = \overline{k}[x,y]_{(x,y)}$, the local equations are $f = y^2 - x^3$ and $g = y$, and

$$\mathcal{O}/(f,g) \;=\; \overline{k}[x,y]_{(x,y)}/(y^2 - x^3,\, y) \;\cong\; \overline{k}[x]_{(x)}/(x^3)$$

of dimension 3 over $\overline{k}$, with basis $1, x, x^2$. So all three units of intersection sit at the cusp, and $[C] \cdot [D] = 3\,[P]$. Compare example 2.2.2, where a conic met its tangent line in a single point of multiplicity two: in both cases the multiplicity is the order to which the equation of the line vanishes along the curve, and here the cusp raises it from two to three.

The integer attached to P in the example is not special to curves in a plane. Whenever two subvarieties of a smooth variety meet in the expected dimension, the product distributes over the components of the intersection with well-defined integer coefficients, and those coefficients are the intersection multiplicities.

Definition 5.4.8: Proper Intersection and Intersection Multiplicity

Let X be a smooth variety of pure dimension n and let $V, W \subseteq X$ be closed subvarieties of dimensions r and s. Say that V and W *meet properly* if every irreducible component of $V \cap W$ has dimension exactly $m = r + s - n$. In that case $V \cap W$ is pure of dimension m, so $A_m(V \cap W) = Z_m(V \cap W)$ is free on the classes of its irreducible components $Z_1, \ldots, Z_t$ (example 1.1.6(3)), and the refined intersection class of definition 5.1.10 may be written uniquely as

$$[V] \cdot [W] \;=\; \sum_{j=1}^{t} i(Z_j;\, V \cdot W;\, X)\,[Z_j] \;\in\; A_m(V \cap W)$$

The integer $i(Z_j;\, V \cdot W;\, X)$ is the *intersection multiplicity* of V and W along Z_j.

The definition would be empty without the refinement: the class $[V] \cdot [W]$ read in $A_m(X)$ carries no information about where the intersection happens, and it is the localization on $V \cap W$ supplied by the refined Gysin map (definition 4.3.6, corollary 4.3.11) that makes the coefficients meaningful. The proper-intersection hypothesis is what makes the coefficients unique, since A_m of an m-dimensional scheme is free on components; drop it and the class still exists but no longer decomposes canonically.

When the scheme-theoretic intersection is as good as its dimension suggests, the multiplicities are lengths and nothing else. Proposition 4.3.3 says that if $V \cap W \hookrightarrow V \times W$ is a regular embedding of codimension n, then $[V] \cdot [W] = [V \cap W]$ is the fundamental cycle, so $i(Z_j; V \cdot W; X)$ is the length of the local ring of $V \cap W$ at the generic point of Z_j. That is exactly what lemma 5.4.5 verified for curves on a surface, and it is the source of the numbers computed in example 5.4.7. In general the regularity hypothesis can fail, and then the multiplicity and the naive length part company. The correction is homological.

Theorem 5.4.9: Serre's Tor Formula

Let X be a smooth variety of pure dimension n and let $V, W \subseteq X$ be closed subvarieties meeting properly, with Z an irreducible component of $V \cap W$ and $\mathcal{O} = \mathcal{O}_{X,Z}$ the local ring of X at the generic point of Z. Then

$$i(Z;\, V \cdot W;\, X) \;=\; \sum_{j \geq 0} (-1)^j \operatorname{length}_{\mathcal{O}} \operatorname{Tor}_j^{\mathcal{O}}\big(\mathcal{O}/\mathcal{I}_V,\, \mathcal{O}/\mathcal{I}_W\big)$$

where $\mathcal{I}_V$ and $\mathcal{I}_W$ are the ideals of V and W in $\mathcal{O}$. The sum is finite, each term having finite length and all terms vanishing for $j > n$.

The proof is omitted: it runs entirely through homological algebra over regular local rings, in particular the vanishing, finite-length and additivity theorems for Tor that this book does not develop, and it is genuinely outside the scope of a treatment built on deformation to the normal cone. Serre proved the formula in *Algèbre locale, multiplicités*, and the comparison with the multiplicities of definition 5.4.8 is carried out in Fulton [8]. The formula matters here because it exhibits the multiplicity as a purely local, purely algebraic invariant of the two local rings, with no reference to the deformation that defined it.

The correction terms are invisible in the case the previous results have been using, and that is worth recording, since it says the classical count of plane curves needs no homological algebra at all.

Proposition 5.4.10: Vanishing of the Higher Terms for Divisors

Let X be a smooth variety of pure dimension n and let $C, D \subseteq X$ be closed subvarieties of dimension $n-1$ with $C \neq D$, so that C and D meet properly. For every irreducible component Z of $C \cap D$, with $\mathcal{O} = \mathcal{O}_{X,Z}$ and local equations $f, g \in \mathcal{O}$,

$$\operatorname{Tor}_j^{\mathcal{O}}\big(\mathcal{O}/(f),\, \mathcal{O}/(g)\big) = 0 \quad \text{for } j \geq 1$$

so Serre's formula reduces to $i(Z;\, C \cdot D;\, X) = \operatorname{length}_{\mathcal{O}} \mathcal{O}/(f, g)$.

***Proof*:**
Since X is smooth, $\mathcal{O}$ is a regular local ring and C, D are effective Cartier divisors near Z, so f and g are nonzerodivisors and

$$0 \longrightarrow \mathcal{O} \xrightarrow{\ f\ } \mathcal{O} \longrightarrow \mathcal{O}/(f) \longrightarrow 0$$

is a free resolution of $\mathcal{O}/(f)$ of length one. Hence $\operatorname{Tor}_j(\mathcal{O}/(f), \mathcal{O}/(g)) = 0$ for $j \geq 2$, while Tor_1 is the kernel of multiplication by f on $\mathcal{O}/(g)$. Because $\mathcal{O}$ is regular and g is a nonzerodivisor, $\mathcal{O}/(g)$ is Cohen-Macaulay, hence has no embedded associated primes ([5, A Cohen-Macaulay Ring Has No Embedded Associated Primes]), so its associated primes are the minimal primes of (g), which are the primes corresponding to the components of D passing through Z. As C and D share no component, f lies in none of them, so multiplication by f on $\mathcal{O}/(g)$ is injective and $\operatorname{Tor}_1 = 0$. The remaining term is $\operatorname{Tor}_0(\mathcal{O}/(f), \mathcal{O}/(g)) = \mathcal{O}/(f, g)$, and theorem 5.4.9 therefore reads $i(Z; C \cdot D; X) = \operatorname{length}_{\mathcal{O}} \mathcal{O}/(f, g)$, as we sought to show.

Everything so far has been internal to this book: a product, a ring, a count. The comparison that remains is external. *Algebraic Geometry* [2] builds a pairing on the divisors of a smooth projective surface without any of this machinery, setting $C \cdot D = \chi(\mathcal{O}_S) - \chi(\mathcal{O}_S(-C)) - \chi(\mathcal{O}_S(-D)) + \chi(\mathcal{O}_S(-C-D))$, and derives adjunction, the behaviour of self-intersection, and the classification of surfaces from it. Two constructions with nothing in common beyond the answer they are supposed to give must be shown to give it.

Theorem 5.4.11: The Agreement Theorem for Surfaces

Let S be a smooth projective surface over $\overline{k}$ and let C, D be Cartier divisors on S, with associated classes $[C] = c_1(\mathcal{O}_S(C)) \cap [S]$ and $[D] = c_1(\mathcal{O}_S(D)) \cap [S]$ in $A^1(S) = A_1(S)$. Then

$$\int_S ([C] \cdot [D]) \;=\; \int_S (C \cdot (D \cdot [S])) \;=\; C \cdot D$$

where the first is the intersection product of section 5.1, the second the iterated divisor operator of definition 2.2.1, and the third the cohomological intersection number [2, Intersection Number (Cohomological)].

***Proof*:**
Write $\langle C, D\rangle_1$, $\langle C, D\rangle_2$ and $\langle C, D\rangle_3$ for the three quantities.

Step 1: all three are $\mathbb{Z}$-bilinear and depend only on linear equivalence. For the first, $C \mapsto [C] = C \cdot [S]$ is additive in C by additivity of the first Chern class (proposition 2.4.4(3)) and depends only on $\mathcal{O}_S(C)$ by proposition 2.4.2; the product is $\mathbb{Z}$-bilinear (proposition 5.1.9(1)) and $\int_S$ is a homomorphism (definition 1.2.4). For the second, $D \mapsto D \cdot [S]$ is additive and linear-equivalence invariant for the same two reasons, and $C \cdot -$ is a homomorphism of Chow groups additive in C by proposition 2.4.4(1),(3). For the third, bilinearity and invariance under linear equivalence are [2, Properties of the Intersection Pairing](1),(3).

Step 2: the three agree for distinct prime divisors. Let C and D be distinct one-dimensional closed subvarieties of S, with local equations f, g at a point P. Lemma 5.4.5(3) gives

$$\langle C, D\rangle_1 \;=\; \int_S \Big(\sum_{P \in C \cap D} \dim_{\overline{k}} \left(\mathcal{O}_{S,P}/(f,g)\right) [P] \Big) \;=\; \sum_{P \in C \cap D} \dim_{\overline{k}} \left(\mathcal{O}_{S,P}/(f,g)\right)$$

each P being a closed point with residue field $\overline{k}$. For the second quantity, $S \not\subseteq |D|$ and $D \not\subseteq |C|$, so the first clause of definition 2.2.1 applies twice: $D \cdot [S] = [\mathrm{div}(s_S)] = [D]$, and $C \cdot [D] = [\mathrm{div}(f|_D)] = \sum_P \mathrm{ord}_P(f|_D)\,[P]$. By definition 1.1.1 the coefficient is $\mathrm{ord}_P(f|_D) = \mathrm{length}_{\mathcal{O}_{D,P}} \left(\mathcal{O}_{D,P}/(f|_D)\right)$, and $\mathcal{O}_{D,P}/(f|_D) = \mathcal{O}_{S,P}/(f,g)$, whose length equals its $\overline{k}$-dimension. Hence $\langle C, D\rangle_2$ is the same sum. For the third, [2, Properties of the Intersection Pairing](5) evaluates $C \cdot D$ on effective divisors with no common component as $\sum_P \dim_{\overline{k}} \mathcal{O}_{S,P}/(f,g)$, again the same sum. So the three agree on such a pair.

Step 3: reduction of the general case to Step 2. Fix a very ample divisor H on the projective surface S. By Serre's theorem on global generation after an ample twist ([2, Global Section Generation of Twisted Sheaves]) there is an $m \geq 1$ such that $\mathcal{O}_S(C + mH)$, $\mathcal{O}_S(D + mH)$ and $\mathcal{O}_S(mH)$ are all generated by their global sections. Set

$$A_1 = C + mH, \quad A_2 = mH, \quad B_1 = D + mH, \quad B_2 = mH$$

so that $C = A_1 - A_2$ and $D = B_1 - B_2$ as Cartier divisors, and by Step 1 it suffices to prove the agreement for each of the four pairs (A_i, B_j).
A globally generated line bundle has a nonzero global section, so each A_i is linearly equivalent to an effective divisor A_i'; let $\Gamma_1, \ldots, \Gamma_t$ be the finitely many prime components of A_1' and A_2' and choose a closed point $p_a \in \Gamma_a$ for each. For a fixed j, the sections of $\mathcal{O}_S(B_j)$ vanishing at p_a form a proper linear subspace of the finite-dimensional space $H^0(S, \mathcal{O}_S(B_j))$, since that

bundle is globally generated; a vector space over the infinite field $\overline{k}$ is not a finite union of proper subspaces, so some section τ_j vanishes at no p_a. Its divisor $B'_j = \operatorname{div}(\tau_j)$ is effective, linearly equivalent to B_j, and contains none of the Γ_a.
Writing $A'_i = \sum_a m_a \Gamma_a$ and $B'_j = \sum_b n_b \Lambda_b$ as nonnegative combinations of prime divisors, no Γ_a equals any Λ_b, so bilinearity reduces each $\langle A'_i, B'_j \rangle_\bullet$ to $\sum_{a,b} m_a n_b \langle \Gamma_a, \Lambda_b \rangle_\bullet$, and every term is covered by Step 2. Invariance under linear equivalence transfers the conclusion back from A'_i, B'_j to A_i, B_j, and Step 1 assembles the four pairs into the general case, completing the proof.

The theorem is what connects the two. Every numerical statement about divisors on a smooth projective surface proved in *Algebraic Geometry* [2] out of Euler characteristics, the adjunction formula, the identification of C^2 with the degree of the normal bundle, the Hodge index theorem, is a statement about the product of this chapter, and conversely every general theorem proved here specializes to a surface fact that book had to obtain by hand. The surface theory is subsumed rather than paralleled, and section 5.3 already used that when it recovered $E^2 = -1$ from the self-intersection formula.

The middle term of the theorem is worth isolating. It says that on a smooth projective surface the ring product of two divisor classes computes the same integer as the iterated divisor operator built in chapter 2, so the operator calculus of that chapter, commutativity and the projection formula included, transfers verbatim to the ring. The two constructions were built from different material: one from local equations and orders of vanishing, the other from a degeneration of the diagonal. Nothing forced them to agree except that both were built to count.

Exercise 5.4.1

1. Let $V, W \subseteq \mathbb{P}^3$ be surfaces of degrees 2 and 3 with no common component. Compute $[V] \cdot [W]$ in $A_1(\mathbb{P}^3)$ and read off the degree of the curve $V \cap W$. Then let $V_1, V_2, V_3 \subseteq \mathbb{P}^3$ be surfaces of degrees d_1, d_2, d_3 and compute $\int_{\mathbb{P}^3} \big([V_1] \cdot [V_2] \cdot [V_3]\big)$.

2. Let $L \cong \mathbb{P}^r$ and $M \cong \mathbb{P}^s$ be linear subspaces of $\mathbb{P}^n$ with $r + s < n$. Show that $[L] \cdot [M] = 0$, and explain why this conclusion is *not* a consequence of the observation that L and M can be chosen disjoint. (Consider the case $L = M$.)

3. Keep $C = \{y^2 z = x^3\} \subseteq \mathbb{P}^2$ from example 5.4.7 and take instead $D' = \{x = 0\}$, the line joining the cusp $[0 : 0 : 1]$ to the inflection point $[0 : 1 : 0]$. Compute $C \cap D'$ and the multiplicity of $[C] \cdot [D']$ at each of its points, and verify corollary 5.4.6.

4. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$, with rulings $f_1 = \{a\} \times \mathbb{P}^1$ and $f_2 = \mathbb{P}^1 \times \{b\}$ as in exercise 2.2.1 (5), where $\mathcal{O}_Q(f_1) = \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(1)$ and $\mathcal{O}_Q(f_2) = \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^1}(1)$. Using theorem 5.4.11, write down the three numbers $\int_Q([f_1] \cdot [f_2])$, $\int_Q([f_1] \cdot [f_1])$ and $\int_Q([f_2] \cdot [f_2])$. Then show that the diagonal $\Delta \subseteq Q$ satisfies $\Delta \sim f_1 + f_2$ and compute $\int_Q([\Delta] \cdot [\Delta])$.

5. Let $L \subseteq \mathbb{P}^2$ be a line. Compute $\int_{\mathbb{P}^2}([L] \cdot [L])$ from example 5.2.3, and identify precisely which conclusion of lemma 5.4.5 fails when $C = D = L$. Explain how the intersection product still returns an answer, and reconcile it with example 2.2.3.

Part III

Cycle Classes, Riemann-Roch, and Enumerative Geometry

The Chow ring is built, and with it the question that motivated the construction has an answer: two subvarieties of a smooth variety can be intersected, and the result is a class that behaves. What the ring does not yet do is talk to anything else. A cycle class is an algebraic invariant of an algebraic object, and the classical invariants of a complex variety are topological; until the two are compared, the theory is self-contained in the unhelpful sense.

This part makes the comparison and then spends it. Chapter 6 maps cycles to cohomology and finds that the map is far from injective, which turns a single equivalence relation into a tower of four: rational, algebraic, homological, numerical. Where the tower collapses is, in almost every case, an open problem, and two of the gaps are among the most famous questions in the subject. Chapter 7 pays the comparison forward: the Chern character carries K-theory into the Chow ring, and Grothendieck-Riemann-Roch measures its failure to commute with pushforward by a Todd class, recovering the classical Riemann-Roch theorems as the case of a map to a point. Chapter 8 then uses the whole apparatus for the purpose the subject was invented for, which is counting.

6

Cycle Classes and Adequate Equivalence Relations

Rational equivalence was chosen in chapter 1 for its formal properties, not because it was the only candidate, and it is in fact the finest of a family. Coarsening it produces new groups, each with its own virtues: algebraic equivalence brings in families over curves and makes the quotient finitely generated; homological equivalence is whatever cohomology cannot distinguish; numerical equivalence remembers only intersection numbers. The four sit in a tower, each contained in the next, and the inclusions are the subject of this chapter.

The comparison that generates the tower is the cycle class map. For a smooth complex variety a subvariety carries a fundamental class in homology, and Poincaré duality turns it into a cohomology class; the content of section 6.2 is that this assignment is blind to rational equivalence and respects everything the Chow ring can do. Once it exists, homological equivalence is defined as its kernel, and the question of how much information the map loses becomes precise.

It loses some. Cohomology sees a cycle only through its class, and the Hodge conjecture asks whether every cohomology class that looks algebraic is algebraic. Whether homological and numerical equivalence agree is the standard conjecture D. Neither is proved here or anywhere; what this chapter owes the reader is exact statements, correct attributions, and a clear account of which implications are theorems.

The chapter closes with correspondences. A cycle on $X \times Y$ acts on cycles and on cohomology by pulling back, intersecting, and pushing forward, and composing two such cycles is again the intersection product. That is the operation the theory of motives is built from, and it is the form in which [1] consumes this book.

6.1 Algebraic Equivalence

A cycle rationally equivalent to zero is a sum of divisors of rational functions, and a nonzero rational function on a $(k+1)$-dimensional subvariety W is a morphism from a dense open subset of W to $\mathbb{P}^1$. The parameter along which such a cycle moves is therefore always the same rational curve. Algebraic equivalence allows that parameter to be an arbitrary smooth curve, and the relation it defines is coarser. Its quotient in codimension one is the group that [2, Néron-Severi Group and Picard Number] attaches to divisors, and that quotient, unlike the Chow group itself, is finitely generated.

The data that moves a cycle is a single cycle on $X \times T$ whose components are spread over the parameter curve T rather than sitting inside one fibre. Cutting such a cycle with a fibre produces a cycle on X, and the first point to settle is that the cut has the expected dimension, since only then is there a k-cycle to compare across two parameter values. Throughout, pr_X and pr_T are the two projections of $X \times T$, and for a closed point $t \in T$

$$\iota_t \colon X \longrightarrow X \times T, \qquad x \longmapsto (x, t)$$

is the closed immersion with image $X \times \{t\}$, an isomorphism onto that fibre; cycles on $X \times \{t\}$ are identified with cycles on X along it without further comment.

Lemma 6.1.1: Fibres of a Dominating Subvariety

Let X be an algebraic scheme, T a smooth curve, and $W \subseteq X \times T$ a $(k+1)$-dimensional subvariety whose image under pr_T is dense in T. For a closed point $t \in T$ let $W_t = W \times_T \{t\}$ be the scheme-theoretic fibre, a closed subscheme of $X \times \{t\}$. Then W_t is either empty or of pure dimension k.

***Proof*:**
Write $\pi \colon W \to T$ for the restriction of pr_T. It is a dominant morphism of integral schemes of finite type over $\bar{k}$, and $\dim W - \dim T = (k+1) - 1 = k$. Suppose $W_t \neq \emptyset$, so that $t \in \pi(W)$, and let Z be an irreducible component of W_t.

For the lower bound, the fibre-dimension theorem for a dominant morphism of integral schemes of finite type over a field ([9, Exercise II.3.22]) applies to π at the point $t \in \pi(W)$ and gives $\dim Z \geq k$.

For the upper bound, Z is a closed irreducible subset of the irreducible $(k+1)$-dimensional W, so $\dim Z \leq k+1$, with equality only if $Z = W$. Equality is impossible: from $Z \subseteq W_t \subseteq X \times \{t\}$, the equality $Z = W$ would give $\pi(W) = \{t\}$, a single closed point of the curve T, which is closed and proper in T and hence not dense, contradicting the hypothesis. So $\dim Z = k$ for every component Z of W_t, as we sought to show.

By the lemma, a subvariety spread over T meets each fibre in a scheme of pure dimension k, and the fundamental cycle of that scheme (definition 1.1.2) is a k-cycle on X. A family of cycles is a $\mathbb{Z}$-combination of such subvarieties.

Definition 6.1.2: Family of Cycles over a Curve

Let X be an algebraic scheme and T a *smooth curve*, by which is meant a smooth variety of dimension one, not necessarily complete. A *family of k-cycles on X parametrized by T* is a $(k+1)$-cycle

$$V = \sum_i n_i [V_i] \in Z_{k+1}(X \times T)$$

each of whose components V_i has dense image under pr_T. For a closed point $t \in T$, the *fibre of V over t* is the k-cycle

$$V(t) = \sum_i n_i \left[(V_i)_t\right] \in Z_k(X)$$

where $(V_i)_t$ is the scheme-theoretic fibre of lemma 6.1.1 and $[(V_i)_t]$ its fundamental cycle, read on X through ι_t. A fibre may be zero, since $(V_i)_t$ is empty for t outside the image of V_i.

Two features of the definition are worth separating. The condition on components is what makes the fibres k-cycles rather than cycles of jumping dimension: a component contained in a single fibre would contribute its whole $(k+1)$-dimensional self there and nothing elsewhere. And $t \mapsto V(t)$ is additive in V by construction, so a sum of families over one curve has for fibres the sums of the fibres.

Definition 6.1.3: Algebraic Equivalence

Let X be an algebraic scheme. The subgroup $\mathrm{Alg}_k(X) \subseteq Z_k(X)$ of *cycles algebraically equivalent to zero* is the subgroup generated by all differences

$$V(t_0) - V(t_1)$$

taken over all smooth curves T, all families V of k-cycles on X parametrized by T, and all pairs of closed points $t_0, t_1 \in T$. Two cycles $\alpha, \beta \in Z_k(X)$ are *algebraically equivalent*, written $\alpha \sim_{\mathrm{alg}} \beta$, if $\alpha - \beta \in \mathrm{Alg}_k(X)$.

Proposition 6.1.4: Algebraic Equivalence is a Graded Equivalence Relation

Let X be an algebraic scheme. Then $\sim_{\mathrm{alg}}$ is an equivalence relation on $Z_k(X)$ compatible with addition of cycles, and $\mathrm{Alg}_*(X) = \bigoplus_k \mathrm{Alg}_k(X)$ is a graded subgroup of $Z_*(X)$.

***Proof*:**
The relation is defined by membership of a difference in a subgroup, so it is reflexive ($0 \in \mathrm{Alg}_k(X)$), symmetric (subgroups are closed under negation) and transitive (subgroups are closed under addition); and from $\alpha - \beta, \alpha' - \beta' \in \mathrm{Alg}_k(X)$ one gets $(\alpha + \alpha') - (\beta + \beta') \in \mathrm{Alg}_k(X)$, which is compatibility with addition. Each generator $V(t_0) - V(t_1)$ lies in the single summand $Z_k(X)$, because V is a $(k+1)$-cycle and its fibres are k-cycles by definition 6.1.2; hence the subgroup they generate inside $Z_*(X)$ is the direct sum of the $\mathrm{Alg}_k(X)$, as we sought to show.

Example 6.1.5: Points on a Curve

Let C be a smooth projective curve and take $T = C$ as the parameter. The diagonal $\Delta \subseteq C \times C$ is a subvariety of dimension one, and pr_2 carries it isomorphically onto C, so its image is dense and $[\Delta]$ is a family of 0-cycles on C parametrized by C. Since $\mathrm{pr}_2|_\Delta$ is an isomorphism, the scheme-theoretic fibre $\Delta_t = \Delta \times_C \{t\}$ is one reduced point, namely (t,t), so

$$\Delta(t) = [t] \in Z_0(C)$$

Hence $[p] - [q] = \Delta(p) - \Delta(q)$ lies in $\mathrm{Alg}_0(C)$ for every pair of closed points p, q. A zero-cycle $\sum_i n_i [p_i]$ with $\sum_i n_i = 0$ is a $\mathbb{Z}$-combination of such differences, so every zero-cycle of degree zero on C is algebraically equivalent to zero.

The contrast with rational equivalence is visible already here. By proposition 2.1.1, applied to the smooth curve C, the group $A_0(C) = \mathrm{Cl}\, C$ is the full divisor class group, in which a degree-zero class need not vanish; under algebraic equivalence every degree-zero class does vanish.

The example exhibits one relation as coarser than the other in a single case. The general statement is that every cycle rationally equivalent to zero is algebraically equivalent to zero, and its proof is the observation the definition was built around: a rational function is a family of cycles parametrized by $\mathbb{P}^1$, whose two marked fibres are the divisor of zeros and the divisor of poles.

Theorem 6.1.6: Rational Equivalence Implies Algebraic Equivalence

For every algebraic scheme X and every k,

$$\mathrm{Rat}_k(X) \subseteq \mathrm{Alg}_k(X)$$

so $\alpha \sim \beta$ implies $\alpha \sim_{\mathrm{alg}} \beta$, with $\mathbb{P}^1$ as the parameter curve.

***Proof*:**

Step 1: reduction to one function. By definition 1.1.3 the group $\mathrm{Rat}_k(X)$ is generated by the cycles $[\mathrm{div}(r)]$ with $W \subseteq X$ a $(k+1)$-dimensional subvariety and $r \in K(W)^*$, and $\mathrm{Alg}_k(X)$ is a subgroup, so it suffices to prove $[\mathrm{div}(r)] \in \mathrm{Alg}_k(X)$ for one such pair (W, r).

If $r \in \overline{k}^*$, then r is a unit in the local ring $\mathcal{O}_{W,\eta_Y}$ at the generic point of every prime divisor $Y \subseteq W$, so $\mathrm{ord}_Y(r) = 0$ for all Y and $[\mathrm{div}(r)] = 0$, which lies in $\mathrm{Alg}_k(X)$. Assume from now on that r is nonconstant.

Step 2: the graph of r. Let $U \subseteq W$ be the union of the locus where r is regular with the locus where r^{-1} is regular; it is a dense open subset on which r defines a morphism $r|_U : U \to \mathbb{P}^1$. Put

$$\Gamma = \overline{\{(x, r(x)) \mid x \in U\}} \subseteq W \times \mathbb{P}^1 \subseteq X \times \mathbb{P}^1$$

the closure of the graph, with its reduced structure. The graph of $r|_U$ is the image of the immersion $(\mathrm{id}, r|_U) : U \to W \times \mathbb{P}^1$, hence irreducible of dimension $\dim U = k+1$, so Γ is a $(k+1)$-dimensional subvariety of $X \times \mathbb{P}^1$.

The projection $p = \mathrm{pr}_W|_\Gamma : \Gamma \to W$ is proper, being the restriction to a closed subscheme of the projection $W \times \mathbb{P}^1 \to W$, which is proper because $\mathbb{P}^1$ is complete. Its image contains U and is closed, hence is W, so p is surjective; and p restricts to an isomorphism over U, where Γ is the graph of a morphism, so p is birational and $p^*\colon K(W) \to K(\Gamma)$ is an isomorphism.

The projection $g = \mathrm{pr}_{\mathbb{P}^1}|_\Gamma\colon \Gamma \to \mathbb{P}^1$ is a morphism whose image contains $r(U)$, dense in $\mathbb{P}^1$ because r is nonconstant. So Γ has dense image in $\mathbb{P}^1$ and $[\Gamma]$ is a family of k-cycles on X parametrized by $T = \mathbb{P}^1$. As an element of $K(\Gamma)^*$ the function g is p^*r, since the two agree on the dense open $p^{-1}(U)$, where $g(x, r(x)) = r(x)$.

Step 3: the two fibres are the zeros and the poles of g. Write Γ_0 and Γ_∞ for the scheme-theoretic fibres of g over 0 and ∞, each empty or pure of dimension k by lemma 6.1.1. The claim is

$$[\mathrm{div}(g)] = [\Gamma_0] - [\Gamma_\infty] \quad \text{in } Z_k(\Gamma) \tag{6.1}$$

Let $Y \subseteq \Gamma$ be a k-dimensional subvariety with generic point η, and compare the coefficients of $[Y]$ on the two sides. Recall that for a function regular at η the order along Y is the length of the quotient of $\mathcal{O}_{\Gamma,\eta}$ by the ideal it generates, and that ord_Y is multiplicative.

Suppose $g(\eta) = 0$. On the open set $\Gamma \setminus \Gamma_\infty = g^{-1}(\mathbb{A}^1)$ the morphism g is a regular function whose zero scheme is Γ_0, so $\mathcal{O}_{\Gamma_0,\eta} = \mathcal{O}_{\Gamma,\eta}/(g)$. By definition 1.1.2 the coefficient of $[Y]$ in $[\Gamma_0]$ is $\mathrm{length}_{\mathcal{O}_{\Gamma,\eta}}\big(\mathcal{O}_{\Gamma,\eta}/(g)\big) = \mathrm{ord}_Y(g)$, which is the coefficient of $[Y]$ in $[\mathrm{div}(g)]$; and Y is not a component of Γ_∞, so it contributes nothing there.

Suppose $g(\eta) = \infty$. The same computation applied to g^{-1}, a regular function on $\Gamma \setminus \Gamma_0$ with zero scheme Γ_∞, gives the coefficient of $[Y]$ in $[\Gamma_\infty]$ as $\mathrm{ord}_Y(g^{-1}) = -\mathrm{ord}_Y(g)$, by multiplicativity of the order; and Y contributes nothing to $[\Gamma_0]$.

In the remaining case $g(\eta) \notin \{0, \infty\}$, so g is regular and nonvanishing at η, hence a unit in $\mathcal{O}_{\Gamma,\eta}$ and $\mathrm{ord}_Y(g) = 0$; and Y is a component of neither fibre, since g maps a component of Γ_0 to 0 and a component of Γ_∞ to ∞. Every k-dimensional subvariety of Γ falls under one of the three cases, and by lemma 6.1.1 the components of the two fibres are k-dimensional, so (6.1) holds.

Step 4: pushing down to W. The morphism p is proper and surjective between varieties of the same dimension $k+1$, so proposition 1.2.2(2) applies to $g = p^*r \in K(\Gamma)^*$ and gives $p_*[\mathrm{div}(g)] = [\mathrm{div}(N(g))]$, where N is the norm of the extension $p^*\colon K(W) \hookrightarrow K(\Gamma)$. That extension is an isomorphism by Step 2, so multiplication by g on the one-dimensional $K(W)$-vector space $K(\Gamma)$ has determinant $N(g) = r$, and therefore

$$p_*[\mathrm{div}(g)] = [\mathrm{div}(r)]$$

On the other side, $\Gamma_0 = \Gamma \cap (W \times \{0\})$ is a closed subscheme of $W \times \{0\}$, and p restricted to it is that closed immersion followed by the isomorphism $W \times \{0\} \cong W$. A closed immersion has $\deg(V/p(V)) = 1$ for every subvariety V of its source (definition 1.2.1), so $p_*[\Gamma_0]$ is the fibre $[\Gamma](0)$ of definition 6.1.2, and likewise $p_*[\Gamma_\infty] = [\Gamma](\infty)$. Applying p_* to (6.1),

$$[\mathrm{div}(r)] = [\Gamma](0) - [\Gamma](\infty)$$

a generator of $\mathrm{Alg}_k(X)$ attached to the family $[\Gamma]$ over $\mathbb{P}^1$ and the marked points $0, \infty$. With Step 1 this gives $\mathrm{Rat}_k(X) \subseteq \mathrm{Alg}_k(X)$, completing the proof.

The theorem makes algebraic equivalence a relation on Chow classes and not only on cycles, so the quotient below may be formed from either. Step 3 identifies the two marked fibres of the graph as the divisor of zeros and the divisor of poles of r, whose difference is $\mathrm{div}(r)$; the content of the theorem is that rational equivalence permits exactly this one motion, along the fixed curve $\mathbb{P}^1$, while algebraic equivalence permits motion along any smooth curve.

Definition 6.1.7: Néron-Severi Group in Codimension k

Let X be an algebraic scheme. Write $A_k(X)_{\mathrm{alg}}$ for the image of $\mathrm{Alg}_k(X)$ in $A_k(X)$ and set

$$\mathrm{NS}_k(X) = A_k(X)/A_k(X)_{\mathrm{alg}}$$

the *Néron-Severi group of X in dimension k*. When X is a smooth variety of dimension n, so that $A^k(X) = A_{n-k}(X)$ carries the codimension grading of theorem 5.2.7, write $\mathrm{NS}^k(X) = \mathrm{NS}_{n-k}(X) = A^k(X)/A^k(X)_{\mathrm{alg}}$.

Corollary 6.1.8: The Néron-Severi Group Computed from Cycles

For every algebraic scheme X the quotient map $Z_k(X) \to A_k(X)$ induces an isomorphism

$$Z_k(X)/\operatorname{Alg}_k(X) \xrightarrow{\sim} \mathrm{NS}_k(X)$$

***Proof*:**
By theorem 6.1.6 the subgroup $\mathrm{Rat}_k(X)$ is contained in $\mathrm{Alg}_k(X)$, and $A_k(X) = Z_k(X)/\mathrm{Rat}_k(X)$ by definition 1.1.4. Hence the image $A_k(X)_{\mathrm{alg}}$ of $\mathrm{Alg}_k(X)$ in $A_k(X)$ is $\mathrm{Alg}_k(X)/\mathrm{Rat}_k(X)$, and the third isomorphism theorem gives

$$\mathrm{NS}_k(X) = \big(Z_k(X)/\mathrm{Rat}_k(X)\big)/\big(\operatorname{Alg}_k(X)/\mathrm{Rat}_k(X)\big) \cong Z_k(X)/\operatorname{Alg}_k(X)$$

as we sought to show.

For a smooth projective curve C the corollary and example 6.1.5 already determine the answer: every degree-zero zero-cycle lies in $\mathrm{Alg}_0(C)$, so $\mathrm{NS}_0(C)$ is generated by the class of a single closed point.

In codimension one the group carries a name and a history predating the Chow group. A divisor on a smooth projective variety is a line bundle, a family of divisors over a curve is a line bundle on $X \times T$, and the classical Néron-Severi group is Pic modulo the subgroup swept out by such families. The two descriptions define the same subgroup of $A^1(X)$, and the proof of that is not formal: on one side the family is a cycle on $X \times T$, on the other a line bundle, and passing between them is the Weil-Cartier comparison applied on $X \times T$ rather than on X.

Definition 6.1.9: Algebraic Equivalence of Line Bundles

Let X be a smooth projective variety. A line bundle L on X is *algebraically equivalent to zero* if there are a smooth curve T, a line bundle $\mathcal{L}$ on $X \times T$, and closed points $t_0, t_1 \in T$ with

$$\iota_{t_0}^*\mathcal{L} \cong L, \qquad \iota_{t_1}^*\mathcal{L} \cong \mathcal{O}_X$$

The subgroup of $\operatorname{Pic} X$ generated by all such L is written $\operatorname{Pic}^0(X)$.

[2, Néron-Severi Group and Picard Number] writes $\operatorname{Pic}^0(X)$ for the group of line bundles algebraically equivalent to zero, and characterizes it as the group of $\overline{k}$-points of the identity component of the

Picard scheme. That this is the same subgroup as the one just defined is a theorem rather than a matter of notation, and it is the one place in this section where the divisor-level theory is taken on trust rather than rebuilt.

Proposition 6.1.10: The Two Descriptions of Pic^0 Agree

Let X be a smooth projective variety. The subgroup $\operatorname{Pic}^0(X) \subseteq \operatorname{Pic} X$ generated by the line bundles of definition 6.1.9 is the group of $\overline{k}$-points of the identity component of the Picard scheme, that is the $\operatorname{Pic}^0(X)$ of [2, Néron-Severi Group and Picard Number].

***Proof Cited*:**
The content is that a *curve* parameter base suffices: a family of line bundles over an arbitrary connected base may be replaced by a chain of families over smooth curves joining the same two members, after which the generators above already exhaust the group. That reduction, and the representability of the Picard functor that gives the identity component its meaning, are both outside this book (QC #3(a)); they are treated in [8] and in the standard accounts of the Picard scheme.

The dependency is worth naming, because the rest of the section rests on it: corollary 6.1.14 obtains finite generation of $\mathrm{NS}^1(X)$ by transporting [2, Theorem of the Base], which is a statement about the upstream Pic^0, across theorem 6.1.12, which is proved for the curve-generated one. If the two subgroups differed, that transport would fail.

Passing between the two descriptions rests on one computation, made once and used in both directions: restricting the line bundle of a family of divisors to a fibre gives the line bundle of the fibre cycle. Here and below, for a Weil divisor α on a smooth variety Y the notation $\mathcal{O}_Y(\alpha)$ means the line bundle attached to α by the isomorphism $\operatorname{WDiv} Y \cong \operatorname{CDiv} Y$ and the correspondence $\operatorname{CDiv} Y \to \operatorname{Pic} Y$ of proposition 2.1.1(2).

Lemma 6.1.11: Restriction of a Family of Divisors to a Fibre

Let X be a smooth variety of dimension n and T a smooth curve, so that $X \times T$ is a smooth variety of dimension $n+1$: smoothness is stable under products, and a product of varieties over the algebraically closed field $\overline{k}$ is again irreducible. Let D be a family of $(n-1)$-cycles on X parametrized by T, that is, an n-cycle on $X \times T$ each of whose components has dense image in T. Then for every closed point $t \in T$,

$$\iota_t^* \mathcal{O}_{X\times T}(D) \cong \mathcal{O}_X\big(D(t)\big)$$

***Proof*:**
Both sides are homomorphisms in D: the left because $\alpha \mapsto \mathcal{O}_{X\times T}(\alpha)$ is a homomorphism $\operatorname{WDiv}(X \times T) \to \operatorname{Pic}(X \times T)$ by proposition 2.1.1(2) and ι_t^* is a homomorphism of Picard groups, the right because $D \mapsto D(t)$ is additive by definition 6.1.2 and $\alpha \mapsto \mathcal{O}_X(\alpha)$ is a homomorphism. Each component of D is itself a family, so it suffices to treat $D = [W]$ for a single subvariety $W \subseteq X \times T$ of dimension n with dense image in T.

Since $X \times T$ is smooth it is locally factorial, so by proposition 2.1.1(2) the prime divisor W is locally principal: there are an open cover $\{U_i\}$ of $X \times T$ and regular functions $f_i \in \mathcal{O}_{X\times T}(U_i)$

generating the ideal sheaf of W on U_i, and $\{(U_i, f_i)\}$ is Cartier data for a divisor with associated line bundle $\mathcal{O}_{X\times T}(W)$ and associated Weil cycle $[W]$.

Fix a closed point t. The subvariety W is not contained in $X \times \{t\}$: the two have the same dimension n and $X \times \{t\}$ is irreducible, so containment would force $W = X \times \{t\}$ and hence $\mathrm{pr}_T(W) = \{t\}$, contradicting density. Therefore each f_i restricts to a nonzero rational function on the variety $X \times \{t\}$, so $\{(U_i \cap (X \times \{t\}), f_i|)\}$ is Cartier data on $X \times \{t\}$ whose line bundle is $\iota_t^* \mathcal{O}_{X\times T}(W)$. Applying proposition 2.1.1(1) on the smooth variety $X \times \{t\}$, the Weil cycle of that restricted Cartier divisor is

$$\sum_Y \nu_Y(f_i|)\,[Y]$$

the sum over prime divisors $Y \subseteq X \times \{t\}$, with i any index whose U_i contains the generic point η of Y.

It remains to identify this cycle with $[W_t]$. Since $X \times \{t\} \cong X$ is smooth, the local ring $A = \mathcal{O}_{X\times\{t\},\eta}$ is regular of dimension one, hence a discrete valuation ring whose valuation is ν_Y; let ϖ be a uniformizer. Near η the scheme W_t is the zero scheme of $f_i|$, so $\mathcal{O}_{W_t,\eta} = A/(f_i|)$. Writing $m = \nu_Y(f_i|)$ gives $(f_i|) = (\varpi^m)$, and the chain $A \supsetneq (\varpi) \supsetneq \cdots \supsetneq (\varpi^m)$ has m successive quotients, each isomorphic to $A/(\varpi)$, so

$$\mathrm{length}_A\left(A/(f_i|)\right) = m = \nu_Y(f_i|)$$

The left-hand side is the coefficient of $[Y]$ in the fundamental cycle $[W_t]$ (definition 1.1.2), and W_t is empty or pure of dimension $n-1$ by lemma 6.1.1, so the two cycles have the same components with the same multiplicities. Hence $\sum_Y \nu_Y(f_i|)[Y] = [W_t]$, which read on X is $W(t)$, and therefore $\iota_t^*\mathcal{O}_{X\times T}(W) \cong \mathcal{O}_X(W(t))$, completing the proof.

Theorem 6.1.12: Agreement with the Divisor-Level Néron-Severi Group

Let X be a smooth projective variety of dimension n. The class map $c\colon \mathrm{Pic}\,X \to A_{n-1}(X) = A^1(X)$ of proposition 2.1.3 is an isomorphism carrying $\mathrm{Pic}^0(X)$ onto $A^1(X)_{\mathrm{alg}}$, and therefore induces an isomorphism

$$\mathrm{NS}(X) \xrightarrow{\sim} \mathrm{NS}^1(X)$$

from the divisor-level Néron-Severi group of [2, Néron-Severi Group and Picard Number] to the codimension-one group of definition 6.1.7.

***Proof*:**
That c is an isomorphism is the last clause of proposition 2.1.3, whose hypothesis holds because a smooth variety is locally factorial; the same statement, through proposition 2.1.1(2), gives $c(\mathcal{O}_X(E)) = [E]$ in $A_{n-1}(X)$ for every Weil divisor E on X. Granting the isomorphism, the induced map on quotients is an isomorphism as soon as $c(\mathrm{Pic}^0(X)) = A^1(X)_{\mathrm{alg}}$, and since both subgroups are generated by the elements described in definition 6.1.9 and definition 6.1.3, it suffices to match generators in the two directions.

Step 1: a family of cycles produces a family of line bundles. Let V be a family of $(n-1)$-cycles on X parametrized by a smooth curve T, with marked points t_0, t_1; the class of $V(t_0) - V(t_1)$ is the general generator of $A^1(X)_{\mathrm{alg}}$. The product $X \times T$ is a smooth variety of dimension

$n+1$ (lemma 6.1.11), so the n-cycle V has an associated line bundle $\mathcal{O}_{X\times T}(V)$; and $V(t_1)$, an $(n-1)$-cycle on the smooth X, has an associated line bundle $\mathcal{O}_X(V(t_1))$. Set

$$\mathcal{L} = \mathcal{O}_{X\times T}(V) \otimes \mathrm{pr}_X^* \mathcal{O}_X\big(V(t_1)\big)^{-1}$$

Since $\mathrm{pr}_X \circ \iota_t = \mathrm{id}_X$, one has $\iota_t^* \mathrm{pr}_X^* N \cong N$ for every line bundle N on X, so lemma 6.1.11 gives

$$\iota_t^* \mathcal{L} \cong \mathcal{O}_X\big(V(t)\big) \otimes \mathcal{O}_X\big(V(t_1)\big)^{-1} = \mathcal{O}_X\big(V(t) - V(t_1)\big)$$

for every closed $t \in T$. At $t = t_1$ this is $\mathcal{O}_X$, and at $t = t_0$ it is $L := \mathcal{O}_X(V(t_0) - V(t_1))$, so $L \in \mathrm{Pic}^0(X)$ by definition 6.1.9. As c is a homomorphism sending $\mathcal{O}_X(E)$ to $[E]$, its value is $c(L) = [V(t_0)] - [V(t_1)]$. Every generator of $A^1(X)_{\mathrm{alg}}$ therefore lies in $c(\mathrm{Pic}^0(X))$.

Step 2: a family of line bundles produces a family of cycles. Let L be a generator of $\mathrm{Pic}^0(X)$, given by a smooth curve T, a line bundle $\mathcal{L}$ on $X \times T$ and closed points t_0, t_1 with $\iota_{t_0}^* \mathcal{L} \cong L$ and $\iota_{t_1}^* \mathcal{L} \cong \mathcal{O}_X$. Since $X \times T$ is smooth, proposition 2.1.1(2) supplies an n-cycle E on $X \times T$ with $\mathcal{L} \cong \mathcal{O}_{X\times T}(E)$. Separate the components of E with dense image in T, which make up a cycle V, from those without:

$$E = V + \sum_j m_j\, [X \times \{s_j\}]$$

Indeed a component D' of E whose image in T is not dense has image a single closed point s, so $D' \subseteq X \times \{s\}$, and both being n-dimensional with $X \times \{s\}$ irreducible forces $D' = X \times \{s\}$. Thus V is a family of $(n-1)$-cycles on X parametrized by T.

The correction terms restrict trivially to every fibre. Let ϖ_s be a uniformizer of the discrete valuation ring $\mathcal{O}_{T,s}$, so that ϖ_s generates the ideal sheaf of $\{s\}$ near s. Then $\mathrm{pr}_T^* \varpi_s$ generates the ideal sheaf of $\mathrm{pr}_T^{-1}(s) = X \times \{s\}$ near the generic point of $X \times \{s\}$, so the Cartier divisor $\mathrm{pr}_T^*(s)$ has Weil cycle $[X \times \{s\}]$ with multiplicity one and

$$\mathcal{O}_{X\times T}\big([X \times \{s\}]\big) \cong \mathrm{pr}_T^* \mathcal{O}_T(s)$$

Restricting along ι_t, the composite $\mathrm{pr}_T \circ \iota_t$ is the constant morphism $X \to T$ with value t, which factors through $\mathrm{Spec}\,\overline{k}$; a line bundle pulled back along such a morphism is trivial, so $\iota_t^* \mathrm{pr}_T^* \mathcal{O}_T(s) \cong \mathcal{O}_X$. Combining this with lemma 6.1.11 applied to V,

$$\iota_t^* \mathcal{L} \cong \mathcal{O}_X\big(V(t)\big)$$

for every closed $t \in T$. At $t = t_1$ this reads $\mathcal{O}_X(V(t_1)) \cong \mathcal{O}_X$, so $[V(t_1)] = c(\mathcal{O}_X(V(t_1))) = c(\mathcal{O}_X) = 0$ in $A^1(X)$; at $t = t_0$ it reads $L \cong \mathcal{O}_X(V(t_0))$, so

$$c(L) = [V(t_0)] = [V(t_0)] - [V(t_1)]$$

which is a generator of $A^1(X)_{\mathrm{alg}}$.

Steps 1 and 2 give $c(\mathrm{Pic}^0(X)) = A^1(X)_{\mathrm{alg}}$, and an isomorphism carrying one subgroup onto the other descends to an isomorphism of the quotients $\mathrm{NS}(X) \cong \mathrm{NS}^1(X)$, completing the proof.

The theorem lets the divisor theory of a smooth projective variety be read inside the Chow ring without translation: a class in $A^1(X)$ is a line bundle, and the two ways of moving it in a family, by cycles and by line bundles, cut out the same subgroup. It also fixes the meaning of the phrase used at the divisor level, since $\operatorname{Pic}^0(X)$ is now the image of a construction performed on cycles.

Example 6.1.13: Néron-Severi Groups of the Plane and of an Abelian Surface

For $X = \mathbb{P}^2$ the Picard group is $\mathbb{Z}$, generated by the class of a line, and $\operatorname{Pic}^0(\mathbb{P}^2) = 0$; both are computed in [2, Picard Groups of Basic Surfaces]. By theorem 6.1.12 the subgroup $A^1(\mathbb{P}^2)_{\mathrm{alg}} = c(\operatorname{Pic}^0(\mathbb{P}^2))$ is zero, so

$$\mathrm{NS}^1(\mathbb{P}^2) = A^1(\mathbb{P}^2) = \mathbb{Z}$$

the last group being $\mathbb{Z}\,[\mathbb{P}^1]$ by theorem 1.5.2. In codimension one on the plane, algebraic equivalence adds nothing to rational equivalence.

For an abelian surface A the two differ. There $\operatorname{Pic}^0(A)$ is the dual abelian variety $\widehat{A}$, of dimension two, and $\mathrm{NS}(A)$ has rank between 1 and 4, again by [2, Picard Groups of Basic Surfaces]. So $A^1(A)_{\mathrm{alg}} = c(\operatorname{Pic}^0(A))$ is the two-dimensional family $\widehat{A}$ inside $A^1(A) = \operatorname{Pic}(A)$, and $\mathrm{NS}^1(A)$ is the proper quotient by it, of rank at most 4.

Finite generation of the codimension-one quotient is the theorem of the base, proved upstream for the divisor-level group, and theorem 6.1.12 transports it to the Chow-theoretic one.

Corollary 6.1.14: Finite Generation in Codimension One

Let X be a smooth projective variety. Then $\mathrm{NS}^1(X)$ is a finitely generated abelian group whose rank is the Picard number $\rho(X)$, and over $\mathbb{C}$ one has $\rho(X) \leq h^{1,1}(X)$.

***Proof*:**

The scheme X is a smooth projective variety over the algebraically closed field $\overline{k}$, which is the hypothesis of [2, Theorem of the Base]; that result gives that $\mathrm{NS}(X) = \operatorname{Pic}(X)/\operatorname{Pic}^0(X)$ is finitely generated of rank $\rho(X)$, with $\rho(X) \leq h^{1,1}(X)$ when $\overline{k} = \mathbb{C}$. By theorem 6.1.12 the group $\mathrm{NS}^1(X)$ is isomorphic to $\mathrm{NS}(X)$, so it is finitely generated of the same rank and satisfies the same bound, as we sought to show.

Nothing of the sort holds in higher codimension, and the failure is not a gap in the proof. For $2 \leq k \leq n-1$ the group $\mathrm{NS}^k(X)$ can have infinite rank. Clemens, in *Homological equivalence, modulo algebraic equivalence, is not finitely generated* (1983), proved that on a very general quintic threefold $X \subseteq \mathbb{P}^4$ the subgroup of $\mathrm{NS}^2(X)$ of classes killed by the cycle class map of section 6.2 has infinite rank, so $\mathrm{NS}^2(X) \otimes \mathbb{Q}$ is not finite dimensional. What survives in every codimension is the definition and the tower of inclusions it enters, taken up in section 6.3 once that cycle class map is available; for the further theory of algebraic equivalence, including the parametrization of $A^k(X)_{\mathrm{alg}}$ by an abelian variety in the cases where one exists, see [8, Ch. 19].

Exercise 6.1.1

1. Let $p, q \in \mathbb{P}^n$ be distinct closed points, let $L \subseteq \mathbb{P}^n$ be the line through them, and fix an isomorphism $\gamma\colon \mathbb{P}^1 \xrightarrow{\sim} L$. Show that $\Gamma = \{(\gamma(t), t) \,|\, t \in \mathbb{P}^1\} \subseteq \mathbb{P}^n \times \mathbb{P}^1$ is a family of 0-cycles on $\mathbb{P}^n$ parametrized by $\mathbb{P}^1$ with $\Gamma(t) = [\gamma(t)]$, and deduce that every zero-cycle of degree zero on $\mathbb{P}^n$ is algebraically equivalent to zero.
2. Let X be a variety of dimension n and T a smooth curve. Show that the only $(n+1)$-dimensional subvariety of $X \times T$ is $X \times T$ itself, compute its fibres, and conclude that $\mathrm{Alg}_n(X) = 0$ and

hence $\mathrm{NS}_n(X) = A_n(X) = \mathbb{Z}\,[X]$, the last equality by example 1.1.6.

3. Show that the set of differences $V(t_0) - V(t_1)$ generating $\mathrm{Alg}_k(X)$ in definition 6.1.3 is closed under negation, and that the sum of two such differences taken over the same parameter curve at the same pair of marked points is again a difference of the same kind. Identify the step that fails when the two families are parametrized by different curves, and say why the definition therefore takes the subgroup generated rather than the set of differences itself.

4. Let ℓ_0, ℓ_1 be linearly independent linear forms on $\mathbb{P}^2$, with zero loci the lines L_0, L_1, and let $r = \ell_1/\ell_0 \in K(\mathbb{P}^2)^*$. Show that the closure of the graph of r is

$$\Gamma = \left\{(x, [u:v]) \middle| v\,\ell_0(x) = u\,\ell_1(x)\right\} \subseteq \mathbb{P}^2 \times \mathbb{P}^1$$

where $\mathbb{P}^1$ has affine coordinate $t = v/u$, compute the fibres $\Gamma(0)$ and $\Gamma(\infty)$ as cycles on $\mathbb{P}^2$, and verify the identity $[\mathrm{div}(r)] = \Gamma(0) - \Gamma(\infty)$ of Step 4 of theorem 6.1.6 in this case.

5. Let S be a smooth projective surface with $\mathrm{Pic}^0(S) = 0$. Show that $\mathrm{NS}^1(S) = A^1(S)$, and compute both groups for $S = \mathbb{P}^1 \times \mathbb{P}^1$, whose Picard group is $\mathbb{Z}^2$ generated by the classes of the two rulings and whose Pic^0 vanishes by [2, Picard Groups of Basic Surfaces].

6. Let C be a smooth projective curve. Using example 6.1.5 and corollary 6.1.8, show that $\mathrm{NS}_0(C)$ is cyclic, generated by the class of any closed point. Identify it with the divisor-level group $\mathrm{NS}(C)$ through theorem 6.1.12, and contrast the answer with $A_0(C) = \mathrm{Cl}\,C$ of proposition 2.1.1.

6.2 The Cycle Class Map

Over the complex numbers a k-dimensional subvariety V of a variety X is also a compact subset of X of real dimension $2k$, oriented wherever it is smooth by its complex structure. Such a subset carries a fundamental class in H_{2k}, and on a smooth ambient variety Poincaré duality turns that class into a cohomology class whose degree is twice the codimension of V. This section constructs the resulting map, proves that it depends only on the rational equivalence class of a cycle, and computes its behaviour under proper pushforward, flat pullback and the intersection product.

6.2.1 Convention: The Complex Setting For the rest of this section and section 6.3 the ground field is $\overline{k} = \mathbb{C}$. A scheme is identified with its set of closed points carrying the analytic topology, and $H^i(-)$, $H_i(-)$ denote singular cohomology and homology with $\mathbb{Z}$ coefficients unless other coefficients are written. Every scheme entering a homological statement is projective over $\mathbb{C}$. For a smooth variety that is not complete the construction below produces classes in a homology theory adapted to noncompact spaces (Borel-Moore homology), which this book does not develop.

For a smooth projective V of dimension k the set of closed points is a compact oriented $2k$-manifold, and its fundamental class is [10, Existence and Uniqueness of the Fundamental Class]. A subvariety is rarely smooth, and the manifold statement says nothing at a singular point. What repairs this is that the singular locus is small: V_{sing} is a closed subvariety of dimension at most $k-1$, hence of real dimension at most $2k-2$, so it is invisible to chains of dimension $2k$ and $2k-1$. The following theorem records the consequence. Its proof rests on the triangulability of complex algebraic varieties, a result of Łojasiewicz and Hironaka on semianalytic and subanalytic sets that this book does not prove; what is written out is the derivation of the four clauses from a triangulation.

Theorem 6.2.2: Fundamental Class of a Complex Projective Variety

Let V be an irreducible projective variety over $\mathbb{C}$ of dimension k, with smooth locus V_{reg}.

1. $H_i(V) = 0$ for $i > 2k$.
2. $H_{2k}(V)$ is infinite cyclic, and for every $p \in V_{\mathrm{reg}}$ the map $H_{2k}(V) \to H_{2k}(V, V \setminus \{p\}) \cong \mathbb{Z}$ is an isomorphism. The *fundamental class* $\mu_V \in H_{2k}(V)$ is the generator sent to the generator determined by the complex orientation of V_{reg} at p; this description does not depend on p.
3. If V is smooth, μ_V is the fundamental class of the compact oriented $2k$-manifold V in the sense of [10, Fundamental Class].
4. If $\iota\colon V \to X$ is a closed immersion into a smooth projective variety X and ω is a closed C^∞ form of degree $2k$ on X, then ω is absolutely integrable on V_{reg} and
$$\langle \omega, \iota_* \mu_V \rangle = \int_{V_{\mathrm{reg}}} \omega$$
where $\langle -, - \rangle$ is the de Rham pairing between $H^{2k}(X; \mathbb{C})$ and $H_{2k}(X; \mathbb{C})$.

***Proof Key ideas*:**
A complex projective variety admits a triangulation by real-analytic simplices in which every closed subvariety is a subcomplex. Fix one for V, with V_{sing} a subcomplex; the complex has dimension $2k$ and the subcomplex V_{sing} has dimension at most $2k-2$. Since simplicial and singular homology agree, clause (1) is the absence of simplices in dimensions above $2k$.

Let c be the sum of the $2k$-simplices, each oriented by the complex orientation of V_{reg}; the interior of such a simplex lies in V_{reg}, because V_{sing} is a subcomplex of smaller dimension. The same dimension count puts the interior of every $(2k-1)$-simplex τ in V_{reg}, near which V is a $2k$-manifold; so τ is a face of exactly two $2k$-simplices, and the complex orientation induces opposite orientations on τ from the two sides. Hence $\partial c = 0$. If $z = \sum_\sigma a_\sigma \sigma$ is any $2k$-cycle, the same two-sided count forces $a_\sigma = a_{\sigma'}$ whenever σ, σ' share a $(2k-1)$-face, and V_{reg} is connected, being a nonempty Zariski-open subset of the irreducible variety V and hence connected in the analytic topology, and remains connected after deleting a subset of real codimension two, so all the coefficients agree: z is a multiple of c. There are no $(2k+1)$-simplices, so no $2k$-boundaries, and $H_{2k}(V) = \mathbb{Z}c$. The image of c in $H_{2k}(V, V \setminus \{p\})$ for p interior to a top simplex is the local complex orientation, which gives clause (2), and clause (3) because the manifold fundamental class is characterized by the same local condition.

For clause (4), c is a finite sum of compact simplices whose interiors are disjoint and cover V_{reg} outside a set of measure zero, so $\int_{V_{\mathrm{reg}}} \omega$ is a finite sum of integrals of a smooth form over compact simplices, hence absolutely convergent and equal to the de Rham pairing of ω with the chain c, completing the account of the four clauses.

The theorem makes the assignment $V \mapsto \mu_V$ available for every subvariety, so a cycle can be pushed into the homology of the ambient scheme coefficient by coefficient.

Definition 6.2.3: Cycle Class Map

Let T be a projective scheme over $\mathbb{C}$. The *homological cycle class* of a k-cycle $\alpha = \sum_i n_i[V_i]$ on T is

$$\mathrm{cl}^{\mathrm{top}}(\alpha) = \sum_i n_i\,(\iota_i)_*\mu_{V_i} \in H_{2k}(T)$$

where $\iota_i \colon V_i \to T$ is the inclusion; this is a homomorphism $\mathrm{cl}^{\mathrm{top}} \colon Z_k(T) \to H_{2k}(T)$.

Let X be a smooth projective variety over $\mathbb{C}$ of dimension n and let $\mathcal{D}_X \colon H^i(X) \to H_{2n-i}(X)$, $u \mapsto \mu_X \frown u$, be the Poincaré duality isomorphism [10, Poincaré Duality]. The *cycle class map* is

$$\mathrm{cl} = \mathcal{D}_X^{-1} \circ \mathrm{cl}^{\mathrm{top}} \colon Z_k(X) \longrightarrow H^{2(n-k)}(X)$$

In codimension indexing the map reads $\mathrm{cl} \colon Z^k(X) \to H^{2k}(X)$: a subvariety of complex codimension k has real codimension $2k$, and its class sits in cohomology of that degree. The smallest case already shows what the map forgets.

Example 6.2.4: Zero-Cycles on a Curve

Let C be a smooth projective connected curve over $\mathbb{C}$ and $\alpha = \sum_i n_i[p_i]$ a zero-cycle. Each p_i is a point, so μ_{p_i} is the generator of $H_0(\{p_i\}) = \mathbb{Z}$, and its image in $H_0(C)$ is the class of a point. As C is connected, $H_0(C) = \mathbb{Z}$ generated by that class, so

$$\mathrm{cl}^{\mathrm{top}}(\alpha) = \Big(\sum_i n_i\Big) \cdot [\mathrm{point}] \in H_0(C)$$

Applying $\mathcal{D}_C^{-1}$, the cycle class of α in $H^2(C) \cong \mathbb{Z}$ is the degree $\sum_i n_i$ of α. Two distinct points therefore have the same cycle class, while they are rationally equivalent only when $C \cong \mathbb{P}^1$.

On a curve the cycle class map is the degree, and the degree annihilates the divisor of a rational function; whether the same happens in every dimension is what has to be proved. Two steps prepare it. The first is that the homological cycle class is compatible with proper pushforward already at the level of cycles, which is the topological counterpart of definition 1.2.1 and is proved from the local characterization in theorem 6.2.2(2).

Proposition 6.2.5: Pushforward of Homological Cycle Classes

Let $h \colon T \to T'$ be a morphism of projective schemes over $\mathbb{C}$. Then for every $\alpha \in Z_k(T)$,

$$h_*\mathrm{cl}^{\mathrm{top}}(\alpha) = \mathrm{cl}^{\mathrm{top}}(h_*\alpha)$$

in $H_{2k}(T')$, where h_* on the left is the map induced on homology and on the right the pushforward of cycles.

***Proof*:**
Both sides are additive in α, so it suffices to treat $\alpha = [V]$ for $V \subseteq T$ a k-dimensional subvariety. A morphism of projective schemes is proper, so $V' = h(V)$ is a closed subvariety of T'; write $h_V : V \to V'$ for the restriction, so that $h \circ \iota_V = \iota_{V'} \circ h_V$. Functoriality of pushforward in homology gives

$$h_* \mathrm{cl}^{\mathrm{top}}([V]) = h_*(\iota_V)_* \mu_V = (\iota_{V'})_*(h_V)_* \mu_V$$

so, comparing with definition 1.2.1, the claim is that $(h_V)_* \mu_V$ equals $d\,\mu_{V'}$ when $\dim V' = k$, where d is the degree of the function field of V over that of V', and 0 when $\dim V' < k$.

The case $\dim V' < k$. Then $2k > 2 \dim V'$, so $H_{2k}(V') = 0$ by theorem 6.2.2(1) and $(h_V)_* \mu_V = 0$, which is what definition 1.2.1 assigns to $h_*[V]$ in this case.

The case $\dim V' = k$. Here h_V is dominant, hence surjective because its image is closed, and generically finite of degree d. Delete from V', at the outset, the closed subsets V'_{sing} and $h_V(V_{\mathrm{sing}})$, the second closed because h_V is proper and both of dimension $< k$; what is left is a dense open $V'_0 \subseteq V'_{\mathrm{reg}}$ whose preimage $V_0 = h_V^{-1}(V'_0)$ is a nonempty open subset of V_{reg}, hence a smooth variety, mapping onto V'_0. The order matters: generic smoothness is a statement about a morphism with smooth source, and only after the deletion is there one. The restriction of h_V over V'_0 is proper and dominant between varieties of the same dimension, so [2, Generic Finiteness and Étaleness] supplies a dense open $U' \subseteq V'_0 \subseteq V'_{\mathrm{reg}}$ over which it is finite and étale of degree d, with every fibre exactly d reduced points. Then $h_V^{-1}(U') \subseteq V_0 \subseteq V_{\mathrm{reg}}$ and h_V is a d-sheeted covering map there in the analytic topology.

Fix $p \in U'$ and write $h_V^{-1}(p) = \{q_1, \ldots, q_d\}$. Choose an open $B \subseteq U'$ containing p and biholomorphic to a ball, small enough that $h_V^{-1}(B) = B_1 \sqcup \cdots \sqcup B_d$ with $q_j \in B_j$ and $h_V : B_j \to B$ biholomorphic; the covering property provides such a B. Excision identifies

$$H_{2k}\big(V, V \setminus h_V^{-1}(p)\big) \cong \bigoplus_{j=1}^{d} H_{2k}(B_j, B_j \setminus \{q_j\})$$

and the projection to the j-th summand is the map to $H_{2k}(V, V \setminus \{q_j\})$. By theorem 6.2.2(2) applied at each $q_j \in V_{\mathrm{reg}}$, the image of μ_V has the complex orientation generator in every summand. The map of pairs $(V, V \setminus h_V^{-1}(p)) \to (V', V' \setminus \{p\})$ carries the j-th summand isomorphically by the biholomorphism $h_V|_{B_j}$, and a biholomorphism preserves the complex orientation, so each summand contributes the complex orientation generator at p. Naturality of the map $H_{2k}(-) \to H_{2k}(-, - \setminus \text{point})$ therefore sends $(h_V)_* \mu_V$ to d times that generator. By theorem 6.2.2(2) for V' this determines the class: $(h_V)_* \mu_V = d\,\mu_{V'}$, as we sought to show.

The second step is the topological fact that makes rational equivalence invisible. A rational function on a $(k+1)$-dimensional variety is a map to $\mathbb{P}^1$, and $\mathbb{P}^1$ is connected, so its two fibres over 0 and ∞ ought to be joined by a path of fibres. Sweeping the fibre along a path from 0 to ∞ takes place inside the preimage of that path, a compact set because the variety is projective, and inside it the two end fibres carry the same class.

Lemma 6.2.6: Fibres of a Family over $\mathbb{P}^1$

Let Γ be an irreducible projective variety over $\mathbb{C}$ of dimension $m \geq 1$ and $g\colon \Gamma \to \mathbb{P}^1$ a nonconstant morphism. For $t \in \mathbb{P}^1$ let Γ_t be the scheme-theoretic fibre and $[\Gamma_t] \in Z_{m-1}(\Gamma)$ its fundamental cycle. Then

$$\mathrm{cl}^{\mathrm{top}}([\Gamma_s]) = \mathrm{cl}^{\mathrm{top}}([\Gamma_t])$$

in $H_{2m-2}(\Gamma)$ for all $s, t \in \mathbb{P}^1$.

Each fibre is cut out locally by one equation and is not all of Γ, so by Krull's principal ideal theorem every component of Γ_t has dimension $m-1$ and the fundamental cycle in the statement is indeed an $(m-1)$-cycle. Write $\mu_T \in H_{2d}(T)$ for the class of the fundamental cycle of a projective scheme T of pure dimension d, so that $\mathrm{cl}^{\mathrm{top}}([\Gamma_t])$ is the image of μ_{Γ_t} under $H_{2m-2}(\Gamma_t) \to H_{2m-2}(\Gamma)$. Since Γ is projective the morphism g is proper, so the preimage of a compact arc is again compact and the sweep runs in the ordinary homology of box 6.2.1; nothing below leaves the compact setting. One local statement is taken on trust, granted here exactly as the triangulability of varieties is granted for theorem 6.2.2 and belonging to the same stratification theory of algebraic maps: there is a finite set $S \subseteq \mathbb{P}^1$ such that g is a topologically locally trivial fibration over $\mathbb{P}^1 \setminus S$ and Γ_u is reduced for every $u \notin S$, and such that every $t \in \mathbb{P}^1$ has a closed disc neighbourhood D with $D \cap S \subseteq \{t\}$ for which $g^{-1}(D)$ deformation retracts onto Γ_t and the retraction carries μ_{Γ_u} to μ_{Γ_t} for $u \in D \setminus \{t\}$. The last clause is where the multiplicities of the scheme-theoretic fibre enter; it is a statement about a neighbourhood of a single fibre, and it is the only part of the sweep that is not carried out below.

***Proof Key ideas*:**
Two values outside S. Let $s, t \notin S$ and let $\gamma \subseteq \mathbb{P}^1 \setminus S$ be an embedded arc from s to t, which exists because a finite set does not disconnect $\mathbb{P}^1$. The preimage $E = g^{-1}(\gamma)$ is compact, g being proper, and local triviality over the contractible compact arc γ trivializes g over γ; the trivialization supplies a retraction $r\colon E \to \Gamma_s$, which is a homotopy equivalence and restricts on Γ_t to a homeomorphism $\Gamma_t \to \Gamma_s$. That homeomorphism matches the components of the two fibres and their complex orientations, the comparison sign at a smooth point being locally constant along the connected arc and equal to $+1$ at s; both fibres are reduced, so it carries μ_{Γ_t} to μ_{Γ_s}. Hence r_* sends the images of μ_{Γ_s} and of μ_{Γ_t} in $H_{2m-2}(E)$ to the same class, and r_* is an isomorphism, so those two images agree in $H_{2m-2}(E)$ and therefore in $H_{2m-2}(\Gamma)$: $\mathrm{cl}^{\mathrm{top}}([\Gamma_s]) = \mathrm{cl}^{\mathrm{top}}([\Gamma_t])$.

A fibre against a nearby one. Let $t \in \mathbb{P}^1$ be arbitrary, let D be a disc as granted above, and let $u \in D \setminus \{t\}$, so that $u \notin S$. The retraction $\rho\colon g^{-1}(D) \to \Gamma_t$ is a homotopy inverse of the inclusion and carries μ_{Γ_u} to μ_{Γ_t}, so the two classes have the same image in $H_{2m-2}(g^{-1}(D))$ and hence in $H_{2m-2}(\Gamma)$: $\mathrm{cl}^{\mathrm{top}}([\Gamma_t]) = \mathrm{cl}^{\mathrm{top}}([\Gamma_u])$.

Arbitrary s and t. Choose u and u' outside S as in the second step, so that $\mathrm{cl}^{\mathrm{top}}([\Gamma_s]) = \mathrm{cl}^{\mathrm{top}}([\Gamma_u])$ and $\mathrm{cl}^{\mathrm{top}}([\Gamma_t]) = \mathrm{cl}^{\mathrm{top}}([\Gamma_{u'}])$; the first step equates the two right-hand sides, as we sought to show.

The two preceding results combine into the statement that the homological cycle class annihilates rational equivalence. The proof replaces a rational function by a morphism to $\mathbb{P}^1$, at the cost of replacing the variety it lives on by the closure of the graph.

Theorem 6.2.7: The Cycle Class Map on Chow Groups

Let T be a projective scheme over $\mathbb{C}$.

1. If $\alpha \in Z_k(T)$ is rationally equivalent to zero, then $\mathrm{cl}^{\mathrm{top}}(\alpha) = 0$; hence $\mathrm{cl}^{\mathrm{top}}$ descends to a homomorphism $A_k(T) \to H_{2k}(T)$.
2. If X is a smooth projective variety over $\mathbb{C}$ of dimension n, the cycle class map descends to a homomorphism
$$\mathrm{cl}\colon A^k(X) \longrightarrow H^{2k}(X)$$

***Proof*:**

1. By definition 1.1.3 a cycle rationally equivalent to zero is a finite sum of cycles $[\mathrm{div}(f)]$, with $W \subseteq T$ a closed subvariety of dimension $k+1$ and f a nonzero rational function on W; since $\mathrm{cl}^{\mathrm{top}}$ is additive it is enough to kill one such summand. The components of $\mathrm{div}(f)$ are subvarieties of W, so by functoriality of pushforward in homology the class of $[\mathrm{div}(f)]$ in $H_{2k}(T)$ is the image of its class in $H_{2k}(W)$ under the inclusion; it suffices to show the latter vanishes. If f is constant then $\mathrm{div}(f) = 0$, so assume f nonconstant.

   Step 1: the graph and the divisor of g. Apply Step 3 of theorem 6.1.6 to f on W: writing $\Gamma \subseteq W \times \mathbb{P}^1$ for the closure of the graph of f and $p\colon \Gamma \to W$, $g\colon \Gamma \to \mathbb{P}^1$ for the two projections, Γ is an irreducible projective variety of dimension $k+1$, p is proper and birational, g is a nonconstant morphism, and the order of g along the k-dimensional subvarieties of Γ is computed there to give (6.1),
$$[\mathrm{div}(g)] = [\Gamma_0] - [\Gamma_\infty]$$

   Step 2: descent to W and conclusion. Under the isomorphism of function fields induced by the birational morphism p, the function g corresponds to f, so the pushforward formula for the divisor of a rational function along a proper surjective morphism, applied to the birational p, gives $p_*[\mathrm{div}(g)] = [\mathrm{div}(f)]$ by proposition 1.2.2(2). Both Γ and W are projective, so proposition 6.2.5 applies to p and gives
$$\mathrm{cl}^{\mathrm{top}}([\mathrm{div}(f)]) = \mathrm{cl}^{\mathrm{top}}(p_*[\mathrm{div}(g)]) = p_*\mathrm{cl}^{\mathrm{top}}([\Gamma_0] - [\Gamma_\infty])$$
   and the class inside the pushforward vanishes by lemma 6.2.6, whose hypotheses hold: Γ is an irreducible projective variety of dimension $k+1$ and g is nonconstant. Hence $\mathrm{cl}^{\mathrm{top}}([\mathrm{div}(f)]) = 0$ in $H_{2k}(W)$, and part (1) follows.

2. For X smooth projective of dimension n the map cl is $\mathcal{D}_X^{-1} \circ \mathrm{cl}^{\mathrm{top}}$ by definition 6.2.3, and $\mathcal{D}_X^{-1}$ is an isomorphism, so cl kills exactly what $\mathrm{cl}^{\mathrm{top}}$ kills. By part (1) it descends to $A_{n-k}(X) = A^k(X)$, with values in $H^{2k}(X)$ by the degree count of definition 6.2.3, completing the proof.

Every operation built on Chow groups in the preceding chapters has a topological counterpart, and the cycle class map intertwines them: proper pushforward with the umkehr map, pullback with the induced map on cohomology, the intersection product with the cup product. Pushforward comes first, and stating it needs the cohomological form of the pushforward map.

Definition 6.2.8: Umkehr Map

Let $f\colon X \to Y$ be a morphism of smooth projective varieties over $\mathbb{C}$ of dimensions n and m. The *umkehr map* (or Gysin map in cohomology) is

$$f_! = \mathcal{D}_Y^{-1} \circ f_* \circ \mathcal{D}_X \colon H^i(X) \longrightarrow H^{i+2(m-n)}(Y)$$

where f_* is the map induced on homology.

Theorem 6.2.9: Compatibility with Proper Pushforward

Let $f\colon X \to Y$ be a morphism of smooth projective varieties over $\mathbb{C}$. Then $\mathrm{cl}_Y(f_*\alpha) = f_!\,\mathrm{cl}_X(\alpha)$ for every $\alpha \in A_k(X)$.

***Proof*:**
A morphism of projective schemes is proper, so f_* on Chow groups is defined. Both X and Y are projective schemes, so proposition 6.2.5 applies to f and gives $f_*\mathrm{cl}^{\mathrm{top}}(\alpha) = \mathrm{cl}^{\mathrm{top}}(f_*\alpha)$ for a cycle representing α; the two sides depend only on the rational equivalence class by theorem 6.2.7(1). Therefore

$$\mathrm{cl}_Y(f_*\alpha) = \mathcal{D}_Y^{-1}\mathrm{cl}^{\mathrm{top}}(f_*\alpha) = \mathcal{D}_Y^{-1} f_*\mathrm{cl}^{\mathrm{top}}(\alpha) = \mathcal{D}_Y^{-1} f_* \mathcal{D}_X \mathrm{cl}_X(\alpha) = f_!\,\mathrm{cl}_X(\alpha)$$

where the third equality is $\mathrm{cl}^{\mathrm{top}} = \mathcal{D}_X \circ \mathrm{cl}$ from definition 6.2.3 and the last is definition 6.2.8, completing the proof.

Taking Y to be a point turns the theorem into a statement about degrees, and identifies the top cohomology class of a zero-cycle with the integer that definition 1.2.4 attaches to it.

Corollary 6.2.10: Cycle Classes of Zero-Cycles

Let X be a smooth projective variety over $\mathbb{C}$ of dimension n and $\alpha \in A_0(X)$. Then

$$\langle \mathrm{cl}(\alpha), \mu_X \rangle = \int_X \alpha$$

where $\langle -, - \rangle$ is the Kronecker pairing between $H^{2n}(X)$ and $H_{2n}(X)$.

***Proof*:**
Let $f\colon X \to \operatorname{Spec}\mathbb{C}$ be the structure morphism, a morphism of smooth projective varieties with target of dimension 0. By definition 1.2.4, $f_*\alpha = (\int_X \alpha)\,[\operatorname{Spec}\mathbb{C}]$ in $A_0(\operatorname{Spec}\mathbb{C}) = \mathbb{Z}$, and $\mathrm{cl}(f_*\alpha) = \int_X \alpha$ under $H^0(\operatorname{Spec}\mathbb{C}) = \mathbb{Z}$, since the fundamental class of a point is the generator of H_0. On the other side, $\mathcal{D}_X(\mathrm{cl}(\alpha)) = \mu_X \frown \mathrm{cl}(\alpha) \in H_0(X)$, and f_* carries $H_0(X)$ to $H_0(\operatorname{Spec}\mathbb{C}) = \mathbb{Z}$ by the augmentation, which sends $\mu_X \frown u$ to $\langle u, \mu_X \rangle$. Hence $f_!\,\mathrm{cl}(\alpha) = \langle \mathrm{cl}(\alpha), \mu_X \rangle$, and theorem 6.2.9 equates the two, as we sought to show.

Pullback and the product are handled together, because on a smooth variety the product is a pullback: the intersection product is the pullback of an exterior product along the diagonal. The exterior product is the case that the topology computes directly, through the Künneth cross product.

Proposition 6.2.11: Cycle Classes of Exterior Products

Let X and Y be smooth projective varieties over $\mathbb{C}$, $\alpha \in A_k(X)$ and $\beta \in A_l(Y)$. Then

$$\mathrm{cl}_{X\times Y}(\alpha \times \beta) = \mathrm{cl}_X(\alpha) \times \mathrm{cl}_Y(\beta)$$

where $\times$ on the left is the exterior product of definition 5.1.2 and on the right the cohomology cross product.

***Proof*:**
Both sides are bilinear, so take $\alpha = [V]$ and $\beta = [W]$ for subvarieties $V \subseteq X$ and $W \subseteq Y$ of dimensions k and l; then $\alpha \times \beta = [V \times W]$ by definition 5.1.2. The smooth locus of $V \times W$ is $V_{\mathrm{reg}} \times W_{\mathrm{reg}}$, and at a point (p, q) of it the local homology of the product is the tensor product of the local homologies, with the complex orientation of the product corresponding to the product of the complex orientations. By theorem 6.2.2(2), applied to $V \times W$ at (p, q) and to V at p and W at q, this identifies

$$\mu_{V\times W} = \mu_V \times \mu_W$$

and the same computation with $V = X$, $W = Y$ gives $\mu_{X\times Y} = \mu_X \times \mu_Y$. Pushing forward along $V \times W \to X \times Y$ and using that the cross product commutes with maps induced by products of maps, $\mathrm{cl}^{\mathrm{top}}([V \times W]) = \mathrm{cl}^{\mathrm{top}}([V]) \times \mathrm{cl}^{\mathrm{top}}([W])$.

It remains to compare the two Poincaré dualities. For classes $u \in H^i(X)$ and $v \in H^j(Y)$ the cap product satisfies $(a \times b) \frown (u \times v) = (-1)^{|b||u|}(a \frown u) \times (b \frown v)$; here $a = \mu_X$, $b = \mu_Y$, and all four classes have even degree, so the sign is $+1$ and

$$\mathcal{D}_{X\times Y}(u \times v) = (\mu_X \times \mu_Y) \frown (u \times v) = \mathcal{D}_X(u) \times \mathcal{D}_Y(v)$$

Applying this with $u = \mathrm{cl}_X(\alpha)$ and $v = \mathrm{cl}_Y(\beta)$ and comparing with the previous display gives $\mathcal{D}_{X\times Y}(\mathrm{cl}_X(\alpha) \times \mathrm{cl}_Y(\beta)) = \mathrm{cl}^{\mathrm{top}}(\alpha \times \beta)$, which is the claim after applying $\mathcal{D}^{-1}_{X\times Y}$, completing the proof.

Pullback along an arbitrary morphism of smooth projective varieties is the one compatibility this book does not carry. Factoring such a morphism f through its graph splits the question into two cases: the projection, which the results already proved settle and which is the flat case stated below, and the regular embedding, where comparing $\mathrm{cl}(f^*\beta)$ with $f^*\mathrm{cl}(\beta)$ needs a class refined to the support of the pullback cycle. Such a class lives in the homology of a noncompact space, which box 6.2.1 places outside this book, so the theorem below is stated for a projection only.

Lemma 6.2.12: The Class of the Ambient Variety

Let X be a smooth projective variety over $\mathbb{C}$. Then $\mathrm{cl}([X]) = 1$ in $H^0(X; \mathbb{Z})$.

***Proof*:**
The inclusion of X in itself is the identity, so $\mathrm{cl}^{\mathrm{top}}([X]) = \mu_X$ by definition 6.2.3, the class μ_X being the fundamental class of the compact oriented manifold X by theorem 6.2.2(3). Since $\mathcal{D}_X(1) = \mu_X \frown 1 = \mu_X$, applying $\mathcal{D}_X^{-1}$ gives $\mathrm{cl}([X]) = \mathcal{D}_X^{-1}\mathrm{cl}^{\mathrm{top}}([X]) = 1$, as we sought to show.

Theorem 6.2.13: Compatibility with Pullback along a Projection

Let Z and Y be smooth projective varieties over $\mathbb{C}$ and let $\mathrm{pr}_Y : Z \times Y \to Y$ be the projection. Then

$$\mathrm{cl}_{Z\times Y}(\mathrm{pr}_Y^*\beta) = \mathrm{pr}_Y^*\mathrm{cl}_Y(\beta)$$

for every $\beta \in A^k(Y)$, where pr_Y^* on the left is the pullback of definition 5.2.11 and on the right the map induced on cohomology.

***Proof*:**
The projection is flat, and $\mathrm{pr}_Y^{-1}(V) = Z \times V$ for a subvariety $V \subseteq Y$, so $\mathrm{pr}_Y^*\beta = [Z] \times \beta$ by definition 1.3.3 together with the agreement of definition 5.2.11 with flat pullback (theorem 5.2.10). By proposition 6.2.11 and lemma 6.2.12,

$$\mathrm{cl}_{Z\times Y}(\mathrm{pr}_Y^*\beta) = \mathrm{cl}_Z([Z]) \times \mathrm{cl}_Y(\beta) = 1 \times \mathrm{cl}_Y(\beta) = \mathrm{pr}_Y^*\mathrm{cl}_Y(\beta)$$

the last equality being the definition of the cross product with 1, completing the proof.

The corollary that cl is a ring homomorphism therefore needs the excluded case for the diagonal alone, and only in the form

$$\mathrm{cl}_X(\Delta^*\gamma) = \Delta^*\mathrm{cl}_{X\times X}(\gamma) \quad \text{for } \gamma \in A^*(X \times X), \quad \Delta\colon X \to X \times X \text{ the diagonal} \tag{6.2}$$

which is an input here and not a theorem of this book.

Corollary 6.2.14: The Cycle Class Map is a Ring Homomorphism

Let X be a smooth projective variety over $\mathbb{C}$ and grant (6.2). Then

$$\mathrm{cl}(\alpha \cdot \beta) = \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta)$$

for all $\alpha, \beta \in A^*(X)$, and $\mathrm{cl}\colon A^*(X) \to H^{2*}(X)$ is a homomorphism of graded rings carrying $1 = [X]$ to 1.

***Proof*:**
Let $\Delta\colon X \to X \times X$ be the diagonal, a regular embedding by theorem 5.1.6, and recall from theorem 5.2.7 that the product on $A^*(X)$ is computed by reduction to the diagonal, $\alpha \cdot \beta = \Delta^*(\alpha \times \beta)$. Granting (6.2) for Δ and applying proposition 6.2.11 to the exterior product,

$$\mathrm{cl}(\alpha \cdot \beta) = \mathrm{cl}(\Delta^*(\alpha \times \beta)) = \Delta^*\mathrm{cl}_{X\times X}(\alpha \times \beta) = \Delta^*\big(\mathrm{cl}(\alpha) \times \mathrm{cl}(\beta)\big) = \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta)$$

the last equality because the cup product is the pullback of the cross product along the diagonal. That $\mathrm{cl}([X]) = 1$ is lemma 6.2.12, and cl doubles degrees by theorem 6.2.7(2), completing the proof.

On projective space the comparison is an isomorphism, and the verification uses nothing beyond the degree formula and multiplicativity.

Example 6.2.15: Cycle Classes on $\mathbb{P}^n$

Let $h \in A^1(\mathbb{P}^n)$ be the class of a hyperplane, so that $A^*(\mathbb{P}^n) = \mathbb{Z}[h]/(h^{n+1})$ with h^k the class of a linear subspace of codimension k and $\int_{\mathbb{P}^n} h^n = 1$ (theorem 1.5.2). The singular cohomology ring is $H^*(\mathbb{P}^n;\mathbb{Z}) = \mathbb{Z}[\xi]/(\xi^{n+1})$ with ξ a generator of H^2 and ξ^n a generator of H^{2n} [10]. Write $\mathrm{cl}(h) = c\,\xi$ with $c \in \mathbb{Z}$. By corollary 6.2.14, $\mathrm{cl}(h^n) = c^n\xi^n$, and by corollary 6.2.10,

$$\langle \mathrm{cl}(h^n), \mu_{\mathbb{P}^n}\rangle = \int_{\mathbb{P}^n} h^n = 1$$

while $\langle \xi^n, \mu_{\mathbb{P}^n}\rangle = \pm 1$ because ξ^n generates $H^{2n}(\mathbb{P}^n;\mathbb{Z})$ and the Kronecker pairing with $\mu_{\mathbb{P}^n}$ is an isomorphism $H^{2n}(\mathbb{P}^n;\mathbb{Z}) \to \mathbb{Z}$. Hence $c^n = \pm 1$, so $c = \pm 1$ and $\mathrm{cl}(h)$ generates $H^2(\mathbb{P}^n;\mathbb{Z})$. Multiplicativity then makes cl carry the basis $1, h, \ldots, h^n$ of $A^*(\mathbb{P}^n)$ to the basis $1, \mathrm{cl}(h), \ldots, \mathrm{cl}(h)^n$ of $H^{2*}(\mathbb{P}^n;\mathbb{Z})$, so $\mathrm{cl}\colon A^*(\mathbb{P}^n) \to H^{2*}(\mathbb{P}^n;\mathbb{Z})$ is an isomorphism of rings. The odd cohomology of $\mathbb{P}^n$ vanishes, so cl is onto all of $H^*(\mathbb{P}^n;\mathbb{Z})$: on projective space the Chow ring and the cohomology ring carry the same information.

The same construction runs over any base field with ℓ-adic étale cohomology in place of singular cohomology, producing $\mathrm{cl}\colon A^k(X) \to H^{2k}_{\text{ét}}(X, \mathbb{Q}_\ell(k))$; the étale topology it requires is not developed in this book. Over $\mathbb{C}$ there is a constraint on the image that has no counterpart in the purely topological picture, and it comes from the Hodge decomposition of the cohomology of a smooth projective variety.

Theorem 6.2.16: Algebraic Cycles Have Hodge Type (k, k)

Let X be a smooth projective variety over $\mathbb{C}$ of dimension n and $\alpha \in A^k(X)$. Then the image of $\mathrm{cl}(\alpha)$ in $H^{2k}(X;\mathbb{C})$ lies in the summand $H^{k,k}(X)$ of the Hodge decomposition.

***Proof*:**

By linearity take $\alpha = [V]$ with $V \subseteq X$ an irreducible closed subvariety of codimension k, so $\dim V = d = n - k$. Let ω be a closed form on X of type (p, q) with $p + q = 2d$ and $(p, q) \neq (d, d)$. Then $p > d$ or $q > d$, and a form of type (p, q) with $p > d$ or $q > d$ restricts to zero on any complex manifold of complex dimension d, since its restriction involves a wedge of more than d of the dz_i or more than d of the $d\bar{z}_i$. In particular ω restricts to zero on V_{reg}, so $\int_{V_{\mathrm{reg}}} \omega = 0$, and theorem 6.2.2(4) gives

$$\langle \omega, \mathrm{cl}^{\mathrm{top}}([V])\rangle = 0$$

By the Hodge decomposition [2, Hodge Decomposition via GAGA], every class in $H^{p,q}(X)$ is represented by a closed form of type (p, q), so $\mathrm{cl}^{\mathrm{top}}([V])$ pairs to zero with $H^{p,q}(X)$ for every $(p, q) \neq (d, d)$ with $p + q = 2d$.

Rewrite this as a statement about cup products. For $u \in H^{2k}(X;\mathbb{C})$ and $v \in H^{2d}(X;\mathbb{C})$ set $(u, v) = \langle u \cup v, \mu_X\rangle$; with field coefficients Poincaré duality makes this pairing perfect. The paragraph above says $(\mathrm{cl}([V]), v) = 0$ for $v \in H^{p,q}(X)$, $(p, q) \neq (d, d)$, because $\langle \mathrm{cl}([V]) \cup v, \mu_X\rangle = \langle v, \mu_X \frown \mathrm{cl}([V])\rangle = \langle v, \mathrm{cl}^{\mathrm{top}}([V])\rangle$.

Now decompose $\mathrm{cl}([V]) = \sum_{a+b=2k} c^{a,b}$ with $c^{a,b} \in H^{a,b}(X)$. A form of type (a,b) wedged with a form of type (p,q) has type $(a+p, b+q)$, and a form of type (s,t) with $s+t=2n$ vanishes unless $s=t=n$, so $(c^{a,b}, H^{p,q}(X)) = 0$ unless $(p,q) = (n-a, n-b)$. Perfectness of the pairing on $H^{2k} \times H^{2d}$ therefore restricts to a perfect pairing between $H^{a,b}(X)$ and $H^{n-a,n-b}(X)$. Fix $(a,b) \neq (k,k)$ with $a+b=2k$. Then $(n-a, n-b) \neq (d,d)$, so the first paragraph gives $(\mathrm{cl}([V]), H^{n-a,n-b}(X)) = 0$; and among the summands $c^{a',b'}$ only $c^{a,b}$ pairs nontrivially with $H^{n-a,n-b}(X)$, so $(c^{a,b}, H^{n-a,n-b}(X)) = 0$ and perfectness gives $c^{a,b} = 0$. Every summand other than $c^{k,k}$ vanishes, so $\mathrm{cl}([V]) \in H^{k,k}(X)$, as we sought to show.

The theorem cuts the target of the cycle class map down to a subgroup that is in general much smaller than $H^{2k}(X;\mathbb{Z})$.

Definition 6.2.17: Hodge Class

Let X be a smooth projective variety over $\mathbb{C}$. The group of *Hodge classes* of degree $2k$ is

$$\mathrm{Hdg}^k(X) = \left\{ u \in H^{2k}(X;\mathbb{Z}) \;\middle|\; \text{the image of } u \text{ in } H^{2k}(X;\mathbb{C}) \text{ lies in } H^{k,k}(X) \right\}$$

Theorem 6.2.16 says that cl takes values in $\mathrm{Hdg}^k(X)$. Whether it fills that group after tensoring with $\mathbb{Q}$ is the Hodge conjecture, open in general for $2 \le k \le n-2$, where $n = \dim X$; for $k=1$ the answer is yes, by [2, Lefschetz Theorem on (1,1)-Classes], which identifies $\mathrm{Hdg}^1(X)$ with the image of $\mathrm{cl}\colon A^1(X) \to H^2(X;\mathbb{Z})$. The two top codimensions are known as well. For $k=n$ every class in $H^{2n}(X;\mathbb{C})$ has type (n,n), so $\mathrm{Hdg}^n(X) = H^{2n}(X;\mathbb{Z})$, which exercise 6.2.1 (4) identifies with $\mathbb{Z}$ by pairing against μ_X; the class of a point pairs to 1 by corollary 6.2.10 and so generates. For $k = n-1$ the hard Lefschetz isomorphism $H^2(X;\mathbb{Q}) \to H^{2n-2}(X;\mathbb{Q})$ is cup product with $\mathrm{cl}(h)^{n-2}$ for $h \in A^1(X)$ the class of a hyperplane section, so it carries rational Hodge classes onto rational Hodge classes and, by corollary 6.2.14, algebraic classes to algebraic classes; that case therefore reduces to $k=1$. The conjecture has content only in the middle range named above, and none at all when $\dim X \le 3$. On a curve the condition is empty, since H^2 of a curve is all of type $(1,1)$, and cl is then the degree map of example 6.2.4, whose kernel is the group of degree-zero divisor classes.

Exercise 6.2.1

1. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$ and put $a = [\{p\} \times \mathbb{P}^1]$ and $b = [\mathbb{P}^1 \times \{q\}]$ in $A^1(X)$. Using theorem 3.1.9 for $X = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1})$, or otherwise, show that $A^*(X)$ is free with basis $1, a, b, ab$ and that $a^2 = b^2 = 0$. Deduce that $\mathrm{cl}\colon A^*(X) \to H^{2*}(X;\mathbb{Z})$ is a ring isomorphism. The diagonal class $[\Delta] = a+b$ is exercise 5.1.1 (2); check that $\mathrm{cl}[\Delta]$ is the Künneth class of the diagonal, and read off $\deg([\Delta]\cdot[\Delta])$.

2. Let E be a smooth projective curve of genus 1 over $\mathbb{C}$ and let $p \neq q$ be two points of E. Show that $\mathrm{cl}([p]-[q]) = 0$ while $[p]-[q] \neq 0$ in $A^1(E)$, so that cl is not injective. (Use that a degree-zero divisor $p-q$ with $p \neq q$ on a curve of genus at least one is not the divisor of a rational function.)

3. Let X be a smooth projective variety over $\mathbb{C}$ of dimension n, and let $\alpha \in A^k(X)$ and $\beta \in A^{n-k}(X)$. Show that
$$\int_X (\alpha \cdot \beta) = \langle \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta), \mu_X \rangle$$
so that intersection numbers of complementary-dimensional cycles are computed by the cup

product in cohomology.

4. Let X be a smooth projective connected variety over $\mathbb{C}$ of dimension n. Show that $u \mapsto \langle u, \mu_X \rangle$ is an isomorphism $H^{2n}(X;\mathbb{Z}) \to \mathbb{Z}$, and deduce that $\mathrm{cl}(\alpha) = 0$ for $\alpha \in A_0(X)$ if and only if $\int_X \alpha = 0$. Conclude that cl is injective on $A_0(X)$ exactly when every zero-cycle of degree zero is rationally equivalent to zero.

5. Let $Z \subseteq \mathbb{P}^n$ be the hypersurface cut out by an irreducible form F of degree d, and let H be the hyperplane $x_0 = 0$. By computing the divisor of the rational function F/x_0^d, show that $[Z] = d\,h$ in $A^1(\mathbb{P}^n)$, where $h = [H]$, and deduce that $\mathrm{cl}([Z]) = d\,\mathrm{cl}(h)$.

6. Let C be a smooth projective connected curve over $\mathbb{C}$. Show that $H^2(C;\mathbb{C}) = H^{1,1}(C)$, so that $\mathrm{Hdg}^1(C) = H^2(C;\mathbb{Z})$, and show that $\mathrm{cl}\colon A^1(C) \to H^2(C;\mathbb{Z})$ is surjective.

7. Let $i\colon Y \to X$ be a closed immersion of smooth projective varieties over $\mathbb{C}$. Show that $\mathrm{cl}_X([Y]) = i_!(1)$, where $1 \in H^0(Y;\mathbb{Z})$ and $i_!$ is the umkehr map of definition 6.2.8.

8. Let Γ be a smooth projective surface over $\mathbb{C}$ and $g\colon \Gamma \to \mathbb{P}^1$ a nonconstant morphism. Show that the fibres of g all have the same cycle class $\varphi \in H^2(\Gamma;\mathbb{Z})$, and that $\langle \varphi \cup \varphi, \mu_\Gamma \rangle = 0$. (Two distinct fibres are disjoint; use corollary 4.3.11 to evaluate their intersection product, and exercise 6.2.1 (3).)

6.3 Homological and Numerical Equivalence

Section 6.2 sent a cycle class to a cohomology class. Calling a cycle trivial when that image vanishes is one way to coarsen rational equivalence; calling it trivial when every intersection number formed against it vanishes is another, and it asks less. This section defines both, isolates the three properties the four relations of this chapter share, and proves the tower they form.

Throughout, X is a smooth projective variety over $\mathbb{C}$ of dimension n, the setting fixed in section 6.2. Every relation below is a relation on the Chow ring $A^*(X)$ of definition 5.2.2, so each is specified by naming a graded subgroup of $A^*(X)$: the classes it declares trivial. Three properties of the cycle class map are used repeatedly, all established in section 6.2: the map $\mathrm{cl}\colon A^*(X) \to H^{2*}(X;\mathbb{Z})$ is a homomorphism of graded rings; it commutes with proper pushforward and with pullback along a morphism of smooth projective varieties; and on zero-cycles it computes the degree, in the sense that $\int_X \mathrm{cl}(\alpha) = \deg(\alpha)$ for $\alpha \in A^n(X)$, where $\int_X\colon H^{2n}(X;\mathbb{Z}) \to \mathbb{Z}$ is evaluation against the fundamental class of $X(\mathbb{C})$.

Definition 6.3.1: Homological Equivalence

Let X be a smooth projective variety over $\mathbb{C}$. A class $\alpha \in A^k(X)$ is *homologically equivalent to zero*, written $\alpha \sim_{\mathrm{hom}} 0$, if $\mathrm{cl}(\alpha) = 0$ in $H^{2k}(X;\mathbb{Z})$. Two classes are homologically equivalent if their difference is. Write

$$\mathcal{R}_{\mathrm{hom}}(X) \;=\; \ker\left(\mathrm{cl}\colon A^*(X) \to H^{2*}(X;\mathbb{Z})\right) \;\subseteq\; A^*(X)$$

and $A^*_{\mathrm{hom}}(X) = A^*(X)/\mathcal{R}_{\mathrm{hom}}(X)$.

The quotient $A^*_{\mathrm{hom}}(X)$ is by construction the image of the cycle class map, the subring of $H^{2*}(X;\mathbb{Z})$

consisting of the classes of algebraic cycles. Passing to it discards exactly the information the topology of $X(\mathbb{C})$ cannot see, which is what makes it the object a Hodge-theoretic question about cycles is posed in.

Example 6.3.2: Homological Equivalence of Zero-Cycles

Take $k = n$, so that $\alpha \in A^n(X) = A_0(X)$ is a class of zero-cycles. Then $\mathrm{cl}(\alpha) \in H^{2n}(X;\mathbb{Z})$, and evaluation against the fundamental class is the degree: $\int_X \mathrm{cl}(\alpha) = \deg(\alpha)$. Since X is a connected smooth projective variety over $\mathbb{C}$, $X(\mathbb{C})$ is a compact connected oriented $2n$-manifold, so $\int_X$ is injective on $H^{2n}(X;\mathbb{Z})$; hence

$$\alpha \sim_{\mathrm{hom}} 0 \quad \Longleftrightarrow \quad \deg(\alpha) = 0$$

On a smooth projective curve C this says that $\mathcal{R}_{\mathrm{hom}}(C)$ meets $A^1(C) = A_0(C) = \mathrm{Pic}\,(C)$ (proposition 2.1.3) in the divisor classes of degree zero, that is in $\mathrm{Pic}^0(C)$.

Integral cohomology can carry torsion classes that no intersection number detects, so definition 6.3.1 needs one caveat attached to it before the comparisons below.

6.3.3 Warning: Integral versus Rational Coefficients Definition 6.3.1 takes $\mathbb{Z}$ coefficients, so a class whose image in $H^{2k}(X;\mathbb{Z})$ is a nonzero torsion element is not homologically trivial. The comparison with numerical equivalence below (theorem 6.3.11) is insensitive to this, but the converse comparison is not: theorem 6.3.9(4) will show that every torsion class of $A^*(X)$ is numerically trivial, whereas cl need not kill torsion: on an Enriques surface X the canonical class K_X is a nonzero 2-torsion element of $A^1(X)$ with $\mathrm{cl}(K_X) \neq 0$, which is what makes the last inclusion of theorem 6.3.11 strict for that X. The rational variant, $\alpha \sim_{\mathrm{hom},\mathbb{Q}} 0$ when $\mathrm{cl}(\alpha) = 0$ in $H^{2k}(X;\mathbb{Q})$, differs from definition 6.3.1 only on torsion classes, and it is the variant in which **??** 6.3.14 is stated.

Rather than requiring a cycle class to vanish in some receiving group, numerical equivalence requires only that every number the intersection product can extract from it be zero. Since a class of complementary codimension multiplies α into $A^n(X)$, where the degree homomorphism of definition 1.2.4 is defined, those numbers are available without leaving the Chow ring.

Definition 6.3.4: Numerical Equivalence

Let X be a smooth projective variety of dimension n. A class $\alpha \in A^k(X)$ is *numerically equivalent to zero*, written $\alpha \sim_{\mathrm{num}} 0$, if

$$\deg\big(\alpha \cdot \beta\big) \;=\; 0 \qquad \text{for every } \beta \in A^{n-k}(X)$$

Two classes are numerically equivalent if their difference is. Write $\mathcal{R}_{\mathrm{num}}(X) \subseteq A^*(X)$ for the graded subgroup of numerically trivial classes and $A^*_{\mathrm{num}}(X) = A^*(X)/\mathcal{R}_{\mathrm{num}}(X)$; the group $A^k_{\mathrm{num}}(X)$ is also written $\mathrm{Num}^k(X)$.

The definition is exactly the statement that the pairing $A^k(X) \times A^{n-k}(X) \to \mathbb{Z}$, $(\alpha,\beta) \mapsto \deg(\alpha\cdot\beta)$, becomes nondegenerate on the left after passing to $A^*_{\mathrm{num}}(X)$. Numerical equivalence is thus the coarsest identification under which intersection numbers remain well defined, and proposition 6.3.10 will make this precise.

Example 6.3.5: The Two Rulings of a Quadric Surface

Let $S = \mathbb{P}^1 \times \mathbb{P}^1$, and let $f_1 = [\mathbb{P}^1 \times \{q\}]$ and $f_2 = [\{p\} \times \mathbb{P}^1]$ in $A^1(S) = A_1(S)$ be the classes of the two rulings (example 5.1.3). Both are flat pullbacks of point classes: writing $\mathrm{pr}_1, \mathrm{pr}_2 \colon S \to \mathbb{P}^1$ for the projections, which are flat by lemma 5.1.1, proposition 1.3.5 gives $f_1 = \mathrm{pr}_2^*[q]$ and $f_2 = \mathrm{pr}_1^*[p]$, the flat pullback of a fundamental cycle being the fundamental cycle of the preimage.

The self-intersections vanish. Choose $q' \neq q$ in $\mathbb{P}^1$. By example 5.2.3 any two closed points of $\mathbb{P}^1$ have the same class in $A_0(\mathbb{P}^1)$, so $f_1 = [\mathbb{P}^1 \times \{q'\}]$ as well. The subvarieties $\mathbb{P}^1 \times \{q\}$ and $\mathbb{P}^1 \times \{q'\}$ are disjoint, so the refined product $[\mathbb{P}^1 \times \{q\}] \cdot_S [\mathbb{P}^1 \times \{q'\}]$ of definition 5.1.10 lies in the Chow group of the empty scheme, which is zero; by proposition 5.1.11 the product in $A^*(S)$ is the pushforward of that class, so $f_1 \cdot f_1 = 0$, and symmetrically $f_2 \cdot f_2 = 0$.

The cross term is a point. Let $\iota \colon \{p\} \times \mathbb{P}^1 \hookrightarrow S$ be the inclusion, a morphism of smooth projective varieties, so that $\iota_*[\{p\} \times \mathbb{P}^1] = f_2$. The projection formula theorem 5.2.10(2), with α the fundamental class of $\{p\} \times \mathbb{P}^1$, which is the unit of that Chow ring by theorem 5.2.7(4), and with $\beta = f_1$, gives

$$f_2 \cdot f_1 \;=\; \iota_*\big(\iota^* f_1\big)$$

and $\iota^* f_1 = \iota^* \mathrm{pr}_2^*[q] = (\mathrm{pr}_2 \circ \iota)^*[q]$ by theorem 5.2.10(4). Since $\mathrm{pr}_2 \circ \iota$ is the isomorphism $\{p\} \times \mathbb{P}^1 \to \mathbb{P}^1$, this is the class of the point (p, q), whence $f_1 \cdot f_2 = [(p, q)]$ and $\deg(f_1 \cdot f_2) = 1$.

Consequently $\deg\big((af_1 + bf_2) \cdot f_2\big) = a$ and $\deg\big((af_1 + bf_2) \cdot f_1\big) = b$, so no nonzero integer combination of the two ruling classes is numerically equivalent to zero. On this surface the pairing already separates the classes it is fed.

Rational, algebraic, homological and numerical equivalence were each defined by a different mechanism, and nothing so far says they have anything in common beyond coarsening one another. What they share is the list of operations that descend to their quotients, and that list is short enough to be made a definition. It is the list section 6.4 will need, since a correspondence acts by a pullback, a product and a pushforward in succession.

Definition 6.3.6: Adequate Equivalence Relation

An *adequate equivalence relation* $\sim$ assigns to every smooth projective variety X a graded subgroup $\mathcal{R}_\sim(X) \subseteq A^*(X)$, the classes *$\sim$-equivalent to zero*, subject to three conditions. For all smooth projective varieties X and Y and every morphism $f \colon X \to Y$:

1. $\mathcal{R}_\sim(X)$ is an ideal of $A^*(X)$: if $\alpha \in \mathcal{R}_\sim(X)$ and $\gamma \in A^*(X)$ then $\alpha \cdot \gamma \in \mathcal{R}_\sim(X)$;
2. $f^*\mathcal{R}_\sim(Y) \subseteq \mathcal{R}_\sim(X)$;
3. $f_*\mathcal{R}_\sim(X) \subseteq \mathcal{R}_\sim(Y)$.

Write $\alpha \sim \beta$ for $\alpha - \beta \in \mathcal{R}_\sim(X)$ and $A^*_\sim(X) = A^*(X)/\mathcal{R}_\sim(X)$. The relation is *nondegenerate* if $\mathcal{R}_\sim(\operatorname{Spec}\mathbb{C}) = 0$.

Two remarks fix how the definition is read. Pushforward preserves dimension rather than codimension, so in (3) the grading is the one by dimension, $f_* A_k(X) \subseteq A_k(Y)$; a graded subgroup is graded for either index, the two being related by $A^p(X) = A_{\dim X - p}(X)$. And the conditions are imposed on $A^*(X)$, not on cycles, so rational equivalence is built into the definition: every adequate relation is coarser than rational equivalence, and $\mathcal{R}_{\mathrm{rat}}(X) = 0$ is the finest one. In this notation $\mathcal{R}_{\mathrm{alg}}(X) = \bigoplus_k A_k(X)_{\mathrm{alg}}$, the graded subgroup of algebraically trivial classes named degreewise in definition 6.1.7.

The three conditions are exactly what is needed for the quotient to retain the structure $A^*(X)$ has. Condition (1) makes $A^*_\sim(X)$ a graded ring; condition (2) makes $X \mapsto A^*_\sim(X)$ contravariant, with f^* a ring homomorphism because it is one on $A^*(X)$ by theorem 5.2.10(3); condition (3) supplies f_*, and the projection formula $f_*(\alpha \cdot f^*\beta) = f_*\alpha \cdot \beta$ of theorem 5.2.10(2) descends because both sides do. Nondegeneracy says the relation does not identify a point with zero on $\operatorname{Spec}\mathbb{C}$, where $A^*(\operatorname{Spec}\mathbb{C}) = \mathbb{Z}$; exercise 6.3.1 (6) shows it is equivalent to the degree homomorphism surviving the quotient.

Example 6.3.7: The Two Extreme Relations

Two assignments satisfy definition 6.3.6 for trivial reasons, and they bound the range of possibilities. Taking $\mathcal{R}_\sim(X) = 0$ for every X gives rational equivalence: the three conditions are conditions on the zero subgroup, and $\mathcal{R}_\sim(\operatorname{Spec}\mathbb{C}) = 0$, so the relation is nondegenerate. Taking $\mathcal{R}_\sim(X) = A^*(X)$ for every X also satisfies the three conditions, since every operation lands in A^* of its target; here $A^*_\sim(X) = 0$ for all X, and $\mathcal{R}_\sim(\operatorname{Spec}\mathbb{C}) = \mathbb{Z} \neq 0$, so this relation is degenerate. Nondegeneracy is what rules out the second, and proposition 6.3.10 will show that the range it leaves is bounded above by numerical equivalence.

Verifying the conditions for algebraic equivalence needs a way to convert its defining families into pullbacks. The construction is the fibre inclusion of the parameter curve.

Lemma 6.3.8: Fibres of a Family as Pullbacks

Let X be a smooth projective variety, let T be a smooth connected curve, and set $P = T \times X$ with projections $p\colon P \to T$ and $q\colon P \to X$. For a closed point $t \in T$ let $i_t\colon X \to P$ be $x \mapsto (t, x)$, and let $t_0, t_1 \in T$ be closed points. Then:

1. i_t is a regular embedding of codimension 1 whose normal bundle is trivial, and its Gysin map agrees with the pullback i_t^* of definition 5.2.8;

2. if $V \subseteq P$ is a subvariety dominating T, then $i_t^*[V] = [V_t]$, the fundamental cycle of the scheme-theoretic fibre $V_t = V \cap (\{t\} \times X)$ read in X; hence $i_t^*[\Gamma] = \Gamma(t)$ for every cycle Γ on P all of whose components dominate T;

3. for every class $\xi \in A_*(P)$, the difference $i_{t_1}^*\xi - i_{t_0}^*\xi$ is algebraically equivalent to zero on X.

***Proof*:**

1. The projection q is the base change of $T \to \operatorname{Spec}\mathbb{C}$ along $X \to \operatorname{Spec}\mathbb{C}$, hence smooth of relative dimension 1, and i_t is a section of it. By lemma 5.2.4(1), i_t is a regular embedding of codimension 1 with normal bundle $i_t^*T_{P/X}$. The relative tangent bundle $T_{P/X}$ is p^*T_T, so the normal bundle is $(p \circ i_t)^*T_T$; the composite $p \circ i_t$ is the constant morphism $X \to T$ with image t, and the pullback of a line bundle along a morphism factoring through a closed point is trivial, being the pullback of a line bundle on $\operatorname{Spec}\mathbb{C}$. Thus the normal bundle is $\mathcal{O}_X$. Both X and P are smooth varieties, so theorem 5.2.12(3) identifies $i_t^!$ with i_t^*, and theorem 5.2.12(2) identifies $i_t^!$ with the Gysin map of the regular embedding.

2. The subscheme $\{t\} \times X = p^{-1}(t)$ has ideal sheaf generated locally by $p^\#\pi$, where π generates the maximal ideal of the regular one-dimensional local ring $\mathcal{O}_{T,t}$. Its restriction

to V is nonzero: V dominates T, so V is not contained in $p^{-1}(t)$. As V is integral, a nonzero section of $\mathcal{O}_V$ is a nonzerodivisor, so the ideal of V_t in V is generated by a single nonzerodivisor and $V_t \hookrightarrow V$ is a regular embedding of codimension 1. Proposition 4.3.3, applied to the regular embedding i_t of codimension 1 and the subvariety V, therefore gives $i_t^![V] = \iota_*[V_t]$, which is $[V_t]$ under the isomorphism $\{t\} \times X \cong X$; by (1) the left side is $i_t^*[V]$. Extending linearly over the components of Γ gives the last assertion.

3. Represent ξ by a cycle $\sum_j n_j[V_j]$, which is possible because i_t^* is defined on $A_*(P)$ and the value does not depend on the representative. The image $p(V_j)$ is irreducible, being the continuous image of an irreducible space, and closed, because p is proper (X is projective, so p is the base change of a proper morphism); the irreducible closed subsets of the curve T are T itself and the closed points, so each V_j either dominates T or lies in a single fibre.

   Let Γ be the part of the cycle whose components dominate T. By (2), $i_{t_1}^*[\Gamma] - i_{t_0}^*[\Gamma] = \Gamma(t_1) - \Gamma(t_0)$, which is algebraically equivalent to zero by the definition of algebraic equivalence recalled in section 6.1.

   Let V be a component lying in a single fibre $p^{-1}(s)$. If $s \neq t$, then $V \cap (\{t\} \times X) = \emptyset$, and by corollary 4.3.11 the class $i_t^![V]$ is the image of a class on that intersection, hence zero. If $s = t$, write $\beta \in A_*(X)$ for the class of V read in $\{t\} \times X \cong X$, so that $[V] = i_{t*}\beta$; then part (1) and theorem 5.3.2(2) give

$$i_t^*[V] \;=\; i_t^!\big(i_{t*}\beta\big) \;=\; c_1(\mathcal{O}_X) \cap \beta \;=\; 0$$

   the last equality because the constant section 1 of $\mathcal{O}_X$ has empty divisor, so the operator $c_1(\mathcal{O}_X)\cap -$ of definition 2.4.1 is zero. Every component lying in a fibre therefore contributes zero to both $i_{t_1}^*\xi$ and $i_{t_0}^*\xi$, and $i_{t_1}^*\xi - i_{t_0}^*\xi = \Gamma(t_1) - \Gamma(t_0)$, completing the proof.

The lemma says that algebraic triviality on X can be tested with an arbitrary Chow class on $T \times X$ rather than with a cycle whose components are constrained to dominate T. That freedom is what makes the next verification routine, since the operations of definition 6.3.6 act on classes and not on chosen representatives.

Theorem 6.3.9: The Four Relations Are Adequate

Each of the following is an adequate equivalence relation in the sense of definition 6.3.6, on the class of smooth projective varieties over $\mathbb{C}$, and each is nondegenerate:

1. rational equivalence, $\mathcal{R}_{\mathrm{rat}}(X) = 0$;
2. algebraic equivalence, $\mathcal{R}_{\mathrm{alg}}(X)$ as defined in section 6.1;
3. homological equivalence, $\mathcal{R}_{\mathrm{hom}}(X)$ of definition 6.3.1;
4. numerical equivalence, $\mathcal{R}_{\mathrm{num}}(X)$ of definition 6.3.4. Moreover $\mathcal{R}_{\mathrm{num}}(X)$ contains every torsion class of $A^*(X)$, and $A^*_{\mathrm{num}}(X)$ is torsion free.

***Proof*:**
A morphism $f\colon X \to Y$ of smooth projective varieties is proper: it factors as the graph $\gamma_f\colon X \to X \times Y$, a closed immersion by lemma 5.2.4(3), followed by the projection $\mathrm{pr}_Y\colon X \times Y \to Y$, which is proper because X is complete. Both f^* (definition 5.2.8) and f_* are therefore defined

on all of $A^*(X)$ throughout, and theorem 5.2.10 applies to f without further hypothesis.

1. The subgroup is zero, so (1), (2) and (3) of definition 6.3.6 hold vacuously, and nondegeneracy is immediate. What the case records is that the three operations exist on $A^*(X)$ at all: the product is theorem 5.2.7, the pullback is theorem 5.2.10, and pushforward descends to rational equivalence by theorem 1.2.3.

2. Fix $\alpha \in \mathcal{R}_{\mathrm{alg}}(X)$ and a family exhibiting it: a smooth connected curve T, closed points $t_0, t_1 \in T$, and a cycle Γ on $P = T \times X$ with every component dominating T and $\alpha = \Gamma(t_1) - \Gamma(t_0)$. By lemma 6.3.8(2), $\alpha = i_{t_1}^*[\Gamma] - i_{t_0}^*[\Gamma]$. A general element of $\mathcal{R}_{\mathrm{alg}}(X)$ is a finite sum of such, and each condition below is additive, so it suffices to treat one family.

   Ideal. Let $\gamma \in A^*(X)$ and put $\xi = [\Gamma] \cdot q^*\gamma \in A_*(P)$, where $q\colon P \to X$ is the projection. Since i_t^* is a ring homomorphism (theorem 5.2.10(3)) and $q \circ i_t = \mathrm{id}_X$, functoriality (theorem 5.2.10(4)) gives

   $$i_t^*\xi \;=\; i_t^*[\Gamma] \cdot i_t^* q^*\gamma \;=\; \Gamma(t) \cdot \gamma$$

   so $\alpha \cdot \gamma = i_{t_1}^*\xi - i_{t_0}^*\xi$, which is algebraically equivalent to zero by lemma 6.3.8(3).

   Pullback. Let $g\colon X' \to X$ be a morphism of smooth projective varieties and let $G = \mathrm{id}_T \times g\colon T \times X' \to P$, a morphism of smooth varieties. Put $\xi = G^*[\Gamma]$. Writing i'_t for the fibre inclusion of $T \times X'$, the square identity $G \circ i'_t = i_t \circ g$ and theorem 5.2.10(4) give $(i'_t)^*\xi = g^* i_t^*[\Gamma] = g^*\Gamma(t)$, so $g^*\alpha = (i'_{t_1})^*\xi - (i'_{t_0})^*\xi$ is algebraically equivalent to zero by lemma 6.3.8(3).

   Pushforward. Let $f\colon X \to Y$ be a morphism of smooth projective varieties and let $F = \mathrm{id}_T \times f\colon P \to T \times Y$, which is proper, being the base change of the proper f. Put $\xi = F_*[\Gamma]$. The preimage $F^{-1}(\{t\} \times Y)$ is $\{t\} \times X$, a regular embedding of codimension 1 in P by lemma 6.3.8(1), so proposition 5.2.6 identifies the refined Gysin map of i_t^Y along F with the Gysin map of i_t^X; theorem 4.3.10(1) for the proper F then gives

   $$(i_t^Y)^*\big(F_*[\Gamma]\big) \;=\; f_*\big((i_t^X)^*[\Gamma]\big) \;=\; f_*\Gamma(t)$$

   so $f_*\alpha = (i_{t_1}^Y)^*\xi - (i_{t_0}^Y)^*\xi$ is algebraically equivalent to zero by lemma 6.3.8(3).

   Nondegeneracy: on $X = \operatorname{Spec}\mathbb{C}$ the product $T \times \operatorname{Spec}\mathbb{C}$ is T, a cycle all of whose components dominate T is a multiple $m[T]$, and its fibre cycle is m times the class of the point for every t, so $\Gamma(t_1) - \Gamma(t_0) = 0$ and $\mathcal{R}_{\mathrm{alg}}(\operatorname{Spec}\mathbb{C}) = 0$.

3. The kernel of a homomorphism of graded rings is a graded ideal, so $\mathcal{R}_{\mathrm{hom}}(X)$ satisfies condition (1). For (2) and (3), the cycle class map commutes with f^* and with f_* (theorem 6.2.13, theorem 6.2.9), so $\mathrm{cl}(f^*\alpha) = f^*\mathrm{cl}(\alpha)$ vanishes when $\mathrm{cl}(\alpha)$ does, and likewise for f_*. Nondegeneracy: on $\operatorname{Spec}\mathbb{C}$ the map cl is the degree $A^0(\operatorname{Spec}\mathbb{C}) = \mathbb{Z} \to H^0(\mathrm{pt};\mathbb{Z}) = \mathbb{Z}$, which is injective.

4. Write $m = \dim X$ and let $\alpha \in \mathcal{R}_{\mathrm{num}}(X)$.

   Ideal. For $\gamma \in A^*(X)$ and $\beta \in A^*(X)$ of the codimension complementary to $\alpha \cdot \gamma$, associativity (theorem 5.2.7(3)) gives $\deg((\alpha \cdot \gamma) \cdot \beta) = \deg(\alpha \cdot (\gamma \cdot \beta))$, and $\gamma \cdot \beta$ has the codimension complementary to α, so the value is zero.

   Pullback. Let $f\colon X' \to X$ be a morphism of smooth projective varieties and $\beta \in A^*(X')$ of complementary codimension. Since $\deg = \int_{X'}$ agrees with $\int_X \circ f_*$ (definition 1.2.4), the projection formula theorem 5.2.10(2) gives

$$\deg\bigl(f^*\alpha\cdot\beta\bigr) \;=\; \deg\bigl(f_*(f^*\alpha\cdot\beta)\bigr) \;=\; \deg\bigl(\alpha\cdot f_*\beta\bigr) \;=\; 0$$

the last equality because $f_*\beta$ has the codimension complementary to α.

Pushforward. Let $f\colon X\to Y$ and $\beta\in A^*(Y)$ of complementary codimension. The same two ingredients give

$$\deg\bigl(f_*\alpha\cdot\beta\bigr) \;=\; \deg\bigl(f_*(\alpha\cdot f^*\beta)\bigr) \;=\; \deg\bigl(\alpha\cdot f^*\beta\bigr) \;=\; 0$$

Nondegeneracy: on $\operatorname{Spec}\mathbb{C}$ a class is an integer a, and $\deg(a\cdot 1)=a$, so only $a=0$ is numerically trivial.

For the last two assertions, let $\alpha\in A^*(X)$ satisfy $r\alpha=0$ for some integer $r\geq 1$. For every β of complementary codimension, $r\deg(\alpha\cdot\beta)=\deg(r\alpha\cdot\beta)=0$ in $\mathbb{Z}$, so $\deg(\alpha\cdot\beta)=0$ and $\alpha\in\mathcal{R}_{\mathrm{num}}(X)$. The same computation applied to a class α with $r\alpha\in\mathcal{R}_{\mathrm{num}}(X)$ gives $\alpha\in\mathcal{R}_{\mathrm{num}}(X)$, so the quotient $A^*_{\mathrm{num}}(X)$ has no nonzero torsion, completing the proof.

Nondegeneracy is what pins numerical equivalence to the coarse end of the list: an adequate relation that keeps the degree of a zero-cycle cannot identify more than numerical equivalence does.

Proposition 6.3.10: Numerical Equivalence Is the Coarsest Nondegenerate Relation

Let $\sim$ be a nondegenerate adequate equivalence relation. Then $\mathcal{R}_\sim(X)\subseteq\mathcal{R}_{\mathrm{num}}(X)$ for every smooth projective variety X.

***Proof*:**
Let $\alpha\in\mathcal{R}_\sim(X)$ with X of dimension n, and let $\beta\in A^{n-k}(X)$ be of the codimension complementary to the codimension-k component of α; replacing α by that component is harmless because $\mathcal{R}_\sim(X)$ is graded. By condition (1) of definition 6.3.6, $\alpha\cdot\beta\in\mathcal{R}_\sim(X)\cap A^n(X)$. Let $\pi\colon X\to\operatorname{Spec}\mathbb{C}$ be the structure morphism, a morphism of smooth projective varieties. By condition (3), $\pi_*(\alpha\cdot\beta)\in\mathcal{R}_\sim(\operatorname{Spec}\mathbb{C})$, which is zero by nondegeneracy. Definition 1.2.4 identifies π_* on $A_0(X)$ with the degree, so $\deg(\alpha\cdot\beta)=0$. As β was arbitrary, $\alpha\in\mathcal{R}_{\mathrm{num}}(X)$, as we sought to show.

With the two ends of the list identified, only the middle inclusion remains, and it is the one carrying geometric content: a family of cycles over a connected curve has a single cohomology class, because the two fibre inclusions are homotopic.

Theorem 6.3.11: The Tower of Adequate Equivalence Relations

Let X be a smooth projective variety over $\mathbb{C}$. Then

$$0 \;=\; \mathcal{R}_{\mathrm{rat}}(X) \;\subseteq\; \mathcal{R}_{\mathrm{alg}}(X) \;\subseteq\; \mathcal{R}_{\mathrm{hom}}(X) \;\subseteq\; \mathcal{R}_{\mathrm{num}}(X)$$

Equivalently, a class that is rationally trivial is algebraically trivial, one that is algebraically trivial is homologically trivial, and one that is homologically trivial is numerically trivial.

***Proof*:**
Rational implies algebraic. This is the statement that $\mathcal{R}_{\mathrm{alg}}(X)$ is a subgroup of $A^*(X)$ rather than of the group of cycles, which is theorem 6.1.6: a cycle rationally equivalent to zero is algebraically equivalent to zero, the family being parametrized by $\mathbb{P}^1$.

Algebraic implies homological. Let $\alpha \in \mathcal{R}_{\mathrm{alg}}(X)$, and fix a family exhibiting it: a smooth connected curve T, closed points $t_0, t_1 \in T$, and a cycle Γ on $T \times X$ with all components dominating T and $\alpha = \Gamma(t_1) - \Gamma(t_0)$. It suffices to treat one such family, since cl is additive.

Step 1: the parameter curve may be taken projective. By [2, Projective Completion of a Smooth Curve] there are a smooth connected projective curve $\overline{T}$ and an open immersion $T \hookrightarrow \overline{T}$ with dense image, the base field $\mathbb{C}$ being perfect. Let $\overline{\Gamma}$ be the cycle on $\overline{T} \times X$ whose components are the closures of the components of Γ, with the same multiplicities. Each closure dominates $\overline{T}$, since it dominates the dense open T. Moreover $\overline{V} \cap \big(T \times X\big) = V$ for each component V, closure and intersection with an open subscheme being inverse operations on closed subvarieties of that open, so the scheme-theoretic fibres over points of T agree and $\overline{\Gamma}(t_i) = \Gamma(t_i)$ for $i = 0, 1$. Replacing T by $\overline{T}$ and Γ by $\overline{\Gamma}$, assume T is a smooth connected projective curve.

Step 2: the fibre cycle as a pushforward. Write $P = T \times X$, with $p\colon P \to T$ and $q\colon P \to X$ the projections; both are flat by lemma 5.1.1, and q is proper because T is complete. Fix a closed point $t \in T$. By proposition 1.3.5 the flat pullback $p^*[t]$ is the fundamental cycle of $p^{-1}(t) = \{t\} \times X$, which is $i_{t*}[X]$. Applying the projection formula theorem 5.2.10(2) to the morphism i_t of smooth projective varieties, with $[X]$ in the first slot and $[\Gamma]$ in the second, and using that $[X]$ is the unit of $A^*(X)$ (theorem 5.2.7(4)),

$$i_{t*}\big(i_t^*[\Gamma]\big) \;=\; i_{t*}\big([X] \cdot i_t^*[\Gamma]\big) \;=\; i_{t*}[X] \cdot [\Gamma] \;=\; p^*[t] \cdot [\Gamma]$$

Since $q \circ i_t = \mathrm{id}_X$, applying q_* and lemma 6.3.8(2) gives

$$\Gamma(t) \;=\; i_t^*[\Gamma] \;=\; q_*\big([\Gamma] \cdot p^*[t]\big) \tag{6.3}$$

Step 3: the cohomology class does not depend on t. Apply cl to (6.3). The map cl carries the proper pushforward q_* to the umkehr map $q_!$ of definition 6.2.8 (theorem 6.2.9), it commutes with the flat pullback p^* (theorem 6.2.13), and it is a ring homomorphism (corollary 6.2.14), so

$$\mathrm{cl}\big(\Gamma(t)\big) \;=\; q_!\Big(\mathrm{cl}\big([\Gamma]\big) \cup p^*\mathrm{cl}\big([t]\big)\Big)$$

Now T is a smooth connected projective curve over $\mathbb{C}$, so $T(\mathbb{C})$ is a connected compact manifold, hence path connected; any two of its points therefore have the same class in $H_0(T;\mathbb{Z})$, and since cl of a point class is the Poincaré dual of that class (definition 6.2.3), $\mathrm{cl}([t_1]) = \mathrm{cl}([t_0])$ in $H^2(T;\mathbb{Z})$. Subtracting the two instances of the displayed identity and using additivity of $q_!$, p^* and the cup product,

$$\mathrm{cl}(\alpha) \;=\; q_!\Big(\mathrm{cl}\big([\Gamma]\big) \cup p^*\big(\mathrm{cl}([t_1]) - \mathrm{cl}([t_0])\big)\Big) \;=\; 0$$

so $\alpha \in \mathcal{R}_{\mathrm{hom}}(X)$.

Homological implies numerical. Homological equivalence is adequate and nondegenerate by theorem 6.3.9(3), so proposition 6.3.10 gives $\mathcal{R}_{\mathrm{hom}}(X) \subseteq \mathcal{R}_{\mathrm{num}}(X)$, completing the proof.

The last inclusion is also direct: if $\mathrm{cl}(\alpha) = 0$ and β has complementary codimension, then $\deg(\alpha \cdot \beta) = \int_X \mathrm{cl}(\alpha \cdot \beta) = \int_X \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta) = 0$. What proposition 6.3.10 adds is that no adequate relation whatever, not merely the three above it, sits below numerical equivalence while keeping degrees.

Each of the three inclusions is strict for some X, and none is strict for every X: on $\mathbb{P}^m$ all four relations coincide, as exercise 6.3.1 (1) asks the reader to check. The strictness of the first is already visible on a curve.

Example 6.3.12: Divisor Classes on an Elliptic Curve

Let (E, O) be an elliptic curve over $\mathbb{C}$ and let $P \in E$ be a closed point with $P \neq O$. In $A^1(E) = A_0(E)$ consider $\alpha = [P] - [O]$.

α is algebraically trivial. Take $T = E$ and $\Gamma = [\Delta]$, the class of the diagonal in $E \times E$, whose single component dominates the first factor. The scheme-theoretic fibre of Δ over a closed point t is the reduced point (t, t), so $\Gamma(t) = [t]$ in $A_0(E)$, and $\Gamma(P) - \Gamma(O) = \alpha$.

α is not rationally trivial. By proposition 2.1.3 the group $A_0(E)$ is $\mathrm{Pic}\,(E)$, and [2, The Group Law on an Elliptic Curve] says that $Q \mapsto [Q] - [O]$ is a bijection from the closed points of E onto $\mathrm{Pic}^0(E)$. In particular $[P] - [O] = 0$ would force $P = O$.

Hence $\mathcal{R}_{\mathrm{rat}}(E) \subsetneq \mathcal{R}_{\mathrm{alg}}(E)$, and the first inclusion of theorem 6.3.11 is strict. The three coarser relations, by contrast, agree here. Numerical triviality in $A^1(E) = A^n(E)$ with $n = 1$ tests α against $A^0(E) = \mathbb{Z}\,[E]$, and $\deg(\alpha \cdot [E]) = \deg(\alpha)$ since $[E]$ is the unit (theorem 5.2.7(4)), so $\mathcal{R}_{\mathrm{num}}(E) \cap A^1(E) = \mathrm{Pic}^0(E)$; the same group is $\mathcal{R}_{\mathrm{hom}}(E) \cap A^1(E)$ by example 6.3.2. The first paragraph puts every $[Q] - [O]$, hence all of $\mathrm{Pic}^0(E)$, into $\mathcal{R}_{\mathrm{alg}}(E)$, and theorem 6.3.11 confines it to $\mathrm{Pic}^0(E)$.

The second inclusion is strict on a general quintic threefold in $\mathbb{P}^4$, and the example is far less accessible: Griffiths produced there a difference of two lines that is homologically trivial but not algebraically trivial, the obstruction coming from the intermediate Jacobian rather than from any intersection number. The construction is not developed here; see [8, Ch. 19]. The third inclusion is strict on an Enriques surface X, for the reason recorded in box 6.3.3: the canonical class K_X is a nonzero 2-torsion element of $\mathrm{Pic}\,(X) = A^1(X)$ (proposition 2.1.3), so it is numerically trivial by theorem 6.3.9(4), while the kernel of cl on divisor classes is $\mathrm{Pic}^0(X)$, which vanishes because $H^1(X, \mathcal{O}_X) = 0$; hence $\mathrm{cl}(K_X) \neq 0$. The surface theory behind those two facts about K_X and $H^1(X, \mathcal{O}_X)$ is not developed here. Whether homological and numerical equivalence agree after tensoring with $\mathbb{Q}$ is open, and is the subject of **??** 6.3.14.

Numerical equivalence collapses $A^*(X)$ far enough to be finitely generated, which neither rational nor algebraic equivalence achieves: $A^*(X)$ itself need not be finitely generated, and even $\mathrm{NS}^*(X) = A^*(X)/{\sim_{\mathrm{alg}}}$ is only known to be finitely generated in codimension one (section 6.1). Homological equivalence does achieve it, its quotient being a subgroup of the cohomology of a compact manifold, and the proof below extracts the numerical statement from that one.

Theorem 6.3.13: Finiteness of Numerical Equivalence

Let X be a smooth projective variety over $\mathbb{C}$. For each k the group $\mathrm{Num}^k(X) = A^k_{\mathrm{num}}(X)$ is a finitely generated free abelian group, of rank at most the rank of $H^{2k}(X; \mathbb{Z})$.

***Proof*:**
Freeness will follow from finite generation together with the absence of torsion, which is theorem 6.3.9(4).

For finite generation, theorem 6.3.11 gives $\mathcal{R}_{\text{hom}}(X) \subseteq \mathcal{R}_{\text{num}}(X)$, so the identity of $A^k(X)$ induces a surjection

$$A^k_{\text{hom}}(X) \twoheadrightarrow A^k_{\text{num}}(X)$$

By definition 6.3.1 the group $A^k_{\text{hom}}(X)$ is the image of cl, hence a subgroup of $H^{2k}(X;\mathbb{Z})$. The space $X(\mathbb{C})$ is a compact manifold, and the singular cohomology of a compact manifold is a finitely generated abelian group in each degree; that topological input is not proved in this book (for the corresponding statement over an arbitrary base field, where a Weil cohomology theory replaces singular cohomology, see [8, Ch. 19]). A subgroup of a finitely generated abelian group is finitely generated, and a quotient of a finitely generated group is finitely generated, so $A^k_{\text{num}}(X)$ is finitely generated, of rank at most that of $H^{2k}(X;\mathbb{Z})$. Being finitely generated and torsion free, it is free, as we sought to show.

The rank of $\mathrm{Num}^1(X)$ is the Picard number $\rho(X)$ met in section 6.1, and the theorem is the statement that every codimension has such a number. For $k = 1$ the finiteness is the theorem of the base, proved there by different means; for general k the argument above is the only one available, and it runs through cohomology rather than through the geometry of families.

Grothendieck's standard conjectures are a short list of statements about algebraic cycles which, if known, would give the theory of motives its expected structure; the one that concerns the tower directly is the following.

Conjecture 6.3.14: Homological and Numerical Equivalence Agree

Let X be a smooth projective variety over $\mathbb{C}$ and let $k \geq 0$. Then a class in $A^k(X) \otimes \mathbb{Q}$ is homologically equivalent to zero if and only if it is numerically equivalent to zero; that is,

$$\mathcal{R}_{\text{hom}}(X) \otimes \mathbb{Q} \;=\; \mathcal{R}_{\text{num}}(X) \otimes \mathbb{Q} \qquad \text{in } A^k(X) \otimes \mathbb{Q}$$

This is the standard conjecture usually labelled $D(X)$.

One inclusion is theorem 6.3.11, so the content is that a class pairing to zero against every algebraic cycle of complementary dimension already has vanishing cohomology class. The conjecture is known at the two ends of the range and in codimension one, and open in the rest. For $k = 0$ and $k = n$ both subgroups are computed outright, integrally, with no rational coefficients needed: in codimension 0 they vanish, since exercise 6.3.1 (2) gives $\mathcal{R}_{\text{num}}(X) \cap A^0(X) = 0$ and theorem 6.3.11 confines $\mathcal{R}_{\text{hom}}(X) \cap A^0(X)$ inside it, while in codimension n both relations say that the degree vanishes, by example 6.3.2 together with definition 6.3.4. For $k = 1$ it holds as well, but the proof uses the hard Lefschetz theorem, which this book does not develop; see [8, Ch. 19]. For $2 \leq k \leq n-1$ no case is known in general. The conjecture does not stand alone: the standard conjecture $B(X)$, on the algebraicity of the inverse of the hard Lefschetz operator, implies $D(X)$ in characteristic zero, so a proof of B would settle the tower. And D is what would make homological equivalence independent of the cohomology theory used to define it, since numerical equivalence refers to no cohomology at all: the étale cycle class map named in section 6.2 defines a second homological equivalence, and D implies that it agrees with the one above.

Exercise 6.3.1

1. Let $X = \mathbb{P}^m$. Using example 5.2.3, which presents $A^*(\mathbb{P}^m)$ as $\mathbb{Z}[h]/(h^{m+1})$ with h^j the class of a linear subspace of codimension j and $\deg(h^m) = 1$, show that no nonzero class of $A^*(\mathbb{P}^m)$ is numerically equivalent to zero. Deduce that rational, algebraic, homological and numerical equivalence all coincide on $\mathbb{P}^m$.

2. Let X be a smooth projective variety of dimension n. Show that $\mathcal{R}_{\text{num}}(X) \cap A^0(X) = 0$, using $A^0(X) = \mathbb{Z}\,[X]$ and the fact that $[X]$ is the unit of $A^*(X)$. Conclude that all four relations of theorem 6.3.11 agree in codimension 0. Show also that $\mathcal{R}_{\text{num}}(X) \cap A^n(X) = \ker(\deg)$, so that homological and numerical equivalence agree in codimension n as well by example 6.3.2, and explain why example 6.3.12 forbids adding rational equivalence to that agreement.

3. Let $\mathcal{T}(X) \subseteq A^*(X)$ be the subgroup of torsion classes. Show that $X \mapsto \mathcal{T}(X)$ satisfies the three conditions of definition 6.3.6 and is nondegenerate. Deduce from proposition 6.3.10 that every torsion class is numerically equivalent to zero, recovering part of theorem 6.3.9(4) without computing degrees.

4. Let $\sim$ and $\sim'$ be adequate equivalence relations. Show that $X \mapsto \mathcal{R}_\sim(X) \cap \mathcal{R}_{\sim'}(X)$ and $X \mapsto \mathcal{R}_\sim(X) + \mathcal{R}_{\sim'}(X)$ are both adequate. Which of the two is nondegenerate whenever $\sim$ is?

5. Let X be smooth projective of dimension n. Show that $(\alpha, \beta) \mapsto \deg(\alpha \cdot \beta)$ descends to a pairing
$$A^k_{\text{num}}(X) \times A^{n-k}_{\text{num}}(X) \longrightarrow \mathbb{Z}$$
and that this pairing is nondegenerate on both sides. Explain why the same statement for $A^k_{\text{hom}}(X)$ would follow from **??** 6.3.14 after tensoring with $\mathbb{Q}$, but is not available unconditionally.

6. Let $\sim$ be an adequate equivalence relation. Show that $\sim$ is nondegenerate if and only if, for every smooth projective X, the degree homomorphism $\deg\colon A^n(X) \to \mathbb{Z}$ vanishes on $\mathcal{R}_\sim(X) \cap A^n(X)$, where $n = \dim X$; that is, if and only if the degree descends to $A^*_\sim$.

7. Continue with $S = \mathbb{P}^1 \times \mathbb{P}^1$ and the classes f_1, f_2 of example 6.3.5. Show that a class of $A^2(S)$ is numerically equivalent to zero if and only if it has degree zero, and that cl is injective on the subgroup of $A^1(S)$ generated by f_1 and f_2. (For the second part, use theorem 6.3.11 rather than any computation in cohomology.)

8. Let X be smooth projective over $\mathbb{C}$ of dimension n and suppose $H^{2k}(X;\mathbb{Z})$ is torsion free. Show that $A^k_{\text{hom}}(X)$ and $A^k_{\text{num}}(X)$ are both finitely generated free abelian groups and that the surjection $A^k_{\text{hom}}(X) \to A^k_{\text{num}}(X)$ of theorem 6.3.13 is an isomorphism if and only if the two groups have the same rank.

6.4 Correspondences

A morphism $f\colon X \to Y$ moves cycles in both directions, forward by f_* and backward by f^*, and the datum determining it is a single cycle on the product: its graph. Conversely, an arbitrary cycle class on $X \times Y$ produces a homomorphism $A^*(X) \to A^*(Y)$, and the graph of f produces f_* while its transpose produces f^*. This section constructs those homomorphisms, shows that composing two of them is again given by a cycle on a product, and matches the construction against the cycle class map of section 6.2.

Throughout, X, Y, Z and W are smooth projective varieties over $\overline{k}$ of dimensions d_X, d_Y, d_Z and d_W; the base field is $\mathbb{C}$ only in proposition 6.4.12, where the cycle class map appears. Products of smooth projective varieties are again smooth projective varieties, as they were in the diagonal construction of theorem 5.1.6, so $A^*(X \times Y)$ is a ring graded by codimension (theorem 5.2.7) and $\dim(X \times Y) = d_X + d_Y$. Every projection from such a product onto a subproduct of its factors is flat, being a base change of a structure morphism $V \to \operatorname{Spec} \overline{k}$, and is proper as soon as the factors it forgets are projective, again by base change; for a flat morphism of smooth varieties the pullback of definition 5.2.11 is the flat pullback of definition 1.3.3 (theorem 5.2.10), so the two are written p^* without distinction below.

The grading is chosen so that the correspondences of degree 0 from X to Y are the classes of dimension d_Y on $X \times Y$, which is the dimension of the graph of a morphism $Y \to X$.

Definition 6.4.1: Correspondence

For $r \in \mathbb{Z}$, the group of *correspondences of degree r from X to Y* is

$$\operatorname{Corr}^r(X, Y) = A^{d_X + r}(X \times Y)$$

The *transpose* of $\alpha \in \operatorname{Corr}^r(X, Y)$ is ${}^t\alpha = \sigma_*\alpha$, where $\sigma \colon X \times Y \to Y \times X$ interchanges the two factors.

Since σ is an isomorphism it preserves codimension, so ${}^t\alpha$ lies in $A^{d_X+r}(Y \times X) = \operatorname{Corr}^{r+d_X-d_Y}(Y, X)$: transposition changes the degree unless X and Y have the same dimension. A correspondence acts on cycle classes by pulling back along the first projection, multiplying, and pushing forward along the second.

Proposition 6.4.2: Action of a Correspondence

Let $\alpha \in \operatorname{Corr}^r(X, Y)$, and let $\mathrm{pr}_1, \mathrm{pr}_2$ be the projections of $X \times Y$ onto X and Y. The assignment

$$\alpha_*\beta = \mathrm{pr}_{2*}\big(\alpha \cdot \mathrm{pr}_1^*\beta\big)$$

is defined for every $\beta \in A^*(X)$, carries $A^k(X)$ into $A^{k+r}(Y)$, and is additive in α and in β.

***Proof*:**
The projection pr_1 is flat of relative dimension d_Y, so pr_1^* carries $A_m(X)$ into $A_{m+d_Y}(X \times Y)$ (definition 1.3.3); in codimension this reads $A^k(X) \to A^k(X \times Y)$, since $\dim(X \times Y) = d_X + d_Y$. The product with $\alpha \in A^{d_X+r}(X \times Y)$ is defined because $X \times Y$ is smooth (theorem 5.2.7) and lies in $A^{k+d_X+r}(X \times Y)$, that is, in $A_{d_Y-k-r}(X \times Y)$. The projection pr_2 is proper, since X is projective, so pr_{2*} is defined and preserves dimension (definition 1.2.1), landing in $A_{d_Y-k-r}(Y) = A^{k+r}(Y)$. Additivity in β holds because pr_1^*, multiplication by α and pr_{2*} are each homomorphisms of groups, and additivity in α because multiplication in $A^*(X \times Y)$ is biadditive, as we sought to show.

The degree r is the amount by which the action raises codimension, so a correspondence of degree 0 preserves the codimension grading. On a product of two projective lines the group of degree 0 correspondences is free of rank two, and each of its elements can be evaluated on all of $A^*(\mathbb{P}^1)$.

Example 6.4.3: Correspondences from $\mathbb{P}^1$ to $\mathbb{P}^1$

Fix points $p, q \in \mathbb{P}^1$ and set $h_1 = \mathrm{pr}_1^*[p]$ and $h_2 = \mathrm{pr}_2^*[q]$ in $A^1(\mathbb{P}^1 \times \mathbb{P}^1)$. This ring was computed in example 6.3.5, where the same two classes appear as f_2 and f_1: the projective bundle formula gives

$$A^0 = \mathbb{Z} \cdot [\mathbb{P}^1 \times \mathbb{P}^1], \qquad A^1 = \mathbb{Z}h_1 \oplus \mathbb{Z}h_2, \qquad A^2 = \mathbb{Z} \cdot [(p,q)],$$

the classes are independent of the chosen points, and the products are $h_1^2 = h_2^2 = 0$ with $h_1 \cdot h_2 = [(p,q)]$. Nothing about that computation is repeated here; what this example adds is the action.

Now let $\alpha = ah_1 + bh_2$ be a correspondence of degree 0 from $\mathbb{P}^1$ to $\mathbb{P}^1$, so $\alpha \in A^1(\mathbb{P}^1 \times \mathbb{P}^1) = \mathrm{Corr}^0(\mathbb{P}^1, \mathbb{P}^1)$. On $A^0(\mathbb{P}^1) = \mathbb{Z} \cdot [\mathbb{P}^1]$, the class $\mathrm{pr}_1^*[\mathbb{P}^1]$ is the unit of the ring, so $\alpha_*[\mathbb{P}^1] = \mathrm{pr}_{2*}(ah_1 + bh_2)$. Here pr_2 maps $\{p\} \times \mathbb{P}^1$ isomorphically onto $\mathbb{P}^1$ and contracts $\mathbb{P}^1 \times \{q\}$ to the point q, so $\mathrm{pr}_{2*}h_1 = [\mathbb{P}^1]$ and $\mathrm{pr}_{2*}h_2 = 0$ by definition 1.2.1, giving $\alpha_*[\mathbb{P}^1] = a[\mathbb{P}^1]$. On $A^1(\mathbb{P}^1) = \mathbb{Z} \cdot [p]$, one has $\mathrm{pr}_1^*[p] = h_1$ and hence $\alpha \cdot h_1 = ah_1^2 + b\,h_1h_2 = b[(p,q)]$, so $\alpha_*[p] = b[q]$. Thus α acts as multiplication by a in codimension 0 and by b in codimension 1; in particular a correspondence of degree 0 from $\mathbb{P}^1$ to $\mathbb{P}^1$ is determined by its action.

Composing two such actions ought to be the action of a third correspondence, and the cycle producing it is manufactured on the triple product: pull the two correspondences back to $X \times Y \times Z$ from the first two and the last two factors, multiply, and push forward to $X \times Z$.

Definition 6.4.4: Composition of Correspondences

Let $\alpha \in \mathrm{Corr}^r(X,Y)$ and $\beta \in \mathrm{Corr}^s(Y,Z)$, and write p_{12}, p_{23}, p_{13} for the projections of $X \times Y \times Z$ onto the indicated pairs of factors. The *composition* of α and β is

$$\beta \circ \alpha = p_{13*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta\big)$$

Proposition 6.4.5: Composition Computes the Composite Action

With α and β as in definition 6.4.4, the class $\beta \circ \alpha$ lies in $\mathrm{Corr}^{r+s}(X,Z)$, the composition is additive in α and in β, and

$$(\beta \circ \alpha)_* = \beta_* \circ \alpha_* \colon A^k(X) \to A^{k+r+s}(Z)$$

for every k.

***Proof*:**
The degree. The projections p_{12} and p_{23} are flat of relative dimensions d_Z and d_X, so they preserve codimension: $p_{12}^*\alpha \in A^{d_X+r}(X \times Y \times Z)$ and $p_{23}^*\beta \in A^{d_Y+s}(X \times Y \times Z)$. Their product lies in $A^{d_X+d_Y+r+s}$, that is, in A_{d_Z-r-s}. The projection p_{13} is proper because Y is projective, and p_{13*} preserves dimension, so $\beta \circ \alpha \in A_{d_Z-r-s}(X \times Z) = A^{d_X+r+s}(X \times Z) = \mathrm{Corr}^{r+s}(X,Z)$. Additivity in each argument is inherited from the biadditivity of the product and the additivity of p_{12}^*, p_{23}^* and p_{13*}.

Reducing both sides to the triple product. Write pr_i^{XY}, pr_i^{YZ}, pr_i^{XZ} for the projections of the three double products, and p_1, p_2, p_3 for the projections of $X \times Y \times Z$ onto single factors. Fix $\zeta \in A^k(X)$. The square

$$\begin{tikzcd} X \times Y \times Z \arrow[r, "p_{23}"] \arrow[d, "p_{12}"'] & Y \times Z \arrow[d, "\mathrm{pr}_1^{YZ}"] \\ X \times Y \arrow[r, "\mathrm{pr}_2^{XY}"'] & Y \end{tikzcd}$$

is Cartesian, since $(X \times Y) \times_Y (Y \times Z) = X \times Y \times Z$; its lower map is proper (X is projective) and its right-hand map is flat. By the compatibility of proper pushforward with flat pullback in a fibre square (proposition 1.3.6),

$$\mathrm{pr}_1^{YZ*} \circ \mathrm{pr}_{2*}^{XY} = p_{23*} \circ p_{12}^*$$

Applying this to $\alpha \cdot \mathrm{pr}_1^{XY*}\zeta$ and using that p_{12}^* is a ring homomorphism with $\mathrm{pr}_1^{XY} \circ p_{12} = p_1$ (theorem 5.2.10) gives

$$\mathrm{pr}_1^{YZ*}(\alpha_*\zeta) = p_{23*}\big(p_{12}^*\alpha \cdot p_1^*\zeta\big)$$

Therefore

$$\beta_*(\alpha_*\zeta) = \mathrm{pr}_{2*}^{YZ}\Big(\beta \cdot p_{23*}\big(p_{12}^*\alpha \cdot p_1^*\zeta\big)\Big)$$

The morphism p_{23} is proper because X is projective, so the projection formula (theorem 5.2.10) moves β inside the pushforward as $p_{23}^*\beta$, and $\mathrm{pr}_2^{YZ} \circ p_{23} = p_3$ together with the functoriality of proper pushforward (definition 1.2.1) gives

$$\beta_*(\alpha_*\zeta) = p_{3*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta \cdot p_1^*\zeta\big)$$

The other side. Expanding the definition of the action of $\beta \circ \alpha$,

$$(\beta \circ \alpha)_*\zeta = \mathrm{pr}_{2*}^{XZ}\Big(p_{13*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta\big) \cdot \mathrm{pr}_1^{XZ*}\zeta\Big)$$

The morphism p_{13} is proper, so the projection formula moves $\mathrm{pr}_1^{XZ*}\zeta$ inside as $p_{13}^*\mathrm{pr}_1^{XZ*}\zeta = (\mathrm{pr}_1^{XZ} \circ p_{13})^*\zeta = p_1^*\zeta$. Since $\mathrm{pr}_2^{XZ} \circ p_{13} = p_3$, functoriality of proper pushforward gives

$$(\beta \circ \alpha)_*\zeta = p_{3*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta \cdot p_1^*\zeta\big)$$

which is the expression obtained for $\beta_*(\alpha_*\zeta)$, completing the proof.

Composition is therefore compatible with the action, but the action does not determine a correspondence, so associativity is not a consequence of the associativity of composing maps. It has to be proved on the product of all four varieties, where both bracketings collapse to the same expression.

Theorem 6.4.6: Associativity of Composition

Let $\alpha \in \mathrm{Corr}^r(W, X)$, $\beta \in \mathrm{Corr}^s(X, Y)$ and $\gamma \in \mathrm{Corr}^t(Y, Z)$. Then

$$(\gamma \circ \beta) \circ \alpha = \gamma \circ (\beta \circ \alpha)$$

in $\mathrm{Corr}^{r+s+t}(W, Z)$.

***Proof*:**
Write p_{ij} and p_{ijk} for the projections of $W \times X \times Y \times Z$ onto the indicated factors, numbering W, X, Y, Z as $1, 2, 3, 4$; write q_{ij} for the projections of $W \times X \times Y$, r_{ij} for those of $W \times Y \times Z$, s_{ij} for those of $X \times Y \times Z$, and t_{ij} for those of $W \times X \times Z$. Both sides are shown to equal

$$\Theta = p_{14*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta \cdot p_{34}^*\gamma\big)$$

The right-hand bracketing. The square

$$\begin{tikzcd} W \times X \times Y \times Z \arrow[r, "p_{134}"] \arrow[d, "p_{123}"'] & W \times Y \times Z \arrow[d, "r_{12}"] \\ W \times X \times Y \arrow[r, "q_{13}"'] & W \times Y \end{tikzcd}$$

is Cartesian, since $(W \times X \times Y) \times_{W \times Y} (W \times Y \times Z) = W \times X \times Y \times Z$. Its lower map q_{13} is proper because X is projective, and its right-hand map r_{12} is flat, so proposition 1.3.6 gives $r_{12}^* q_{13*} = p_{134*} p_{123}^*$. Since p_{123}^* is a ring homomorphism with $q_{12} \circ p_{123} = p_{12}$ and $q_{23} \circ p_{123} = p_{23}$ (theorem 5.2.10),

$$r_{12}^*(\beta \circ \alpha) = p_{134*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta\big)$$

Hence

$$\gamma \circ (\beta \circ \alpha) = r_{13*}\Big(p_{134*}\big(p_{12}^*\alpha \cdot p_{23}^*\beta\big) \cdot r_{23}^*\gamma\Big)$$

The morphism p_{134} is proper because X is projective, so the projection formula (theorem 5.2.10) moves $r_{23}^*\gamma$ inside the pushforward as $p_{134}^* r_{23}^*\gamma = (r_{23} \circ p_{134})^*\gamma = p_{34}^*\gamma$. Finally $r_{13} \circ p_{134} = p_{14}$, so functoriality of proper pushforward (definition 1.2.1) gives $\gamma \circ (\beta \circ \alpha) = \Theta$.

The left-hand bracketing. The square

$$\begin{tikzcd} W \times X \times Y \times Z \arrow[r, "p_{234}"] \arrow[d, "p_{124}"'] & X \times Y \times Z \arrow[d, "s_{13}"] \\ W \times X \times Z \arrow[r, "t_{23}"'] & X \times Z \end{tikzcd}$$

is Cartesian for the same reason, its lower map t_{23} is flat, and its right-hand map s_{13} is proper because Y is projective. By proposition 1.3.6, $t_{23}^* s_{13*} = p_{124*} p_{234}^*$, and since $s_{12} \circ p_{234} = p_{23}$ and $s_{23} \circ p_{234} = p_{34}$,

$$t_{23}^*(\gamma \circ \beta) = p_{124*}\big(p_{23}^*\beta \cdot p_{34}^*\gamma\big)$$

Therefore

$$(\gamma \circ \beta) \circ \alpha = t_{13*}\Big(t_{12}^*\alpha \cdot p_{124*}\big(p_{23}^*\beta \cdot p_{34}^*\gamma\big)\Big)$$

The morphism p_{124} is proper because Y is projective, so the projection formula moves $t_{12}^*\alpha$ inside as $p_{124}^* t_{12}^*\alpha = (t_{12} \circ p_{124})^*\alpha = p_{12}^*\alpha$, and $t_{13} \circ p_{124} = p_{14}$ gives $(\gamma \circ \beta) \circ \alpha = \Theta$. The two bracketings agree, as we sought to show.

Only three properties of the projections entered that argument: flat base change, the projection formula, and functoriality. The same three properties identify the correspondences attached to an actual morphism.

Proposition 6.4.7: The Graph of a Morphism and its Transpose

Let $f\colon X \to Y$ be a morphism and let $\gamma_f = (\mathrm{id}_X, f)\colon X \to X \times Y$, a closed immersion with image the graph Γ_f. Then $[\Gamma_f] \in \mathrm{Corr}^{d_Y - d_X}(X, Y)$ and ${}^t[\Gamma_f] \in \mathrm{Corr}^0(Y, X)$, and for all $\beta \in A^*(X)$ and $\beta' \in A^*(Y)$,

$$[\Gamma_f]_*\beta = f_*\beta, \qquad {}^t[\Gamma_f]_*\beta' = f^*\beta'$$

***Proof*:**
The morphism γ_f is a section of $\mathrm{pr}_1\colon X \times Y \to X$, and a section of a morphism with separated target is a closed immersion, so Γ_f is a closed subvariety of $X \times Y$ isomorphic to X by γ_f. Hence $[\Gamma_f] = \gamma_{f*}[X]$ (definition 1.2.1), the extension of function fields being trivial, and $\dim \Gamma_f = d_X$, so $[\Gamma_f] \in A^{d_Y}(X \times Y) = \mathrm{Corr}^{d_Y - d_X}(X, Y)$. Transposing preserves codimension, so ${}^t[\Gamma_f] \in A^{d_Y}(Y \times X) = \mathrm{Corr}^0(Y, X)$.

The action of the graph. A closed immersion is proper, so the projection formula (theorem 5.2.10) applies to γ_f and gives, for $\beta \in A^*(X)$,

$$[\Gamma_f] \cdot \mathrm{pr}_1^*\beta = \gamma_{f*}\big([X] \cdot \gamma_f^*\mathrm{pr}_1^*\beta\big) = \gamma_{f*}\big((\mathrm{pr}_1 \circ \gamma_f)^*\beta\big) = \gamma_{f*}\beta$$

where $[X]$ was dropped because it is the unit of $A^*(X)$ (theorem 5.2.7) and $\mathrm{pr}_1 \circ \gamma_f = \mathrm{id}_X$. Pushing forward along pr_2 and using $\mathrm{pr}_2 \circ \gamma_f = f$ with the functoriality of proper pushforward gives $[\Gamma_f]_*\beta = f_*\beta$.

The action of the transpose. Let $\sigma\colon X \times Y \to Y \times X$ be the interchange, so ${}^t[\Gamma_f] = \sigma_*\gamma_{f*}[X] = \tilde{\gamma}_{f*}[X]$ with $\tilde{\gamma}_f = \sigma \circ \gamma_f = (f, \mathrm{id}_X)\colon X \to Y \times X$, again a closed immersion. Writing $\mathrm{pr}_1', \mathrm{pr}_2'$ for the projections of $Y \times X$, the same two steps give, for $\beta' \in A^*(Y)$,

$${}^t[\Gamma_f] \cdot \mathrm{pr}_1'^*\beta' = \tilde{\gamma}_{f*}\big((\mathrm{pr}_1' \circ \tilde{\gamma}_f)^*\beta'\big) = \tilde{\gamma}_{f*}(f^*\beta')$$

and then, since $\mathrm{pr}_2' \circ \tilde{\gamma}_f = \mathrm{id}_X$, ${}^t[\Gamma_f]_*\beta' = f^*\beta'$, completing the proof.

The graph therefore carries both directions of f: the class itself acts as the pushforward, its transpose as the pullback, and the degrees match, since f_* raises codimension by $d_Y - d_X$ while f^* preserves it. For a self-map of $\mathbb{P}^1$ the degree of the map determines the graph class.

Example 6.4.8: The Graph of a Self-Map of $\mathbb{P}^1$

Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be a nonconstant morphism, so that f is finite of some degree $d = [k(\mathbb{P}^1) : f^*k(\mathbb{P}^1)]$. Both source and target have dimension 1, so $[\Gamma_f] \in \mathrm{Corr}^0(\mathbb{P}^1, \mathbb{P}^1) = A^1(\mathbb{P}^1 \times \mathbb{P}^1)$, and by example 6.4.3 it can be written $[\Gamma_f] = ah_1 + bh_2$ with the action of a in codimension 0 and b in codimension 1. By proposition 6.4.7 that action is f_*. Now $f_*[\mathbb{P}^1] = d[\mathbb{P}^1]$ by definition 1.2.1, since f maps $\mathbb{P}^1$ onto $\mathbb{P}^1$ with function-field extension of degree d; and $f_*[x] = [f(x)]$ for a point x, since f restricts to an isomorphism $\{x\} \to \{f(x)\}$. Comparing with the action computed in example 6.4.3 gives $a = d$ and $b = 1$, that is,

$$[\Gamma_f] = d\,h_1 + h_2$$

For $f = \mathrm{id}$ this reads $[\Delta_{\mathbb{P}^1}] = h_1 + h_2$, and the action is multiplication by 1 in both codimensions.

Graphs compose the way the morphisms do, and the diagonal is a unit for composition. Both are proved the same way: rewrite one of the two pulled-back classes as a pushforward from a copy of a smaller product, then follow where the remaining projections send that copy.

Proposition 6.4.9: Graphs and the Diagonal under Composition

Let $f\colon X \to Y$ and $g\colon Y \to Z$ be morphisms, and let $\Delta_X \subseteq X \times X$ be the diagonal.

1. $[\Gamma_g] \circ [\Gamma_f] = [\Gamma_{g\circ f}]$.
2. $\alpha \circ [\Delta_X] = \alpha$ and $[\Delta_Y] \circ \alpha = \alpha$ for every $\alpha \in \mathrm{Corr}^r(X, Y)$.

***Proof*:**

1. Let $u = \mathrm{id}_X \times \gamma_g\colon X \times Y \to X \times Y \times Z$, so $u(x, y) = (x, y, g(y))$, a closed immersion. The square

$$\begin{tikzcd} X \times Y \arrow[r, "\mathrm{pr}_2^{XY}"] \arrow[d, "u"'] & Y \arrow[d, "\gamma_g"] \\ X \times Y \times Z \arrow[r, "p_{23}"'] & Y \times Z \end{tikzcd}$$

is Cartesian: a point of $(X \times Y \times Z) \times_{Y\times Z} Y$ is a triple (x, y, z) together with a point y' of Y satisfying $(y, z) = (y', g(y'))$, so $y' = y$ and $z = g(y)$. Its right-hand map γ_g is a closed immersion, hence proper, and its lower map p_{23} is flat, so proposition 1.3.6 gives $p_{23}^*\gamma_{g*} = u_*\mathrm{pr}_2^{XY*}$. Since $\mathrm{pr}_2^{XY*}[Y] = [X \times Y]$ (definition 1.3.3) and $[\Gamma_g] = \gamma_{g*}[Y]$,

$$p_{23}^*[\Gamma_g] = u_*[X \times Y]$$

The morphism u is proper, so the projection formula (theorem 5.2.10) and $p_{12} \circ u = \mathrm{id}_{X\times Y}$ give

$$p_{12}^*[\Gamma_f] \cdot u_*[X \times Y] = u_*\big(u^*p_{12}^*[\Gamma_f] \cdot [X \times Y]\big) = u_*[\Gamma_f]$$

Finally $p_{13} \circ u = \mathrm{id}_X \times g$, so by the functoriality of proper pushforward

$$[\Gamma_g] \circ [\Gamma_f] = (\mathrm{id}_X \times g)_*[\Gamma_f]$$

The morphism $\mathrm{id}_X \times g$ carries Γ_f onto $\Gamma_{g\circ f}$, and it does so isomorphically, because on both graphs the first projection to X is an isomorphism and $\mathrm{id}_X \times g$ commutes with it. Hence the pushforward is $[\Gamma_{g\circ f}]$ by definition 1.2.1.

2. Let $\delta\colon X \to X \times X$ be the diagonal immersion, so $[\Delta_X] = \delta_*[X]$, and let $v = \delta \times \mathrm{id}_Y : X \times Y \to X \times X \times Y$, $v(x, y) = (x, x, y)$. The maps v and $\mathrm{pr}_1^{XY} : X \times Y \to X$ present $X \times Y$ as the fibre product $(X \times X \times Y) \times_{X\times X} X$ taken over p_{12} and δ, since a point of that fibre product is a triple (x_1, x_2, y) together with a point x satisfying $(x_1, x_2) = (x, x)$. The immersion δ is proper and p_{12} is flat, so proposition 1.3.6 together with $\mathrm{pr}_1^{XY*}[X] = [X\times Y]$ gives $p_{12}^*[\Delta_X] = v_*[X \times Y]$, as in part (1). Since $p_{23} \circ v = \mathrm{id}_{X\times Y}$, the projection formula gives
$$p_{12}^*[\Delta_X] \cdot p_{23}^*\alpha = v_*\big(v^* p_{23}^*\alpha\big) = v_*\alpha$$
and since $p_{13} \circ v = \mathrm{id}_{X\times Y}$ as well, pushing forward gives $\alpha \circ [\Delta_X] = \alpha$. For the other identity, let $w = \mathrm{id}_X \times \delta_Y : X \times Y \to X \times Y \times Y$, $w(x, y) = (x, y, y)$, where $\delta_Y : Y \to Y \times Y$ is the diagonal. The maps w and pr_2^{XY} present $X \times Y$ as $(X \times Y \times Y) \times_{Y\times Y} Y$ over p_{23} and δ_Y, so the same argument gives $p_{23}^*[\Delta_Y] = w_*[X \times Y]$; then $p_{12} \circ w = p_{13} \circ w = \mathrm{id}_{X\times Y}$ turns the projection formula and the pushforward into $[\Delta_Y] \circ \alpha = \alpha$, completing the proof.

Transposition reverses composition, which is what makes the assignment $f \mapsto {}^t[\Gamma_f]$ contravariant.

Proposition 6.4.10: Transposition Reverses Composition

For $\alpha \in \mathrm{Corr}^r(X, Y)$ and $\beta \in \mathrm{Corr}^s(Y, Z)$,
$${}^t(\beta \circ \alpha) = {}^t\alpha \circ {}^t\beta$$

***Proof*:**
Let $\tau\colon X \times Y \times Z \to Z \times Y \times X$ reverse the three factors, let q_{ij} be the projections of $Z \times Y \times X$, and let $\sigma_{XY}, \sigma_{YZ}, \sigma_{XZ}$ be the interchanges of the corresponding double products. Composing maps gives $q_{12} \circ \tau = \sigma_{YZ} \circ p_{23}$, $q_{23} \circ \tau = \sigma_{XY} \circ p_{12}$ and $q_{13} \circ \tau = \sigma_{XZ} \circ p_{13}$.

For an isomorphism h, the flat pullback h^* and the pushforward $(h^{-1})_*$ send the class of a subvariety V to the class of $h^{-1}(V)$ with multiplicity one (definition 1.3.3, definition 1.2.1), so $h^* = (h^{-1})_*$ and hence $h^*h_* = \mathrm{id}$ and $h_*h^* = \mathrm{id}$ by functoriality of proper pushforward. Applying this to σ_{YZ} and σ_{XY} and using functoriality of pullback,
$$\tau^* q_{12}^*({}^t\beta) = p_{23}^*\sigma_{YZ}^*\sigma_{YZ*}\beta = p_{23}^*\beta, \qquad \tau^* q_{23}^*({}^t\alpha) = p_{12}^*\alpha$$
Since τ^* is a ring homomorphism (theorem 5.2.10) and $\tau_*\tau^* = \mathrm{id}$,
$$q_{12}^*({}^t\beta) \cdot q_{23}^*({}^t\alpha) = \tau_*\big(p_{23}^*\beta \cdot p_{12}^*\alpha\big)$$
Pushing forward along q_{13} and using $q_{13} \circ \tau = \sigma_{XZ} \circ p_{13}$ with the functoriality of proper pushforward,
$${}^t\alpha \circ {}^t\beta = q_{13*}\tau_*\big(p_{12}^*\alpha \cdot p_{23}^*\beta\big) = \sigma_{XZ*}\big(\beta \circ \alpha\big) = {}^t(\beta \circ \alpha)$$
as we sought to show.

Composition preserves degree 0, the transposed graphs have degree 0, and proposition 6.4.9 supplies identities, so these correspondences form a category.

Corollary 6.4.11: The Category of Correspondences

There is a category $\mathcal{C}$ whose objects are the smooth projective varieties over $\overline{k}$, with $\mathrm{Hom}_{\mathcal{C}}(X,Y) = \mathrm{Corr}^0(X,Y)$, composition as in definition 6.4.4, and identity morphism $[\Delta_X]$ at X. The assignment sending X to itself and a morphism $f\colon X \to Y$ to ${}^t[\Gamma_f] \in \mathrm{Hom}_{\mathcal{C}}(Y,X)$ is a contravariant functor from smooth projective varieties to $\mathcal{C}$.

***Proof*:**
Composition carries $\mathrm{Corr}^0(X,Y) \times \mathrm{Corr}^0(Y,Z)$ into $\mathrm{Corr}^0(X,Z)$ by proposition 6.4.5, and is associative by theorem 6.4.6. The class $[\Delta_X]$ lies in $A^{d_X}(X \times X) = \mathrm{Corr}^0(X,X)$ and is a two-sided identity by proposition 6.4.9(2), so $\mathcal{C}$ is a category.

For the functor, proposition 6.4.7 places ${}^t[\Gamma_f]$ in $\mathrm{Corr}^0(Y,X) = \mathrm{Hom}_{\mathcal{C}}(Y,X)$. Given $f\colon X \to Y$ and $g\colon Y \to Z$, proposition 6.4.10 and proposition 6.4.9(1) give

$$ {}^t[\Gamma_{g\circ f}] = {}^t\big([\Gamma_g] \circ [\Gamma_f]\big) = {}^t[\Gamma_f] \circ {}^t[\Gamma_g] $$

and $\Gamma_{\mathrm{id}_X} = \Delta_X$ is carried to itself by the interchange of factors, so ${}^t[\Delta_X] = [\Delta_X]$ is the identity of X in $\mathcal{C}$, completing the proof.

Over $\mathbb{C}$ the construction is compatible with the cycle class map, because it is built from flat pullback, the intersection product and proper pushforward, and section 6.2 establishes compatibility with each of the three.

Proposition 6.4.12: Correspondences and the Cycle Class Map

Let X and Y be smooth projective varieties over $\mathbb{C}$ and $\alpha \in \mathrm{Corr}^r(X,Y)$. Define

$$ \mathrm{cl}(\alpha)_*\colon H^{2k}(X;\mathbb{Z}) \to H^{2(k+r)}(Y;\mathbb{Z}), \qquad u \mapsto \mathrm{pr}_{2!}\big(\mathrm{cl}(\alpha) \cup \mathrm{pr}_1^* u\big) $$

where $\mathrm{pr}_{2!}$ is the Gysin homomorphism through which section 6.2 states the compatibility of cl with proper pushforward, obtained from pushforward on homology by [10, Poincaré Duality]. Then for every $\beta \in A^k(X)$,

$$ \mathrm{cl}\big(\alpha_*\beta\big) = \mathrm{cl}(\alpha)_*\mathrm{cl}(\beta) $$

***Proof*:**
The product $X \times Y$ is a smooth projective variety over $\mathbb{C}$, so cl is defined on $A^*(X \times Y)$ and is a ring homomorphism there (section 6.2). The projection pr_1 is flat, being a base change of $Y \to \operatorname{Spec}\mathbb{C}$, and pr_2 is proper, being a base change of the proper morphism $X \to \operatorname{Spec}\mathbb{C}$, so the compatibilities of cl with flat pullback and with proper pushforward established in section 6.2 apply to them. Reading the definition of $\alpha_*\beta$ through those three compatibilities in turn,

$$ \mathrm{cl}\big(\mathrm{pr}_{2*}(\alpha \cdot \mathrm{pr}_1^*\beta)\big) = \mathrm{pr}_{2!}\,\mathrm{cl}\big(\alpha \cdot \mathrm{pr}_1^*\beta\big) = \mathrm{pr}_{2!}\big(\mathrm{cl}(\alpha) \cup \mathrm{cl}(\mathrm{pr}_1^*\beta)\big) = \mathrm{pr}_{2!}\big(\mathrm{cl}(\alpha) \cup \mathrm{pr}_1^*\mathrm{cl}(\beta)\big) $$

and the right-hand side is $\mathrm{cl}(\alpha)_*\mathrm{cl}(\beta)$ by definition, completing the proof.

The degrees on the two sides agree: $\mathrm{cl}(\alpha)$ lies in $H^{2(d_X+r)}(X \times Y; \mathbb{Z})$ and $\mathrm{pr}_{2!}$ lowers cohomological degree by $2d_X$, so $\mathrm{cl}(\alpha)_*$ raises it by $2r$, matching the shift of α_* by r in codimension. One consequence is that the action of a correspondence on classes modulo homological equivalence depends only on its cycle class.

Corollary 6.4.13: Correspondences Preserve Homological Equivalence

Let X and Y be smooth projective over $\mathbb{C}$ and $\alpha \in \mathrm{Corr}^r(X, Y)$. If $\beta \in A^k(X)$ is homologically equivalent to zero, then so is $\alpha_*\beta$. Consequently α_* descends to a homomorphism from $A^k(X)$ modulo homological equivalence to $A^{k+r}(Y)$ modulo homological equivalence.

***Proof*:**
Homological equivalence to zero means $\mathrm{cl}(\beta) = 0$ (section 6.3). By proposition 6.4.12, $\mathrm{cl}(\alpha_*\beta) = \mathrm{pr}_{2!}(\mathrm{cl}(\alpha) \cup \mathrm{pr}_1^* 0) = 0$, so $\alpha_*\beta$ is homologically equivalent to zero. Since α_* is additive (proposition 6.4.2), it therefore carries the subgroup of classes homologically equivalent to zero into the corresponding subgroup on Y and descends to the quotients, as we sought to show.

The proofs of theorem 6.4.6 and proposition 6.4.9 used only proper pushforward, pullback and the intersection product, and an adequate equivalence relation in the sense of section 6.3 is by definition compatible with all three. Replacing $A^*(X \times Y)$ throughout by its quotient modulo such a relation therefore yields a category of correspondences for each relation in the tower of section 6.3, with the same objects, identities and composition. Formally adjoining an image for each idempotent $p \in \mathrm{Corr}^0(X, X)$, that is, each p with $p \circ p = p$, produces the category of pure motives attached to the relation: its objects are pairs consisting of a variety and such an idempotent, and a single variety typically decomposes into several of them.

Exercise 6.4.1

1. Let $\alpha \in \mathrm{Corr}^r(X, Y)$ for smooth projective varieties X and Y. Show that ${}^t\alpha$ lies in $\mathrm{Corr}^{r+d_X-d_Y}(Y, X)$, that ${}^t({}^t\alpha) = \alpha$, and hence that transposition is an isomorphism of groups $\mathrm{Corr}^r(X, Y) \to \mathrm{Corr}^{r+d_X-d_Y}(Y, X)$.

2. By example 6.4.3 a correspondence of degree 0 from $\mathbb{P}^1$ to $\mathbb{P}^1$ is determined by its action. Use this together with proposition 6.4.5 to show that
$$(a'h_1 + b'h_2) \circ (ah_1 + bh_2) = aa'\, h_1 + bb'\, h_2$$
Conclude that $\mathrm{Corr}^0(\mathbb{P}^1, \mathbb{P}^1)$, with composition as multiplication, is isomorphic as a ring to $\mathbb{Z} \times \mathbb{Z}$.

3. Let X and Y be smooth projective, $\xi \in A^a(X)$ and $\eta \in A^b(Y)$, and set $\alpha = \mathrm{pr}_1^*\xi \cdot \mathrm{pr}_2^*\eta$. Show that $\alpha \in \mathrm{Corr}^{a+b-d_X}(X, Y)$ and that, for $\beta \in A^k(X)$,
$$\alpha_*\beta = \Big(\int_X \xi \cdot \beta\Big)\,\eta \quad \text{if } a + k = d_X, \qquad \alpha_*\beta = 0 \quad \text{otherwise}$$
where $\int_X$ is the degree of a zero cycle (definition 1.2.4). (For the vanishing, treat $a + k > d_X$ and $a + k < d_X$ separately.)

4. Let X and Y be smooth projective and let $c\colon X \to Y$ be the constant morphism with image a point y_0. Show that $[\Gamma_c] = \mathrm{pr}_2^*[y_0]$, and compute the action of $[\Gamma_c]$ on $A^k(X)$ for every k.

5. Let X and Y be smooth projective over $\mathbb{C}$ and let $\alpha, \alpha' \in \mathrm{Corr}^r(X, Y)$ be homologically equivalent. Show that $\alpha_*\beta$ and $\alpha'_*\beta$ are homologically equivalent for every $\beta \in A^k(X)$, so that the induced map on classes modulo homological equivalence depends only on the class of α modulo homological equivalence.

6. Let X be a smooth projective variety. Show that composition makes $\mathrm{Corr}^0(X, X)$ an associative ring with identity $[\Delta_X]$, and that $\alpha \mapsto \alpha_*$ is a homomorphism of rings from $\mathrm{Corr}^0(X, X)$ to the ring of additive endomorphisms of $A^*(X)$ preserving the codimension grading.

7. Verify from example 6.4.8 that $[\Delta_{\mathbb{P}^1}] = h_1 + h_2$, and check using the composition formula of exercise 6.4.1 (2) that this class is idempotent. Which other classes in $\mathrm{Corr}^0(\mathbb{P}^1, \mathbb{P}^1)$ are idempotent?

7

Riemann-Roch

Riemann-Roch computes a cohomological invariant from a geometric one. In its oldest form it says that for a divisor D on a curve of genus g the difference $\ell(D) - \ell(K - D)$ is $\deg D + 1 - g$: a count of sections, which is hard, expressed through a degree and a genus, which are not. Every later version keeps that shape. Something defined by cohomology on the left, something defined by intersection numbers on the right, and an equality that lets each be computed from the other.

Grothendieck's formulation makes the shape exact and, in doing so, changes what the theorem is about. It is not a statement about one variety but about a *morphism*: pushing a class forward in K-theory and then taking its Chern character differs from taking the Chern character first and pushing forward in the Chow ring, and the discrepancy is a universal correction, the Todd class of the relative tangent bundle. Set Y to a point and the classical statements fall out.

Section 7.1 builds the two power series the correction is made of. The Chern character converts the multiplicative bookkeeping of K-theory into the additive bookkeeping of the Chow ring, and the Todd class measures its failure to commute with pushforward. Section 7.2 proves the theorem for a proper morphism of smooth varieties by factoring it into a closed immersion and a projection, and states the extension to local complete intersection morphisms. Section 7.3 spends it: Hirzebruch-Riemann-Roch, then the curve and surface statements that [2] proved by hand, recovered here as instances.

7.1 The Chern Character and the Todd Class

Two power series in the Chern roots are what Riemann-Roch is written in, and both are produced by the calculus of proposition 3.2.22: substitute the roots, verify a symmetric-function identity, descend by injectivity. The first is the exponential.

Definition 7.1.1: The Chern Character

Let E be a vector bundle of rank r on an algebraic scheme X. The *Chern character* of E is the operator on $A_*(X)_{\mathbb{Q}}$ obtained from the symmetric power series

$$\operatorname{ch}(a_1, \ldots, a_r) = \sum_{p=1}^{r} e^{a_p} = \sum_{p=1}^{r} \sum_{k \geq 0} \frac{a_p^k}{k!}$$

by proposition 3.2.22. It is written $\operatorname{ch}(E)$, and its component lowering dimension by k is written $\operatorname{ch}_k(E)$, so that $\operatorname{ch}(E) = \sum_{k \geq 0} \operatorname{ch}_k(E)$ with the sum finite by lemma 3.2.20.

The definition is stated through the roots for brevity, but no roots are used: what proposition 3.2.22 returns is a specific polynomial in $c_1(E), \ldots, c_r(E)$ with rational coefficients, and the next proposition writes it out. In every positive degree that polynomial is the same for all ranks, once the convention $c_i(E) = 0$ for $i > \operatorname{rank} E$ from section 3.2 is in force; the rank enters only in degree zero, where the constant term is the rank itself. That is what allows bundles of different ranks to be compared in a single identity: the positive-degree components match term by term and the ranks do their own bookkeeping in degree zero.

Proposition 7.1.2: Components of the Chern Character

Let E be a vector bundle of rank r on X, and write $c_i = c_i(E)$. Then $\operatorname{ch}_k(E) = \frac{1}{k!}\, p_k$, where $p_0 = r$ and the operators p_k are determined by Newton's recursion

$$p_k - c_1 p_{k-1} + c_2 p_{k-2} - \cdots + (-1)^{k-1} c_{k-1} p_1 + (-1)^k k\, c_k = 0 \qquad (k \geq 1) \tag{7.1}$$

In particular

$$\operatorname{ch}(E) = r + c_1 + \tfrac{1}{2}\big(c_1^2 - 2c_2\big) + \tfrac{1}{6}\big(c_1^3 - 3c_1c_2 + 3c_3\big) + \cdots$$

so $\operatorname{ch}_0(E) = \operatorname{rank} E$ and $\operatorname{ch}_1(E) = c_1(E)$. For a line bundle L, $\operatorname{ch}(L) = e^{c_1(L)}$.

***Proof*:**
Work first with formal symbols. In $\mathbb{Q}[a_1, \ldots, a_r][[t]]$ put $E(t) = \prod_{p=1}^{r}(1 + a_p t) = \sum_{j \geq 0} e_j t^j$, where $e_j = e_j(a)$ and $e_j = 0$ for $j > r$. Each factor $1 + a_p t$ has constant term 1, hence is invertible, and the product rule gives

$$E'(t) = \sum_{p=1}^{r} a_p \prod_{q \neq p} (1 + a_q t) = E(t) \sum_{p=1}^{r} \frac{a_p}{1 + a_p t}$$

Expanding $\frac{a_p}{1+a_p t} = \sum_{k \geq 1} (-1)^{k-1} a_p^k t^{k-1}$ and summing over p turns this into

$$E'(t) = E(t) \sum_{k \geq 1} (-1)^{k-1} p_k t^{k-1}, \qquad p_k := \sum_{p=1}^{r} a_p^k$$

Comparing coefficients of t^{k-1} on both sides gives $k e_k = \sum_{m=1}^{k} (-1)^{m-1} e_{k-m}\, p_m$, which on multiplying by $(-1)^{k-1}$ and transposing is exactly (7.1) with e_i in place of c_i. Solving the recursion in low degrees, $p_1 = e_1$, then $p_2 = e_1 p_1 - 2e_2 = e_1^2 - 2e_2$, then $p_3 = e_1 p_2 - e_2 p_1 + 3e_3 = e_1^3 - 3e_1 e_2 + 3e_3$.

The power sums determine the Chern character, since interchanging the two finite-in-each-degree summations gives

$$\sum_{p=1}^{r} e^{a_p} = \sum_{k\geq 0} \frac{1}{k!} \sum_{p=1}^{r} a_p^k = r + \sum_{k\geq 1} \frac{p_k}{k!}$$

Each p_k is symmetric, and (7.1) exhibits it as an isobaric polynomial of weight k in $e_1, \ldots, e_k$; by the uniqueness in [6, Symmetric Power Series in the Chern Roots] that polynomial is the one proposition 3.2.22 substitutes into. Substituting $c_i = c_i(E)$ therefore gives $\mathrm{ch}_k(E) = \frac{1}{k!}p_k$ with p_k satisfying (7.1), and the displayed expansion follows from the three computed cases. Finally $\mathrm{ch}_0(E) = p_0 = r$ and $\mathrm{ch}_1(E) = p_1 = c_1$. For $r = 1$ the series is e^{a_1} and $e_1 = a_1$, so $\widetilde{\mathrm{ch}}(y_1) = e^{y_1}$ and $\mathrm{ch}(L) = e^{c_1(L)} = \sum_{k\geq 0} c_1(L)^k/k!$, a finite sum by lemma 3.2.20, as we sought to show.

The denominators do not cancel. Already the factor $\frac{1}{2}$ in $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ survives on projective space, where $c_1^2 - 2c_2$ is an odd multiple of the point class for suitable bundles, so ch_2 is not the image of any operator on $A_*(X)$ with integer coefficients. The next example exhibits the smallest such case.

Example 7.1.3: The Chern Character of a Line Bundle on $\mathbb{P}^n$

Let $h = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, so that $h^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for $0 \leq k \leq n$ by example 2.4.7, and $h^{n+1} = 0$ as an operator by lemma 3.2.20. For the line bundle $\mathcal{O}_{\mathbb{P}^n}(d)$ one has $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,h$ by proposition 2.4.4(3), so

$$\mathrm{ch}\big(\mathcal{O}_{\mathbb{P}^n}(d)\big) = e^{dh} = 1 + dh + \frac{d^2}{2}h^2 + \cdots + \frac{d^n}{n!}h^n$$

the series terminating because $h^{n+1} = 0$. Capping with the fundamental class on $\mathbb{P}^2$,

$$\mathrm{ch}\big(\mathcal{O}_{\mathbb{P}^2}(d)\big) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + d\,[\mathbb{P}^1] + \frac{d^2}{2}\,[\mathrm{pt}]$$

For odd d the last coefficient is not an integer, so this class lies in $A_0(\mathbb{P}^2)_{\mathbb{Q}}$ and in no subgroup coming from $A_0(\mathbb{P}^2)$. Rational coefficients are not a convenience here; without them the Chern character of $\mathcal{O}_{\mathbb{P}^2}(1)$ does not exist.

The point of the exponential is the identity $e^{a+b} = e^a e^b$, which converts both operations on bundles into operations on operators. Both of the proofs below compare two bundles at once, so they need a single splitting map that works for several bundles simultaneously.

Theorem 7.1.4: Additivity and Multiplicativity of the Chern Character

Let X be an algebraic scheme and let E, F be vector bundles on X.

1. If $0 \to E' \to E \to E'' \to 0$ is an exact sequence of vector bundles on X, then

$$\mathrm{ch}(E) = \mathrm{ch}(E') + \mathrm{ch}(E'')$$

as operators on $A_*(X)_{\mathbb{Q}}$. In particular $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$.

2. $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$, the product on the right being composition of operators.

***Proof*:**
Write $r = \operatorname{rank} E$, $r' = \operatorname{rank} E'$, $r'' = \operatorname{rank} E''$ and $s = \operatorname{rank} F$; in part (1) the ranks satisfy $r = r' + r''$.

1. *Additivity.* By theorem 3.2.14 choose a flat proper $f\colon X' \to X$ with f^* injective that is a splitting map for both E' and E'', with Chern roots $a_1, \dots, a_{r'}$ and $b_1, \dots, b_{r''}$ respectively.

   The same f splits E. Pulling the exact sequence back gives $0 \to f^*E' \xrightarrow{\iota} f^*E \xrightarrow{\pi} f^*E'' \to 0$, again exact since pullback of vector bundles is exact. Let $0 = E'_0 \subset \cdots \subset E'_{r'} = f^*E'$ and $0 = E''_0 \subset \cdots \subset E''_{r''} = f^*E''$ be the two filtrations, with line-bundle quotients L_p and M_q, and define

   $$H_p = \iota(E'_p) \quad (0 \le p \le r'), \qquad H_{r'+q} = \pi^{-1}\big(E''_q\big) \quad (1 \le q \le r'')$$

   This is a filtration of f^*E by subbundles: the first stretch is the filtration of the subbundle f^*E', the second consists of preimages of subbundles under a surjection of bundles. Its successive quotients are $L_1, \dots, L_{r'}$ followed by $H_{r'+q}/H_{r'+q-1} \cong E''_q/E''_{q-1} = M_q$, the isomorphism holding because π carries $H_{r'+q}$ onto E''_q with kernel $f^*E' = H_{r'}$ contained in $H_{r'+q-1}$. So f is a splitting map for E with Chern roots $a_1, \dots, a_{r'}, b_1, \dots, b_{r''}$.

   Now apply proposition 3.2.22(4) three times. For every $\alpha \in A_*(X)_{\mathbb{Q}}$,

   $$f^*\big(\operatorname{ch}(E) \cap \alpha\big) = \Big(\sum_{p=1}^{r'} e^{a_p} + \sum_{q=1}^{r''} e^{b_q}\Big) \cap f^*\alpha = f^*\big(\operatorname{ch}(E') \cap \alpha\big) + f^*\big(\operatorname{ch}(E'') \cap \alpha\big)$$

   the first equality reading the roots of E off the spliced filtration and the second applying the proposition to E' and to E'' separately. Since f^* is injective on $A_*(X)_{\mathbb{Q}}$, this gives $\operatorname{ch}(E) \cap \alpha = \operatorname{ch}(E') \cap \alpha + \operatorname{ch}(E'') \cap \alpha$ for every α, which is the asserted identity of operators. The direct sum is the case of a split exact sequence.

2. *Multiplicativity.* First a statement about filtrations: if a bundle A has a filtration with line-bundle quotients $N_1, \dots, N_a$ and B one with quotients $P_1, \dots, P_b$, then $A \otimes B$ has a filtration with quotients the $N_i \otimes P_j$. Induct on a. For $a = 1$ the bundle $A = N_1$ is a line bundle, and tensoring the filtration B_j of B by it gives a filtration of $N_1 \otimes B$ with quotients $N_1 \otimes P_j$, since tensoring by a line bundle is exact and carries subbundles to subbundles. For $a > 1$, tensoring $0 \to A_{a-1} \to A \to N_a \to 0$ with the locally free B keeps it exact; the subbundle $A_{a-1} \otimes B$ has the required filtration by induction, and taking preimages of the filtration of $N_a \otimes B$ under the surjection $A \otimes B \to N_a \otimes B$ extends it, exactly as in part (1).

   By theorem 3.2.14 choose a flat proper $g\colon X'' \to X$ with g^* injective that is a splitting map for both E and F, with Chern roots $w_1, \dots, w_r$ and $u_1, \dots, u_s$. Applying the previous paragraph to $g^*E \otimes g^*F = g^*(E \otimes F)$ produces a filtration whose quotients are the rs tensor products of line bundles from the two filtrations, and the first Chern class of such a tensor product is $w_p + u_j$ by proposition 2.4.4(3). So g is a splitting map for $E \otimes F$ with Chern roots the operators $w_p + u_j$. By proposition 3.2.22(4) and the formal identity $e^{w+u} = e^w e^u$,

   $$g^*\big(\operatorname{ch}(E \otimes F) \cap \alpha\big) = \Big(\sum_{p,j} e^{w_p + u_j}\Big) \cap g^*\alpha = \Big(\sum_p e^{w_p}\Big)\Big(\sum_j e^{u_j}\Big) \cap g^*\alpha$$

and the right-hand side is $g^*\big(\mathrm{ch}(E)\,\mathrm{ch}(F)\cap\alpha\big)$, again by proposition 3.2.22(4) applied twice, the two operators commuting by proposition 3.2.22(1). Injectivity of g^* gives $\mathrm{ch}(E\otimes F)=\mathrm{ch}(E)\,\mathrm{ch}(F)$, completing the proof.

The theorem says that ch converts the two ways of building bundles into the two operations of a commutative ring: extensions become sums, tensor products become composites. That is the structure of the Grothendieck group of vector bundles, in which an exact sequence is declared to be a sum and the tensor product is the multiplication, and the Chern character is the map out of it that the Riemann-Roch theorem later in this book computes with. A reader who has met Chern classes topologically [10] will recognize the axioms driving the argument, the Whitney formula and naturality being the same there; the topological Chern character satisfies the same two identities. The construction here is independent of that one, valid over any algebraically closed field and on singular schemes, and landing in rational Chow groups rather than in cohomology.

Additivity comes at a price. The total Chern class is multiplicative, so ch cannot also be, and the class that Riemann-Roch attaches to the tangent bundle, correcting the Chern character of a bundle to its Euler characteristic, has to be multiplicative. The Todd class is that multiplicative partner. The specific power series below is not derivable from anything in this chapter; it is the one that makes Riemann-Roch true, and the first evidence for it is the computation at the end of this section, where the top Todd component of the tangent bundle of $\mathbb{P}^n$ integrates to $1=\chi(\mathcal{O}_{\mathbb{P}^n})$.

Definition 7.1.5: The Todd Class

Write $Q(t)=\dfrac{t}{1-e^{-t}}\in\mathbb{Q}[[t]]$, which is a power series because $1-e^{-t}=t\big(1-\frac{t}{2}+\frac{t^2}{6}-\cdots\big)$ and the second factor has constant term 1, hence is invertible. For a vector bundle E of rank r on an algebraic scheme X, the *Todd class* of E is the operator on $A_*(X)_{\mathbb{Q}}$ obtained from the symmetric power series

$$\mathrm{td}(a_1,\dots,a_r)=\prod_{p=1}^{r}\frac{a_p}{1-e^{-a_p}}$$

by proposition 3.2.22, written $\mathrm{td}(E)$, with $\mathrm{td}_k(E)$ its component lowering dimension by k. When X is smooth, $\mathrm{td}(X)$ abbreviates $\mathrm{td}(T_X)$ for the tangent bundle T_X.

Proposition 7.1.6: Components of the Todd Class

Let E be a vector bundle on X and write $c_i=c_i(E)$. Then

$$\mathrm{td}(E)=1+\tfrac{1}{2}c_1+\tfrac{1}{12}\big(c_1^2+c_2\big)+\tfrac{1}{24}c_1c_2+\cdots$$

In particular $\mathrm{td}_0(E)=1$, so $\mathrm{td}(E)$ is an invertible operator on $A_*(X)_{\mathbb{Q}}$, and for a line bundle L,

$$\mathrm{td}(L)=Q\big(c_1(L)\big)=1+\tfrac{1}{2}c_1(L)+\tfrac{1}{12}c_1(L)^2+\cdots$$

***Proof*:**
Expand Q first. From $1-e^{-t} = t\big(1-\frac{t}{2}+\frac{t^2}{6}-\frac{t^3}{24}+\cdots\big)$ one gets $Q(t) = (1-u)^{-1}$ with $u = \frac{t}{2}-\frac{t^2}{6}+\frac{t^3}{24}-\cdots$, so $Q(t) = 1+u+u^2+u^3+\cdots$. Collecting terms,

$$Q(t) = 1+\tfrac{t}{2}+\Big(\tfrac{1}{4}-\tfrac{1}{6}\Big)t^2+\Big(\tfrac{1}{8}-\tfrac{1}{6}+\tfrac{1}{24}\Big)t^3+\cdots = 1+\tfrac{t}{2}+\tfrac{t^2}{12}+0\cdot t^3+\cdots$$

Now write $q_1 = \frac{1}{2}$, $q_2 = \frac{1}{12}$, $q_3 = 0$ and expand $\prod_{p=1}^{r}\big(1+q_1a_p+q_2a_p^2+q_3a_p^3+\cdots\big)$ in degrees at most three, using the power sums $p_k = \sum_p a_p^k$ of proposition 7.1.2. In degree one the only contribution is $q_1p_1 = \frac{1}{2}e_1$. In degree two the contributions are q_2p_2 from a single factor and $q_1^2e_2$ from two distinct factors, giving

$$\tfrac{1}{12}\big(e_1^2-2e_2\big)+\tfrac{1}{4}e_2 = \tfrac{1}{12}\big(e_1^2+e_2\big)$$

In degree three the contributions are $q_3p_3 = 0$, then $q_1q_2\sum_{p\neq q}a_p^2a_q$ from two distinct factors, then $q_1^3e_3$ from three distinct factors. Since $\sum_{p\neq q}a_p^2a_q = p_1p_2-p_3 = e_1e_2-3e_3$, the degree-three term is

$$\tfrac{1}{24}\big(e_1e_2-3e_3\big)+\tfrac{1}{8}e_3 = \tfrac{1}{24}e_1e_2$$

Substituting $e_i = c_i(E)$ through proposition 3.2.22 gives the stated expansion. Invertibility follows from $\mathrm{td}(E) = \mathrm{id}+N$ with $N = \sum_{k\geq 1}\mathrm{td}_k(E)$ a sum of dimension-lowering operators, so $N^{\dim X+1} = 0$ by lemma 3.2.20 and $\mathrm{td}(E)^{-1} = \sum_{k\geq 0}(-N)^k$ is a finite sum. For $r=1$ the series is $Q(a_1)$ with $e_1 = a_1$, so $\mathrm{td}(L) = Q(c_1(L))$, as we sought to show.

The first component $\frac{1}{2}c_1$ already forces rational coefficients, exactly as ch_2 did, and the degree-three coefficient of Q vanishes because $Q(t)-\frac{t}{2}$ is even, so Q has no odd-degree terms beyond the first. The next example computes the class in the smallest case, where a single Chern class controls everything.

Example 7.1.7: The Todd Class of a Line Bundle on $\mathbb{P}^2$
With $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ as in example 7.1.3 and $h^3 = 0$, proposition 7.1.6 gives

$$\mathrm{td}\big(\mathcal{O}_{\mathbb{P}^2}(d)\big) = 1+\frac{d}{2}h+\frac{d^2}{12}h^2$$

so that, capping with the fundamental class, $\mathrm{td}(\mathcal{O}_{\mathbb{P}^2}(d))\cap[\mathbb{P}^2] = [\mathbb{P}^2]+\frac{d}{2}[\mathbb{P}^1]+\frac{d^2}{12}[\mathrm{pt}]$. For $d=1$ the zero-dimensional component is $\frac{1}{12}[\mathrm{pt}]$, of degree $\frac{1}{12}$ under $\int_{\mathbb{P}^2}$. Individual Todd components carry denominators that no bundle removes; it is only the full expression appearing in Riemann-Roch, a Todd class multiplied by a Chern character, that is forced to have integral degree.

Proposition 7.1.8: Multiplicativity of the Todd Class

Let $0\to E'\to E\to E''\to 0$ be an exact sequence of vector bundles on an algebraic scheme X. Then

$$\mathrm{td}(E) = \mathrm{td}(E')\,\mathrm{td}(E'')$$

as operators on $A_*(X)_{\mathbb{Q}}$. In particular $\mathrm{td}(E\oplus F) = \mathrm{td}(E)\,\mathrm{td}(F)$, and $\mathrm{td}(E\oplus\mathcal{O}_X^{\oplus m}) = \mathrm{td}(E)$.

***Proof*:**
Choose, by theorem 3.2.14, a flat proper $f\colon X' \to X$ with f^* injective splitting both E' and E'', with roots $a_1, \dots, a_{r'}$ and $b_1, \dots, b_{r''}$. Part (1) of the proof of theorem 7.1.4 shows that f then splits E, with roots the union of the two lists. Applying proposition 3.2.22(4) three times, to E, to E' and to E'', and using the factorization of the defining product over the union of the two lists,

$$f^*\big(\mathrm{td}(E) \cap \alpha\big) = \Big(\prod_p Q(a_p) \prod_q Q(b_q)\Big) \cap f^*\alpha = f^*\big(\mathrm{td}(E')\,\mathrm{td}(E'') \cap \alpha\big)$$

for every $\alpha \in A_*(X)_{\mathbb{Q}}$, the two operators on the right commuting by proposition 3.2.22(1). Injectivity of f^* gives $\mathrm{td}(E) = \mathrm{td}(E')\mathrm{td}(E'')$. The direct sum is the split case, and a trivial bundle contributes $Q(0)^m = 1$ because $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, completing the proof.

Additivity of ch and multiplicativity of td are what make both computable: any bundle presented by an exact sequence with known ends has both classes determined. The tangent bundle of projective space is presented in exactly that way, and the computation below is the one Riemann-Roch is checked against.

Example 7.1.9: The Tangent Bundle of $\mathbb{P}^n$
Example 3.2.17 has already imported the Euler sequence from [2, The Euler Sequence], dualized it to

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

and read off $c(T_{\mathbb{P}^n}) = (\mathrm{id} + H)^{n+1}$. What is new here is the Chern character and Todd class the same sequence yields. Here $\mathbb{P}^n$ is projective n-space with its standard twisting sheaf, so the convention of box 0.0.2 does not intervene and no further dualization is needed. Write $h = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$.

The Chern character. By theorem 7.1.4(1), $\mathrm{ch}(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}) = \mathrm{ch}(\mathcal{O}_{\mathbb{P}^n}) + \mathrm{ch}(T_{\mathbb{P}^n})$, and the left side is $(n+1)e^h$ while $\mathrm{ch}(\mathcal{O}_{\mathbb{P}^n}) = 1$. Hence

$$\mathrm{ch}(T_{\mathbb{P}^n}) = (n+1)e^h - 1$$

The Todd class. By proposition 7.1.8 applied to the same sequence, $\mathrm{td}(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}) = \mathrm{td}(\mathcal{O}_{\mathbb{P}^n})\,\mathrm{td}(T_{\mathbb{P}^n}) = \mathrm{td}(T_{\mathbb{P}^n})$, so

$$\mathrm{td}(T_{\mathbb{P}^n}) = Q(h)^{n+1}$$

with Q the series of definition 7.1.5. Both classes are computed from the single operator h, whose powers are known: $h^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ and $h^{n+1} = 0$.

The case $n = 1$. Here $h^2 = 0$, so $\mathrm{ch}(T_{\mathbb{P}^1}) = 2(1+h) - 1 = 1 + 2h$ and $\mathrm{td}(T_{\mathbb{P}^1}) = (1 + \frac{h}{2})^2 = 1 + h$. The first agrees with the identification $T_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(2)$, whose Chern character is $e^{2h} = 1 + 2h$ by example 7.1.3. The dimension-lowering component of $\mathrm{td}(T_{\mathbb{P}^1})$ caps to $h \cap [\mathbb{P}^1] = [\mathrm{pt}]$, of degree $\int_{\mathbb{P}^1}[\mathrm{pt}] = 1$.

The case $n = 2$. Here $h^3 = 0$ and

$$\mathrm{ch}(T_{\mathbb{P}^2}) = 3\Big(1 + h + \tfrac{h^2}{2}\Big) - 1 = 2 + 3h + \tfrac{3}{2}h^2$$

which cross-checks against proposition 7.1.2: the Whitney formula applied to the Euler sequence gives $c(T_{\mathbb{P}^2}) = (1+h)^3 = 1 + 3h + 3h^2$, so $c_1 = 3h$ and $c_2 = 3h^2$, and then $\mathrm{ch}_0 = 2 = \mathrm{rank} T_{\mathbb{P}^2}$,

$\mathrm{ch}_1 = c_1 = 3h$, and $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2) = \frac{1}{2}(9h^2 - 6h^2) = \frac{3}{2}h^2$. For the Todd class,

$$\mathrm{td}(T_{\mathbb{P}^2}) = \left(1 + \tfrac{h}{2} + \tfrac{h^2}{12}\right)^3 = 1 + \tfrac{3}{2}h + \left(3 \cdot \tfrac{1}{12} + 3 \cdot \tfrac{1}{4}\right)h^2 = 1 + \tfrac{3}{2}h + h^2$$

agreeing with $\mathrm{td}_2 = \frac{1}{12}(c_1^2 + c_2) = \frac{1}{12}(9+3)h^2 = h^2$ from proposition 7.1.6. Capping with the fundamental class,

$$\mathrm{ch}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2] = 2[\mathbb{P}^2] + 3[\mathbb{P}^1] + \tfrac{3}{2}[\mathrm{pt}], \qquad \mathrm{td}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + \tfrac{3}{2}[\mathbb{P}^1] + [\mathrm{pt}]$$

in $A_*(\mathbb{P}^2)_{\mathbb{Q}}$.

The check. In both cases the zero-dimensional component of $\mathrm{td}(T_{\mathbb{P}^n}) \cap [\mathbb{P}^n]$ has degree 1 under $\int_{\mathbb{P}^n}$, and $\chi(\mathcal{O}_{\mathbb{P}^n}) = 1$ by [2, Twisted Sheaf Cohomology]. The agreement of these two numbers is the simplest instance of the Riemann-Roch theorem proved later in this book, which computes, for a vector bundle E on a smooth projective X, the integer $\chi(E) = \int_X \mathrm{ch}(E)\,\mathrm{td}(X) \cap [X]$; and it is the reason $t/(1-e^{-t})$, rather than some other multiplicative series, is the one attached to the tangent bundle.

7.1.10 Remark: Operators Now, Ring Elements Later Both classes have been defined as operators on $A_*(X)_{\mathbb{Q}}$, and every identity above is an identity of operators, composition serving as multiplication. This is the only option available so far, since $A_*(X)$ carries no product; chapter 4 and the chapter after it supply one for smooth X, at which point $A^*(X)_{\mathbb{Q}}$ becomes a graded ring, capping with the fundamental class turns each of these operators into a class, and $\mathrm{ch}(E)$ and $\mathrm{td}(E)$ are elements of that ring. No statement in this section changes when that happens: an identity of operators on $A_*(X)_{\mathbb{Q}}$ specializes to an identity of classes by applying it to $[X]$.

So far ch and td are operators attached to a *bundle*. Riemann-Roch needs them attached to a class in K-theory, because the pushforward it compares against lives there and not on bundles: a proper morphism sends a coherent sheaf to an alternating sum of higher direct images, which is a K-theory class and nothing smaller. The passage costs nothing, and the reason is theorem 7.1.4.

Proposition 7.1.11: The Chern Character on K-Theory

Let X be an algebraic scheme. The assignment $E \mapsto \mathrm{ch}(E)$ descends to a homomorphism

$$\mathrm{ch}\colon K^\circ(X) \longrightarrow \mathrm{End}\left(A_*(X)_{\mathbb{Q}}\right)$$

from the Grothendieck group of vector bundles ([3, K_0 of a Scheme]), and it carries the product of $K^\circ(X)$ to composition of operators. If X is a smooth variety, then under proposition 5.3.13 the image lies in $A^*(X)_{\mathbb{Q}}$ and

$$\mathrm{ch}\colon K^\circ(X) \longrightarrow A^*(X)_{\mathbb{Q}}$$

is a homomorphism of rings.

***Proof*:**
$K^\circ(X)$ is the free abelian group on isomorphism classes of vector bundles modulo the relations $[E] = [E'] + [E'']$ for every short exact sequence $0 \to E' \to E \to E'' \to 0$ ([3, K_0 of a Scheme]). Theorem 7.1.4(1) says exactly that ch respects those relations, so it factors through the quotient; the resulting map is a homomorphism of groups because $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$. Multiplicativity is theorem 7.1.4(2), and the product on $K^\circ(X)$ is induced by tensor product ([3, Ring and Module Structures]), so ch is multiplicative. For X smooth, proposition 5.3.13 identifies the operator $\mathrm{ch}(\alpha)$ with multiplication by $\mathrm{ch}(\alpha) \cap [X]$, and both structures match, as we sought to show.

On a smooth variety the distinction between bundles and coherent sheaves also disappears, which is what lets the right-hand side of Riemann-Roch be written at all.

Corollary 7.1.12: The Chern Character of a Coherent Sheaf

Let X be a smooth separated variety. Then $K^\circ(X) \to K_\circ(X)$ is an isomorphism ([3, The Duality Map]), so every coherent sheaf $\mathscr{F}$ has a well-defined Chern character

$$\mathrm{ch}(\mathscr{F}) \;=\; \sum_i (-1)^i \mathrm{ch}(E_i) \;\in\; A^*(X)_{\mathbb{Q}}$$

computed from any finite locally free resolution $E_\bullet \to \mathscr{F}$, and independent of it.

***Proof*:**
A smooth variety is regular, so the duality map is an isomorphism with the stated inverse, and ch of the resulting class is computed by proposition 7.1.11. Independence of the resolution is independence of the class in $K^\circ(X)$, which the cited isomorphism supplies.

Exercise 7.1.1

1. Let $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ and let $L = \mathcal{O}_{\mathbb{P}^2}(d)$. Compute $\mathrm{ch}(L) \cap [\mathbb{P}^2]$ and $\mathrm{td}(L) \cap [\mathbb{P}^2]$ in $A_*(\mathbb{P}^2)_{\mathbb{Q}}$, and compute the degree $\int_{\mathbb{P}^2}$ of the zero-dimensional component of $\mathrm{ch}(L)\,\mathrm{td}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2]$. Determine for which d this last number is an integer.

2. Show that $\mathrm{ch}_k(E^\vee) = (-1)^k \mathrm{ch}_k(E)$ for every vector bundle E and every k. (Dualize a filtration by subbundles and use proposition 2.4.4(3).) Deduce that $\mathrm{ch}(E \otimes E^\vee)$ has $\mathrm{ch}_1 = 0$ for every E.

3. Let E have rank r and let $\det E = \Lambda^r E$. Show that $c_1(\det E) = c_1(E)$ and hence $\mathrm{ch}(\det E) = e^{c_1(E)}$. (Use that the top exterior power of a filtered bundle is the tensor product of the line-bundle quotients.)

4. Compute $\mathrm{ch}(T_{\mathbb{P}^3})$ and $\mathrm{td}(T_{\mathbb{P}^3})$ as polynomials in $h = c_1(\mathcal{O}_{\mathbb{P}^3}(1))$, and verify that $\int_{\mathbb{P}^3}$ of the zero-dimensional component of $\mathrm{td}(T_{\mathbb{P}^3}) \cap [\mathbb{P}^3]$ equals 1. Check the degree-two and degree-three Todd components against proposition 7.1.6 using $c(T_{\mathbb{P}^3}) = (1 + h)^4$.

5. Let S be a smooth projective surface and E a vector bundle of rank r on S. Write out the zero-dimensional component of $\mathrm{ch}(E)\,\mathrm{td}(T_S) \cap [S]$ in terms of $c_1(E), c_2(E), c_1(T_S), c_2(T_S)$, and specialize to $E = \mathcal{O}_S$ and to E a line bundle.

6. Show that $\mathrm{ch}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{ch}(E) + m$ and $\mathrm{td}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{td}(E)$, and explain why no bundle of positive rank can have $\mathrm{ch}(E) = 0$.

7.2 Grothendieck-Riemann-Roch

This section defines the Todd class of the relative tangent bundle of a morphism, proves Riemann-Roch for a proper morphism of smooth quasi-projective varieties, and states the extension to local complete intersection morphisms.

Two conventions are in force throughout. First, every variety appearing is smooth and quasi-projective over $\overline{k}$, hence regular and separated of finite Krull dimension. So $K^\circ = K_\circ$ on it by corollary 7.1.12, and the pushforward $f_*[\mathscr{F}] = \sum_i (-1)^i [R^i f_* \mathscr{F}]$ of [3, Proper Pushforward] transports along that identification to a homomorphism $f_* \colon K^\circ(X) \to K^\circ(Y)$ for every proper $f \colon X \to Y$. Second, since the varieties are smooth, proposition 5.3.13 and box 7.1.10 identify each of the operators $\mathrm{ch}(\alpha)$ and $\mathrm{td}(E)$ with the class it produces from the fundamental class, and all products below are taken in the graded ring $A^*(X)_\mathbb{Q}$ of theorem 5.2.7.

Definition 7.2.1: The Todd Class of a Morphism

Let $f \colon X \to Y$ be a morphism of smooth varieties, with tangent bundles T_X and T_Y ([2, Tangent Sheaf]). The *relative tangent class* of f is

$$T_f \;=\; [T_X] - [f^* T_Y] \;\in\; K^\circ(X)$$

and the *Todd class of f* is

$$\mathrm{td}(T_f) \;=\; \mathrm{td}(T_X)\,\mathrm{td}(f^* T_Y)^{-1} \;\in\; A^*(X)_\mathbb{Q}$$

The second formula is forced by the first once td is known to be defined on K-theory. The group $K^\circ(X)$ is the free abelian group on isomorphism classes of vector bundles modulo the relation $[E] = [E'] + [E'']$ for each exact sequence $0 \to E' \to E \to E'' \to 0$ ([3, K_0 of a Scheme]). The assignment $E \mapsto \mathrm{td}(E)$ takes values in the units of $A^*(X)_\mathbb{Q}$, since $\mathrm{td}_0(E) = 1$ by proposition 7.1.6, and converts those relations into products by proposition 7.1.8. So it extends uniquely to a homomorphism from the additive group $K^\circ(X)$ to the multiplicative group of units of $A^*(X)_\mathbb{Q}$, and $\mathrm{td}(T_f)$ is the value of that homomorphism at T_f. In particular $\mathrm{td}(T_f)$ depends only on the class T_f, not on any presentation of it.

Three cases fix what the definition computes. If $Y = \operatorname{Spec} \overline{k}$ then $T_Y = 0$ and $\mathrm{td}(T_f) = \mathrm{td}(T_X)$, so the relative class is the absolute one. If $i \colon X \to Y$ is a closed immersion of smooth varieties with normal bundle N, then dualizing the conormal sequence of [2, Nonsingular Closed Subschemes and Differentials] gives the exact sequence $0 \to T_X \to i^* T_Y \to N \to 0$, so $\mathrm{td}(i^* T_Y) = \mathrm{td}(T_X)\mathrm{td}(N)$ by proposition 7.1.8 and

$$T_i \;=\; -[N], \qquad \mathrm{td}(T_i) \;=\; \mathrm{td}(N)^{-1}$$

Finally let $p \colon Y \times \mathbb{P}^m \to Y$ be the projection, with ρ the projection to $\mathbb{P}^m$. Over an affine open $\operatorname{Spec} R \subseteq Y$ and an affine open $\operatorname{Spec} S \subseteq \mathbb{P}^m$, a $\overline{k}$-derivation of $R \otimes_{\overline{k}} S$ is determined by, and can be prescribed arbitrarily on, the two factors separately, so $\Omega_{R \otimes S} \cong (\Omega_R \otimes S) \oplus (R \otimes \Omega_S)$. These isomorphisms are compatible on overlaps and glue, giving $T_{Y \times \mathbb{P}^m} = p^* T_Y \oplus \rho^* T_{\mathbb{P}^m}$. Hence $T_p = [\rho^* T_{\mathbb{P}^m}]$, and writing $\zeta = c_1(\rho^* \mathcal{O}_{\mathbb{P}^m}(1))$ and $Q(t) = t/(1 - e^{-t})$ as in definition 7.1.5, example 7.1.9

together with the flat naturality of proposition 3.2.22(2) gives

$$\operatorname{td}(T_p) \;=\; Q(\zeta)^{m+1}$$

Proper pushforward commutes with any operator built as a polynomial in the Chern classes of bundles on the target and pulled back to the source. This is used repeatedly below, so it is isolated first. Let $g\colon X' \to X$ be a proper morphism, let $G_1, \dots, G_t$ be vector bundles on X, and let Θ be an operator on $A_*(X)_{\mathbb{Q}}$ whose component in each degree is a fixed polynomial with rational coefficients in the Chern classes $c_l(G_r)$. Write $g^*\Theta$ for the operator on $A_*(X')_{\mathbb{Q}}$ given by the same polynomials in the $c_l(g^*G_r)$. Then

$$g_*\big((g^*\Theta) \cap \gamma\big) \;=\; \Theta \cap g_*\gamma \qquad \text{for all } \gamma \in A_*(X')_{\mathbb{Q}} \tag{7.2}$$

Each monomial in the $c_l(g^*G_r)$ is a composite of Chern class operators of pullback bundles, and proposition 3.2.10(2) moves them across g_* one factor at a time. Summing over the monomials of a fixed degree and then over degrees, both sums being finite on a given class by lemma 3.2.20(1), gives (7.2). Every operator used below is of this kind: proposition 3.2.22(1) exhibits $\operatorname{ch}(G)$ and $\operatorname{td}(G)$ in each degree as such a polynomial, the same applies to $\operatorname{td}(G)^{-1}$ because the reciprocal series $\prod_p (1 - e^{-a_p})/a_p$ is again symmetric with constant term 1, and ch of a difference $[G_1] - [G_2]$ is the corresponding difference.

What Riemann-Roch asserts is that a single correction of this kind repairs the failure of ch to commute with pushforward, and that the correction is the Todd class of the tangent bundle at each end.

Theorem 7.2.2: Grothendieck-Riemann-Roch

Let $f\colon X \to Y$ be a proper morphism of smooth quasi-projective varieties over $\overline{k}$. Then for every $\alpha \in K^\circ(X)$,

$$\operatorname{ch}\big(f_*\alpha\big)\operatorname{td}(T_Y) \;=\; f_*\big(\operatorname{ch}(\alpha)\operatorname{td}(T_X)\big)$$

in $A^*(Y)_{\mathbb{Q}}$, the pushforward on the left being that of [3, Proper Pushforward] and the one on the right the proper pushforward of cycle classes of definition 1.2.1.

Equivalently, writing $\tau_X(\alpha) = \operatorname{ch}(\alpha)\operatorname{td}(T_X) \cap [X]$ for the *Riemann-Roch map* of a smooth variety, the square

$$\begin{tikzcd} K^\circ(X) \arrow[r, "\tau_X"] \arrow[d, "f_*"'] & A_*(X)_{\mathbb{Q}} \arrow[d, "f_*"] \\ K^\circ(Y) \arrow[r, "\tau_Y"'] & A_*(Y)_{\mathbb{Q}} \end{tikzcd}$$

commutes: τ intertwines proper pushforward of K-classes with proper pushforward of cycle classes. The Todd factor is what makes it commute, and without it the square already fails for $X = \mathbb{P}^1$ over a point.

Both sides are homomorphisms $K^\circ(X) \to A^*(Y)_{\mathbb{Q}}$, and neither is a ring homomorphism, so the content is an equality of two additive maps and nothing more can be extracted by multiplying the identity by itself. Taking $Y = \operatorname{Spec}\overline{k}$ turns the left side into the Euler characteristic $\chi(X, \alpha) = \sum_i (-1)^i \dim_{\overline{k}} H^i(X, \alpha)$ and the right side into the degree of a zero-cycle, which is the form the next section works with. The smallest instance is $X = \mathbb{P}^1$ with $\alpha = [\mathcal{O}_{\mathbb{P}^1}(n)]$: the left side is $\chi(\mathbb{P}^1, \mathcal{O}(n)) = n+1$, and the right side is the degree of the zero-dimensional part of $(1+nh)(1+h) \cap [\mathbb{P}^1]$

with $h = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$, using $\mathrm{td}(T_{\mathbb{P}^1}) = 1 + h$ from example 7.1.9, which is $n + 1$ as well. Without the Todd factor the right side would be n, so the correction is not optional even in dimension one.

Rewriting the identity as $\mathrm{ch}(f_*\alpha) = f_*\big(\mathrm{ch}(\alpha)\mathrm{td}(T_f)\big)$ is legitimate and is how the theorem is usually quoted: by proposition 7.1.8 and (7.2),

$$f_*\big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_f)\big)\,\mathrm{td}(T_Y) = f_*\big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_X)\,\mathrm{td}(f^*T_Y)^{-1}\,\mathrm{td}(f^*T_Y)\big) = f_*\big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_X)\big)$$

and $\mathrm{td}(T_Y)$ is invertible by proposition 7.1.6, so the two forms are equivalent.

The proof factors the morphism, and the square of theorem 7.2.2 is what makes that legitimate. The Riemann-Roch map lands in $A_*(X)_{\mathbb{Q}}$, the theorem reads $\tau_Y(f_*\alpha) = f_*\tau_X(\alpha)$, and in this form it is stable under composition: if it holds for $f\colon X \to Y$ and $g\colon Y \to Z$ then $\tau_Z((gf)_*\alpha) = \tau_Z(g_*f_*\alpha) = g_*\tau_Y(f_*\alpha) = g_*f_*\tau_X(\alpha) = (gf)_*\tau_X(\alpha)$, using functoriality of pushforward in K-theory ([3, Proper Pushforward]) and on Chow groups (theorem 1.2.3). A proper morphism from a quasi-projective source factors as a closed immersion followed by a projection $Y \times \mathbb{P}^m \to Y$, so two cases suffice, and they are the next two lemmas.

The projection is the case where the theorem becomes a computation on $\mathbb{P}^m$ and nothing else. The reduction to that computation rests on the description of $K^\circ(Y \times \mathbb{P}^m)$ by twisted pullbacks, which is K-theory rather than intersection theory and is cited from [3].

Lemma 7.2.3: Riemann-Roch for a Projection

Let Y be a smooth quasi-projective variety, $P = Y \times \mathbb{P}^m$ and $p\colon P \to Y$ the projection. Then for every $\alpha \in K^\circ(P)$,

$$\mathrm{ch}\big(p_*\alpha\big)\,\mathrm{td}(T_Y) \;=\; p_*\big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_P)\big)$$

in $A^*(Y)_{\mathbb{Q}}$.

***Proof*:**
Write $\rho\colon P \to \mathbb{P}^m$ for the second projection and $\zeta = c_1(\rho^*\mathcal{O}_{\mathbb{P}^m}(1)) \in A^1(P)$.

Step 1: reduction to twisted pullbacks. Both sides are additive in α, so it suffices to treat α in a generating set of the group $K^\circ(P)$. The generators used are the classes

$$p^*\beta \cdot \rho^*\big[\mathcal{O}_{\mathbb{P}^m}(n)\big], \qquad \beta \in K^\circ(Y),\ n \in \mathbb{Z}$$

These generate $K^\circ(Y \times \mathbb{P}^m)$: by [3, K_0 of a Relative Projective Space] the group $K^\circ(Y \times \mathbb{P}^m)$ is free over $K^\circ(Y)$ on $1, t, \ldots, t^m$ for $t = [\rho^*\mathcal{O}(-1)]$, and t is a unit there, so the classes above already exhaust it. The hypotheses of that theorem hold: Y is a smooth variety, hence Noetherian and separated, and carries an ample line bundle.

Step 2: the left-hand side. Fix β and n and put $\alpha = p^*\beta \cdot \rho^*[\mathcal{O}(n)]$. The projection formula in K-theory ([3, Projection Formula]) gives $p_*\alpha = \beta \cdot p_*\rho^*[\mathcal{O}(n)]$, so the remaining task is to identify $p_*\rho^*[\mathcal{O}(n)]$. Over an affine open $U = \operatorname{Spec} R \subseteq Y$ one has $p^{-1}(U) = \mathbb{P}^m_R$ and $\rho^*\mathcal{O}(n)$ restricts to $\mathcal{O}_{\mathbb{P}^m_R}(n)$, so [2, Twisted Sheaf Cohomology] computes $H^i\big(p^{-1}(U), \rho^*\mathcal{O}(n)\big)$ as a free R-module: the degree-n part of $R[x_0, \ldots, x_m]$ for $i = 0$, the degree-n part of $(x_0 \cdots x_m)^{-1}R[x_0^{-1}, \ldots, x_m^{-1}]$ for $i = m$, and zero otherwise. These modules are compatible with restriction to smaller affines and have R-rank equal to the $\overline{k}$-dimension of the corresponding space for $R = \overline{k}$, so the sheaves $R^ip_*\rho^*\mathcal{O}(n)$, being the sheafifications of $U \mapsto H^i(p^{-1}(U), -)$, are free of rank $h^i(\mathbb{P}^m, \mathcal{O}_{\mathbb{P}^m}(n))$.

Counting monomials in the two displayed modules gives

$$\chi\big(\mathbb{P}^m, \mathcal{O}_{\mathbb{P}^m}(n)\big) \;=\; \binom{n+m}{m} \;:=\; \frac{(n+1)(n+2)\cdots(n+m)}{m!}$$

for every $n \in \mathbb{Z}$. For $n \geq 0$ only H^0 survives, of dimension the number of monomials of degree n in $m+1$ variables, which is the displayed product. For $-m \leq n \leq -1$ both displayed modules vanish, and the displayed product has the factor $n+j=0$ with $j=-n$. For $n \leq -m-1$ only H^m survives, of dimension the number of tuples $(a_0, \dots, a_m)$ of integers $a_l \geq 1$ with $\sum_l a_l = -n$, that is $\binom{-n-1}{m}$, so that $\chi = (-1)^m\binom{-n-1}{m}$. Writing $n = -m-1-s$ with $s \geq 0$ turns the displayed product into $(-1)^m\binom{m+s}{m}$, the same number. Hence $p_*\rho^*[\mathcal{O}(n)] = \binom{n+m}{m}[\mathcal{O}_Y]$ and

$$\mathrm{ch}\big(p_*\alpha\big)\,\mathrm{td}(T_Y) \;=\; \binom{n+m}{m}\,\mathrm{ch}(\beta)\,\mathrm{td}(T_Y)$$

Step 3: the right-hand side. By proposition 7.1.11 and the computation of T_P preceding theorem 7.2.2,

$$\mathrm{ch}(\alpha)\,\mathrm{td}(T_P) \;=\; \Big(\mathrm{ch}(p^*\beta)\,\mathrm{td}(p^*T_Y)\Big)\cdot\Big(e^{n\zeta}\,Q(\zeta)^{m+1}\Big)$$

where $\mathrm{ch}(\rho^*\mathcal{O}(n)) = e^{n\zeta}$ because $c_1(\rho^*\mathcal{O}(n)) = n\zeta$ by proposition 2.4.4(3) and (4), and $\mathrm{td}(\rho^*T_{\mathbb{P}^m}) = Q(\zeta)^{m+1}$ as computed there. The first bracket is a polynomial in Chern classes of bundles pulled back from Y, so (7.2) applies to the proper morphism p and gives

$$p_*\big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_P)\big) \;=\; \mathrm{ch}(\beta)\,\mathrm{td}(T_Y)\cdot p_*\Big(e^{n\zeta}Q(\zeta)^{m+1}\cap[P]\Big)$$

Expand $e^{n\zeta}Q(\zeta)^{m+1} = \sum_{k\geq 0}\lambda_k\zeta^k$ with $\lambda_k \in \mathbb{Q}$. The bundle $\mathcal{O}_Y^{\oplus(m+1)}$ has $\mathbb{P}(\mathcal{O}_Y^{\oplus(m+1)}) = P$ with tautological class ζ and Segre classes $s_0 = 1$, $s_i = 0$ for $i \neq 0$ (example 3.1.7), so definition 3.1.3 with $r = m$ and $[P] = p^*[Y]$ (proposition 1.3.5(2)) gives $p_*(\zeta^k \cap [P]) = s_{k-m}(\mathcal{O}_Y^{\oplus(m+1)}) \cap [Y]$, which is $[Y]$ for $k = m$ and zero otherwise. Hence

$$p_*\big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_P)\big) \;=\; \lambda_m\,\mathrm{ch}(\beta)\,\mathrm{td}(T_Y), \qquad \lambda_m = \big[\zeta^m\big]\Big(e^{n\zeta}Q(\zeta)^{m+1}\Big)$$

Step 4: the coefficient. Comparing Steps 2 and 3, the lemma is the numerical identity

$$\big[\zeta^m\big]\Big(e^{n\zeta}\,Q(\zeta)^{m+1}\Big) \;=\; \binom{n+m}{m} \tag{7.3}$$

for all $m \geq 0$ and $n \in \mathbb{Z}$. Work in the field $\mathbb{Q}((\zeta))$ of formal Laurent series and write $\mathrm{res}(g\,d\zeta)$ for the coefficient of ζ^{-1} in g. If $\varphi \in \mathbb{Q}[[\zeta]]$ has $\varphi(0) = 0$ and $\varphi'(0) \neq 0$, then substituting φ for u carries $\mathbb{Q}((u))$ into $\mathbb{Q}((\zeta))$ and

$$\mathrm{res}\,\big(G(\varphi)\,\varphi'\,d\zeta\big) \;=\; \mathrm{res}\,\big(G\,du\big) \qquad \text{for all } G \in \mathbb{Q}((u))$$

By linearity it is enough to check $G = u^k$. For $k \neq -1$ one has $\varphi^k\varphi' = \frac{1}{k+1}\big(\varphi^{k+1}\big)'$, and the derivative of a formal Laurent series has no term in ζ^{-1}, so both sides are zero. For $k = -1$, writing $\varphi = \zeta\psi$ with $\psi(0) \neq 0$ gives $\varphi'/\varphi = \zeta^{-1} + \psi'/\psi$ with $\psi'/\psi \in \mathbb{Q}[[\zeta]]$, so both sides are 1.

Now $e^{n\zeta}Q(\zeta)^{m+1} = \zeta^{m+1}g(\zeta)$ with $g(\zeta) = e^{n\zeta}(1-e^{-\zeta})^{-(m+1)}$, so the coefficient of ζ^m on the left of (7.3) is $\operatorname{res}(g\,d\zeta)$. Put $u = \varphi(\zeta) = 1-e^{-\zeta}$, which lies in $\zeta + \zeta^2\mathbb{Q}[[\zeta]]$ and so satisfies the hypothesis, and note $\varphi' = e^{-\zeta} = 1-u$ and $e^{n\zeta} = (1-u)^{-n}$. Then $g = G(\varphi)\varphi'$ with $G(u) = (1-u)^{-n-1}u^{-m-1}$, whence

$$\operatorname{res}(g\,d\zeta) \;=\; \operatorname{res}\big((1-u)^{-n-1}u^{-m-1}\,du\big) \;=\; \big[u^m\big](1-u)^{-n-1}$$

Finally, differentiating m times at $u=0$,

$$\big[u^m\big](1-u)^{-n-1} \;=\; \frac{1}{m!}\,(n+1)(n+2)\cdots(n+m) \;=\; \binom{n+m}{m}$$

which is (7.3), as we sought to show.

The identity (7.3) is the reason the series $t/(1-e^{-t})$ appears in the theorem rather than some other multiplicative series: it is exactly the substitution $u = 1-e^{-\zeta}$ that turns the coefficient extraction into the binomial coefficient counting monomials on $\mathbb{P}^m$. Setting $n=0$ recovers the check made at the end of section 7.1, that the top Todd component of $T_{\mathbb{P}^m}$ has degree $1 = \chi(\mathcal{O}_{\mathbb{P}^m})$.

The closed immersion is the other case, and it is where the Todd class enters through a resolution rather than through a coefficient. The case of the zero section of a vector bundle is proved first, and the transfer to a general closed immersion is made by resolving on the deformation space of chapter 4 and restricting the resolution to its two fibres.

Lemma 7.2.4: Riemann-Roch for a Closed Immersion

Let $i\colon X \to Y$ be a closed immersion of smooth quasi-projective varieties. Then for every $\alpha \in K^\circ(X)$,

$$\operatorname{ch}\big(i_*\alpha\big)\,\operatorname{td}(T_Y) \;=\; i_*\big(\operatorname{ch}(\alpha)\,\operatorname{td}(T_X)\big)$$

in $A^*(Y)_{\mathbb{Q}}$.

***Proof*:**
Step 1: reduction to the normal bundle. Let $d = \dim Y - \dim X$. By [2, Nonsingular Closed Subschemes and Differentials] the ideal sheaf $\mathcal{I}$ of X is locally generated by d elements and $\mathcal{I}/\mathcal{I}^2$ is locally free of rank d. That ideal has height d at every point of X, since X has pure codimension d in the variety Y ([2, Dimension Formula for Finite Type k-Schemes]). The local rings of Y are regular, hence Cohen-Macaulay, and d elements generating an ideal of height d in such a ring form a regular sequence ([5, Height-Many Elements in a Cohen-Macaulay Ring Form a Regular Sequence]), so i is a regular embedding of codimension d ([2, Regular Embedding]) with normal bundle $N = (\mathcal{I}/\mathcal{I}^2)^\vee$. Dualizing the conormal sequence of the same theorem gives $0 \to T_X \to i^*T_Y \to N \to 0$, so $\operatorname{td}(i^*T_Y) = \operatorname{td}(T_X)\operatorname{td}(N)$ by proposition 7.1.8, and therefore

$$i_*\big(\operatorname{ch}(\alpha)\,\operatorname{td}(T_X)\big) = i_*\Big(\operatorname{ch}(\alpha)\,\operatorname{td}(N)^{-1}\,\operatorname{td}(i^*T_Y)\Big) = \operatorname{td}(T_Y)\cdot i_*\Big(\operatorname{ch}(\alpha)\,\operatorname{td}(N)^{-1}\Big)$$

the second equality by (7.2) applied to the proper morphism i and the bundle T_Y on Y. Since $\operatorname{td}(T_Y)$ is invertible (proposition 7.1.6), the lemma is equivalent to

$$\operatorname{ch}\big(i_*\alpha\big) \;=\; i_*\Big(\operatorname{ch}(\alpha)\,\operatorname{td}(N)^{-1}\Big) \tag{7.4}$$

Step 2: the zero section, inside the projective completion. Let E be a vector bundle of rank d on a smooth variety X and set

$$P \;=\; \mathbb{P}(E \oplus 1), \qquad \varrho\colon P \to X,$$

the projective completion of definition 3.3.3, a smooth X-scheme, proper over X, containing the total space of E as a dense open. Let $\mathcal{Q}$ be the universal quotient bundle on P, of rank d, and let $f\colon X \to P$ be the zero section of E read inside P. It is a section of ϱ, which is a projective bundle and so smooth of relative dimension d; so f is a regular embedding of codimension d ([2, A Section of a Smooth Morphism Is a Regular Embedding]). Its normal bundle is $f^*\mathcal{Q}$: the total space of E is an open subscheme of P containing $f(X)$, normal bundles are local on the ambient, and the normal bundle of a zero section is the bundle itself, which is the computation at (4.6). And $f^*\mathcal{Q} = E$, because along the zero section the tautological subbundle of $\varrho^*(E \oplus 1)$ is the trivial summand, whose quotient is E. So $N_f = E$ and this step proves (7.4) for f. It is stated on P rather than on the total space E deliberately: Step 3 below consumes it on the projective completion, and restriction $A_*(P) \to A_*(E)$ is only surjective (theorem 1.4.1), so an identity proved on E need not transfer back.

Write $\sigma \in \Gamma(P, \mathcal{Q})$ for the section obtained by projecting the trivial summand of $\varrho^*(E \oplus 1)$ onto $\mathcal{Q}$. Its zero scheme is exactly the zero section: at a point of P the fibre of $\mathcal{Q}$ is the quotient of $E \oplus 1$ by the line the point names, and the image of $(0, 1)$ vanishes there precisely when that line is the trivial summand, which is the zero section of E. The comparison is scheme-theoretic, not merely pointwise: on the chart of $P|_U$ used below, where the trivial coordinate is invertible, σ becomes the tuple of affine coordinates, which cuts out the zero section with multiplicity one; on the complementary charts σ is nowhere zero. So $Z(\sigma) = f(X)$, of pure codimension d in P.

First a resolution. Contraction against σ makes

$$0 \longrightarrow \Lambda^d \mathcal{Q}^\vee \longrightarrow \cdots \longrightarrow \Lambda^1 \mathcal{Q}^\vee \longrightarrow \mathcal{O}_P \longrightarrow f_*\mathcal{O}_X \longrightarrow 0$$

a complex, and it is exact. Exactness is local, and over an affine open $U = \operatorname{Spec} A \subseteq X$ trivializing E the section σ is, on the chart of $P|_U = U \times \mathbb{P}^d$ where the trivial coordinate is invertible, the tuple $(t_1, \ldots, t_d)$ of affine coordinates on that chart, so the displayed complex becomes the Koszul complex of that tuple ([6, Koszul Complex]); on the complementary charts σ is nowhere vanishing, so contraction against it is a homotopy equivalence and the complex is exact there as well. On the first chart $t_1, \ldots, t_d$ is a regular sequence, t_j being a nonzerodivisor on $A[t_1, \ldots, t_d]/(t_1, \ldots, t_{j-1}) = A[t_j, \ldots, t_d]$, with quotient $A = f_*\mathcal{O}_X$, so [6, Exactness of the Koszul Complex] applies. A closed immersion has exact pushforward on quasi-coherent sheaves, being extension by zero on the underlying space, so $R^q f_*\mathcal{O}_X = 0$ for $q > 0$ and the K-theoretic pushforward of [3, Proper Pushforward] is the single class $[f_*\mathcal{O}_X]$. Taking the alternating sum of the classes of the terms,

$$\bigl[f_*\mathcal{O}_X\bigr] \;=\; \sum_{j=0}^{d} (-1)^j \bigl[\Lambda^j \mathcal{Q}^\vee\bigr] \qquad \text{in } K^\circ(P) \tag{7.5}$$

Next the Chern character of that alternating sum. Let G be a vector bundle of rank d on an algebraic scheme W. The claim is

$$\sum_{j=0}^{d} (-1)^j \operatorname{ch}\bigl(\Lambda^j G^\vee\bigr) \;=\; c_d(G)\operatorname{td}(G)^{-1} \tag{7.6}$$

as operators on $A_*(W)_{\mathbb{Q}}$. Choose by theorem 3.2.14 a splitting map $g\colon W' \to W$ for G, with g^*G filtered by subbundles with line-bundle quotients $L_1, \dots, L_d$ and $a_p = c_1(L_p)$. Dualizing that filtration term by term gives a filtration of $(g^*G)^\vee$ by subbundles with line-bundle quotients $L_1^\vee, \dots, L_d^\vee$ in the reverse order, dualization preserving exactness because a short exact sequence of vector bundles is locally split. If a bundle H sits in an exact sequence $0 \to M \to H \to H' \to 0$ with M a line bundle, then for each j the sequence

$$0 \longrightarrow M \otimes \Lambda^{j-1}H' \longrightarrow \Lambda^j H \longrightarrow \Lambda^j H' \longrightarrow 0$$

is exact: the right-hand map is Λ^j of $H \to H'$, the left-hand one sends $m \otimes \omega$ to $m \wedge \widetilde{\omega}$ for any lift $\widetilde{\omega}$ of ω to $\Lambda^{j-1}H$, which is well defined because two lifts differ by a section of $M \otimes \Lambda^{j-2}H$ and $\Lambda^2 M = 0$, and exactness is a local statement, checked where the sequence splits. Iterating this along the filtration of $(g^*G)^\vee$ shows that $\Lambda^j(g^*G)^\vee = g^*\Lambda^j G^\vee$ is filtered by subbundles whose line-bundle quotients are the $\bigotimes_{p \in S} L_p^\vee$ with $S \subseteq \{1, \dots, d\}$ of size j. By proposition 2.4.4(3) their first Chern classes are the $-\sum_{p\in S} a_p$. So g is a splitting map for $\Lambda^j G^\vee$ with those Chern roots, and proposition 3.2.22(4) gives, for every $\gamma \in A_*(W)_{\mathbb{Q}}$,

$$g^*\Big(\sum_j (-1)^j \mathrm{ch}(\Lambda^j G^\vee) \cap \gamma\Big) = \Big(\sum_j (-1)^j \sum_{|S|=j} e^{-\sum_{p\in S} a_p}\Big) \cap g^*\gamma = \Big(\prod_{p=1}^{d} \big(1 - e^{-a_p}\big)\Big) \cap g^*\gamma$$

the last equality being the expansion of the product. The same proposition applied to $c_d(G)$ and to the symmetric series $\prod_p (1 - e^{-a_p})/a_p$ defining $\mathrm{td}(G)^{-1}$ gives $g^*\big(c_d(G)\mathrm{td}(G)^{-1} \cap \gamma\big) = \prod_p a_p \prod_p \frac{1-e^{-a_p}}{a_p} \cap g^*\gamma$, which is the same operator. Injectivity of g^* yields (7.6).

Now combine. Applying ch to (7.5) and using (7.6) for $G = \mathcal{Q}$,

$$\mathrm{ch}\big(f_*[\mathcal{O}_X]\big) \;=\; c_d(\mathcal{Q})\,\mathrm{td}(\mathcal{Q})^{-1}$$

For a general $\alpha \in K^\circ(X)$ one has $\alpha = f^*\varrho^*\alpha$, since $\varrho f = \mathrm{id}_X$, so the projection formula in K-theory ([3, Projection Formula]) gives $f_*\alpha = \varrho^*\alpha \cdot f_*[\mathcal{O}_X]$ and hence, ch being multiplicative (proposition 7.1.11),

$$\mathrm{ch}\big(f_*\alpha\big) \;=\; \mathrm{ch}(\varrho^*\alpha)\, c_d(\mathcal{Q})\,\mathrm{td}(\mathcal{Q})^{-1}$$

On the other side, $f^*(\varrho^*\alpha) = \alpha$ and $f^*\mathcal{Q} = E$, so (7.2) applied to the proper morphism f and the classes $\varrho^*\alpha$ and $\mathcal{Q}$ on P gives

$$f_*\Big(\mathrm{ch}(\alpha)\,\mathrm{td}(E)^{-1} \cap [X]\Big) = \mathrm{ch}(\varrho^*\alpha)\,\mathrm{td}(\mathcal{Q})^{-1} \cap f_*[X]$$

and $f_*[X] = c_d(\mathcal{Q}) \cap [P]$ by lemma 5.3.14, applied to the bundle $\mathcal{Q}$ on the smooth variety P and its section σ, whose zero scheme is $f(X)$ of pure codimension d. The two displayed right-hand sides agree, which is (7.4) for f.

Step 3: from the zero section to a general closed immersion. Both sides are additive in α and $K^\circ(X)$ is generated by classes of vector bundles, so we may take $\alpha = [E]$. The transfer is made by resolving on the deformation space and restricting the resolution to its two fibres.

Let $M = M_X Y$ be the deformation space of definition 4.1.1, the blow-up of $Y \times \mathbb{P}^1$ along $X \times \{\infty\}$, with blow-down p and projection $\pi\colon M \to \mathbb{P}^1$. Four properties of it are used, and each is recorded before it is needed.

(a) M is a nonsingular quasi-projective variety. The centre $X \times \{\infty\}$ is a nonsingular closed subvariety of the nonsingular variety $Y \times \mathbb{P}^1$, so [2, Blow-up Along a Nonsingular Centre] makes M a nonsingular variety of dimension $\dim Y + 1$, with p projective and birational. It is quasi-projective: p is projective, so M is a closed subscheme of $Y \times \mathbb{P}^1 \times \mathbb{P}^r$; writing Y as an open subscheme of a projective variety $\overline{Y}$, the Segre embedding of [2, Products of Projective Varieties] makes $\overline{Y} \times \mathbb{P}^1 \times \mathbb{P}^r$ projective, so the open subscheme $Y \times \mathbb{P}^1 \times \mathbb{P}^r$ is quasi-projective and so is its closed subscheme M. By [2, Quasi-Projective Schemes Carry an Ample Sheaf] it therefore carries an ample invertible sheaf.

(b) M is flat over $\mathbb{P}^1$. Each of the two standard charts of $\mathbb{P}^1$ has coordinate ring $\overline{k}[u]$, a principal ideal domain, over which a module is flat exactly when it is torsion-free. An affine chart of M over such a chart is a domain by (a) into which $\overline{k}[u]$ injects, π being dominant, so it is torsion-free and hence flat. Consequently every locally free sheaf on M is flat over $\mathbb{P}^1$. Record also, for use in the conclusion, that the composite $q\colon M \to Y \times \mathbb{P}^1 \to Y$ is proper, both factors being so.

(c) The constant subscheme is closed in M. Apply lemma 4.2.6 to the closed immersion $i\colon X \to Y$ with $Z = X$. Here $i^{-1}(X) = X$, so the centre is $X \times \{\infty\}$ inside $X \times \mathbb{P}^1$, an effective Cartier divisor; blowing up an invertible ideal changes nothing, by the universal property [2, Basic Properties of Blow-ups](4) applied to the identity, so $M_X X = X \times \mathbb{P}^1$. The deleted piece $\mathrm{Bl}_X X$ is empty, the Rees algebra of the zero ideal being $\mathcal{O}_X$ in degree zero and nothing above, so $M^\circ_X X = X \times \mathbb{P}^1$ too. The lemma produces $F\colon X \times \mathbb{P}^1 \to M$ over $i \times \mathrm{id}_{\mathbb{P}^1}$ with $F^{-1}(M^\circ_X Y) = M^\circ_X X = X \times \mathbb{P}^1$, and by (2) of that lemma F is proper because i is; so F has closed image and lands in the open subscheme $M^\circ_X Y$.

It agrees with the closed immersion of theorem 4.1.5(4). Both morphisms are defined on $X \times \mathbb{P}^1$, which is reduced, both land in the separated M, and over $\mathbb{A}^1$ both restrict to $i \times \mathrm{id}$: for F this is lemma 4.2.6(1), and for the immersion of theorem 4.1.5(4) it is that part's own description of its restriction over $\mathbb{A}^1$. Two morphisms from a reduced scheme to a separated scheme that agree on a dense open of the source are equal, and $X \times \mathbb{A}^1$ is dense in $X \times \mathbb{P}^1$. Hence F is a closed immersion onto its image in $M^\circ_X Y$, and a closed immersion into an open subscheme whose image is closed in the ambient is a closed immersion into the ambient. Put $\bar{E} = \mathrm{pr}_X^* E$, so $F_* \bar{E}$ is coherent on M and flat over $\mathbb{P}^1$, being pulled back from X along a projection and pushed forward along a closed immersion over $\mathbb{P}^1$.

(d) The fibre at infinity. Let $\operatorname{Spec} A \subseteq Y$ be an affine open meeting X in the ideal I, and let $U \subseteq \mathbb{P}^1$ be the chart around ∞ with coordinate u. Write $B = A[u]$, so the ideal of the centre is $J = IB + uB$. Because B is free over A on the powers of u, an element of IB divisible by u lies in uIB, whence $IB \cap uB = uIB \subseteq J^2$; comparing u-degrees in $J^2 = I^2B + uIB + u^2B$ likewise gives $J^2 \cap IB = I^2B + uIB$ and $J^2 \cap uB = uIB + u^2B$. So the conormal module splits,

$$J/J^2 \;=\; IB/(I^2B + uIB) \;\oplus\; uB/(uIB + u^2B) \;=\; I/I^2 \;\oplus\; A/I,$$

and these identifications agree on overlaps: the decomposition of the ideal sheaf of the centre into the two pulled-back factors is global, and the splitting is read off from it by the same formula on every chart. The centre is a nonsingular closed subvariety of the nonsingular $Y \times \mathbb{P}^1$, hence regularly embedded, of codimension $d+1$ and with normal bundle $N \oplus 1$ by the computation just made; so lemma 5.3.5(1) identifies the exceptional divisor of M with $\mathbb{P}(N \oplus 1)$, the projective completion of definition 3.3.3, exactly the ambient of Step 2 for the bundle N. The fibre M_∞ is the pullback of the Cartier divisor $Y \times \{\infty\}$, hence the total transform, so

$$[M_\infty] \;=\; \big[\mathbb{P}(N \oplus 1)\big] \;+\; \big[\mathrm{Bl}_X Y\big] \tag{7.7}$$

both multiplicities being one. The computation is chart by chart. Shrink $\operatorname{Spec} A$ so that I is generated by a regular sequence $b_1, \dots, b_d$, available from Step 1; then J is generated by $b_1, \dots, b_d, u$ and the blow-up is covered by the charts on which $J\mathcal{O}_M$ is generated by one of them. On the chart where the generator is u, every b_i becomes u times a regular function, so the pullback of $\{u = 0\}$ is cut out by u alone: the exceptional divisor occurs with multiplicity one and the strict transform does not meet this chart. On the chart where the generator is b_j, one has $u = b_j u'$ with b_j a local equation for the exceptional divisor and u' a regular function. The exceptional coefficient is already known to be one from the u-chart, which meets the generic point of the irreducible $\mathbb{P}(N \oplus 1)$, so u' carries no exceptional component and cuts out exactly the closure of the locus where u vanishes and the b_i do not, which is the strict transform. It occurs to the first power because p is an isomorphism away from the centre ([2, Basic Properties of Blow-ups](2)), in particular at the generic point of that closure, where $Y \times \{\infty\}$ is reduced. So as effective Cartier divisors on M,

$$p^*\big(Y \times \{\infty\}\big) \;=\; \mathbb{P}(N \oplus 1) \;+\; \widetilde{Y_\infty},$$

$\widetilde{Y_\infty}$ denoting the strict transform, and (7.7) is the associated identity of cycles. The strict transform is the deleted piece $\mathrm{Bl}_X Y$ of lemma 4.1.2(2): both are the closure in M of $p^{-1}\big((Y \setminus X) \times \{\infty\}\big)$, and the lemma's final clause makes it closed in M. Because the exceptional multiplicity is one, the closed subschemes M_∞ and $\mathbb{P}(N \oplus 1)$ agree on the open set $M \setminus \mathrm{Bl}_X Y$, which is the fact used below.

Restricting the resolution. By (a) the variety M is regular of finite dimension and carries an ample invertible sheaf, so [2, Finite Locally Free Resolutions on a Regular Scheme] supplies a finite resolution $G_\bullet \to F_*\bar{E}$ by locally free sheaves on M. Since M, every G_i and $F_*\bar{E}$ are flat over $\mathbb{P}^1$ by (b) and (c), [5, Restriction of a Resolution to a Fibre] keeps $G_\bullet$ a resolution after restriction to any fibre of π. Over a point of $\mathbb{A}^1$ the fibre is Y: the deleted piece $\mathrm{Bl}_X Y$ lies over $Y \times \{\infty\}$ by lemma 4.1.2(2), so M and $M^\circ_X Y$ agree over $\mathbb{A}^1$, where theorem 4.1.5(1) identifies the family with $Y \times \mathbb{A}^1$; and F restricts there to i by lemma 4.2.6(1). A closed immersion is affine, and pushforward along an affine morphism commutes with arbitrary base change; F lies over $\mathbb{P}^1$, so $F^{-1}(M_t) = X \times \{t\}$ scheme-theoretically and $F_*\bar{E}|_Y = i_*E$. Hence $G_\bullet|_Y$ resolves i_*E.

Restricting further to a *component* of M_∞ is not covered by that statement, and one observation settles both components. Write $\bar{f}\colon X \to \mathbb{P}(N \oplus 1)$ for the zero section and $S \subseteq M_\infty$ for the support of $F_*\bar{E}|_{M_\infty}$. By (c) and theorem 4.1.5(4) that support is the vertex of $C_X Y$, which is N by proposition 3.3.11; the vertex of a vector bundle regarded as a cone is its zero section (definition 3.3.1), so $S = \bar{f}(X)$, disjoint from $\mathrm{Bl}_X Y$. As over $\mathbb{A}^1$, base change along the affine morphism F identifies $F_*\bar{E}|_{M_\infty}$ with the pushforward of E along the restriction of F over ∞, which by theorem 4.1.5(4) is $\bar{f}$. On the open set $M_\infty \setminus S$ the complex $G_\bullet|_{M_\infty}$ is an exact complex of vector bundles, hence locally split exact, hence stays exact under pullback along any morphism into $M_\infty \setminus S$. Since $\mathrm{Bl}_X Y \subseteq M_\infty \setminus S$, the restriction $G_\bullet|_{\mathrm{Bl}_X Y}$ is acyclic. For the other component, cover $\mathbb{P}(N \oplus 1)$ by the two opens $\mathbb{P}(N \oplus 1) \setminus S$, where the same splitness argument applies and $\bar{f}_*E$ vanishes, and $M_\infty \setminus \mathrm{Bl}_X Y$, which contains S, is open in M_∞ and coincides with the corresponding open subscheme of $\mathbb{P}(N \oplus 1)$ by the divisor identity of (d); there $G_\bullet$ restricts to a resolution of $F_*\bar{E}$ already. Exactness being local, the restriction $G_\bullet|_{\mathbb{P}(N\oplus 1)}$ resolves $\bar{f}_*E$, which is Step 2's situation for the bundle N.

Conclusion. Write $\mathrm{ch}(G_\bullet) = \sum_i (-1)^i \mathrm{ch}(G_i)$ and let $j_0\colon Y \hookrightarrow M$, $k\colon \mathbb{P}(N \oplus 1) \hookrightarrow M$ and $l\colon \mathrm{Bl}_X Y \hookrightarrow M$ be the inclusions of (7.7) and of the fibre over 0. Both fibres are Cartier divisors whose associated line bundle is $\pi^*\mathcal{O}_{\mathbb{P}^1}(1)$, so $M_0 \cdot [M] = M_\infty \cdot [M]$ in $A_*(M)$ by proposition 2.4.2;

neither contains M, so definition 2.2.1(1) evaluates both sides as the associated cycles and gives $[M_0] = [M_\infty]$. Applying (7.2) to each inclusion,

$$j_{0*}\Big(\mathrm{ch}\big(i_*E\big) \cap [Y]\Big) \;=\; \mathrm{ch}(G_\bullet) \cap [M_0] \;=\; \mathrm{ch}(G_\bullet) \cap [M_\infty] \;=\; k_*\Big(\mathrm{ch}\big(\bar{f}_*E\big) \cap [\mathbb{P}(N \oplus 1)]\Big)$$

the $\mathrm{Bl}_X Y$ term vanishing because an acyclic bounded complex of vector bundles is locally split exact, so its syzygies are locally free and its alternating sum is zero already in $K^\circ(\mathrm{Bl}_X Y)$, hence so is its Chern character. Step 2 evaluates the right-hand side as $k_*\bar{f}_*\big(\mathrm{ch}(E)\mathrm{td}(N)^{-1} \cap [X]\big)$.

Now apply q_*, legitimate because q is proper by (b). On the left $q \circ j_0 = \mathrm{id}_Y$; on the right $q \circ k \circ \bar{f} = i$, by construction of the deformation. Hence

$$\mathrm{ch}\big(i_*E\big) \;=\; i_*\Big(\mathrm{ch}(E)\,\mathrm{td}(N)^{-1}\Big)$$

which is (7.4) for i, as we sought to show.

***Proof of theorem 7.2.2*:**
Step 1: the factorization. Since X is quasi-projective there is a locally closed immersion $j\colon X \to \mathbb{P}^m$ for some m. Put $P = Y \times \mathbb{P}^m$, let $p\colon P \to Y$ be the projection and set $i = (f, j)\colon X \to P$, so that $f = p \circ i$. The morphism i is the composite of the graph $X \to Y \times X$ of f, which is a closed immersion because Y is separated, with $\mathrm{id}_Y \times j$, which is an immersion because immersions are stable under base change, so i is an immersion. It is proper: it factors as $X \to X \times_Y P \to P$, where the first map is the graph of i over Y, a closed immersion because p is separated, and the second is the base change of the proper morphism f. A proper morphism has closed image, and an immersion whose image is closed is a closed immersion, so i is a closed immersion. Finally P is smooth and quasi-projective, being a product of such varieties.

Step 2: composing the two cases. With $\tau_X(\alpha) = \mathrm{ch}(\alpha)\mathrm{td}(T_X) \cap [X]$ as above, lemma 7.2.4 applied to i gives $\tau_P(i_*\alpha) = i_*\tau_X(\alpha)$, and lemma 7.2.3 applied to p and to the class $i_*\alpha \in K^\circ(P)$ gives $\tau_Y(p_*i_*\alpha) = p_*\tau_P(i_*\alpha)$. Both lemmas apply because X, P and Y are smooth and quasi-projective. Combining them,

$$\tau_Y\big(f_*\alpha\big) = \tau_Y\big(p_*i_*\alpha\big) = p_*\tau_P\big(i_*\alpha\big) = p_*i_*\tau_X(\alpha) = f_*\tau_X(\alpha)$$

where the first and last equalities use $f_* = p_* \circ i_*$, in K-theory by [3, Proper Pushforward] and on Chow groups by theorem 1.2.3. The first and last terms of this chain are the two sides of the asserted identity, as we sought to show.

The two cases contribute differently. In the projection case the Todd class is a power series whose m-th coefficient is forced by (7.3). In the closed immersion case it is the Chern character of a Koszul complex, appearing because a sheaf supported on a subvariety is resolved on the ambient one by exterior powers of the conormal bundle. The factorization argument shows that these two are the only inputs, everything else being the behaviour of τ under composition.

The example below runs the theorem end to end for $Y = \operatorname{Spec} \overline{k}$, where the closed immersion of Step 1 is an isomorphism and the whole content is lemma 7.2.3 over a point. It is checked against a cohomology computation that does not use the theorem.

Example 7.2.5: The Euler Characteristic of $\mathcal{O}_{\mathbb{P}^m}(n)$

Let $f\colon \mathbb{P}^m \to \operatorname{Spec}\overline{k}$ be the structure morphism and $\alpha = [\mathcal{O}_{\mathbb{P}^m}(n)]$. Write $h = c_1(\mathcal{O}_{\mathbb{P}^m}(1))$, so that $h^m \cap [\mathbb{P}^m]$ is the class of a point and $h^{m+1} = 0$ (example 7.1.3).

The left-hand side of theorem 7.2.2 is a number. Since $A^*(\operatorname{Spec}\overline{k})_{\mathbb{Q}} = \mathbb{Q}$ and $T_{\operatorname{Spec}\overline{k}} = 0$, it is $\operatorname{ch}(f_*\alpha) = \chi(\mathbb{P}^m, \mathcal{O}(n))$, the alternating sum of the dimensions of the cohomology groups.

The right-hand side is the degree of a zero-cycle: by definition 1.2.4, f_* of a class in $A_*(\mathbb{P}^m)_{\mathbb{Q}}$ is the degree of its zero-dimensional part, so the right-hand side is

$$\int_{\mathbb{P}^m} \operatorname{ch}\big(\mathcal{O}(n)\big)\operatorname{td}(T_{\mathbb{P}^m}) \cap [\mathbb{P}^m] \;=\; \big[h^m\big]\Big(e^{nh}\, Q(h)^{m+1}\Big)$$

using $\operatorname{ch}(\mathcal{O}(n)) = e^{nh}$ and $\operatorname{td}(T_{\mathbb{P}^m}) = Q(h)^{m+1}$ from example 7.1.9, and $\int_{\mathbb{P}^m} h^m \cap [\mathbb{P}^m] = 1$.

For $m = 2$ the two sides read as follows. From example 7.1.9, $\operatorname{td}(T_{\mathbb{P}^2}) = 1 + \frac{3}{2}h + h^2$, and $\operatorname{ch}(\mathcal{O}(n)) = 1 + nh + \frac{n^2}{2}h^2$, so the coefficient of h^2 in the product is

$$\frac{n^2}{2} + \frac{3n}{2} + 1 \;=\; \frac{(n+1)(n+2)}{2}$$

which is $\dim_{\overline{k}}$ of the space of degree-n forms in three variables for $n \geq 0$, that is $\chi(\mathbb{P}^2, \mathcal{O}(n))$. For $m = 3$, with $\operatorname{td}(T_{\mathbb{P}^3}) = 1 + 2h + \frac{11}{6}h^2 + h^3$ and $\operatorname{ch}(\mathcal{O}(n)) = 1 + nh + \frac{n^2}{2}h^2 + \frac{n^3}{6}h^3$, the coefficient of h^3 is

$$\frac{n^3}{6} + \frac{n^2}{2}\cdot 2 + n\cdot\frac{11}{6} + 1 \;=\; \frac{n^3 + 6n^2 + 11n + 6}{6} \;=\; \frac{(n+1)(n+2)(n+3)}{6}$$

again $\chi(\mathbb{P}^3, \mathcal{O}(n))$. Both agree with (7.3), which Step 4 of lemma 7.2.3 proved for all m and n at once. Neither factor has integral coefficients on its own, the Todd coefficients $\frac{3}{2}$ and $\frac{11}{6}$ and the Chern character coefficients $\frac{n^2}{2}$ and $\frac{n^3}{6}$ all being fractions. Only the product has integral degree.

Smoothness of X enters the statement of the theorem only through $\operatorname{td}(T_X)$, and that much survives a weaker hypothesis. A local complete intersection morphism carries a virtual relative tangent class even when its source is singular, built from a factorization rather than from a tangent bundle, and the theorem holds in that generality.

Corollary 7.2.6: Riemann-Roch for a Local Complete Intersection Morphism

Let Y be a smooth quasi-projective variety, X a variety, and $f\colon X \to Y$ a proper local complete intersection morphism ([2, Local Complete Intersection Morphism]) admitting on all of X a factorization $f = p \circ i$ with $i\colon X \to P$ a regular embedding with normal bundle N_i and $p\colon P \to Y$ smooth, as in definition 5.2.11. Set

$$T_f \;=\; \big[i^*T_{P/Y}\big] - \big[N_i\big] \;\in\; K^\circ(X)$$

Then for every $\alpha \in K^\circ(X)$,

$$\mathrm{ch}\big(f_*\alpha\big) \cap [Y] \;=\; f_*\Big(\mathrm{ch}(\alpha)\,\mathrm{td}(T_f) \cap [X]\Big) \qquad \text{in } A_*(Y)_{\mathbb{Q}}$$

When X is smooth, T_f agrees with the relative tangent class of definition 7.2.1 and the identity is theorem 7.2.2.

***Proof Cited*:**
The proof is not given here (QC #3(a)). An l.c.i. morphism need not have smooth source, so the argument of theorem 7.2.2 does not apply: what replaces it is a Riemann-Roch transformation defined on the K-theory of coherent sheaves on a possibly singular X, which this book does not construct. That transformation is built, and the theorem in this generality proved, in [8, the Grothendieck-Riemann-Roch theorem]. The case of a proper morphism of smooth varieties, theorem 7.2.2, is proved above. The present extension is stated only.

The comparison in the last sentence is a computation and is given. If X is smooth, then $0 \to T_X \to i^*T_P \to N_i \to 0$ is exact by [2, Nonsingular Closed Subschemes and Differentials], and $0 \to T_{P/Y} \to T_P \to p^*T_Y \to 0$ is exact because p is smooth. Pulling the second back along i and subtracting in $K^\circ(X)$,

$$\big[i^*T_{P/Y}\big] - \big[N_i\big] = \Big(\big[i^*T_P\big] - \big[f^*T_Y\big]\Big) - \Big(\big[i^*T_P\big] - \big[T_X\big]\Big) = \big[T_X\big] - \big[f^*T_Y\big]$$

which is T_f as defined in definition 7.2.1. The displayed identity then becomes $\mathrm{ch}(f_*\alpha) = f_*\big(\mathrm{ch}(\alpha)\mathrm{td}(T_X)\mathrm{td}(f^*T_Y)^{-1}\big)$, which multiplied by $\mathrm{td}(T_Y)$ and rearranged by (7.2) is the statement of theorem 7.2.2, completing the comparison.

The class T_f is what replaces the tangent bundle of the source. Only the difference $[i^*T_{P/Y}] - [N_i]$ enters the identity, never either term on its own, and this is the same shape as definition 5.2.11: the Gysin pullback $f^! = i^! \circ p^*$ is also assembled from a factorization, and by theorem 5.2.12(1) neither it nor the virtual relative dimension, which is the rank of T_f, depends on which factorization is chosen.

Exercise 7.2.1

1. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be morphisms of smooth varieties. Show that $T_{g\circ f} = T_f + f^*T_g$ in $K^\circ(X)$, where f^* is pullback of bundles extended to K-theory, and deduce that $\mathrm{td}(T_{g\circ f}) = \mathrm{td}(T_f)\,\mathrm{td}(f^*T_g)$.
2. Compute the coefficient of h^4 in $Q(h)^5$, where $Q(t) = t/(1-e^{-t})$ and $h = c_1(\mathcal{O}_{\mathbb{P}^4}(1))$, using $Q(t) = 1 + \frac{t}{2} + \frac{t^2}{12} + 0\cdot t^3 - \frac{t^4}{720} + \cdots$. Check the answer against (7.3) with $n = 0$, and say

what the check asserts about $\chi(\mathcal{O}_{\mathbb{P}^4})$.

3. Fix $m \geq 1$. Show that $[\zeta^m]\big(e^{n\zeta}Q(\zeta)^{m+1}\big) = 0$ for every integer n with $-m \leq n \leq -1$, and identify the cohomological statement this corresponds to under lemma 7.2.3.

4. Use theorem 7.2.2 for $\mathbb{P}^m \to \operatorname{Spec} \overline{k}$ and the classes computed in example 7.1.9 to evaluate $\chi(\mathbb{P}^2, T_{\mathbb{P}^2})$ and $\chi(\mathbb{P}^3, T_{\mathbb{P}^3})$. Verify both against the Euler sequence, which expresses $T_{\mathbb{P}^m}$ through $\mathcal{O}(1)^{\oplus(m+1)}$ and $\mathcal{O}$.

5. Specialize the Koszul resolution in Step 2 of lemma 7.2.4 to a line bundle L on a smooth variety X, with $\pi\colon L \to X$ the total space and s the zero section. Write down the resulting two-term resolution of $s_*\mathcal{O}_X$, compute $\operatorname{ch}(s_*[\mathcal{O}_X])$ from it, and check directly that the answer equals $c_1(\pi^*L)\operatorname{td}(\pi^*L)^{-1}$.

6. Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree e with inclusion i, so that C is the zero scheme of a section of $\mathcal{O}_{\mathbb{P}^2}(e)$ and its normal bundle is $N = \mathcal{O}_{\mathbb{P}^2}(e)|_C$. Compute $\operatorname{ch}(i_*[\mathcal{O}_C]) \in A^*(\mathbb{P}^2)_{\mathbb{Q}}$ from the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(-e) \to \mathcal{O}_{\mathbb{P}^2} \to i_*\mathcal{O}_C \to 0$, compute $i_*\big(\operatorname{td}(N)^{-1} \cap [C]\big)$ using proposition 3.2.10(2), and check that the two agree as (7.4) predicts.

7. Let $i\colon X \to Y$ be a closed immersion of smooth quasi-projective varieties of codimension d and let $\alpha \in K^\circ(X)$. Using lemma 7.2.4 in the form (7.4), show that $\operatorname{ch}_k(i_*\alpha) = 0$ for $k < d$ and that $\operatorname{ch}_d(i_*\alpha) = \operatorname{rk}(\alpha)\, i_*[X]$.

8. For $f\colon \mathbb{P}^1 \to \operatorname{Spec} \overline{k}$ and $\alpha = [\mathcal{O}_{\mathbb{P}^1}(n)]$, compute $\chi(\mathbb{P}^1, \mathcal{O}(n))$, the degree $\int_{\mathbb{P}^1} \operatorname{ch}(\mathcal{O}(n)) \cap [\mathbb{P}^1]$, and the degree $\int_{\mathbb{P}^1} \operatorname{ch}(\mathcal{O}(n))\operatorname{td}(T_{\mathbb{P}^1}) \cap [\mathbb{P}^1]$. Deduce that ch does not commute with proper pushforward, and that the discrepancy in this case is exactly one.

7.3 Consequences

Setting $Y = \operatorname{Spec} \overline{k}$ in theorem 7.2.2 leaves a statement about a single variety rather than about a morphism: the pushforward in K-theory becomes an alternating sum of dimensions of cohomology groups, and the pushforward on Chow groups becomes the degree of a zero-cycle. What survives is a formula for the Euler characteristic of a coherent sheaf in terms of Chern classes. This section proves that formula and then derives the classical curve and surface statements from the cases $\dim X = 1$ and $\dim X = 2$.

Throughout, X is a smooth projective variety of dimension n over $\overline{k}$, and $T_X = (\Omega^1_{X/\overline{k}})^\vee$ is its tangent bundle, locally free of rank n because X is smooth ([2, Tangent Sheaf]). On a smooth variety the Chern character and the Todd class take values in the Chow ring: the operator $\operatorname{ch}(\alpha)$ is multiplication by an element of $A^*(X)_{\mathbb{Q}}$ and $\operatorname{ch}\colon K^\circ(X) \to A^*(X)_{\mathbb{Q}}$ is a ring homomorphism (proposition 7.1.11), so both are written as elements from here on. For $\beta \in A^*(X)_{\mathbb{Q}}$ the symbol β_p denotes the component in $A^p(X)_{\mathbb{Q}} = A_{n-p}(X)_{\mathbb{Q}}$, and $\deg\colon A_0(X)_{\mathbb{Q}} \to \mathbb{Q}$ is the $\mathbb{Q}$-linear extension of the degree of a zero-cycle (definition 1.2.4).

The invariant on the cohomological side is the Euler characteristic

$$\chi(X, \mathscr{F}) \;=\; \sum_{i \geq 0} (-1)^i \dim_{\overline{k}} H^i(X, \mathscr{F})$$

of a coherent sheaf $\mathscr{F}$, a sum in which every dimension is finite ([2, Finite-Dimensionality of Cohomology]) and all but finitely many vanish because X is proper over $\overline{k}$. That is the alternating sum defining the K-theoretic pushforward along the structure morphism ([3, Proper Pushforward]), so Grothendieck-Riemann-Roch for that morphism already has χ on its left side. Extracting the formula is then a matter of computing what K-theory, the Chow ring and the Todd class do on a point.

Theorem 7.3.1: Hirzebruch-Riemann-Roch

Let X be a smooth projective variety of dimension n over $\overline{k}$ and let $\mathscr{F}$ be a coherent sheaf on X. Then

$$\chi(X, \mathscr{F}) \;=\; \deg\big(\mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X)\big)_n$$

the degree of the component of $\mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X)$ in $A^n(X)_{\mathbb{Q}} = A_0(X)_{\mathbb{Q}}$.

***Proof*:**

Step 1: Grothendieck-Riemann-Roch for the structure morphism. Write $P = \operatorname{Spec} \overline{k}$ and let $f\colon X \to P$ be the structure morphism. Then P is a smooth variety of dimension zero, X is a smooth variety, and X is projective over $\overline{k}$, so X is quasi-projective and f is proper. Both X and P are regular and separated, so $K^\circ \to K_\circ$ is an isomorphism on each ([3, The Duality Map]); in particular $\mathscr{F}$ determines a class $\alpha \in K^\circ(X)$ whose Chern character is $\mathrm{ch}(\mathscr{F})$ (corollary 7.1.12). Applying theorem 7.2.2 to f and α gives

$$\mathrm{ch}(f_*\alpha)\,\mathrm{td}(T_P) \;=\; f_*\big(\mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X)\big)$$

in $A^*(P)_{\mathbb{Q}}$.

Step 2: the Chow ring of a point, and $\mathrm{td}(T_P) = 1$. The scheme P is a single point, whose only subvariety is P itself, so $A_0(P) = Z_0(P) = \mathbb{Z}\,[P]$ and $A_k(P) = 0$ for $k \neq 0$ (definition 1.2.4). Hence $A^*(P)_{\mathbb{Q}} = A^0(P)_{\mathbb{Q}} = \mathbb{Q}$, the class $[P]$ corresponding to 1. Every Todd class has $\mathrm{td}_0 = 1$ (proposition 7.1.6), and its components in positive codimension lie in $A^{>0}(P)_{\mathbb{Q}} = 0$, so $\mathrm{td}(T_P) = 1$ and the left side of the displayed identity is $\mathrm{ch}(f_*\alpha)$.

Step 3: the left side is $\chi(X, \mathscr{F})$. A coherent sheaf on P is a finite-dimensional $\overline{k}$-vector space V, and $V \cong \overline{k}^{\dim V}$, so $[V] = (\dim V)\,[\overline{k}]$ in $K_\circ(P)$; conversely $\dim_{\overline{k}}$ is additive on short exact sequences of vector spaces, so it descends to a homomorphism $K_\circ(P) \to \mathbb{Z}$ inverse to $m \mapsto m\,[\overline{k}]$. Thus $\dim_{\overline{k}}$ identifies $K_\circ(P)$, and with it $K^\circ(P)$, with $\mathbb{Z}$. Proper pushforward in K-theory is $f_*[\mathscr{F}] = \sum_i (-1)^i [R^i f_* \mathscr{F}]$ ([3, Proper Pushforward]), and $R^i f_* \mathscr{F}$ is the sheaf on P attached to the vector space $H^i(X, \mathscr{F})$, since $R^i f_*$ is the sheaf associated to $U \mapsto H^i(f^{-1}U, \mathscr{F})$ and P has exactly one nonempty open set. Therefore $\dim_{\overline{k}} f_*\alpha = \chi(X, \mathscr{F})$. Finally $\mathrm{ch}_0(E) = \mathrm{rk} E$ (proposition 7.1.2) while ch_k lands in $A^k(P)_{\mathbb{Q}} = 0$ for $k > 0$, so $\mathrm{ch}\colon K^\circ(P) \to A^*(P)_{\mathbb{Q}} = \mathbb{Q}$ is exactly the map $\dim_{\overline{k}}$, giving $\mathrm{ch}(f_*\alpha) = \chi(X, \mathscr{F})$.

Step 4: the right side is a degree. Proper pushforward of cycles preserves dimension (definition 1.2.1), so f_* annihilates $A_k(X)_{\mathbb{Q}}$ for $k > 0$ because $A_k(P)_{\mathbb{Q}} = 0$ there, and on $A_0(X)_{\mathbb{Q}}$ it is deg (definition 1.2.4). Writing $\beta = \mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X)$, the component β_p lies in $A_{n-p}(X)_{\mathbb{Q}}$, so only β_n contributes and $f_*\beta = \deg(\beta_n)$. Combining this with Step 3 and Step 2 gives $\chi(X, \mathscr{F}) = \deg(\mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X))_n$, as we sought to show.

The left side of the formula is built from the cohomology of $\mathscr{F}$, and the right side is a polynomial in the Chern classes of $\mathscr{F}$ and of T_X evaluated on the fundamental class. Two consequences follow. First, $\chi(X, \mathscr{F})$ depends on $\mathscr{F}$ only through its Chern character, so two coherent sheaves with the same Chern character have the same Euler characteristic however different their individual cohomology groups. Second, only the top component of the product contributes, so on an n-dimensional X the Chern classes c_i with $i > n$ never enter, and the formula is a finite computation with $n+1$ terms. What the theorem does not do is compute the individual dimensions $h^i(X, \mathscr{F})$; separating them out of the alternating sum requires vanishing theorems, which are not part of this book.

In dimension one only two terms survive, and the identity they produce is the theorem for curves. Recall that the genus of a smooth projective curve C over $\overline{k}$ is $g = \dim_{\overline{k}} H^1(C, \mathcal{O}_C)$; Serre duality identifies this with $\dim_{\overline{k}} H^0(C, \omega_C)$ ([2, Serre Duality]), which is the other standard normalization. For a line bundle L on C the class $c_1(L)$ lies in $A^1(C) = A_0(C)$, and $\deg L$ means $\deg c_1(L)$; when $L \cong \mathcal{O}_C(D)$ this is the degree of the divisor D, since $c_1(L) \cap [C] = D \cdot [C] = [D]$ (definition 2.4.1).

Corollary 7.3.2: Riemann-Roch for Curves

Let C be a smooth projective curve of genus g over $\overline{k}$, with canonical bundle $\omega_C = \Omega^1_{C/\overline{k}} = T_C^\vee$ and canonical class $K_C = c_1(\omega_C) \in A^1(C)$. Then:

1. $\deg K_C = 2g - 2$;
2. $\chi(C, L) = \deg L + 1 - g$ for every line bundle L on C.

***Proof*:**
Since $\dim C = 1$, $A^p(C) = A_{1-p}(C) = 0$ for $p \geq 2$. A line bundle has the single Chern root $c_1(L)$, so $\mathrm{ch}(L) = e^{c_1(L)} = 1 + c_1(L)$ in $A^*(C)_{\mathbb{Q}}$ (definition 7.1.1), the terms $c_1(L)^k/k!$ with $k \geq 2$ lying in $A^{\geq 2}(C)_{\mathbb{Q}} = 0$; for the same reason $\mathrm{td}(T_C) = 1 + \frac{1}{2} c_1(T_C)$ (proposition 7.1.6). Multiplying and taking the component in $A^1(C)_{\mathbb{Q}}$,

$$\big(\mathrm{ch}(L)\,\mathrm{td}(T_C)\big)_1 \;=\; c_1(L) + \tfrac{1}{2} c_1(T_C)$$

so theorem 7.3.1 with $n = 1$ gives $\chi(C, L) = \deg L + \frac{1}{2} \deg c_1(T_C)$ for every L.

1. Take $L = \mathcal{O}_C = \mathcal{O}_C(0)$, whose first Chern class is the operator attached to the zero divisor and so vanishes (definition 2.4.1), giving $\deg L = 0$. The displayed formula then reads $\chi(C, \mathcal{O}_C) = \frac{1}{2} \deg c_1(T_C)$. On the other hand $H^0(C, \mathcal{O}_C) = \overline{k}$ by [2, Global Sections of Proper Schemes], whose hypotheses a smooth projective curve over an algebraically closed field meets, being reduced and connected. Hence $\chi(C, \mathcal{O}_C) = 1 - g$ and $\deg c_1(T_C) = 2 - 2g$. Since $\omega_C = T_C^\vee$ gives $K_C = -c_1(T_C)$ (proposition 2.4.4(3)), $\deg K_C = 2g - 2$.
2. Substituting $\frac{1}{2} \deg c_1(T_C) = 1 - g$ from (1) into the displayed formula gives $\chi(C, L) = \deg L + 1 - g$, as we sought to show.

This is the statement [2, Riemann-Roch Theorem] proves cohomologically, in a different notation. There the theorem reads $h^0(C, \mathcal{O}_C(D)) - h^0(C, \mathcal{O}_C(K - D)) = \deg D + 1 - g$ for a divisor D and a canonical divisor K. Serre duality gives $H^1(C, L) \cong H^0(C, \omega_C \otimes L^\vee)^*$ ([2, Serre Duality]), and $\omega_C \otimes L^\vee \cong \mathcal{O}_C(K - D)$ for $L = \mathcal{O}_C(D)$, so the left side there is $h^0(C, L) - h^1(C, L) = \chi(C, L)$ and the two statements agree term by term. The difference is in what each proof uses: the cohomological argument is internal to the curve, whereas corollary 7.3.2 is the one-dimensional case of a formula holding in every dimension, and the genus enters it only as $\frac{1}{2}\deg c_1(T_C)$.

Dimension two brings in the second Todd component, which involves c_2. Stating the result in the classical shape needs the translation between the tangent bundle and the canonical class. For a vector bundle E of rank r,

$$c_1(\det E) = c_1(E), \qquad c_1(E^\vee) = -c_1(E)$$

The second is the case $i = 1$ of corollary 3.2.16, and the first follows from the splitting principle in the same way: on a flat $\pi\colon F \to X$ with π^* injective and π^*E filtered with line-bundle quotients $L_1, \dots, L_r$ one has $\det \pi^*E \cong L_1 \otimes \cdots \otimes L_r$, so proposition 2.4.4(3) and the Whitney formula (theorem 3.2.15) give $c_1(\det \pi^*E) = \sum_i c_1(L_i) = c_1(\pi^*E)$, and π^* commutes with c_1 and with det (proposition 3.2.10(3)). Applied to $\Omega^1_{S/\overline{k}} = T_S^\vee$ on a smooth projective surface S, they give the canonical class

$$K_S \;=\; c_1(\omega_S) \;=\; -c_1(T_S), \qquad \omega_S = \det \Omega^1_{S/\overline{k}}$$

Following the standard surface notation, juxtaposition denotes the product in $A^*(S)$ and a dot between two classes in $A^1(S)$ denotes the degree of their product, $\alpha \cdot \beta = \deg(\alpha\beta) \in \mathbb{Z}$; thus $K_S^2 = K_S \cdot K_S$ is an integer, and for line bundles L, M the number $L \cdot M$ means $c_1(L) \cdot c_1(M)$.

Corollary 7.3.3: Noether's Formula

Let S be a smooth projective surface over $\overline{k}$, and write $c_2(S) = \deg c_2(T_S)$. Then

$$\chi(S, \mathcal{O}_S) \;=\; \frac{1}{12}\big(K_S^2 + c_2(S)\big)$$

***Proof*:**
The structure sheaf is a line bundle with $c_1(\mathcal{O}_S) = 0$ (definition 2.4.1, taking the divisor to be 0), so $\operatorname{ch}(\mathcal{O}_S) = e^0 = 1$ and $\operatorname{ch}(\mathcal{O}_S)\operatorname{td}(T_S) = \operatorname{td}(T_S)$. By proposition 7.1.6 the component of $\operatorname{td}(T_S)$ in $A^2(S)_{\mathbb{Q}}$ is $\frac{1}{12}\big(c_1(T_S)^2 + c_2(T_S)\big)$, so theorem 7.3.1 with $n = 2$ gives

$$\chi(S, \mathcal{O}_S) \;=\; \frac{1}{12}\deg\big(c_1(T_S)^2 + c_2(T_S)\big)$$

Now $c_1(T_S) = -K_S$, so $c_1(T_S)^2 = K_S K_S$ and $\deg c_1(T_S)^2 = K_S \cdot K_S = K_S^2$, while $\deg c_2(T_S) = c_2(S)$ by definition. Hence $\chi(S, \mathcal{O}_S) = \frac{1}{12}(K_S^2 + c_2(S))$, completing the proof.

The left side is a sum of three dimensions of cohomology groups and the right side is a pair of intersection numbers, so the formula constrains which pairs $(K_S^2, c_2(S))$ can occur: their sum is divisible by 12 for every smooth projective surface. Over $\mathbb{C}$ the number $c_2(S)$ is the topological Euler characteristic of $S(\mathbb{C})$, which is the form in which [2, Noether's Formula] is stated there; the identification of the degree of the top Chern class with that Euler characteristic is a topological input, and it is not proved in this book. The version above is the algebraic one and needs no base change to $\mathbb{C}$: $c_2(S)$ is a degree of a zero-cycle over any algebraically closed $\overline{k}$.

Riemann-Roch for a general line bundle on a surface follows by the same computation with $\operatorname{ch}(L)$ in place of 1.

Example 7.3.4: Riemann-Roch for a Line Bundle on a Surface

Let S be a smooth projective surface and L a line bundle on S, and abbreviate $\ell = c_1(L)$. As in corollary 7.3.2, L has the single Chern root ℓ, so $\mathrm{ch}(L) = e^{\ell} = 1 + \ell + \frac{1}{2}\ell^2$, the terms of codimension at least 3 vanishing because $A^p(S) = 0$ for $p \geq 3$. By proposition 7.1.6 the Todd class has $\mathrm{td}_1(T_S) = \frac{1}{2}c_1(T_S) = -\frac{1}{2}K_S$, so the component of the product in $A^2(S)_{\mathbb{Q}}$ is

$$\big(\mathrm{ch}(L)\,\mathrm{td}(T_S)\big)_2 \;=\; \tfrac{1}{2}\ell^2 - \tfrac{1}{2}\ell K_S + \mathrm{td}_2(T_S)$$

the three summands coming from $\mathrm{ch}_2(L)\mathrm{td}_0$, $\mathrm{ch}_1(L)\mathrm{td}_1$ and $\mathrm{ch}_0(L)\mathrm{td}_2$. The proof of corollary 7.3.3 computed $\deg \mathrm{td}_2(T_S) = \frac{1}{12}(K_S^2 + c_2(S))$, which that corollary identifies with $\chi(S, \mathcal{O}_S)$, so taking degrees and applying theorem 7.3.1 on the left gives

$$\chi(S, L) \;=\; \chi(S, \mathcal{O}_S) + \tfrac{1}{2}\, L \cdot (L - K_S)$$

where $L \cdot (L - K_S)$ abbreviates $\deg\big(\ell(\ell - K_S)\big) = L \cdot L - L \cdot K_S$.

Take $S = \mathbb{P}^2$ and $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$, so that h^2 is the class of a point and $\deg h^2 = 1$ (example 7.1.3). The Euler sequence computation recorded in example 7.1.9 gives

$$c(T_{\mathbb{P}^2}) \;=\; (1+h)^3 \;=\; 1 + 3h + 3h^2$$

so $c_1(T_{\mathbb{P}^2}) = 3h$, $K_{\mathbb{P}^2} = -3h$ and $c_2(T_{\mathbb{P}^2}) = 3h^2$. Hence $K_{\mathbb{P}^2}^2 = \deg(9h^2) = 9$ and $c_2(\mathbb{P}^2) = \deg(3h^2) = 3$, and corollary 7.3.3 gives $\chi(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) = \frac{1}{12}(9+3) = 1$. For $L = \mathcal{O}_{\mathbb{P}^2}(d)$ one has $c_1(L) = d\,h$ (example 7.1.3), so $L \cdot L = d^2$ and $L \cdot K_{\mathbb{P}^2} = -3d$, and the formula above reads

$$\chi\big(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)\big) \;=\; 1 + \tfrac{1}{2}\big(d^2 + 3d\big) \;=\; \tfrac{1}{2}(d+1)(d+2)$$

For $d \geq 0$ this is $\binom{d+2}{2}$, the number of monomials of degree d in three variables, and it agrees with the count $\chi(\mathbb{P}^m, \mathcal{O}(n)) = \binom{m+n}{m}$ obtained directly in example 7.2.5.

The surface formula shows that $\chi(S, L)$ depends on L only through the two intersection numbers $L \cdot L$ and $L \cdot K_S$, so two line bundles agreeing in both have the same Euler characteristic. It also constrains those numbers: $\chi(S, L)$ and $\chi(S, \mathcal{O}_S)$ are integers, so $L \cdot (L - K_S)$ is even for every line bundle on every smooth projective surface.

7.3.5 Note: Singular Riemann-Roch Every statement in this section assumes X smooth, and the assumption is used twice: to have a tangent bundle whose Todd class can be formed, and to have $K^\circ(X) = K_\circ(X)$ so that a coherent sheaf has a Chern character. On a singular X neither is available, and $\mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X)$ has no meaning. Baum, Fulton and MacPherson replace the product by a single homomorphism $\tau_X : K_\circ(X) \to A_*(X)_{\mathbb{Q}}$, defined for every quasi-projective scheme X over a field and covariant for proper morphisms in the sense that $\tau_Y \circ f_* = f_* \circ \tau_X$; when X is smooth, $\tau_X([\mathscr{F}]) = \mathrm{ch}(\mathscr{F})\,\mathrm{td}(T_X) \cap [X]$, so that naturality specializes to theorem 7.2.2. The class $\tau_X([\mathcal{O}_X])$ plays the role of the Todd class of a singular variety, its top-dimensional component being $[X]$, and for X proper the degree of $\tau_X([\mathscr{F}])$ is $\chi(X, \mathscr{F})$. The construction rests on localized Chern characters and on MacPherson's graph construction, neither of which is developed here; see [8].

Exercise 7.3.1

1. Let C be a smooth projective curve of genus g over $\overline{k}$ and let E be a vector bundle of rank r on C, with $\deg E = \deg c_1(E)$. Show that
$$\chi(C, E) \;=\; \deg E + r(1-g)$$
and conclude that $\chi(C, E)$ depends on E only through its rank and degree. (Compute $\operatorname{ch}(E)$ in $A^*(C)_{\mathbb{Q}}$ using proposition 7.1.2, and use corollary 7.3.2(1) for $\deg c_1(T_C)$.)

2. Let S be a smooth projective surface and let $C \subseteq S$ be a smooth curve of genus g, regarded as an effective divisor with associated line bundle $\mathcal{O}_S(C)$ and class $c_1(\mathcal{O}_S(C)) \in A^1(S)$. Using the exact sequence
$$0 \longrightarrow \mathcal{O}_S(-C) \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_C \longrightarrow 0$$
the additivity of χ on short exact sequences of coherent sheaves ([2, Additivity of the Euler Characteristic]), the equality $\chi(S, \mathcal{O}_C) = \chi(C, \mathcal{O}_C)$ for a closed immersion, and example 7.3.4, prove the adjunction formula
$$2g - 2 \;=\; C \cdot (C + K_S)$$

3. Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d, so that $\mathcal{O}_{\mathbb{P}^2}(C) \cong \mathcal{O}_{\mathbb{P}^2}(d)$. Using exercise 7.3.1 (2) and the Chern data of $T_{\mathbb{P}^2}$ computed in example 7.3.4, show that the genus of C is $g = \frac{1}{2}(d-1)(d-2)$. Check the answer for $d = 1, 2, 3$ against the genera of a line, a smooth conic and a smooth plane cubic.

4. Let $S = \mathbb{P}^1 \times \mathbb{P}^1$ with projections p_1, p_2, and let $f_i = c_1(p_i^*\mathcal{O}_{\mathbb{P}^1}(1)) \in A^1(S)$, so that $f_1^2 = f_2^2 = 0$, $f_1 f_2$ is the class of a point and $\deg(f_1 f_2) = 1$ (theorem 3.1.9 applied to $S = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1})$). Using $T_S \cong p_1^* T_{\mathbb{P}^1} \oplus p_2^* T_{\mathbb{P}^1}$, the Whitney formula (theorem 3.2.15) and the flat pullback compatibility of Chern classes (proposition 3.2.10(3)), compute K_S^2 and $c_2(S)$, and deduce $\chi(S, \mathcal{O}_S) = 1$ from corollary 7.3.3. Then show that the line bundle L with $c_1(L) = af_1 + bf_2$ has $\chi(S, L) = (a+1)(b+1)$.

5. Let S be a smooth projective surface and L a line bundle on S. Show from example 7.3.4 that
$$\chi\big(S, \omega_S \otimes L^\vee\big) \;=\; \chi(S, L)$$
and explain why Serre duality ([2, Serre Duality]) predicts this equality. (Use proposition 2.4.4(3) to compute $c_1(\omega_S \otimes L^\vee)$.)

6. Let S be a smooth projective surface.
    1. Deduce from corollary 7.3.3 that $K_S^2 + c_2(S)$ is divisible by 12.
    2. Deduce from example 7.3.4 that $L \cdot (L - K_S)$ is even for every line bundle L on S.
    3. Verify both statements for $S = \mathbb{P}^2$ and $L = \mathcal{O}_{\mathbb{P}^2}(d)$.

8

Enumerative Geometry and Degeneracy Loci

Enumerative geometry asks how many geometric objects satisfy a list of conditions, and it is the oldest motivation for everything in this book. How many lines meet four given lines in space; how many conics are tangent to five given conics; how many lines lie on a smooth cubic surface. The nineteenth century answered such questions by a principle of conservation of number, moving the given data into special position, counting there, and asserting that the count was unchanged. The arguments were right often enough to build a subject and wrong often enough to discredit it, and Hilbert's fifteenth problem asked for the foundation that would separate the two.

The Chow ring is that foundation, and this chapter spends it. A counting problem becomes a product of classes in the Chow ring of a parameter space, and the answer is the degree of that product. What the classical arguments could not supply, and what chapter 5 now does, is the guarantee that the product is well defined and independent of the position of the data.

Two parameter spaces carry almost all of the classical problems, and section 8.1 builds the first. The Grassmannian of r-planes has a cell decomposition whose closures, the Schubert varieties, give a free basis of its Chow ring; the multiplication in that basis is Schubert calculus, and section 8.2 makes it explicit enough to compute with. Section 8.3 then treats the general mechanism behind such counts: a condition on a map of bundles cuts out a locus whose class is a determinant in the Chern classes, the Thom–Porteous formula, of which the Schubert classes are the case of a Grassmannian. Section 8.4 returns to the classical questions and answers them, including the twenty-seven lines that [7] found by hand.

A word on what is not here. The five-conic problem is famous partly because the naive parameter space gives the wrong answer, and the repair is a better parameter space rather than a better count. That repair is the beginning of moduli theory, and the chapter says where it goes without building it.

8.1 The Chow Ring of the Grassmannian

Fix an n-dimensional vector space V over $\overline{k}$ and an integer r with $0 < r < n$. This section decomposes $\mathrm{Gr}(r, n)$ into affine cells indexed by the partitions fitting in an $r \times (n-r)$ box, produces from that decomposition a free basis of $A^*(\mathrm{Gr}(r, n))$, and identifies the one-row basis elements with the Chern classes of the tautological quotient bundle.

The Grassmannian $\mathrm{Gr}(r, n)$ parametrizes the r-dimensional subspaces $W \subseteq V$. It is an irreducible projective variety of dimension $r(n-r)$, embedded in $\mathbb{P}(\Lambda^r V)$ by the Plücker map [7, Plücker Embedding], [7, The Grassmannian Is a Projective Variety]. Its construction [7, Grassmannian] provides affine charts, one family for each basis of V: given a basis $e_1, \dots, e_n$ of V and an r-element subset $a = \{a_1 < \cdots < a_r\} \subseteq \{1, \dots, n\}$, put $E_a = \langle e_j \big| j \notin a\rangle$ and

$$U_a = \{W \in \mathrm{Gr}(r, n) \big| W \oplus E_a = V\}$$

an open subset of $\mathrm{Gr}(r, n)$. Every $W \in U_a$ has a unique basis $w_1, \dots, w_r$ of the form $w_i = e_{a_i} + \sum_{j \notin a} c_{ij} e_j$, and $(c_{ij})_{i \le r,\, j \notin a}$ identifies U_a with $\mathbb{A}^{r(n-r)}$. For a fixed basis the $\binom{n}{r}$ charts U_a cover $\mathrm{Gr}(r, n)$, so $\mathrm{Gr}(r, n)$ is smooth of dimension $r(n-r)$. Write $N = r(n-r)$ for the dimension, $\mathcal{O} = \mathcal{O}_{\mathrm{Gr}(r,n)}$, and let

$$0 \to S \to V \otimes \mathcal{O} \to Q \to 0$$

be the tautological sequence, in which the fibre of S at W is W itself and Q has rank $n - r$ ([2, The Tautological Sequence on a Grassmannian]).

Since $\mathrm{Gr}(r, n)$ is smooth, theorem 5.2.7 and proposition 5.3.13 give the intersection product on $A^*(\mathrm{Gr}(r, n)) = \bigoplus_p A^p(\mathrm{Gr}(r, n))$ with $A^p = A_{N-p}$, and Chern classes may be treated as ring elements: $c_i(E)$ denotes $c_i(E) \cap [\mathrm{Gr}(r, n)] \in A^i(\mathrm{Gr}(r, n))$, and capping a class with $c_i(E)$ is multiplication by that element.

A subvariety of $\mathrm{Gr}(r, n)$ can be cut out by conditions on how W meets a fixed collection of subspaces of V. Taking that collection to be a complete flag and prescribing the pattern of intersection dimensions produces a locally closed subset that is an affine space, and letting the pattern vary covers the whole Grassmannian exactly once.

Definition 8.1.1: Schubert Cell and Schubert Variety

A *complete flag* in V is a chain of subspaces $F_\bullet : 0 = F_0 \subset F_1 \subset \cdots \subset F_n = V$ with $\dim F_j = j$. A partition $\lambda = (\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_r \ge 0)$ *lies in the* $r \times (n-r)$ *box* if $\lambda_1 \le n - r$; its size is $|\lambda| = \lambda_1 + \cdots + \lambda_r$, and for such a λ set

$$a_i(\lambda) = n - r + i - \lambda_i, \qquad i = 1, \dots, r$$

Given a complete flag $F_\bullet$ and a partition λ in the box, the *Schubert cell* is

$$\Omega^\circ_\lambda(F_\bullet) = \left\{W \in \mathrm{Gr}(r, n) \;\middle|\; \dim\left(W \cap F_{a_i(\lambda)}\right) = i \text{ and } \dim\left(W \cap F_{a_i(\lambda)-1}\right) = i - 1 \text{ for } i = 1, \dots, r\right\}$$

and the *Schubert variety* $\Omega_\lambda(F_\bullet)$ is its closure in $\mathrm{Gr}(r, n)$.

Because λ is weakly decreasing, $a_i(\lambda) - a_{i-1}(\lambda) = 1 + \lambda_{i-1} - \lambda_i \ge 1$, and $\lambda_1 \le n - r$ forces $a_1(\lambda) \ge 1$ while $\lambda_r \ge 0$ forces $a_r(\lambda) \le n$; so $1 \le a_1(\lambda) < a_2(\lambda) < \cdots < a_r(\lambda) \le n$. The defining conditions say that the function $j \mapsto \dim(W \cap F_j)$, which climbs from 0 to r in steps of 0 or 1, takes its steps exactly at $a_1(\lambda), \dots, a_r(\lambda)$.

For $r = 2$ and $n = 4$ the box is 2×2 and the point of $\mathrm{Gr}(2,4)$ is a line in $\mathbb{P}^3 = \mathbb{P}(V)$. The partition $\lambda = (1,0)$ has $a(\lambda) = (2,4)$, so its cell consists of the W with $\dim(W \cap F_1) = 0$, $\dim(W \cap F_2) = 1$ and $\dim(W \cap F_3) = 1$: the lines that meet the line $\mathbb{P}(F_2)$ in exactly one point, miss the point $\mathbb{P}(F_1)$, and are not contained in the plane $\mathbb{P}(F_3)$. The partition $(2,2)$ has $a(\lambda) = (1,2)$ and its cell is the single point $W = F_2$. The remaining four partitions in the box are $(0,0)$, $(2,0)$, $(1,1)$ and $(2,1)$, and example 8.1.6 works out the whole list with its geometry.

Three properties of these sets carry the rest of the section: each cell is an affine space of dimension $r(n-r) - |\lambda|$, the cells cover $\mathrm{Gr}(r,n)$ without overlap, and the closure of a cell is again cut out by intersection conditions, namely the inequalities obtained by relaxing the equalities in definition 8.1.1.

Proposition 8.1.2: Structure of the Schubert Cells

Let $F_\bullet$ be a complete flag and λ a partition in the $r \times (n-r)$ box, and abbreviate $a_i = a_i(\lambda)$ and $a = \{a_1, \dots, a_r\}$.

1. $\Omega^\circ_\lambda(F_\bullet)$ is a locally closed subvariety of $\mathrm{Gr}(r,n)$ isomorphic to $\mathbb{A}^{N-|\lambda|}$. More precisely, let $e_1, \dots, e_n$ be a basis of V *adapted* to the flag, meaning that $F_j = \langle e_1, \dots, e_j \rangle$ for every j, and let U_a be the chart of that basis attached to a. Then $\Omega^\circ_\lambda(F_\bullet) \subseteq U_a$ and
$$\Omega^\circ_\lambda(F_\bullet) = \{W \in U_a \big| c_{ij} = 0 \text{ for } j > a_i\}$$
the coordinate subspace on the coordinates c_{ij} with $j < a_i$, $j \notin a$.
2. $\mathrm{Gr}(r,n)$ is the disjoint union of the cells $\Omega^\circ_\mu(F_\bullet)$ over the partitions μ in the box.
3. Writing $\lambda \subseteq \mu$ for $\lambda_i \le \mu_i$ for all i,
$$\Omega_\lambda(F_\bullet) = \big\{W \in \mathrm{Gr}(r,n) \big| \dim(W \cap F_{a_i}) \ge i \text{ for } i = 1, \dots, r\big\} = \bigsqcup_{\mu \supseteq \lambda} \Omega^\circ_\mu(F_\bullet)$$
It is an irreducible closed subvariety of codimension $|\lambda|$, and $\Omega^\circ_\lambda(F_\bullet)$ is open and dense in it.

***Proof*:**
Fix a basis $e_1, \dots, e_n$ of V adapted to the flag, that is with $F_j = \langle e_1, \dots, e_j \rangle$ for every j; such a basis exists, since choosing $e_j \in F_j \setminus F_{j-1}$ for each j makes $\langle e_1, \dots, e_j \rangle$ a j-dimensional subspace of F_j. All charts below are the charts of this basis, and adaptedness is the only property of it that the argument uses, so the description in (1) holds for every adapted basis.

For $W \in \mathrm{Gr}(r,n)$ let $J(W) = \{j \big| \dim(W \cap F_j) > \dim(W \cap F_{j-1})\}$. Since $\dim(W \cap F_j)$ increases from 0 at $j = 0$ to r at $j = n$ and each step raises it by at most 1 (the quotient $(W \cap F_j)/(W \cap F_{j-1})$ injects into F_j/F_{j-1}, which is one-dimensional), $J(W)$ has exactly r elements and $\dim(W \cap F_j) = \#\{c \in J(W) \big| c \le j\}$ for every j. Comparing with definition 8.1.1, $W \in \Omega^\circ_\lambda(F_\bullet)$ if and only if $J(W) = \{a_1, \dots, a_r\}$. Moreover $\lambda \mapsto (a_1(\lambda), \dots, a_r(\lambda))$ is a bijection from the partitions in the box to the r-element subsets of $\{1, \dots, n\}$: it lands there by the monotonicity computed after definition 8.1.1, and $\lambda_i = n - r + i - a_i$ recovers λ from a strictly increasing sequence $1 \le a_1 < \cdots < a_r \le n$, the resulting λ being weakly decreasing with $\lambda_1 \le n - r$ and $\lambda_r \ge 0$ precisely because $a_i - a_{i-1} \ge 1$, $a_1 \ge 1$ and $a_r \le n$.

1. Work in the chart U_a of the adapted basis just fixed; since $F_{a_i} = \langle e_1, \ldots, e_{a_i}\rangle$ and $F_{a_i-1} = \langle e_1, \ldots, e_{a_i-1}\rangle$, a vector of F_{a_i} lies outside F_{a_i-1} exactly when its e_{a_i}-coordinate is nonzero. If $W \in \Omega^\circ_\lambda(F_\bullet)$, choose for each i a vector $w_i \in (W \cap F_{a_i}) \setminus F_{a_i-1}$, scale it so that its e_{a_i}-coordinate is 1, and subtract from it the multiples $c\,w_k$ $(k < i)$ that clear its e_{a_k}-coordinates; the resulting vectors satisfy
$$w_i = e_{a_i} + \sum_{j<a_i,\ j\notin a} c_{ij} e_j$$
and are a basis of W, having distinct top indices $a_1 < \cdots < a_r$. In particular $W \oplus E_a = V$, so $W \in U_a$ and (w_i) is the basis of W attached to the chart, whose coordinates satisfy $c_{ij} = 0$ for $j > a_i$. Conversely, if $W \in U_a$ has chart coordinates with $c_{ij} = 0$ for $j > a_i$, then its chart basis has top indices $a_1 < \cdots < a_r$, so any nonzero combination $\sum_i \alpha_i w_i$ has top index $\max\{a_i | \alpha_i \neq 0\}$; hence $\dim(W \cap F_j) = \#\{i | a_i \leq j\}$ and $J(W) = a$. Thus $\Omega^\circ_\lambda(F_\bullet)$ is the closed subvariety of $U_a \cong \mathbb{A}^{r(n-r)}$ cut out by the vanishing of the coordinates c_{ij} with $j > a_i$, that is, a coordinate subspace with coordinates c_{ij} for $j < a_i$, $j \notin a$. Their number is $\sum_{i=1}^r \big((a_i - 1) - (i-1)\big) = \sum_{i=1}^r (n - r - \lambda_i) = N - |\lambda|$, so $\Omega^\circ_\lambda(F_\bullet) \cong \mathbb{A}^{N-|\lambda|}$, locally closed in $\mathrm{Gr}(r, n)$ as a closed subvariety of an open one.

2. Every W has a jump set $J(W)$ of size r, and by the bijection above $J(W) = a(\mu)$ for exactly one partition μ in the box; by the first paragraph this says that W lies in $\Omega^\circ_\mu(F_\bullet)$ and in no other cell.

3. Write $\Sigma_\lambda = \{W \big| \dim(W \cap F_{a_i}) \geq i \text{ for all } i\}$. It is closed: in a chart U_b the composite $W \to V \to V/F_{a_i}$ is given, in the chart basis of W and a basis of V/F_{a_i}, by a matrix whose entries are the chart coordinates or constants, and $\dim(W \cap F_{a_i}) \geq i$ says that this matrix has rank at most $r - i$, which is the vanishing of its $(r-i+1) \times (r-i+1)$ minors. Next, $\Sigma_\lambda = \bigsqcup_{\mu \supseteq \lambda} \Omega^\circ_\mu(F_\bullet)$: by (2) it suffices to see, for $W \in \Omega^\circ_\mu(F_\bullet)$, that $W \in \Sigma_\lambda$ if and only if $\mu \supseteq \lambda$; and indeed $\dim(W \cap F_{a_i(\lambda)}) = \#\{k | a_k(\mu) \leq a_i(\lambda)\}$, which is $\geq i$ if and only if $a_i(\mu) \leq a_i(\lambda)$ (the $a_k(\mu)$ being increasing), if and only if $\mu_i \geq \lambda_i$.

   It remains to prove $\Sigma_\lambda = \overline{\Omega^\circ_\lambda(F_\bullet)}$. The inclusion $\supseteq$ holds because Σ_λ is closed and contains the cell. For $\subseteq$, take $\mu \supseteq \lambda$ in the box; by induction on $|\mu| - |\lambda|$ it suffices to treat the case where μ is obtained from λ by adding one box, since if $\mu \supsetneq \lambda$ and k is the largest index with $\mu_k > \lambda_k$, then $\mu_{k+1} = \lambda_{k+1} \leq \lambda_k \leq \mu_k - 1$ (with $\mu_{r+1} = 0$), so removing one box from row k of μ leaves a partition in the box still containing λ. So let $\mu_k = \lambda_k + 1$ and $\mu_i = \lambda_i$ for $i \neq k$, and let $W \in \Omega^\circ_\mu(F_\bullet)$ have a basis $w_1, \ldots, w_r$ with w_i of top index $a_i(\mu)$, as in (1). Here $a_i(\mu) = a_i(\lambda)$ for $i \neq k$ and $a_k(\mu) = a_k(\lambda) - 1$. For $t \in \mathbb{A}^1$ set
$$W_t = \langle w_1, \ldots, w_{k-1},\ t\, e_{a_k(\lambda)} + w_k,\ w_{k+1}, \ldots, w_r\rangle$$
   For every t the listed vectors are independent: for $t \neq 0$ their top indices are $a_1(\lambda), \ldots, a_r(\lambda)$, which are distinct, and for $t = 0$ they are the w_i. Hence $t \mapsto W_t$ is a morphism $\mathbb{A}^1 \to \mathrm{Gr}(r, n)$, its Plücker coordinates being the $r \times r$ minors of a matrix with entries linear in t, not all vanishing for any t. By the criterion in (1), $W_t \in \Omega^\circ_\lambda(F_\bullet)$ for $t \neq 0$, while $W_0 = W$. The preimage of the closed set $\overline{\Omega^\circ_\lambda(F_\bullet)}$ under this morphism is closed in $\mathbb{A}^1$ and contains the dense subset $\mathbb{A}^1 \setminus \{0\}$, so it is all of $\mathbb{A}^1$ and $W = W_0 \in \overline{\Omega^\circ_\lambda(F_\bullet)}$.

   Finally, $\Omega_\lambda(F_\bullet)$ is the closure of the irreducible set $\Omega^\circ_\lambda(F_\bullet) \cong \mathbb{A}^{N-|\lambda|}$, hence irreducible of dimension $N - |\lambda|$, that is, of codimension $|\lambda|$; the cell is dense in it by construction, and open in it because its complement $\bigsqcup_{\mu\supsetneq\lambda}\Omega^\circ_\mu(F_\bullet)$ is the union of the closed sets Σ_μ over the partitions μ obtained from λ by adding a single box, as we sought to show.

The proposition converts a question about $\mathrm{Gr}(r,n)$ into a question about partitions. Each Schubert variety is the closure of one cell, indexed by a partition whose size is its codimension, and the containment order on partitions is the containment order on Schubert varieties. At the two ends of that order, $\lambda=(0,\dots,0)$ gives the open cell and $\Omega_\lambda=\mathrm{Gr}(r,n)$, while the full box $\lambda=(n-r,\dots,n-r)$ has $a_i(\lambda)=i$ and gives the single point $W=F_r$.

Proposition 8.1.3: Duality of Schubert Classes

Call two complete flags $F_\bullet$ and $G_\bullet$ *transverse* if $F_a\cap G_b=0$ whenever $a+b\le n$. For a partition λ in the $r\times(n-r)$ box let

$$\lambda^\vee=(n-r-\lambda_r,\ n-r-\lambda_{r-1},\ \dots,\ n-r-\lambda_1)$$

the *complementary partition*, again in the box, with $|\lambda|+|\lambda^\vee|=N$ and $(\lambda^\vee)^\vee=\lambda$. Let $F_\bullet,G_\bullet$ be transverse and let λ,μ be partitions in the box with $|\lambda|+|\mu|\ge N$. Then $\Omega_\lambda(F_\bullet)\cap\Omega_\mu(G_\bullet)$ is empty unless $|\lambda|+|\mu|=N$ and $\mu=\lambda^\vee$, in which case it is a single reduced point at which both Schubert varieties are smooth with complementary tangent spaces. Consequently, for $|\lambda|+|\mu|=N$,

$$\deg\big([\Omega_\lambda(F_\bullet)]\cdot[\Omega_\mu(G_\bullet)]\big)=\begin{cases}1 & \mu=\lambda^\vee\\ 0 & \mu\ne\lambda^\vee\end{cases}$$

***Proof*:**
Step 1: an adapted basis. For each i the space $F_i\cap G_{n-i+1}$ is one-dimensional: it is nonzero because $\dim F_i+\dim G_{n-i+1}=n+1$, and it meets the hyperplane G_{n-i} of G_{n-i+1} in $F_i\cap G_{n-i}=0$, so it has dimension at most 1. Choose a generator e_i. If $\sum_{k\le i}\alpha_ke_k=0$ with some $\alpha_k\ne0$ and k_0 is the largest index with $\alpha_{k_0}\ne0$, then $e_{k_0}\in\langle e_1,\dots,e_{k_0-1}\rangle\subseteq F_{k_0-1}$ while $e_{k_0}\in G_{n-k_0+1}$, and $F_{k_0-1}\cap G_{n-k_0+1}=0$ forces $e_{k_0}=0$, a contradiction. So $e_1,\dots,e_n$ is a basis, $F_j=\langle e_1,\dots,e_j\rangle$ by dimension count, and $G_j\supseteq\langle e_{n-j+1},\dots,e_n\rangle$ with equality for the same reason.

Step 2: the partitions must be complementary. Let $W\in\Omega_\lambda(F_\bullet)\cap\Omega_\mu(G_\bullet)$. Put $A_i=F_{a_i(\lambda)}$ and $B_i=G_{a_i(\mu)}$, so that by proposition 8.1.2(3) $\dim(W\cap A_i)\ge i$ and $\dim(W\cap B_i)\ge i$ for all i. Fix i and apply this to A_i and B_{r+1-i}: inside the r-dimensional space W the subspaces $W\cap A_i$ and $W\cap B_{r+1-i}$ have dimensions summing to at least $r+1$, so they meet nontrivially, whence $A_i\cap B_{r+1-i}\ne0$. By Step 1, $F_c\cap G_d=0$ as soon as $c+d\le n$, so

$$a_i(\lambda)+a_{r+1-i}(\mu)\ge n+1$$

which upon substituting $a_i(\lambda)=n-r+i-\lambda_i$ and $a_{r+1-i}(\mu)=n+1-i-\mu_{r+1-i}$ reads $\lambda_i+\mu_{r+1-i}\le n-r$. Summing over i gives $|\lambda|+|\mu|\le N$. So the intersection is empty when $|\lambda|+|\mu|>N$; and when $|\lambda|+|\mu|=N$ all r of the inequalities are equalities, $\mu_{r+1-i}=n-r-\lambda_i$ for every i, which is $\mu=\lambda^\vee$.

Step 3: the intersection is one point. Assume $\mu=\lambda^\vee$ and write $a_i=a_i(\lambda)$. Then $a_i+a_{r+1-i}(\mu)=n+1$, so with the basis of Step 1, $A_i\cap B_{r+1-i}=\langle e_1,\dots,e_{a_i}\rangle\cap\langle e_{a_i},\dots,e_n\rangle=\langle e_{a_i}\rangle$. The argument of Step 2 produced a nonzero vector of W inside $A_i\cap B_{r+1-i}$, so $e_{a_i}\in W$ for every i; as the

a_i are distinct, W is the single subspace $W_\lambda := \langle e_{a_1}, \ldots, e_{a_r}\rangle$. Conversely W_λ does lie in both Schubert varieties, since $\dim(W_\lambda \cap F_j) = \#\{i \big| a_i \le j\}$ shows $W_\lambda \in \Omega^\circ_\lambda(F_\bullet)$, and the computation of Step 4 below identifies W_λ as a point of $\Omega^\circ_{\lambda^\vee}(G_\bullet)$ as well.

Step 4: transversality at that point. Work in the chart U_a of the basis $e_\bullet$ of Step 1, with $a = \{a_1, \ldots, a_r\}$; that basis is adapted to $F_\bullet$, and the chart contains W_λ, the origin of its coordinates c_{ij}. By proposition 8.1.2(1),

$$\Omega^\circ_\lambda(F_\bullet) \cap U_a = \{c_{ij} = 0 \text{ for } j > a_i\}$$

Reindex by the basis $f_j = e_{n+1-j}$, for which $G_b = \langle f_1, \ldots, f_b\rangle$. The subset of $\{1, \ldots, n\}$ attached to W_λ in the f-basis is $a' = \{n+1-a_i\}$, that is, $a'_k = n+1-a_{r+1-k}$, and $n+1-a_{r+1-k} = k + \lambda_{r+1-k} = a_k(\lambda^\vee)$; so $W_\lambda \in \Omega^\circ_{\lambda^\vee}(G_\bullet)$, completing Step 3, and $U_{a'} = U_a$ because $\langle f_j \big| j \notin a'\rangle = \langle e_m \big| m \notin a\rangle$. The chart bases match up as $w'_k = w_{r+1-k}$, so $c'_{kj} = c_{r+1-k,\, n+1-j}$, and proposition 8.1.2(1), applied to $\lambda^\vee$ and $G_\bullet$ in the f-basis, which is adapted to $G_\bullet$, gives

$$\Omega^\circ_{\lambda^\vee}(G_\bullet) \cap U_a = \{c_{ij} = 0 \text{ for } j < a_i\}$$

The two cells are therefore complementary coordinate subspaces of $U_a \cong \mathbb{A}^N$, meeting exactly at the origin and with tangent spaces there intersecting in 0. Each cell is open in its Schubert variety (proposition 8.1.2(3)), so $\Omega_\lambda(F_\bullet)$ and $\Omega_{\lambda^\vee}(G_\bullet)$ are smooth at W_λ with complementary tangent spaces, and their codimensions $|\lambda|$ and $|\lambda^\vee|$ sum to N. Proposition 5.3.12 now gives $\deg([\Omega_\lambda(F_\bullet)] \cdot [\Omega_{\lambda^\vee}(G_\bullet)]) = 1$, while for $\mu \neq \lambda^\vee$ with $|\lambda| + |\mu| = N$ the intersection is empty and the product vanishes, completing the proof.

Taking the degree of the product with $[\Omega_{\lambda^\vee}(G_\bullet)]$ therefore extracts the coefficient of $[\Omega_\lambda(F_\bullet)]$ from a class, one coefficient at a time, and this is what turns the cell decomposition into a basis. Generation comes from the cells and the localization sequence; independence comes from the pairing just computed; and the same pairing shows that the class of a Schubert variety does not depend on the flag used to define it, so the notation σ_λ below is unambiguous.

Theorem 8.1.4: The Chow Ring of the Grassmannian

For a partition λ in the $r \times (n-r)$ box, the class $[\Omega_\lambda(F_\bullet)] \in A^{|\lambda|}(\mathrm{Gr}(r,n))$ is independent of the complete flag $F_\bullet$; write σ_λ for it, the *Schubert class* of λ. For each p the group $A^p(\mathrm{Gr}(r,n))$ is free abelian with basis $\{\sigma_\lambda \big| |\lambda| = p\}$, so $A^*(\mathrm{Gr}(r,n))$ is free abelian of total rank $\binom{n}{r}$. Moreover $\sigma_{(0,\ldots,0)} = 1$, the class $\sigma_{(n-r,\ldots,n-r)}$ is the class of a point, and $\deg\colon A^N(\mathrm{Gr}(r,n)) \to \mathbb{Z}$ is an isomorphism.

***Proof*:**
Fix a complete flag $F_\bullet$ and abbreviate $\Omega_\mu = \Omega_\mu(F_\bullet)$, $\Omega^\circ_\mu = \Omega^\circ_\mu(F_\bullet)$.

Step 1: the classes $[\Omega_\mu]$ generate. For $0 \le p \le N+1$ set $Y_p = \bigcup_{|\mu|=p} \Omega_\mu$, a closed subset by proposition 8.1.2(3), with $Y_0 = \mathrm{Gr}(r,n)$ and $Y_{N+1} = \emptyset$. By proposition 8.1.2(2) and (3),

$$Y_p = \bigsqcup_{|\nu| \ge p} \Omega^\circ_\nu$$

since Ω_μ is the union of the cells indexed by partitions containing μ, and conversely every ν with $|\nu| \geq p$ contains a partition of size exactly p (remove boxes one at a time from the last nonempty row). In particular $Y_{p+1} \subseteq Y_p$ and $Y_p \setminus Y_{p+1} = \bigsqcup_{|\mu|=p} \Omega^\circ_\mu$, a disjoint union of copies of $\mathbb{A}^{N-p}$ by proposition 8.1.2(1). A cycle on a disjoint union is the sum of its restrictions, so $A_k(Y_p \setminus Y_{p+1})$ is 0 for $k \neq N-p$ and is free on the classes $[\Omega^\circ_\mu]$, $|\mu| = p$, for $k = N-p$ (proposition 1.5.1).

Claim: $A_k(Y_p)$ is generated by the classes $[\Omega_\mu]$ with $|\mu| = N-k$ and $|\mu| \geq p$, an empty list (so that $A_k(Y_p) = 0$) when $N-k < p$. This holds for $p = N+1$. Assume it for $p+1$ and apply the localization sequence (theorem 1.4.1)

$$A_k(Y_{p+1}) \to A_k(Y_p) \to A_k(Y_p \setminus Y_{p+1}) \to 0$$

in which the first map is pushforward along a closed immersion and the second is restriction of cycles. If $k \neq N-p$ the third group vanishes, so $A_k(Y_p)$ is the image of $A_k(Y_{p+1})$, which by induction is generated by the $[\Omega_\mu]$ with $|\mu| = N-k \geq p+1$, and these classes push forward to the classes of the same subvarieties. If $k = N-p$, then $N-k = p < p+1$, so $A_k(Y_{p+1}) = 0$ by induction and the restriction map is an isomorphism; each generator $[\Omega^\circ_\mu]$ with $|\mu| = p$ is the restriction of $[\Omega_\mu]$, because $\Omega_\mu \setminus Y_{p+1} = \Omega^\circ_\mu$ by the displayed description of Y_{p+1} together with proposition 8.1.2(3). This proves the claim, and $p = 0$ gives that $A_k(\mathrm{Gr}(r,n))$ is generated by the classes $[\Omega_\mu]$ with $|\mu| = N-k$.

Step 2: they are independent. A flag transverse to a given flag $F_\bullet$ exists: build $G_\bullet$ by choosing $u_k \notin F_{n-k}+G_{k-1}$ and setting $G_k = G_{k-1}+\langle u_k\rangle$, which is possible since $\dim(F_{n-k}+G_{k-1}) \leq n-1$; then $G_k \cap F_{n-k} = 0$, for if $g+\alpha u_k \in F_{n-k}$ with $g \in G_{k-1}$ then $\alpha \neq 0$ would put $u_k \in F_{n-k}+G_{k-1}$, while $\alpha = 0$ gives $g \in G_{k-1} \cap F_{n-k} \subseteq G_{k-1} \cap F_{n-k+1} = 0$ by induction. Fix such a $G_\bullet$ and suppose $\sum_{|\lambda|=p} c_\lambda[\Omega_\lambda] = 0$ in $A^p(\mathrm{Gr}(r,n))$. Multiplying by $[\Omega_{\lambda_0^\vee}(G_\bullet)]$ for a partition λ_0 of size p and taking degrees, proposition 8.1.3 leaves the single term c_{λ_0}, so every coefficient vanishes. With Step 1 this shows that $A^p(\mathrm{Gr}(r,n)) = A_{N-p}(\mathrm{Gr}(r,n))$ is free with basis $\{[\Omega_\lambda(F_\bullet)] \,|\, |\lambda| = p\}$.

Step 3: independence of the flag. Let $F_\bullet$ and $F'_\bullet$ be complete flags. A flag $G_\bullet$ transverse to both exists by the construction of Step 2, run with $u_k \notin (F_{n-k}+G_{k-1}) \cup (F'_{n-k}+G_{k-1})$: a vector space is never the union of two proper subspaces, since for $u \in A \setminus B$ and $v \in B \setminus A$ the vector $u+v$ lies in neither. By Step 2 applied to $F'_\bullet$, write $[\Omega_\lambda(F_\bullet)] = \sum_{|\mu|=|\lambda|} c_\mu[\Omega_\mu(F'_\bullet)]$. Multiplying by $[\Omega_\nu(G_\bullet)]$ with $|\nu| = N-|\lambda|$ and taking degrees, proposition 8.1.3 gives $\delta_{\nu,\lambda^\vee} = c_{\nu^\vee}$ on the two sides. As $\nu \mapsto \nu^\vee$ is a bijection of the partitions of size $N-|\lambda|$ onto those of size $|\lambda|$, with $\nu = \lambda^\vee$ exactly when $\nu^\vee = \lambda$, this says $c_\mu = \delta_{\mu\lambda}$, that is, $[\Omega_\lambda(F_\bullet)] = [\Omega_\lambda(F'_\bullet)]$.

Step 4: the remaining assertions. The partitions in the box biject with the r-element subsets of $\{1,\ldots,n\}$ (proof of proposition 8.1.2), so the total rank is $\binom{n}{r}$. For $\lambda = (0,\ldots,0)$ the Schubert variety is $\mathrm{Gr}(r,n)$ itself, whose class is the identity of $A^*(\mathrm{Gr}(r,n))$, and for $\lambda = (n-r,\ldots,n-r)$ it is a single point. Hence $A^N(\mathrm{Gr}(r,n))$ is free of rank one on the class of a point, whose degree is 1 (definition 1.2.4), so deg is an isomorphism onto $\mathbb{Z}$, completing the proof.

The Chow ring of $\mathrm{Gr}(r,n)$ is thus free of rank $\binom{n}{r}$ on classes indexed by combinatorial data, graded so that σ_λ sits in codimension $|\lambda|$, and equipped with a perfect pairing $A^p \times A^{N-p} \to \mathbb{Z}$ whose matrix in the Schubert bases is a permutation matrix. An enumerative problem posed as an intersection of Schubert conditions is therefore a finite computation as soon as the products $\sigma_\lambda \cdot \sigma_\mu$ are known in the basis. The next proposition supplies the first family of such products indirectly, by identifying the classes of the one-row partitions with Chern classes, which are already subject to the Whitney formula.

Write $\sigma_m = \sigma_{(m,0,\dots,0)}$ for $0 \le m \le n-r$; these are the *special* Schubert classes. Only the first of the defining conditions is a restriction: for $\lambda = (m, 0, \dots, 0)$ and $i \ge 2$ one has $a_i(\lambda) = n - r + i$ and $\dim(W \cap F_{n-r+i}) \ge r + (n-r+i) - n = i$ automatically, so proposition 8.1.2(3) gives

$$\Omega_{(m)}(F_\bullet) = \{W \big| W \cap F_{n-r+1-m} \ne 0\}$$

a condition on a single subspace of V.

Proposition 8.1.5: Special Schubert Classes as Chern Classes

In $A^*(\mathrm{Gr}(r,n))$ the tautological sub-bundle and quotient bundle satisfy $c(S)\,c(Q) = 1$, and

$$c_m(Q) = \sigma_m \qquad \text{for } 0 \le m \le n-r$$

so that $c(Q) = 1 + \sigma_1 + \cdots + \sigma_{n-r}$ and $c(S) = (1 + \sigma_1 + \cdots + \sigma_{n-r})^{-1}$.

***Proof*:**
Step 1: the two total Chern classes are inverse. The trivial bundle $V \otimes \mathcal{O}$ has $c_i(V \otimes \mathcal{O}) = 0$ for $i > 0$ (example 3.2.12), so the Whitney formula (theorem 3.2.15) applied to the tautological sequence gives $c(S)\,c(Q) = c(V \otimes \mathcal{O}) = 1$.

Step 2: a resolution of the special Schubert variety. Let $\pi\colon \mathbb{P}(S) \to \mathrm{Gr}(r,n)$ be the bundle of lines in S (box 3.2.5; [2, Projective Bundle] uses the quotient convention, so the import dualizes, box 0.0.2), whose points are the pairs $(W, [v])$ with $0 \ne v \in W$, and let $\mathcal{O}_{\mathbb{P}(S)}(-1) \subseteq \pi^* S$ be the tautological sub-line-bundle ([2, The Tautological Quotient]); put $\xi = c_1(\mathcal{O}_{\mathbb{P}(S)}(1))$. Composing $S \hookrightarrow V \otimes \mathcal{O}$ with the tautological inclusion gives a morphism

$$q\colon \mathbb{P}(S) \to \mathbb{P}(V), \qquad (W, [v]) \mapsto [v]$$

with $q^*\mathcal{O}_{\mathbb{P}(V)}(-1) = \mathcal{O}_{\mathbb{P}(S)}(-1)$, both being the line $\langle v \rangle$; hence $\xi = q^* h$ for $h = c_1(\mathcal{O}_{\mathbb{P}(V)}(1))$. The map q is flat: fixing $\psi \in V^\vee$ and writing $V' = \ker\psi$, every $[v]$ with $\psi(v) \ne 0$ has the normalized representative $v/\psi(v)$, which depends algebraically on $[v]$ and splits $V = \langle v \rangle \oplus V'$, so that $W \mapsto W/\langle v \rangle \subseteq V/\langle v \rangle \cong V'$ identifies $q^{-1}(\{\psi \ne 0\})$ with $\{\psi \ne 0\} \times \mathrm{Gr}(r-1, V')$ over $\{\psi \ne 0\}$.

For a nonzero subspace $L \subseteq V$ set $\Sigma_L = q^{-1}(\mathbb{P}(L)) = \{(W,[v]) \big| v \in L \cap W\}$. By the local description just given it is a Zariski-locally trivial bundle over $\mathbb{P}(L)$ with fibre $\mathrm{Gr}(r-1, n-1)$, hence an irreducible variety of dimension $(\dim L - 1) + (r-1)(n-r)$, and $q^*[\mathbb{P}(L)] = [\Sigma_L]$ by flat pullback of cycles (definition 1.3.3). Furthermore $[\mathbb{P}(L)] = h^{n-\dim L} \cap [\mathbb{P}(V)]$: for a hyperplane $L' \subset L$, choose $\psi \in V^\vee$ with $\ker\psi \cap L = L'$; then ψ is a section of $\mathcal{O}_{\mathbb{P}(V)}(1)$ whose restriction to $\mathbb{P}(L)$ is a section whose divisor of zeros is $\mathbb{P}(L')$ with multiplicity one, since in each standard affine chart of $\mathbb{P}(L)$ it is a nonzero linear polynomial and so generates a prime ideal, so $h \cap [\mathbb{P}(L)] = [\mathbb{P}(L')]$ by the construction of the first Chern class operator (definition 2.4.1), and $[\mathbb{P}(V)] = h^0 \cap [\mathbb{P}(V)]$ starts the induction (theorem 1.5.2). Since Chern class operators commute with flat pullback (proposition 3.2.10) and $q^*[\mathbb{P}(V)] = [\mathbb{P}(S)]$,

$$\xi^{n-\dim L} \cap [\mathbb{P}(S)] = q^*\big(h^{n-\dim L} \cap [\mathbb{P}(V)]\big) = [\Sigma_L] \tag{8.1}$$

Step 3: pushing forward. Fix $0 \le m \le n-r$, put $\ell = n-r+1-m$ and $L = F_\ell$ for a complete flag $F_\bullet$, so that $n - \dim L = r-1+m$ and, by the description of $\Omega_{(m)}(F_\bullet)$ preceding this proposition, $\pi(\Sigma_L) = \Omega_{(m)}(F_\bullet)$. The map π is proper, so $\pi|_{\Sigma_L}$ is proper; the two varieties have the same dimension, namely $(\ell-1)+(r-1)(n-r) = N-m$; and over a point W of the dense open cell $\Omega^\circ_{(m)}(F_\bullet)$, where $\dim(W \cap F_\ell) = 1$, the fibre is the reduced point $\mathbb{P}(W \cap F_\ell)$, of length one. Hence $\pi|_{\Sigma_L}$ is birational onto its image and

$$\pi_*[\Sigma_L] = [\Omega_{(m)}(F_\bullet)] = \sigma_m$$

by definition 1.2.1 and theorem 8.1.4. Combining with (8.1), $\sigma_m = \pi_*\big(\xi^{\,r-1+m} \cap [\mathbb{P}(S)]\big)$ for $0 \le m \le n-r$.

Step 4: identifying the pushforwards. For $j \in \mathbb{Z}$ set $p_j = \pi_*\big(\xi^{\,r-1+j} \cap [\mathbb{P}(S)]\big)$, with $p_j = 0$ for $j < 0$: indeed for $0 \le k < r-1$ the class $\xi^k \cap [\mathbb{P}(S)]$ has dimension $\dim \mathbb{P}(S) - k = N+r-1-k > N$, and $A_d(\mathrm{Gr}(r,n)) = 0$ for $d > N$. Step 3 gives $p_m = \sigma_m$ for $0 \le m \le n-r$, and in particular $p_0 = \sigma_0 = 1$. Being a projective bundle over a smooth variety, $\mathbb{P}(S)$ is itself smooth, so theorem 5.2.7 supplies its Chow ring as well. The tautological sequence on $\mathbb{P}(S)$ reads $0 \to \mathcal{O}_{\mathbb{P}(S)}(-1) \to \pi^*S \to Q' \to 0$ with Q' of rank $r-1$, so by theorem 3.2.15 and $c(\mathcal{O}_{\mathbb{P}(S)}(-1)) = 1-\xi$ (proposition 2.4.4),

$$c(Q') = \pi^*c(S)\cdot(1-\xi)^{-1} = \pi^*c(S)\cdot(1+\xi+\xi^2+\cdots)$$

the inverse making sense because ξ is nilpotent in $A^*(\mathbb{P}(S))$. The rank of Q' is $r-1$, so $c_r(Q') = 0$ (theorem 3.2.9), which is the relation

$$\sum_{i=0}^{r} \pi^*c_i(S)\cdot \xi^{\,r-i} = 0$$

Multiplying it by $\xi^{\,j-1}$ for $j \ge 1$, capping with $[\mathbb{P}(S)]$ and pushing forward by π, the projection formula (proposition 3.2.10) gives

$$0 = \sum_{i=0}^{r} c_i(S)\cdot \pi_*\big(\xi^{\,r-i+j-1} \cap [\mathbb{P}(S)]\big) = \sum_{i=0}^{r} c_i(S)\cdot p_{j-i}$$

since $r-i+j-1 = r-1+(j-i)$. Together with $p_0 = 1$ these relations say that the class $p = \sum_{j\ge 0} p_j$ satisfies $c(S)\cdot p = 1$, and $c(S)$ is invertible in $A^*(\mathrm{Gr}(r,n))$ because it is 1 plus a nilpotent class. Step 1 gives $c(S)\cdot c(Q) = 1$ as well, so $p = c(Q)$ and $p_m = c_m(Q)$ for every m. With Step 3, $\sigma_m = c_m(Q)$ for $0 \le m \le n-r$, and since $c_m(Q) = 0$ for $m > n-r$ the total class is $c(Q) = 1 + \sigma_1 + \cdots + \sigma_{n-r}$, completing the proof.

The single-condition Schubert classes are therefore characteristic classes of the tautological bundles, and as such they obey the Whitney formula: $c(S)c(Q) = 1$ is a family of relations among Schubert classes, and it determines all of them once the products of the special classes with arbitrary Schubert classes are known. In $\mathrm{Gr}(2,4)$ the whole ring is already visible from what has been proved.

Example 8.1.6: Lines in Projective Three-Space

Take $r = 2$ and $n = 4$, so that $N = 4$ and a point of $\mathrm{Gr}(2, 4)$ is a line in $\mathbb{P}^3 = \mathbb{P}(V)$. The 2×2 box holds six partitions, and proposition 8.1.2(3) turns each into an incidence condition on a line $\mathbb{P}(W) \subseteq \mathbb{P}^3$ relative to the flag $\mathbb{P}(F_1) \subset \mathbb{P}(F_2) \subset \mathbb{P}(F_3)$, a point on a line in a plane.

1. $\lambda = (0, 0)$: $\Omega_\lambda = \mathrm{Gr}(2, 4)$, and $\sigma_{(0,0)} = 1$.
2. $\lambda = (1, 0)$, $a = (2, 4)$: the conditions are $\dim(W \cap F_2) \geq 1$ and $\dim(W \cap F_4) \geq 2$, the second vacuous, so this is the locus of lines meeting $\mathbb{P}(F_2)$; codimension 1.
3. $\lambda = (2, 0)$, $a = (1, 4)$: the conditions reduce to $F_1 \subseteq W$, the lines through the point $\mathbb{P}(F_1)$; codimension 2.
4. $\lambda = (1, 1)$, $a = (2, 3)$: the conditions are $\dim(W \cap F_2) \geq 1$ and $\dim(W \cap F_3) \geq 2$, the second of which says $W \subseteq F_3$ and implies the first: the lines inside the plane $\mathbb{P}(F_3)$; codimension 2.
5. $\lambda = (2, 1)$, $a = (1, 3)$: the conditions are $F_1 \subseteq W \subseteq F_3$, the pencil of lines through $\mathbb{P}(F_1)$ inside $\mathbb{P}(F_3)$, a copy of $\mathbb{P}^1$; codimension 3.
6. $\lambda = (2, 2)$, $a = (1, 2)$: the conditions force $W = F_2$, the single line $\mathbb{P}(F_2)$; codimension 4.

By theorem 8.1.4 the Chow ring is free of rank $6 = \binom{4}{2}$ on these classes, with $A^0 = \mathbb{Z}$, $A^1 = \mathbb{Z}\sigma_1$, $A^2 = \mathbb{Z}\sigma_2 \oplus \mathbb{Z}\sigma_{1,1}$, $A^3 = \mathbb{Z}\sigma_{2,1}$ and $A^4 = \mathbb{Z}\sigma_{2,2}$. Complementation acts by $(0, 0)^\vee = (2, 2)$, $(1, 0)^\vee = (2, 1)$, $(2, 0)^\vee = (2, 0)$ and $(1, 1)^\vee = (1, 1)$, so proposition 8.1.3 gives $\deg(\sigma_1\sigma_{2,1}) = 1$, $\deg(\sigma_2^2) = \deg(\sigma_{1,1}^2) = 1$ and $\deg(\sigma_2\sigma_{1,1}) = 0$. Since A^4 is free on the class of a point, this already determines $\sigma_2^2 = \sigma_{1,1}^2 = \sigma_{2,2}$ and $\sigma_2\sigma_{1,1} = 0$.

The product $\sigma_1^2 = a\,\sigma_2 + b\,\sigma_{1,1}$ is fixed by the two numbers $a = \deg(\sigma_1^2\sigma_2)$ and $b = \deg(\sigma_1^2\sigma_{1,1})$, again by proposition 8.1.3. For the first, choose F_2 and G_2 with $V = F_2 \oplus G_2$, bases $F_2 = \langle u_0, u_1\rangle$ and $G_2 = \langle v_0, v_1\rangle$, and the point $p = [u_0 + v_0]$. The loci $\{W' \big| W' \cap F_2 \neq 0\}$, $\{W' \big| W' \cap G_2 \neq 0\}$ and $\{W' \big| u_0 + v_0 \in W'\}$ are Schubert varieties for complete flags through F_2, through G_2 and through $\langle u_0 + v_0\rangle$, so by theorem 8.1.4 their classes are σ_1, σ_1 and σ_2. A line $\mathbb{P}(W)$ meeting $\mathbb{P}(F_2)$ and $\mathbb{P}(G_2)$ and passing through p has $\dim(W \cap F_2) = 1$, since $\dim(W \cap F_2) = 2$ would give $W = F_2$ and then $v_0 \in F_2 \cap G_2 = 0$, because $u_0 + v_0 \in F_2$; writing $W \cap F_2 = \langle u\rangle$ and $W \cap G_2 = \langle v\rangle$, these are independent and $W = \langle u, v\rangle$. Expanding $u_0 + v_0 = xu + yv$ and comparing components in $V = F_2 \oplus G_2$ gives $xu = u_0$ and $yv = v_0$, so $W = \langle u_0, v_0\rangle$ is the unique solution. At that point, use the chart attached to the basis u_0, u_1, v_0, v_1 and the index set picking out u_0 and v_0: it identifies the subspaces complementary to $E = \langle u_1, v_1\rangle$ with $\mathrm{Hom}(W, E)$, a homomorphism φ corresponding to its graph $W_\varphi = \langle u_0 + \varphi(u_0),\ v_0 + \varphi(v_0)\rangle$ and W itself to $\varphi = 0$. Write $\varphi(u_0) = \alpha u_1 + \beta v_1$ and $\varphi(v_0) = \gamma u_1 + \delta v_1$. In this chart each of the three conditions is a system of linear equations in $\alpha, \beta, \gamma, \delta$. A combination $s(u_0 + \varphi(u_0)) + t(v_0 + \varphi(v_0))$ lies in $F_2 = \langle u_0, u_1\rangle$ exactly when $t = 0$ and $s\beta = 0$, so $W_\varphi \cap F_2 \neq 0$ if and only if $\beta = 0$; exchanging the roles of the u and v vectors gives $W_\varphi \cap G_2 \neq 0$ if and only if $\gamma = 0$; and $u_0 + v_0$ lies in W_φ only for $s = t = 1$, hence exactly when $\alpha + \gamma = 0$ and $\beta + \delta = 0$. The three loci are therefore linear subspaces of the chart, so they are smooth at W and equal to their own tangent spaces there, and the equations $\beta = 0$, $\gamma = 0$ and $\alpha + \gamma = \beta + \delta = 0$ have only the solution $\varphi = 0$, so the three tangent spaces meet in zero. As the three codimensions $1 + 1 + 2$ sum to 4, the transversality proposition 5.3.12 gives $a = 1$.

For the second, keep F_2 and G_2 and choose the plane $F_3' = \langle u_0, v_0, u_1 + v_1\rangle$, which meets F_2 in

$\langle u_0\rangle$ and G_2 in $\langle v_0\rangle$. If $W \subseteq F'_3$ meets both F_2 and G_2, then $W \cap F_2 \subseteq F'_3 \cap F_2 = \langle u_0\rangle$ and likewise $W \cap G_2 = \langle v_0\rangle$, so again $W = \langle u_0, v_0\rangle$ is the unique solution, and $\{W' \big| W' \subseteq F'_3\}$ is a Schubert variety for a complete flag through F'_3, of class $\sigma_{1,1}$. In the chart above, $W_\varphi \subseteq F'_3$ if and only if $\varphi(u_0), \varphi(v_0) \in F'_3 \cap E = \langle u_1 + v_1\rangle$, that is $\alpha = \beta$ and $\gamma = \delta$, a linear subspace again; intersecting it with $\beta = 0$ and $\gamma = 0$ leaves only $\varphi = 0$, and the codimensions $1 + 1 + 2$ again sum to 4, so $b = 1$ and

$$\sigma_1^2 = \sigma_2 + \sigma_{1,1}$$

The relation says that the lines meeting two general lines of $\mathbb{P}^3$ break, as a class, into the lines through a point and the lines in a plane. Finally proposition 8.1.5 reads $c(Q) = 1 + \sigma_1 + \sigma_2$ here, so $c(S) = (1 + \sigma_1 + \sigma_2)^{-1} = 1 - \sigma_1 + (\sigma_1^2 - \sigma_2) - \cdots = 1 - \sigma_1 + \sigma_{1,1} - \cdots$, whose terms in codimensions 3 and 4 vanish because S has rank 2.

Exercise 8.1.1

1. Show that $\lambda \mapsto (a_1(\lambda), \ldots, a_r(\lambda))$ is a bijection from the partitions in the $r \times (n-r)$ box onto the strictly increasing sequences $1 \le a_1 < \cdots < a_r \le n$, with inverse $\lambda_i = n - r + i - a_i$. Deduce that the number of partitions in the box is $\binom{n}{r}$, and check directly that for $r = 2$, $n = 5$ the ranks of $A^p(\mathrm{Gr}(2,5))$ for $p = 0, \ldots, 6$ are $1, 1, 2, 2, 2, 1, 1$.

2. Take $r = 1$, so that $\mathrm{Gr}(1,n) = \mathbb{P}(V) = \mathbb{P}^{n-1}$ and the box is $1 \times (n-1)$. Show that the Schubert cells for a complete flag $F_\bullet$ are the sets $\mathbb{P}(F_{n-m}) \setminus \mathbb{P}(F_{n-m-1})$ for $0 \le m \le n-1$, that $\Omega_{(m)}(F_\bullet) = \mathbb{P}(F_{n-m})$, and that theorem 8.1.4 recovers theorem 1.5.2. Then verify proposition 8.1.5 in this case by computing $c(Q) = c(S)^{-1}$ from $S = \mathcal{O}_{\mathbb{P}(V)}(-1)$ and comparing with the classes of the linear subspaces.

3. Show that $\Omega_{(n-r,0,\ldots,0)}(F_\bullet) = \{W \big| F_1 \subseteq W\}$ and that $\Omega_{(1,1,\ldots,1)}(F_\bullet) = \{W \big| W \subseteq F_{n-1}\}$, and identify the two with $\mathrm{Gr}(r-1, n-1)$ and $\mathrm{Gr}(r, n-1)$ respectively. Check that their codimensions in $\mathrm{Gr}(r,n)$ are $n - r$ and r, in agreement with proposition 8.1.2(3).

4. Show that the pairing $A^p(\mathrm{Gr}(r,n)) \times A^{N-p}(\mathrm{Gr}(r,n)) \to \mathbb{Z}$ sending (α, β) to $\deg(\alpha \cdot \beta)$ is a perfect pairing of free abelian groups, in the sense that its matrix in the Schubert bases is invertible over $\mathbb{Z}$. Deduce that A^p and A^{N-p} have the same rank, and that a class $\alpha \in A^p$ is determined by the integers $\deg(\alpha \cdot \sigma_\mu)$ with $|\mu| = N - p$.

5. For two partitions λ, μ in the box let $\nu_i = \max(\lambda_i, \mu_i)$. Show that ν is again a partition in the box and that, for a single flag $F_\bullet$,

$$\Omega_\lambda(F_\bullet) \cap \Omega_\mu(F_\bullet) = \Omega_\nu(F_\bullet)$$

   Explain why this does not compute $\sigma_\lambda \cdot \sigma_\mu$, by comparing the codimension of Ω_ν with $|\lambda| + |\mu|$ in the case $\lambda = \mu = (1, 0, \ldots, 0)$.

6. Let λ be a partition in the box and let $i \colon \Omega_\lambda(F_\bullet) \hookrightarrow \mathrm{Gr}(r,n)$ be the inclusion. Show that $A_*(\Omega_\lambda(F_\bullet))$ is free abelian with basis the classes $[\Omega_\mu(F_\bullet)]$ for $\mu \supseteq \lambda$. (For generation, repeat Step 1 of the proof of theorem 8.1.4 with the stratification of Ω_λ by the cells it contains; for independence, push forward to $\mathrm{Gr}(r,n)$.)

7. On $\mathrm{Gr}(2,4)$, use example 8.1.6 to show that $c(S) = 1 - \sigma_1 + \sigma_{1,1}$. Deduce from the vanishing of $c_3(S)$ and $c_4(S)$ the two relations

$$\sigma_1\sigma_{1,1} = \sigma_1\sigma_2, \qquad \sigma_2\sigma_{1,1} = 0$$

and check the second against proposition 8.1.3.

8. Let $F_\bullet$ be a complete flag in a four-dimensional V and consider $\Omega_{(1,0)}(F_\bullet) \subseteq \mathrm{Gr}(2,4)$. In the chart U_a with $a = \{1,2\}$ for a basis with $F_2 = \langle e_1, e_2\rangle$, show that $\Omega_{(1,0)}(F_\bullet) \cap U_a$ is the affine quadric $\{c_{13}c_{24} - c_{14}c_{23} = 0\}$, and conclude that $\Omega_{(1,0)}(F_\bullet)$ is singular at the point $W = F_2$. Which point of which Schubert cell is $W = F_2$, and how does the answer square with proposition 8.1.2(1)?

8.2 Schubert Calculus

Section 8.1 produced the additive structure of $A^*(\mathrm{Gr}(r,n))$ and a pairing recovering the coefficients of a class in the Schubert basis, but nothing so far computes a product of two Schubert classes, and every enumerative count is such a product. This section supplies the multiplication table in three forms: Pieri's rule, which multiplies an arbitrary Schubert class by a special one; Giambelli's formula, which writes every Schubert class as a determinant in the special classes; and the presentation of $A^*(\mathrm{Gr}(r,n))$ by generators and relations that the two together force. The four lines problem is settled on the way.

Throughout, V is a fixed vector space of dimension n over $\overline{k}$, and $\mathrm{Gr}(r,n)$ parametrizes its r-dimensional subspaces W. *The box* is the set of partitions $\lambda = (\lambda_1 \geq \cdots \geq \lambda_r \geq 0)$ with $\lambda_1 \leq n-r$; these index the Schubert classes $\sigma_\lambda = [\Omega_\lambda(F_\bullet)]$ of theorem 8.1.4, where by definition 8.1.1

$$\Omega_\lambda(F_\bullet) \;=\; \left\{ W \in \mathrm{Gr}(r,n) \;\middle|\; \dim(W \cap F_{n-r+i-\lambda_i}) \geq i \;\text{ for } i = 1,\ldots,r \right\}$$

and $\Omega^\circ_\lambda(F_\bullet)$ is its open cell, on which each of these dimensions equals i and, in addition, $\dim(W \cap F_{n-r+i-\lambda_i-1}) = i-1$; equivalently, the function $j \mapsto \dim(W \cap F_j)$ takes its r jumps exactly at the indices $n-r+i-\lambda_i$. Write $\lambda \subseteq \mu$ for $\lambda_i \leq \mu_i$ for all i, $|\lambda| = \sum_i \lambda_i$, and $\lambda^\vee = (n-r-\lambda_r, \ldots, n-r-\lambda_1)$ for the complementary partition, so that $(\lambda^\vee)^\vee = \lambda$ and $|\lambda| + |\lambda^\vee| = r(n-r)$. Two conventions make every formula below uniform: σ_j denotes $\sigma_{(j)}$ for $1 \leq j \leq n-r$, with $\sigma_0 = [\mathrm{Gr}(r,n)]$ the unit of the ring and $\sigma_j = 0$ for $j < 0$ and for $j > n-r$; and $\sigma_\lambda = 0$ for any partition λ outside the box, that is, one with $\lambda_1 > n-r$ or with more than r nonzero parts.

The special classes have a description free of flags. For $1 \leq m \leq n-r$ put $a = n-r+1-m$ and, for a subspace $A \subseteq V$ of dimension a,

$$\Sigma_A \;=\; \{ W \in \mathrm{Gr}(r,n) \big| W \cap A \neq 0 \}$$

Any complete flag $F_\bullet$ with $F_a = A$ has $\Omega_{(m)}(F_\bullet) = \Sigma_A$: the condition indexed by $i = 1$ is $\dim(W \cap A) \geq 1$, and the conditions indexed by $i \geq 2$ read $\dim(W \cap F_{n-r+i}) \geq i$, which every W satisfies because $\dim(W \cap F_j) \geq r+j-n$. Hence

$$\sigma_m \;=\; [\Sigma_A] \qquad \text{for every } A \text{ with } \dim A = n-r+1-m \tag{8.2}$$

and the class does not depend on which A is chosen, since σ_m does not depend on the flag.

Multiplying by σ_m adds m boxes to a Young diagram, and the rule says exactly which ways of adding them are allowed. Call μ obtained from λ by adding a *horizontal m-strip* if $\lambda \subseteq \mu$, $|\mu| = |\lambda| + m$, and no two of the added boxes lie in the same column; in inequalities this is the interlacing condition

$$\mu_1 \;\geq\; \lambda_1 \;\geq\; \mu_2 \;\geq\; \lambda_2 \;\geq\; \cdots \;\geq\; \mu_r \;\geq\; \lambda_r$$

since $\mu_{i+1} \leq \lambda_i$ says precisely that row $i+1$ of μ stops before row i of λ ends, which is what forbids two added boxes in one column.

Theorem 8.2.1: Pieri's Formula

Let λ be a partition in the box and $m \geq 1$. Then in $A^*(\mathrm{Gr}(r,n))$

$$\sigma_\lambda \cdot \sigma_m \;=\; \sum_\mu \sigma_\mu$$

the sum being over the partitions μ in the box obtained from λ by adding a horizontal m-strip, that is, those μ with $|\mu| = |\lambda| + m$ and $\mu_1 \geq \lambda_1 \geq \mu_2 \geq \cdots \geq \mu_r \geq \lambda_r$.

***Proof*:**
For $m > n-r$ both sides vanish: the left side because $\sigma_m = 0$, and the right side because a horizontal m-strip occupies at least m columns, so every μ in the sum has $\mu_1 \geq m > n-r$ and lies outside the box. Assume therefore $1 \leq m \leq n-r$ and set $a = n-r+1-m$.

Step 1: the local form of a rank condition. Fix $W_0 \in \mathrm{Gr}(r,n)$ and a complement C, so $V = W_0 \oplus C$. The set $U_C = \{W \big| W \cap C = 0\}$ is open in $\mathrm{Gr}(r,n)$ and

$$\mathrm{Hom}(W_0, C) \longrightarrow U_C, \qquad \varphi \longmapsto \Gamma_\varphi = \{\, w + \varphi(w) \big| w \in W_0 \,\}$$

is an isomorphism of varieties carrying 0 to W_0; this is the standard affine chart of the Grassmannian, [7, Grassmannian]. Identify $T_{W_0}\,\mathrm{Gr}(r,n)$ with $\mathrm{Hom}(W_0, C)$, and C with V/W_0, through it.

Let $B \subseteq V$ be a subspace with $\dim(W_0 \cap B) = c$ exactly, let $\rho\colon V \to V/B$ be the quotient map, and let $M(\varphi) = \rho \circ (\mathrm{id}_{W_0} + \varphi)$, a linear map $W_0 \to V/B$ depending affinely on φ. Since $w + \varphi(w) \in B$ if and only if $M(\varphi)(w) = 0$, the locus $\{W \big| \dim(W \cap B) \geq c\}$ meets U_C in $\{\varphi \big| \mathrm{rk} M(\varphi) \leq r - c\}$. Split $W_0 = K' \oplus K$ with $K = W_0 \cap B$, and $V/B = M(0)(W_0) \oplus T$; in the resulting blocks

$$M(\varphi) \;=\; \begin{pmatrix} P(\varphi) & Q(\varphi) \\ R(\varphi) & S(\varphi) \end{pmatrix}, \qquad P(0) \text{ invertible}, \quad Q(0) = R(0) = S(0) = 0$$

because $M(0)$ kills K and maps K' isomorphically onto $M(0)(W_0)$. For φ near 0 the block $P(\varphi)$ is invertible, and the row and column operations

$$\begin{pmatrix} I & 0 \\ -RP^{-1} & I \end{pmatrix} M \begin{pmatrix} I & -P^{-1}Q \\ 0 & I \end{pmatrix} \;=\; \begin{pmatrix} P & 0 \\ 0 & S - RP^{-1}Q \end{pmatrix}$$

show that $\mathrm{rk} M(\varphi) \leq r - c$ if and only if $S(\varphi) - R(\varphi)P(\varphi)^{-1}Q(\varphi) = 0$. These are $c \cdot \dim T$ regular functions near $\varphi = 0$; as Q and R vanish at $\varphi = 0$ and depend affinely on φ, the term $RP^{-1}Q$ vanishes to order two, so the differential at $\varphi = 0$ of the system is $\varphi \mapsto S'(\varphi)$, the composite

$$\mathrm{Hom}(W_0, C) \longrightarrow \mathrm{Hom}(K, T), \qquad \varphi \longmapsto \big(K \xrightarrow{\varphi} C \xrightarrow{\rho} V/B \longrightarrow T\big)$$

Restriction $\mathrm{Hom}(W_0, C) \to \mathrm{Hom}(K, C)$ is surjective, and $C \to T$ is surjective because $V = W_0 + C$ while W_0 maps into $M(0)(W_0)$, which dies in T; so the differential is surjective.

Consequently the locus $\{\dim(W \cap B) \geq c\}$ is smooth at W_0 of codimension $c \cdot \dim T$, with tangent space the kernel of that differential, namely

$$T_{W_0}\{\dim(W \cap B) \geq c\} \;=\; \big\{\, \varphi \in \mathrm{Hom}(W_0, V/W_0) \;\big|\; \varphi(W_0 \cap B) \subseteq (W_0 + B)/W_0 \,\big\} \tag{8.3}$$

Any closed subscheme of $\mathrm{Gr}(r,n)$ contained in this locus and passing through W_0 has its tangent space at W_0 contained in the right-hand side of (8.3).

Step 3: reduction to a triple intersection number. By theorem 8.1.4 write $\sigma_\lambda \cdot \sigma_m = \sum_\mu c_\mu \sigma_\mu$ over the partitions μ in the box with $|\mu| = |\lambda| + m$. Pairing with a complementary class and using proposition 8.1.3 in the form $\deg(\sigma_\theta \cdot \sigma_{\mu^\vee}) = \delta_{\theta\mu}$,

$$c_\mu \;=\; \deg\big(\sigma_\lambda \cdot \sigma_m \cdot \sigma_{\mu^\vee}\big) \tag{8.4}$$

so the theorem is the assertion that this triple intersection number is 1 when μ/λ is a horizontal m-strip and 0 otherwise.

Fix a basis $e_1, \ldots, e_n$ of V and the two opposite flags $F_i = \langle e_1, \ldots, e_i\rangle$ and $G_i = \langle e_{n-i+1}, \ldots, e_n\rangle$, and abbreviate $G'_l = \langle e_l, \ldots, e_n\rangle = G_{n-l+1}$. Represent the three classes by $\Omega_\lambda(F_\bullet)$, by $\Omega_{\mu^\vee}(G_\bullet)$, and by Σ_A as in (8.2), with A to be chosen. Put

$$j_i \;=\; n - r + i - \lambda_i, \qquad l_i \;=\; n - r + i - \mu_i, \qquad E_i \;=\; \langle e_{l_i}, \ldots, e_{j_i}\rangle$$

so that $j_1 < \cdots < j_r$ and $l_1 < \cdots < l_r$, all lying in $[1,n]$. Rewriting the conditions cutting out $\Omega_{\mu^\vee}(G_\bullet)$ in terms of the index $i' = r+1-i$, using $(\mu^\vee)_i = (n-r) - \mu_{r+1-i}$, they read $\dim(W \cap G'_{l_{i'}}) \geq r + 1 - i'$. So

$$Y \;:=\; \Omega_\lambda(F_\bullet) \cap \Omega_{\mu^\vee}(G_\bullet) \;=\; \big\{\, W \;\big|\; \dim(W \cap F_{j_i}) \geq i \;\text{ and }\; \dim(W \cap G'_{l_i}) \geq r+1-i \;\text{ for all } i \,\big\} \tag{8.5}$$

Step 4: the geometry of Y. For any $W \in Y$ and any i, the two subspaces $W \cap F_{j_i}$ and $W \cap G'_{l_i}$ of W have dimensions at least i and $r+1-i$, so they meet in a subspace of dimension at least 1; as $F_{j_i} \cap G'_{l_i} = \langle e_{l_i}, \ldots, e_{j_i}\rangle$ when $l_i \leq j_i$ and 0 otherwise, this gives two consequences.

First, if $\mu \not\supseteq \lambda$, say $\mu_i < \lambda_i$, then $l_i > j_i$ and the intersection $F_{j_i} \cap G'_{l_i}$ is zero, so $Y = \emptyset$. The refined product of definition 5.1.10 then lies in $A_*(Y) = 0$, and proposition 5.1.11 gives $\sigma_\lambda \cdot \sigma_{\mu^\vee} = 0$, whence $c_\mu = 0$ by (8.4). Assume from now on $\lambda \subseteq \mu$, so $l_i \leq j_i$ and $E_i \neq 0$ for every i; then $W \cap E_i \neq 0$ for every $W \in Y$ and every i.

Second, let $P = E_1 + \cdots + E_r$, spanned by the e_s with s in the union $T = \bigcup_i [l_i, j_i]$. Every $W \in Y$ satisfies $W \subseteq P$. To see this, let $g \in [1,n] \setminus T$ be a gap. For each i exactly one of $j_i < g$ and $l_i > g$ holds; write α and β for the number of indices of each kind, so $\alpha + \beta = r$. The indices with $j_i < g$ are $1, \ldots, \alpha$ and the indices with $l_i > g$ are $r - \beta + 1, \ldots, r$, since $j_\bullet$ and $l_\bullet$ increase. Taking $i = \alpha$ in the first family of conditions of (8.5) gives $\dim(W \cap F_{g-1}) \geq \alpha$, since $F_{j_\alpha} \subseteq F_{g-1}$, and taking $i = r - \beta + 1$ in the second gives $\dim(W \cap G'_{g+1}) \geq \beta$, since $G'_{l_{r-\beta+1}} \subseteq G'_{g+1}$; both bounds are vacuous when the count in question is zero. These two subspaces of W intersect in $W \cap F_{g-1} \cap G'_{g+1} = 0$ and their dimensions sum to at least $r = \dim W$, so

$$W \;=\; (W \cap F_{g-1}) \oplus (W \cap G'_{g+1}) \;\subseteq\; \langle e_s \,|\, s \neq g\rangle$$

As g was an arbitrary gap, $W \subseteq P$.

The intervals $[l_i, j_i]$ have lengths $j_i - l_i + 1 = \mu_i - \lambda_i + 1$, summing to $m+r$, and consecutive ones are disjoint exactly when $j_i < l_{i+1}$, that is when $\mu_{i+1} \le \lambda_i$. So the intervals are pairwise disjoint precisely when μ/λ is a horizontal m-strip, in which case $\dim P = m+r$ and $P = E_1 \oplus \cdots \oplus E_r$; otherwise $\dim P \le m+r-1$.

In the horizontal-strip case Y is exactly the set of W of the form $\langle w_1, \ldots, w_r\rangle$ with $0 \ne w_i \in E_i$. One inclusion holds because such a W has $w_1, \ldots, w_i \in F_{j_i}$ and $w_i, \ldots, w_r \in G'_{l_i}$, the w's being independent as the E_i are spanned by disjoint sets of basis vectors. Conversely for $W \in Y$ choose $0 \ne w_i \in W \cap E_i$; these are independent, hence a basis of W. The resulting bijection is the morphism

$$\mathbb{P}(E_1) \times \cdots \times \mathbb{P}(E_r) \longrightarrow \mathrm{Gr}(r,n) \subseteq \mathbb{P}(\textstyle\bigwedge^r V), \qquad (\langle w_1\rangle, \ldots, \langle w_r\rangle) \longmapsto [w_1 \wedge \cdots \wedge w_r]$$

which is the Segre embedding followed by the linear map $E_1 \otimes \cdots \otimes E_r \to \bigwedge^r V$ sending $e_{s_1} \otimes \cdots \otimes e_{s_r} \mapsto e_{s_1} \wedge \cdots \wedge e_{s_r}$; distinct basis tensors of the source go to distinct nonzero basis vectors of the target because the intervals are disjoint, so that linear map is injective and the composite is a closed immersion. Thus $Y \cong \mathbb{P}(E_1) \times \cdots \times \mathbb{P}(E_r)$ is a smooth irreducible variety of dimension $\sum_i (\mu_i - \lambda_i) = m$.

Step 5: choosing A. Call $A \in \mathrm{Gr}(a,n)$ *admissible* if, for every one of the finitely many partitions μ in the box with $|\mu| = |\lambda| + m$ and $\lambda \subseteq \mu$, writing P_μ and E_i^μ for the subspaces just constructed: A meets no one of the subspaces

$$P_\mu \text{ (when } \mu/\lambda \text{ is not a horizontal strip)}, \qquad P_\mu \cap \{x_{l_i} = 0\}, \qquad P_\mu \cap \{x_{j_i} = 0\}$$

nontrivially, and $\dim(A \cap P_\mu) \le 1$ for the remaining μ. Each listed subspace has dimension at most $m+r-1$, and $a + (m+r-1) = n$, so for each of them the set of A meeting it nontrivially is a proper closed subset of $\mathrm{Gr}(a,n)$: closed because it is cut out by a rank condition, and proper because a basis of the subspace extends to a basis of V whose remaining a vectors span a complementary A. Likewise $\{A \,|\, \dim(A \cap P_\mu) \ge 2\}$ is a proper closed subset. Finitely many conditions are imposed, so an admissible A exists; fix one.

Let $\mu \supseteq \lambda$ with μ/λ not a horizontal strip. Then $A \cap P_\mu = 0$, so $Y \cap \Sigma_A = \emptyset$ because every $W \in Y$ lies in P_μ. By definition 5.1.10 the refined product of $[\Omega_\lambda(F_\bullet)]$ and $[\Omega_{\mu^\vee}(G_\bullet)]$ is a class $\gamma \in A_m(Y)$ whose image in $A_m(\mathrm{Gr}(r,n))$ is $\sigma_\lambda \cdot \sigma_{\mu^\vee}$ (proposition 5.1.11); applying the same two results to γ and $[\Sigma_A]$, the product $(\sigma_\lambda \cdot \sigma_{\mu^\vee}) \cdot \sigma_m$ is the image of a class in $A_0(Y \cap \Sigma_A) = 0$. Hence $c_\mu = 0$ by (8.4).

Step 6: the horizontal-strip case. Let μ/λ be a horizontal m-strip, so $P = \bigoplus_i E_i$ has dimension $m+r$ and $\dim(A \cap P) = 1$, admissibility giving both ≤ 1 and, since $a + \dim P = n+1$, also ≥ 1. Write $A \cap P = \langle v\rangle$ and $v = v_1 + \cdots + v_r$ with $v_i \in E_i$. Admissibility says v has nonzero coefficient on e_{l_i} and on e_{j_i} for every i; in particular every $v_i \ne 0$. Set $W_0 = \langle v_1, \ldots, v_r\rangle \in Y$.

First, $\sigma_\lambda \cdot \sigma_{\mu^\vee} = [Y]$. The two Schubert varieties have dimensions $r(n-r) - |\lambda|$ and $r(n-r) - |\mu^\vee| = |\mu|$, whose sum minus $r(n-r)$ is $m = \dim Y$, and Y is irreducible, so proposition 5.3.11 applies once a suitable point is exhibited. Take W_0. Its intersections with the flag $F_\bullet$ are governed by the largest index s with e_s occurring in a given vector: that index is j_i for v_i, and distinct v_i involve disjoint sets of basis vectors, so $\dim(W_0 \cap F_p) = \#\{i \,|\, j_i \le p\}$ for every p. That function jumps exactly at $j_1, \ldots, j_r$, which are the indices $n - r + i - \lambda_i$ attached to λ; symmetrically the smallest occurring index of v_i is l_i, so $\dim(W_0 \cap G_q) = \#\{i \,|\, l_i \ge n - q + 1\}$ for every q, jumping exactly at the indices $n+1-l_i$, which are those attached to $\mu^\vee$. Both jump patterns are the ones prescribed by definition 8.1.1, so W_0 lies in both open cells, where the Schubert varieties are smooth by proposition 8.1.2. For the tangent spaces, $W_0 \cap F_{j_i} = \langle v_1, \ldots, v_i\rangle$ and $W_0 \cap G'_{l_i} = \langle v_i, \ldots, v_r\rangle$, so (8.3) applied to each rank condition bounds

$$T_{W_0}\Omega_\lambda(F_\bullet) \cap T_{W_0}\Omega_{\mu^\vee}(G_\bullet) \;\subseteq\; \left\{ \varphi \;\middle|\; \varphi(v_i) \in \left[(F_{j_i}+W_0)\cap(G'_{l_i}+W_0)\right]/W_0 \text{ for all } i \right\}$$

the condition indexed by k constraining φ on $\langle v_1, \ldots, v_k\rangle$ and on $\langle v_k, \ldots, v_r\rangle$ respectively, so that the vector v_i is constrained by the conditions with $k \geq i$ in the first family and $k \leq i$ in the second, of which the binding ones are $k = i$ in both, the spaces F_{j_k} increasing and the spaces G'_{l_k} decreasing with k. Now $\dim(F_{j_i}+W_0) = j_i + r - i$ and $\dim(G'_{l_i}+W_0) = (n-l_i+1)+r-(r+1-i) = n-l_i+i$, while $F_{j_i}+G'_{l_i} = V$ because $l_i \leq j_i$; so the two sum to V and their intersection has dimension $(j_i+r-i)+(n-l_i+i)-n = j_i - l_i + r$. That is the dimension of $E_i + W_0$, which is contained in it, so the two agree and the displayed bound reads $\varphi(v_i) \in (E_i+W_0)/W_0$ for all i. Since $v_1, \ldots, v_r$ is a basis of W_0 and $\dim(E_i+W_0)/W_0 = j_i - l_i$, that space of φ has dimension $\sum_i (j_i - l_i) = m$. Proposition 5.3.11 applies and gives $\sigma_\lambda \cdot \sigma_{\mu^\vee} = [Y]$; incidentally the bound is an equality on $T_{W_0}Y$, which has dimension m and is contained in it, so

$$T_{W_0}Y \;=\; \left\{ \varphi \;\middle|\; \varphi(v_i) \in (E_i+W_0)/W_0 \text{ for all } i \right\} \tag{8.6}$$

Second, $\deg([Y]\cdot\sigma_m) = 1$. Set-theoretically $Y \cap \Sigma_A$ is the single point W_0: any $W \in Y$ meeting A meets $P \cap A = \langle v\rangle$, hence contains v, and writing $W = \langle w_1, \ldots, w_r\rangle$ with $w_i \in E_i$ and comparing E_i-components of $v \in W$ forces $\langle w_i\rangle = \langle v_i\rangle$ for every i, so $W = W_0$. Both Y and Σ_A are smooth at W_0, the former being a product of projective spaces and the latter by Step 1 applied to $B = A$ and $c = 1$, which is legitimate because $\dim(W_0 \cap A) = 1$; the same step gives $T_{W_0}\Sigma_A = \{\varphi | \varphi(v) \in (A+W_0)/W_0\}$. Let φ lie in both tangent spaces and write $\varphi(v_i) = u_i \bmod W_0$ with $u_i \in E_i$, which (8.6) permits. Then $\varphi(v) = \sum_i u_i \bmod W_0$ lies in $(A+W_0)/W_0$, so $\sum_i u_i \in P \cap (A+W_0)$. That intersection is W_0: if $p = \alpha + w$ with $p \in P$, $\alpha \in A$ and $w \in W_0 \subseteq P$, then $\alpha = p - w \in A \cap P = \langle v\rangle \subseteq W_0$, so $p \in W_0$. Hence $\sum_i u_i \in W_0 = \bigoplus_i \langle v_i\rangle$, and comparing components in $P = \bigoplus_i E_i$ gives $u_i \in \langle v_i\rangle$, that is $\varphi = 0$. So the tangent spaces meet in 0, which is the expected dimension $\dim Y + \dim \Sigma_A - r(n-r) = 0$, and proposition 5.3.11 applied to Y and Σ_A gives $[Y]\cdot[\Sigma_A] = [\{W_0\}]$, a reduced point, of degree 1 by definition 1.2.4.

Combining the two, $c_\mu = \deg(\sigma_\lambda \cdot \sigma_{\mu^\vee} \cdot \sigma_m) = \deg([Y]\cdot[\Sigma_A]) = 1$. With Steps 4 and 5 this evaluates every coefficient in (8.4), as we sought to show.

The rule is an algorithm: every product of special classes can be expanded by applying it repeatedly, and theorem 8.2.4 will show that this exhausts the ring. The combinatorial side deserves one remark. The box constrains the answer twice, once by discarding μ with more than r rows and once by discarding μ with $\mu_1 > n-r$; both discards are invisible in the interlacing inequalities and are what make the ring finite rather than a ring of symmetric functions.

The conventions fixed above extend the formula to every partition, which is what the inductions below use. If λ lies outside the box then so does every $\mu \supseteq \lambda$, and both sides of Pieri's formula vanish. If λ lies in the box but $m > n-r$, the left side vanishes and every μ occurring on the right has $\mu_1 \geq m > n-r$, so the right side vanishes too. And for $m = 0$ the only horizontal 0-strip is the empty one, so the formula reads $\sigma_\lambda \cdot \sigma_0 = \sigma_\lambda$. Hence for all partitions λ and all $m \geq 0$,

$$\sigma_\lambda \cdot \sigma_m \;=\; \sum_\mu \sigma_\mu \tag{8.7}$$

summed over all partitions μ obtained from λ by adding a horizontal m-strip, with $\sigma_\theta = 0$ for θ outside the box.

Example 8.2.2: Lines Meeting Four General Lines in $\mathbb{P}^3$

Take $r = 2$ and $n = 4$, so $\mathrm{Gr}(2,4)$ parametrizes the lines of $\mathbb{P}^3$ and has dimension four, with box the 2×2 square. Here $\sigma_1 = [\Sigma_A]$ with $\dim A = 2$ by (8.2): the condition $W \cap A \neq 0$ says that the line of $\mathbb{P}^3$ determined by W meets the line determined by A. So $\deg(\sigma_1^4)$ is the intersection number attached to the classical problem of lines meeting four given lines.

The computation. Pieri's formula gives, in the 2×2 box,

$$\sigma_1^2 \;=\; \sigma_2 + \sigma_{1,1}$$

since one box may be added to the single box of (1) either beside it or below it. Multiplying again, $\sigma_2 \cdot \sigma_1 = \sigma_{2,1}$, the partition (3) falling outside the box, and $\sigma_{1,1} \cdot \sigma_1 = \sigma_{2,1}$, the partition $(1,1,1)$ having three rows; hence

$$\sigma_1^3 \;=\; 2\,\sigma_{2,1}, \qquad \sigma_1^4 \;=\; 2\,\sigma_{2,1} \cdot \sigma_1 \;=\; 2\,\sigma_{2,2}$$

because the only box addable to $(2,1)$ inside the square is the one completing it. Now $\sigma_{2,2}$ is the class of a point, $(2,2)$ being the full box, so $\deg(\sigma_1^4) = 2$. Duality gives the same answer from the second line: $\deg(\sigma_1^4) = \deg\left((\sigma_2 + \sigma_{1,1})^2\right) = \deg(\sigma_2^2) + 2\deg(\sigma_2\sigma_{1,1}) + \deg(\sigma_{1,1}^2) = 1 + 0 + 1$, using proposition 8.1.3 and $(2)^\vee = (2)$, $(1,1)^\vee = (1,1)$.

The geometry. Let $L_1, \ldots, L_4 \subseteq \mathbb{P}^3$ be four general lines. The first three are pairwise skew and lie on a unique smooth quadric surface Q, which they meet as three members of one of its two rulings; a line meets all three exactly when it belongs to the other ruling. The fourth line meets Q in two distinct points, and through each point passes exactly one line of the second ruling. So exactly two lines meet all four, and the number two is attained by two distinct honest solutions rather than by one solution counted twice.

Example 8.2.3: Incidence Counts for Lines in $\mathbb{P}^4$

Take $r = 2$ and $n = 5$, so $\mathrm{Gr}(2,5)$ parametrizes the lines of $\mathbb{P}^4$, has dimension six, and has the 2×3 rectangle as its box. By (8.2), $\sigma_1 = [\Sigma_A]$ with $\dim A = 3$: the lines meeting a fixed plane of $\mathbb{P}^4$. Likewise $\sigma_2 = [\Sigma_A]$ with $\dim A = 2$: the lines meeting a fixed line.

Repeated application of theorem 8.2.1 inside the 2×3 box gives

$$\begin{aligned}
\sigma_1^2 &= \sigma_2 + \sigma_{1,1} \\
\sigma_1^3 &= (\sigma_3 + \sigma_{2,1}) + \sigma_{2,1} \;=\; \sigma_3 + 2\,\sigma_{2,1} \\
\sigma_1^4 &= \sigma_{3,1} + 2(\sigma_{3,1} + \sigma_{2,2}) \;=\; 3\,\sigma_{3,1} + 2\,\sigma_{2,2} \\
\sigma_1^5 &= 3\,\sigma_{3,2} + 2\,\sigma_{3,2} \;=\; 5\,\sigma_{3,2} \\
\sigma_1^6 &= 5\,\sigma_{3,3}
\end{aligned}$$

where at each stage the partitions with a first part exceeding 3 or with three rows are discarded. For instance $\sigma_{2,1} \cdot \sigma_1 = \sigma_{3,1} + \sigma_{2,2}$, the third candidate $(2,1,1)$ having three rows. Since $\sigma_{3,3}$ is the point class, $\deg(\sigma_1^6) = 5$: five lines of $\mathbb{P}^4$ meet six general planes.

The same table answers a mixed problem. From $\sigma_1^4 = 3\sigma_{3,1} + 2\sigma_{2,2}$ and $(3,1)^\vee = (2)$, $(2,2)^\vee = (1,1)$,

$$\deg\left(\sigma_2 \cdot \sigma_1^4\right) \;=\; 3\deg(\sigma_2 \cdot \sigma_{3,1}) + 2\deg(\sigma_2 \cdot \sigma_{2,2}) \;=\; 3$$

by proposition 8.1.3: three lines of $\mathbb{P}^4$ meet four general planes and one general line.
The first number has a second reading. For $\dim A = n - r$ the condition $W \cap A \neq 0$ is equivalent to $W + A \neq V$, hence to the vanishing of $w_1 \wedge \cdots \wedge w_r \wedge a_1 \wedge \cdots \wedge a_{n-r} \in \bigwedge^n V \cong \overline{k}$, which is a linear condition on the Plücker coordinates of W. So σ_1 is the hyperplane class of the Plücker embedding ([7, Plücker Embedding]), and $\deg(\sigma_1^{r(n-r)})$ is the degree of $\mathrm{Gr}(r, n)$ in the projective space it embeds into: $\mathrm{Gr}(2, 4) \subseteq \mathbb{P}^5$ has degree 2 and $\mathrm{Gr}(2, 5) \subseteq \mathbb{P}^9$ has degree 5.
These numbers are intersection numbers. That a general choice of the incidence data makes the intersection transverse, so that the number counts distinct solutions, is a separate assertion; it was verified by hand for four lines in example 8.2.2, and in general it rests on a transversality theorem for the transitive action of $\mathrm{GL}(V)$ on $\mathrm{Gr}(r, n)$, which this book does not prove (see [8]).

Pieri's formula computes with one special factor at a time, which is enough to multiply out any polynomial in the special classes but says nothing about a product such as $\sigma_{2,1} \cdot \sigma_{2,1}$. Giambelli's formula removes the gap by expressing every Schubert class as a polynomial in the special ones, after which every product is computable by iterated Pieri. The polynomial is a determinant, and the determinant is the one that computes a Schur polynomial from the complete homogeneous symmetric polynomials.

Theorem 8.2.4: Giambelli's Formula

Let λ be a partition with at most ℓ nonzero parts. Then

$$\det\left(\sigma_{\lambda_p+q-p}\right)_{1\leq p,q\leq \ell} \;=\; \sigma_\lambda$$

in $A^*(\mathrm{Gr}(r, n))$: the determinant equals the Schubert class σ_λ when λ lies in the box, and vanishes otherwise. In particular the value of the determinant does not depend on the choice of $\ell \geq \ell(\lambda)$, and the special classes $\sigma_1, \ldots, \sigma_{n-r}$ generate $A^*(\mathrm{Gr}(r, n))$ as a ring.

***Proof*:**
Write $D_\ell(\lambda)$ for the determinant, entries being read with the conventions $\sigma_0 = 1$ and $\sigma_j = 0$ for $j < 0$ or $j > n - r$. Independence of ℓ is immediate: a row indexed by $p > \ell(\lambda)$ has entries σ_{q-p}, which vanish for $q < p$ and equal 1 for $q = p$, so expanding along the last row of an $\ell \times \ell$ determinant with $\ell > \ell(\lambda)$ returns the $(\ell - 1) \times (\ell - 1)$ one. The proof of the identity is by induction on ℓ, the case $\ell = 0$ being the empty determinant $1 = \sigma_\emptyset$.

Step 1: expansion along the first column. Let λ have at most $\ell \geq 1$ parts. The entry in position $(p, 1)$ is σ_{λ_p-p+1}, and deleting row p and column 1 leaves the matrix with entries σ_{λ_i+q-i} for $i \neq p$ and $q \geq 2$. Re-indexing rows by $i' \in \{1, \ldots, \ell - 1\}$ and columns by $q' = q - 1$, the entry becomes $\sigma_{(\lambda_{i'}+1)+q'-i'}$ for $i' < p$ and $\sigma_{\lambda_{i'+1}+q'-i'}$ for $i' \geq p$. So that minor is $D_{\ell-1}(\beta(p))$ for the sequence

$$\beta(p) \;=\; (\lambda_1 + 1, \ldots, \lambda_{p-1} + 1, \lambda_{p+1}, \ldots, \lambda_\ell)$$

which is a partition with $\ell - 1$ parts, being weakly decreasing at the junction because $\lambda_{p-1} + 1 > \lambda_{p-1} \geq \lambda_{p+1}$. Laplace expansion and the inductive hypothesis give

$$D_\ell(\lambda) \;=\; \sum_{p=1}^{\ell} (-1)^{p-1}\, \sigma_{\lambda_p-p+1}\, \sigma_{\beta(p)}$$

and (8.7) expands each summand as a sum of Schubert classes. Writing T_p for the set of partitions μ with at most ℓ parts obtained from $\beta(p)$ by adding a horizontal strip of size $\lambda_p - p + 1$, and noting that every such μ has $|\mu| = |\beta(p)| + \lambda_p - p + 1 = |\lambda|$,

$$D_\ell(\lambda) \;=\; \sum_\mu c_\mu \sigma_\mu, \qquad c_\mu = \sum_{p\,:\,\mu\in T_p} (-1)^{p-1}$$

the outer sum running over partitions μ of $|\lambda|$ with at most ℓ parts. It remains to prove that $c_\mu = \delta_{\mu\lambda}$.

Step 2: the membership conditions. Interlacing for the pair $\beta(p) \subseteq \mu$ says $\mu_i \geq \beta(p)_i \geq \mu_{i+1}$ for $1 \leq i \leq \ell - 1$. Substituting the entries of $\beta(p)$, and pairing each inequality with the index at which it constrains μ, the condition $\mu \in T_p$ is the conjunction of $|\mu| = |\lambda|$ with

(A) $\mu_i \geq \lambda_i{+}1$ $(i < p)$; (B) $\mu_q \leq \lambda_{q-1}{+}1$ $(2 \leq q \leq p)$; (C) $\mu_i \geq \lambda_{i+1}$ $(p \leq i \leq \ell{-}1)$; (D) $\mu_q \leq \lambda_q$ $(q > p)$

Let $S = \{i \in [1,\ell] \big| \mu_i > \lambda_i\}$. Conditions (A) and (D) force $[1, p-1] \subseteq S \subseteq [1,p]$, so T_p can contain μ only if S is an initial segment $[1,\alpha]$ and $p \in \{\alpha, \alpha+1\}$.

If $\mu = \lambda$ then $S = \emptyset$, so only $p = 1$ can contribute; and $\lambda \in T_1$, since (A) and (B) are vacuous, (D) holds with equality, and (C) is $\lambda_i \geq \lambda_{i+1}$. Hence $c_\lambda = 1$.

If $\mu \neq \lambda$ then $S \neq \emptyset$, because $|\mu| = |\lambda|$ forbids $\mu_i \leq \lambda_i$ for all i unless $\mu = \lambda$; and $S \neq [1,\ell]$ for the same reason. So either S is not an initial segment, in which case no p contributes and $c_\mu = 0$, or $S = [1,\alpha]$ with $1 \leq \alpha \leq \ell - 1$ and only $p = \alpha$ and $p = \alpha + 1$ can contribute. For those two values, (A) and (D) hold automatically by the definition of S, and the remaining conditions differ in exactly two inequalities: $p = \alpha$ additionally requires (C) at $i = \alpha$, namely $\mu_\alpha \geq \lambda_{\alpha+1}$, and $p = \alpha+1$ additionally requires (B) at $q = \alpha+1$, namely $\mu_{\alpha+1} \leq \lambda_\alpha + 1$. Both hold automatically: $\mu_\alpha \geq \lambda_\alpha + 1 > \lambda_{\alpha+1}$ because $\alpha \in S$, and $\mu_{\alpha+1} \leq \lambda_{\alpha+1} \leq \lambda_\alpha + 1$ because $\alpha + 1 \notin S$. So μ belongs to T_α if and only if it belongs to $T_{\alpha+1}$, and the two contributions $(-1)^{\alpha-1}$ and $(-1)^\alpha$ cancel, giving $c_\mu = 0$.

Hence $D_\ell(\lambda) = \sigma_\lambda$. For the last assertion, the formula writes each σ_λ as a polynomial with integer coefficients in $\sigma_1, \ldots, \sigma_{n-r}$, and the classes σ_λ with λ in the box span $A^*(\mathrm{Gr}(r,n))$ by theorem 8.1.4, completing the proof.

The determinant is the Jacobi-Trudi determinant. In the ring Λ_d of symmetric polynomials in d variables over $\mathbb{Z}$, the Schur polynomial s_λ satisfies the two Jacobi-Trudi formulas [6, Jacobi–Trudi Formulas]

$$s_\lambda \;=\; \det\left(h_{\lambda_p+q-p}\right)_{1\leq p,q\leq \ell(\lambda)}, \qquad s_\lambda \;=\; \det\left(e_{\lambda'_p+q-p}\right)_{1\leq p,q\leq \lambda_1}$$

where h_j and e_j are the complete homogeneous and elementary symmetric polynomials and λ' is the conjugate partition, the first valid for $\ell(\lambda) \leq d$ and the second for $\lambda_1 \leq d$. Theorem 8.2.4 says that the assignment $h_j \mapsto \sigma_j$ turns the first of these into the Schubert basis, truncated by the box. The truncation is what the next result makes precise: the relations of $A^*(\mathrm{Gr}(r,n))$ are exactly the vanishing of the Schur classes that fall out of the box.

Corollary 8.2.5: Presentation of the Chow Ring of the Grassmannian

Let $d = n - r$ and let $R = \mathbb{Z}[t_1, \ldots, t_d]$ be graded by $\deg t_i = i$. Put $t_0 = 1$, $t_i = 0$ for $i < 0$ and for $i > d$, and

$$Y_j \;=\; \det\left(t_{1+q-p}\right)_{1\le p,q\le j} \;\in\; R$$

Then $t_i \mapsto \sigma_i$ induces an isomorphism of graded rings

$$R/(Y_{r+1}, Y_{r+2}, \ldots, Y_n) \;\xrightarrow{\sim}\; A^*(\mathrm{Gr}(r,n))$$

***Proof*:**
Write $\varphi\colon R \to A^*(\mathrm{Gr}(r,n))$ for the graded ring homomorphism with $\varphi(t_i) = \sigma_i$, which exists and is unique because R is a polynomial ring. For a partition λ put $S_\lambda = \det(t_{\lambda_p+q-p})_{1\le p,q\le \ell(\lambda)}$, so that $Y_j = S_{(1^j)}$ and, by theorem 8.2.4, $\varphi(S_\lambda) = \sigma_\lambda$, this being 0 exactly when λ is outside the box.

Step 1: a symmetric-function model for R. The elementary symmetric polynomials $e_1, \ldots, e_d$ are algebraically independent and generate Λ_d, by the fundamental theorem on symmetric polynomials [6, Fundamental Theorem On Symmetric Functions], so $\psi\colon R \to \Lambda_d$ with $\psi(t_i) = e_i$ is an isomorphism of graded rings. For a partition λ with $\lambda_1 \le d$ the second Jacobi-Trudi formula applied to λ', whose conjugate is λ and whose first part is $\ell(\lambda)$, gives $\psi(S_\lambda) = s_{\lambda'}$; in particular $\psi(Y_j) = s_{(j)} = h_j$. Since the s_μ with $\ell(\mu) \le d$ form a $\mathbb{Z}$-basis of Λ_d [6, Schur Polynomials Are a Basis], and $\lambda \mapsto \lambda'$ is a bijection between the partitions with $\lambda_1 \le d$ and those with $\ell(\mu) \le d$, the elements S_λ with $\lambda_1 \le d$ form a $\mathbb{Z}$-basis of R.

Step 2: the kernel. On that basis, $\varphi(S_\lambda) = \sigma_\lambda$ is nonzero exactly when λ lies in the box, that is when $\ell(\lambda) \le r$, the constraint $\lambda_1 \le d$ being already imposed; and distinct such λ give distinct basis elements σ_λ of $A^*(\mathrm{Gr}(r,n))$. Hence φ is surjective and

$$\ker\varphi \;=\; \bigoplus_{\lambda_1\le d,\ \ell(\lambda)>r} \mathbb{Z}\, S_\lambda$$

This kernel is the ideal generated by the Y_j with $j > r$. One containment holds because $Y_j = S_{(1^j)}$ has $\ell = j > r$, so lies in $\ker\varphi$, and $\ker\varphi$ is an ideal. For the other, let $\lambda_1 \le d$ and $\ell(\lambda) > r$. Applying ψ and the first Jacobi-Trudi formula to λ', which has $\ell(\lambda') = \lambda_1 \le d$ parts and first part $\lambda'_1 = \ell(\lambda)$,

$$\psi(S_\lambda) \;=\; s_{\lambda'} \;=\; \det\left(h_{\lambda'_p+q-p}\right)_{1\le p,q\le \lambda_1}$$

whose first row consists of the entries $h_{\lambda'_1+q-1}$ with indices at least $\lambda'_1 = \ell(\lambda) > r$. Expanding along that row exhibits $s_{\lambda'}$ in the ideal generated by the h_j with $j > r$, so S_λ lies in the ideal generated by the Y_j with $j > r$.

Step 3: finitely many relations suffice. Expand Y_j along its first row. Its entries are t_q for $q = 1, \ldots, j$, and the minor obtained by deleting row 1 and column q is block lower triangular: its entries t_{1+c-p} vanish for $c < p - 1$, so the rows $2, \ldots, q$ meet the columns $1, \ldots, q-1$ in a lower triangular block with 1's on the diagonal $c = p - 1$ and the rows $q+1, \ldots, j$ vanish on those columns, while the rows and columns from $q+1$ to j form the matrix of Y_{j-q}. Hence

$$Y_j \;=\; \sum_{q=1}^{j}(-1)^{q-1}\,t_q\,Y_{j-q} \;=\; \sum_{q=1}^{d}(-1)^{q-1}\,t_q\,Y_{j-q}$$

the second equality because $t_q = 0$ for $q > d$. For $j > n$ every index appearing on the right satisfies $j - q \geq j - d > n - d = r$, so induction on j puts every Y_j with $j > n$ in the ideal generated by $Y_{r+1}, \ldots, Y_n$. Therefore $\ker\varphi = (Y_{r+1}, \ldots, Y_n)$ and φ induces the stated isomorphism, completing the proof.

The presentation identifies the relations with a statement about the tautological bundles. Since $\sigma_j = c_j(Q)$ and $c(S)c(Q) = 1$ by proposition 8.1.5, the class $(-1)^j Y_j$ evaluated at $t_i = \sigma_i$ is the degree j part of the inverse of the total Chern class of Q, that is $c_j(S)$; and S has rank r, so $c_j(S) = 0$ for $j > r$. The relations of corollary 8.2.5 are exactly these vanishings, and the content of the corollary is that they are the only ones. Two rank counts confirm the shape of the answer: the presentation has $n - r$ generators and $n - r$ relations, and the quotient is free of rank $\binom{n}{r}$, the number of partitions in the box.

Exercise 8.2.1

1. Let λ be a partition in the box. Using theorem 8.2.1, show that

$$\sigma_\lambda \cdot \sigma_{n-r} \;=\; \begin{cases} \sigma_{(n-r,\,\lambda_1,\ldots,\lambda_{r-1})} & \text{if } \lambda_r = 0 \\ 0 & \text{if } \lambda_r > 0 \end{cases}$$

   (Compare the sizes of $\mu_2 + \cdots + \mu_r$ and $\lambda_1 + \cdots + \lambda_{r-1}$ for any μ occurring in the sum.)

2. Take $r = 1$, so that $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$. Show that in the presentation of corollary 8.2.5 the relations $Y_2 = \cdots = Y_{n-1} = 0$ force $t_j = t_1^j$ for $j \leq n-1$, and that Y_n then becomes $(-1)^n t_1^n$; conclude that the presentation reads $\mathbb{Z}[\sigma_1]/(\sigma_1^n)$ and identify σ_1 with the hyperplane class of theorem 1.5.2.

3. With R, t_i and Y_j as in corollary 8.2.5, prove that in $R[[u]]$

$$\Big(\sum_{i\geq 0} t_i u^i\Big)^{-1} \;=\; \sum_{j\geq 0}(-1)^j Y_j u^j$$

   using the recursion established in the proof of that corollary. Deduce, from $\sigma_j = c_j(Q)$ and $c(S)c(Q) = 1$ (proposition 8.1.5), that the relations of the presentation are the vanishing of $c_j(S)$ for $j > r$.

4. In $\mathrm{Gr}(2, 4)$ compute $\deg(\sigma_2^2)$, $\deg(\sigma_2 \cdot \sigma_{1,1})$ and $\deg(\sigma_{1,1}^2)$, using theorem 8.2.1 where one factor is special and proposition 8.1.3 otherwise. Then interpret each number, using (8.2) to see that σ_2 is the class of the lines through a fixed point of $\mathbb{P}^3$, and definition 8.1.1 to see that the conditions defining $\Omega_{(1,1)}(F_\bullet)$ amount to $W \subseteq F_3$, so that $\sigma_{1,1}$ is the class of the lines contained in a fixed plane.

5. Verify theorem 8.2.4 in $\mathrm{Gr}(2, 4)$ by computing the determinants for $\lambda = (1, 1)$, $(2, 1)$ and $(2, 2)$, and check each answer against the products computed in example 8.2.2.

6. Write out the presentation of corollary 8.2.5 for $\mathrm{Gr}(2,4)$, computing Y_3 and Y_4 explicitly as polynomials in σ_1, σ_2. Then verify directly, degree by degree, that the quotient ring is free of rank 6, and that the images of 1, σ_1, σ_1^2, σ_2, $\sigma_1\sigma_2$, σ_2^2 span it.
7. In $\mathrm{Gr}(2,5)$ compute σ_2^2 by theorem 8.2.1 and then $\deg(\sigma_2^2 \cdot \sigma_1^2)$, using the expansion of σ_1^2 from example 8.2.3 together with proposition 8.1.3. State the enumerative problem in $\mathbb{P}^4$ whose intersection number this is.
8. Let λ and μ be partitions in the box with $\lambda_i + \mu_{r+1-i} > n - r$ for some i. Adapting Step 4 of the proof of theorem 8.2.1, show that $\Omega_\lambda(F_\bullet) \cap \Omega_\mu(G_\bullet) = \emptyset$ for opposite flags $F_\bullet$ and $G_\bullet$, and deduce that $\sigma_\lambda \cdot \sigma_\mu = 0$. Check that for $|\lambda| + |\mu| = r(n-r)$ this recovers the vanishing half of proposition 8.1.3.

8.3 Degeneracy Loci

The Schubert conditions of section 8.1 constrain a subspace to meet the members of a fixed flag in prescribed dimensions. Read through the tautological subbundle, each such condition says that a map of vector bundles drops rank. This section treats that phenomenon in general: for an arbitrary map $\varphi\colon E \to F$ of vector bundles on a smooth variety, the locus where $\mathrm{rk}\varphi \leq k$ carries a class which is a determinant in the Chern classes of E and F, provided the locus has the codimension one expects. Schubert classes of rectangular shape return at the end of the section as the case of the tautological map on a Grassmannian.

A rank condition is a condition on minors, and minors generate an ideal, so the locus where the rank drops is a subscheme and not merely a subset. Fixing that scheme structure first is what makes the class of the locus well defined.

Definition 8.3.1: Degeneracy Locus

Let X be a scheme, let $\varphi\colon E \to F$ be a morphism of vector bundles of ranks e and f on X, and let $k \geq 0$ be an integer. The *k-th degeneracy locus* $D_k(\varphi)$ is the closed subscheme of X whose ideal sheaf is generated, over any open set on which E and F are trivialized, by the $(k+1) \times (k+1)$ minors of the $f \times e$ matrix of regular functions representing φ in those trivializations. Its underlying set is $\{x \in X \,|\, \mathrm{rk}\varphi(x) \leq k\}$, and by convention $D_{-1}(\varphi) = \emptyset$. The integer $(e-k)(f-k)$ is the *expected codimension* of $D_k(\varphi)$.

The definition is independent of the trivializations. Changing them replaces the matrix A by PAQ with P and Q invertible matrices of regular functions. Each row of PA is a linear combination of rows of A, so by multilinearity in the rows every $(k+1)$-minor of PA is a linear combination of $(k+1)$-minors of A; the same argument in the columns handles the right factor. Hence the ideal of $(k+1)$-minors of PAQ is contained in that of A, and applying the argument to $P^{-1}(PAQ)Q^{-1}$ gives the reverse containment. The two ideals agree, so the local definitions glue.

The name "expected codimension" is justified by the universal case, where the rank condition is imposed on the space of all matrices at once.

Example 8.3.2: The Generic Determinantal Variety

Let $\mathbb{A}^{ef}$ be the space of $f \times e$ matrices over $\overline{k}$, let φ be the tautological matrix (its entries are the coordinate functions), and write $M_k = D_k(\varphi)$. Fix $0 \le k < \min(e, f)$ and let $U \subseteq \mathbb{A}^{ef}$ be the open set where the upper left $k \times k$ block A_{11} is invertible. Writing

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

in blocks of sizes k and $e-k$ down the columns and k and $f-k$ down the rows, the identity

$$\begin{pmatrix} I & 0 \\ -A_{21}A_{11}^{-1} & I \end{pmatrix} A \begin{pmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ 0 & \beta \end{pmatrix}, \qquad \beta := A_{22} - A_{21}A_{11}^{-1}A_{12} \tag{8.8}$$

holds on U, with both outer matrices invertible over $\mathcal{O}(U)$. By the invariance proved above, the ideal of $(k+1)$-minors of A on U equals that of the block diagonal matrix on the right. A $(k+1)$-minor of the latter is a product of a minor of A_{11} with a minor of β of positive size, hence lies in the ideal generated by the entries of β; conversely $\det(A_{11})\, \beta_{ij}$ is such a minor and $\det(A_{11})$ is a unit on U. So $M_k \cap U$ is the zero scheme of the $(e-k)(f-k)$ entries of β.

The map $(A_{11}, A_{12}, A_{21}, A_{22}) \mapsto (A_{11}, A_{12}, A_{21}, \beta)$ is an automorphism of U, being a morphism with an inverse of the same shape. Under it $M_k \cap U$ becomes the product of $\{A_{11}$ invertible$\}$ with the affine spaces of A_{12} and A_{21} and the origin in the β coordinates, so $M_k \cap U$ is smooth and irreducible of dimension

$$k^2 + k(e-k) + k(f-k) = ef - (e-k)(f-k)$$

Every matrix of rank exactly k has some invertible $k \times k$ submatrix, and permuting rows and columns is an automorphism of $\mathbb{A}^{ef}$ carrying M_k to itself, so the locus of matrices of rank exactly k is covered by copies of $M_k \cap U$ and is smooth of that dimension. A matrix A of rank $j < k$ is the limit as $t \to 0$ of $A + tC$, where C kills a chosen complement W of $\ker A$ and maps a $(k-j)$-dimensional subspace of $\ker A$ isomorphically onto a complement of the image of A; for $t \neq 0$ that sum has rank k. Hence M_k is the closure of the rank-k locus, irreducible of codimension exactly $(e-k)(f-k)$ in $\mathbb{A}^{ef}$.

The Schur complement in (8.8) is a local statement about matrices, so it globalizes to bundles at once. The next lemma is what every later argument in this section uses: near a point where the rank is exactly k, a degeneracy locus is the zero scheme of a section of a bundle of rank equal to its expected codimension.

Lemma 8.3.3: Local Form at a Point of Exact Rank

Let $\varphi\colon E \to F$ be as in definition 8.3.1 and let $x \in X$ satisfy $\mathrm{rk}\varphi(x) = k$. There is an open neighbourhood U of x, direct sum decompositions $E|_U = E_1 \oplus E_2$ and $F|_U = F_1 \oplus F_2$ with $\mathrm{rk}E_1 = \mathrm{rk}F_1 = k$, and an identification $F_1 \cong E_1$, under which

$$\varphi|_U = \begin{pmatrix} \mathrm{id}_{E_1} & 0 \\ 0 & \beta \end{pmatrix}, \qquad \beta \in \Gamma(U, \mathrm{Hom}(E_2, F_2)), \quad \beta(x) = 0$$

Consequently $D_k(\varphi) \cap U$ is the zero scheme of β, a section of a bundle of rank $(e-k)(f-k)$, and $\ker\varphi$ restricted to $D_k(\varphi) \cap U$ is the subbundle E_2.

***Proof*:**
Choose trivializations of E and F near x and let A be the resulting matrix. Since $\mathrm{rk}A(x) = k$, some $k \times k$ minor of $A(x)$ is nonzero; after permuting the frames we may assume it is the upper left one, and $\det A_{11}$ is then invertible on an open $U \ni x$. On U the two outer matrices in (8.8) have entries in $\mathcal{O}(U)$ and are invertible there, so they are automorphisms of $F|_U$ and $E|_U$; in the new frames φ is the block diagonal matrix with blocks A_{11} and β. Absorbing A_{11} into the frame of the first summand of $F|_U$ makes the first block the identity. Since $\mathrm{rk}\varphi(x) = k$ and the first block already has rank k at x, the second block satisfies $\beta(x) = 0$.

The computation of the ideal of $(k+1)$-minors carried out in example 8.3.2 applies verbatim to the block diagonal form, giving that $D_k(\varphi) \cap U$ is the zero scheme of the entries of β, that is, of the section β of $\mathrm{Hom}(E_2, F_2)$, a bundle of rank $(e-k)(f-k)$. On that zero scheme φ restricts to $\mathrm{id}_{E_1} \oplus 0$, whose kernel is E_2, completing the proof.

The formula to be proved expresses the class of $D_k(\varphi)$ through the Chern classes of E and F combined in one formal expression, which the following notation records.

Definition 8.3.4: Chern Classes of a Difference

Let X be a smooth variety and let E and F be vector bundles on X, either of which may be the zero bundle, with total Chern class 1 in that case. In the ring $A^*(X)$ of theorem 5.2.7 the total Chern class $c(E) = 1 + c_1(E) + \cdots$ is a unit, since $A^i(X) = 0$ for $i > \dim X$ makes $\sum_{j \geq 0}(1 - c(E))^j$ a finite sum inverting it. Set

$$c(F - E) := c(F)\, c(E)^{-1} \in A^*(X)$$

and write $c_i(F - E)$ for its component in $A^i(X)$, so that $c_0(F - E) = 1$ and $c_i(F - E) = 0$ for $i < 0$.

Two consequences of the definition are used constantly. First, $c(F - E)$ is multiplicative in the evident sense: $c((F \oplus F') - E) = c(F - E)c(F')$ by theorem 3.2.15. Second, the components of $c(-E) = c(E)^{-1}$ satisfy

$$\sum_{i=0}^{j} c_i(E)\, c_{j-i}(-E) = 0 \qquad (j \geq 1) \tag{8.9}$$

which is the degree j component of $c(E)c(-E) = 1$.

Theorem 8.3.5: Thom-Porteous Formula

Let X be a smooth variety, let $\varphi\colon E \to F$ be a morphism of vector bundles of ranks e and f, and let $0 \le k < \min(e, f)$. Assume

1. $D_k(\varphi)$ is empty or of pure codimension $(e-k)(f-k)$ in X; and
2. $D_{k-1}(\varphi)$ is empty or of codimension greater than $(e-k)(f-k)$.

Then, in $A^{(e-k)(f-k)}(X)$,

$$[D_k(\varphi)] = \det\left(c_{f-k+j-i}(F-E)\right)_{1\le i,j\le e-k}$$

where $[D_k(\varphi)]$ is the class of the fundamental cycle of the determinantal scheme (definition 1.1.2), each component counted with the length of the local ring of $D_k(\varphi)$ at its generic point.

Hypothesis (2) holds whenever $D_{k-1}(\varphi)$ has its own expected codimension, since $(e-k+1)(f-k+1) > (e-k)(f-k)$; in the applications below both hypotheses are verified directly. Fulton proves the formula assuming only (1), at the cost of constructing a class supported on $D_k(\varphi)$ before comparing it with the fundamental cycle ([8, Chapter 14]).

The determinant is an $(e-k)\times(e-k)$ matrix whose entries increase along rows, and each of its terms $\prod_i c_{f-k+\sigma(i)-i}(F-E)$ has total degree $(e-k)(f-k)$, so the right-hand side does live in the codimension the left-hand side occupies. The class of the locus depends on φ only through the bundles: any two maps $E \to F$ whose degeneracy loci have the expected codimension produce the same class. The answer is a Schur-type determinant in the classes $c_i(F-E)$, the same shape of expression as Giambelli's formula in theorem 8.2.4; corollary 8.3.7 shows the two agree where they overlap.

The proof replaces the rank condition, which is not linear, by a linear condition on a larger space: instead of asking that $\ker\varphi(x)$ have dimension at least $e-k$, one records a flag of subspaces of E_x of dimension $e-k$ inside the kernel. The locus of such records is the zero scheme of a section of a bundle, and its class is a top Chern class. Two ingredients are needed: the identification of the class of a zero scheme, and the computation of the pushforward from the flag bundle back to X.

The space on which the rank condition becomes linear is the bundle of flags of subbundles of E, built by iterating the projective bundle construction.

Definition 3.2.13 already builds this tower: $\mathrm{Fl}_m(E)$ is the m-step flag bundle of E, with tautological subbundles $S_1 \subset \cdots \subset S_m \subseteq q^*E$ and line quotients $\Lambda_i = S_i/S_{i-1}$. Write $\eta_i = c_1(\Lambda_i^\vee)$ and $S = S_m$, of rank m. One point of the construction bears repeating because getting it wrong negates every odd Chern class below: $\mathbb{P}(E^{(i-1)})$ is the bundle of *lines*, so importing a formula from [2, Projective Bundle], which uses the quotient convention, requires dualizing (box 0.0.2).

A point of Y_i over $y \in Y_{i-1}$ is a line in $E^{(i-1)}_y = E_x/(S_{i-1})_y$, that is a subspace S_i of E_x of dimension i containing S_{i-1}; inductively, a point of $\mathrm{Fl}_m(E)$ over $x \in X$ is a flag $S_1 \subset \cdots \subset S_m \subseteq E_x$ with $\dim S_i = i$, which is the description used below. If X is a smooth variety then so is $\mathrm{Fl}_m(E)$: each step is locally $U \times \mathbb{P}^{e-i}$ over a trivializing open U, hence smooth, and irreducible because any two such opens have preimages meeting over the intersection of two dense open subsets of an irreducible base. Its dimension is $\dim X + \sum_{i=1}^m (e-i)$.

Lemma 8.3.6: Pushforward from the Flag Bundle

Let X be a smooth variety, let E and F be bundles of ranks e and f, let $1 \le m \le e$, and let $q\colon Y = \mathrm{Fl}_m(E) \to X$ with tautological subbundle $S \subseteq q^*E$ of rank m. Put $\ell = f - e + m$ and

$$\gamma_m = \prod_{i=1}^{m-1} \eta_i^{m-i}$$

(an empty product, equal to 1, when $m = 1$). Then

$$q_*\Big(\gamma_m \cdot c_{mf}\big(S^\vee \otimes q^*F\big)\Big) = \det\big(c_{\ell+j-i}(F-E)\big)_{1\le i,j\le m}$$

***Proof*:**
Both sides lie in $A^{m\ell}(X)$: the class pushed forward has degree $mf + \binom{m}{2}$ and q has relative dimension $me - \binom{m+1}{2}$, and $mf + \binom{m}{2} - me + \binom{m+1}{2} = mf - me + m^2 = m\ell$, while each term of the determinant has degree $m\ell$. The proof is by induction on m, and the first three steps record what the projective bundle $\mathbb{P}(E)$ contributes at each stage of the tower. Throughout, $p\colon \mathbb{P}(E) \to X$ is the first step, $\Lambda = \Lambda_1$ and $\eta = \eta_1$.

Step 1: twisting by a line bundle. For a bundle F of rank f and a line bundle L on any smooth variety,

$$c_f(L \otimes F) = \sum_{i=0}^{f} c_i(F)\, c_1(L)^{f-i} \tag{8.10}$$

By theorem 3.2.14 it suffices to prove this after pulling back to a variety on which F has a filtration with line bundle quotients $L_1, \dots, L_f$ and on which pullback of Chern classes is injective. There $c_f(L \otimes F) = \prod_{i=1}^{f}(c_1(L) + c_1(L_i))$ by theorem 3.2.15, and expanding the product and collecting the elementary symmetric functions of the $c_1(L_i)$, which are the $c_i(F)$, gives (8.10).

Step 2: the tautological relation. The inclusion $\Lambda \subseteq p^*E$ is a nowhere-vanishing section of $\Lambda^\vee \otimes p^*E$, so that bundle contains a trivial line subbundle with locally free quotient W of rank $e-1$; by theorem 3.2.15 its total Chern class equals $c(W)$, whose component in degree e vanishes because W has rank $e-1$ (theorem 3.2.9). Applying (8.10) with $L = \Lambda^\vee$ and $F = p^*E$,

$$\sum_{i=0}^{e} c_i(E)\, \eta^{e-i} = 0 \tag{8.11}$$

where the Chern classes of E are written for their pullbacks.

Step 3: pushforward of powers of η. The class $p_*(\eta^j)$ lies in $A^{j-e+1}(X)$, hence vanishes for $j < e-1$. For $j = e-1$ it lies in $A^0(X) = \mathbb{Z}[X]$ and may be computed after restricting to an open U trivializing E, since p_* commutes with flat pullback (proposition 1.3.6) and $A^0(X) \to A^0(U)$ is an isomorphism. Over such a U one has $\mathbb{P}(E)|_U \cong U \times \mathbb{P}^{e-1}$ with η the pullback of the hyperplane class, and $\eta^{e-1} \cap [U \times \mathbb{P}^{e-1}]$ is the flat pullback of $\eta^{e-1} \cap [\mathbb{P}^{e-1}] = [\text{point}]$ (theorem 1.5.2, proposition 3.2.10), namely $[U \times \text{point}]$, which p maps isomorphically onto U.

Hence $p_*(\eta^{e-1}) = 1$. For $j \geq 1$, multiplying (8.11) by η^{j-1} and pushing forward gives

$$p_*(\eta^{e-1+j}) = -\sum_{i=1}^{e} c_i(E)\, p_*(\eta^{e-1+j-i})$$

by the projection formula (proposition 1.3.6), which is the recursion (8.9) for the components of $c(-E)$ with the same initial value. Therefore $p_*(\eta^{e-1+j}) = c_j(-E)$ for all $j \geq 0$. Combining this with (8.10) and the projection formula, for every $a \geq 0$,

$$p_*\Big(\eta^a\, c_f(\Lambda^\vee \otimes p^*F)\Big) = \sum_{i=0}^{f} c_i(F)\, p_*(\eta^{a+f-i}) = \sum_{i=0}^{f} c_i(F)\, c_{a+f-i-e+1}(-E) = c_{a+f-e+1}(F-E) \tag{8.12}$$

the last equality being the degree $a+f-e+1$ component of $c(F-E) = c(F)c(-E)$.

Step 4: the base case. For $m = 1$ one has $Y = \mathbb{P}(E)$, $S = \Lambda$, $\gamma_1 = 1$ and $\ell = f-e+1$, so (8.12) with $a = 0$ gives $q_*(c_f(\Lambda^\vee \otimes q^*F)) = c_\ell(F-E)$, which is the asserted 1×1 determinant.

Step 5: the inductive step. Let $m \geq 2$ and assume the statement for $m-1$ over every smooth base. By construction $\mathrm{Fl}_m(E) = \mathrm{Fl}_{m-1}(E^{(1)})$ over the base $\mathbb{P}(E)$, where $E^{(1)} = p^*E/\Lambda$ has rank $e-1$; write $b\colon Y \to \mathbb{P}(E)$ for that flag bundle, so $q = p \circ b$, and let $S' \subseteq b^*E^{(1)}$ be its tautological subbundle of rank $m-1$. Under the identification of points as flags, $S' = S/\Lambda$, and the classes $\eta_2, \ldots, \eta_{m-1}$ of Y over $\mathbb{P}(E)$ are those of the smaller tower, so

$$\gamma_m = \eta^{m-1} \cdot \gamma'_{m-1}, \qquad \gamma'_{m-1} = \prod_{i=1}^{m-2} \eta_{i+1}^{m-1-i}$$

The exact sequence $0 \to \Lambda \to S \to S' \to 0$ dualizes and tensors with q^*F to an exact sequence with sub $S'^\vee \otimes q^*F$ and quotient $\Lambda^\vee \otimes q^*F$, so theorem 3.2.15 gives

$$c_{mf}(S^\vee \otimes q^*F) = c_{(m-1)f}\big(S'^\vee \otimes b^*(p^*F)\big) \cdot c_f(\Lambda^\vee \otimes q^*F)$$

The second factor and η^{m-1} are pulled back from $\mathbb{P}(E)$, so by the projection formula and the inductive hypothesis applied over the smooth variety $\mathbb{P}(E)$ to the bundles $E^{(1)}$ and p^*F, whose associated integer is $f-(e-1)+(m-1) = \ell$,

$$q_*\Big(\gamma_m\, c_{mf}(S^\vee \otimes q^*F)\Big) = p_*\Big(\eta^{m-1}\, c_f(\Lambda^\vee \otimes p^*F) \cdot \det\big(c'_{\ell+j-i}\big)_{1\leq i,j\leq m-1}\Big)$$

where $c' = c(p^*F - E^{(1)})$. From $c(E^{(1)}) = c(p^*E)c(\Lambda)^{-1}$ and $c(\Lambda) = 1-\eta$ one gets $c' = c(F-E)(1-\eta)$, that is

$$c'_r = c_r - \eta\, c_{r-1}, \qquad c_r := c_r(F-E) \tag{8.13}$$

Step 6: the determinant expansion. Let M be the $m \times m$ matrix with $M_{ij} = c_{\ell+j-i}$, and let $M^{(1,t)}$ denote the matrix obtained from M by deleting the first row and the t-th column. Let R be the $(m-1)\times m$ matrix consisting of the last $m-1$ rows of M, so $R_{ij} = c_{\ell+j-i-1}$, and let N be the $(m-1)\times(m-1)$ matrix with entries $c'_{\ell+j-i}$. By (8.13) the j-th column of N is $R_{\bullet,j+1} - \eta R_{\bullet,j}$. Expanding $\det N$ by multilinearity in the columns produces one term for each subset $T \subseteq \{1, \dots, m-1\}$ of columns at which the second alternative is taken, with coefficient $(-\eta)^{|T|}$ and columns $R_{\bullet,j}$ for $j \in T$ and $R_{\bullet,j+1}$ for $j \notin T$. If some $j \in T$ has $j \geq 2$ and $j-1 \notin T$ then the column $R_{\bullet,j}$ occurs twice, once from $j \in T$ and once as $R_{\bullet,(j-1)+1}$ from $j-1 \notin T$, and the term vanishes. The subsets with no such j are exactly the sets $T = \{1, \dots, t\}$ for $0 \leq t \leq m-1$, the empty set being the case $t = 0$, so only those contribute; for those the columns are $R_{\bullet,1}, \dots, R_{\bullet,t}, R_{\bullet,t+2}, \dots, R_{\bullet,m}$ in increasing order, whose determinant is $\det M^{(1,t+1)}$. Hence

$$\det \left(c'_{\ell+j-i}\right)_{1 \leq i,j \leq m-1} = \sum_{t \geq 0} (-\eta)^t \det M^{(1,t+1)} \tag{8.14}$$

Step 7: assembly. Substituting (8.14) into the expression of Step 5, moving the classes $\det M^{(1,t+1)}$ (which are pulled back from X) out by the projection formula, and evaluating each remaining pushforward by (8.12) with $a = m-1+t$, which returns $c_{\ell+t}$ because $(m-1+t)+f-e+1 = \ell+t$,

$$q_*\Big(\gamma_m \, c_{mf}(S^\vee \otimes q^*F)\Big) = \sum_{t \geq 0} (-1)^t \, c_{\ell+t} \det M^{(1,t+1)}$$

The right-hand side is the Laplace expansion of $\det M$ along its first row, whose entries are $M_{1,t+1} = c_{\ell+t}$ with cofactor sign $(-1)^{1+(t+1)} = (-1)^t$. So the pushforward equals $\det M$, completing the induction and the proof.

***Proof of theorem 8.3.5*:**
Write $m = e-k \geq 1$ and $\nu = f-k \geq 1$, so the expected codimension is $m\nu$ and the asserted determinant is the one produced by lemma 8.3.6, whose integer $\ell = f-e+m$ equals ν.

Step 1: the resolution. Let $q\colon Y = \mathrm{Fl}_m(E) \to X$ with tautological subbundle $S \subseteq q^*E$, and let

$$\Psi\colon S \hookrightarrow q^*E \xrightarrow{q^*\varphi} q^*F$$

regarded as a section of $G := S^\vee \otimes q^*F$, a bundle of rank mf. Write $Z = Z(\Psi)$ for its zero scheme: a point of Y lies in Z exactly when the flag it records has its top member inside $\ker\varphi$.

Step 2: the resolution over the locus of rank at least k. Put $X^\circ = X \setminus D_{k-1}(\varphi)$, an open subvariety, nonempty because hypothesis (2) makes $D_{k-1}(\varphi)$ of positive codimension. Write $Y^\circ = q^{-1}(X^\circ)$, $D^\circ = D_k(\varphi) \cap X^\circ$ and $Z^\circ = Z \cap Y^\circ$. Cover X° by open sets of two kinds: those meeting $D_k(\varphi)$ not at all, over which Z is empty, and neighbourhoods U of points of exact rank k, on which lemma 8.3.3 puts φ in the block form $\mathrm{id}_{E_1} \oplus \beta$. On such a U, a point of Z is a flag whose top member S satisfies $\varphi(S) = 0$; since the first block is the identity, the composite $S \to E|_U \to E_1$ vanishes, so $S \subseteq E_2$, and the induced map $S \to E_2$ of bundles of rank m is injective on fibres, hence an isomorphism. Thus $S = E_2$ on $Z \cap q^{-1}(U)$, the remaining data is a complete flag inside E_2, and the equations that survive are the entries of β, which cut out $D_k(\varphi) \cap U$. So

$$Z \cap q^{-1}(U) \cong \big(D_k(\varphi) \cap U\big) \times_U \mathrm{Fl}_m(E_2)$$

as schemes. Globally over X° this says $Z^\circ \cong \mathrm{Fl}_m(K)$, where $K = \ker(\varphi)|_{D^\circ}$ is locally free of rank m by lemma 8.3.3, and $\mathrm{Fl}_m(K)$ is the bundle of complete flags in K.

Step 3: purity. By hypothesis (1) the scheme D° is empty or of pure dimension $\dim X - m\nu$, so by Step 2, Z° is empty or of pure dimension $\dim X - m\nu + \binom{m}{2}$, the second summand being the dimension of a complete flag variety in m variables. Since $\dim Y = \dim X + \sum_{i=1}^{m}(e-i) = \dim X + me - \binom{m+1}{2}$, the codimension of Z° in Y° is

$$me - \binom{m+1}{2} + m\nu - \binom{m}{2} = me + m\nu - m^2 = m(k+\nu) = mf$$

using $e - m = k$ and $k + \nu = f$. So Z° is empty or of pure codimension mf, the rank of G.

Step 4: the two pushforwards. Lemma 5.3.14, applied on the smooth variety Y° to the section Ψ of G, gives $[Z^\circ] = c_{mf}(G) \cap [Y^\circ]$ in $A_{\dim Y - mf}(Y^\circ)$. Multiply by γ_m and push forward by q. On the right, lemma 8.3.6 applied over the smooth variety X° evaluates the result as the determinant $\Delta := \det(c_{\nu+j-i}(F-E))_{1\le i,j\le m}$ restricted to X°, Chern classes being compatible with restriction to an open subset (proposition 3.2.10). On the left, use Step 2: the morphism $Z^\circ \to D^\circ$ is the flag bundle of K, which is locally a product, so the multiplicity of each component of Z° in its fundamental cycle equals that of the component of D° below it, and γ_m restricts on Z° to the corresponding product of the classes η_i for the tower $\mathrm{Fl}_m(K) \to D^\circ$. That tower has $m-1$ nontrivial steps, the i-th being a projective bundle of a rank-$(m-i+1)$ bundle, and γ_m contributes $\eta_i^{(m-i+1)-1}$ at that step; pushing down one step at a time with the projection formula and $p_*(\eta^{r-1}) = 1$ from Step 3 of the proof of lemma 8.3.6 yields

$$q_*\big(\gamma_m \cap [Z^\circ]\big) = [D^\circ]$$

Hence $[D^\circ] = \Delta \cap [X^\circ]$ in $A_{\dim X - m\nu}(X^\circ)$.

Step 5: from X° to X. Set $d = \dim X - m\nu$. Restriction to the open subvariety X° carries $[D_k(\varphi)]$ to $[D^\circ]$ and $\Delta \cap [X]$ to $\Delta \cap [X^\circ]$, so the difference of the two classes on X lies in the kernel of $A_d(X) \to A_d(X^\circ)$, which by theorem 1.4.1 is the image of $A_d(D_{k-1}(\varphi))$. Hypothesis (2) makes $\dim D_{k-1}(\varphi) < d$, and a scheme carries no cycles of dimension exceeding its own, so that group vanishes. Therefore $[D_k(\varphi)] = \Delta \cap [X]$, completing the proof.

The formula returns familiar statements in its two extreme cases. For $e = 1$ and $k = 0$ the map φ is a section of $E^\vee \otimes F$ and the determinant is the single class $c_f(F-E)$, which is $c_f(E^\vee \otimes F)$; this is lemma 5.3.14 again. For $e = f$ and $k = e-1$ it reads $[D_{e-1}(\varphi)] = c_1(F) - c_1(E)$, the class of the divisor cut by $\det \varphi$, a section of $\mathrm{Hom}(\det E, \det F)$.

The case that connects this section to section 8.1 is the tautological map on a Grassmannian, where a single rank condition cuts out a Schubert variety of rectangular shape.

Corollary 8.3.7: Rectangular Schubert Classes as Determinants

Let $1 \le a \le r$ and $1 \le b \le n-r$, and let $\lambda = (b^a)$ be the partition with a parts equal to b, a rectangle inside the $r \times (n-r)$ box. On $\mathrm{Gr}(r,n)$,

$$\sigma_\lambda = \det\left(\sigma_{b+j-i}\right)_{1\le i,j\le a}$$

where $\sigma_j = c_j(Q)$ are the special Schubert classes and $\sigma_j = 0$ for $j < 0$.

***Proof*:**
Fix the complete flag $F_\bullet$ of definition 8.1.1 in the n-dimensional space V, put $t = n - r + a - b$, and let

$$\varphi\colon S \longrightarrow (V/F_t) \otimes \mathcal{O}$$

be the tautological subbundle followed by the quotient map, a morphism of bundles of ranks $e = r$ and $f = n - t = r - a + b$. For a point W of $\mathrm{Gr}(r, n)$ the fibre $\varphi(W)$ is the composite $W \subseteq V \to V/F_t$, whose kernel is $W \cap F_t$. Set $k = r - a$, so that $e - k = a$ and $f - k = b$, and the expected codimension is $ab = |\lambda|$.

Step 1: the support is the Schubert variety. A point W lies in $D_k(\varphi)$ exactly when $\dim(W \cap F_t) \geq a$. That single condition implies the others defining $\Omega_\lambda(F_\bullet)$: for $i < a$ the flag index attached to $\lambda_i = b$ is $n - r + i - b = t - (a - i)$, and lowering a flag index by one lowers the dimension of the intersection by at most one, so $\dim(W \cap F_{t-(a-i)}) \geq a - (a - i) = i$. Conversely the condition at $i = a$ is one of the defining conditions. Hence $D_k(\varphi)$ and $D_{k-1}(\varphi)$ have supports $\Omega_{(b^a)}(F_\bullet)$ and $\Omega_{((b+1)^{a+1})}(F_\bullet)$ respectively, the latter being empty when $((b + 1)^{a+1})$ leaves the box. By proposition 8.1.2 these have codimensions ab and $(a + 1)(b + 1)$, and each is irreducible; so hypothesis (1) of theorem 8.3.5 holds once the codimension statement is known for the scheme, which it is since a scheme and its support have the same dimension, and hypothesis (2) holds because $(a + 1)(b + 1) > ab$.

Step 2: the determinantal scheme is generically reduced. Let W_0 be a point of the open cell $\Omega^\circ_\lambda(F_\bullet)$, so $K_0 = W_0 \cap F_t$ has dimension exactly a. Choose a complement U_0 of W_0 in V and a splitting $W_0 = K_0 \oplus K_0'$. The subspaces W with $W \cap U_0 = 0$ are exactly the graphs $W_u = \{w + u(w) \big| w \in W_0\}$ for $u \in \mathrm{Hom}(W_0, U_0)$, and $u \mapsto W_u$ is a bijection onto an open neighbourhood of W_0; identifying W_u with W_0 by $w \mapsto w + u(w)$, the map φ becomes $\varphi_u(w) = \overline{w + u(w)} \in V/F_t$, whose entries are affine functions of u. Decompose $V/F_t = \varphi_0(K_0') \oplus C$ with $\dim C = f - k = b$. In the resulting block form of φ_u, the $K_0' \to \varphi_0(K_0')$ block is invertible near $u = 0$ and depends only on $u|_{K_0'}$; the $K_0' \to C$ block vanishes at $u = 0$ and depends only on $u|_{K_0'}$; the $K_0 \to \varphi_0(K_0')$ block is linear in $u|_{K_0}$; and the $K_0 \to C$ block is the linear map $u|_{K_0}$ followed by the projection $U_0 \to V/(F_t + W_0) \cong C$. By (8.8) the ideal of $D_k(\varphi)$ near W_0 is generated by the entries of the Schur complement, which by the shape just listed is of the form $\beta(u) = \Phi_{u|_{K_0'}}(u|_{K_0})$ with Φ_{u_2} linear in the first argument and regular in u_2. At $u = 0$ the map Φ_0 is the restriction-then-project map $\mathrm{Hom}(W_0, U_0) \to \mathrm{Hom}(K_0, V/(F_t + W_0))$, which is surjective because $U_0 + W_0 = V$ forces $U_0 \to V/(F_t + W_0)$ to be surjective; surjectivity of a matrix of regular functions is an open condition, so Φ_{u_2} stays surjective for u_2 near 0. Its kernel is then a subbundle, and choosing a local trivialization splitting it makes the entries of β into ab of the coordinates. Hence $D_k(\varphi)$ is smooth of codimension ab near W_0, in particular reduced there, and its fundamental cycle is $[\Omega_\lambda(F_\bullet)] = \sigma_\lambda$.

Step 3: the Chern classes. The target of φ is a trivial bundle, so its total Chern class is 1 (example 3.2.12), and $c(S)c(Q) = 1$ by proposition 8.1.5, the tautological sequence being the one recalled in [2, The Tautological Sequence on a Grassmannian]. Hence

$$c\big((V/F_t) \otimes \mathcal{O} - S\big) = c(S)^{-1} = c(Q)$$

so c_j of the difference is σ_j. Theorem 8.3.5 with $e - k = a$ and $f - k = b$ now reads $\sigma_\lambda = \det(\sigma_{b+j-i})_{1 \leq i,j \leq a}$, as we sought to show.

The corollary is Giambelli's formula (theorem 8.2.4) for rectangular λ, obtained without Pieri's formula and without the symmetric function identity that section 8.2 used: the determinant appears here as the shape of the Thom-Porteous class rather than as a Jacobi-Trudi expansion. A Schubert variety attached to a non-rectangular partition is cut out by more than one rank condition, and one bundle map does not see it; the determinantal formula for several conditions at once is the Kempf-Laksov formula, of which theorem 8.3.5 is the one-condition case.

Away from Grassmannians the formula computes degrees of classically studied curves and surfaces from the shape of the matrix defining them.

Example 8.3.8: The Twisted Cubic as a Degeneracy Locus

On $X = \mathbb{P}^3$ with homogeneous coordinates $x_0, \dots, x_3$, let

$$\varphi\colon \mathcal{O}^{\oplus 2} \longrightarrow \mathcal{O}(1)^{\oplus 3}, \qquad \varphi = \begin{pmatrix} x_0 & x_1 \\ x_1 & x_2 \\ x_2 & x_3 \end{pmatrix}$$

so $e = 2$, $f = 3$ and the interesting locus is $D_1(\varphi)$, of expected codimension $(2-1)(3-1) = 2$: a curve. Its ideal is generated by the three 2×2 minors $x_0x_2 - x_1^2$, $x_0x_3 - x_1x_2$, $x_1x_3 - x_2^2$. A point of D_1 is one where the two columns are proportional; parametrizing the proportionality by $(1 : s)$ or $(s : 1)$ shows the support of D_1 is the image C of $\mathbb{P}^1 \to \mathbb{P}^3$, $(u : v) \mapsto (u^3 : u^2v : uv^2 : v^3)$, the twisted cubic. In particular D_1 has pure codimension 2, and $D_0(\varphi)$, where all six entries vanish, is empty. Both hypotheses of theorem 8.3.5 hold.

With h the hyperplane class, $c(E) = 1$ and $c(F) = (1+h)^3$, so $c(F - E) = 1 + 3h + 3h^2 + h^3$. Here $e - k = 1$ and $f - k = 2$, so the determinant is the single entry

$$[D_1(\varphi)] = c_2(F - E) = 3h^2$$

Independently, $A^2(\mathbb{P}^3)$ is generated by the class of a line (theorem 1.5.2), and a curve of degree d has class dh^2; the twisted cubic meets the plane $\{a_0x_0 + \cdots + a_3x_3 = 0\}$ where $a_0u^3 + a_1u^2v + a_2uv^2 + a_3v^3 = 0$, which for a general plane has three distinct roots in $\mathbb{P}^1$, so $\deg C = 3$ and $[C] = 3h^2$. The two computations agree, and since $[D_1(\varphi)] = \mu[C]$ for the multiplicity μ of D_1 along C, comparing them gives $\mu = 1$: the ideal of 2×2 minors above defines the twisted cubic with its reduced structure.

Exercise 8.3.1

1. Let $\varphi\colon \mathcal{O}_{\mathbb{P}^2}^{\oplus 2} \to \mathcal{O}_{\mathbb{P}^2}(1)^{\oplus 2}$ be given by a 2×2 matrix of linear forms whose determinant is not identically zero and whose four entries have no common zero. Show that $D_1(\varphi)$ satisfies the hypotheses of theorem 8.3.5, compute its class by the formula, and check the answer against the description of $D_1(\varphi)$ as the zero divisor of $\det \varphi$.
2. Fix $n \geq 2$ and let $\varphi\colon \mathcal{O}_{\mathbb{P}^n}^{\oplus 2} \to \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus n}$ have i-th row $(x_{i-1} \;\; x_i)$ for $1 \leq i \leq n$. Show that $D_1(\varphi)$ is the rational normal curve of degree n, that $D_0(\varphi)$ is empty, and that theorem 8.3.5 assigns $D_1(\varphi)$ the class nh^{n-1}. Verify the degree independently from the parametrization $(u : v) \mapsto (u^n : u^{n-1}v : \cdots : v^n)$.
3. Let F be a vector bundle of rank f on a smooth variety X and let $\psi \in \Gamma(X, F)$ have zero

scheme of pure codimension f. Deduce from theorem 8.3.5, applied to $\varphi\colon \mathcal{O} \to F$, that $[Z(\psi)] = c_f(F)$, and identify which hypothesis of the theorem is discharged by the convention $D_{-1}(\varphi) = \emptyset$.

4. Let $\varphi\colon E \to F$ be a morphism of bundles of ranks e and f on a smooth variety. Show that $D_k(\varphi^\vee) = D_k(\varphi)$ as subschemes, where $\varphi^\vee\colon F^\vee \to E^\vee$. Then check, for $e = 2$, $f = 3$ and $k = 1$, that the two determinants theorem 8.3.5 produces from φ and from $\varphi^\vee$ agree, using $c_i(E^\vee) = (-1)^i c_i(E)$.

5. In $A^*(\mathrm{Gr}(2,5))$ use corollary 8.3.7 to express $\sigma_{2,2}$ as a polynomial in the special classes, and confirm the answer by expanding the two products that occur with Pieri's formula (theorem 8.2.1).

6. Let X be a smooth projective surface and $\varphi\colon E \to F$ a morphism of rank-2 bundles whose determinant $\det\varphi$ is a nonzero section of $\mathrm{Hom}(\det E, \det F)$ and whose vanishing locus $D_0(\varphi)$ is finite. Show that $D_1(\varphi)$ is the zero divisor of $\det\varphi$, that the hypotheses of theorem 8.3.5 hold, and that the formula returns $c_1(F) - c_1(E)$ in both descriptions.

7. Let $E = \mathcal{O} \oplus \mathcal{O}(1)$ on $\mathbb{P}^1$ and let $p\colon \mathbb{P}(E) \to \mathbb{P}^1$ with $\eta = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$. Using the tautological relation (8.11), compute $p_*(\eta^2)$ and compare it with the component $c_1(-E)$ of $c(E)^{-1}$.

8.4 Classical Enumerative Problems

Two counts are carried out here with the classes of sections 8.1 to 8.3. The first is the number of lines on a smooth cubic surface in $\mathbb{P}^3$: lines in $\mathbb{P}^3$ are parametrized by $\mathrm{Gr}(2,4)$, the condition of lying on a fixed cubic surface is the vanishing of a section of a rank-four bundle on that Grassmannian, and the count is the degree of the top Chern class of the bundle, once the section is known to vanish at finitely many points and to vanish at each of them simply. The second is the five-conic problem, where the same procedure on the space $\mathbb{P}^5$ of plane conics returns the wrong number. Between the two counts, the classical principle of conservation of number is stated precisely enough to identify which of its hypotheses fails.

Fix a four-dimensional vector space V over $\overline{k}$ and write $\mathbb{P}^3 = \mathbb{P}(V)$. A line in $\mathbb{P}^3$ is $\mathbb{P}(W)$ for a unique two-dimensional subspace $W \subseteq V$, so $\mathrm{Gr}(2,4)$ is the parameter space of lines in $\mathbb{P}^3$, of dimension 4. Let $S \subseteq V \otimes \mathcal{O}$ be the tautological subbundle, whose fibre at $[W]$ is W, and let Q be the tautological quotient, so that

$$0 \to S \to V \otimes \mathcal{O} \to Q \to 0$$

is exact ([2, The Tautological Sequence on a Grassmannian]).

A cubic form $F \in \mathrm{Sym}^3 V^\vee$ restricts on each two-plane to $F|_W \in \mathrm{Sym}^3 W^\vee$, and $\mathbb{P}(W)$ is contained in the surface defined by F precisely when $F|_W = 0$. These restrictions are the values of a single global section. Dualizing the tautological inclusion gives a surjection $V^\vee \otimes \mathcal{O} \to S^\vee$, hence a surjection $\mathrm{Sym}^3 V^\vee \otimes \mathcal{O} \to \mathrm{Sym}^3 S^\vee$; the image of the constant section F of the trivial bundle $\mathrm{Sym}^3 V^\vee \otimes \mathcal{O}$ is a global section σ_F of $\mathrm{Sym}^3 S^\vee$ with $\sigma_F([W]) = F|_W$. A rank-two bundle has Sym^3 of rank 4, since locally Sym^3 of a free module on u, v is free on u^3, u^2v, uv^2, v^3. So $\mathrm{Sym}^3 S^\vee$ has rank $4 = \dim \mathrm{Gr}(2,4)$, and the zero scheme of σ_F has expected dimension zero.

Theorem 8.4.1: The Twenty-Seven Lines on a Smooth Cubic Surface

Let $F \in \mathrm{Sym}^3 V^\vee$ be a nonzero cubic form and let $X \subseteq \mathbb{P}^3$ be the cubic surface it defines. Assume X is smooth, so that the four partial derivatives F_0, F_1, F_2, F_3 of F have no common zero on X ([2, Smooth Scheme Over a Field]). Then X contains exactly 27 lines.

***Proof*:**

Step 1: the zero scheme parametrizes the lines on X. Write Z for the zero scheme of σ_F, that is, the degeneracy locus $D_0(\sigma_F)$ of the bundle map $\sigma_F \colon \mathcal{O} \to \mathrm{Sym}^3 S^\vee$, whose determinantal structure is cut by the 1×1 minors, namely by the components of σ_F in any local trivialization. A point $[W] \in \mathrm{Gr}(2,4)$ lies in Z if and only if $F|_W = 0$, if and only if $\mathbb{P}(W) \subseteq X$. So the closed points of Z are the lines on X, each occurring once. What remains is to show that Z is finite, that it is reduced, and that its degree is 27.

Step 2: $\deg c_4(\mathrm{Sym}^3 S^\vee) = 27$. By section 8.1, the open Schubert cells $\Omega^\circ_{(0,\ldots,0)}(F_\bullet) = \{W' \big| W' \cap F_2 = 0\}$ cover $\mathrm{Gr}(2,4)$ by copies of $\mathbb{A}^4$ (proposition 8.1.2), so $\mathrm{Gr}(2,4)$ is smooth and $A^*(\mathrm{Gr}(2,4))$ is a graded ring (theorem 5.2.7), in which the Chern classes may be multiplied.

By the splitting principle (theorem 3.2.14) there is a morphism $f \colon Y \to \mathrm{Gr}(2,4)$ with f^* injective on Chow groups and with f^*S carrying a line subbundle L whose quotient $M = f^*S/L$ is a line bundle. Set $x = c_1(L)$ and $y = c_1(M)$, so that $c(f^*S) = (1+x)(1+y)$ by the Whitney formula (theorem 3.2.15). Dualizing gives $0 \to M^\vee \to f^*S^\vee \to L^\vee \to 0$, and $c_1(L^\vee) = -x$, $c_1(M^\vee) = -y$, because $L \otimes L^\vee \cong \mathcal{O}$ and the first Chern class is additive on tensor products of line bundles and vanishes on $\mathcal{O}$ (proposition 2.4.4). Write $E = f^*S^\vee$, with line subbundle $M^\vee$ and quotient $L^\vee$.

The bundle $\mathrm{Sym}^3 E$ carries the filtration $\mathrm{Sym}^3 E = G^0 \supseteq G^1 \supseteq G^2 \supseteq G^3 \supseteq 0$, where G^j is the image of $(M^\vee)^{\otimes j} \otimes \mathrm{Sym}^{3-j} E \to \mathrm{Sym}^3 E$. Locally, if u spans $M^\vee$ and u, v span E, then G^j is spanned by the monomials in u, v of degree 3 with at least j factors of u, so G^j/G^{j+1} is a line bundle with generator the image of $u^j v^{3-j}$, that is, $G^j/G^{j+1} \cong (M^\vee)^{\otimes j} \otimes (L^\vee)^{\otimes(3-j)}$. Its first Chern class is $-jy - (3-j)x$. Applying theorem 3.2.15 to the three short exact sequences $0 \to G^{j+1} \to G^j \to G^j/G^{j+1} \to 0$ gives

$$c(\mathrm{Sym}^3 E) = (1-3x)(1-2x-y)(1-x-2y)(1-3y)$$

so that

$$c_4(\mathrm{Sym}^3 E) = 3x(2x+y)(x+2y)3y = 9xy\,(2x^2+5xy+2y^2)$$

Writing $e_1 = x+y = c_1(f^*S)$ and $e_2 = xy = c_2(f^*S)$ and substituting $2x^2+5xy+2y^2 = 2e_1^2+e_2$ gives $c_4(\mathrm{Sym}^3 E) = 9e_2(2e_1^2+e_2)$. Both sides are pullbacks along f, and f^* is injective, so

$$c_4(\mathrm{Sym}^3 S^\vee) = 9c_2(S)\big(2c_1(S)^2 + c_2(S)\big)$$

in $A^*(\mathrm{Gr}(2,4))$.

Now express $c_1(S)$ and $c_2(S)$ in the Schubert basis. By proposition 8.1.5, $c(S)c(Q) = 1$ with $c(Q) = 1 + \sigma_1 + \sigma_2$, and $c_i(S) = 0$ for $i > 2$ (theorem 3.2.9); comparing terms in degrees 1 and 2 gives $c_1(S) = -\sigma_1$ and $c_2(S) = \sigma_1^2 - \sigma_2$, which is $\sigma_{1,1}$ by the relation $\sigma_1^2 = \sigma_2 + \sigma_{1,1}$ of example 8.1.6. Therefore

$$c_4(\mathrm{Sym}^3 S^\vee) = 9\sigma_{1,1}(2\sigma_1^2 + \sigma_{1,1}) = 18\sigma_1^2\sigma_{1,1} + 9\sigma_{1,1}^2$$

The two degrees are read from Schubert duality (proposition 8.1.3): in the 2×2 box the complement of $\lambda = (2,0)$ is $(2,0)$ and the complement of $\lambda = (1,1)$ is $(1,1)$, so $\deg(\sigma_2\sigma_{1,1}) = 0$ and $\deg(\sigma_{1,1}^2) = 1$. Since $\sigma_1^2\sigma_{1,1} = \sigma_2\sigma_{1,1} + \sigma_{1,1}^2$, this gives $\deg(\sigma_1^2\sigma_{1,1}) = 1$, and hence

$$\deg c_4(\mathrm{Sym}^3 S^\vee) = 18 + 9 = 27$$

Step 3: Z is nonempty. Suppose σ_F vanished nowhere. Fix a point p and a local frame of $\mathrm{Sym}^3 S^\vee$ near p; since $\sigma_F(p) \neq 0$ in the fibre at p, one may complete $\sigma_F(p)$ to a basis of that fibre and lift the remaining basis vectors to local sections s_2, s_3, s_4. By Nakayama's lemma σ_F, s_2, s_3, s_4 generate $\mathrm{Sym}^3 S^\vee$ near p, and a surjection of free modules of equal rank is an isomorphism, so $\mathrm{Sym}^3 S^\vee \cong \mathcal{O}\sigma_F \oplus \mathcal{O}^3$ near p. Hence $\mathcal{O} \to \mathrm{Sym}^3 S^\vee$ is a subbundle inclusion with locally free quotient G of rank 3, and theorem 3.2.15 gives $c(\mathrm{Sym}^3 S^\vee) = c(\mathcal{O})c(G) = c(G)$. But $c_4(G) = 0$ because G has rank 3 (theorem 3.2.9), contradicting $\deg c_4(\mathrm{Sym}^3 S^\vee) = 27$ of Step 2. So σ_F vanishes somewhere: X contains at least one line.

Step 4: σ_F vanishes simply at each of its zeros. Let $[W] \in Z$, so that $\ell = \mathbb{P}(W) \subseteq X$. Choose a basis e_0, e_1, e_2, e_3 of V with $W = \langle e_0, e_1 \rangle$ and let $U_0 = \{W' \,|\, W' \cap \langle e_2, e_3 \rangle = 0\}$, the open Schubert cell containing $[W]$ from Step 2. Each $W' \in U_0$ is the row space of a unique matrix

$$\begin{pmatrix} 1 & 0 & a_0 & a_1 \\ 0 & 1 & b_0 & b_1 \end{pmatrix}$$

which identifies U_0 with $\mathbb{A}^4$ with coordinates (a_0, a_1, b_0, b_1) and $[W]$ with the origin. Over U_0 the bundle S is framed by $u = e_0 + a_0 e_2 + a_1 e_3$ and $v = e_1 + b_0 e_2 + b_1 e_3$, so $\mathrm{Sym}^3 S^\vee$ is framed by the monomials of degree 3 in the dual frame, and the section σ_F has as its four components the coefficients c_0, c_1, c_2, c_3 of the binary cubic

$$\Psi(a, b; s, t) = F(su + tv) = F(s,\ t,\ sa_0 + tb_0,\ sa_1 + tb_1)$$

in s^3, s^2t, st^2, t^3. Thus $Z \cap U_0$ is the closed subscheme of $\mathbb{A}^4$ defined by c_0, c_1, c_2, c_3.

Differentiating Ψ and evaluating at the origin, where the last two arguments of F vanish, gives

$$\left.\frac{\partial \Psi}{\partial a_j}\right|_0 = s\, G_{2+j}(s,t), \qquad \left.\frac{\partial \Psi}{\partial b_j}\right|_0 = t\, G_{2+j}(s,t), \qquad j = 0, 1$$

where $G_m(s,t) = F_m(s,t,0,0)$ is the restriction of the m-th partial derivative of F to ℓ, a binary quadratic. So the differential of (c_0, c_1, c_2, c_3) at the origin, viewed as a linear map from $\mathbb{A}^4$ to the space of binary cubics, is

$$(\alpha_0, \alpha_1, \beta_0, \beta_1) \longmapsto s(\alpha_0 G_2 + \alpha_1 G_3) + t(\beta_0 G_2 + \beta_1 G_3)$$

The two quadratics G_2, G_3 have no common zero. Indeed F vanishes identically on W, so $F(s,t,0,0) = 0$ in $\overline{k}[s,t]$; differentiating this identity with respect to s and to t gives $G_0 = G_1 = 0$. At a point $p \in \ell$ the gradient of F is therefore $(0, 0, G_2(p), G_3(p))$, and it is nonzero because X is smooth, so G_2 and G_3 do not both vanish at p.

Suppose now that $(\alpha_0, \alpha_1, \beta_0, \beta_1)$ lies in the kernel of the differential. Put $P = \alpha_0 G_2 + \alpha_1 G_3$ and $R = \beta_0 G_2 + \beta_1 G_3$, so that $sP = -tR$ in the unique factorization domain $\overline{k}[s,t]$. Then s divides R, say $R = sR_1$ with R_1 linear, and cancelling s gives $P = -tR_1$. Two cases arise.

If $R_1 = 0$ then $P = R = 0$. The quadratics G_2, G_3 are linearly independent: a relation $\alpha_0 G_2 + \alpha_1 G_3 = 0$ with, say, $\alpha_0 \neq 0$ makes G_2 a multiple of G_3, so every zero of G_3 is a zero of G_2, and G_3 has a zero in $\mathbb{P}^1$ unless $G_3 = 0$, in which case $G_2 = 0$ as well; either way G_2 and G_3 acquire a common zero, which they do not have. Hence $\alpha_0 = \alpha_1 = \beta_0 = \beta_1 = 0$.

If $R_1 \neq 0$, set $D = \alpha_0\beta_1 - \alpha_1\beta_0$. Eliminating G_3 from the two equations $P = -tR_1$ and $R = sR_1$ gives $\beta_1 P - \alpha_1 R = D\,G_2$, that is, $D\,G_2 = -(\alpha_1 s + \beta_1 t)R_1$; eliminating G_2 gives $\beta_0 P - \alpha_0 R = -D\,G_3$, that is, $D\,G_3 = (\alpha_0 s + \beta_0 t)R_1$. If $D \neq 0$ then the nonconstant form R_1 divides both G_2 and G_3, so they share a zero, which is excluded. If $D = 0$ then $(\alpha_1 s + \beta_1 t)R_1 = 0$ and $(\alpha_0 s + \beta_0 t)R_1 = 0$; since $R_1 \neq 0$ and $\overline{k}[s,t]$ is a domain, all four coefficients vanish, whence $P = 0 = -tR_1$ and $R_1 = 0$, contrary to assumption.

So the kernel is zero, and the differential is an isomorphism of four-dimensional spaces. Consequently the images of c_0, c_1, c_2, c_3 in $\mathfrak{m}/\mathfrak{m}^2$ are a basis, where $\mathfrak{m}$ is the maximal ideal of the local ring $\mathcal{O}_{\mathrm{Gr}(2,4),[W]}$; by Nakayama's lemma they generate $\mathfrak{m}$, and the local ring of Z at $[W]$ is $\mathcal{O}_{\mathrm{Gr}(2,4),[W]}/(c_0, c_1, c_2, c_3) = \mathcal{O}_{\mathrm{Gr}(2,4),[W]}/\mathfrak{m} = \overline{k}$. So $[W]$ is an isolated point of Z and Z is reduced there.

Step 5: the count. By Step 4 every point of Z is isolated, so Z is a zero-dimensional closed subscheme of the projective variety $\mathrm{Gr}(2,4)$, hence finite, and it is reduced, so its fundamental cycle is $[Z] = \sum_{p \in Z}[p]$ (definition 1.1.2). By Step 3 it is nonempty, so its codimension is 4, which is the expected codimension $(e-k)(f-k) = (1-0)(4-0)$ of the locus $D_0(\sigma_F)$ for $\sigma_F\colon \mathcal{O} \to \mathrm{Sym}^3 S^\vee$. The Thom-Porteous formula (theorem 8.3.5) applies and its determinant is 1×1:

$$[Z] = c_4(\mathrm{Sym}^3 S^\vee - \mathcal{O}) \cap [\mathrm{Gr}(2,4)] = c_4(\mathrm{Sym}^3 S^\vee) \cap [\mathrm{Gr}(2,4)]$$

the second equality because $c(\mathrm{Sym}^3 S^\vee - \mathcal{O}) = c(\mathrm{Sym}^3 S^\vee)/c(\mathcal{O}) = c(\mathrm{Sym}^3 S^\vee)$. Taking degrees (definition 1.2.4), the number of points of Z is $\deg c_4(\mathrm{Sym}^3 S^\vee) = 27$ by Step 2, and by Step 1 those points are the lines on X, as we sought to show.

The count holds for every smooth cubic surface, not merely for a general one: Step 4 verifies the simplicity of each zero using only the smoothness of X along the line in question. Nothing in the argument exhibits a line, and Step 3 shows that the nonvanishing of the Chern number is itself what forbids the empty answer. The configuration of the 27 lines, their 45 triples of coplanar lines and the incidence graph they form, is classical.

Smoothness is used twice and cannot be dropped. A cubic cone contains a one-parameter family of lines, so its zero scheme is a curve rather than 27 points, while $\deg c_4(\mathrm{Sym}^3 S^\vee)$ is still 27; the class is unchanged and it is the geometric hypothesis that has failed. This is the situation the classical enumerative tradition had no way to control.

8.4.2 Conservation of Number The principle of conservation of number asserted that the number of solutions of an enumerative problem is unchanged when the data imposing the conditions are varied continuously, provided the number of solutions remains finite. It was used to compute a count in a convenient special position and transport the answer back to the general position.

The proviso fails in two ways. A position can make the solution set infinite: the five tangency conditions of example 8.4.4 all contain the same surface of double lines, whatever the five conics are, so there is no position in which those conditions meet in a finite set. And a position can make a solution appear with multiplicity greater than one, so that the number of points drops while the number counted with multiplicity does not.

What takes its place is the pair of facts used in theorem 8.4.1. The locus imposed by a condition is a cycle class on a parameter space, and its class is independent of the datum imposing it: $[T_C] = 6h$ for every smooth conic C (proposition 8.4.3). The product of such classes is defined whether or not the loci meet properly (theorem 5.2.7), so no special position is needed to form it, and rational equivalence (definition 1.1.4) is what makes the product depend on the classes alone. The degree of the product is the number of solutions exactly when the solution set is finite and each solution occurs with multiplicity one, which is what Steps 4 and 5 of theorem 8.4.1 establish for the cubic surface. When a component of excess dimension occurs instead, theorem 5.3.9 is what accounts for its contribution to the same degree.

The conic problems are where the second failure mode is visible, and they need one computation: the class of a single tangency condition. Plane conics are parametrized by $\mathbb{P}^5 = \mathbb{P}(W)$, where $W = \mathrm{Sym}^2 V_3^\vee$ is the six-dimensional space of quadratic forms on a three-dimensional vector space V_3, and $\mathbb{P}^2 = \mathbb{P}(V_3)$; write h for the hyperplane class of $\mathbb{P}^5$. Two loci in $\mathbb{P}^5$ matter: the tangency locus of a fixed smooth conic, and the surface

$$\mathcal{V} = \{[\ell^2] \big| \ell \in V_3^\vee \setminus \{0\}\} \subseteq \mathbb{P}^5$$

of double lines. The following computes the first and shows that it contains the second. The computation assumes $\operatorname{char} \overline{k} = 0$.[1]

[1] What fails in characteristic two is the symmetric-matrix criterion of Step 1 below, not the normal form. The polar form of a ternary quadratic form is alternating in characteristic two, hence degenerate in odd dimension, so no invertible symmetric A represents a smooth conic; indeed $x^\mathsf{T} A x$ with A symmetric is there a sum of squares of the coordinates and cannot produce the monomial $x_0 x_2$ at all. The conic $\{x_0 x_2 = x_1^2\}$ is itself smooth in every characteristic, being the image of $[s : t] \mapsto [s^2 : st : t^2]$, and it remains the normal form for a smooth plane conic; what would need another proof is the passage from smoothness to that normal form. Characteristic zero is used a second time in Step 2, for the separability of the field extension induced by a dominant morphism.

Proposition 8.4.3: The Tangency Condition Is a Sextic Hypersurface

Assume char $\overline{k} = 0$. Let $C \subseteq \mathbb{P}^2$ be a smooth conic, defined by $q_C \in W$, and fix an isomorphism $\nu\colon \mathbb{P}^1 \to C$. Let $T_C \subseteq \mathbb{P}^5$ be the closure of the set of $[q]$ with q not proportional to q_C for which the degree-four divisor $\operatorname{div}(\nu^*q)$ on $\mathbb{P}^1$ is non-reduced. Since ν identifies $\operatorname{div}(\nu^*q)$ with the intersection divisor of C and the conic $\{q = 0\}$, a repeated point of that divisor is a point at which the two curves meet with intersection multiplicity at least two; so T_C is the locus of conics tangent to C. The locus T_C is an irreducible hypersurface with

$$[T_C] = 6h$$

and T_C contains the surface $\mathcal{V}$ of double lines.

***Proof*:**

Step 1: normal form and restriction to C. Writing $q_C(x) = x^\mathsf{T} A x$ with A symmetric, a singular point of C is a nonzero x with $Ax = 0$ and $q_C(x) = 0$; the first condition implies the second, so C is smooth exactly when A is invertible. Over an algebraically closed field of characteristic other than two any two invertible symmetric matrices of the same size are congruent, so after a linear change of coordinates $q_C = x_0x_2 - x_1^2$ and $\nu[s : t] = [s^2 : st : t^2]$. Restriction along ν is the linear map

$$\rho\colon W \to U, \qquad \rho(q)(s,t) = q(s^2, st, t^2)$$

where U is the five-dimensional space of binary quartics. It is surjective, since the six monomials $x_0^2, x_0x_1, x_0x_2, x_1^2, x_1x_2, x_2^2$ go to $s^4, s^3t, s^2t^2, s^2t^2, st^3, t^4$, which span U; so its kernel is one-dimensional, and it contains q_C, so it is spanned by q_C.

Step 2: the discriminant hypersurface in $\mathbb{P}(U) = \mathbb{P}^4$. Let $\Delta \subseteq \mathbb{P}^4$ be the set of $[f]$ such that f has a repeated root. A binary quartic has a repeated root exactly when it is divisible by the square of a linear form, so Δ is the image of

$$\Phi\colon \mathbb{P}^1 \times \mathbb{P}^2 \to \mathbb{P}^4, \qquad ([p],[g]) \mapsto [\ell_p^2 g]$$

where $\mathbb{P}^2 = \mathbb{P}(U_2)$ is the space of binary quadratics g and $\ell_p = p_1s - p_0t$ is the linear form vanishing at $p = [p_0 : p_1]$. The five coefficients of $\ell_p^2 g$ are bihomogeneous of bidegree $(2,1)$ in the coordinates of p and of g, and they do not vanish simultaneously, since $\ell_p^2 g \neq 0$. Hence Φ is a morphism and $\Phi^*\mathcal{O}_{\mathbb{P}^4}(1) \cong \mathcal{O}_{\mathbb{P}^1}(2) \boxtimes \mathcal{O}_{\mathbb{P}^2}(1)$, so that

$$\Phi^* h_4 = 2\theta + \eta$$

where h_4 is the hyperplane class of $\mathbb{P}^4$, θ the pullback of the class of a point of $\mathbb{P}^1$ and η the pullback of the hyperplane class of $\mathbb{P}^2$; the identification of c_1 of a tensor product with the sum is proposition 2.4.4.

The map Φ is injective over the quartics with exactly one repeated root, that root having multiplicity two: for such an f, the form ℓ is determined up to scalar as the linear form vanishing at the double root and then $g = f/\ell^2$. Such f exist, for instance $s^2t(s+t)$. Since $\mathbb{P}^1 \times \mathbb{P}^2$ is complete, the image Δ is closed, and it is irreducible as the image of an irreducible variety. In characteristic zero the function field extension induced by a dominant morphism is separable, so a morphism injective over a nonempty open subset of its image has degree one over that image and is birational onto it; hence Δ has dimension 3 and $\Phi_*[\mathbb{P}^1 \times \mathbb{P}^2] = [\Delta]$ (definition 1.2.1).

Step 3: the degree of Δ. Since $\mathbb{P}^1 \times \mathbb{P}^2 = \mathbb{P}(\mathcal{O}^{\oplus 3})$ over $\mathbb{P}^1$, the projective bundle formula (theorem 3.1.9) makes $A^*(\mathbb{P}^1 \times \mathbb{P}^2)$ free over $A^*(\mathbb{P}^1)$ on $1, \eta, \eta^2$ with $\eta^3 = 0$, the Chern classes of a trivial bundle being zero, while $\theta^2 = 0$ in $A^*(\mathbb{P}^1)$ (theorem 1.5.2) and $\theta\eta^2$ is the class of a point. Hence

$$(2\theta + \eta)^3 = \eta^3 + 6\theta\eta^2 + 12\theta^2\eta + 8\theta^3 = 6\theta\eta^2$$

By the projection formula (proposition 3.2.10) applied to Φ,

$$\deg\left(h_4^3 \cap [\Delta]\right) = \deg\left(h_4^3 \cap \Phi_*[\mathbb{P}^1 \times \mathbb{P}^2]\right) = \deg\left((\Phi^* h_4)^3 \cap [\mathbb{P}^1 \times \mathbb{P}^2]\right) = 6$$

Since $A_3(\mathbb{P}^4)$ is generated by the class of a hyperplane (theorem 1.5.2), on which h_4^3 has degree 1, this says $[\Delta]$ is 6 times that class. Writing Δ as the zero locus of an irreducible form G of degree m, the rational function G/y_0^m on $\mathbb{P}^4$ has divisor $\Delta - m\{y_0 = 0\}$, so $[\Delta]$ is m times the hyperplane class and $m = 6$.

Step 4: from Δ to T_C. Choose coordinates $y_0, \dots, y_5$ on $\mathbb{P}^5$ so that y_5 is the coordinate of q_C and $y_0, \dots, y_4$ are pulled back along ρ from coordinates on $\mathbb{P}(U)$; this is possible by Step 1. For $[q] \neq [q_C]$ the divisor $\operatorname{div}(\nu^* q)$ is non-reduced exactly when $[\rho(q)] \in \Delta$, that is, when $G(y_0, \dots, y_4) = 0$. Hence T_C is the zero locus in $\mathbb{P}^5$ of $G(y_0, \dots, y_4)$, an irreducible form of degree 6 in six variables, and the divisor computation of Step 3, run on $\mathbb{P}^5$, gives $[T_C] = 6h$.

Step 5: the double lines. Let $\ell \in V_3^\vee$ be nonzero. The line $\{\ell = 0\}$ does not contain the irreducible conic C, so $\nu^*\ell$ is a nonzero binary quadratic and $\nu^*(\ell^2) = (\nu^*\ell)^2$ is a nonzero binary quartic every root of which is repeated. Also $[\ell^2] \neq [q_C]$, since q_C is irreducible and ℓ^2 is not. So $[\ell^2] \in T_C$, and as ℓ varies $\mathcal{V} \subseteq T_C$, completing the proof.

The proposition supplies the invariance of the class and the containment $\mathcal{V} \subseteq T_C$. The class $6h$ is the same for every smooth conic C, which is the invariance the classical principle wanted, and it is here a theorem about classes rather than an assumption about counts. The containment $\mathcal{V} \subseteq T_C$ is the obstruction: the double lines are conics in the sense of $\mathbb{P}^5$, they meet every conic in a non-reduced divisor, and they cannot be removed by moving the five conics into general position, since the containment holds for all of them.

Example 8.4.4: The Five-Conic Problem

Let $C_1, \dots, C_5$ be smooth conics in $\mathbb{P}^2$, over a field of characteristic zero, and ask for the conics tangent to all five. Each condition is the hypersurface T_{C_i} of proposition 8.4.3, of class $6h$, and the product of the five classes has degree

$$\deg\left((6h)^5\right) = 6^5 \deg(h^5) = 7776$$

This is the number Steiner published in 1848.

It is not the number of conics tangent to five general conics. By proposition 8.4.3 each T_{C_i} contains the surface $\mathcal{V}$ of double lines, so

$$\mathcal{V} \subseteq T_{C_1} \cap \dots \cap T_{C_5}$$

and the intersection has a component of dimension 2 where the expected dimension is 0, for every choice of the five conics. The number 7776 is the degree of a product of five classes whose loci do not meet in a finite set; theorem 5.3.9 is what governs the contribution the excess component makes to that degree, and the hypotheses of theorem 8.4.1, Step 5, on which the passage from a degree to a count depends, are not available here.

The remedy is a different parameter space. A double line ℓ^2 records the line ℓ but no tangency data, and the conics tangent to the C_i degenerate onto $\mathcal{V}$ in a way that remembers more; the space of complete conics, the blow-up of $\mathbb{P}^5$ along $\mathcal{V}$, is the parameter space on which the five tangency conditions do meet in a finite reduced set. Its count is 3264, the number Chasles obtained in 1864. That construction is moduli theory and is not carried out here; the space of complete conics and the verification of the count are in [8].

Exercise 8.4.1

1. The lines contained in a quadric surface $X \subseteq \mathbb{P}^3$ defined by $F \in \operatorname{Sym}^2 V^\vee$ are the zeros of the section σ_F of $\operatorname{Sym}^2 S^\vee$ on $\operatorname{Gr}(2,4)$, a bundle of rank 3, so they are expected to form a curve rather than a finite set. Compute $c_3(\operatorname{Sym}^2 S^\vee)$ in the Schubert basis by the method of Step 2 of theorem 8.4.1. Then, assuming X smooth and taking for granted that its lines form two rulings, each meeting a general line of $\mathbb{P}^3$ in as many members as that line meets X, compute the class of a single ruling in $A_1(\operatorname{Gr}(2,4))$ and check that the two rulings account for $c_3(\operatorname{Sym}^2 S^\vee)$.

2. Lines in $\mathbb{P}^4$ contained in a quintic hypersurface are the zeros of a section of $\operatorname{Sym}^5 S^\vee$ on $\operatorname{Gr}(2,5)$, a bundle whose rank equals $\dim \operatorname{Gr}(2,5) = 6$. Show that
$$\deg c_6(\operatorname{Sym}^5 S^\vee) = 2875$$
Use the splitting principle to express c_6 in $c_1(S)$ and $c_2(S)$, then proposition 8.1.5 and theorem 8.2.1 to rewrite the answer in the Schubert basis of $\operatorname{Gr}(2,5)$, and proposition 8.1.3 to take degrees in the 2×3 box.

3. Let $d \geq 1$ and $N = \binom{d+3}{3} - 1$, so that $\mathbb{P}^N$ parametrizes the surfaces of degree d in $\mathbb{P}^3$. Show that $\operatorname{Sym}^d S^\vee$ has rank $d+1$ and that the restriction map $H^0(\mathbb{P}^3, \mathcal{O}(d)) \to H^0(\ell, \mathcal{O}(d))$ is surjective for every line ℓ. Deduce that the incidence variety $I = \{(\ell, Y) \big| \ell \subseteq Y\} \subseteq \operatorname{Gr}(2,4) \times \mathbb{P}^N$ is a projective bundle over $\operatorname{Gr}(2,4)$ with fibre $\mathbb{P}^{N-d-1}$, so that $\dim I = N + 3 - d$. Conclude that for $d \geq 4$ the image of I in $\mathbb{P}^N$ is a proper closed subset, so a general surface of degree at least 4 in $\mathbb{P}^3$ contains no line.

4. Let F be a cubic form in x_0, x_1, x_2 alone, defining a cone $X \subseteq \mathbb{P}^3$ with vertex $v = [0:0:0:1]$ over a plane cubic curve. Show that X is singular at v, and that the lines joining v to the points of the plane cubic all lie on X, so that the zero scheme of σ_F contains a curve. Which step of the proof of theorem 8.4.1 fails, and what does the number $\deg c_4(\operatorname{Sym}^3 S^\vee) = 27$ compute in this case?

5. Assume $\operatorname{char} \overline{k} \neq 3$ and let $X \subseteq \mathbb{P}^3$ be the surface $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$. Show that X is smooth. For each of the three ways of splitting $\{0,1,2,3\}$ into two pairs and each choice of ζ, ξ with $\zeta^3 = \xi^3 = -1$, write down a line on X; show that the 27 lines so obtained are pairwise distinct, and explain why theorem 8.4.1 shows that X has no others.

6. Assume $\operatorname{char} \overline{k} = 0$ and fix a line $L \subseteq \mathbb{P}^2$. Let $T_L \subseteq \mathbb{P}^5$ be the closure of the set of $[q]$ with q not divisible by the equation of L for which the degree-two divisor $q|_L$ on L is non-reduced. Show that $[T_L] = 2h$, by restricting to L as in Step 1 of proposition 8.4.3 and using the discriminant of a binary quadratic. Show that every double line $[\ell^2]$ with ℓ not proportional to the equation of L lies in T_L, hence $\mathcal{V} \subseteq T_L$, and conclude that $2^5 = 32$ is not the number of conics tangent to five general lines.

7. Let $p_1, \ldots, p_4 \in \mathbb{P}^2$ be four points, no three collinear. Show that the conics through all four form a line $P \subseteq \mathbb{P}^5$, that $P \cap \mathcal{V} = \emptyset$, and that $\deg([T_C] \cap [P]) = 6$ for every smooth conic C. So the count of conics through four general points and tangent to a general conic is not obstructed by the double lines, unlike the count in example 8.4.4.

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Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Abelian Varieties*. 1st ed. 2026. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_abelian_varieties.pdf.

[2] Nathanael Chwojko-Srawley. *Everything You Need to Know about Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic K-Theory*. 1st ed.

[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Commutative Algebra*.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Homological Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_homological_algebra.pdf.

[6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.

[7] Nathanael Chwojko-Srawley. *EYNTKA Classical Algebraic Geometry*.

[8] William Fulton. *Intersection Theory*. Springer-Verlag, 1984. 492 pp. ISBN: 978-0-387-12176-5.

[9] Robin Hartshorne. *Algebraic Geometry*. Graduate Texts in Mathematics 52. New York: Springer-Verlag, 1977, p. 496. ISBN: 0-387-90244-9.

[10] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology*. 1st ed. 126 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.

[11] Ravi Vakil. “INTERSECTION THEORY CLASS 9”. In: ().